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Tamura et al.

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(54) **MICROORGANISM IDENTIFICATION METHOD**

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(51) **Int. Cl.**

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C12Q 1/04	(2006.01)
G01N 27/64	(2006.01)
G01N 33/68	(2006.01)

(52) **U.S. Cl.**

CPC ***G01N 33/6851*** (2013.01); ***C12Q 1/04*** (2013.01); ***G01N 27/64*** (2013.01); *G01N 2333/255* (2013.01)

(58) **Field of Classification Search**

CPC . G01N 33/6851; G01N 2333/255; C12Q 1/04

See application file for complete search history.

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(57) **ABSTRACT**

Provided is a method of identifying a serovar of *Salmonella* bacteria including a step of subjecting a sample containing microorganisms to mass spectrometry to obtain a mass spectrum, a step of reading a mass-to-charge ratio m/z of a peak derived from a marker protein from the mass spectrum, and an identification step of identifying a serovar of *Salmonella* bacteria in the sample, based on the mass-to-charge ratio m/z, wherein the serovars of *Salmonella* bacteria are classified using cluster analysis using as an index the mass-to-charge ratio m/z derived from at least 12 types of ribosomal proteins S8, L15, L17, L21, L25, S7, SODa, peptidylprolyl isomerase, gns, YibT, YaiA and YciF as the marker proteins.

2 Claims, 80 Drawing Sheets

Specification includes a Sequence Listing.

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Fig. 1

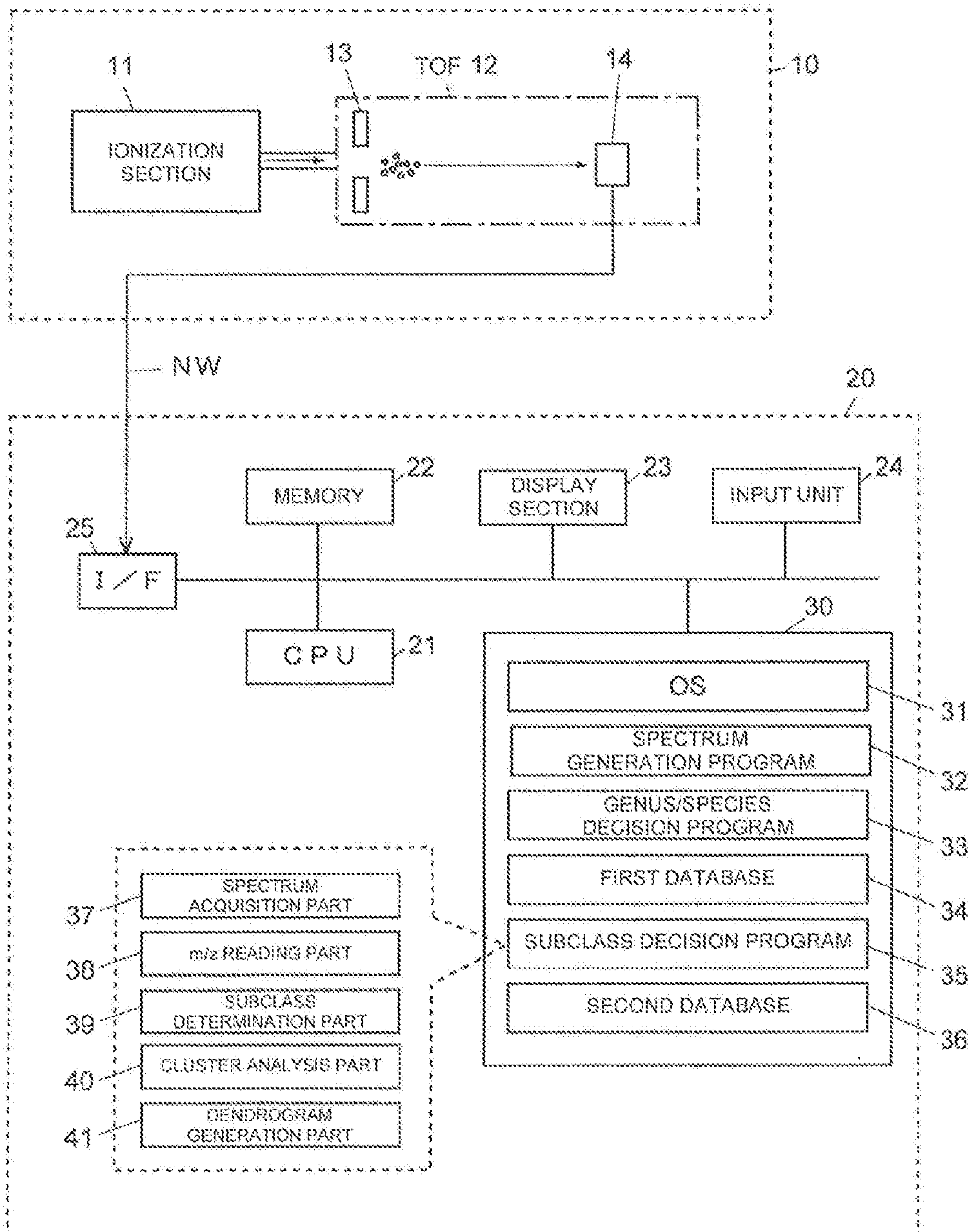

11
13
TOF 12
14
10
IONIZATION SECTION
NW
20
MEMORY
22
DISPLAY SECTION
23
INPUT UNIT
24
25
I / F
CPU
21
30
OS
31
SPECTRUM GENERATION PROGRAM
32
GENUS/SPECIES DECISION PROGRAM
33
FIRST DATABASE
34
SUBCLASS DECISION PROGRAM
35
SECOND DATABASE
36
37
SPECTRUM ACQUISITION PART
38
m/z READING PART
39
SUBCLASS DETERMINATION PART
40
CLUSTER ANALYSIS PART
41
DENDROGRAM GENERATION PART

Fig. 2

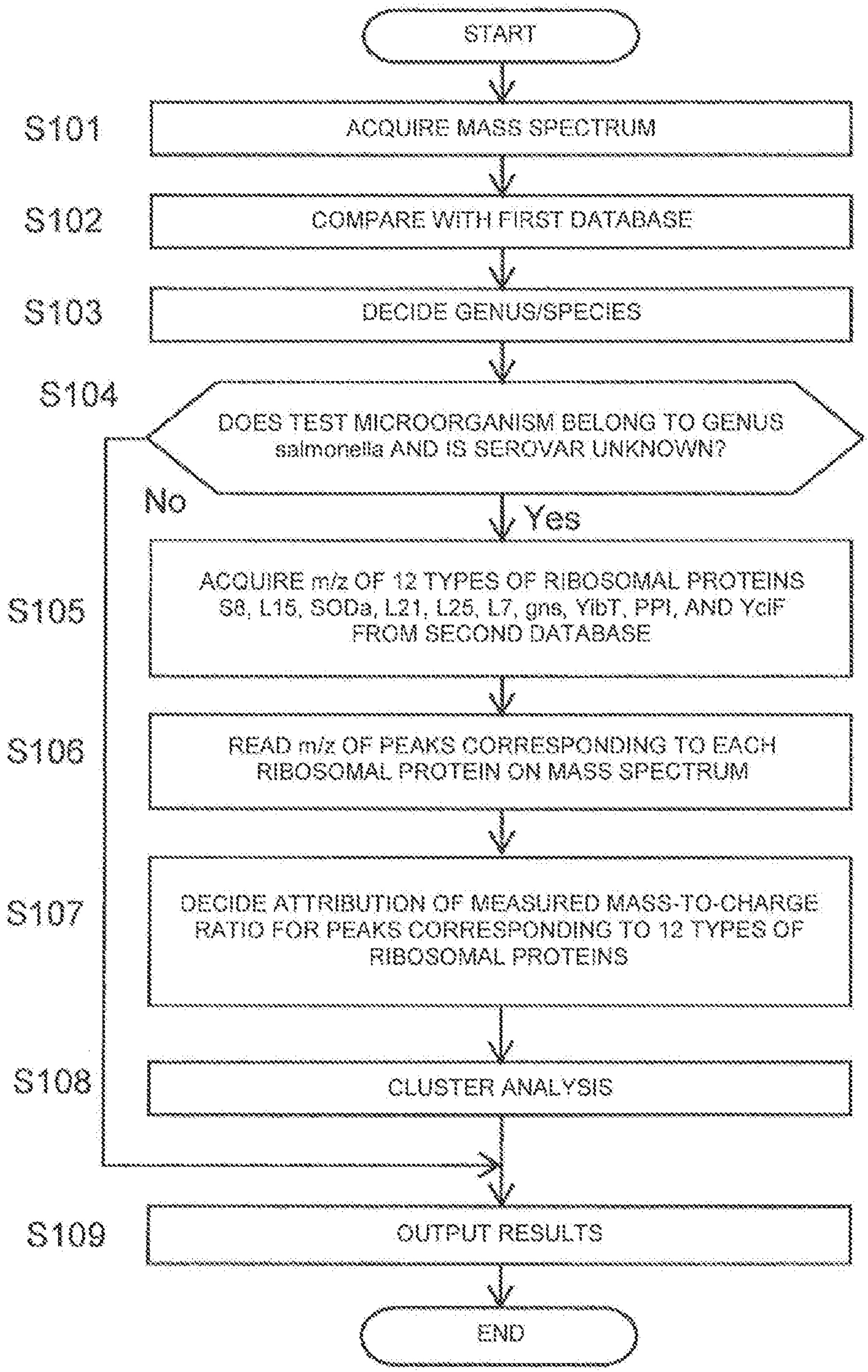

START
S101
ACQUIRE MASS SPECTRUM
S102
COMPARE WITH FIRST DATABASE
S103
DECIDE GENUS/SPECIES
S104
DOES TEST MICROORGANISM BELONG TO GENUS salmonella AND IS SEROVAR UNKNOWN?
No
Yes
S105
ACQUIRE m/z OF 12 TYPES OF RIBOSOMAL PROTEINS S8, L15, SODa, L21, L25, L7, gns, YibT, PPI, AND YciF FROM SECOND DATABASE
S106
READ m/z OF PEAKS CORRESPONDING TO EACH RIBOSOMAL PROTEIN ON MASS SPECTRUM
S107
DECIDE ATTRIBUTION OF MEASURED MASS-TO-CHARGE RATIO FOR PEAKS CORRESPONDING TO 12 TYPES OF RIBOSOMAL PROTEINS
S108
CLUSTER ANALYSIS
S109
OUTPUT RESULTS
END

Fig. 3

No.	Genus	Species	Subspecies	Serovar	Agglutinated O serum	Name of strain	Acquisition source
1	*Salmonella*	*enterica*	*enterica*	Typhimurium	O4	NBRC13245T	NBRC
2	*Salmonella*	*enterica*	*enterica*	Enteritidis	O9	GTC00131	NBRP
3	*Salmonella*	*enterica*	*enterica*	Enteritidis	O9	GTC03491	NBRP
4	*Salmonella*	*enterica*	*enterica*	Enteritidis	O9	GTC03838	NBRP
5	*Salmonella*	*enterica*	*enterica*	Enteritidis	O9	GTC08914	NBRP
6	*Salmonella*	*enterica*	*enterica*	Enteritidis	O9	GTC09421	NBRP
7	*Salmonella*	*enterica*	*enterica*	Enteritidis	O9	GTC09488	NBRP
8	*Salmonella*	*enterica*	*enterica*	Gallinarum Pullorum	O9	NBRC3163	NBRC
9	*Salmonella*	*enterica*	*enterica*	Enteritidis	O9	NBRC3313	NBRC
10	*Salmonella*	*enterica*	*enterica*	Typhimurium	O4	NBRC12529	NBRC
11	*Salmonella*	*enterica*	*enterica*	Typhimurium	O4	NBRC14193	NBRC
12	*Salmonella*	*enterica*	*enterica*	Typhimurium	O4	NBRC14194	NBRC
13	*Salmonella*	*enterica*	*enterica*	Typhimurium	O4	NBRC14209	NBRC
14	*Salmonella*	*enterica*	*enterica*	Typhimurium	O4	NBRC14210	NBRC
15	*Salmonella*	*enterica*	*enterica*	Typhimurium	O4	NBRC14211	NBRC
16	*Salmonella*	*enterica*	*enterica*	Typhimurium	O4	NBRC14212	NBRC
17	*Salmonella*	*enterica*	*enterica*	Typhimurium	O4	NBRC15181	NBRC
18	*Salmonella*	*enterica*	*enterica*	Minnesota	O21	NBRC15182	NBRC
19	*Salmonella*	*enterica*	*enterica*	Minnesota	O21	NBRC15183	NBRC
20	*Salmonella*	*enterica*	*enterica*	Minnesota	O21	NBRC15184	NBRC
21	*Salmonella*	*enterica*	*enterica*	Minnesota	O21	NBRC15185	NBRC
22	*Salmonella*	*enterica*	*enterica*	Minnesota	O21	NBRC15186	NBRC
23	*Salmonella*	*enterica*	*enterica*	Minnesota	O21	NBRC15187	NBRC
24	*Salmonella*	*enterica*	*enterica*	Abony	O4	NBRC100797	NBRC
25	*Salmonella*	*enterica*	*enterica*	Choleraesuis	O7	NBRC105684	NBRC
26	*Salmonella*	*enterica*	*enterica*	Typhimurium	O4	NBRC105726	NBRC
27	*Salmonella*	*enterica*	*enterica*	UN_O7	O7	JCM3819	JCM
28	*Salmonella*	*enterica*	*enterica*	Minnesota	O21	NBRC15335	NBRC
29	*Salmonella*	*enterica*	*enterica*	Enteritidis	O9	GTC03490	NBRP
30	*Salmonella*	*enterica*	*enterica*	Braenderup	O7	GTC03492	NBRP
31	*Salmonella*	*enterica*	*enterica*	Pakistan	O8	GTC03493	NBRP
32	*Salmonella*	*enterica*	*enterica*	Typhimurium*	O4	GTC09548	NBRP
33	*Salmonella*	*enterica*	*enterica*	Infantis	O7	ATCC BAA-1675	ATCC
34	*Salmonella*	*enterica*	*enterica*	Thompson	O7	ATCC BAA-1738	ATCC
35	*Salmonella*	*enterica*	*enterica*	Saintpaul	O4	ATCC 9712	ATCC
36	*Salmonella*	*enterica*	*enterica*	Infantis	O7	jfrlSe1402-1	Japan Food Research Laboratories
38	*Salmonella*	*enterica*	*enterica*	Brandenburg	O4	jfrlSe1402-3	Japan Food Research Laboratories
39	*Salmonella*	*enterica*	*enterica*	Infantis	O7	jfrlSe1402-4	Japan Food Research Laboratories
40	*Salmonella*	*enterica*	*enterica*	Brandenburg	O4	jfrlSe1402-5	Japan Food Research Laboratories
41	*Salmonella*	*enterica*	*enterica*	Rissen	O7	jfrlSe1402-6	Japan Food Research Laboratories
42	*Salmonella*	*enterica*	*enterica*	Orion	O3,10	jfrlSe1402-7	Japan Food Research Laboratories
43	*Salmonella*	*enterica*	*enterica*	Rissen	O7	jfrlSe1402-8	Japan Food Research Laboratories
44	*Salmonella*	*enterica*	*enterica*	Rissen	O7	jfrlSe1402-9	Japan Food Research Laboratories
45	*Salmonella*	*enterica*	*enterica*	Rissen	O7	jfrlSe1402-10	Japan Food Research Laboratories
46	*Salmonella*	*enterica*	*enterica*	Rissen	O7	jfrlSe1402-11	Japan Food Research Laboratories
47	*Salmonella*	*enterica*	*enterica*	Rissen	O7	jfrlSe1402-12	Japan Food Research Laboratories
48	*Salmonella*	*enterica*	*enterica*	Mbandaka	O7	jfrlSe1402-13	Japan Food Research Laboratories
49	*Salmonella*	*enterica*	*enterica*	Mbandaka	O7	jfrlSe1402-14	Japan Food Research Laboratories
50	*Salmonella*	*enterica*	*enterica*	Orion	O3,10	jfrlSe1402-15	Japan Food Research Laboratories
51	*Salmonella*	*enterica*	*enterica*	Altona	O8	jfrlSe1409-1	Japan Food Research Laboratories
52	*Salmonella*	*enterica*	*enterica*	Istanbul	O8	jfrlSe1409-2	Japan Food Research Laboratories
53	*Salmonella*	*enterica*	*enterica*	Senftenberg	O1,3,19	jfrlSe1409-3	Japan Food Research Laboratories
54	*Salmonella*	*enterica*	*enterica*	UN_O13	O13	jfrlSe1409-4	Japan Food Research Laboratories
55	*Salmonella*	*enterica*	*enterica*	UN_O1,3,19	O1,3,19	jfrlSe1409-5	Japan Food Research Laboratories
56	*Salmonella*	*enterica*	*enterica*	Montevideo	O7	jfrlSe1409-6	Japan Food Research Laboratories
57	*Salmonella*	*enterica*	*enterica*	UN_O1,3,19	O1,3,19	jfrlSe1409-7	Japan Food Research Laboratories
58	*Salmonella*	*enterica*	*enterica*	UN_O18	O18	jfrlSe1409-8	Japan Food Research Laboratories
61	*Salmonella*	*enterica*	*enterica*	UN_O1,3,19	O1,3,19	jfrlSe1409-11	Japan Food Research Laboratories
67	*Salmonella*	*enterica*	*enterica*	UN_O7	O7	jfrlSe1409-17	Japan Food Research Laboratories
71	*Salmonella*	*enterica*	*enterica*	Amsterdam	O3,10	jfrlSe1409-21	Japan Food Research Laboratories
80_HySe09	*Salmonella*	*enterica*	*enterica*	Manhattan O6,8:d:1,5	O8	Hyogo SO11001	Hyogo Prefectural Institute of Public Health Science
100_HySe29	*Salmonella*	*enterica*	*enterica*	Schwarzengrund O4:d:1,7	O4	Hyogo SO12004	Hyogo Prefectural Institute of Public Health Science
103_HySe32	*Salmonella*	*enterica*	*enterica*	Schwarzengrund O4:d:1,7	O4	Hyogo SO13003	Hyogo Prefectural Institute of Public Health Science
106_HySe35	*Salmonella*	*enterica*	*enterica*	Schwarzengrund O4:d:1,7	O4	Hyogo SO14002	Hyogo Prefectural Institute of Public Health Science

ATCC: American Type Culture Collection
JCM: Riken Microbe Division / Japan Collection of Microorganisms
NBRC : Biological Resource Center, NITE
NBRP: National Bioresource Project

Fig. 4

Agglutinated_O serum	Serovar	Number of strains
O1,3,19	Senftenberg	1
O1,3,19	UN	3
O3,10	Amsterdam	1
O3,10	Orion	2
O4	Abony	1
O4	Brandenburg	2
O4	Saintpaul	1
O4	Schwarzengrund	3
O4	Typhimurium	10
O4	UN	1
O7	Braenderup	1
O7	Choleraesuis	1
O7	Infantis	3
O7	Mbandaka	2
O7	Montevideo	1
O7	Rissen	6
O7	Thompson	1
O7	UN	2
O8	Altona	1
O8	Istanbul	1
O8	Manhattan	1
O8	Pakistan	1
O9	Enteritidis	8
O9	Gallinarum Pullorum	1
O13	UN	1
O18	UN	1
O21	Minnesota	7 (Confirmed by PCR)
	22 serovars	64 strains

Fig. 5

Name	Sequence (5'-3')	Use
Ec-S10-F	AAGAACGGTTACACTCTCCC (SEQ ID NO:1)	Amplification and sequence analysis of S10 region
Se-S10-R	ATGTCGCCTACGCCTGCGTAGCG (SEQ ID NO:2)	Amplification and sequence analysis of S10 region
EcW3110-S10-10	GCTGGCATGATTCGTGAAGAACG (SEQ ID NO:3)	Sequence analysis of S10 region
EcW3110-S10-3	TGCTGAAGTAACTGGTTCCGG (SEQ ID NO:4)	Sequence analysis of S10 region
EcW3110-S10-5	CATAACGTAGAAATGAAACCAGG (SEQ ID NO:5)	Sequence analysis of S10 region
EcW3110-S10-8	AGCGTCGCTGATGTTACAAC (SEQ ID NO:6)	Sequence analysis of S10 region
Se-S10-1	ATCAATCGTAATGGGTGTGAG (SEQ ID NO:7)	Sequence analysis of S10 region
Se-S10-2	AAGCCGGAAGCGGGTCACTTTG (SEQ ID NO:8)	Sequence analysis of S10 region
Se-S10-3r	CTTTTTGACTGTGGTCCTGC (SEQ ID NO:9)	Sequence analysis of S10 region
Se-S10-5f	CGTTCTCTCAAGAAAGGTCC (SEQ ID NO:10)	Sequence analysis of S10 region
Se-S10-6r	GCCGATTACACCATAAGTGG (SEQ ID NO:11)	Sequence analysis of S10 region
Se-S10-7r	TAGCACGACGTGCTGCTTCGAT (SEQ ID NO:12)	Sequence analysis of S10 region
Se-S10-9	TCCGTACTCTGCAAGGTCGCGTG (SEQ ID NO:13)	Sequence analysis of S10 region
Se-S10-11	ACTCCGGTGCACGTCGCGTAATG (SEQ ID NO:14)	Amplification and sequence analysis of spc region
EcW3110-spc-R	AGCAGTCTGCGTTTCAGCTC (SEQ ID NO:15)	Amplification and amplification of sequence analysis region of spc region
EcW3110-spc-2	ATTGTTGAAGGTATCAACCTG (SEQ ID NO:16)	Sequence analysis of spc region
EcW3110-spc-2r	TTACCGGTTAACACGATAAC (SEQ ID NO:17)	Sequence analysis of spc region
EcW3110-spc-3	TCGTGGTAACTACAGCATG (SEQ ID NO:18)	Sequence analysis of spc region
EcW3110-spc-4	ACCATGCCTTCCTCCAAG CT (SEQ ID NO:19)	Sequence analysis of spc region
EcW3110-spc-6	ATGCTGCCCGTGAAGCTGGC (SEQ ID NO:20)	Sequence analysis of spc region
EcW3110-spc-7	ATCGGTCGTCTGCCGAAACAC (SEQ ID NO:21)	Sequence analysis of spc region and amplification of alpha region
EcW3110-spc-9	GTCACCATGCCTTCCTCCAAG (SEQ ID NO:22)	Sequence analysis of spc region
Se-spc-1	AATGTGTATCAAGGTTCTGGG (SEQ ID NO:23)	Sequence analysis of spc region
Se-spc-8	CTTCACTAAGAAGCTGCAGCTG (SEQ ID NO:24)	Sequence analysis of spc region
Se-spc-3r2	ATATCCAAACCACGAACGCG (SEQ ID NO:25)	Sequence analysis of spc region
Se-spc-8r	GATGTGAGCCATCTTACACCTCT (SEQ ID NO:26)	Sequence analysis of spc region
EcW3110-alpha-5	TGCGGACATTAACGAACACCTG (SEQ ID NO:27)	Sequence analysis of alpha region
EcW3110-alpha-6	TGCCTACAATGTTGAAGCAGCG (SEQ ID NO:28)	Sequence analysis of alpha region
Se_alpha-7	AATCATCAAGACGACCCTGCC (SEQ ID NO:29)	Sequence analysis of alpha region
Se-al-3r	CGCGATAGCAACCAAGATCCA (SEQ ID NO:30)	Sequence analysis of alpha region
Se-al-4r	CGCGAACGCCAGACTTAAGGA (SEQ ID NO:31)	Sequence analysis of alpha region
Se-al-5r	ACTCTGTCACAGAACCCTGCA (SEQ ID NO:32)	Sequence analysis of alpha region
Se-alpha-2r	TGTACTGGCTGATCTTACGACG (SEQ ID NO:33)	Sequence analysis of alpha region
Se-alpha-8	GGTTCATCCAACGCTGGCAG (SEQ ID NO:34)	Sequence analysis of alpha region
Se-al-R2	ATATGACGCTCTGCCCACTG (SEQ ID NO:35)	Amplification and sequence analysis of alpha region
Se-SODa-F	GCCGTAACGTTTATAACCCTGG (SEQ ID NO:36)	Amplification of SOD gene
Se-SODa-R	ACACGTCAACCGGATAATGCA (SEQ ID NO:37)	Amplification of SOD gene
Se-SODa-1f	ATGAACCAACTGCTTACGC (SEQ ID NO:38)	Sequence analysis of SOD gene
Se-L21-F	CGTTACCGTATGCGTTGTGT (SEQ ID NO:39)	Amplification of L21 gene
Se-L21-R	GCACAGCCACACGGGTATAT (SEQ ID NO:40)	Amplification of L21 gene
Se-L21-1f	CACTTGTCAAAGCCGTGCAA (SEQ ID NO:41)	Sequence analysis of L21 gene
Se-L21-1r	GTAAAAAGCCCCGCAACAGG (SEQ ID NO:42)	Sequence analysis of L21 gene
Se-S7-F	GCGATTGAAGGTAACCGCTT (SEQ ID NO:43)	Amplification and sequence analysis of S7 gene
Se-S7-R	ATGTGCGCACTGATACCGAT (SEQ ID NO:44)	Amplification and sequence analysis of S7 gene
Se-S7-f	GTCGTCAGGGTTGACTATACT (SEQ ID NO:45)	Amplification and sequence analysis of S7 gene
Se-STM1513-F	GCACAGGGCGACTAGATTTAG (SEQ ID NO:46)	Amplification and sequence analysis of STM1513 gene
Se-STM1513-R	GCCCTCGCTGAATTGCTTTTTC (SEQ ID NO:47)	Amplification and sequence analysis of STM1513 gene
Se-gns-2-F	CATGACGACACTGTCTTATTGC (SEQ ID NO:48)	Amplification and sequence analysis of gns gene
Se-gns-2-R	TCGGTAAACCAGTCACCACT (SEQ ID NO:49)	Amplification and sequence analysis of gns gene
Se-YbT-F	TAAACTCAAATAAGCGRCCGCG (SEQ ID NO:50)	Amplification and sequence analysis of YbT gene
Se-YbT-R	TACGCCATGCAAATTCAGCGC (SEQ ID NO:51)	Amplification and sequence analysis of YbT gene
Se-ppi-F	AAAATCAAGCAGACGATGTAGGC (SEQ ID NO:52)	Amplification and sequence analysis of peptidylprolyl isomerase
Se-ppi-R	GTCTGACGGCCTAAGGAGTGAT (SEQ ID NO:53)	Amplification and sequence analysis of peptidylprolyl isomerase

Fig. 6

Amino acid	Mass
A	71.079
R	156.188
N	114.103
D	115.088
C	103.145
Q	128.13
E	129.114
G	57.052
H	137.141
I	113.159
L	113.159
K	128.174
M	131.198
F	147.176
P	97.116
S	87.078
T	101.104
W	186.213
Y	163.175
V	99.132

Fig. 7A

	O_antigen		1	2	3	4	5	6	7	8	9	10
			Typhimurium	Enteritidis	Enteritidis	Enteritidis	Enteritidis	Enteritidis	Enteritidis	Pullorum	Enteritidis	Typhimurium
	protein	Attr.	NBRC 13245T	GTC 00131	GTC 03491	GTC 03878	GTC 06914	GTC 06421	GTC 03466	NBRC 3163	NBRC 3313	NBRC 12529
	S10	O	11787.58	11787.58	11787.58	11787.58	11787.58	11787.58	11787.58	11787.58	11787.58	11787.58
	L3	X	22248.48	22248.48	22248.48	22248.48	22248.48	22248.48	22248.48	22248.48	22248.48	22248.48
	L4	X	22087.49	22087.49	22087.49	22087.49	22087.49	22087.49	22087.49	22087.49	22087.49	22087.49
	L23	Δ	11214.13	11214.13	11214.13	11214.13	11214.13	11214.13	11214.13	11214.13	11214.13	11214.13
	L2	X	29690.16	29690.16	29690.16	29690.16	29690.16	29690.16	29690.16	29690.16	29690.16	29690.16
S10	S19	O	10286.07	10286.07	10286.07	10286.07	10286.07	10286.07	10286.07	10286.07	10286.07	10286.07
	L22	O	12227.29	12227.29	12227.29	12227.29	12227.29	12227.29	12227.29	12227.29	12227.29	12227.29
	S3	X	25853.00	25853.00	25853.00	25853.00	25853.00	25853.00	25853.00	25853.00	25853.00	25853.00
	L18	X	15195.13	15195.13	15195.13	15195.13	15195.13	15195.13	15195.13	[illegible]	15195.13	15195.13
	L29	O	7261.45	7261.45	7261.45	7261.45	7261.45	7261.45	7261.45	7261.45	7261.45	7261.45
	S17	X	9592.29	9592.29	9592.29	9592.29	9592.29	9592.29	9592.29	9592.29	9592.29	9592.29
	L14	O	13569.05	13569.05	13569.05	13569.05	13569.05	13569.05	13569.05	13569.05	13569.05	13569.05
	L24	X	11186.00	11186.00	11186.00	11186.00	11186.00	11186.00	11186.00	11186.00	11186.00	11186.00
	L5	X	20187.32	20187.32	20187.32	20187.32	20187.32	20187.32	20187.32	20187.32	20187.32	20187.32
	S14	O	11478.35	11478.35	11478.35	11478.35	11478.35	11478.35	11478.35	11478.35	11478.35	11478.35
	S8	O	13998.36	13998.36	13998.36	13998.36	13998.36	13998.36	13998.36	13998.36	13998.36	13998.36
spc	L6	X	18729.50	18729.50	18729.50	18729.50	18729.50	18729.50	18729.50	18729.50	18729.50	18729.50
	L19	O	12770.66	12770.66	12770.66	12770.66	12770.66	12770.66	12770.66	12770.66	12770.66	12770.66
	S5	Δ	17473.18	17473.18	17473.18	17473.18	17473.18	17473.18	17473.18	17468.15	17473.18	17473.18
	L30	O	8383.55	8383.55	8383.55	8383.55	8383.55	8383.55	8383.55	8383.55	8383.55	8383.55
	L15	O	14967.36	14967.36	14967.36	14967.36	14967.36	14967.36	14967.36	14967.36	14967.36	14967.36
	L36	O	4395.35	4395.35	4395.35	4395.35	4395.35	4395.35	4395.35	4395.35	4395.35	4395.35
alpha	S13	O	13031.26	13031.26	13031.26	13031.26	13031.26	13031.26	13031.26	13031.26	13031.26	13031.26
+Mc	S11	O	13700.71	13700.71	13700.71	13700.71	13700.71	13700.71	13700.71	13700.71	13700.71	13700.71
	S4	O	23354.87	23354.87	23354.87	23354.87	23354.87	23354.87	23354.87	23354.87	23354.87	23354.87
	L17	O	14395.61	14395.61	14395.61	14395.61	14395.61	14395.61	14395.61	14381.59	14395.61	14395.61
	SODa	O	22348.92	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	22348.92
	L21	O	11579.36	11579.36	11579.36	11579.36	11579.36	11579.36	11579.36	11579.36	11579.36	11579.36
	L25	O	10542.19	10542.19	10542.19	10542.19	10542.19	10542.19	10542.19	10542.19	10542.19	10542.19
	S7	O	17460.15	17474.18	17474.18	17474.18	17474.18	17474.18	17474.18	17474.18	17474.18	17460.15
	pts	Δ	8483.508				8483.508	8483.508		8483.508		
	YbT	Δ	7993.051				7993.051	7993.051		7993.051		
	Peptidylpro	O	10198.068				10198.068	10198.068		10198.068		
	YaiA	Δ	7096.821				[illegible]	[illegible]		[illegible]	[illegible]	7096.821
	YaiF	Δ	18653.164				18643.126	18643.126				18653.164

Fig. 7B

11	12	13	14	15	16	17	18	19	20
Typhimurium	Typhimurium	Typhimurium	Typhimurium	Typhimurium	Typhimurium	Typhimurium	Minnesota	Minnesota	Minnesota
NBRC 14193	NBRC 14194	NBRC 14209	NBRC 14210	NBRC 14211	NBRC 14212	NBRC 15181	NBRC 15182	NBRC 15183	NBRC 15184
11787.58	11787.58	11787.58	11787.58	11787.58	11787.58	11787.58	11787.58	11787.58	11787.58
22248.48	22248.48	22248.48	22248.48	22248.48	22248.48	22248.48	22248.48	22248.48	22248.48
22087.492	22087.49	22087.49	22087.49	22087.49	22087.49	22087.49	22087.49	22087.49	22087.49
11214.13	11214.13	11214.13	11214.13	11214.13	11214.13	11214.13	11214.13	11214.13	11214.13
29690.16	29690.16	29690.16	29690.16	29690.16	29690.16	29690.16	29690.16	29690.16	29690.16
10286.07	10286.07	10286.07	10286.07	10286.07	10286.07	10286.07	10286.07	10286.07	10286.07
12227.29	12227.29	12227.29	12227.29	12227.29	12227.29	12227.29	12227.29	12227.29	12227.29
25853.00	25853.00	25853.00	25853.00	25853.00	25853.00	25853.00	25853.00	25853.00	25853.00
15195.13	15195.13	15195.13	15195.13	15195.13	15195.13	15195.13	15195.13	15195.13	15195.13
7261.45	7261.45	7261.45	7261.45	7261.45	7261.45	7261.45	7261.45	7261.45	7261.45
9592.29	9592.29	9592.29	9592.29	9592.29	9592.29	9592.29	9592.29	9592.29	9592.29
13569.05	13569.05	13569.05	13569.05	13569.05	13569.05	13569.05	13569.05	13569.05	13569.05
11186	11186.00	11186.00	11186.00	11186.00	11186.00	11186.00	11186.00	11186.00	11186.00
20187.32	20187.32	20187.32	20187.32	20187.32	20187.32	20187.32	20187.32	20187.32	20187.32
11478.35	11478.35	11478.35	11478.35	11478.35	11478.35	11478.35	11478.35	11478.35	11478.35
13998.36	13998.36	13998.36	13998.36	13998.36	13998.36	13998.36	13998.36	13998.36	13998.36
18729.5	18729.50	18729.50	18729.50	18729.50	18729.50	18729.50	18729.50	18729.50	18729.50
12770.66	12770.66	12770.66	12770.66	12770.66	12770.66	12770.66	12770.66	12770.66	12770.66
17473.18	17473.18	17473.18	17473.18	17473.18	17473.18	17473.18	17473.18	17473.18	17473.18
8383.55	8383.55	8383.55	8383.55	8383.55	8383.55	8383.55	8383.55	8383.55	8383.55
14967.36	14967.36	14967.36	14967.36	14967.36	14967.36	14967.36	14967.36	14967.36	14967.36
4395.35	4395.35	4395.35	4395.35	4395.35	4395.35	4395.35	4395.35	4395.35	4395.35
13031.26	13031.26	13031.26	13031.26	13031.26	13031.26	13031.26	13031.26	13031.26	13031.26
13700.71	13700.71	13700.71	13700.71	13700.71	13700.71	13700.71	13700.71	13700.71	13700.71
23354.87	23354.87	23354.87	23354.87	23354.87	23354.87	23354.87	23354.87	23354.87	23354.87
14395.61	14395.61	14395.61	14395.61	14395.61	14395.61	14395.61	14395.61	14395.61	14395.61
22348.92	22348.92	22348.92	22348.92	22348.92	22348.92	22348.92	[illegible]	[illegible]	[illegible]
11579.36	11579.36	11579.36	11579.36	11579.36	11579.36	11579.36	[illegible]	[illegible]	[illegible]
10542.19	10542.19	10542.19	10542.19	10542.19	10542.19	10542.19	10542.19	10542.19	10542.19
17460.15	17460.15	17460.15	17460.15	17460.15	17460.15	17460.15	17460.15	17460.15	17460.15
						8511.581	8483.51		
						7993.051	7993.051		
						10198.068	10198.068		
7096.821	7096.821	7096.821	7096.821	7096.821	7096.821	7096.821	[illegible]		

Fig. 7C

21	22	23	24	25	26	27	28	29	30
Minnesota	Minnesota	Minnesota	Abony	Chloeraesuis	Typhimurium		Minnesota?	Enteritidis	Brandenburg
NBRC 15185	NBRC 15186	NBRC 15187	NBRC 100797	NBRC 105684	NBRC 105726	JCM 3919	NBRC 15335	GTC 09490	GTC 09492
11767.58	11767.58	11767.58	11767.58	11767.58	11767.58	11767.58	11767.58	11767.581	11767.581
22248.48	22248.48	22248.48	22248.48	22248.48	22248.48	22248.48	22248.48	22248.479	22248.479
22087.49	22087.49	22087.49	22087.49	22087.49	22087.49	22087.49	22087.49	22087.492	22087.492
11214.13	11214.13	11214.13	11214.13	11214.13	[illegible]	11214.13	11214.13	11214.13	11214.13
29690.16	29690.16	29690.16	29690.16	29690.16	29690.16	29690.16	29690.16	29690.16	29690.16
10286.07	10286.07	10286.07	10286.07	10286.07	10286.07	10286.07	10286.07	10286.065	10286.065
12227.29	12227.29	12227.29	12227.29	12227.29	12227.29	12227.29	12227.29	12227.285	12227.285
25853.00	25853.00	25853.00	25853.00	25853.00	25853.00	25853.00	25853.00	25853.001	25853.001
15195.13	15195.13	15195.13	15195.13	15195.13	15195.13	15195.13	15195.13	15195.126	15195.126
7261.45	7261.45	7261.45	7261.45	7261.45	7261.45	7261.45	7261.45	7261.445	7261.445
9592.29	9592.29	9592.29	9592.29	9592.29	9592.29	9592.29	9592.29	9592.291	9592.291
13569.05	13569.05	13569.05	13569.05	13569.05	13569.05	13569.05	13569.05	13569.051	13569.051
11186.00	11186.00	11186.00	11186.00	11186.00	11186.00	[illegible]	11186.00	11186.003	11186.003
20187.32	20187.32	20187.32	20187.32	20187.32	20187.32	20187.32	20187.32	20187.318	20187.318
11478.35	11478.35	11478.35	11478.35	11478.35	11478.35	11478.35	11478.35	11478.354	11478.354
13996.36	13996.36	13996.36	13996.36	13996.36	13996.36	13996.36	13996.36	13996.359	13996.359
18729.50	18729.50	18729.50	18729.50	18729.50	18729.50	18729.50	18729.50	18729.502	18729.502
12770.66	12770.66	12770.66	12770.66	12770.66	12770.66	12770.66	12770.66	12770.66	12770.66
17473.18	17473.18	17473.18	17473.18	17473.18	17473.18	17473.18	17473.18	17473.179	17473.179
6383.55	6383.55	6383.55	6383.55	6383.55	6383.55	6383.55	6383.55	6383.552	6383.552
14987.38	14987.38	14987.38	14987.38	14987.38	14981.41	14987.38	14987.38	14987.378	14987.378
4365.35	4365.35	4365.35	4365.35	4365.35	4365.35	4365.35	4365.35	4365.352	4365.352
13031.26	13031.26	13031.26	13031.26	13031.26	13031.26	13031.26	13031.26	13031.261	13031.261
13700.71	13700.71	13700.71	13700.71	13700.71	13700.71	13700.71	13700.71	13700.705	13700.705
23354.87	23354.87	23354.87	23354.87	23354.87	23354.87	23354.87	23354.87	23354.866	23354.866
14395.61	14395.61	14395.61	14395.61	14395.61	14395.61	14395.61	14395.61	14395.611	14395.611
[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]	11579.36	11579.36	11579.36	11579.36	[illegible]	11579.367	11579.367
10542.19	10542.19	10542.19	10542.19	10542.19	10542.19	10542.19	10542.19	10542.19	[illegible]
17460.15	17460.15	17460.15	17460.15	[illegible]	17460.15	17460.15	17460.15	17474.18	17474.18
			6483.508	6483.508	6483.508	6483.508			6483.508
			7993.051	7993.051	7993.051	7993.051			7993.051
			10198.068	10198.068	10198.068	10198.068			10198.068
			[illegible]	[illegible]	7098.831	[illegible]			[illegible]
					18855.138				

Fig. 7D

UB_04					7	4	7	4
31	32	33	34	35	36	38	39	40
Pakistan(8)/Litchf	Typhimurium	Infantis	Thomson	Saintpaul	Infantis	Brandenburg	Infantis	Brandenburg
GTC 08493	GTC 08548	ATCC BAA-1675	ATCC BAA-1779	ATCC 9712	jfdSe1402-1	jfdSe1402-3	jfdSe1402-4	jfdSe1402-5
11767.581	11767.581	11767.58	11767.58	11767.58	11767.581	11767.581	11767.581	11767.581
22248.479	22248.479	22248.48	22248.48	22248.48	22248.479	22248.479	22248.479	22248.479
22087.492	22087.492	22087.49	22087.49	22087.49	22087.492	22087.492	22087.492	22087.492
11214.13	11214.13	11214.13	11214.13	11214.13	11214.13	11214.13	11214.13	11214.13
29690.16	29690.16	29690.16	29690.16	[illegible]	29690.16	29690.16	29690.16	29690.16
10286.065	10286.065	10286.07	10286.07	10286.07	10286.065	10286.065	10286.065	10286.065
12227.285	12227.285	12227.29	12227.29	12227.29	12227.285	12227.285	12227.285	12227.285
25853.001	25853.001	25853.00	25853.00	25853.00	25853.001	25853.001	25853.001	25853.001
15195.126	15195.126	15195.13	15195.13	15195.13	15195.126	15195.126	15195.126	15195.126
7261.445	7261.445	7261.45	7261.45	7261.45	7261.445	7261.445	7261.445	7261.445
9592.291	9592.291	9592.29	9592.29	9592.29	9592.291	9592.291	9592.291	9592.291
13569.051	13569.051	13569.05	13569.05	13569.05	13569.051	13569.051	13569.051	13569.051
[illegible]	11186.003	11186.00	11186.00	11186.00	11186.003	11186.003	11186.003	11186.003
20187.318	20187.318	20187.32	20187.32	20187.32	20187.318	20187.318	20187.318	20187.32
11478.354	11478.354	11478.35	11478.35	11478.35	11478.354	11478.354	11478.354	11478.35
13996.359	13996.359	13996.36	13996.36	13996.36	13996.359	13996.359	13996.359	13996.36
18729.502	18729.502	18729.50	18729.50	18729.50	18729.502	18729.502	18729.502	18729.5
12770.66	12770.66	12770.66	12770.66	12770.66	12770.66	12770.66	12770.66	12770.66
17473.179	17473.179	17473.18	17473.18	17473.18	17473.179	17501.237	17501.237	17473.18
6383.552	6383.552	6383.55	6383.55	6383.55	6383.552	6383.552	6383.552	6383.552
14987.378	14987.378	14987.38	14987.38	14987.38	14987.378	14987.378	14987.378	14987.38
4365.352	4365.352	4365.35	4365.35	4365.35	4365.352	4365.352	4365.352	4365.352
13031.261	13031.261	13031.26	13031.26	13031.26	13031.261	13031.261	13031.261	13031.26
13700.705	13700.705	13700.71	13700.71	13700.71	13700.705	13700.705	13700.705	13700.71
23354.866	23354.866	23354.87	23354.87	23354.87	23354.866	23354.866	23354.866	23354.87
14395.611	14395.611	14395.61	14395.61	14395.61	14395.611	14395.611	14395.611	14395.61
[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
11579.357	11579.357	[illegible]	11579.36	11579.36	11579.357	[illegible]	11579.36	[illegible]
10542.19	10542.19	10542.19	10542.19	10542.19	10542.19	10542.19	10542.19	10542.19
17460.15	17460.15	17460.15	17474.18	17460.15	17460.15	17460.15	17460.15	17460.15
6483.508	6483.508	6483.508	6483.508	6483.51	6511.561		6511.561	
	7993.051	7993.051	[illegible]	[illegible]	7993.051		7993.051	
10198.068	10198.068	10198.068	10198.068	10198.068	10198.068		10198.068	
[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]		[illegible]	
		18643.126						

Fig. 7E

	7,3,10	7	7	7	7	7	7		7,3,10
41	42	43	44	45	46	47	48	49	50
	Oron	Rissen	Rissen	Rissen	Rissen	Rissen	Minnesota	Minnesota	Orion
jfdSe1402-6	jfdSe1402-7	jfdSe1402-8	jfdSe1402-9	jfdSe1402-10	jfdSe1402-11	jfdSe1402-12	jfdSe1402-13	jfdSe1402-14	jfdSe1402-15
11767.581	11767.581	11767.581	11767.581	11767.581	11767.581	11767.581	11767.581	11767.581	11767.581
22248.479	22248.479	22248.479	22248.479	22248.479	22248.479	22248.479	22248.479	22248.479	22248.479
22087.492	22087.492	22087.492	22087.492			22087.492	22087.492	22087.492	22087.492
11214.13	11214.13	11214.13	11214.13	11214.13	11214.13	11214.13	11214.13	11214.13	11214.13
29690.16	29690.16	29690.16	29690.16	29690.16	29690.16	29690.16	29690.16	29690.16	29690.16
10286.085	10286.085	10286.085	10286.085	10286.085	10286.085	10286.085	10286.085	10286.085	10286.085
12227.285	12227.285	12227.285	12227.285	12227.285	12227.285	12227.285	12227.285	12227.285	12227.285
25853.001	25853.001	25853.001	25853.001	25853.001		25853.001	25853.001	25853.001	25853.001
15195.126	15195.126	15195.126	15195.126	15195.126	15195.126	15195.126	15195.126	15195.126	15195.126
7281.445	7281.445	7281.445	7281.445	7281.445	7281.445	7281.445	7281.445	7281.445	7281.445
9592.291	9592.291		9592.291	9592.291	9592.291	9592.291	9592.291	9592.291	9592.291
13569.051	13569.051	13569.051	13569.051	13569.051	13569.051	13569.051	13569.051	13569.051	13569.051
11186.003	11186.003	11186.003	11186.003	11186.003	11186.003	11186.003	11186.003	11186.003	11186.003
20187.318	20187.318	20187.318	20187.318	20187.318	20187.318	20187.318	20187.318	20187.318	20187.318
11478.354	11478.354	11478.354	11478.354	11478.354	11478.354	11478.354	11205.063	11478.354	11478.354
	13996.359						13996.359	13996.359	13996.359
18729.502	18729.502	18729.502	18729.502	18729.502	18729.502	18729.502	18729.502	18729.502	18729.5
12770.66	12770.66	12770.66	12770.66	12770.66	12770.66	12770.66	12770.66	12770.66	12770.66
17473.179	17473.179	17473.179	17473.179	17473.179	17473.179	17473.179	17473.179	17473.179	17473.179
6383.552	6383.552	6383.552	6383.552	6383.552	6383.552	6383.552	6383.552	6383.552	6383.552
14967.376	14967.376	14967.376	14967.376	14967.376	14967.376	14967.376	14967.376	14967.376	14967.376
4385.352	4385.352	4385.352	4385.352	4385.352	4385.352	4385.352	4385.352	4385.352	4385.352
13031.261	13031.261	13031.261	13031.261	13031.261	13031.261	13031.261	13031.261	13031.261	13031.261
13700.705	13700.705		13700.705	13700.705	13700.705	13700.705	13700.705	13700.705	13700.705
23354.866	23354.866	23354.866	23354.866	23354.866	23354.866	23354.866	23354.866	23354.866	23354.866
14395.611	14395.611	14395.611	14395.611	14395.611	14395.611	14395.611	14395.611	14395.611	14395.611
11579.357	11579.357	11579.357	11579.357	11579.357	11579.357	11579.357	11579.357	11579.357	11579.357
10542.19	10542.19	10542.19	10542.19	10542.19	10542.19	10542.19	10542.19	10542.19	10542.19
17474.18	17460.15	17474.18	17474.18	17474.18	17474.18	17474.18	17474.18	17474.18	17460.15
	6483.51						6511.56	6511.56	6483.51
	7993.051						8	7993.051	7993.051
							10196.068	10196.068	
							18653.194		

Fig. 7F

O8	O8	O1,3,19	O13	O1,3,19	O7	O1,3,19	O18	O1,3,19	O7
51	52	53	54	55	56	57	58	61	62
Altona	Istanbul	Senftenberg	UN_O13	UN_O1,3,19	Montevideo	UN_O1,3,19	UN_O18	UN_O1,3,19	UN_O7
jfdSe1409-1	jfdSe1409-2	jfdSe1409-3	jfdSe1409-4	jfdSe1409-5	jfdSe1409-6	jfdSe1409-7	jfdSe1409-8	jfdSe1409-11	jfdSe1409-12
11767.581	11767.581	11767.581	11767.581	11767.581	11767.581	11767.581	11767.581	11767.581	11767.58
22248.479	22248.479	22248.479	22248.479	22248.479	22248.479	22248.479	22248.479	22248.479	
22087.492	22087.492	22087.492	22087.492	22087.492	22087.492	22087.492	22087.492	22087.492	22087.49
11214.13	11214.13	11214.13	11214.13	11214.13	11214.13	11214.13	11214.13	11214.13	11214.13
29690.16		29690.16	29690.16	29690.16	29690.16	29690.16	29690.16	29690.16	29690.16
10286.085	10286.085	10286.085	10286.085	10286.085	10286.085	10286.085	10286.085	10286.085	10286.07
12227.285	12227.285	12227.285	12227.285	12227.285	12227.285	12227.285	12227.285	12227.285	12227.29
25853.001	25853.001	25853.001	25853.001	25853.001	25853.001	25853.001	25853.001	25853.001	25853
15195.126	15195.126	15195.126	15195.126	15195.126	15195.126	15195.126	15195.126	15195.126	15195.13
7281.445	7281.445	7281.445	7281.445	7281.445	7281.445	7281.445	7281.445	7281.445	7281.445
9592.291	9592.291	9592.291	9592.291	9592.291	9592.291	9592.291	9592.291	9592.291	9592.291
13569.051		13569.051	13569.051	13569.051	13569.051	13569.051	13569.051	13569.051	13569.05
11186.003	11186.003	11186.003	11186.003	11186.003	11186.003	11186.003	11126.003	11126.003	11186
20187.318	20187.318	20187.318	20187.318	20155.301	20173.292		20187.318	20187.318	20187.32
11063.885	11063.885	11063.885	11063.885	11063.885	11063.885		11063.885	11063.885	11063.89
13996.359	13996.359	13996.359	13996.359	13996.359	13996.359	13996.359	13996.359	13996.359	
		18729.502	18729.502	18729.502	18729.502	18729.502	18729.502	18729.502	18729.5
17473.179	17473.179	17473.179	17473.179	17473.179	17473.179	17473.179	17473.179	17473.179	17473.18
6383.552	6383.552	6383.552	6383.552	6383.552	6383.552	6383.552	6383.552	6383.552	6383.552
	14967.376	14967.376	14967.376	14967.376	14967.376	14967.376	14967.376	14967.376	14967.38
	4385.352	4385.352	4385.352	4385.352		4385.352	4385.352	4385.352	4385.353
	13031.261	13031.261	13031.261	13031.261		13031.261	13031.261	13031.261	13031.26
	13700.705	13700.705	13700.705	13700.705		13700.705	13700.705	13700.705	13700.71
	23354.866	23354.866	23354.866	23354.866		23354.866	23354.866	23354.866	23354.87
	14395.611	14395.611		14395.611		14395.611	14395.611	14395.611	14395.61
22978.826							22848.216		
	11579.357	11579.357	11579.357	11579.357		11579.357	11579.357	11579.357	
17474.18	17474.18	17474.18	17474.18	17474.18	17474.18	17474.18	17474.18	17474.18	17474.18
6483.508	6483.508	6511.581	6483.508	6511.581	6483.508	6511.581	6483.508	6511.581	6483.508
8420.515	8178.313		7993.051	7993.104	7993.051		7993.051		7993.051
10198.068	10198.068	10198.068	10198.068	10198.068	10198.068	10198.068	10198.068	10198.068	10198.07
		18605.099							

Fig. 7G

Q3.10	Q8	Q4	Q4	Q4
[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
Amsterdam	Manhattan	Schwarzengrund O4:d1,7	Schwarzengrund O4:d1,7	Schwarzengrund O4:d1,7
[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
11767.581	11767.58	11767.581	11767.581	11767.581
22248.479	22248.48	22248.479	22248.479	22248.479
22087.492	22087.49	22087.492	22087.492	22087.492
11214.13	11214.13	11214.13	11214.13	11214.13
29690.18	29690.18	29690.18	29690.18	29690.18
10288.065	10288.07	10288.065	10288.065	10288.065
12227.285	12227.29	12227.285	12227.285	12227.285
25853.001	25853	25853.001	25853.001	25853.001
15195.128	15195.13	15195.128	15195.128	15195.128
7281.445	7281.445	7281.445	7281.445	7281.445
[illegible]	9592.291	9592.291	9592.291	9592.291
[illegible]	13589.05	13589.051	13589.051	13589.051
11186.003	[illegible]	11186.003	11186.003	11186.003
20187.318	20187.32	20187.318	20187.318	20187.318
11063.885	11063.89	11063.885	10727.849	11063.885
13996.359	13996.36	13996.359	13996.359	13996.359
	18729.5	18729.502	18729.502	18729.502
	12770.86	12770.86	12770.86	12770.86
17473.179	17473.18	17473.179	17473.179	17473.179
6383.552	6383.552	6383.552	6383.552	6383.552
14967.378	14967.38	14967.378	14967.378	14967.378
4365.352	4365.352	4365.352	4365.352	4365.352
13031.261	13031.26	13031.261	13031.261	13031.261
13700.705	13700.71	13700.705	13700.705	13700.705
23354.868	23354.87	23354.868	23354.868	23354.868
14395.611	14395.61	14395.611	14395.611	14395.611
[illegible]	[illegible]	[illegible]	[illegible]	
11579.357	11579.36	[illegible]		[illegible]
[illegible]	[illegible]			
13606.873	17474.18	17474.18	17474.18	17474.18
6511.561	6511.561	6483.508	6483.508	6483.508
7893.051	7893.051			
10198.058	10198.07	10198.058	10198.058	10198.058
[illegible]	[illegible]	[illegible]	[illegible]	[illegible]

Fig. 8A

			Identifying Orion	Identifying Thompson					Showing mutation Manhattan, Amsterdam Branderup			Identifying Typhimurium
	Difference<800ppm				Diff. 850ppm			Diff. 1200ppm				
0			No Peak	No Peak						No Peak	No Peak	No Peak
1	17460.15	22948.82	10198.07	8023.08	13986.36	14967.38	14395.61	11579.36	10542.19	6463.51	18643.128	7110.89
2	17474.18	23010.84	10216.11	7993.05	14008.41	14981.41	14381.59	11565.33	10528.17	6511.581	18653.164	7098.82
3	17432.10	22976.83				14948.30					18605.1	
4		22918.79										
5		22962.8										
6		22986.817										
7		23004.89										
	S7	SOD	ppis	YbT	S8	L15	L17	L21	L25	gns	YoiF	YaiA
01_NBRC13245T_Typhimurium	1	1	1	2	1	1	1	1	1	1	0	2
10_NBRC12529_Typhimurium	1	1	1	2	1	1	1	1	1	1	2	2
11_NBRC14193_Typhimurium	1	1	1	2	1	1	1	1	1	1	0	0
12_NBRC14194_Typhimurium	1	1	1	2	1	1	1	1	1	1	0	0
13_NBRC14209_Typhimurium	1	1	1	0	1	1	1	1	1	1	0	0
14_NBRC14210_Typhimurium	1	1	1	2	1	1	1	1	1	1	0	2
15_NBRC14211_Typhimurium	1	1	1	2	1	1	1	1	1	1	0	2
16_NBRC14212_Typhimurium	1	1	1	2	1	1	1	1	1	1	0	2
17_NBRC15181_Typhimurium	1	1	1	2	1	1	1	1	1	1	0	2
112_jfrlSe1507-1_Typhimurium	1	1	1	2	1	1	1	1	1	1	0	0
32_GTC09548_Typhimurium *	1	1	1	0	1	1	1	1	1	1	0	2
101_HySe30_Typhimurium	1	1	1	2	1	1	1	1	1	1	2	0
26_NBRC105726_Typhimurium	1	1	1	2	1	2	1	1	1	1	0	2
117_jfrlSe1507-6_UN_O4_	1	1	1	2	1	1	1	2	1	1	2	0
116_jfrlSe1507-5_UN_O4_	1	2	1	2	1	1	1	1	1	1	0	0
35_ATCC9712_Saintpaul	1	1	1	0	1	1	1	2	1	1	2	1
58_jfrlSe1409-2_UN_O18_	1	1	1	2	1	1	1	1	1	1	2	1

Fig. 8B

43_jfrlSe1402-7_Orion	1	1	2	2	1	1	1	1	1	1	1	1
50_jfrlSe1402-15_Orion	1	1	2	2	1	1	1	1	1	1	1	1
66_jfrlSe1409-16_Orion	1	1	3	2	1	1	1	1	1	1	1	1
30_GTC09492_Braenderup	2	3	1	2	1	1	1	1	2	1	2	1
56_jfrlSe1409-6_Montevideo	2	1	1	2	1	1	1	2	1	1	1	1
62_jfrlSe1409-12_Montevideo	2	1	1	2	1	1	1	2	1	1	1	1
103_HySe32_Schwarzengrund	2	1	1	2	1	1	1	2	1	1	1	1
106_HySe35_Schwarzengrund	2	1	1	2	1	1	1	2	1	1	1	1
113_jfrlSe1507-2_Schwarzengrund	2	1	1	0	1	1	1	2	1	1	0	1
114_jfrlSe1507-3_Schwarzengrund	2	1	1	0	1	1	1	2	1	1	0	1
24_NBRC100797_Abony	1	2	1	0	1	1	1	1	1	1	2	1
31_GTC09493_Pakistan	1	3	1	0	1	1	1	1	1	1	0	1
02_GTC00131_Enteritidis	2	2	1	0	1	1	1	1	1	1	0	1
03_GTC09491_Enteritidis	2	2	1	0	1	1	1	1	1	1	0	1
04_GTC03838_Enteritidis	2	2	1	2	1	1	1	1	1	1	0	1
07_GTC09489_Enteritidis	2	2	1	0	1	1	1	1	1	1	0	1
29_GTC09490_Enteritidis	2	2	1	2	1	1	1	1	1	1	0	1
72_HySe01_Enteritidis	2	2	1	2	1	1	1	1	1	1	0	1
73_HySe02_Enteritidis	2	2	1	0	1	1	1	1	1	1	0	1
74_HySe03_Enteritidis	2	2	1	0	1	1	1	1	1	1	0	1
79_HySe08_Enteritidis	2	2	1	0	1	1	1	1	1	1	0	1
05_GTC08914_Enteritidis	2	2	1	0	1	1	1	1	1	1	1	1
06_GTC09421_Enteritidis	2	2	1	0	1	1	1	1	1	1	1	1
09_NBRC3313_Enteritidis	2	2	1	2	1	1	1	1	1	1	1	1
75_HySe04_Enteritidis	2	2	1	2	1	1	1	1	1	1	1	1
76_HySe05_Enteritidis	2	2	1	2	1	1	1	1	1	1	1	1
77_HySe06_Enteritidis	2	2	1	2	1	1	1	1	1	1	1	1
78_HySe07_Enteritidis	2	2	1	2	1	1	1	1	1	1	1	1
85_HySe14_Enteritidis	2	2	1	2	1	1	1	1	1	1	1	1
87_HySe16_Enteritidis	2	2	1	2	1	1	1	1	1	1	1	1
88_HySe17_Enteritidis	2	2	1	2	1	1	1	1	1	1	1	1
89_HySe18_Enteritidis	2	2	1	2	1	1	1	1	1	1	1	1

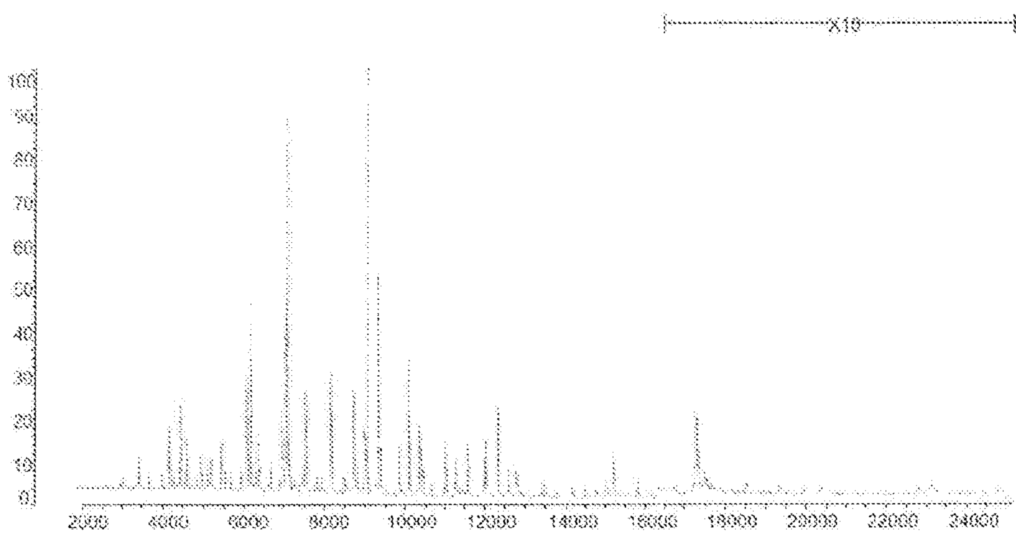
Fig. 8C

90_HySe19_Enteritidis	2	2	1	2	1	1	1	1	1	1	1	1
91_HySe20_Enteritidis	2	2	1	2	1	1	1	1	1	1	1	1
92_HySe21_Enteritidis	2	2	1	2	1	1	1	1	1	1	1	1
93_HySe22_Enteritidis	2	2	1	2	1	1	1	1	1	1	1	1
94_HySe23_Enteritidis	2	2	1	2	1	1	1	1	1	1	1	1
95_HySe24_Enteritidis	2	2	1	2	1	1	1	1	1	1	1	1
96_HySe25_Enteritidis	2	2	1	2	1	1	1	1	1	1	1	1
97_HySe26_Enteritidis	2	2	1	2	1	1	1	1	1	1	1	1
98_HySe27_Enteritidis	2	2	1	2	1	1	1	1	1	1	1	1
99_HySe28_Enteritidis	2	2	1	2	1	1	1	1	1	1	1	1
107_HySe36_Enteritidis	2	2	1	2	1	1	1	1	1	1	1	1
108_HySe37_Enteritidis	2	2	1	2	1	1	1	1	1	1	1	1
109_HySe38_Enteritidis	2	2	1	2	1	1	1	1	1	1	1	1
111_HySe40_Enteritidis	2	2	1	2	1	1	1	1	1	1	1	1
110_HySe39_Enteritidis	2	2	1	2	1	1	1	1	1	1	1	1
43_jfrlSe1402-8_Rissen	2	3	1	2	2	1	1	1	1	1	1	1
44_jfrlSe1402-9_Rissen	2	3	1	2	2	1	1	1	1	1	1	1
45_jfrlSe1402-10_Rissen	2	3	1	2	2	1	1	1	1	1	1	1
46_jfrlSe1402-11_Rissen	2	3	1	2	2	1	1	1	1	1	1	1
47_jfrlSe1402-12_Rissen	2	3	1	2	2	1	1	1	1	1	1	1
41_jfrlSe1402-6_Rissen	2	3	1	2	2	1	1	1	1	1	1	1
63_jfrlSe1409-13_Rissen	2	3	1	2	2	1	1	1	1	1	1	1
65_jfrlSe1409-15_Rissen	2	3	1	2	2	1	1	1	1	1	1	1
67_jfrlSe1409-17_Rissen	2	3	1	2	2	1	1	1	1	1	1	1
69_jfrlSe1409-19_Rissen	2	3	1	2	2	1	1	1	1	1	1	1
08_NBRC3163_Pullorum_Gallinarum	2	2	1	2	1	1	2	1	1	1	0	1
51_jfrlSe1409-1_Altona	2	3	1	2	1	3	1	1	1	1	1	1
59_jfrlSe1409-9_Altona	2	3	1	2	1	3	1	1	1	1	1	1
71_jfrlSe1409-21_Amsterdam	1	2	1	2	1	1	1	1	2	2	2	1
36_jfrlSe1402-1_Infantis	1	3	1	2	1	1	1	1	1	2	1	1

Fig. 8D

39_jfrlSe1402-4_Infantis	1	3	1	2	1	1	1	1	1	2	1	1
81_HySe10_Infantis	1	3	1	2	1	1	1	1	1	2	1	1
84_HySe13_Infantis	1	3	1	2	1	1	1	1	1	2	1	1
86_HySe15_Infantis	1	3	1	2	1	1	1	1	1	2	1	1
33_ATCCBAA-1675_Infantis	1	3	1	0	1	1	1	2	1	1	1	1
52_jfrlSe1409-2_Istanbul	2	3	1	2	1	1	1	1	1	1	0	1
54_jfrlSe1409-4_UN_O13_	2	3	1	2	1	1	1	1	1	1	1	1
80_HySe09_Manhattan	2	3	1	2	1	1	1	1	2	2	0	1
48_jfrlSe1402-13_Mbandaka	2	3	1	2	1	1	1	1	1	2	2	1
49_jfrlSe1402-14_Mbandaka	2	3	1	2	1	1	1	1	1	2	2	1
60_jfrlSe1409-10_Mbandaka	2	3	1	2	1	1	1	1	1	2	2	1
64_jfrlSe1409-14_Mbandaka	2	3	1	2	1	1	1	1	1	2	2	1
70_jfrlSe1409-20_Mbandaka	2	3	1	2	1	1	1	1	1	2	2	1
53_jfrlSe1409-3_Senftenberg	2	3	1	2	1	1	1	1	1	2	3	1
55_jfrlSe1409-5_UN_O1,3,19_	2	3	1	2	1	1	1	1	1	2	3	1
57_jfrlSe1409-7_UN_O1,3,19_	2	3	1	2	1	1	1	1	1	2	3	1
61_jfrlSe1409-11_UN_O1,3,19_	2	3	1	2	1	1	1	1	1	2	3	1
34_ATCCBAA-1738_Thompson	2	3	1	1	1	1	1	1	1	1	2	1
68_jfrlSe1409-18_Thompson	2	3	1	1	1	1	1	1	1	1	2	1
82_HySe11_Thompson	2	3	1	1	1	1	1	1	1	1	2	1
102_HySe31_Thompson	2	3	1	1	1	1	1	1	1	1	2	1
105_HySe34_Thompson	2	3	1	1	1	1	1	1	1	1	2	1
115_jfrlSe1507-4_UN_O4_	1	3	1	2	1	1	1	1	1	2	0	1
27_JCM3919_UN_O7_	1	3	1	0	1	1	1	1	1	1	0	1
38_jfrlSe1402-3_Brandenburg	1	3	1	2	1	1	1	2	1	1	1	1
18_NBRC15192_Minnesota	1	3	1	2	1	1	1	2	1	1	0	1
19_NBRC15183_Minnesota	1	3	1	0	1	1	1	2	1	1	0	1
20_NBRC15184_Minnesota	1	3	1	2	1	1	1	2	1	1	0	1
21_NBRC15185_Minnesota	1	3	1	0	1	1	1	2	1	1	0	1
22_NBRC15186_Minnesota	1	3	1	0	1	1	1	2	1	1	0	1
23_NBRC15187_Minnesota	1	3	1	0	1	1	1	2	1	1	0	1
26_NBRC15335_Minnesota	1	3	1	0	1	1	1	2	1	1	0	1
83_HySe12_Saintpaul	1	3	1	2	1	1	1	2	1	1	0	1
40_jfrlSe1402-5_Brandenburg	1	3	1	2	1	1	1	2	1	1	2	1
104_HySe33_Saintpaul	1	3	1	2	1	1	1	2	1	1	2	1
25_NBRC105684_Choleraesuis	3	4	1	0	1	1	1	1	1	1	0	1

Fig. 9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60
50
40
30
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10
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2000
4000
6000
8000
10000
12000
14000
16000
18000
20000
22000
24000

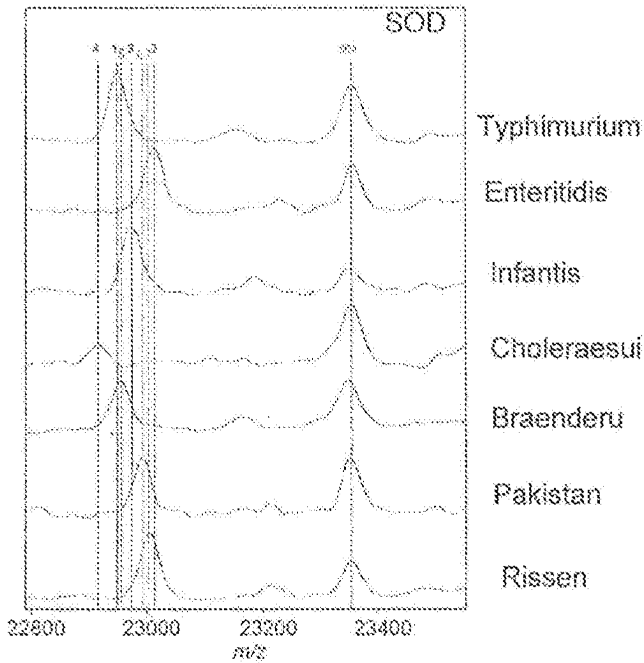
Fig. 10A

sample_name	%	family	genus	species
01_NBRC13245T_Typhimurium	99.9	Family I Enterobacteriaceae	Salmonella	sp.
02_GTC00131_Enteritidis	99.9	Family I Enterobacteriaceae	Salmonella	enterica subsp. enterica
03_GTC09481_Enteritidis	99.9	Family I Enterobacteriaceae	Salmonella	enterica subsp. enterica
04_GTC03838_Enteritidis	97.9	Family I Enterobacteriaceae	Salmonella	enterica subsp. enterica
05_GTC09914_Enteritidis	99.9	Family I Enterobacteriaceae	Salmonella	enterica subsp. enterica
06_GTC09421_Enteritidis	99.9	Family I Enterobacteriaceae	Salmonella	enterica subsp. enterica
07_GTC09489_Enteritidis	99.9	Family I Enterobacteriaceae	Salmonella	enterica subsp. enterica
08_NBRC3163_PullorumGallinarum	97.6	Family I Enterobacteriaceae	Salmonella	enterica subsp. enterica
09_NBRC3313_Enteritidis	99.9	Family I Enterobacteriaceae	Salmonella	enterica subsp. enterica
10_NBRC12529_Typhimurium	99.9	Family I Enterobacteriaceae	Salmonella	enterica subsp. enterica
11_NBRC14193_Typhimurium	99.9	Family I Enterobacteriaceae	Salmonella	enterica subsp. enterica
12_NBRC14194_Typhimurium	99.9	Family I Enterobacteriaceae	Salmonella	enterica subsp. enterica
13_NBRC14209_Typhimurium	99.9	Family I Enterobacteriaceae	Salmonella	enterica subsp. enterica
14_NBRC14210_Typhimurium	99.9	Family I Enterobacteriaceae	Salmonella	enterica subsp. enterica
15_NBRC14211_Typhimurium	85.4	Family I Enterobacteriaceae	Salmonella	enterica subsp. enterica
16_NBRC14212_Typhimurium	99.9	Family I Enterobacteriaceae	Salmonella	enterica subsp. enterica
17_NBRC15181_Typhimurium	99.9	Family I Enterobacteriaceae	Salmonella	enterica subsp. enterica
18_NBRC15182_Minnesota	97.7	Family I Enterobacteriaceae	Salmonella	enterica subsp. enterica
19_NBRC15183_Minnesota	94.9	Family I Enterobacteriaceae	Salmonella	enterica subsp. enterica
20_NBRC15184_Minnesota	97.8	Family I Enterobacteriaceae	Salmonella	enterica subsp. enterica
21_NBRC15185_Minnesota	99.9	Family I Enterobacteriaceae	Salmonella	enterica subsp. enterica
22_NBRC15186_Minnesota	91.7	Family I Enterobacteriaceae	Salmonella	enterica subsp. enterica
23_NBRC15187_Minnesota	99.9	Family I Enterobacteriaceae	Salmonella	enterica subsp. enterica
24_NBRC100797_Abony	99.9	Family I Enterobacteriaceae	Salmonella	enterica subsp. enterica
25_NBRC105684_Choleraesuis	99.9	Family I Enterobacteriaceae	Salmonella	enterica subsp. enterica
26_NBRC105726_Typhimurium	99.9	Family I Enterobacteriaceae	Salmonella	enterica subsp. enterica
27_JCM3919_UN_O7	99.9	Family I Enterobacteriaceae	Salmonella	enterica subsp. enterica
28_NBRC15335_Minnesota	99.9	Family I Enterobacteriaceae	Salmonella	enterica subsp. enterica
29_GTC09490_Enteritidis	99.9	Family I Enterobacteriaceae	Salmonella	enterica subsp. enterica
30_GTC09492_Braenderup	95	Family I Enterobacteriaceae	Salmonella	enterica subsp. enterica
31_GTC09493_Pakistan	92	Family I Enterobacteriaceae	Salmonella	enterica subsp. enterica
32_GTC09549_Typhimurium	97.4	Family I Enterobacteriaceae	Salmonella	enterica subsp. enterica
33-ATCCBAA-1675	99.9	Family I Enterobacteriaceae	Salmonella	enterica subsp. enterica
34-ATCCBAA-1738	99.9	Family I Enterobacteriaceae	Salmonella	enterica subsp. enterica
35-ATCC9712	99.9	Family I Enterobacteriaceae	Salmonella	enterica subsp. enterica
36_JfrlSe1402-01	98.2	Family I Enterobacteriaceae	Salmonella	enterica subsp. enterica
38_JfrlSe1402-03	99.9	Family I Enterobacteriaceae	Salmonella	enterica subsp. enterica
39_JfrlSe1402-04	99.9	Family I Enterobacteriaceae	Salmonella	enterica subsp. enterica
40_JfrlSe1402-05	99.9	Family I Enterobacteriaceae	Salmonella	enterica subsp. enterica
41_JfrlSe1402-06	99.9	Family I Enterobacteriaceae	Salmonella	enterica subsp. enterica
42_JfrlSe1402-07	99.9	Family I Enterobacteriaceae	Salmonella	enterica subsp. enterica
43_JfrlSe1402-08	99.9	Family I Enterobacteriaceae	Salmonella	enterica subsp. enterica
44_JfrlSe1402-09	99.9	Family I Enterobacteriaceae	Salmonella	enterica subsp. enterica
45_JfrlSe1402-10	99.9	Family I Enterobacteriaceae	Salmonella	enterica subsp. enterica
46_JfrlSe1402-11	99.9	Family I Enterobacteriaceae	Salmonella	enterica subsp. enterica
47_JfrlSe1402-12	99.9	Family I Enterobacteriaceae	Salmonella	enterica subsp. enterica
48_JfrlSe1402-13	99.9	Family I Enterobacteriaceae	Salmonella	enterica subsp. enterica
49_JfrlSe1402-14	99.9	Family I Enterobacteriaceae	Salmonella	enterica subsp. enterica
50_JfrlSe1402-15_	99.9	Family I Enterobacteriaceae	Salmonella	enterica subsp. enterica
51_jfrlSe1409-1_Altona	99.9	Family I Enterobacteriaceae	Salmonella	enterica subsp. enterica
52_jfrlSe1409-2_Istanbul	99.9	Family I Enterobacteriaceae	Salmonella	enterica subsp. enterica
52_jfrlSe1409-2_Istanbul	97.7	Family I Enterobacteriaceae	Salmonella	enterica subsp. enterica
53_jfrlSe1409-3_Senftenberg	99.9	Family I Enterobacteriaceae	Salmonella	enterica subsp. enterica
54_jfrlSe1409-4_UN_O13	97.2	Family I Enterobacteriaceae	Salmonella	enterica subsp. enterica
55_jfrlSe1409-5_UN_O1_3_19	99.9	Family I Enterobacteriaceae	Salmonella	enterica subsp. enterica
56_jfrlSe1409-6_Montevideo	99.9	Family I Enterobacteriaceae	Salmonella	enterica subsp. enterica
57_jfrlSe1409-7_UN_O1_3_19	99.9	Family I Enterobacteriaceae	Salmonella	enterica subsp. enterica
58_jfrlSe1409-8_UN_O18	94.8	Family I Enterobacteriaceae	Salmonella	enterica subsp. enterica
59_jfrlSe1409-9_Altona	94.6	Family I Enterobacteriaceae	Salmonella	enterica subsp. enterica
60_jfrlSe1409-10_Mbandaka	97.8	Family I Enterobacteriaceae	Salmonella	enterica subsp. enterica

Fig. 10B

61_jfrlSe1409-11_UN_O1_3_19	99.9	Family I Enterobacteriaceae	Salmonella	enterica subsp. enterica
62_jfrlSe1409-12_Montevideo	99.9	Family I Enterobacteriaceae	Salmonella	enterica subsp. enterica
63_jfrlSe1409-13_Rissen	97.7	Family I Enterobacteriaceae	Salmonella	enterica subsp. enterica
64_jfrlSe1409-14_Mbandaka	99.9	Family I Enterobacteriaceae	Salmonella	enterica subsp. enterica
65_jfrlSe1409-15_Rissen	99.9	Family I Enterobacteriaceae	Salmonella	enterica subsp. enterica
66_jfrlSe1409-16_Orion	99.9	Family I Enterobacteriaceae	Salmonella	enterica subsp. enterica
67_jfrlSe1409-17_UN_O7	91.7	Family I Enterobacteriaceae	Salmonella	enterica subsp. enterica
68_jfrlSe1409-18_Thompson	97.3	Family I Enterobacteriaceae	Salmonella	enterica subsp. enterica
69_jfrlSe1409-19_Rissen	99.9	Family I Enterobacteriaceae	Salmonella	enterica subsp. enterica
70_jfrlSe1409-20_Mbandaka	99.9	Family I Enterobacteriaceae	Salmonella	enterica subsp. enterica
71_jfrlSe1409-21_Amsterdam	99.8	Family I Enterobacteriaceae	Salmonella	enterica subsp. enterica
72_HySe01_Enteritidis_O9	99.9	Family I Enterobacteriaceae	Salmonella	enterica subsp. enterica
73_HySe02_Enteritidis_O9	99.9	Family I Enterobacteriaceae	Salmonella	enterica subsp. enterica
74_HySe03_Enteritidis_O9	99.9	Family I Enterobacteriaceae	Salmonella	enterica subsp. enterica
75_HySe04_Enteritidis_O9	99.9	Family I Enterobacteriaceae	Salmonella	enterica subsp. enterica
76_HySe05_Enteritidis_O9	99.9	Family I Enterobacteriaceae	Salmonella	enterica subsp. enterica
77_HySe06_Enteritidis_O9	99.9	Family I Enterobacteriaceae	Salmonella	enterica subsp. enterica
78_HySe07_Enteritidis_O9	99.9	Family I Enterobacteriaceae	Salmonella	enterica subsp. enterica
79_HySe08_Enteritidis_O9	99.9	Family I Enterobacteriaceae	Salmonella	enterica subsp. enterica
80_HySe09_Manhattan_O8	99.9	Family I Enterobacteriaceae	Salmonella	enterica subsp. enterica
81_HySe10_Infantis_O7	99.9	Family I Enterobacteriaceae	Salmonella	enterica subsp. enterica
82_HySe11_Thompson_O7	99.9	Family I Enterobacteriaceae	Salmonella	enterica subsp. enterica
83_HySe12_Saintpaul_O4	99.8	Family I Enterobacteriaceae	Salmonella	enterica subsp. enterica
84_HySe13_Infantis_O7	99.9	Family I Enterobacteriaceae	Salmonella	enterica subsp. enterica
85_HySe14_Enteritidis_O9	99.9	Family I Enterobacteriaceae	Salmonella	enterica subsp. enterica
86_HySe15_Infantis_O7	99.9	Family I Enterobacteriaceae	Salmonella	enterica subsp. enterica
87_HySe16_Enteritidis_O9	99.9	Family I Enterobacteriaceae	Salmonella	enterica subsp. enterica
88_HySe17_Enteritidis_O9	99.9	Family I Enterobacteriaceae	Salmonella	enterica subsp. enterica
89_HySe18_Enteritidis_O9	99.9	Family I Enterobacteriaceae	Salmonella	enterica subsp. enterica
90_HySe19_Enteritidis_O9	99.9	Family I Enterobacteriaceae	Salmonella	enterica subsp. enterica
91_HySe20_Enteritidis_O9	99.9	Family I Enterobacteriaceae	Salmonella	enterica subsp. enterica
92_HySe21_Enteritidis_O9	99.9	Family I Enterobacteriaceae	Salmonella	enterica subsp. enterica
93_HySe22_Enteritidis_O9	99.9	Family I Enterobacteriaceae	Salmonella	enterica subsp. enterica
94_HySe23_Enteritidis_O9	99.9	Family I Enterobacteriaceae	Salmonella	enterica subsp. enterica
95_HySe24_Enteritidis_O9	99.9	Family I Enterobacteriaceae	Salmonella	enterica subsp. enterica
96_HySe25_Enteritidis_O9	99.9	Family I Enterobacteriaceae	Salmonella	enterica subsp. enterica
97_HySe26_Enteritidis_O9	99.9	Family I Enterobacteriaceae	Salmonella	enterica subsp. enterica
98_HySe27_Enteritidis_O9	99.9	Family I Enterobacteriaceae	Salmonella	enterica subsp. enterica
99_HySe28_Enteritidis_O9	99.9	Family I Enterobacteriaceae	Salmonella	enterica subsp. enterica
100_HySe29_Schwarzengrund_O4	99.9	Family I Enterobacteriaceae	Salmonella	enterica subsp. enterica
101_HySe30_Typhimurium_O4	99.9	Family I Enterobacteriaceae	Salmonella	enterica subsp. enterica
102_HySe31_Thompson_O7	99.9	Family I Enterobacteriaceae	Salmonella	enterica subsp. enterica
103_HySe32_Schwarzengrund_O4	99.9	Family I Enterobacteriaceae	Salmonella	enterica subsp. enterica
104_HySe33_Saintpaul_O4	99.9	Family I Enterobacteriaceae	Salmonella	enterica subsp. enterica
105_HySe34_Thompson_O7	99.9	Family I Enterobacteriaceae	Salmonella	enterica subsp. enterica
106_HySe35_Schwarzengrund_O4	99.9	Family I Enterobacteriaceae	Salmonella	enterica subsp. enterica
107_HySe36_Enteritidis_O9	99.9	Family I Enterobacteriaceae	Salmonella	enterica subsp. enterica
108_HySe37_Enteritidis_O9	99.9	Family I Enterobacteriaceae	Salmonella	enterica subsp. enterica
109_HySe38_Enteritidis_O9	99.9	Family I Enterobacteriaceae	Salmonella	enterica subsp. enterica
110_HySe39_Enteritidis_O9	99.9	Family I Enterobacteriaceae	Salmonella	enterica subsp. enterica
111_HySe40_Enteritidis_O9	99.9	Family I Enterobacteriaceae	Salmonella	enterica subsp. enterica
112_jfrlSe1507-01_Typhimurium	99.9	Family I Enterobacteriaceae	Salmonella	enterica subsp. enterica
113_jfrlSe1507-02_Schwarzengrund	99.9	Family I Enterobacteriaceae	Salmonella	enterica subsp. enterica
114_jfrlSe1507-03_Schwarzengrund	99.9	Family I Enterobacteriaceae	Salmonella	enterica subsp. enterica
115_jfrlSe1507-04_UN_O4	99.9	Family I Enterobacteriaceae	Salmonella	enterica subsp. enterica
116_jfrlSe1507-05_UN_O4	97.3	Family I Enterobacteriaceae	Salmonella	enterica subsp. enterica
117_jfrlSe1507-06_UN_O4	99.9	Family I Enterobacteriaceae	Salmonella	enterica subsp. enterica

Fig. 11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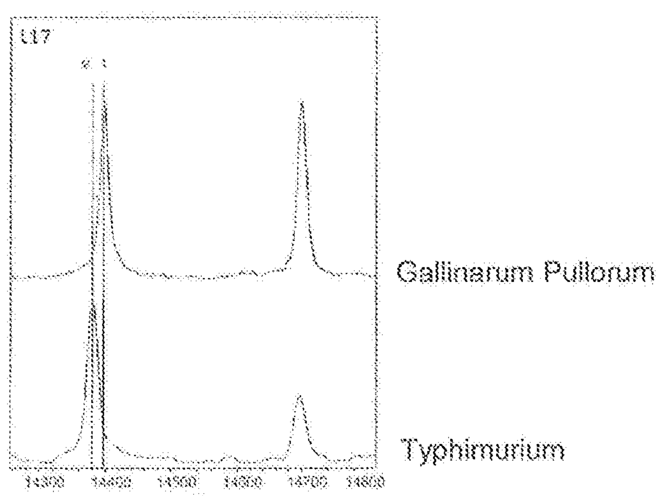
Enteritidis
Infantis
Choleraesui
Braenderu
Pakistan
Rissen
22800
23000
23200
23400
m/z

Fig. 12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Typhimurium
14300
14400
14500
14600
14700
14800
m/z

Fig. 13

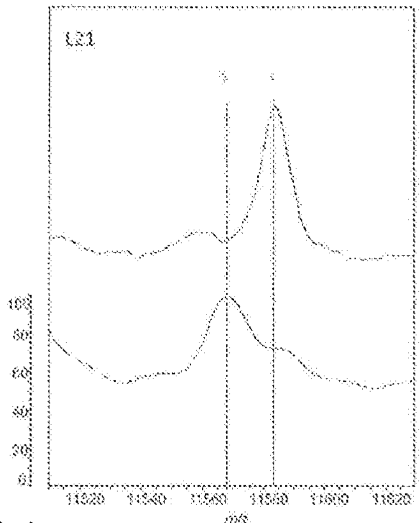

Fig. 14

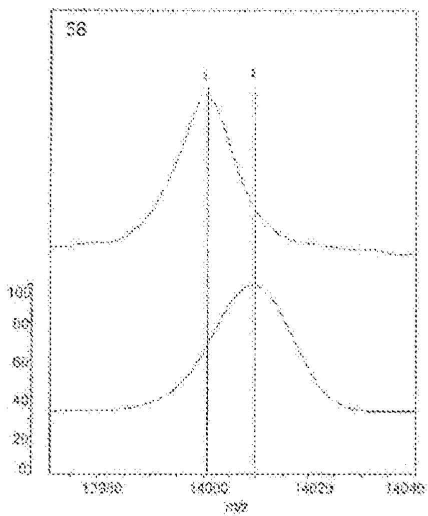

Fig. 15

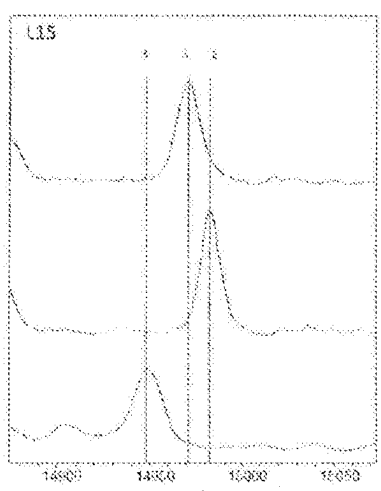

Fig. 16

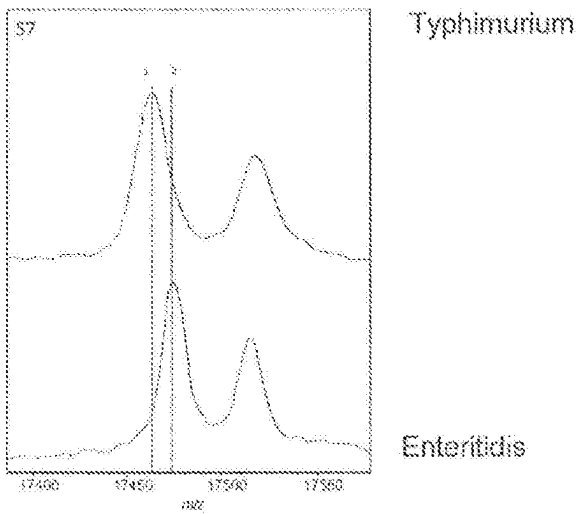

Fig. 17

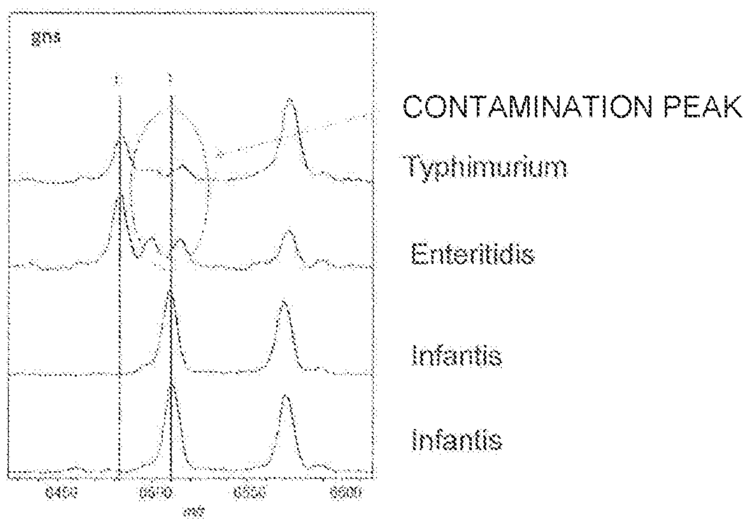

Fig. 18

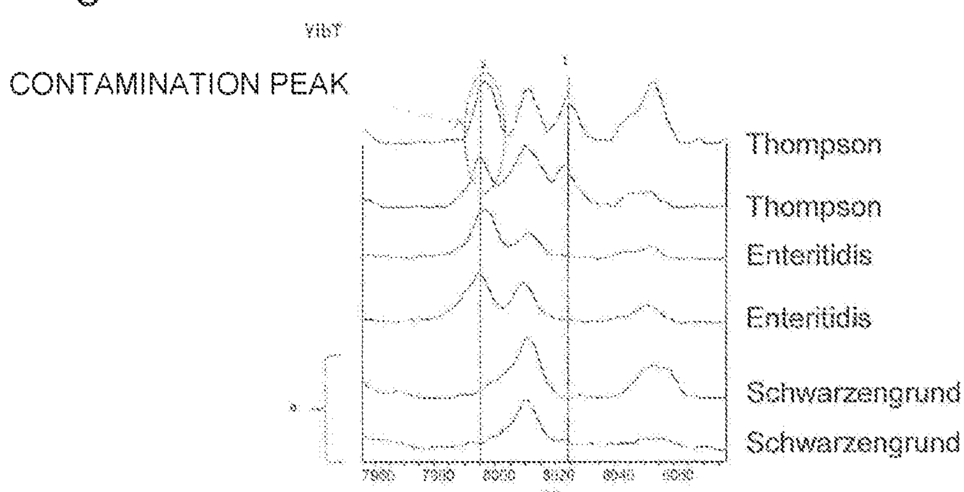

Fig. 19

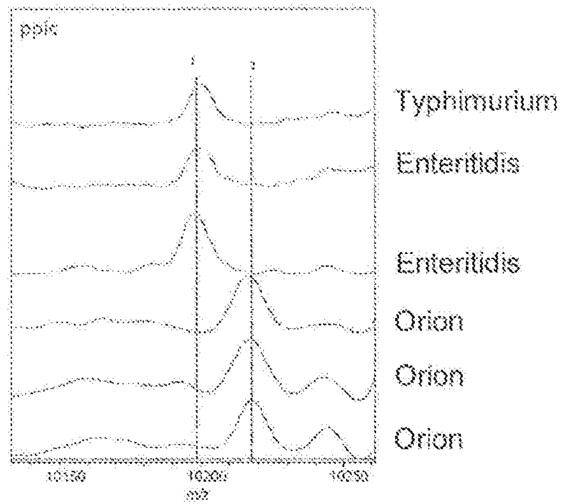

Fig. 20

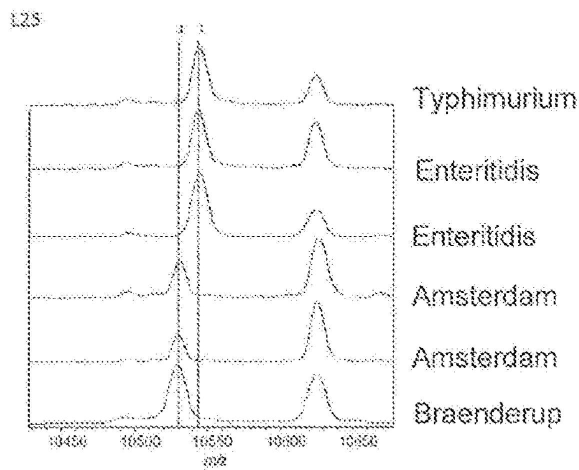

Fig. 21

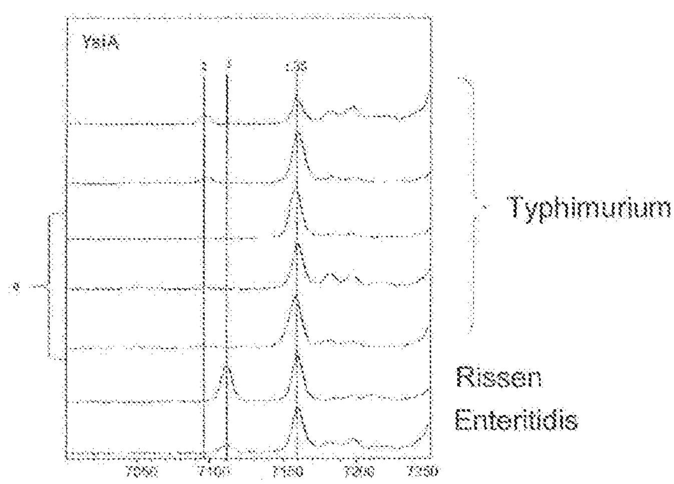

Fig. 22

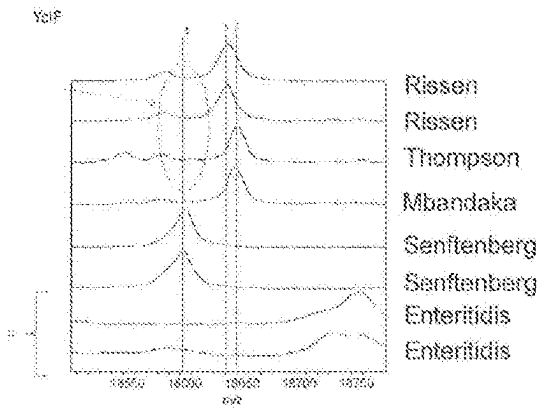
Rissen
Rissen
Thompson
Mbandaka
Senftenberg
Senftenberg
Enteritidis
Enteritidis

Fig. 23

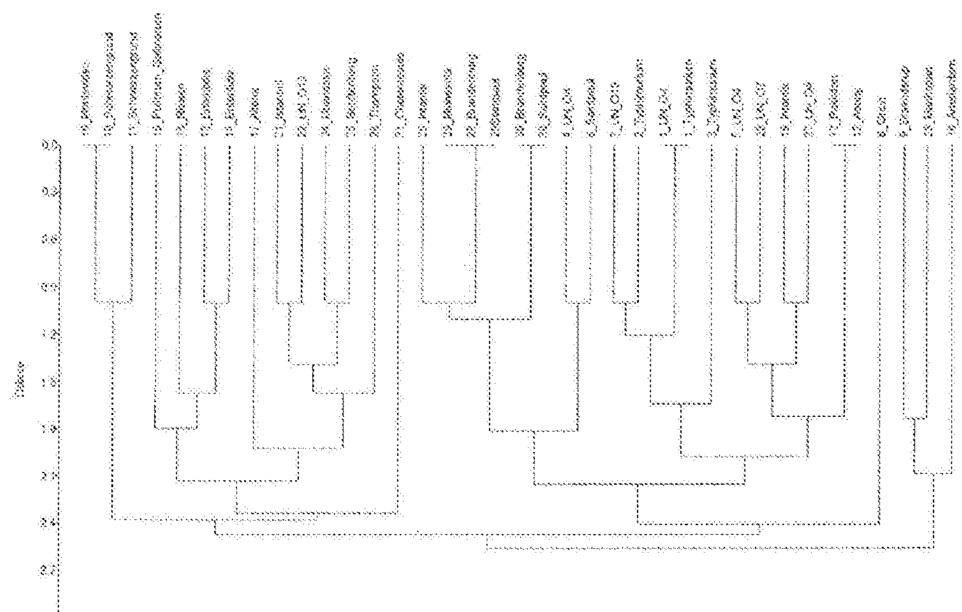

Fig. 24A

	S8
01_NBRC13245T_Typhimurium	ATGAGCATGCAAGATCCGATCGCGGATATGCTGACCCGTATCCGTAACGGTCAGGCCGCGAACAAAGCTGCGGTCACCATGCC TTCCTCCAAGCTGAAAGTGGCAATTGCCAACGTGCTGAAGGAAGAAGGTTTTATTGAAGATTTTAAAGTTGAAGGCGACACCAA GCCGGAACTGGAACTGACTCTTAAGTATTTCCAGGGTAAAGCTGTTGTAGAAAGCATTCAGCGTGTCAGCCGCCCAGGTCTGC GCATCTACAAACGTAAAGATGAGCTGCCGAAAGTTATGGCCGGTCTGGGTATCGCAGTTGTTTCTACCTCTAAAGGTGTTATGA CTGATCGTGCAGCGCGCCAGGCTGGTCTTGGTGGCGAAATTATCTGCTACGTAGCCTAA (SEQ ID NO: 54)
02_GTC00131_Enteritidis	ATGAGCATGCAAGATCCGATCGCGGATATGCTGACCCGTATCCGTAACGGTCAGGCCGCGAACAAAGCTGCGGTCACCATGCC TTCCTCCAAGCTGAAAGTGGCAATTGCCAACGTGCTGAAGGAAGAAGGTTTTATTGAAGATTTTAAAGTTGAAGGCGACACCAA GCCGGAACTGGAACTGACTCTTAAGTATTTCCAGGGTAAAGCTGTTGTAGAAAGCATTCAGCGTGTCAGCCGCCCAGGTCTGC GCATCTACAAACGTAAAGATGAGCTGCCGAAAGTTATGGCCGGTCTGGGTATCGCAGTTGTTTCTACCTCTAAAGGTGTTATGA CTGATCGTGCAGCGCGCCAGGCTGGTCTTGGTGGCGAAATTATCTGCTACGTAGCCTAA (SEQ ID NO: 66)
03_GTC09491_Enteritidis	ATGAGCATGCAAGATCCGATCGCGGATATGCTGACCCGTATCCGTAACGGTCAGGCCGCGAACAAAGCTGCGGTCACCATGCC TTCCTCCAAGCTGAAAGTGGCAATTGCCAACGTGCTGAAGGAAGAAGGTTTTATTGAAGATTTTAAAGTTGAAGGCGACACCAA GCCGGAACTGGAACTGACTCTTAAGTATTTCCAGGGTAAAGCTGTTGTAGAAAGCATTCAGCGTGTCAGCCGCCCAGGTCTGC GCATCTACAAACGTAAAGATGAGCTGCCGAAAGTTATGGCCGGTCTGGGTATCGCAGTTGTTTCTACCTCTAAAGGTGTTATGA CTGATCGTGCAGCGCGCCAGGCTGGTCTTGGTGGCGAAATTATCTGCTACGTAGCCTAA (SEQ ID NO: 75)
04_GTC03838_Enteritidis	ATGAGCATGCAAGATCCGATCGCGGATATGCTGACCCGTATCCGTAACGGTCAGGCCGCGAACAAAGCTGCGGTCACCATGCC TTCCTCCAAGCTGAAAGTGGCAATTGCCAACGTGCTGAAGGAAGAAGGTTTTATTGAAGATTTTAAAGTTGAAGGCGACACCAA GCCGGAACTGGAACTGACTCTTAAGTATTTCCAGGGTAAAGCTGTTGTAGAAAGCATTCAGCGTGTCAGCCGCCCAGGTCTGC GCATCTACAAACGTAAAGATGAGCTGCCGAAAGTTATGGCCGGTCTGGGTATCGCAGTTGTTTCTACCTCTAAAGGTGTTATGA CTGATCGTGCAGCGCGCCAGGCTGGTCTTGGTGGCGAAATTATCTGCTACGTAGCCTAA (SEQ ID NO: 82)
05_GTC08914_Enteritidis	ATGAGCATGCAAGATCCGATCGCGGATATGCTGACCCGTATCCGTAACGGTCAGGCCGCGAACAAAGCTGCGGTCACCATGCC TTCCTCCAAGCTGAAAGTGGCAATTGCCAACGTGCTGAAGGAAGAAGGTTTTATTGAAGATTTTAAAGTTGAAGGCGACACCAA GCCGGAACTGGAACTGACTCTTAAGTATTTCCAGGGTAAAGCTGTTGTAGAAAGCATTCAGCGTGTCAGCCGCCCAGGTCTGC GCATCTACAAACGTAAAGATGAGCTGCCGAAAGTTATGGCCGGTCTGGGTATCGCAGTTGTTTCTACCTCTAAAGGTGTTATGA CTGATCGTGCAGCGCGCCAGGCTGGTCTTGGTGGCGAAATTATCTGCTACGTAGCCTAA (SEQ ID NO: 89)
06_GTC09421_Enteritidis	ATGAGCATGCAAGATCCGATCGCGGATATGCTGACCCGTATCCGTAACGGTCAGGCCGCGAACAAAGCTGCGGTCACCATGCC TTCCTCCAAGCTGAAAGTGGCAATTGCCAACGTGCTGAAGGAAGAAGGTTTTATTGAAGATTTTAAAGTTGAAGGCGACACCAA GCCGGAACTGGAACTGACTCTTAAGTATTTCCAGGGTAAAGCTGTTGTAGAAAGCATTCAGCGTGTCAGCCGCCCAGGTCTGC GCATCTACAAACGTAAAGATGAGCTGCCGAAAGTTATGGCCGGTCTGGGTATCGCAGTTGTTTCTACCTCTAAAGGTGTTATGA CTGATCGTGCAGCGCGCCAGGCTGGTCTTGGTGGCGAAATTATCTGCTACGTAGCCTAA (SEQ ID NO: 99)
07_GTC09489_Enteritidis	ATGAGCATGCAAGATCCGATCGCGGATATGCTGACCCGTATCCGTAACGGTCAGGCCGCGAACAAAGCTGCGGTCACCATGCC TTCCTCCAAGCTGAAAGTGGCAATTGCCAACGTGCTGAAGGAAGAAGGTTTTATTGAAGATTTTAAAGTTGAAGGCGACACCAA GCCGGAACTGGAACTGACTCTTAAGTATTTCCAGGGTAAAGCTGTTGTAGAAAGCATTCAGCGTGTCAGCCGCCCAGGTCTGC GCATCTACAAACGTAAAGATGAGCTGCCGAAAGTTATGGCCGGTCTGGGTATCGCAGTTGTTTCTACCTCTAAAGGTGTTATGA CTGATCGTGCAGCGCGCCAGGCTGGTCTTGGTGGCGAAATTATCTGCTACGTAGCCTAA (SEQ ID NO: 109)
08_NBRC3163_Pullorum_Gallinarum	ATGAGCATGCAAGATCCGATCGCGGATATGCTGACCCGTATCCGTAACGGTCAGGCCGCGAACAAAGCTGCGGTCACCATGCC TTCCTCCAAGCTGAAAGTGGCAATTGCCAACGTGCTGAAGGAAGAAGGTTTTATTGAAGATTTTAAAGTTGAAGGCGACACCAA GCCGGAACTGGAACTGACTCTTAAGTATTTCCAGGGTAAAGCTGTTGTAGAAAGCATTCAGCGTGTCAGCCGCCCAGGTCTGC GCATCTACAAACGTAAAGATGAGCTGCCGAAAGTTATGGCCGGTCTGGGTATCGCAGTTGTTTCTACCTCTAAAGGTGTTATGA CTGATCGTGCAGCGCGCCAGGCTGGTCTTGGTGGCGAAATTATCTGCTACGTAGCCTAA (SEQ ID NO: 116)
09_NBRC3313_Enteritidis	ATGAGCATGCAAGATCCGATCGCGGATATGCTGACCCGTATCCGTAACGGTCAGGCCGCGAACAAAGCTGCGGTCACCATGCC TTCCTCCAAGCTGAAAGTGGCAATTGCCAACGTGCTGAAGGAAGAAGGTTTTATTGAAGATTTTAAAGTTGAAGGCGACACCAA GCCGGAACTGGAACTGACTCTTAAGTATTTCCAGGGTAAAGCTGTTGTAGAAAGCATTCAGCGTGTCAGCCGCCCAGGTCTGC GCATCTACAAACGTAAAGATGAGCTGCCGAAAGTTATGGCCGGTCTGGGTATCGCAGTTGTTTCTACCTCTAAAGGTGTTATGA CTGATCGTGCAGCGCGCCAGGCTGGTCTTGGTGGCGAAATTATCTGCTACGTAGCCTAA (SEQ ID NO: 126)
10_NBRC12529_Typhimurium	ATGAGCATGCAAGATCCGATCGCGGATATGCTGACCCGTATCCGTAACGGTCAGGCCGCGAACAAAGCTGCGGTCACCATGCC TTCCTCCAAGCTGAAAGTGGCAATTGCCAACGTGCTGAAGGAAGAAGGTTTTATTGAAGATTTTAAAGTTGAAGGCGACACCAA GCCGGAACTGGAACTGACTCTTAAGTATTTCCAGGGTAAAGCTGTTGTAGAAAGCATTCAGCGTGTCAGCCGCCCAGGTCTGC GCATCTACAAACGTAAAGATGAGCTGCCGAAAGTTATGGCCGGTCTGGGTATCGCAGTTGTTTCTACCTCTAAAGGTGTTATGA CTGATCGTGCAGCGCGCCAGGCTGGTCTTGGTGGCGAAATTATCTGCTACGTAGCCTAA (SEQ ID NO: 133)
11_NBRC14193_Typhimurium	ATGAGCATGCAAGATCCGATCGCGGATATGCTGACCCGTATCCGTAACGGTCAGGCCGCGAACAAAGCTGCGGTCACCATGCC TTCCTCCAAGCTGAAAGTGGCAATTGCCAACGTGCTGAAGGAAGAAGGTTTTATTGAAGATTTTAAAGTTGAAGGCGACACCAA GCCGGAACTGGAACTGACTCTTAAGTATTTCCAGGGTAAAGCTGTTGTAGAAAGCATTCAGCGTGTCAGCCGCCCAGGTCTGC GCATCTACAAACGTAAAGATGAGCTGCCGAAAGTTATGGCCGGTCTGGGTATCGCAGTTGTTTCTACCTCTAAAGGTGTTATGA CTGATCGTGCAGCGCGCCAGGCTGGTCTTGGTGGCGAAATTATCTGCTACGTAGCCTAA (SEQ ID NO: 140)
12_NBRC14194_Typhimurium	ATGAGCATGCAAGATCCGATCGCGGATATGCTGACCCGTATCCGTAACGGTCAGGCCGCGAACAAAGCTGCGGTCACCATGCC TTCCTCCAAGCTGAAAGTGGCAATTGCCAACGTGCTGAAGGAAGAAGGTTTTATTGAAGATTTTAAAGTTGAAGGCGACACCAA GCCGGAACTGGAACTGACTCTTAAGTATTTCCAGGGTAAAGCTGTTGTAGAAAGCATTCAGCGTGTCAGCCGCCCAGGTCTGC GCATCTACAAACGTAAAGATGAGCTGCCGAAAGTTATGGCCGGTCTGGGTATCGCAGTTGTTTCTACCTCTAAAGGTGTTATGA CTGATCGTGCAGCGCGCCAGGCTGGTCTTGGTGGCGAAATTATCTGCTACGTAGCCTAA (SEQ ID NO: 147)
13_NBRC14209_Typhimurium	ATGAGCATGCAAGATCCGATCGCGGATATGCTGACCCGTATCCGTAACGGTCAGGCCGCGAACAAAGCTGCGGTCACCATGCC TTCCTCCAAGCTGAAAGTGGCAATTGCCAACGTGCTGAAGGAAGAAGGTTTTATTGAAGATTTTAAAGTTGAAGGCGACACCAA GCCGGAACTGGAACTGACTCTTAAGTATTTCCAGGGTAAAGCTGTTGTAGAAAGCATTCAGCGTGTCAGCCGCCCAGGTCTGC GCATCTACAAACGTAAAGATGAGCTGCCGAAAGTTATGGCCGGTCTGGGTATCGCAGTTGTTTCTACCTCTAAAGGTGTTATGA CTGATCGTGCAGCGCGCCAGGCTGGTCTTGGTGGCGAAATTATCTGCTACGTAGCCTAA (SEQ ID NO: 154)
14_NBRC14210_Typhimurium	ATGAGCATGCAAGATCCGATCGCGGATATGCTGACCCGTATCCGTAACGGTCAGGCCGCGAACAAAGCTGCGGTCACCATGCC TTCCTCCAAGCTGAAAGTGGCAATTGCCAACGTGCTGAAGGAAGAAGGTTTTATTGAAGATTTTAAAGTTGAAGGCGACACCAA GCCGGAACTGGAACTGACTCTTAAGTATTTCCAGGGTAAAGCTGTTGTAGAAAGCATTCAGCGTGTCAGCCGCCCAGGTCTGC GCATCTACAAACGTAAAGATGAGCTGCCGAAAGTTATGGCCGGTCTGGGTATCGCAGTTGTTTCTACCTCTAAAGGTGTTATGA CTGATCGTGCAGCGCGCCAGGCTGGTCTTGGTGGCGAAATTATCTGCTACGTAGCCTAA (SEQ ID NO: 161)
15_NBRC14211_Typhimurium	ATGAGCATGCAAGATCCGATCGCGGATATGCTGACCCGTATCCGTAACGGTCAGGCCGCGAACAAAGCTGCGGTCACCATGCC TTCCTCCAAGCTGAAAGTGGCAATTGCCAACGTGCTGAAGGAAGAAGGTTTTATTGAAGATTTTAAAGTTGAAGGCGACACCAA GCCGGAACTGGAACTGACTCTTAAGTATTTCCAGGGTAAAGCTGTTGTAGAAAGCATTCAGCGTGTCAGCCGCCCAGGTCTGC GCATCTACAAACGTAAAGATGAGCTGCCGAAAGTTATGGCCGGTCTGGGTATCGCAGTTGTTTCTACCTCTAAAGGTGTTATGA CTGATCGTGCAGCGCGCCAGGCTGGTCTTGGTGGCGAAATTATCTGCTACGTAGCCTAA (SEQ ID NO: 168)
16_NBRC14212_Typhimurium	ATGAGCATGCAAGATCCGATCGCGGATATGCTGACCCGTATCCGTAACGGTCAGGCCGCGAACAAAGCTGCGGTCACCATGCC TTCCTCCAAGCTGAAAGTGGCAATTGCCAACGTGCTGAAGGAAGAAGGTTTTATTGAAGATTTTAAAGTTGAAGGCGACACCAA GCCGGAACTGGAACTGACTCTTAAGTATTTCCAGGGTAAAGCTGTTGTAGAAAGCATTCAGCGTGTCAGCCGCCCAGGTCTGC GCATCTACAAACGTAAAGATGAGCTGCCGAAAGTTATGGCCGGTCTGGGTATCGCAGTTGTTTCTACCTCTAAAGGTGTTATGA CTGATCGTGCAGCGCGCCAGGCTGGTCTTGGTGGCGAAATTATCTGCTACGTAGCCTAA (SEQ ID NO: 174)

Fig. 24B

17_NBRC15181_Typhimurium	ATGAGCATGCAAGATCCGATCGCGGATATGCTGACCCGTATCCGTAACGGTCAGGCCGCGAACAAAGCTGCGGTCACCATGCC TTCCTCCAAGCTGAAAGTGGCAATTGCCAACGTGCTGAAGGAAGAAGGTTTTATTGAAGATTTTAAAGTTGAAGGCGACACCAA GCCGGAACTGGAACTGACTCTTAAGTATTTCCAGGGTAAAGCTGTTGTAGAAAGCATTCAGCGTGTCAGCCGCCCAGGTCTGC GCATCTACAAACGTAAAGATGAGCTGCCGAAAGTTATGGCCGGTCTGGGTATCGCAGTTGTTTCTACCTCTAAAGGTGTTATGA CTGATCGTGCAGCGCGCCAGGCTGGTCTTGGTGGCGAAATTATCTGCTACGTAGCCTAA (SEQ ID NO: 181)
18_NBRC15182_Minnesota	ATGAGCATGCAAGATCCGATCGCGGATATGCTGACCCGTATCCGTAACGGTCAGGCCGCGAACAAAGCTGCGGTCACCATGCC TTCCTCCAAGCTGAAAGTGGCAATTGCCAACGTGCTGAAGGAAGAAGGTTTTATTGAAGATTTTAAAGTTGAAGGCGACACCAA GCCGGAACTGGAACTGACTCTTAAGTATTTCCAGGGTAAAGCTGTTGTAGAAAGCATTCAGCGTGTCAGCCGCCCAGGTCTGC GCATCTACAAACGTAAAGATGAGCTGCCGAAAGTTATGGCCGGTCTGGGTATCGCAGTTGTTTCTACCTCTAAAGGTGTTATGA CTGATCGTGCAGCGCGCCAGGCTGGTCTTGGTGGCGAAATTATCTGCTACGTAGCCTAA (SEQ ID NO: 191)
19_NBRC15183_Minnesota	ATGAGCATGCAAGATCCGATCGCGGATATGCTGACCCGTATCCGTAACGGTCAGGCCGCGAACAAAGCTGCGGTCACCATGCC TTCCTCCAAGCTGAAAGTGGCAATTGCCAACGTGCTGAAGGAAGAAGGTTTTATTGAAGATTTTAAAGTTGAAGGCGACACCAA GCCGGAACTGGAACTGACTCTTAAGTATTTCCAGGGTAAAGCTGTTGTAGAAAGCATTCAGCGTGTCAGCCGCCCAGGTCTGC GCATCTACAAACGTAAAGATGAGCTGCCGAAAGTTATGGCCGGTCTGGGTATCGCAGTTGTTTCTACCTCTAAAGGTGTTATGA CTGATCGTGCAGCGCGCCAGGCTGGTCTTGGTGGCGAAATTATCTGCTACGTAGCCTAA (SEQ ID NO: 201)
20_NBRC15184_Minnesota	ATGAGCATGCAAGATCCGATCGCGGATATGCTGACCCGTATCCGTAACGGTCAGGCCGCGAACAAAGCTGCGGTCACCATGCC TTCCTCCAAGCTGAAAGTGGCAATTGCCAACGTGCTGAAGGAAGAAGGTTTTATTGAAGATTTTAAAGTTGAAGGCGACACCAA GCCGGAACTGGAACTGACTCTTAAGTATTTCCAGGGTAAAGCTGTTGTAGAAAGCATTCAGCGTGTCAGCCGCCCAGGTCTGC GCATCTACAAACGTAAAGATGAGCTGCCGAAAGTTATGGCCGGTCTGGGTATCGCAGTTGTTTCTACCTCTAAAGGTGTTATGA CTGATCGTGCAGCGCGCCAGGCTGGTCTTGGTGGCGAAATTATCTGCTACGTAGCCTAA (SEQ ID NO: 208)
21_NBRC15185_Minnesota	ATGAGCATGCAAGATCCGATCGCGGATATGCTGACCCGTATCCGTAACGGTCAGGCCGCGAACAAAGCTGCGGTCACCATGCC TTCCTCCAAGCTGAAAGTGGCAATTGCCAACGTGCTGAAGGAAGAAGGTTTTATTGAAGATTTTAAAGTTGAAGGCGACACCAA GCCGGAACTGGAACTGACTCTTAAGTATTTCCAGGGTAAAGCTGTTGTAGAAAGCATTCAGCGTGTCAGCCGCCCAGGTCTGC GCATCTACAAACGTAAAGATGAGCTGCCGAAAGTTATGGCCGGTCTGGGTATCGCAGTTGTTTCTACCTCTAAAGGTGTTATGA CTGATCGTGCAGCGCGCCAGGCTGGTCTTGGTGGCGAAATTATCTGCTACGTAGCCTAA (SEQ ID NO: 215)
22_NBRC15186_Minnesota	ATGAGCATGCAAGATCCGATCGCGGATATGCTGACCCGTATCCGTAACGGTCAGGCCGCGAACAAAGCTGCGGTCACCATGCC TTCCTCCAAGCTGAAAGTGGCAATTGCCAACGTGCTGAAGGAAGAAGGTTTTATTGAAGATTTTAAAGTTGAAGGCGACACCAA GCCGGAACTGGAACTGACTCTTAAGTATTTCCAGGGTAAAGCTGTTGTAGAAAGCATTCAGCGTGTCAGCCGCCCAGGTCTGC GCATCTACAAACGTAAAGATGAGCTGCCGAAAGTTATGGCCGGTCTGGGTATCGCAGTTGTTTCTACCTCTAAAGGTGTTATGA CTGATCGTGCAGCGCGCCAGGCTGGTCTTGGTGGCGAAATTATCTGCTACGTAGCCTAA (SEQ ID NO: 222)
23_NBRC15187_Minnesota	ATGAGCATGCAAGATCCGATCGCGGATATGCTGACCCGTATCCGTAACGGTCAGGCCGCGAACAAAGCTGCGGTCACCATGCC TTCCTCCAAGCTGAAAGTGGCAATTGCCAACGTGCTGAAGGAAGAAGGTTTTATTGAAGATTTTAAAGTTGAAGGCGACACCAA GCCGGAACTGGAACTGACTCTTAAGTATTTCCAGGGTAAAGCTGTTGTAGAAAGCATTCAGCGTGTCAGCCGCCCAGGTCTGC GCATCTACAAACGTAAAGATGAGCTGCCGAAAGTTATGGCCGGTCTGGGTATCGCAGTTGTTTCTACCTCTAAAGGTGTTATGA CTGATCGTGCAGCGCGCCAGGCTGGTCTTGGTGGCGAAATTATCTGCTACGTAGCCTAA (SEQ ID NO: 229)
24_NBRC100797_Abony	ATGAGCATGCAAGATCCGATCGCGGATATGCTGACCCGTATCCGTAACGGTCAGGCCGCGAACAAAGCTGCGGTCACCATGCC TTCCTCCAAGCTGAAAGTGGCAATTGCCAACGTGCTGAAGGAAGAAGGTTTTATTGAAGATTTTAAAGTTGAAGGCGACACCAA GCCGGAACTGGAACTGACTCTTAAGTATTTCCAGGGTAAAGCTGTTGTAGAAAGCATTCAGCGTGTCAGCCGCCCAGGTCTGC GCATCTACAAACGTAAAGATGAGCTGCCGAAAGTTATGGCCGGTCTGGGTATCGCAGTTGTTTCTACCTCTAAAGGTGTTATGA CTGATCGTGCAGCGCGCCAGGCTGGTCTTGGTGGCGAAATTATCTGCTACGTAGCCTAA (SEQ ID NO: 236)
25_NBRC105684_Choleraesuis	ATGAGCATGCAAGATCCGATCGCGGATATGCTGACCCGTATCCGTAACGGTCAGGCCGCGAACAAAGCTGCGGTCACCATGCC TTCCTCCAAGCTGAAAGTGGCAATTGCCAACGTGCTGAAGGAAGAAGGTTTTATTGAAGATTTTAAAGTTGAAGGCGACACCAA GCCGGAACTGGAACTGACTCTTAAGTATTTCCAGGGTAAAGCTGTTGTAGAAAGCATTCAGCGTGTCAGCCGCCCAGGTCTGC GCATCTACAAACGTAAAGATGAGCTGCCGAAAGTTATGGCCGGTCTGGGTATCGCAGTTGTTTCTACCTCTAAAGGTGTTATGA CTGATCGTGCAGCGCGCCAGGCTGGTCTTGGTGGCGAAATTATCTGCTACGTAGCCTAA (SEQ ID NO: 246)
26_NBRC105726_Typhimurium	ATGAGCATGCAAGATCCGATCGCGGATATGCTGACCCGTATCCGTAACGGTCAGGCCGCGAACAAAGCTGCGGTCACCATGCC TTCCTCCAAGCTGAAAGTGGCAATTGCCAACGTGCTGAAGGAAGAAGGTTTTATTGAAGATTTTAAAGTTGAAGGCGACACCAA GCCGGAACTGGAACTGACTCTTAAGTATTTCCAGGGTAAAGCTGTTGTAGAAAGCATTCAGCGTGTCAGCCGCCCAGGTCTGC GCATCTACAAACGTAAAGATGAGCTGCCGAAAGTTATGGCCGGTCTGGGTATCGCAGTTGTTTCTACCTCTAAAGGTGTTATGA CTGATCGTGCAGCGCGCCAGGCTGGTCTTGGTGGCGAAATTATCTGCTACGTAGCCTAA (SEQ ID NO: 256)
27_JCM3919_UN_O7_	ATGAGCATGCAAGATCCGATCGCGGATATGCTGACCCGTATCCGTAACGGTCAGGCCGCGAACAAAGCTGCGGTCACCATGCC TTCCTCCAAGCTGAAAGTGGCAATTGCCAACGTGCTGAAGGAAGAAGGTTTTATTGAAGATTTTAAAGTTGAAGGCGACACCAA GCCGGAACTGGAACTGACTCTTAAGTATTTCCAGGGTAAAGCTGTTGTAGAAAGCATTCAGCGTGTCAGCCGCCCAGGTCTGC GCATCTACAAACGTAAAGATGAGCTGCCGAAAGTTATGGCCGGTCTGGGTATCGCAGTTGTTTCTACCTCTAAAGGTGTTATGA CTGATCGTGCAGCGCGCCAGGCTGGTCTTGGTGGCGAAATTATCTGCTACGTAGCCTAA (SEQ ID NO: 265)
28_NBRC15335_Minnesota	ATGAGCATGCAAGATCCGATCGCGGATATGCTGACCCGTATCCGTAACGGTCAGGCCGCGAACAAAGCTGCGGTCACCATGCC TTCCTCCAAGCTGAAAGTGGCAATTGCCAACGTGCTGAAGGAAGAAGGTTTTATTGAAGATTTTAAAGTTGAAGGCGACACCAA GCCGGAACTGGAACTGACTCTTAAGTATTTCCAGGGTAAAGCTGTTGTAGAAAGCATTCAGCGTGTCAGCCGCCCAGGTCTGC GCATCTACAAACGTAAAGATGAGCTGCCGAAAGTTATGGCCGGTCTGGGTATCGCAGTTGTTTCTACCTCTAAAGGTGTTATGA CTGATCGTGCAGCGCGCCAGGCTGGTCTTGGTGGCGAAATTATCTGCTACGTAGCCTAA (SEQ ID NO: 272)
29_GTC09490_Enteritidis	ATGAGCATGCAAGATCCGATCGCGGATATGCTGACCCGTATCCGTAACGGTCAGGCCGCGAACAAAGCTGCGGTCACCATGCC TTCCTCCAAGCTGAAAGTGGCAATTGCCAACGTGCTGAAGGAAGAAGGTTTTATTGAAGATTTTAAAGTTGAAGGCGACACCAA GCCGGAACTGGAACTGACTCTTAAGTATTTCCAGGGTAAAGCTGTTGTAGAAAGCATTCAGCGTGTCAGCCGCCCAGGTCTGC GCATCTACAAACGTAAAGATGAGCTGCCGAAAGTTATGGCCGGTCTGGGTATCGCAGTTGTTTCTACCTCTAAAGGTGTTATGA CTGATCGTGCAGCGCGCCAGGCTGGTCTTGGTGGCGAAATTATCTGCTACGTAGCCTAA (SEQ ID NO: 279)
30_GTC09492_Braenderup	ATGAGCATGCAAGATCCGATCGCGGATATGCTGACCCGTATCCGTAACGGTCAGGCCGCGAACAAAGCTGCGGTCACCATGCC TTCCTCCAAGCTGAAAGTGGCAATTGCCAACGTGCTGAAGGAAGAAGGTTTTATTGAAGATTTTAAAGTTGAAGGCGACACCAA GCCGGAACTGGAACTGACTCTTAAGTATTTCCAGGGTAAAGCTGTTGTAGAAAGCATTCAGCGTGTCAGCCGCCCAGGTCTGC GCATCTACAAACGTAAAGATGAGCTGCCGAAAGTTATGGCCGGTCTGGGTATCGCAGTTGTTTCTACCTCTAAAGGTGTTATGA CTGATCGTGCAGCGCGCCAGGCTGGTCTTGGTGGCGAAATTATCTGCTACGTAGCCTAA (SEQ ID NO: 286)
31_GTC09493_Pakistan	ATGAGCATGCAAGATCCGATCGCGGATATGCTGACCCGTATCCGTAACGGTCAGGCCGCGAACAAAGCTGCGGTCACCATGCC TTCCTCCAAGCTGAAAGTGGCAATTGCCAACGTGCTGAAGGAAGAAGGTTTTATTGAAGATTTTAAAGTTGAAGGCGACACCAA GCCGGAACTGGAACTGACTCTTAAGTATTTCCAGGGTAAAGCTGTTGTAGAAAGCATTCAGCGTGTCAGCCGCCCAGGTCTGC GCATCTACAAACGTAAAGATGAGCTGCCGAAAGTTATGGCCGGTCTGGGTATCGCAGTTGTTTCTACCTCTAAAGGTGTTATGA CTGATCGTGCAGCGCGCCAGGCTGGTCTTGGTGGCGAAATTATCTGCTACGTAGCCTAA (SEQ ID NO: 293)
32_GTC09549_Typhimurium	ATGAGCATGCAAGATCCGATCGCGGATATGCTGACCCGTATCCGTAACGGTCAGGCCGCGAACAAAGCTGCGGTCACCATGCC TTCCTCCAAGCTGAAAGTGGCAATTGCCAACGTGCTGAAGGAAGAAGGTTTTATTGAAGATTTTAAAGTTGAAGGCGACACCAA GCCGGAACTGGAACTGACTCTTAAGTATTTCCAGGGTAAAGCTGTTGTAGAAAGCATTCAGCGTGTCAGCCGCCCAGGTCTGC GCATCTACAAACGTAAAGATGAGCTGCCGAAAGTTATGGCCGGTCTGGGTATCGCAGTTGTTTCTACCTCTAAAGGTGTTATGA CTGATCGTGCAGCGCGCCAGGCTGGTCTTGGTGGCGAAATTATCTGCTACGTAGCCTAA (SEQ ID NO: 300)
33_ATCCBAA-1675_Infantis	ATGAGCATGCAAGATCCGATCGCGGATATGCTGACCCGTATCCGTAACGGTCAGGCCGCGAACAAAGCTGCGGTCACCATGCC TTCCTCCAAGCTGAAAGTGGCAATTGCCAACGTGCTGAAGGAAGAAGGTTTTATTGAAGATTTTAAAGTTGAAGGCGACACCAA GCCGGAACTGGAACTGACTCTTAAGTATTTCCAGGGTAAAGCTGTTGTAGAAAGCATTCAGCGTGTCAGCCGCCCAGGTCTGC GCATCTACAAACGTAAAGATGAGCTGCCGAAAGTTATGGCCGGTCTGGGTATCGCAGTTGTTTCTACCTCTAAAGGTGTTATGA CTGATCGTGCAGCGCGCCAGGCTGGTCTTGGTGGCGAAATTATCTGCTACGTAGCCTAA (SEQ ID NO: 307)
34_ATCCBAA-1738_Thompson	ATGAGCATGCAAGATCCGATCGCGGATATGCTGACCCGTATCCGTAACGGTCAGGCCGCGAACAAAGCTGCGGTCACCATGCC TTCCTCCAAGCTGAAAGTGGCAATTGCCAACGTGCTGAAGGAAGAAGGTTTTATTGAAGATTTTAAAGTTGAAGGCGACACCAA GCCGGAACTGGAACTGACTCTTAAGTATTTCCAGGGTAAAGCTGTTGTAGAAAGCATTCAGCGTGTCAGCCGCCCAGGTCTGC GCATCTACAAACGTAAAGATGAGCTGCCGAAAGTTATGGCCGGTCTGGGTATCGCAGTTGTTTCTACCTCTAAAGGTGTTATGA CTGATCGTGCAGCGCGCCAGGCTGGTCTTGGTGGCGAAATTATCTGCTACGTAGCCTAA (SEQ ID NO: 318)
35_ATCC9712_Saintpaul	ATGAGCATGCAAGATCCGATCGCGGATATGCTGACCCGTATCCGTAACGGTCAGGCCGCGAACAAAGCTGCGGTCACCATGCC TTCCTCCAAGCTGAAAGTGGCAATTGCCAACGTGCTGAAGGAAGAAGGTTTTATTGAAGATTTTAAAGTTGAAGGCGACACCAA GCCGGAACTGGAACTGACTCTTAAGTATTTCCAGGGTAAAGCTGTTGTAGAAAGCATTCAGCGTGTCAGCCGCCCAGGTCTGC GCATCTACAAACGTAAAGATGAGCTGCCGAAAGTTATGGCCGGTCTGGGTATCGCAGTTGTTTCTACCTCTAAAGGTGTTATGA CTGATCGTGCAGCGCGCCAGGCTGGTCTTGGTGGCGAAATTATCTGCTACGTAGCCTAA (SEQ ID NO: 328)

Fig. 24C

36_jfrlSe1402-1_Infantis	ATGAGCATGCAAGATCCGATCGCGGATATGCTGACCCGTATCCGTAACGGTCAGGCCGCGAACAAAGCTGCGGTCACCATGCC TTCCTCCAAGCTGAAAGTGGCAATTGCCAACGTGCTGAAGGAAGAAGGTTTTATTGAAGATTTTAAAGTTGAAGGCGACACCAA GCCGGAACTGGAACTGACTCTTAAGTATTTCCAGGGTAAAGCTGTTGTAGAAAGCATTCAGCGTGTCAGCCGCCCAGGTCTGC GCATCTACAAACGTAAAGATGAGCTGCCGAAAGTTATGGCCGGTCTGGGTATCGCAGTTGTTTCTACCTCTAAAGGTGTTATGA CTGATCGTGCAGCGCGCCAGGCTGGTCTTGGTGGCGAAATTATCTGCTACGTAGCCTAA (SEQ ID NO: 339)
38_jfrlSe1402-3_Brandenburg	ATGAGCATGCAAGATCCGATCGCGGATATGCTGACCCGTATCCGTAACGGTCAGGCCGCGAACAAAGCTGCGGTCACCATGCC TTCCTCCAAGCTGAAAGTGGCAATTGCCAACGTGCTGAAGGAAGAAGGTTTTATTGAAGATTTTAAAGTTGAAGGCGACACCAA GCCGGAACTGGAACTGACTCTTAAGTATTTCCAGGGTAAAGCTGTTGTAGAAAGCATTCAGCGTGTCAGCCGCCCAGGTCTGC GCATCTACAAACGTAAAGATGAGCTGCCGAAAGTTATGGCCGGTCTGGGTATCGCAGTTGTTTCTACCTCTAAAGGTGTTATGA CTGATCGTGCAGCGCGCCAGGCTGGTCTTGGTGGCGAAATTATCTGCTACGTAGCCTAA (SEQ ID NO: 349)
39_jfrlSe1402-4_Infantis	ATGAGCATGCAAGATCCGATCGCGGATATGCTGACCCGTATCCGTAACGGTCAGGCCGCGAACAAAGCTGCGGTCACCATGCC TTCCTCCAAGCTGAAAGTGGCAATTGCCAACGTGCTGAAGGAAGAAGGTTTTATTGAAGATTTTAAAGTTGAAGGCGACACCAA GCCGGAACTGGAACTGACTCTTAAGTATTTCCAGGGTAAAGCTGTTGTAGAAAGCATTCAGCGTGTCAGCCGCCCAGGTCTGC GCATCTACAAACGTAAAGATGAGCTGCCGAAAGTTATGGCCGGTCTGGGTATCGCAGTTGTTTCTACCTCTAAAGGTGTTATGA CTGATCGTGCAGCGCGCCAGGCTGGTCTTGGTGGCGAAATTATCTGCTACGTAGCCTAA (SEQ ID NO: 356)
40_jfrlSe1402-5_Brandenburg	ATGAGCATGCAAGATCCGATCGCGGATATGCTGACCCGTATCCGTAACGGTCAGGCCGCGAACAAAGCTGCGGTCACCATGCC TTCCTCCAAGCTGAAAGTGGCAATTGCCAACGTGCTGAAGGAAGAAGGTTTTATTGAAGATTTTAAAGTTGAAGGCGACACCAA GCCGGAACTGGAACTGACTCTTAAGTATTTCCAGGGTAAAGCTGTTGTAGAAAGCATTCAGCGTGTCAGCCGCCCAGGTCTGC GCATCTACAAACGTAAAGATGAGCTGCCGAAAGTTATGGCCGGTCTGGGTATCGCAGTTGTTTCTACCTCTAAAGGTGTTATGA CTGATCGTGCAGCGCGCCAGGCTGGTCTTGGTGGCGAAATTATCTGCTACGTAGCCTAA (SEQ ID NO: 366)
41_jfrlSe1402-6_Rissen	ATGAGCATGCAAGATCCGATCGCGGATATGCTGACCCGTATCCGTAACGGTCAGGCCGCGAACAAAGCTGCGGTCACCATGCC TTCCTCCAAGCTGAAAGTGGCAATTGCCAACGTGCTGAAGGAAGAAGGTTTTATTGAAGATTTTAAAGTTGAAGGCGACACCAA GCCGGAACTGGAACTGATTCTTAAGTATTTCCAGGGTAAAGCTGTTGTAGAAAGCATTCAGCGTGTCAGCCGCCCAGGTCTGC GCATCTACAAACGTAAAGATGAGCTGCCGAAAGTTATGGCCGGTCTGGGTATCGCAGTTGTTTCTACCTCTAAAGGTGTTATGA CTGATCGTGCAGCGCGCCAGGCTGGTCTTGGTGGCGAAATTATCTGCTACGTAGCCTAA (SEQ ID NO: 373)
42_jfrlSe1402-7_Orion	ATGAGCATGCAAGATCCGATCGCGGATATGCTGACCCGTATCCGTAACGGTCAGGCCGCGAACAAAGCTGCGGTCACCATGCC TTCCTCCAAGCTGAAAGTGGCAATTGCCAACGTGCTGAAGGAAGAAGGTTTTATTGAAGATTTTAAAGTTGAAGGCGACACCAA GCCGGAACTGGAACTGACTCTTAAGTATTTCCAGGGTAAAGCTGTTGTAGAAAGCATTCAGCGTGTCAGCCGCCCAGGTCTGC GCATCTACAAACGTAAAGATGAGCTGCCGAAAGTTATGGCCGGTCTGGGTATCGCAGTTGTTTCTACCTCTAAAGGTGTTATGA CTGATCGTGCAGCGCGCCAGGCTGGTCTTGGTGGCGAAATTATCTGCTACGTAGCCTAA (SEQ ID NO: 380)
43_jfrlSe1402-8_Rissen	ATGAGCATGCAAGATCCGATCGCGGATATGCTGACCCGTATCCGTAACGGTCAGGCCGCGAACAAAGCTGCGGTCACCATGCC TTCCTCCAAGCTGAAAGTGGCAATTGCCAACGTGCTGAAGGAAGAAGGTTTTATTGAAGATTTTAAAGTTGAAGGCGACACCAA GCCGGAACTGGAACTGATTCTTAAGTATTTCCAGGGTAAAGCTGTTGTAGAAAGCATTCAGCGTGTCAGCCGCCCAGGTCTGC GCATCTACAAACGTAAAGATGAGCTGCCGAAAGTTATGGCCGGTCTGGGTATCGCAGTTGTTTCTACCTCTAAAGGTGTTATGA CTGATCGTGCAGCGCGCCAGGCTGGTCTTGGTGGCGAAATTATCTGCTACGTAGCCTAA (SEQ ID NO: 390)
44_jfrlSe1402-9_Rissen	ATGAGCATGCAAGATCCGATCGCGGATATGCTGACCCGTATCCGTAACGGTCAGGCCGCGAACAAAGCTGCGGTCACCATGCC TTCCTCCAAGCTGAAAGTGGCAATTGCCAACGTGCTGAAGGAAGAAGGTTTTATTGAAGATTTTAAAGTTGAAGGCGACACCAA GCCGGAACTGGAACTGATTCTTAAGTATTTCCAGGGTAAAGCTGTTGTAGAAAGCATTCAGCGTGTCAGCCGCCCAGGTCTGC GCATCTACAAACGTAAAGATGAGCTGCCGAAAGTTATGGCCGGTCTGGGTATCGCAGTTGTTTCTACCTCTAAAGGTGTTATGA CTGATCGTGCAGCGCGCCAGGCTGGTCTTGGTGGCGAAATTATCTGCTACGTAGCCTAA (SEQ ID NO: 397)
45_jfrlSe1402-10_Rissen	ATGAGCATGCAAGATCCGATCGCGGATATGCTGACCCGTATCCGTAACGGTCAGGCCGCGAACAAAGCTGCGGTCACCATGCC TTCCTCCAAGCTGAAAGTGGCAATTGCCAACGTGCTGAAGGAAGAAGGTTTTATTGAAGATTTTAAAGTTGAAGGCGACACCAA GCCGGAACTGGAACTGATTCTTAAGTATTTCCAGGGTAAAGCTGTTGTAGAAAGCATTCAGCGTGTCAGCCGCCCAGGTCTGC GCATCTACAAACGTAAAGATGAGCTGCCGAAAGTTATGGCCGGTCTGGGTATCGCAGTTGTTTCTACCTCTAAAGGTGTTATGA CTGATCGTGCAGCGCGCCAGGCTGGTCTTGGTGGCGAAATTATCTGCTACGTAGCCTAA (SEQ ID NO: 404)
46_jfrlSe1402-11_Rissen	ATGAGCATGCAAGATCCGATCGCGGATATGCTGACCCGTATCCGTAACGGTCAGGCCGCGAACAAAGCTGCGGTCACCATGCC TTCCTCCAAGCTGAAAGTGGCAATTGCCAACGTGCTGAAGGAAGAAGGTTTTATTGAAGATTTTAAAGTTGAAGGCGACACCAA GCCGGAACTGGAACTGATTCTTAAGTATTTCCAGGGTAAAGCTGTTGTAGAAAGCATTCAGCGTGTCAGCCGCCCAGGTCTGC GCATCTACAAACGTAAAGATGAGCTGCCGAAAGTTATGGCCGGTCTGGGTATCGCAGTTGTTTCTACCTCTAAAGGTGTTATGA CTGATCGTGCAGCGCGCCAGGCTGGTCTTGGTGGCGAAATTATCTGCTACGTAGCCTAA (SEQ ID NO: 411)
47_jfrlSe1402-12_Rissen	ATGAGCATGCAAGATCCGATCGCGGATATGCTGACCCGTATCCGTAACGGTCAGGCCGCGAACAAAGCTGCGGTCACCATGCC TTCCTCCAAGCTGAAAGTGGCAATTGCCAACGTGCTGAAGGAAGAAGGTTTTATTGAAGATTTTAAAGTTGAAGGCGACACCAA GCCGGAACTGGAACTGATTCTTAAGTATTTCCAGGGTAAAGCTGTTGTAGAAAGCATTCAGCGTGTCAGCCGCCCAGGTCTGC GCATCTACAAACGTAAAGATGAGCTGCCGAAAGTTATGGCCGGTCTGGGTATCGCAGTTGTTTCTACCTCTAAAGGTGTTATGA CTGATCGTGCAGCGCGCCAGGCTGGTCTTGGTGGCGAAATTATCTGCTACGTAGCCTAA (SEQ ID NO: 418)
48_jfrlSe1402-13_Mbandaka	ATGAGCATGCAAGATCCGATCGCGGATATGCTGACCCGTATCCGTAACGGTCAGGCCGCGAACAAAGCTGCGGTCACCATGCC TTCCTCCAAGCTGAAAGTGGCAATTGCCAACGTGCTGAAGGAAGAAGGTTTTATTGAAGATTTTAAAGTTGAAGGCGACACCAA GCCGGAACTGGAACTGACTCTTAAGTATTTCCAGGGCAAAGCTGTTGTAGAAAGCATTCAGCGTGTCAGCCGCCCAGGTCTGC GCATCTACAAACGTAAAGATGAGCTGCCGAAAGTTATGGCCGGTCTGGGTATCGCAGTTGTTTCTACCTCTAAAGGTGTTATGA CTGATCGTGCAGCGCGCCAGGCTGGTCTTGGTGGCGAAATTATCTGCTACGTAGCCTAA (SEQ ID NO: 425)
49_jfrlSe1402-14_Mbandaka	ATGAGCATGCAAGATCCGATCGCGGATATGCTGACCCGTATCCGTAACGGTCAGGCCGCGAACAAAGCTGCGGTCACCATGCC TTCCTCCAAGCTGAAAGTGGCAATTGCCAACGTGCTGAAGGAAGAAGGTTTTATTGAAGATTTTAAAGTTGAAGGCGACACCAA GCCGGAACTGGAACTGACTCTTAAGTATTTCCAGGGCAAAGCTGTTGTAGAAAGCATTCAGCGTGTCAGCCGCCCAGGTCTGC GCATCTACAAACGTAAAGATGAGCTGCCGAAAGTTATGGCCGGTCTGGGTATCGCAGTTGTTTCTACCTCTAAAGGTGTTATGA CTGATCGTGCAGCGCGCCAGGCTGGTCTTGGTGGCGAAATTATCTGCTACGTAGCCTAA (SEQ ID NO: 435)
50_jfrlSe1402-15_Orion	ATGAGCATGCAAGATCCGATCGCGGATATGCTGACCCGTATCCGTAACGGTCAGGCCGCGAACAAAGCTGCGGTCACCATGCC TTCCTCCAAGCTGAAAGTGGCAATTGCCAACGTGCTGAAGGAAGAAGGTTTTATTGAAGATTTTAAAGTTGAAGGCGACACCAA GCCGGAACTGGAACTGACTCTTAAGTATTTCCAGGGTAAAGCTGTTGTAGAAAGCATTCAGCGTGTCAGCCGCCCAGGTCTGC GCATCTACAAACGTAAAGATGAGCTGCCGAAAGTTATGGCCGGTCTGGGTATCGCAGTTGTTTCTACCTCTAAAGGTGTTATGA CTGATCGTGCAGCGCGCCAGGCTGGTCTTGGTGGCGAAATTATCTGCTACGTAGCCTAA (SEQ ID NO: 445)
51_jfrlSe1409-1_Altona	ATGAGCATGCAAGATCCGATCGCGGATATGCTGACCCGTATCCGTAACGGTCAGGCCGCGAACAAAGCTGCGGTCACCATGCC TTCCTCCAAGCTGAAAGTGGCAATTGCCAACGTGCTGAAGGAAGAAGGTTTTATTGAAGATTTTAAAGTTGAAGGCGACACCAA GCCGGAACTGGAACTGACTCTTAAGTATTTCCAGGGTAAAGCTGTTGTAGAAAGCATTCAGCGTGTCAGCCGCCCAGGTCTGC GCATCTACAAACGTAAAGATGAGCTGCCGAAAGTTATGGCCGGTCTGGGTATCGCAGTTGTTTCTACCTCTAAAGGTGTTATGA CTGATCGTGCAGCGCGCCAGGCTGGTCTTGGTGGCGAAATTATCTGCTACGTAGCCTAA (SEQ ID NO: 455)
52_jfrlSe1409-2_Istanbul	ATGAGCATGCAAGATCCGATCGCGGATATGCTGACCCGTATCCGTAACGGTCAGGCCGCGAACAAAGCTGCGGTCACCATGCC TTCCTCCAAGCTGAAAGTGGCAATTGCCAACGTGCTGAAGGAAGAAGGTTTTATTGAAGATTTTAAAGTTGAAGGCGACACCAA GCCGGAACTGGAACTGACTCTTAAGTATTTCCAGGGTAAAGCTGTTGTAGAAAGCATTCAGCGTGTCAGCCGCCCAGGCCTGC GCATCTACAAACGTAAAGATGAGCTGCCGAAAGTTATGGCCGGTCTGGGTATCGCAGTTGTTTCTACCTCTAAAGGTGTTATGA CTGATCGTGCAGCGCGCCAGGCTGGTCTTGGTGGCGAAATTATCTGCTACGTAGCCTAA (SEQ ID NO: 462)
53_jfrlSe1409-3_Senftenberg	

Fig. 24D

54_jfrlSe1409-4_UN_O13_	ATGAGCATGCAAGATCCGATCGCGGATATGCTGACCCGTATCCGTAACGGTCAGGCCGCGAACAAAGCTGCGGTCACCATGCC TTCCTCCAAGCTGAAAGTGGCAATTGCCAACGTGCTGAAGGAAGAAGGTTTTATTGAAGATTTTAAAGTTGAAGGCGACACCAA GCCGGAACTGGAACTGACTCTTAAGTATTTCCAGGGTAAAGCTGTTGTAGAAAGCATTCAGCGTGTCAGCCGCCCAGGTCTGC GCATCTACAAACGTAAAGATGAGCTGCCGAAAGTTATGGCCGGTCTGGGTATCGCAGTTGTTTCTACCTCTAAAGGTGTTATGA CTGATCGTGCAGCGCGCCAGGCTGGTCTTGGTGGCGAAATTATCTGCTACGTAGCCTAA (SEQ ID NO: 475)
55_jfrlSe1409-5_UN_O1,3,19_	ATGAGCATGCAAGATCCGATCGCGGATATGCTGACCCGTATCCGTAACGGTCAGGCCGCGAACAAAGCTGCGGTCACCATGCC TTCCTCCAAGCTGAAAGTGGCAATTGCCAACGTGCTGAAGGAAGAAGGTTTTATTGAAGATTTTAAAGTTGAAGGCGACACCAA GCCGGAACTGGAACTGACTCTTAAGTATTTCCAGGGTAAAGCTGTTGTAGAAAGCATTCAGCGTGTCAGCCGCCCAGGTCTGC GCATCTACAAACGTAAAGATGAGCTGCCGAAAGTTATGGCCGGTCTGGGTATCGCAGTTGTTTCTACCTCTAAAGGTGTTATGA CTGATCGTGCAGCGCGCCAGGCTGGTCTTGGTGGCGAAATTATCTGCTACGTAGCCTAA (SEQ ID NO: 483)
56_jfrlSe1409-6_Montevideo	ATGAGCATGCAAGATCCGATCGCGGATATGCTGACCCGTATCCGTAACGGTCAGGCCGCGAACAAAGCTGCGGTCACCATGCC TTCCTCCAAGCTGAAAGTGGCAATTGCCAACGTGCTGAAGGAAGAAGGTTTTATTGAAGATTTTAAAGTTGAAGGCGACACCAA GCCGGAACTGGAACTGACTCTTAAGTATTTCCAGGGTAAAGCTGTTGTAGAAAGCATTCAGCGTGTCAGCCGCCCAGGCCTGC GCATCTACAAACGTAAAGATGAGCTGCCGAAAGTTATGGCCGGTCTGGGTATCGCAGTTGTTTCTACCTCTAAAGGTGTTATGA CTGATCGTGCAGCGCGCCAGGCTGGTCTTGGTGGCGAAATTATCTGCTACGTAGCCTAA (SEQ ID NO: 492)
57_jfrlSe1409-7_UN_O1,3,19_	ATGAGCATGCAAGATCCGATCGCGGATATGCTGACCCGTATCCGTAACGGTCAGGCCGCGAACAAAGCTGCGGTCACCATGCC TTCCTCCAAGCTGAAAGTGGCAATTGCCAACGTGCTGAAGGAAGAAGGTTTTATTGAAGATTTTAAAGTTGAAGGCGACACCAA GCCGGAACTGGAACTGACTCTTAAGTATTTCCAGGGTAAAGCTGTTGTAGAAAGCATTCAGCGTGTCAGCCGCCCAGGTCTGC GCATCTACAAACGTAAAGATGAGCTGCCGAAAGTTATGGCCGGTCTGGGTATCGCAGTTGTTTCTACCTCTAAAGGTGTTATGA CTGATCGTGCAGCGCGCCAGGCTGGTCTTGGTGGCGAAATTATCTGCTACGTAGCCTAA (SEQ ID NO: 499)
58_jfrlSe1409-8_UN_O18_	ATGAGCATGCAAGATCCGATCGCGGATATGCTGACCCGTATCCGTAACGGTCAGGCCGCGAACAAAGCTGCGGTCACCATGCC TTCCTCCAAGCTGAAAGTGGCAATTGCCAACGTGCTGAAGGAAGAAGGTTTTATTGAAGATTTTAAAGTTGAAGGCGACACCAA GCCGGAACTGGAACTGACTCTTAAGTATTTCCAGGGTAAAGCTGTTGTAGAAAGCATTCAGCGTGTCAGCCGCCCAGGTCTGC GCATCTACAAACGTAAAGATGAGCTGCCGAAAGTTATGGCCGGTCTGGGTATCGCAGTTGTTTCTACCTCTAAAGGTGTTATGA CTGATCGTGCAGCGCGCCAGGCTGGTCTTGGTGGCGAAATTATCTGCTACGTAGCCTAA (SEQ ID NO: 508)
61_jfrlSe1409-11_UN_O1,3,19_	ATGAGCATGCAAGATCCGATCGCGGATATGCTGACCCGTATCCGTAACGGTCAGGCCGCGAACAAAGCTGCGGTCACCATGCC TTCCTCCAAGCTGAAAGTGGCAATTGCCAACGTGCTGAAGGAAGAAGGTTTTATTGAAGATTTTAAAGTTGAAGGCGACACCAA GCCGGAACTGGAACTGACTCTTAAGTATTTCCAGGGTAAAGCTGTTGTAGAAAGCATTCAGCGTGTCAGCCGCCCAGGTCTGC GCATCTACAAACGTAAAGATGAGCTGCCGAAAGTTATGGCCGGTCTGGGTATCGCAGTTGTTTCTACCTCTAAAGGTGTTATGA CTGATCGTGCAGCGCGCCAGGCTGGTCTTGGTGGCGAAATTATCTGCTACGTAGCCTAA (SEQ ID NO: 517)
67_jfrlSe1409-17_Rissen	ATGAGCATGCAAGATCCGATCGCGGATATGCTGACCCGTATCCGTAACGGTCAGGCCGCGAACAAAGCTGCGGTCACCATGCC TTCCTCCAAGCTGAAAGTGGCAATTGCCAACGTGCTGAAGGAAGAAGGTTTTATTGAAGATTTTAAAGTTGAAGGCGACACCAA GCCGGAACTGGAACTGATTCTTAAGTATTTCCAGGGTAAAGCTGTTGTAGAAAGCATTCAGCGTGTCAGCCGCCCAGGTCTGC GCATCTACAAACGTAAAGATGAGCTGCCGAAAGTTATGGCCGGTCTGGGTATCGCAGTTGTTTCTACCTCTAAAGGTGTTATGA CTGATCGTGCAGCGCGCCAGGCTGGTCTTGGTGGCGAAATTATCTGCTACGTAGCCTAA (SEQ ID NO: 526)
71_jfrlSe1409-21_Amsterdam	ATGAGCATGCAAGATCCGATCGCGGATATGCTGACCCGTATCCGTAACGGTCAGGCCGCGAACAAAGCTGCGGTCACCATGCC TTCCTCCAAGCTGAAAGTGGCAATTGCCAACGTGCTGAAGGAAGAAGGTTTTATTGAAGATTTTAAAGTTGAAGGCGACACCAA GCCGGAACTGGAACTGACTCTTAAGTATTTCCAGGGTAAAGCTGTTGTAGAAAGCATTCAGCGTGTCAGCCGCCCAGGCCTGC GCATCTACAAACGTAAAGATGAGCTGCCGAAAGTTATGGCCGGTCTGGGTATCGCAGTTGTTTCTACCTCTAAAGGTGTTATGA CTGATCGTGCAGCGCGCCAGGCTGGTCTTGGTGGCGAAATTATCTGCTACGTAGCCTAA (SEQ ID NO: 534)
80_HySe09_Enteritidis	ATGAGCATGCAAGATCCGATCGCGGATATGCTGACCCGTATCCGTAACGGTCAGGCCGCGAACAAAGCTGCGGTCACCATGCC TTCCTCCAAGCTGAAAGTGGCAATTGCCAACGTGCTGAAGGAAGAAGGTTTTATTGAAGATTTTAAAGTTGAAGGCGACACCAA GCCGGAACTGGAACTGACTCTTAAGTATTTCCAGGGTAAAGCTGTTGTAGAAAGCATTCAGCGTGTCAGCCGCCCAGGTCTGC GCATCTACAAACGTAAAGATGAGCTGCCGAAAGTTATGGCCGGTCTGGGTATCGCAGTTGTTTCTACCTCTAAAGGTGTTATGA CTGATCGTGCAGCGCGCCAGGCTGGTCTTGGTGGCGAAATTATCTGCTACGTAGCCTAA (SEQ ID NO: 544)
100_HySe29_Schwarzengrund	ATGAGCATGCAAGATCCGATCGCGGATATGCTGACCCGTATCCGTAACGGTCAGGCCGCGAACAAAGCTGCGGTCACCATGCC TTCCTCCAAGCTGAAAGTGGCAATTGCCAACGTGCTGAAGGAAGAAGGTTTTATTGAAGATTTTAAAGTTGAAGGCGACACCAA GCCGGAACTGGAACTGACTCTTAAGTATTTCCAGGGTAAAGCTGTTGTAGAAAGCATTCAGCGTGTCAGCCGCCCAGGTCTGC GCATCTACAAACGTAAAGATGAGCTGCCGAAAGTTATGGCCGGTCTGGGTATCGCAGTTGTTTCTACCTCTAAAGGTGTTATGA CTGATCGTGCAGCGCGCCAGGCTGGTCTTGGTGGCGAAATTATCTGCTACGTAGCCTAA (SEQ ID NO: 554)
103_HySe32_Schwarzengrund	ATGAGCATGCAAGATCCGATCGCGGATATGCTGACCCGTATCCGTAACGGTCAGGCCGCGAACAAAGCTGCGGTCACCATGCC TTCCTCCAAGCTGAAAGTGGCAATTGCCAACGTGCTGAAGGAAGAAGGTTTTATTGAAGATTTTAAAGTTGAAGGCGACACCAA GCCGGAACTGGAACTGACTCTTAAGTATTTCCAGGGTAAAGCTGTTGTAGAAAGCATTCAGCGTGTCAGCCGCCCAGGTCTGC GCATCTACAAACGTAAAGATGAGCTGCCGAAAGTTATGGCCGGTCTGGGTATCGCAGTTGTTTCTACCTCTAAAGGTGTTATGA CTGATCGTGCAGCGCGCCAGGCTGGTCTTGGTGGCGAAATTATCTGCTACGTAGCCTAA (SEQ ID NO: 562)
106_HySe35_Schwarzengrund	ATGAGCATGCAAGATCCGATCGCGGATATGCTGACCCGTATCCGTAACGGTCAGGCCGCGAACAAAGCTGCGGTCACCATGCC TTCCTCCAAGCTGAAAGTGGCAATTGCCAACGTGCTGAAGGAAGAAGGTTTTATTGAAGATTTTAAAGTTGAAGGCGACACCAA GCCGGAACTGGAACTGACTCTTAAGTATTTCCAGGGTAAAGCTGTTGTAGAAAGCATTCAGCGTGTCAGCCGCCCAGGTCTGC GCATCTACAAACGTAAAGATGAGCTGCCGAAAGTTATGGCCGGTCTGGGTATCGCAGTTGTTTCTACCTCTAAAGGTGTTATGA CTGATCGTGCAGCGCGCCAGGCTGGTCTTGGTGGCGAAATTATCTGCTACGTAGCCTAA (SEQ ID NO: 569)

Fig. 25A

	L15
01_NBRC13245T_Typhimurium	ATGCGTTTAAATACTCTGTCTCCGGCCGAAGGCTCCAAAAAGGCGGGTAAACGCCTGGGTCGTGGTATCGGTTCTGGCCTCGG TAAAACCGGTGGTCGTGGTCACAAAGGTCAGAAGTCTCGTTCTGGCGGTGGCGTACGTCGCGGTTTCGAGGGTGGTCAGATGC CTCTGTACCGTCGTCTGCCGAAATTCGGCTTCACTTCTCGCAAAGCAGCGATTACAGCCGAAGTTCGTCTGTCTGACCTGGCG AAAGTAGAAGGCGGCGTTGTAGACCTGAACACGCTGAAAGCGGCAAACATTATCGGTATCCAGATCGAGTTCGCGAAAGTGATC CTGGCTGGCGAAGTCACTACTCCGGTAACTGTTCGTGGCCTGCGTGTTACTAAAGGCGCTCGTGCTGCTATCGAAGCTGCTGG CGGTAAAATCGAGGAATAA (SEQ ID NO: 55)
02_GTC00131_Enteritidis	ATGCGTTTAAATACTCTGTCTCCGGCCGAAGGCTCCAAAAAGGCGGGTAAACGCCTGGGTCGTGGTATCGGTTCTGGCCTCGG TAAAACCGGTGGTCGTGGTCACAAAGGTCAGAAGTCTCGTTCTGGCGGTGGCGTACGTCGCGGTTTCGAGGGTGGTCAGATGC CTCTGTACCGTCGTCTGCCGAAATTCGGCTTCACTTCTCGCAAAGCAGCGATTACAGCCGAAGTTCGTCTGTCTGACTTGGCGA AAGTAGAAGGCGGCGTTGTAGACCTGAACACGCTGAAAGCGGCAAACATTATCGGTATCCAGATCGAGTTCGCGAAAGTGATCT TGGCTGGCGAAGTCACTACTCCGGTAACTGTTCGTGGCCTGCGTGTTACTAAAGGCGCTCGTGCTGCTATCGAAGCTGCTGGC GGTAAAATCGAGGAATAA (SEQ ID NO: 67)
03_GTC09491_Enteritidis	ATGCGTTTAAATACTCTGTCTCCGGCCGAAGGCTCCAAAAAGGCGGGTAAACGCCTGGGTCGTGGTATCGGTTCTGGCCTCGG TAAAACCGGTGGTCGTGGTCACAAAGGTCAGAAGTCTCGTTCTGGCGGTGGCGTACGTCGCGGTTTCGAGGGTGGTCAGATGC CTCTGTACCGTCGTCTGCCGAAATTCGGCTTCACTTCTCGCAAAGCAGCGATTACAGCCGAAGTTCGTCTGTCTGACTTGGCGA AAGTAGAAGGCGGCGTTGTAGACCTGAACACGCTGAAAGCGGCAAACATTATCGGTATCCAGATCGAGTTCGCGAAAGTGATCT TGGCTGGCGAAGTCACTACTCCGGTAACTGTTCGTGGCCTGCGTGTTACTAAAGGCGCTCGTGCTGCTATCGAAGCTGCTGGC GGTAAAATCGAGGAATAA (SEQ ID NO: 76)
04_GTC03838_Enteritidis	ATGCGTTTAAATACTCTGTCTCCGGCCGAAGGCTCCAAAAAGGCGGGTAAACGCCTGGGTCGTGGTATCGGTTCTGGCCTCGG TAAAACCGGTGGTCGTGGTCACAAAGGTCAGAAGTCTCGTTCTGGCGGTGGCGTACGTCGCGGTTTCGAGGGTGGTCAGATGC CTCTGTACCGTCGTCTGCCGAAATTCGGCTTCACTTCTCGCAAAGCAGCGATTACAGCCGAAGTTCGTCTGTCTGACTTGGCGA AAGTAGAAGGCGGCGTTGTAGACCTGAACACGCTGAAAGCGGCAAACATTATCGGTATCCAGATCGAGTTCGCGAAAGTGATCT TGGCTGGCGAAGTCACTACTCCGGTAACTGTTCGTGGCCTGCGTGTTACTAAAGGCGCTCGTGCTGCTATCGAAGCTGCTGGC GGTAAAATCGAGGAATAA (SEQ ID NO: 83)
05_GTC08914_Enteritidis	ATGCGTTTAAATACTCTGTCTCCGGCCGAAGGCTCCAAAAAGGCGGGTAAACGCCTGGGTCGTGGTATCGGTTCTGGCCTCGG TAAAACCGGTGGTCGTGGTCACAAAGGTCAGAAGTCTCGTTCTGGCGGTGGCGTACGTCGCGGTTTCGAGGGTGGTCAGATGC CTCTGTACCGTCGTCTGCCGAAATTCGGCTTCACTTCTCGCAAAGCAGCGATTACAGCCGAAGTTCGTCTGTCTGACTTGGCGA AAGTAGAAGGCGGCGTTGTAGACCTGAACACGCTGAAAGCGGCAAACATTATCGGTATCCAGATCGAGTTCGCGAAAGTGATCT TGGCTGGCGAAGTCACTACTCCGGTAACTGTTCGTGGCCTGCGTGTTACTAAAGGCGCTCGTGCTGCTATCGAAGCTGCTGGC GGTAAAATCGAGGAATAA (SEQ ID NO: 90)
06_GTC09421_Enteritidis	ATGCGTTTAAATACTCTGTCTCCGGCCGAAGGCTCCAAAAAGGCGGGTAAACGCCTGGGTCGTGGTATCGGTTCTGGCCTCGG TAAAACCGGTGGTCGTGGTCACAAAGGTCAGAAGTCTCGTTCTGGCGGTGGCGTACGTCGCGGTTTCGAGGGTGGTCAGATGC CTCTGTACCGTCGTCTGCCGAAATTCGGCTTCACTTCTCGCAAAGCAGCGATTACAGCCGAAGTTCGTCTGTCTGACTTGGCGA AAGTAGAAGGCGGCGTTGTAGACCTGAACACGCTGAAAGCGGCAAACATTATCGGTATCCAGATCGAGTTCGCGAAAGTGATCT TGGCTGGCGAAGTCACTACTCCGGTAACTGTTCGTGGCCTGCGTGTTACTAAAGGCGCTCGTGCTGCTATCGAAGCTGCTGGC GGTAAAATCGAGGAATAA (SEQ ID NO: 100)
07_GTC09489_Enteritidis	ATGCGTTTAAATACTCTGTCTCCGGCCGAAGGCTCCAAAAAGGCGGGTAAACGCCTGGGTCGTGGTATCGGTTCTGGCCTCGG TAAAACCGGTGGTCGTGGTCACAAAGGTCAGAAGTCTCGTTCTGGCGGTGGCGTACGTCGCGGTTTCGAGGGTGGTCAGATGC CTCTGTACCGTCGTCTGCCGAAATTCGGCTTCACTTCTCGCAAAGCAGCGATTACAGCCGAAGTTCGTCTGTCTGACTTGGCGA AAGTAGAAGGCGGCGTTGTAGACCTGAACACGCTGAAAGCGGCAAACATTATCGGTATCCAGATCGAGTTCGCGAAAGTGATCT TGGCTGGCGAAGTCACTACTCCGGTAACTGTTCGTGGCCTGCGTGTTACTAAAGGCGCTCGTGCTGCTATCGAAGCTGCTGGC GGTAAAATCGAGGAATAA (SEQ ID NO: 110)
08_NBRC3163_Pullorum_Gallinarum	ATGCGTTTAAATACTCTGTCTCCGGCCGAAGGCTCCAAAAAGGCGGGTAAACGCCTGGGTCGTGGTATCGGTTCTGGCCTCGG TAAAACCGGTGGTCGTGGTCACAAAGGTCAGAAGTCTCGTTCTGGCGGTGGCGTACGTCGCGGTTTCGAGGGTGGTCAGATGC CTCTGTACCGTCGTCTGCCGAAATTCGGCTTCACTTCTCGCAAAGCAGCGATTACAGCCGAAGTTCGTCTGTCTGACTTGGCGA AAGTAGAAGGCGGCGTTGTAGACCTGAACACGCTGAAAGCGGCAAACATTATCGGTATCCAGATCGAGTTCGCGAAAGTGATCC TGGCTGGCGAAGTCACTACTCCGGTAACTGTTCGTGGCCTGCGTGTTACTAAAGGCGCTCGTGCTGCTATCGAAGCTGCTGGC GGTAAAATCGAGGAATAA (SEQ ID NO: 117)
09_NBRC3313_Enteritidis	ATGCGTTTAAATACTCTGTCTCCGGCCGAAGGCTCCAAAAAGGCGGGTAAACGCCTGGGTCGTGGTATCGGTTCTGGCCTCGG TAAAACCGGTGGTCGTGGTCACAAAGGTCAGAAGTCTCGTTCTGGCGGTGGCGTACGTCGCGGTTTCGAGGGTGGTCAGATGC CTCTGTACCGTCGTCTGCCGAAATTCGGCTTCACTTCTCGCAAAGCAGCGATTACAGCCGAAGTTCGTCTGTCTGACTTGGCGA AAGTAGAAGGCGGCGTTGTAGACCTGAACACGCTGAAAGCGGCAAACATTATCGGTATCCAGATCGAGTTCGCGAAAGTGATCT TGGCTGGCGAAGTCACTACTCCGGTAACTGTTCGTGGCCTGCGTGTTACTAAAGGCGCTCGTGCTGCTATCGAAGCTGCTGGC GGTAAAATCGAGGAATAA (SEQ ID NO: 127)
10_NBRC12529_Typhimurium	ATGCGTTTAAATACTCTGTCTCCGGCCGAAGGCTCCAAAAAGGCGGGTAAACGCCTGGGTCGTGGTATCGGTTCTGGCCTCGG TAAAACCGGTGGTCGTGGTCACAAAGGTCAGAAGTCTCGTTCTGGCGGTGGCGTACGTCGCGGTTTCGAGGGTGGTCAGATGC CTCTGTACCGTCGTCTGCCGAAATTCGGCTTCACTTCTCGCAAAGCAGCGATTACAGCCGAAGTTCGTCTGTCTGACCTGGCG AAAGTAGAAGGCGGCGTTGTAGACCTGAACACGCTGAAAGCGGCAAACATTATCGGTATCCAGATCGAGTTCGCGAAAGTGATC CTGGCTGGCGAAGTCACTACTCCGGTAACTGTTCGTGGCCTGCGTGTTACTAAAGGCGCTCGTGCTGCTATCGAAGCTGCTGG CGGTAAAATCGAGGAATAA (SEQ ID NO: 134)
11_NBRC14193_Typhimurium	ATGCGTTTAAATACTCTGTCTCCGGCCGAAGGCTCCAAAAAGGCGGGTAAACGCCTGGGTCGTGGTATCGGTTCTGGCCTCGG TAAAACCGGTGGTCGTGGTCACAAAGGTCAGAAGTCTCGTTCTGGCGGTGGCGTACGTCGCGGTTTCGAGGGTGGTCAGATGC CTCTGTACCGTCGTCTGCCGAAATTCGGCTTCACTTCTCGCAAAGCAGCGATTACAGCCGAAGTTCGTCTGTCTGACCTGGCG AAAGTAGAAGGCGGCGTTGTAGACCTGAACACGCTGAAAGCGGCAAACATTATCGGTATCCAGATCGAGTTCGCGAAAGTGATC CTGGCTGGCGAAGTCACTACTCCGGTAACTGTTCGTGGCCTGCGTGTTACTAAAGGCGCTCGTGCTGCTATCGAAGCTGCTGG CGGTAAAATCGAGGAATAA (SEQ ID NO: 141)
12_NBRC14194_Typhimurium	ATGCGTTTAAATACTCTGTCTCCGGCCGAAGGCTCCAAAAAGGCGGGTAAACGCCTGGGTCGTGGTATCGGTTCTGGCCTCGG TAAAACCGGTGGTCGTGGTCACAAAGGTCAGAAGTCTCGTTCTGGCGGTGGCGTACGTCGCGGTTTCGAGGGTGGTCAGATGC CTCTGTACCGTCGTCTGCCGAAATTCGGCTTCACTTCTCGCAAAGCAGCGATTACAGCCGAAGTTCGTCTGTCTGACCTGGCG AAAGTAGAAGGCGGCGTTGTAGACCTGAACACGCTGAAAGCGGCAAACATTATCGGTATCCAGATCGAGTTCGCGAAAGTGATC CTGGCTGGCGAAGTCACTACTCCGGTAACTGTTCGTGGCCTGCGTGTTACTAAAGGCGCTCGTGCTGCTATCGAAGCTGCTGG CGGTAAAATCGAGGAATAA (SEQ ID NO: 148)
13_NBRC14209_Typhimurium	ATGCGTTTAAATACTCTGTCTCCGGCCGAAGGCTCCAAAAAGGCGGGTAAACGCCTGGGTCGTGGTATCGGTTCTGGCCTCGG TAAAACCGGTGGTCGTGGTCACAAAGGTCAGAAGTCTCGTTCTGGCGGTGGCGTACGTCGCGGTTTCGAGGGTGGTCAGATGC CTCTGTACCGTCGTCTGCCGAAATTCGGCTTCACTTCTCGCAAAGCAGCGATTACAGCCGAAGTTCGTCTGTCTGACCTGGCG AAAGTAGAAGGCGGCGTTGTAGACCTGAACACGCTGAAAGCGGCAAACATTATCGGTATCCAGATCGAGTTCGCGAAAGTGATC CTGGCTGGCGAAGTCACTACTCCGGTAACTGTTCGTGGCCTGCGTGTTACTAAAGGCGCTCGTGCTGCTATCGAAGCTGCTGG CGGTAAAATCGAGGAATAA (SEQ ID NO: 155)
14_NBRC14210_Typhimurium	ATGCGTTTAAATACTCTGTCTCCGGCCGAAGGCTCCAAAAAGGCGGGTAAACGCCTGGGTCGTGGTATCGGTTCTGGCCTCGG TAAAACCGGTGGTCGTGGTCACAAAGGTCAGAAGTCTCGTTCTGGCGGTGGCGTACGTCGCGGTTTCGAGGGTGGTCAGATGC CTCTGTACCGTCGTCTGCCGAAATTCGGCTTCACTTCTCGCAAAGCAGCGATTACAGCCGAAGTTCGTCTGTCTGACCTGGCG AAAGTAGAAGGCGGCGTTGTAGACCTGAACACGCTGAAAGCGGCAAACATTATCGGTATCCAGATCGAGTTCGCGAAAGTGATC CTGGCTGGCGAAGTCACTACTCCGGTAACTGTTCGTGGCCTGCGTGTTACTAAAGGCGCTCGTGCTGCTATCGAAGCTGCTGG CGGTAAAATCGAGGAATAA (SEQ ID NO: 162)
15_NBRC14211_Typhimurium	
16_NBRC14212_Typhimurium	ATGCGTTTAAATACTCTGTCTCCGGCCGAAGGCTCCAAAAAGGCGGGTAAACGCCTGGGTCGTGGTATCGGTTCTGGCCTCGG TAAAACCGGTGGTCGTGGTCACAAAGGTCAGAAGTCTCGTTCTGGCGGTGGCGTACGTCGCGGTTTCGAGGGTGGTCAGATGC CTCTGTACCGTCGTCTGCCGAAATTCGGCTTCACTTCTCGCAAAGCAGCGATTACAGCCGAAGTTCGTCTGTCTGACCTGGCG AAAGTAGAAGGCGGCGTTGTAGACCTGAACACGCTGAAAGCGGCAAACATTATCGGTATCCAGATCGAGTTCGCGAAAGTGATC CTGGCTGGCGAAGTCACTACTCCGGTAACTGTTCGTGGCCTGCGTGTTACTAAAGGCGCTCGTGCTGCTATCGAAGCTGCTGG CGGTAAAATCGAGGAATAA (SEQ ID NO: 175)
17_NBRC15181_Typhimurium	ATGCGTTTAAATACTCTGTCTCCGGCCGAAGGCTCCAAAAAGGCGGGTAAACGCCTGGGTCGTGGTATCGGTTCTGGCCTCGG TAAAACCGGTGGTCGTGGTCACAAAGGTCAGAAGTCTCGTTCTGGCGGTGGCGTACGTCGCGGTTTCGAGGGTGGTCAGATGC CTCTGTACCGTCGTCTGCCGAAATTCGGCTTCACTTCTCGCAAAGCAGCGATTACAGCCGAAGTTCGTCTGTCTGACCTGGCG AAAGTAGAAGGCGGCGTTGTAGACCTGAACACGCTGAAAGCGGCAAACATTATCGGTATCCAGATCGAGTTCGCGAAAGTGATC CTGGCTGGCGAAGTCACTACTCCGGTAACTGTTCGTGGCCTGCGTGTTACTAAAGGCGCTCGTGCTGCTATCGAAGCTGCTGG CGGTAAAATCGAGGAATAA (SEQ ID NO: 182)

Fig. 25B

18_NBRC15182_Minnesota	ATGCGTTTAAATACTCTGTCTCCGGCCGAAGGCTCCAAAAAGGCGGGTAAACGCCTGGGTCGTGGTATCGGTTCTGGCCTCGG TAAAACCGGTGGTCGTGGTCACAAAGGTCAGAAGTCTCGTTCTGGCGGTGGCGTACGTCGCGGTTTCGAGGGTGGTCAGATGC CTCTGTACCGTCGTCTGCCGAAATTCGGCTTCACTTCTCGCAAAGCAGCGATTACAGCCGAAGTTCGTCTGTCTGACCTGGCG AAAGTAGAAGGCGGCGTTGTAGACCTGAACACGCTGAAAGCGGCAAACATTATCGGTATCCAGATCGAGTTCGCGAAAGTGATC CTGGCTGGCGAAGTCACTACTCCGGTAACTGTTCGTGGCCTGCGTGTTACTAAAGGCGCTCGTGCTGCTATCGAAGCTGCTGG CGGTAAAATCGAGGAATAA (SEQ ID NO: 192)
19_NBRC15183_Minnesota	ATGCGTTTAAATACTCTGTCTCCGGCCGAAGGCTCCAAAAAGGCGGGTAAACGCCTGGGTCGTGGTATCGGTTCTGGCCTCGG TAAAACCGGTGGTCGTGGTCACAAAGGTCAGAAGTCTCGTTCTGGCGGTGGCGTACGTCGCGGTTTCGAGGGTGGTCAGATGC CTCTGTACCGTCGTCTGCCGAAATTCGGCTTCACTTCTCGCAAAGCAGCGATTACAGCCGAAGTTCGTCTGTCTGACCTGGCG AAAGTAGAAGGCGGCGTTGTAGACCTGAACACGCTGAAAGCGGCAAACATTATCGGTATCCAGATCGAGTTCGCGAAAGTGATC CTGGCTGGCGAAGTCACTACTCCGGTAACTGTTCGTGGCCTGCGTGTTACTAAAGGCGCTCGTGCTGCTATCGAAGCTGCTGG CGGTAAAATCGAGGAATAA (SEQ ID NO: 202)
20_NBRC15184_Minnesota	ATGCGTTTAAATACTCTGTCTCCGGCCGAAGGCTCCAAAAAGGCGGGTAAACGCCTGGGTCGTGGTATCGGTTCTGGCCTCGG TAAAACCGGTGGTCGTGGTCACAAAGGTCAGAAGTCTCGTTCTGGCGGTGGCGTACGTCGCGGTTTCGAGGGTGGTCAGATGC CTCTGTACCGTCGTCTGCCGAAATTCGGCTTCACTTCTCGCAAAGCAGCGATTACAGCCGAAGTTCGTCTGTCTGACCTGGCG AAAGTAGAAGGCGGCGTTGTAGACCTGAACACGCTGAAAGCGGCAAACATTATCGGTATCCAGATCGAGTTCGCGAAAGTGATC CTGGCTGGCGAAGTCACTACTCCGGTAACTGTTCGTGGCCTGCGTGTTACTAAAGGCGCTCGTGCTGCTATCGAAGCTGCTGG CGGTAAAATCGAGGAATAA (SEQ ID NO: 209)
21_NBRC15185_Minnesota	ATGCGTTTAAATACTCTGTCTCCGGCCGAAGGCTCCAAAAAGGCGGGTAAACGCCTGGGTCGTGGTATCGGTTCTGGCCTCGG TAAAACCGGTGGTCGTGGTCACAAAGGTCAGAAGTCTCGTTCTGGCGGTGGCGTACGTCGCGGTTTCGAGGGTGGTCAGATGC CTCTGTACCGTCGTCTGCCGAAATTCGGCTTCACTTCTCGCAAAGCAGCGATTACAGCCGAAGTTCGTCTGTCTGACCTGGCG AAAGTAGAAGGCGGCGTTGTAGACCTGAACACGCTGAAAGCGGCAAACATTATCGGTATCCAGATCGAGTTCGCGAAAGTGATC CTGGCTGGCGAAGTCACTACTCCGGTAACTGTTCGTGGCCTGCGTGTTACTAAAGGCGCTCGTGCTGCTATCGAAGCTGCTGG CGGTAAAATCGAGGAATAA (SEQ ID NO: 216)
22_NBRC15186_Minnesota	ATGCGTTTAAATACTCTGTCTCCGGCCGAAGGCTCCAAAAAGGCGGGTAAACGCCTGGGTCGTGGTATCGGTTCTGGCCTCGG TAAAACCGGTGGTCGTGGTCACAAAGGTCAGAAGTCTCGTTCTGGCGGTGGCGTACGTCGCGGTTTCGAGGGTGGTCAGATGC CTCTGTACCGTCGTCTGCCGAAATTCGGCTTCACTTCTCGCAAAGCAGCGATTACAGCCGAAGTTCGTCTGTCTGACCTGGCG AAAGTAGAAGGCGGCGTTGTAGACCTGAACACGCTGAAAGCGGCAAACATTATCGGTATCCAGATCGAGTTCGCGAAAGTGATC CTGGCTGGCGAAGTCACTACTCCGGTAACTGTTCGTGGCCTGCGTGTTACTAAAGGCGCTCGTGCTGCTATCGAAGCTGCTGG CGGTAAAATCGAGGAATAA (SEQ ID NO: 223)
23_NBRC15187_Minnesota	ATGCGTTTAAATACTCTGTCTCCGGCCGAAGGCTCCAAAAAGGCGGGTAAACGCCTGGGTCGTGGTATCGGTTCTGGCCTCGG TAAAACCGGTGGTCGTGGTCACAAAGGTCAGAAGTCTCGTTCTGGCGGTGGCGTACGTCGCGGTTTCGAGGGTGGTCAGATGC CTCTGTACCGTCGTCTGCCGAAATTCGGCTTCACTTCTCGCAAAGCAGCGATTACAGCCGAAGTTCGTCTGTCTGACCTGGCG AAAGTAGAAGGCGGCGTTGTAGACCTGAACACGCTGAAAGCGGCAAACATTATCGGTATCCAGATCGAGTTCGCGAAAGTGATC CTGGCTGGCGAAGTCACTACTCCGGTAACTGTTCGTGGCCTGCGTGTTACTAAAGGCGCTCGTGCTGCTATCGAAGCTGCTGG CGGTAAAATCGAGGAATAA (SEQ ID NO: 230)
24_NBRC100797_Abony	ATGCGTTTAAATACTCTGTCTCCGGCCGAAGGCTCCAAAAAGGCGGGTAAACGCCTGGGTCGTGGTATCGGTTCTGGCCTCGG TAAAACCGGTGGTCGTGGTCACAAAGGTCAGAAGTCTCGTTCTGGCGGTGGCGTACGTCGCGGTTTCGAGGGTGGTCAGATGC CTCTGTACCGTCGTCTGCCGAAATTCGGCTTCACTTCTCGCAAAGCAGCGATTACAGCCGAAGTTCGTCTGTCTGACCTGGCG AAAGTAGAAGGCGGCGTTGTAGACCTGAACACGCTGAAAGCGGCAAACATTATCGGTATCCAGATCGAGTTCGCGAAAGTGATC CTGGCTGGCGAAGTCACTACTCCGGTAACTGTTCGTGGCCTGCGTGTTACTAAAGGCGCTCGTGCTGCTATCGAAGCTGCTGG CGGTAAAATCGAGGAATAA (SEQ ID NO: 237)
25_NBRC105684_Choleraesuis	ATGCGTTTAAATACTCTGTCTCCGGCCGAAGGCTCCAAAAAGGCGGGTAAACGCCTGGGTCGTGGTATCGGTTCTGGCCTCGG TAAAACCGGTGGTCGTGGTCACAAAGGTCAGAAGTCTCGTTCTGGCGGTGGCGTACGTCGCGGTTTCGAGGGTGGTCAGATGC CTCTGTACCGTCGTCTGCCGAAATTCGGCTTCACTTCTCGCAAAGCAGCGATTACAGCCGAAGTTCGTCTGTCTGACCTGGCG AAAGTAGAAGGCGGCGTTGTAGACCTGAACACGCTGAAAGCGGCAAACATTATCGGTATCCAGATCGAGTTCGCGAAAGTGATC CTGGCTGGCGAAGTCACTACTCCGGTAACTGTTCGTGGCCTGCGTGTTACTAAAGGCGCTCGTGCTGCTATCGAAGCTGCTGG CGGTAAAATCGAGGAATAA (SEQ ID NO: 247)
26_NBRC105726_Typhimurium	ATGCGTTTAAATACTCTGTCTCCGGCCGAAGGCTCCAAAAAGGCGGGTAAACGCCTGGGTCGTGGTATCGGTTCTGGCCTCGG TAAAACCGGTGGTCGTGGTCACAAAGGTCAGAAGTCTCGTTCTGGCGGTGGCGTACGTCGCGGTTTCGAGGGTGGTCAGATGC CTCTGTACCGTCGTCTGCCGAAATTCGGCTTCACTTCTCGCAAAGCAGCGATTACAGCCGAAGTTCGTCTGTCTGACCTGGCG AAAGTAGAAGGCGGCGTTGTAGACCTGAACACGCTGAAAGCGGCAAACATTATCGGTATCCAGATCGAGTTCGCGAAAGTGATC CTGGCTGGCGAAGTCACTACTCCGGTAACTATTCGTGGCCTGCGTGTTACTAAAGGCGCTCGTGCTGCTATCGAAGCTGCTGG CGGTAAAATCGAGGAATAA (SEQ ID NO: 257)
27_JCM3919_UN_O7_	ATGCGTTTAAATACTCTGTCTCCGGCCGAAGGCTCCAAAAAGGCGGGTAAACGCCTGGGTCGTGGTATCGGTTCTGGCCTCGG TAAAACCGGTGGTCGTGGTCACAAAGGTCAGAAGTCTCGTTCTGGCGGTGGCGTACGTCGCGGTTTCGAGGGTGGTCAGATGC CTCTGTACCGTCGTCTGCCGAAATTCGGCTTCACTTCTCGCAAAGCAGCGATTACAGCCGAAGTTCGTCTGTCTGACCTGGCG AAAGTAGAAGGCGGCGTTGTAGACCTGAACACGCTGAAAGCGGCAAACATTATCGGTATCCAGATCGAGTTCGCGAAAGTGATC CTGGCTGGCGAAGTCACTACTCCGGTAACTGTTCGTGGCCTGCGTGTTACTAAAGGCGCTCGTGCTGCTATCGAAGCTGCTGG CGGTAAAATCGAGGAATAA (SEQ ID NO: 266)
28_NBRC15335_Minnesota	ATGCGTTTAAATACTCTGTCTCCGGCCGAAGGCTCCAAAAAGGCGGGTAAACGCCTGGGTCGTGGTATCGGTTCTGGCCTCGG TAAAACCGGTGGTCGTGGTCACAAAGGTCAGAAGTCTCGTTCTGGCGGTGGCGTACGTCGCGGTTTCGAGGGTGGTCAGATGC CTCTGTACCGTCGTCTGCCGAAATTCGGCTTCACTTCTCGCAAAGCAGCGATTACAGCCGAAGTTCGTCTGTCTGACCTGGCG AAAGTAGAAGGCGGCGTTGTAGACCTGAACACGCTGAAAGCGGCAAACATTATCGGTATCCAGATCGAGTTCGCGAAAGTGATC CTGGCTGGCGAAGTCACTACTCCGGTAACTGTTCGTGGCCTGCGTGTTACTAAAGGCGCTCGTGCTGCTATCGAAGCTGCTGG CGGTAAAATCGAGGAATAA (SEQ ID NO: 273)
29_GTC09490_Enteritidis	ATGCGTTTAAATACTCTGTCTCCGGCCGAAGGCTCCAAAAAGGCGGGTAAACGCCTGGGTCGTGGTATCGGTTCTGGCCTCGG TAAAACCGGTGGTCGTGGTCACAAAGGTCAGAAGTCTCGTTCTGGCGGTGGCGTACGTCGCGGTTTCGAGGGTGGTCAGATGC CTCTGTACCGTCGTCTGCCGAAATTCGGCTTCACTTCTCGCAAAGCAGCGATTACAGCCGAAGTTCGTCTGTCTGACTTGGCGA AAGTAGAAGGCGGCGTTGTAGACCTGAACACGCTGAAAGCGGCAAACATTATCGGTATCCAGATCGAGTTCGCGAAAGTGATCT TGGCTGGCGAAGTCACTACTCCGGTAACTGTTCGTGGCCTGCGTGTTACTAAAGGCGCTCGTGCTGCTATCGAAGCTGCTGGC GGTAAAATCGAGGAATAA (SEQ ID NO: 280)
30_GTC09492_Braenderup	ATGCGTTTAAATACTCTGTCTCCGGCCGAAGGCTCCAAAAAGGCGGGTAAACGCCTGGGTCGTGGTATCGGTTCTGGCCTCGG TAAAACCGGTGGTCGTGGTCACAAAGGTCAGAAGTCTCGTTCTGGCGGTGGCGTACGTCGCGGTTTCGAGGGTGGTCAGATGC CTCTGTACCGTCGTCTGCCGAAATTCGGCTTCACTTCTCGCAAAGCAGCGATTACAGCCGAAGTTCGTCTGTCTGACCTGGCG AAAGTAGAAGGCGGCGTTGTAGACCTGAACACGCTGAAAGCGGCAAACATTATCGGTATCCAGATCGAGTTCGCGAAAGTGATC CTGGCTGGCGAAGTCACTACTCCGGTAACTGTTCGTGGCCTGCGTGTTACTAAAGGCGCTCGTGCTGCTATCGAAGCTGCTGG CGGTAAAATCGAGGAATAA (SEQ ID NO: 287)
31_GTC09493_Pakistan	ATGCGTTTAAATACTCTGTCTCCGGCCGAAGGCTCCAAAAAGGCGGGTAAACGCCTGGGTCGTGGTATCGGTTCTGGCCTCGG TAAAACCGGTGGTCGTGGTCACAAAGGTCAGAAGTCTCGTTCTGGCGGTGGCGTACGTCGCGGTTTCGAGGGTGGTCAGATGC CTCTGTACCGTCGTCTGCCGAAATTCGGCTTCACTTCTCGCAAAGCAGCGATTACAGCCGAAGTTCGTCTGTCTGACCTGGCG AAAGTAGAAGGCGGCGTTGTAGACCTGAACACGCTGAAAGCGGCAAACATTATCGGTATCCAGATCGAGTTCGCGAAAGTGATC CTGGCTGGCGAAGTCACTACTCCGGTAACTGTTCGTGGCCTGCGTGTTACTAAAGGCGCTCGTGCTGCTATCGAAGCTGCTGG CGGTAAAATCGAGGAATAA (SEQ ID NO: 294)

Fig. 25C

32_GTC09549_Typhimurium	ATGCGTTTAAATACTCTGTCTCCGGCCGAAGGCTCCAAAAAGGCGGGTAAACGCCTGGGTCGTGGTATCGGTTCTGGCCTCGG TAAAACCGGTGGTCGTGGTCACAAAGGTCAGAAGTCTCGTTCTGGCGGTGGCGTACGTCGCGGTTTCGAGGGTGGTCAGATGC CTCTGTACCGTCGTCTGCCGAAATTCGGCTTCACTTCTCGCAAAGCAGCGATTACAGCCGAAGTTCGTCTGTCTGACCTGGCG AAAGTAGAAGGCGGCGTTGTAGACCTGAACACGCTGAAAGCGGCAAACATTATCGGTATCCAGATCGAGTTCGCGAAAGTGATC CTGGCTGGCGAAGTCACTACTCCGGTAACTGTTCGTGGCCTGCGTGTTACTAAAGGCGCTCGTGCTGCTATCGAAGCTGCTGG CGGTAAAATCGAGGAATAA (SEQ ID NO: 301)
33_ATCCBAA-1675_Infantis	ATGCGTTTAAATACTCTGTCTCCGGCCGAAGGCTCCAAAAAGGCGGGTAAACGCCTGGGTCGTGGTATCGGTTCTGGCCTCGG TAAAACCGGTGGTCGTGGTCACAAAGGTCAGAAGTCTCGTTCTGGCGGTGGCGTACGTCGCGGTTTCGAGGGTGGTCAGATGC CTCTGTACCGTCGTCTGCCGAAATTCGGCTTCACTTCTCGCAAAGCAGCGATTACAGCCGAAGTTCGTCTGTCTGACCTGGCG AAAGTAGAAGGCGGCGTTGTAGACCTGAACACGCTGAAAGCGGCAAACATTATCGGTATCCAGATCGAGTTCGCGAAAGTGATC CTGGCTGGCGAAGTCACTACTCCGGTAACTGTTCGTGGCCTGCGTGTTACTAAAGGCGCTCGTGCTGCTATCGAAGCTGCTGG CGGTAAAATCGAGGAATAA (SEQ ID NO: 308)
34_ATCCBAA-1738_Thompson	ATGCGTTTAAATACTCTGTCTCCGGCCGAAGGCTCCAAAAAGGCGGGTAAACGCCTGGGTCGTGGTATCGGTTCTGGCCTCGG TAAAACCGGTGGTCGTGGTCACAAAGGTCAGAAGTCTCGTTCTGGCGGTGGCGTACGTCGCGGTTTCGAGGGTGGTCAGATGC CTCTGTACCGTCGTCTGCCGAAATTCGGCTTCACTTCTCGCAAAGCAGCGATTACAGCCGAAGTTCGTCTGTCTGACCTGGCG AAAGTAGAAGGCGGCGTTGTAGACCTGAACACGCTGAAAGCGGCAAACATTATCGGTATCCAGATCGAGTTCGCGAAAGTGATC CTGGCTGGCGAAGTCACTACTCCGGTAACTGTTCGTGGCCTGCGTGTTACTAAAGGCGCTCGTGCTGCTATCGAAGCTGCTGG CGGTAAAATCGAGGAATAA (SEQ ID NO: 319)
35_ATCC9712_Saintpaul	ATGCGTTTAAATACTCTGTCTCCGGCCGAAGGCTCCAAAAAGGCGGGTAAACGCCTGGGTCGTGGTATCGGTTCTGGCCTCGG TAAAACCGGTGGTCGTGGTCACAAAGGTCAGAAGTCTCGTTCTGGCGGTGGCGTACGTCGTGGTTTCGAGGGTGGTCAGATGC CTCTGTACCGTCGTCTGCCGAAATTCGGCTTCACTTCTCGCAAAGCAGCGATTACAGCCGAAGTTCGTCTGTCTGACCTGGCG AAAGTAGAAGGCGGCGTTGTAGACCTGAACACGCTGAAAGCGGCAAACATTATCGGTATCCAGATCGAGTTCGCGAAAGTGATC CTGGCTGGCGAAGTCACTACTCCGGTAACTGTTCGTGGCCTGCGTGTTACTAAAGGCGCTCGTGCTGCTATCGAAGCTGCTGG CGGTAAAATCGAGGAATAA (SEQ ID NO: 329)
36_jfrlSe1402-1_Infantis	ATGCGTTTAAATACTCTGTCTCCGGCCGAAGGCTCCAAAAAGGCGGGTAAACGCCTGGGTCGTGGTATCGGTTCTGGCCTCGG TAAAACCGGTGGTCGTGGTCACAAAGGTCAGAAGTCTCGTTCTGGCGGTGGCGTACGTCGCGGTTTCGAGGGTGGTCAGATGC CTCTGTACCGTCGTCTGCCGAAATTCGGCTTCACTTCTCGCAAAGCAGCGATTACAGCCGAAGTTCGTCTGTCTGACCTGGCG AAAGTAGAAGGCGGCGTTGTAGACCTGAACACGCTGAAAGCGGCAAACATTATCGGTATCCAGATCGAGTTCGCGAAAGTGATC CTGGCTGGCGAAGTCACTACTCCGGTAACTGTTCGTGGCCTGCGTGTTACTAAAGGCGCTCGTGCTGCTATCGAAGCTGCTGG CGGTAAAATCGAGGAATAA (SEQ ID NO: 340)
38_jfrlSe1402-3_Brandenburg	ATGCGTTTAAATACTCTGTCTCCGGCCGAAGGCTCCAAAAAGGCGGGTAAACGCCTGGGTCGTGGTATCGGTTCTGGCCTCGG TAAAACCGGTGGTCGTGGTCACAAAGGTCAGAAGTCTCGTTCTGGCGGTGGCGTACGTCGCGGTTTCGAGGGTGGTCAGATGC CTCTGTACCGTCGTCTGCCGAAATTCGGCTTCACTTCTCGCAAAGCAGCGATTACAGCCGAAGTTCGTCTGTCTGACCTGGCG AAAGTAGAAGGCGGCGTTGTAGACCTGAACACGCTGAAAGCGGCAAACATTATCGGTATCCAGATCGAGTTCGCGAAAGTGATC CTGGCTGGCGAAGTCACTACTCCGGTAACTGTTCGTGGCCTGCGTGTTACTAAAGGCGCTCGTGCTGCTATCGAAGCTGCTGG CGGTAAAATCGAGGAATAA (SEQ ID NO: 350)
39_jfrlSe1402-4_Infantis	ATGCGTTTAAATACTCTGTCTCCGGCCGAAGGCTCCAAAAAGGCGGGTAAACGCCTGGGTCGTGGTATCGGTTCTGGCCTCGG TAAAACCGGTGGTCGTGGTCACAAAGGTCAGAAGTCTCGTTCTGGCGGTGGCGTACGTCGCGGTTTCGAGGGTGGTCAGATGC CTCTGTACCGTCGTCTGCCGAAATTCGGCTTCACTTCTCGCAAAGCAGCGATTACAGCCGAAGTTCGTCTGTCTGACCTGGCG AAAGTAGAAGGCGGCGTTGTAGACCTGAACACGCTGAAAGCGGCAAACATTATCGGTATCCAGATCGAGTTCGCGAAAGTGATC CTGGCTGGCGAAGTCACTACTCCGGTAACTGTTCGTGGCCTGCGTGTTACTAAAGGCGCTCGTGCTGCTATCGAAGCTGCTGG CGGTAAAATCGAGGAATAA (SEQ ID NO: 357)
40_jfrlSe1402-5_Brandenburg	ATGCGTTTAAATACTCTGTCTCCGGCCGAAGGCTCCAAAAAGGCGGGTAAACGCCTGGGTCGTGGTATCGGTTCTGGCCTCGG TAAAACCGGTGGTCGTGGTCACAAAGGTCAGAAGTCTCGTTCTGGCGGTGGCGTACGTCGCGGTTTCGAGGGTGGTCAGATGC CTCTGTACCGTCGTCTGCCGAAATTCGGCTTCACTTCTCGCAAAGCAGCGATTACAGCCGAAGTTCGTCTGTCTGACCTGGCG AAAGTAGAAGGCGGCGTTGTAGACCTGAACACGCTGAAAGCGGCAAACATTATCGGTATCCAGATCGAGTTCGCGAAAGTGATC CTGGCTGGCGAAGTCACTACTCCGGTAACTGTTCGTGGCCTGCGTGTTACTAAAGGCGCTCGTGCTGCTATCGAAGCTGCTGG CGGTAAAATCGAGGAATAA (SEQ ID NO: 367)
41_jfrlSe1402-6_Rissen	ATGCGTTTAAATACTCTGTCTCCGGCCGAAGGCTCCAAAAAGGCGGGTAAACGCCTGGGTCGTGGTATCGGTTCTGGCCTCGG TAAAACCGGTGGTCGTGGTCACAAAGGTCAGAAGTCTCGTTCTGGCGGTGGCGTACGTCGCGGTTTCGAGGGTGGTCAGATGC CTCTGTACCGTCGTCTGCCGAAATTCGGCTTCACTTCTCGCAAAGCAGCGATTACAGCCGAAGTTCGTCTGTCTGACCTGGCG AAAGTAGAAGGCGGCGTTGTAGACCTGAACACGCTGAAAGCGGCAAACATTATCGGTATCCAGATCGAGTTCGCGAAAGTGATC CTGGCTGGCGAAGTCACTACTCCGGTAACTGTTCGTGGCCTGCGTGTTACTAAAGGCGCTCGTGCTGCTATCGAAGCTGCTGG CGGTAAAATCGAGGAATAA (SEQ ID NO: 374)
42_jfrlSe1402-7_Orion	ATGCGTTTAAATACTCTGTCTCCGGCCGAAGGCTCCAAAAAGGCGGGTAAACGCCTGGGTCGTGGTATCGGTTCTGGCCTCGG TAAAACCGGTGGTCGTGGTCACAAAGGTCAGAAGTCTCGTTCTGGCGGTGGCGTACGTCGCGGTTTCGAGGGTGGTCAGATGC CTCTGTACCGTCGTCTGCCGAAATTCGGCTTCACTTCTCGCAAAGCAGCGATTACAGCCGAAGTTCGTCTGTCTGACCTGGCG AAAGTAGAAGGCGGCGTTGTAGACCTGAACACGCTGAAAGCGGCAAACATTATCGGTATCCAGATCGAGTTCGCGAAAGTGATC CTGGCTGGCGAAGTCACTACTCCGGTAACTGTTCGTGGCCTGCGTGTTACTAAAGGCGCTCGTGCTGCTATCGAAGCTGCTGG CGGTAAAATCGAGGAATAA (SEQ ID NO: 381)
43_jfrlSe1402-8_Rissen	ATGCGTTTAAATACTCTGTCTCCGGCCGAAGGCTCCAAAAAGGCGGGTAAACGCCTGGGTCGTGGTATCGGTTCTGGCCTCGG TAAAACCGGTGGTCGTGGTCACAAAGGTCAGAAGTCTCGTTCTGGCGGTGGCGTACGTCGCGGTTTCGAGGGTGGTCAGATGC CTCTGTACCGTCGTCTGCCGAAATTCGGCTTCACTTCTCGCAAAGCAGCGATTACAGCCGAAGTTCGTCTGTCTGACCTGGCG AAAGTAGAAGGCGGCGTTGTAGACCTGAACACGCTGAAAGCGGCAAACATTATCGGTATCCAGATCGAGTTCGCGAAAGTGATC CTGGCTGGCGAAGTCACTACTCCGGTAACTGTTCGTGGCCTGCGTGTTACTAAAGGCGCTCGTGCTGCTATCGAAGCTGCTGG CGGTAAAATCGAGGAATAA (SEQ ID NO: 391)
44_jfrlSe1402-9_Rissen	ATGCGTTTAAATACTCTGTCTCCGGCCGAAGGCTCCAAAAAGGCGGGTAAACGCCTGGGTCGTGGTATCGGTTCTGGCCTCGG TAAAACCGGTGGTCGTGGTCACAAAGGTCAGAAGTCTCGTTCTGGCGGTGGCGTACGTCGCGGTTTCGAGGGTGGTCAGATGC CTCTGTACCGTCGTCTGCCGAAATTCGGCTTCACTTCTCGCAAAGCAGCGATTACAGCCGAAGTTCGTCTGTCTGACCTGGCG AAAGTAGAAGGCGGCGTTGTAGACCTGAACACGCTGAAAGCGGCAAACATTATCGGTATCCAGATCGAGTTCGCGAAAGTGATC CTGGCTGGCGAAGTCACTACTCCGGTAACTGTTCGTGGCCTGCGTGTTACTAAAGGCGCTCGTGCTGCTATCGAAGCTGCTGG CGGTAAAATCGAGGAATAA (SEQ ID NO: 398)
45_jfrlSe1402-10_Rissen	ATGCGTTTAAATACTCTGTCTCCGGCCGAAGGCTCCAAAAAGGCGGGTAAACGCCTGGGTCGTGGTATCGGTTCTGGCCTCGG TAAAACCGGTGGTCGTGGTCACAAAGGTCAGAAGTCTCGTTCTGGCGGTGGCGTACGTCGCGGTTTCGAGGGTGGTCAGATGC CTCTGTACCGTCGTCTGCCGAAATTCGGCTTCACTTCTCGCAAAGCAGCGATTACAGCCGAAGTTCGTCTGTCTGACCTGGCG AAAGTAGAAGGCGGCGTTGTAGACCTGAACACGCTGAAAGCGGCAAACATTATCGGTATCCAGATCGAGTTCGCGAAAGTGATC CTGGCTGGCGAAGTCACTACTCCGGTAACTGTTCGTGGCCTGCGTGTTACTAAAGGCGCTCGTGCTGCTATCGAAGCTGCTGG CGGTAAAATCGAGGAATAA (SEQ ID NO: 405)
46_jfrlSe1402-11_Rissen	ATGCGTTTAAATACTCTGTCTCCGGCCGAAGGCTCCAAAAAGGCGGGTAAACGCCTGGGTCGTGGTATCGGTTCTGGCCTCGG TAAAACCGGTGGTCGTGGTCACAAAGGTCAGAAGTCTCGTTCTGGCGGTGGCGTACGTCGCGGTTTCGAGGGTGGTCAGATGC CTCTGTACCGTCGTCTGCCGAAATTCGGCTTCACTTCTCGCAAAGCAGCGATTACAGCCGAAGTTCGTCTGTCTGACCTGGCG AAAGTAGAAGGCGGCGTTGTAGACCTGAACACGCTGAAAGCGGCAAACATTATCGGTATCCAGATCGAGTTCGCGAAAGTGATC CTGGCTGGCGAAGTCACTACTCCGGTAACTGTTCGTGGCCTGCGTGTTACTAAAGGCGCTCGTGCTGCTATCGAAGCTGCTGG CGGTAAAATCGAGGAATAA (SEQ ID NO: 412)

Fig. 25D

47_jfrlSe1402-12_Rissen	ATGCGTTTAAATACTCTGTCTCCGGCCGAAGGCTCCAAAAAGGCGGGTAAACGCCTGGGTCGTGGTATCGGTTCTGGCCTCGG TAAAACCGGTGGTCGTGGTCACAAAGGTCAGAAGTCTCGTTCTGGCGGTGGCGTACGTCGCGGTTTCGAGGGTGGTCAGATGC CTCTGTACCGTCGTCTGCCGAAATTCGGCTTCACTTCTCGCAAAGCAGCGATTACAGCCGAAGTTCGTCTGTCTGACCTGGCG AAAGTAGAAGGCGGCGTTGTAGACCTGAACACGCTGAAAGCGGCAAACATTATCGGTATCCAGATCGAGTTCGCGAAAGTGATC CTGGCTGGCGAAGTCACTACTCCGGTAACTGTTCGTGGCCTGCGTGTTACTAAAGGCGCTCGTGCTGCTATCGAAGCTGCTGG CGGTAAAATCGAGGAATAA (SEQ ID NO: 419)
48_jfrlSe1402-13_Mbandaka	ATGCGTTTAAATACTCTGTCTCCGGCCGAAGGCTCCAAAAAGGCGGGTAAACGCCTGGGTCGTGGTATCGGTTCTGGCCTCGG TAAAACCGGTGGTCGTGGTCACAAAGGTCAGAAGTCTCGTTCTGGCGGTGGCGTACGTCGCGGTTTCGAGGGTGGTCAGATGC CTCTGTACCGTCGTCTGCCGAAATTCGGCTTCACTTCTCGCAAAGCAGCGATTACAGCCGAAGTTCGTCTGTCTGACCTGGCG AAAGTAGAAGGCGGCGTTGTAGACCTGAACACGCTGAAAGCGGCAAACATTATCGGTATCCAGATCGAGTTCGCGAAAGTGATC CTGGCTGGCGAAGTCACTACTCCGGTAACTGTTCGTGGCCTGCGTGTTACTAAAGGCGCTCGTGCTGCTATCGAAGCTGCTGG CGGTAAAATCGAGGAATAA (SEQ ID NO: 426)
49_jfrlSe1402-14_Mbandaka	ATGCGTTTAAATACTCTGTCTCCGGCCGAAGGCTCCAAAAAGGCGGGTAAACGCCTGGGTCGTGGTATCGGTTCTGGCCTCGG TAAAACCGGTGGTCGTGGTCACAAAGGTCAGAAGTCTCGTTCTGGCGGTGGCGTACGTCGCGGTTTCGAGGGTGGTCAGATGC CTCTGTACCGTCGTCTGCCGAAATTCGGCTTCACTTCTCGCAAAGCAGCGATTACAGCCGAAGTTCGTCTGTCTGACCTGGCG AAAGTAGAAGGCGGCGTTGTAGACCTGAACACGCTGAAAGCGGCAAACATTATCGGTATCCAGATCGAGTTCGCGAAAGTGATC CTGGCTGGCGAAGTCACTACTCCGGTAACTGTTCGTGGCCTGCGTGTTACTAAAGGCGCTCGTGCTGCTATCGAAGCTGCTGG CGGTAAAATCGAGGAATAA (SEQ ID NO: 436)
50_jfrlSe1402-15_Orion	ATGCGTTTAAATACTCTGTCTCCGGCCGAAGGCTCCAAAAAGGCGGGTAAACGCCTGGGTCGTGGTATCGGTTCTGGCCTCGG TAAAACCGGTGGTCGTGGTCACAAAGGTCAGAAGTCTCGTTCTGGCGGTGGCGTACGTCGCGGTTTCGAGGGTGGTCAGATGC CTCTGTACCGTCGTCTGCCGAAATTCGGCTTCACTTCTCGCAAAGCAGCGATTACAGCCGAAGTTCGTCTGTCTGACCTGGCG AAAGTAGAAGGCGGCGTTGTAGACCTGAACACGCTGAAAGCGGCAAACATTATCGGTATCCAGATCGAGTTCGCGAAAGTGATC CTGGCTGGCGAAGTCACTACTCCGGTAACTGTTCGTGGCCTGCGTGTTACTAAAGGCGCTCGTGCTGCTATCGAAGCTGCTGG CGGTAAAATCGAGGAATAA (SEQ ID NO: 446)
51_jfrlSe1409-1_Altona	ATGCATTTAAATACTCTGTCTCCGGCCGAAGGCTCCAAAAAGGCGGGTAAACGCCTGGGTCGTGGTATCGGTTCTGGCCTCGG TAAAACCGGTGGTCGTGGTCACAAAGGTCAGAAGTCTCGTTCTGGCGGTGGCGTACGTCGCGGTTTCGAGGGTGGTCAGATGC CTCTGTACCGTCGTCTGCCGAAATTCGGCTTCACTTCTCGCAAAGCAGCGATTACAGCCGAAGTTCGTCTGTCTGACCTGGCG AAAGTAGAAGGCGGCGTTGTAGACCTGAACACGCTGAAAGCGGCAAACATTATCGGTATTCAGATCGAGTTCGCGAAAGTGATC CTGGCTGGCGAAGTCACTACTCCGGTAACTGTTCGTGGCCTGCGTGTTACTAAAGGCGCTCGTGCTGCTATCGAAGCTGCTGG CGGTAAAATCGAGGAATAA (SEQ ID NO: 456)
52_jfrlSe1409-2_Istanbul	ATGCGTTTAAATACTCTGTCTCCGGCCGAAGGCTCCAAAAAGGCGGGTAAACGCCTGGGTCGTGGTATCGGTTCTGGCCTCGG TAAAACCGGTGGTCGTGGTCACAAAGGTCAGAAGTCTCGTTCTGGCGGTGGCGTACGTCGCGGTTTCGAGGGTGGTCAGATGC CTCTGTACCGTCGTCTGCCGAAATTCGGCTTCACTTCTCGCAAAGCAGCGATTACAGCCGAAGTTCGTCTGTCTGACCTGGCG AAAGTAGAAGGCGGCGTTGTAGACCTGAACACGCTGAAAGCGGCAAACATTATCGGTATCCAGATCGAGTTCGCGAAAGTGATC CTGGCTGGCGAAGTCACTACTCCGGTAACTGTTCGTGGCCTGCGTGTTACTAAAGGCGCTCGTGCTGCTATCGAAGCTGCTGG CGGTAAAATCGAGGAATAA (SEQ ID NO: 463)
53_jfrlSe1409-3_Senftenberg	
54_jfrlSe1409-4_UN_O13_	ATGCGTTTAAATACTCTGTCTCCGGCCGAAGGCTCCAAAAAGGCGGGTAAACGCCTGGGTCGTGGTATCGGTTCTGGCCTCGG TAAAACCGGTGGTCGTGGTCACAAAGGTCAGAAGTCTCGTTCTGGCGGTGGCGTACGTCGCGGTTTCGAGGGTGGTCAGATGC CTCTGTACCGTCGTCTGCCGAAATTCGGCTTCACTTCTCGCAAAGCAGCGATTACAGCCGAAGTTCGTCTGTCTGACCTGGCG AAAGTAGAAGGCGGCGTTGTAGACCTGAACACGCTGAAAGCGGCAAACATTATCGGTATCCAGATCGAGTTCGCGAAAGTGATC CTGGCTGGCGAAGTCACTACTCCGGTAACTGTTCGTGGCCTGCGTGTTACTAAAGGCGCTCGTGCTGCTATCGAAGCTGCTGG CGGTAAAATCGAGGAATAA (SEQ ID NO: 476)
55_jfrlSe1409-5_UN_O1,3,19_	ATGCGTTTAAATACTCTGTCTCCGGCCGAAGGCTCCAAAAAGGCGGGTAAACGCCTGGGTCGTGGTATCGGTTCTGGCCTCGG TAAAACCGGTGGTCGTGGTCACAAAGGTCAGAAGTCTCGTTCTGGCGGTGGCGTACGTCGCGGTTTCGAGGGTGGTCAGATGC CTCTGTACCGTCGTCTGCCGAAATTCGGCTTCACTTCTCGCAAAGCAGCGATTACAGCCGAAGTTCGTCTGTCTGACCTGGCG AAAGTAGAAGGCGGCGTTGTAGACCTGAACACGCTGAAAGCGGCAAACATTATCGGTATCCAGATCGAGTTCGCGAAAGTGATC CTGGCTGGCGAAGTCACTACTCCGGTAACTGTTCGTGGCCTGCGTGTTACTAAAGGCGCTCGTGCTGCTATCGAAGCTGCTGG CGGTAAAATCGAGGAATAA (SEQ ID NO: 484)
56_jfrlSe1409-6_Montevideo	ATGCGTTTAAATACTCTGTCTCCGGCCGAAGGCTCCAAAAAGGCGGGTAAACGCCTGGGTCGTGGTATCGGTTCTGGCCTCGG TAAAACCGGTGGTCGTGGTCACAAAGGTCAGAAGTCTCGTTCTGGCGGTGGCGTACGTCGCGGTTTCGAGGGTGGTCAGATGC CTCTGTACCGTCGTCTGCCGAAATTCGGCTTCACTTCTCGCAAAGCAGCGATTACAGCCGAAGTTCGTCTGTCTGACCTGGCG AAAGTAGAAGGCGGCGTTGTAGACCTGAACACGCTGAAAGCGGCAAACATTATCGGTATCCAGATCGAGTTCGCGAAAGTGATC CTGGCTGGCGAAGTCACTACTCCGGTAACTGTTCGTGGCCTGCGTGTTACTAAAGGCGCTCGTGCTGCTATCGAAGCTGCTGG CGGTAAAATCGAGGAATAA (SEQ ID NO: 493)
57_jfrlSe1409-7_UN_O1,3,19_	ATGCGTTTAAATACTCTGTCTCCGGCCGAAGGCTCCAAAAAGGCGGGTAAACGCCTGGGTCGTGGTATCGGTTCTGGCCTCGG TAAAACCGGTGGTCGTGGTCACAAAGGTCAGAAGTCTCGTTCTGGCGGTGGCGTACGTCGCGGTTTCGAGGGTGGTCAGATGC CTCTGTACCGTCGTCTGCCGAAATTCGGCTTCACTTCTCGCAAAGCAGCGATTACAGCCGAAGTTCGTCTGTCTGACCTGGCG AAAGTAGAAGGCGGCGTTGTAGACCTGAACACGCTGAAAGCGGCAAACATTATCGGTATCCAGATCGAGTTCGCGAAAGTGATC CTGGCTGGCGAAGTCACTACTCCGGTAACTGTTCGTGGCCTGCGTGTTACTAAAGGCGCTCGTGCTGCTATCGAAGCTGCTGG CGGTAAAATCGAGGAATAA (SEQ ID NO: 500)
58_jfrlSe1409-8_UN_O18_	ATGCGTTTAAATACTCTGTCTCCGGCCGAAGGCTCCAAAAAGGCGGGTAAACGCCTGGGTCGTGGTATCGGTTCTGGCCTCGG TAAAACCGGTGGTCGTGGTCACAAAGGTCAGAAGTCTCGTTCTGGCGGTGGCGTACGTCGCGGTTTCGAGGGTGGTCAGATGC CTCTGTACCGTCGTCTGCCGAAATTCGGCTTCACTTCTCGCAAAGCAGCGATTACAGCCGAAGTTCGTCTGTCTGACCTGGCG AAAGTAGAAGGCGGCGTTGTAGACCTGAACACGCTGAAAGCGGCAAACATTATCGGTATCCAGATCGAGTTCGCGAAAGTGATC CTGGCTGGCGAAGTCACTACTCCGGTAACTGTTCGTGGCCTGCGTGTTACTAAAGGCGCTCGTGCTGCTATCGAAGCTGCTGG CGGTAAAATCGAGGAATAA (SEQ ID NO: 509)

Fig. 25E

61_jfrlSe1409-11_UN_O1,3,19_	ATGCGTTTAAATACTCTGTCTCCGGCCGAAGGCTCCAAAAAGGCGGGTAAACGCCTGGGTCGTGGTATCGGTTCTGGCCTCGG TAAAACCGGTGGTCGTGGTCACAAAGGTCAGAAGTCTCGTTCTGGCGGTGGCGTACGTCGCGGTTTCGAGGGTGGTCAGATGC CTCTGTACCGTCGTCTGCCGAAATTCGGCTTCACTTCTCGCAAAGCAGCGATTACAGCCGAAGTTCGTCTGTCTGACCTGGCG AAAGTAGAAGGCGGCGTTGTAGACCTGAACACGCTGAAAGCGGCAAACATTATCGGTATCCAGATCGAGTTCGCGAAAGTGATC CTGGCTGGCGAAGTCACTACTCCGGTAACTGTTCGTGGCCTGCGTGTTACTAAAGGCGCTCGTGCTGCTATCGAAGCTGCTGG CGGTAAAATCGAGGAATAA (SEQ ID NO: 518)
67_jfrlSe1409-17_Rissen	ATGCGTTTAAATACTCTGTCTCCGGCCGAAGGCTCCAAAAAGGCGGGTAAACGCCTGGGTCGTGGTATCGGTTCTGGCCTCGG TAAAACCGGTGGTCGTGGTCACAAAGGTCAGAAGTCTCGTTCTGGCGGTGGCGTACGTCGCGGTTTCGAGGGTGGTCAGATGC CTCTGTACCGTCGTCTGCCGAAATTCGGCTTCACTTCTCGCAAAGCAGCGATTACAGCCGAAGTTCGTCTGTCTGACCTGGCG AAAGTAGAAGGCGGCGTTGTAGACCTGAACACGCTGAAAGCGGCAAACATTATCGGTATCCAGATCGAGTTCGCGAAAGTGATC CTGGCTGGCGAAGTCACTACTCCGGTAACTGTTCGTGGCCTGCGTGTTACTAAAGGCGCTCGTGCTGCTATCGAAGCTGCTGG CGGTAAAATCGAGGAATAA (SEQ ID NO: 527)
71_jfrlSe1409-21_Amsterdam	ATGCGTTTAAATACTCTGTCTCCGGCCGAAGGCTCCAAAAAGGCGGGTAAACGCCTGGGTCGTGGTATCGGTTCTGGCCTCGG TAAAACCGGTGGTCGTGGTCACAAAGGTCAGAAGTCTCGTTCTGGCGGTGGCGTACGTCGCGGTTTCGAGGGTGGTCAGATGC CTCTGTACCGTCGTCTGCCGAAATTCGGCTTCACTTCTCGCAAAGCAGCGATTACAGCCGAAGTTCGTCTGTCTGACCTGGCG AAAGTAGAAGGCGGCGTTGTAGACCTGAACACGCTGAAAGCGGCAAACATTATCGGTATCCAGATCGAGTTCGCGAAAGTGATC CTGGCTGGCGAAGTCACTACTCCGGTAACTGTTCGTGGCCTGCGTGTTACTAAAGGCGCTCGTGCTGCTATCGAAGCTGCTGG CGGTAAAATCGAGGAATAA (SEQ ID NO: 535)
80_HySe09_Enteritidis	ATGCGTTTAAATACTCTGTCTCCGGCCGAAGGCTCCAAAAAGGCGGGTAAACGCCTGGGTCGTGGTATCGGTTCTGGCCTCGG TAAAACCGGTGGTCGTGGTCACAAAGGTCAGAAGTCTCGTTCTGGCGGTGGCGTACGTCGCGGTTTCGAGGGTGGTCAGATGC CTCTGTACCGTCGTCTGCCGAAATTCGGCTTCACTTCTCGCAAAGCAGCGATTACAGCCGAAGTTCGTCTGTCTGACCTGGCG AAAGTAGAAGGCGGCGTTGTAGACCTGAACACGCTGAAAGCGGCAAACATTATCGGTATCCAGATCGAGTTCGCGAAAGTGATC CTGGCTGGCGAAGTCACTACTCCGGTAACTGTTCGTGGCCTGCGTGTTACTAAAGGCGCTCGTGCTGCTATCGAAGCTGCTGG CGGTAAAATCGAGGAATAA (SEQ ID NO: 545)
100_HySe29_Schwarzengrund	ATGCGTTTAAATACTCTGTCTCCGGCCGAAGGCTCCAAAAAGGCGGGTAAACGCCTGGGTCGTGGTATCGGTTCTGGCCTCGG TAAAACCGGTGGTCGTGGTCACAAAGGTCAGAAGTCTCGTTCTGGCGGTGGCGTACGTCGCGGTTTCGAGGGTGGTCAGATGC CTCTGTACCGTCGTCTGCCGAAATTCGGCTTCACTTCTCGCAAAGCAGCGATTACAGCCGAAGTTCGTCTGTCTGACCTGGCG AAAGTAGAAGGCGGCGTTGTAGACCTGAACACGCTGAAAGCGGCAAACATTATCGGTATCCAGATCGAGTTCGCGAAAGTGATC CTGGCTGGCGAAGTCACTACTCCGGTAACTGTTCGTGGCCTGCGTGTTACTAAAGGCGCTCGTGCTGCTATCGAAGCTGCTGG CGGTAAAATCGAGGAATAA (SEQ ID NO: 555)
103_HySe32_Schwarzengrund	ATGCGTTTAAATACTCTGTCTCCGGCCGAAGGCTCCAAAAAGGCGGGTAAACGCCTGGGTCGTGGTATCGGTTCTGGCCTCGG TAAAACCGGTGGTCGTGGTCACAAAGGTCAGAAGTCTCGTTCTGGCGGTGGCGTACGTCGCGGTTTCGAGGGTGGTCAGATGC CTCTGTACCGTCGTCTGCCGAAATTCGGCTTCACTTCTCGCAAAGCAGCGATTACAGCCGAAGTTCGTCTGTCTGACCTGGCG AAAGTAGAAGGCGGCGTTGTAGACCTGAACACGCTGAAAGCGGCAAACATTATCGGTATCCAGATCGAGTTCGCGAAAGTGATC CTGGCTGGCGAAGTCACTACTCCGGTAACTGTTCGTGGCCTGCGTGTTACTAAAGGCGCTCGTGCTGCTATCGAAGCTGCTGG CGGTAAAATCGAGGAATAA (SEQ ID NO: 563)
106_HySe35_Schwarzengrund	ATGCGTTTAAATACTCTGTCTCCGGCCGAAGGCTCCAAAAAGGCGGGTAAACGCCTGGGTCGTGGTATCGGTTCTGGCCTCGG TAAAACCGGTGGTCGTGGTCACAAAGGTCAGAAGTCTCGTTCTGGCGGTGGCGTACGTCGCGGTTTCGAGGGTGGTCAGATGC CTCTGTACCGTCGTCTGCCGAAATTCGGCTTCACTTCTCGCAAAGCAGCGATTACAGCCGAAGTTCGTCTGTCTGACCTGGCG AAAGTAGAAGGCGGCGTTGTAGACCTGAACACGCTGAAAGCGGCAAACATTATCGGTATCCAGATCGAGTTCGCGAAAGTGATC CTGGCTGGCGAAGTCACTACTCCGGTAACTGTTCGTGGCCTGCGTGTTACTAAAGGCGCTCGTGCTGCTATCGAAGCTGCTGG CGGTAAAATCGAGGAATAA (SEQ ID NO: 570)

Fig. 26A

	L17
01_NBRC13245T_Typhimurium	ATGCGCCATCGTAAGAGTGGTCGTCAACTGAACCGCAACAGCAGCCATCGCCAGGCTATGTTCCGCAACATGGCAGGTTCACT GGTTCGTCATGAAATCATCAAGACGACCCTGCCTAAAGCGAAAGAGCTGCGTCGCGTAGTTGAGCCGCTGATTACTCTTGCCA AGACTGATAGCGTTGCTAATCGTCGTCTGGCATTCGCCCGTACTCGTGATAACGAGATCGTGGCAAAACTGTTTAACGAGCTG GGCCCGCGTTTCGCGAGCCGCGCCGGTGGTTACACTCGCATTCTGAAGTGTGGCTTCCGTGCAGGCGACAACGCGCCGATGG CATACATCGAGCTGGTTGATCGTTCAGAGAAAACAGAAGCTGCTGCAGAGTAA (SEQ ID NO: 56)
02_GTC00131_Enteritidis	ATGCGCCATCGTAAGAGTGGTCGTCAACTGAACCGCAACAGCAGCCATCGCCAGGCTATGTTCCGCAACATGGCAGGTTCACT GGTTCGTCATGAAATCATCAAGACGACCCTGCCTAAAGCGAAAGAGCTGCGTCGCGTAGTTGAGCCGCTGATTACTCTTGCCA AGACTGATAGCGTTGCTAATCGTCGTCTGGCATTCGCCCGTACTCGTGATAACGAGATCGTGGCAAAACTGTTTAACGAGCTG GGCCCGCGTTTCGCGAGCCGCGCCGGTGGTTACACTCGCATTCTGAAGTGTGGCTTCCGTGCAGGCGACAACGCGCCGATGG CATACATCGAGCTGGTTGATCGTTCAGAGAAAACAGAAGCTGCTGCAGAGTAA (SEQ ID NO: 68)
03_GTC09491_Enteritidis	ATGCGCCATCGTAAGAGTGGTCGTCAACTGAACCGCAACAGCAGCCATCGCCAGGCTATGTTCCGCAACATGGCAGGTTCACT GGTTCGTCATGAAATCATCAAGACGACCCTGCCTAAAGCGAAAGAGCTGCGTCGCGTAGTTGAGCCGCTGATTACTCTTGCCA AGACTGATAGCGTTGCTAATCGTCGTCTGGCATTCGCCCGTACTCGTGATAACGAGATCGTGGCAAAACTGTTTAACGAGCTG GGCCCGCGTTTCGCGAGCCGCGCCGGTGGTTACACTCGCATTCTGAAGTGTGGCTTCCGTGCAGGCGACAACGCGCCGATGG CATACATCGAGCTGGTTGATCGTTCAGAGAAAACAGAAGCTGCTGCAGAGTAA (SEQ ID NO: 77)
04_GTC03838_Enteritidis	ATGCGCCATCGTAAGAGTGGTCGTCAACTGAACCGCAACAGCAGCCATCGCCAGGCTATGTTCCGCAACATGGCAGGTTCACT GGTTCGTCATGAAATCATCAAGACGACCCTGCCTAAAGCGAAAGAGCTGCGTCGCGTAGTTGAGCCGCTGATTACTCTTGCCA AGACTGATAGCGTTGCTAATCGTCGTCTGGCATTCGCCCGTACTCGTGATAACGAGATCGTGGCAAAACTGTTTAACGAGCTG GGCCCGCGTTTCGCGAGCCGCGCCGGTGGTTACACTCGCATTCTGAAGTGTGGCTTCCGTGCAGGCGACAACGCGCCGATGG CATACATCGAGCTGGTTGATCGTTCAGAGAAAACAGAAGCTGCTGCAGAGTAA (SEQ ID NO: 84)
05_GTC08914_Enteritidis	ATGCGCCATCGTAAGAGTGGTCGTCAACTGAACCGCAACAGCAGCCATCGCCAGGCTATGTTCCGCAACATGGCAGGTTCACT GGTTCGTCATGAAATCATCAAGACGACCCTGCCTAAAGCGAAAGAGCTGCGTCGCGTAGTTGAGCCGCTGATTACTCTTGCCA AGACTGATAGCGTTGCTAATCGTCGTCTGGCATTCGCCCGTACTCGTGATAACGAGATCGTGGCAAAACTGTTTAACGAGCTG GGCCCGCGTTTCGCGAGCCGCGCCGGTGGTTACACTCGCATTCTGAAGTGTGGCTTCCGTGCAGGCGACAACGCGCCGATGG CATACATCGAGCTGGTTGATCGTTCAGAGAAAACAGAAGCTGCTGCAGAGTAA (SEQ ID NO: 91)
06_GTC09421_Enteritidis	ATGCGCCATCGTAAGAGTGGTCGTCAACTGAACCGCAACAGCAGCCATCGCCAGGCTATGTTCCGCAACATGGCAGGTTCACT GGTTCGTCATGAAATCATCAAGACGACCCTGCCTAAAGCGAAAGAGCTGCGTCGCGTAGTTGAGCCGCTGATTACTCTTGCCA AGACTGATAGCGTTGCTAATCGTCGTCTGGCATTCGCCCGTACTCGTGATAACGAGATCGTGGCAAAACTGTTTAACGAGCTG GGCCCGCGTTTCGCGAGCCGCGCCGGTGGTTACACTCGCATTCTGAAGTGTGGCTTCCGTGCAGGCGACAACGCGCCGATGG CATACATCGAGCTGGTTGATCGTTCAGAGAAAACAGAAGCTGCTGCAGAGTAA (SEQ ID NO: 101)
07_GTC09489_Enteritidis	ATGCGCCATCGTAAGAGTGGTCGTCAACTGAACCGCAACAGCAGCCATCGCCAGGCTATGTTCCGCAACATGGCAGGTTCACT GGTTCGTCATGAAATCATCAAGACGACCCTGCCTAAAGCGAAAGAGCTGCGTCGCGTAGTTGAGCCGCTGATTACTCTTGCCA AGACTGATAGCGTTGCTAATCGTCGTCTGGCATTCGCCCGTACTCGTGATAACGAGATCGTGGCAAAACTGTTTAACGAGCTG GGCCCGCGTTTCGCGAGCCGCGCCGGTGGTTACACTCGCATTCTGAAGTGTGGCTTCCGTGCAGGCGACAACGCGCCGATGG CATACATCGAGCTGGTTGATCGTTCAGAGAAAACAGAAGCTGCTGCAGAGTAA (SEQ ID NO: 111)
08_NBRC3163_Pullorum_Gallinarum	ATGCGCCATCGTAAGAGTGGTCGTCAACTGAACCGCAACAGCAGCCATCGCCAGGCTATGTTCCGCAACATGGCAGGTTCACT GGTTCGTCATGAAATCATCAAGACGACCCTGCCTAAAGCGAAAGAGCTGCGTCGCGTAGTTGAGCCGCTGATTACTCTTGCCA AGACTGATAGCGTTGCTAATCGTCGTCTGGCATTCGCCCGTACTCGTGATAACGAGATCGTGGCAAAACTGTTTAACGATCTG GGCCCGCGTTTCGCGAGCCGCGCCGGTGGTTACACTCGCATTCTGAAGTGTGGCTTCCGTGCAGGCGACAACGCGCCGATGG CATACATCGAGCTGGTTGATCGCTCAGAGAAAACAGAAGCTGCTGCAGAGTAA (SEQ ID NO: 118)

Fig. 26B

09_NBRC3313_Enteritidis	ATGCGCCATCGTAAGAGTGGTCGTCAACTGAACCGCAACAGCAGCCATCGCCAGGCTATGTTCCGCAACATGGCAGGTTCACT GGTTCGTCATGAAATCATCAAGACGACCCTGCCTAAAGCGAAAGAGCTGCGTCGCGTAGTTGAGCCGCTGATTACTCTTGCCA AGACTGATAGCGTTGCTAATCGTCGTCTGGCATTCGCCCGTACTCGTGATAACGAGATCGTGGCAAAACTGTTTAACGAGCTG GGCCCGCGTTTCGCGAGCCGCGCCGGTGGTTACACTCGCATTCTGAAGTGTGGCTTCCGTGCAGGCGACAACGCGCCGATGG CATACATCGAGCTGGTTGATCGTTCAGAGAAAACAGAAGCTGCTGCAGAGTAA (SEQ ID NO: 128)
10_NBRC12529_Typhimurium	ATGCGCCATCGTAAGAGTGGTCGTCAACTGAACCGCAACAGCAGCCATCGCCAGGCTATGTTCCGCAACATGGCAGGTTCACT GGTTCGTCATGAAATCATCAAGACGACCCTGCCTAAAGCGAAAGAGCTGCGTCGCGTAGTTGAGCCGCTGATTACTCTTGCCA AGACTGATAGCGTTGCTAATCGTCGTCTGGCATTCGCCCGTACTCGTGATAACGAGATCGTGGCAAAACTGTTTAACGAGCTG GGCCCGCGTTTCGCGAGCCGCGCCGGTGGTTACACTCGCATTCTGAAGTGTGGCTTCCGTGCAGGCGACAACGCGCCGATGG CATACATCGAGCTGGTTGATCGTTCAGAGAAAACAGAAGCTGCTGCAGAGTAA (SEQ ID NO: 135)
11_NBRC14193_Typhimurium	ATGCGCCATCGTAAGAGTGGTCGTCAACTGAACCGCAACAGCAGCCATCGCCAGGCTATGTTCCGCAACATGGCAGGTTCACT GGTTCGTCATGAAATCATCAAGACGACCCTGCCTAAAGCGAAAGAGCTGCGTCGCGTAGTTGAGCCGCTGATTACTCTTGCCA AGACTGATAGCGTTGCTAATCGTCGTCTGGCATTCGCCCGTACTCGTGATAACGAGATCGTGGCAAAACTGTTTAACGAGCTG GGCCCGCGTTTCGCGAGCCGCGCCGGTGGTTACACTCGCATTCTGAAGTGTGGCTTCCGTGCAGGCGACAACGCGCCGATGG CATACATCGAGCTGGTTGATCGTTCAGAGAAAACAGAAGCTGCTGCAGAGTAA (SEQ ID NO: 142)
12_NBRC14194_Typhimurium	ATGCGCCATCGTAAGAGTGGTCGTCAACTGAACCGCAACAGCAGCCATCGCCAGGCTATGTTCCGCAACATGGCAGGTTCACT GGTTCGTCATGAAATCATCAAGACGACCCTGCCTAAAGCGAAAGAGCTGCGTCGCGTAGTTGAGCCGCTGATTACTCTTGCCA AGACTGATAGCGTTGCTAATCGTCGTCTGGCATTCGCCCGTACTCGTGATAACGAGATCGTGGCAAAACTGTTTAACGAGCTG GGCCCGCGTTTCGCGAGCCGCGCCGGTGGTTACACTCGCATTCTGAAGTGTGGCTTCCGTGCAGGCGACAACGCGCCGATGG CATACATCGAGCTGGTTGATCGTTCAGAGAAAACAGAAGCTGCTGCAGAGTAA (SEQ ID NO: 149)
13_NBRC14209_Typhimurium	ATGCGCCATCGTAAGAGTGGTCGTCAACTGAACCGCAACAGCAGCCATCGCCAGGCTATGTTCCGCAACATGGCAGGTTCACT GGTTCGTCATGAAATCATCAAGACGACCCTGCCTAAAGCGAAAGAGCTGCGTCGCGTAGTTGAGCCGCTGATTACTCTTGCCA AGACTGATAGCGTTGCTAATCGTCGTCTGGCATTCGCCCGTACTCGTGATAACGAGATCGTGGCAAAACTGTTTAACGAGCTG GGCCCGCGTTTCGCGAGCCGCGCCGGTGGTTACACTCGCATTCTGAAGTGTGGCTTCCGTGCAGGCGACAACGCGCCGATGG CATACATCGAGCTGGTTGATCGTTCAGAGAAAACAGAAGCTGCTGCAGAGTAA (SEQ ID NO: 156)
14_NBRC14210_Typhimurium	ATGCGCCATCGTAAGAGTGGTCGTCAACTGAACCGCAACAGCAGCCATCGCCAGGCTATGTTCCGCAACATGGCAGGTTCACT GGTTCGTCATGAAATCATCAAGACGACCCTGCCTAAAGCGAAAGAGCTGCGTCGCGTAGTTGAGCCGCTGATTACTCTTGCCA AGACTGATAGCGTTGCTAATCGTCGTCTGGCATTCGCCCGTACTCGTGATAACGAGATCGTGGCAAAACTGTTTAACGAGCTG GGCCCGCGTTTCGCGAGCCGCGCCGGTGGTTACACTCGCATTCTGAAGTGTGGCTTCCGTGCAGGCGACAACGCGCCGATGG CATACATCGAGCTGGTTGATCGTTCAGAGAAAACAGAAGCTGCTGCAGAGTAA (SEQ ID NO: 163)
15_NBRC14211_Typhimurium	ATGCGCCATCGTAAGAGTGGTCGTCAACTGAACCGCAACAGCAGCCATCGCCAGGCTATGTTCCGCAACATGGCAGGTTCACT GGTTCGTCATGAAATCATCAAGACGACCCTGCCTAAAGCGAAAGAGCTGCGTCGCGTAGTTGAGCCGCTGATTACTCTTGCCA AGACTGATAGCGTTGCTAATCGTCGTCTGGCATTCGCCCGTACTCGTGATAACGAGATCGTGGCAAAACTGTTTAACGAGCTG GGCCCGCGTTTCGCGAGCCGCGCCGGTGGTTACACTCGCATTCTGAAGTGTGGCTTCCGTGCAGGCGACAACGCGCCGATGG CATACATCGAGCTGGTTGATCGTTCAGAGAAAACAGAAGCTGCTGCAGAGTAA (SEQ ID NO: 169)
16_NBRC14212_Typhimurium	ATGCGCCATCGTAAGAGTGGTCGTCAACTGAACCGCAACAGCAGCCATCGCCAGGCTATGTTCCGCAACATGGCAGGTTCACT GGTTCGTCATGAAATCATCAAGACGACCCTGCCTAAAGCGAAAGAGCTGCGTCGCGTAGTTGAGCCGCTGATTACTCTTGCCA AGACTGATAGCGTTGCTAATCGTCGTCTGGCATTCGCCCGTACTCGTGATAACGAGATCGTGGCAAAACTGTTTAACGAGCTG GGCCCGCGTTTCGCGAGCCGCGCCGGTGGTTACACTCGCATTCTGAAGTGTGGCTTCCGTGCAGGCGACAACGCGCCGATGG CATACATCGAGCTGGTTGATCGTTCAGAGAAAACAGAAGCTGCTGCAGAGTAA (SEQ ID NO: 176)
17_NBRC15181_Typhimurium	ATGCGCCATCGTAAGAGTGGTCGTCAACTGAACCGCAACAGCAGCCATCGCCAGGCTATGTTCCGCAACATGGCAGGTTCACT GGTTCGTCATGAAATCATCAAGACGACCCTGCCTAAAGCGAAAGAGCTGCGTCGCGTAGTTGAGCCGCTGATTACTCTTGCCA AGACTGATAGCGTTGCTAATCGTCGTCTGGCATTCGCCCGTACTCGTGATAACGAGATCGTGGCAAAACTGTTTAACGAGCTG GGCCCGCGTTTCGCGAGCCGCGCCGGTGGTTACACTCGCATTCTGAAGTGTGGCTTCCGTGCAGGCGACAACGCGCCGATGG CATACATCGAGCTGGTTGATCGTTCAGAGAAAACAGAAGCTGCTGCAGAGTAA (SEQ ID NO: 183)
18_NBRC15182_Minnesota	ATGCGCCATCGTAAGAGTGGTCGTCAACTGAACCGCAACAGCAGCCATCGCCAGGCTATGTTCCGCAACATGGCAGGTTCACT GGTTCGTCATGAAATCATCAAGACGACCCTGCCTAAAGCGAAAGAGCTGCGTCGCGTAGTTGAGCCGCTGATTACTCTTGCCA AGACTGATAGCGTTGCTAATCGTCGTCTGGCATTCGCCCGTACTCGTGATAACGAGATCGTGGCAAAACTGTTTAACGAGCTG GGCCCGCGTTTCGCGAGCCGCGCCGGTGGTTACACTCGCATTCTGAAGTGTGGCTTCCGTGCAGGCGACAACGCGCCGATGG CATACATCGAGCTGGTTGATCGTTCAGAGAAAACAGAAGCTGCTGCAGAGTAA (SEQ ID NO: 193)
19_NBRC15183_Minnesota	ATGCGCCATCGTAAGAGTGGTCGTCAACTGAACCGCAACAGCAGCCATCGCCAGGCTATGTTCCGCAACATGGCAGGTTCACT GGTTCGTCATGAAATCATCAAGACGACCCTGCCTAAAGCGAAAGAGCTGCGTCGCGTAGTTGAGCCGCTGATTACTCTTGCCA AGACTGATAGCGTTGCTAATCGTCGTCTGGCATTCGCCCGTACTCGTGATAACGAGATCGTGGCAAAACTGTTTAACGAGCTG GGCCCGCGTTTCGCGAGCCGCGCCGGTGGTTACACTCGCATTCTGAAGTGTGGCTTCCGTGCAGGCGACAACGCGCCGATGG CATACATCGAGCTGGTTGATCGTTCAGAGAAAACAGAAGCTGCTGCAGAGTAA (SEQ ID NO: 203)
20_NBRC15184_Minnesota	ATGCGCCATCGTAAGAGTGGTCGTCAACTGAACCGCAACAGCAGCCATCGCCAGGCTATGTTCCGCAACATGGCAGGTTCACT GGTTCGTCATGAAATCATCAAGACGACCCTGCCTAAAGCGAAAGAGCTGCGTCGCGTAGTTGAGCCGCTGATTACTCTTGCCA AGACTGATAGCGTTGCTAATCGTCGTCTGGCATTCGCCCGTACTCGTGATAACGAGATCGTGGCAAAACTGTTTAACGAGCTG GGCCCGCGTTTCGCGAGCCGCGCCGGTGGTTACACTCGCATTCTGAAGTGTGGCTTCCGTGCAGGCGACAACGCGCCGATGG CATACATCGAGCTGGTTGATCGTTCAGAGAAAACAGAAGCTGCTGCAGAGTAA (SEQ ID NO: 210)
21_NBRC15185_Minnesota	ATGCGCCATCGTAAGAGTGGTCGTCAACTGAACCGCAACAGCAGCCATCGCCAGGCTATGTTCCGCAACATGGCAGGTTCACT GGTTCGTCATGAAATCATCAAGACGACCCTGCCTAAAGCGAAAGAGCTGCGTCGCGTAGTTGAGCCGCTGATTACTCTTGCCA AGACTGATAGCGTTGCTAATCGTCGTCTGGCATTCGCCCGTACTCGTGATAACGAGATCGTGGCAAAACTGTTTAACGAGCTG GGCCCGCGTTTCGCGAGCCGCGCCGGTGGTTACACTCGCATTCTGAAGTGTGGCTTCCGTGCAGGCGACAACGCGCCGATGG CATACATCGAGCTGGTTGATCGTTCAGAGAAAACAGAAGCTGCTGCAGAGTAA (SEQ ID NO: 217)
22_NBRC15186_Minnesota	ATGCGCCATCGTAAGAGTGGTCGTCAACTGAACCGCAACAGCAGCCATCGCCAGGCTATGTTCCGCAACATGGCAGGTTCACT GGTTCGTCATGAAATCATCAAGACGACCCTGCCTAAAGCGAAAGAGCTGCGTCGCGTAGTTGAGCCGCTGATTACTCTTGCCA AGACTGATAGCGTTGCTAATCGTCGTCTGGCATTCGCCCGTACTCGTGATAACGAGATCGTGGCAAAACTGTTTAACGAGCTG GGCCCGCGTTTCGCGAGCCGCGCCGGTGGTTACACTCGCATTCTGAAGTGTGGCTTCCGTGCAGGCGACAACGCGCCGATGG CATACATCGAGCTGGTTGATCGTTCAGAGAAAACAGAAGCTGCTGCAGAGTAA (SEQ ID NO: 224)
23_NBRC15187_Minnesota	ATGCGCCATCGTAAGAGTGGTCGTCAACTGAACCGCAACAGCAGCCATCGCCAGGCTATGTTCCGCAACATGGCAGGTTCACT GGTTCGTCATGAAATCATCAAGACGACCCTGCCTAAAGCGAAAGAGCTGCGTCGCGTAGTTGAGCCGCTGATTACTCTTGCCA AGACTGATAGCGTTGCTAATCGTCGTCTGGCATTCGCCCGTACTCGTGATAACGAGATCGTGGCAAAACTGTTTAACGAGCTG GGCCCGCGTTTCGCGAGCCGCGCCGGTGGTTACACTCGCATTCTGAAGTGTGGCTTCCGTGCAGGCGACAACGCGCCGATGG CATACATCGAGCTGGTTGATCGTTCAGAGAAAACAGAAGCTGCTGCAGAGTAA (SEQ ID NO: 231)
24_NBRC100797_Abony	ATGCGCCATCGTAAGAGTGGTCGTCAACTGAACCGCAACAGCAGCCATCGCCAGGCTATGTTCCGCAACATGGCAGGTTCACT GGTTCGTCATGAAATCATCAAGACGACCCTGCCTAAAGCGAAAGAGCTGCGTCGCGTAGTTGAGCCGCTGATTACTCTTGCCA AGACTGATAGCGTTGCTAATCGTCGTCTGGCATTCGCCCGTACTCGTGATAACGAGATCGTGGCAAAACTGTTTAACGAGCTG GGCCCGCGTTTCGCGAGCCGCGCCGGTGGTTACACTCGCATTCTGAAGTGTGGCTTCCGTGCAGGCGACAACGCGCCGATGG CATACATCGAGCTGGTTGATCGTTCAGAGAAAACAGAAGCTGCTGCAGAGTAA (SEQ ID NO: 238)
25_NBRC105684_Choleraesuis	ATGCGCCATCGTAAGAGTGGTCGTCAACTGAACCGCAACAGCAGCCATCGCCAGGCTATGTTCCGCAACATGGCAGGTTCACT GGTTCGTCATGAAATCATCAAGACGACCCTGCCTAAAGCGAAAGAGCTGCGTCGCGTAGTTGAGCCGCTGATTACTCTTGCCA AGACTGATAGCGTTGCTAATCGTCGTCTGGCATTCGCCCGTACTCGTGATAACGAGATCGTGGCAAAACTGTTTAACGAGCTG GGCCCGCGTTTCGCGAGCCGCGCCGGTGGTTACACTCGCATTCTGAAGTGTGGCTTCCGTGCAGGCGACAACGCGCCGATGG CATACATCGAGCTGGTTGATCGTTCAGAGAAAACAGAAGCTGCTGCAGAGTAA (SEQ ID NO: 248)

Fig. 26C

26_NBRC105726_Typhimurium	ATGCGCCATCGTAAGAGTGGTCGTCAACTGAACCGCAACAGCAGCCATCGCCAGGCTATGTTCCGCAACATGGCAGGTTCACT GGTTCGTCATGAAATCATCAAGACGACCCTGCCTAAAGCGAAAGAGCTGCGTCGCGTAGTTGAGCCGCTGATTACTCTTGCCA AGACTGATAGCGTTGCTAATCGTCGTCTGGCATTCGCCCGTACTCGTGATAACGAGATCGTGGCAAAACTGTTTAACGAGCTG GGCCCGCGTTTCGCGAGCCGCGCCGGTGGTTACACTCGCATTCTGAAGTGTGGCTTCCGTGCAGGCGACAACGCGCCGATGG CATACATCGAGCTGGTTGATCGTTCAGAGAAAACAGAAGCTGCTGCAGAGTAA (SEQ ID NO: 258)
27_JCM3919_UN_O7_	ATGCGCCATCGTAAGAGTGGTCGTCAACTGAACCGCAACAGCAGCCATCGCCAGGCTATGTTCCGCAACATGGCAGGTTCACT GGTTCGTCATGAAATCATCAAGACGACCCTGCCTAAAGCGAAAGAGCTGCGTCGCGTAGTTGAGCCGCTGATTACTCTTGCCA AGACTGATAGCGTTGCTAATCGTCGTCTGGCATTCGCCCGTACTCGTGATAACGAGATCGTGGCAAAACTGTTTAACGAGCTG GGCCCGCGTTTCGCGAGCCGCGCCGGTGGTTACACTCGCATTCTGAAGTGTGGCTTCCGTGCAGGCGACAACGCGCCGATGG CATACATCGAGCTGGTTGATCGTTCAGAGAAAACAGAAGCTGCTGCAGAGTAA (SEQ ID NO: 267)
28_NBRC15335_Minnesota	ATGCGCCATCGTAAGAGTGGTCGTCAACTGAACCGCAACAGCAGCCATCGCCAGGCTATGTTCCGCAACATGGCAGGTTCACT GGTTCGTCATGAAATCATCAAGACGACCCTGCCTAAAGCGAAAGAGCTGCGTCGCGTAGTTGAGCCGCTGATTACTCTTGCCA AGACTGATAGCGTTGCTAATCGTCGTCTGGCATTCGCCCGTACTCGTGATAACGAGATCGTGGCAAAACTGTTTAACGAGCTG GGCCCGCGTTTCGCGAGCCGCGCCGGTGGTTACACTCGCATTCTGAAGTGTGGCTTCCGTGCAGGCGACAACGCGCCGATGG CATACATCGAGCTGGTTGATCGTTCAGAGAAAACAGAAGCTGCTGCAGAGTAA (SEQ ID NO: 274)
29_GTC09490_Enteritidis	ATGCGCCATCGTAAGAGTGGTCGTCAACTGAACCGCAACAGCAGCCATCGCCAGGCTATGTTCCGCAACATGGCAGGTTCACT GGTTCGTCATGAAATCATCAAGACGACCCTGCCTAAAGCGAAAGAGCTGCGTCGCGTAGTTGAGCCGCTGATTACTCTTGCCA AGACTGATAGCGTTGCTAATCGTCGTCTGGCATTCGCCCGTACTCGTGATAACGAGATCGTGGCAAAACTGTTTAACGAGCTG GGCCCGCGTTTCGCGAGCCGCGCCGGTGGTTACACTCGCATTCTGAAGTGTGGCTTCCGTGCAGGCGACAACGCGCCGATGG CATACATCGAGCTGGTTGATCGTTCAGAGAAAACAGAAGCTGCTGCAGAGTAA (SEQ ID NO: 281)
30_GTC09492_Braenderup	ATGCGCCATCGTAAGAGTGGTCGTCAACTGAACCGCAACAGCAGCCATCGCCAGGCTATGTTCCGCAACATGGCAGGTTCACT GGTTCGTCATGAAATCATCAAGACGACCCTGCCTAAAGCGAAAGAGCTGCGTCGCGTAGTTGAGCCGCTGATTACTCTTGCCA AGACTGATAGCGTTGCTAATCGTCGTCTGGCATTCGCCCGTACTCGTGATAACGAGATCGTGGCAAAACTGTTTAACGAGCTG GGCCCGCGTTTCGCGAGCCGCGCCGGTGGTTACACTCGCATTCTGAAGTGTGGCTTCCGTGCAGGCGACAACGCGCCGATGG CATACATCGAGCTGGTTGATCGTTCAGAGAAAACAGAAGCTGCTGCAGAGTAA (SEQ ID NO: 288)
31_GTC09493_Pakistan	ATGCGCCATCGTAAGAGTGGTCGTCAACTGAACCGCAACAGCAGCCATCGCCAGGCTATGTTCCGCAACATGGCAGGTTCACT GGTTCGTCATGAAATCATCAAGACGACCCTGCCTAAAGCGAAAGAGCTGCGTCGCGTAGTTGAGCCGCTGATTACTCTTGCCA AGACTGATAGCGTTGCTAATCGTCGTCTGGCATTCGCCCGTACTCGTGATAACGAGATCGTGGCAAAACTGTTTAACGAGCTG GGCCCGCGTTTCGCGAGCCGCGCCGGTGGTTACACTCGCATTCTGAAGTGTGGCTTCCGTGCAGGCGACAACGCGCCGATGG CATACATCGAGCTGGTTGATCGTTCAGAGAAAACAGAAGCTGCTGCAGAGTAA (SEQ ID NO: 295)
32_GTC09549_Typhimurium	ATGCGCCATCGTAAGAGTGGTCGTCAACTGAACCGCAACAGCAGCCATCGCCAGGCTATGTTCCGCAACATGGCAGGTTCACT GGTTCGTCATGAAATCATCAAGACGACCCTGCCTAAAGCGAAAGAGCTGCGTCGCGTAGTTGAGCCGCTGATTACTCTTGCCA AGACTGATAGCGTTGCTAATCGTCGTCTGGCATTCGCCCGTACTCGTGATAACGAGATCGTGGCAAAACTGTTTAACGAGCTG GGCCCGCGTTTCGCGAGCCGCGCCGGTGGTTACACTCGCATTCTGAAGTGTGGCTTCCGTGCAGGCGACAACGCGCCGATGG CATACATCGAGCTGGTTGATCGTTCAGAGAAAACAGAAGCTGCTGCAGAGTAA (SEQ ID NO: 302)
33_ATCCBAA-1675_Infantis	ATGCGCCATCGTAAGAGTGGTCGTCAACTGAACCGCAACAGCAGCCATCGCCAGGCTATGTTCCGCAACATGGCAGGTTCACT GGTTCGTCATGAAATCATCAAGACGACCCTGCCTAAAGCGAAAGAGCTGCGTCGCGTAGTTGAGCCGCTGATTACTCTTGCCA AGACTGATAGCGTTGCTAATCGTCGTCTGGCATTCGCCCGTACTCGTGATAACGAGATCGTGGCAAAACTGTTTAACGAGCTG GGCCCGCGTTTCGCGAGCCGCGCCGGTGGTTACACTCGCATTCTGAAGTGTGGCTTCCGTGCAGGCGACAACGCGCCGATGG CATACATCGAGCTGGTTGATCGTTCAGAGAAAACAGAAGCTGCTGCAGAGTAA (SEQ ID NO: 309)
34_ATCCBAA-1738_Thompson	ATGCGCCATCGTAAGAGTGGTCGTCAACTGAACCGCAACAGCAGCCATCGCCAGGCTATGTTCCGCAATATGGCAGGTTCACT GGTTCGTCATGAAATCATCAAGACGACCCTGCCTAAAGCGAAAGAGCTGCGTCGCGTAGTTGAGCCGCTGATTACTCTTGCCA AGACTGATAGCGTTGCTAATCGTCGTCTGGCATTCGCCCGTACTCGTGATAACGAGATCGTGGCAAAACTGTTTAACGAGCTG GGCCCGCGTTTCGCGAGCCGCGCCGGTGGTTACACTCGCATTCTGAAGTGTGGCTTCCGTGCAGGCGACAACGCGCCGATGG CATACATCGAGCTGGTTGATCGTTCAGAGAAAACAGAAGCTGCTGCAGAGTAA (SEQ ID NO: 320)
35_ATCC9712_Saintpaul	ATGCGCCATCGTAAGAGTGGTCGTCAACTGAACCGCAACAGCAGCCATCGCCAGGCTATGTTCCGCAACATGGCAGGTTCACT GGTTCGTCATGAAATCATCAAGACGACCCTGCCTAAAGCGAAAGAGCTGCGTCGCGTAGTTGAGCCGCTGATTACTCTTGCCA AGACTGATAGCGTTGCTAATCGTCGTCTGGCATTCGCCCGTACTCGTGATAACGAGATCGTGGCAAAACTGTTTAACGAGCTG GGCCCGCGTTTCGCGAGCCGCGCCGGTGGTTACACTCGCATTCTGAAGTGTGGCTTCCGTGCAGGCGACAACGCGCCGATGG CATACATCGAGCTGGTTGATCGTTCAGAGAAAACAGAAGCTGCTGCAGAGTAA (SEQ ID NO: 330)
36_jfrlSe1402-1_Infantis	ATGCGCCATCGTAAGAGTGGTCGTCAACTGAACCGCAACAGCAGCCATCGCCAGGCTATGTTCCGCAACATGGCAGGTTCACT GGTTCGTCATGAAATCATCAAGACGACCCTGCCTAAAGCGAAAGAGCTGCGTCGCGTAGTTGAGCCGCTGATTACTCTTGCCA AGACTGATAGCGTTGCTAATCGTCGTCTGGCATTCGCCCGTACTCGTGATAACGAGATCGTGGCAAAACTGTTTAACGAGCTG GGCCCGCGTTTCGCGAGCCGCGCCGGTGGTTACACTCGCATTCTGAAGTGTGGCTTCCGTGCAGGCGACAACGCGCCGATGG CATACATCGAGCTGGTTGATCGTTCAGAGAAAACAGAAGCTGCTGCAGAGTAA (SEQ ID NO: 341)
38_jfrlSe1402-3_Brandenburg	ATGCGCCATCGTAAGAGTGGTCGTCAACTGAACCGCAACAGCAGCCATCGCCAGGCTATGTTCCGCAACATGGCAGGTTCACT GGTTCGTCATGAAATCATCAAGACGACCCTGCCTAAAGCGAAAGAGCTGCGTCGCGTAGTTGAGCCGCTGATTACTCTTGCCA AGACTGATAGCGTTGCTAATCGTCGTCTGGCATTCGCCCGTACTCGTGATAACGAGATCGTGGCAAAACTGTTTAACGAGCTG GGCCCGCGTTTCGCGAGCCGCGCCGGTGGTTACACTCGCATTCTGAAGTGTGGCTTCCGTGCAGGCGACAACGCGCCGATGG CATACATCGAGCTGGTTGATCGTTCAGAGAAAACAGAAGCTGCTGCAGAGTAA (SEQ ID NO: 351)
39_jfrlSe1402-4_Infantis	ATGCGCCATCGTAAGAGTGGTCGTCAACTGAACCGCAACAGCAGCCATCGCCAGGCTATGTTCCGCAACATGGCAGGTTCACT GGTTCGTCATGAAATCATCAAGACGACCCTGCCTAAAGCGAAAGAGCTGCGTCGCGTAGTTGAGCCGCTGATTACTCTTGCCA AGACTGATAGCGTTGCTAATCGTCGTCTGGCATTCGCCCGTACTCGTGATAACGAGATCGTGGCAAAACTGTTTAACGAGCTG GGCCCGCGTTTCGCGAGCCGCGCCGGTGGTTACACTCGCATTCTGAAGTGTGGCTTCCGTGCAGGCGACAACGCGCCGATGG CATACATCGAGCTGGTTGATCGTTCAGAGAAAACAGAAGCTGCTGCAGAGTAA (SEQ ID NO: 358)
40_jfrlSe1402-5_Brandenburg	ATGCGCCATCGTAAGAGTGGTCGTCAACTGAACCGCAACAGCAGCCATCGCCAGGCTATGTTCCGCAACATGGCAGGTTCACT GGTTCGTCATGAAATCATCAAGACGACCCTGCCTAAAGCGAAAGAGCTGCGTCGCGTAGTTGAGCCGCTGATTACTCTTGCCA AGACTGATAGCGTTGCTAATCGTCGTCTGGCATTCGCCCGTACTCGTGATAACGAGATCGTGGCAAAACTGTTTAACGAGCTG GGCCCGCGTTTCGCGAGCCGCGCCGGTGGTTACACTCGCATTCTGAAGTGTGGCTTCCGTGCAGGCGACAACGCGCCGATGG CATACATCGAGCTGGTTGATCGTTCAGAGAAAACAGAAGCTGCTGCAGAGTAA (SEQ ID NO: 368)
41_jfrlSe1402-6_Rissen	ATGCGCCATCGTAAGAGTGGTCGTCAACTGAACCGCAACAGCAGCCATCGCCAGGCTATGTTCCGCAACATGGCAGGTTCACT GGTTCGTCATGAAATCATCAAGACGACCCTGCCTAAAGCGAAAGAGCTGCGTCGCGTAGTTGAGCCGCTGATTACTCTTGCCA AGACTGATAGCGTTGCTAATCGTCGTCTGGCATTCGCCCGTACTCGTGATAACGAGATCGTGGCAAAACTGTTTAACGAGCTG GGCCCGCGTTTCGCGAGCCGCGCCGGTGGTTACACTCGCATTCTGAAGTGTGGCTTCCGTGCAGGCGACAACGCGCCGATGG CATACATCGAGCTGGTTGATCGTTCAGAGAAAACAGAAGCTGCTGCAGAGTAA (SEQ ID NO: 375)
42_jfrlSe1402-7_Orion	ATGCGCCATCGTAAGAGTGGTCGTCAACTGAACCGCAACAGCAGCCATCGCCAGGCTATGTTCCGCAACATGGCAGGTTCACT GGTTCGTCATGAAATCATCAAGACGACCCTGCCTAAAGCGAAAGAGCTGCGTCGCGTAGTTGAGCCGCTGATTACTCTTGCCA AGACTGATAGCGTTGCTAATCGTCGTCTGGCATTCGCCCGTACTCGTGATAACGAGATCGTGGCAAAACTGTTTAACGAGCTG GGCCCGCGTTTCGCGAGCCGCGCCGGTGGTTACACTCGCATTCTGAAGTGTGGCTTCCGTGCAGGCGACAACGCGCCGATGG CATACATCGAGCTGGTTGATCGTTCAGAGAAAACAGAAGCTGCTGCAGAGTAA (SEQ ID NO: 382)
43_jfrlSe1402-8_Rissen	ATGCGCCATCGTAAGAGTGGTCGTCAACTGAACCGCAACAGCAGCCATCGCCAGGCTATGTTCCGCAACATGGCAGGTTCACT GGTTCGTCATGAAATCATCAAGACGACCCTGCCTAAAGCGAAAGAGCTGCGTCGCGTAGTTGAGCCGCTGATTACTCTTGCCA AGACTGATAGCGTTGCTAATCGTCGTCTGGCATTCGCCCGTACTCGTGATAACGAGATCGTGGCAAAACTGTTTAACGAGCTG GGCCCGCGTTTCGCGAGCCGCGCCGGTGGTTACACTCGCATTCTGAAGTGTGGCTTCCGTGCAGGCGACAACGCGCCGATGG CATACATCGAGCTGGTTGATCGTTCAGAGAAAACAGAAGCTGCTGCAGAGTAA (SEQ ID NO: 392)

Fig. 26D

44_jfrlSe1402-9_Rissen	ATGCGCCATCGTAAGAGTGGTCGTCAACTGAACCGCAACAGCAGCCATCGCCAGGCTATGTTCCGCAACATGGCAGGTTCACT GGTTCGTCATGAAATCATCAAGACGACCCTGCCTAAAGCGAAAGAGCTGCGTCGCGTAGTTGAGCCGCTGATTACTCTTGCCA AGACTGATAGCGTTGCTAATCGTCGTCTGGCATTCGCCCGTACTCGTGATAACGAGATCGTGGCAAAACTGTTTAACGAGCTG GGCCCGCGTTTCGCGAGCCGCGCCGGTGGTTACACTCGCATTCTGAAGTGTGGCTTCCGTGCAGGCGACAACGCGCCGATGG CATACATCGAGCTGGTTGATCGTTCAGAGAAAACAGAAGCTGCTGCAGAGTAA (SEQ ID NO: 399)
45_jfrlSe1402-10_Rissen	ATGCGCCATCGTAAGAGTGGTCGTCAACTGAACCGCAACAGCAGCCATCGCCAGGCTATGTTCCGCAACATGGCAGGTTCACT GGTTCGTCATGAAATCATCAAGACGACCCTGCCTAAAGCGAAAGAGCTGCGTCGCGTAGTTGAGCCGCTGATTACTCTTGCCA AGACTGATAGCGTTGCTAATCGTCGTCTGGCATTCGCCCGTACTCGTGATAACGAGATCGTGGCAAAACTGTTTAACGAGCTG GGCCCGCGTTTCGCGAGCCGCGCCGGTGGTTACACTCGCATTCTGAAGTGTGGCTTCCGTGCAGGCGACAACGCGCCGATGG CATACATCGAGCTGGTTGATCGTTCAGAGAAAACAGAAGCTGCTGCAGAGTAA (SEQ ID NO: 406)
46_jfrlSe1402-11_Rissen	ATGCGCCATCGTAAGAGTGGTCGTCAACTGAACCGCAACAGCAGCCATCGCCAGGCTATGTTCCGCAACATGGCAGGTTCACT GGTTCGTCATGAAATCATCAAGACGACCCTGCCTAAAGCGAAAGAGCTGCGTCGCGTAGTTGAGCCGCTGATTACTCTTGCCA AGACTGATAGCGTTGCTAATCGTCGTCTGGCATTCGCCCGTACTCGTGATAACGAGATCGTGGCAAAACTGTTTAACGAGCTG GGCCCGCGTTTCGCGAGCCGCGCCGGTGGTTACACTCGCATTCTGAAGTGTGGCTTCCGTGCAGGCGACAACGCGCCGATGG CATACATCGAGCTGGTTGATCGTTCAGAGAAAACAGAAGCTGCTGCAGAGTAA (SEQ ID NO: 413)
47_jfrlSe1402-12_Rissen	ATGCGCCATCGTAAGAGTGGTCGTCAACTGAACCGCAACAGCAGCCATCGCCAGGCTATGTTCCGCAACATGGCAGGTTCACT GGTTCGTCATGAAATCATCAAGACGACCCTGCCTAAAGCGAAAGAGCTGCGTCGCGTAGTTGAGCCGCTGATTACTCTTGCCA AGACTGATAGCGTTGCTAATCGTCGTCTGGCATTCGCCCGTACTCGTGATAACGAGATCGTGGCAAAACTGTTTAACGAGCTG GGCCCGCGTTTCGCGAGCCGCGCCGGTGGTTACACTCGCATTCTGAAGTGTGGCTTCCGTGCAGGCGACAACGCGCCGATGG CATACATCGAGCTGGTTGATCGTTCAGAGAAAACAGAAGCTGCTGCAGAGTAA (SEQ ID NO: 420)
48_jfrlSe1402-13_Mbandaka	ATGCGCCATCGTAAGAGTGGTCGTCAACTGAACCGCAACAGCAGCCATCGCCAGGCTATGTTCCGCAACATGGCAGGTTCACT GGTTCGTCATGAAATCATCAAGACGACCCTGCCTAAAGCGAAAGAGCTGCGTCGCGTAGTTGAGCCGCTGATTACTCTTGCCA AGACTGATAGCGTTGCTAATCGTCGTCTGGCATTCGCCCGTACTCGTGATAACGAGATCGTGGCAAAACTGTTTAACGAGCTG GGCCCGCGTTTCGCGAGCCGCGCCGGTGGTTACACTCGCATTCTGAAGTGTGGCTTCCGTGCAGGCGACAACGCGCCGATGG CATACATCGAGCTGGTTGATCGTTCAGAGAAAACAGAAGCTGCTGCAGAGTAA (SEQ ID NO: 427)
49_jfrlSe1402-14_Mbandaka	ATGCGCCATCGTAAGAGTGGTCGTCAACTGAACCGCAACAGCAGCCATCGCCAGGCTATGTTCCGCAACATGGCAGGTTCACT GGTTCGTCATGAAATCATCAAGACGACCCTGCCTAAAGCGAAAGAGCTGCGTCGCGTAGTTGAGCCGCTGATTACTCTTGCCA AGACTGATAGCGTTGCTAATCGTCGTCTGGCATTCGCCCGTACTCGTGATAACGAGATCGTGGCAAAACTGTTTAACGAGCTG GGCCCGCGTTTCGCGAGCCGCGCCGGTGGTTACACTCGCATTCTGAAGTGTGGCTTCCGTGCAGGCGACAACGCGCCGATGG CATACATCGAGCTGGTTGATCGTTCAGAGAAAACAGAAGCTGCTGCAGAGTAA (SEQ ID NO: 437)
50_jfrlSe1402-15_Orion	ATGCGCCATCGTAAGAGTGGTCGTCAACTGAACCGCAACAGCAGCCATCGCCAGGCTATGTTCCGCAACATGGCAGGTTCACT GGTTCGTCATGAAATCATCAAGACGACCCTGCCTAAAGCGAAAGAGCTGCGTCGCGTAGTTGAGCCGCTGATTACTCTTGCCA AGACTGATAGCGTTGCTAATCGTCGTCTGGCATTCGCCCGTACTCGTGATAACGAGATCGTGGCAAAACTGTTTAACGAGCTG GGCCCGCGTTTCGCGAGCCGCGCCGGTGGTTACACTCGCATTCTGAAGTGTGGCTTCCGTGCAGGCGACAACGCGCCGATGG CATACATCGAGCTGGTTGATCGTTCAGAGAAAACAGAAGCTGCTGCAGAGTAA (SEQ ID NO: 447)
51_jfrlSe1409-1_Altona	
52_jfrlSe1409-2_Istanbul	ATGCGCCATCGTAAGAGTGGTCGTCAACTGAACCGCAACAGCAGCCATCGCCAGGCTATGTTCCGCAACATGGCAGGTTCACT GGTTCGTCATGAAATCATCAAGACGACCCTGCCTAAAGCGAAAGAGCTGCGTCGCGTAGTTGAGCCGCTGATTACTCTTGCCA AGACTGATAGCGTTGCTAATCGTCGTCTGGCATTCGCCCGTACTCGTGATAACGAGATCGTGGCAAAACTGTTTAACGAGCTG GGCCCGCGTTTCGCGAGCCGCGCCGGTGGTTACACTCGCATTCTGAAGTGTGGCTTCCGTGCAGGCGACAACGCGCCGATGG CATACATCGAGCTGGTTGATCGTTCAGAGAAAACAGAAGCTGCTGCAGAGTAA (SEQ ID NO: 464)
53_jfrlSe1409-3_Senftenberg	
54_jfrlSe1409-4_UN_O13_	
55_jfrlSe1409-5_UN_O1,3,19_	ATGCGCCATCGTAAGAGTGGTCGTCAACTGAACCGCAACAGCAGCCATCGCCAGGCTATGTTCCGCAACATGGCAGGTTCACT GGTTCGTCATGAAATCATCAAGACGACCCTGCCTAAAGCGAAAGAGCTGCGTCGCGTAGTTGAGCCGCTGATTACTCTTGCCA AGACTGATAGCGTTGCTAATCGTCGTCTGGCATTCGCCCGTACTCGTGATAACGAGATCGTGGCAAAACTGTTTAACGAGCTG GGCCCGCGTTTCGCGAGCCGCGCCGGTGGTTACACTCGCATTCTGAAGTGTGGCTTCCGTGCAGGCGACAACGCGCCGATGG CATACATCGAGCTGGTTGATCGTTCAGAGAAAACAGAAGCTGCTGCAGAGTAA (SEQ ID NO: 485)
56_jfrlSe1409-6_Montevideo	
57_jfrlSe1409-7_UN_O1,3,19_	ATGCGCCATCGTAAGAGTGGTCGTCAACTGAACCGCAACAGCAGCCATCGCCAGGCTATGTTCCGCAACATGGCAGGTTCACT GGTTCGTCATGAAATCATCAAGACGACCCTGCCTAAAGCGAAAGAGCTGCGTCGCGTAGTTGAGCCGCTGATTACTCTTGCCA AGACTGATAGCGTTGCTAATCGTCGTCTGGCATTCGCCCGTACTCGTGATAACGAGATCGTGGCAAAACTGTTTAACGAGCTG GGCCCGCGTTTCGCGAGCCGCGCCGGTGGTTACACTCGCATTCTGAAGTGTGGCTTCCGTGCAGGCGACAACGCGCCGATGG CATACATCGAGCTGGTTGATCGTTCAGAGAAAACAGAAGCTGCTGCAGAGTAA (SEQ ID NO: 501)
58_jfrlSe1409-8_UN_O18_	ATGCGCCATCGTAAGAGTGGTCGTCAACTGAACCGCAACAGCAGCCATCGCCAGGCTATGTTCCGCAACATGGCAGGTTCACT GGTTCGTCATGAAATCATCAAGACGACCCTGCCTAAAGCGAAAGAGCTGCGTCGCGTAGTTGAGCCGCTGATTACTCTTGCCA AGACTGATAGCGTTGCTAATCGTCGTCTGGCATTCGCCCGTACTCGTGATAACGAGATCGTGGCAAAACTGTTTAACGAGCTG GGCCCGCGTTTCGCGAGCCGCGCCGGTGGTTACACTCGCATTCTGAAGTGTGGCTTCCGTGCAGGCGACAACGCGCCGATGG CATACATCGAGCTGGTTGATCGTTCAGAGAAAACAGAAGCTGCTGCAGAGTAA (SEQ ID NO: 510)
61_jfrlSe1409-11_UN_O1,3,19_	ATGCGCCATCGTAAGAGTGGTCGTCAACTGAACCGCAACAGCAGCCATCGCCAGGCTATGTTCCGCAACATGGCAGGTTCACT GGTTCGTCATGAAATCATCAAGACGACCCTGCCTAAAGCGAAAGAGCTGCGTCGCGTAGTTGAGCCGCTGATTACTCTTGCCA AGACTGATAGCGTTGCTAATCGTCGTCTGGCATTCGCCCGTACTCGTGATAACGAGATCGTGGCAAAACTGTTTAACGAGCTG GGCCCGCGTTTCGCGAGCCGCGCCGGTGGTTACACTCGCATTCTGAAGTGTGGCTTCCGTGCAGGCGACAACGCGCCGATGG CATACATCGAGCTGGTTGATCGTTCAGAGAAAACAGAAGCTGCTGCAGAGTAA (SEQ ID NO: 519)
67_jfrlSe1409-17_Rissen	ATGCGCCATCGTAAGAGTGGTCGTCAACTGAACCGCAACAGCAGCCATCGCCAGGCTATGTTCCGCAACATGGCAGGTTCACT GGTTCGTCATGAAATCATCAAGACGACCCTGCCTAAAGCGAAAGAGCTGCGTCGCGTAGTTGAGCCGCTGATTACTCTTGCCA AGACTGATAGCGTTGCTAATCGTCGTCTGGCATTCGCCCGTACTCGTGATAACGAGATCGTGGCAAAACTGTTTAACGAGCTG GGCCCGCGTTTCGCGAGCCGCGCCGGTGGTTACACTCGCATTCTGAAGTGTGGCTTCCGTGCAGGCGACAACGCGCCGATGG CATACATCGAGCTGGTTGATCGTTCAGAGAAAACAGAAGCTGCTGCAGAGTAA (SEQ ID NO: 528)
71_jfrlSe1409-21_Amsterdam	ATGCGCCATCGTAAGAGTGGTCGTCAACTGAACCGCAACAGCAGCCATCGCCAGGCTATGTTCCGCAACATGGCAGGTTCACT GGTTCGTCATGAAATCATCAAGACGACCCTGCCTAAAGCGAAAGAGCTGCGTCGCGTAGTTGAGCCGCTGATTACTCTTGCCA AGACTGATAGCGTTGCTAATCGTCGTCTGGCATTCGCCCGTACTCGTGATAACGAGATCGTGGCAAAACTGTTTAACGAGCTG GGCCCGCGTTTCGCGAGCCGCGCCGGTGGTTACACTCGCATTCTGAAGTGTGGCTTCCGTGCAGGCGACAACGCGCCGATGG CATACATCGAGCTGGTTGATCGTTCAGAGAAAACAGAAGCTGCTGCAGAGTAA (SEQ ID NO: 536)
80_HySe09_Enteritidis	ATGCGCCATCGTAAGAGTGGTCGTCAACTGAACCGCAACAGCAGCCATCGCCAGGCTATGTTCCGCAACATGGCAGGTTCACT GGTTCGTCATGAAATCATCAAGACGACCCTGCCTAAAGCGAAAGAGCTGCGTCGCGTAGTTGAGCCGCTGATTACTCTTGCCA AGACTGATAGCGTTGCTAATCGTCGTCTGGCATTCGCCCGTACTCGTGATAACGAGATCGTGGCAAAACTGTTTAACGAGCTG GGCCCGCGTTTCGCGAGCCGCGCCGGTGGTTACACTCGCATTCTGAAGTGTGGCTTCCGTGCAGGCGACAACGCGCCGATGG CATACATCGAGCTGGTTGATCGTTCAGAGAAAACAGAAGCTGCTGCAGAGTAA (SEQ ID NO: 546)

Fig. 26E

100_HySe29_Schwarzengrund	GTGAAAGAAGAGAAACCAGAATTCGATCCGATCTGCTGCGCCTGTTGACGATCTGGAATTTACTGGTCCGCTCTGCTAACTGC CTCAAGGCAGAAGCTATCCACTATATCGGTGATCTGGTACAGCGTACCGAGGTTGAGCTTCTTAAGACGCCTAACCTGGGTAA AAAATCTCTTACCGAGATTAAAGACGTGCTGGCTTCCCGTGGACTGTCTCTGGGTATGCGCCTGGAAAACTGGCCACCGGCAA GCATCGCTGACGAGTAACCGGATCACAGGTTAAGGTTTTACTGAGAAGGATAAGGTCATGCGCCATCGTAAGAGTGGTCGTCA ACTGAACCGCAACAGCAGCCATCGCCAGGCTATGTTCCGCAACATGGCAGGTTCACTGGTTCGTCATGAAATCATCAAGACGA CCCTGCCTAAAGCGAAAGAGCTGCGTCGCGTAGTTGAGCCGCTGATTACTCTTGCCAAGACTGATAGCGTTGCTAATCGTCGT CTGGCATTCGCCCGTACTCGTGATAACGAGATCGTGGCAAAACTGTTTAACGAGCTGGGCCCGCGTTTCGCGAGCCGCGCCG GTGGTTACACTCGCATTCTGAAGTGTGGCTTCCGTGCAGGCGACAACGCGCCGATGGCATACATCGAGCTGGTTGATCGTTCA GAGAAAACAGAAGCTGCTGCAGAGTAATCTGTAGTAACGTAAAAAAACCCGCTTCGGCGGGTTTTTTTATACCCTCCTGAACCC CATGTATCTACAATAATTGTATTCTTTTCGTT (SEQ ID NO: 556)
103_HySe32_Schwarzengrund	ATGCGCCATCGTAAGAGTGGTCGTCAACTGAACCGCAACAGCAGCCATCGCCAGGCTATGTTCCGCAACATGGCAGGTTCACT GGTTCGTCATGAAATCATCAAGACGACCCTGCCTAAAGCGAAAGAGCTGCGTCGCGTAGTTGAGCCGCTGATTACTCTTGCCA AGACTGATAGCGTTGCTAATCGTCGTCTGGCATTCGCCCGTACTCGTGATAACGAGATCGTGGCAAAACTGTTTAACGAGCTG GGCCCGCGTTTCGCGAGCCGCGCCGGTGGTTACACTCGCATTCTGAAGTGTGGCTTCCGTGCAGGCGACAACGCGCCGATGG CATACATCGAGCTGGTTGATCGTTCAGAGAAAACAGAAGCTGCTGCAGAGTAA (SEQ ID NO: 564)
106_HySe35_Schwarzengrund	ATGCGCCATCGTAAGAGTGGTCGTCAACTGAACCGCAACAGCAGCCATCGCCAGGCTATGTTCCGCAACATGGCAGGTTCACT GGTTCGTCATGAAATCATCAAGACGACCCTGCCTAAAGCGAAAGAGCTGCGTCGCGTAGTTGAGCCGCTGATTACTCTTGCCA AGACTGATAGCGTTGCTAATCGTCGTCTGGCATTCGCCCGTACTCGTGATAACGAGATCGTGGCAAAACTGTTTAACGAGCTG GGCCCGCGTTTCGCGAGCCGCGCCGGTGGTTACACTCGCATTCTGAAGTGTGGCTTCCGTGCAGGCGACAACGCGCCGATGG CATACATCGAGCTGGTTGATCGTTCAGAGAAAACAGAAGCTGCTGCAGAGTAA (SEQ ID NO: 571)

Fig. 27A

	sodA
01_NBRC13245T_Typhimurium	ATGAGTTATACACTGCCATCCCTGCCGTACGCTTATGATGCACTGGAACCGCACTTCGATAAGCAGACGATGGAGATTCACCA CACCAAACACCATCAAACCTATGTCAACAACGCTAACGCGGCGCTGGAAAACCTGCCTGAGTTTGCCAGCCTGCCGGTTGAAG AACTGATTACTAAACTGGACCAGGTGCCAGCGGACAAAAAACTGTGCTGCGTAACAACGCGGGCGGCCATGCTAACCACAGC CTGTTCTGGAAAGGGCTGAAAAAAGGCACCACTCTGCAGGGCGATCTGAAAGCGGCTATCGAGCGTGACTTCGGTTCCGTTGA CAACTTCAAAGCTGAATTCGAAAAAGCAGCAGCAACCCGTTTCGGCTCCGGCTGGGCGTGGCTGGTGCTGAAAGGCGACAAA CTGGCTGTGGTTTCTACCGCAAACCAGGATTCCCCGCTGATGGGTGAAGCCATTTCCGGCGCTTCCGGCTTCCCGATCCTGG GCCTGGACGTGTGGGAACACGCTTACTACCTGAAATTCCAGAACCGCCGCCCGGACTACATCAAAGAGTTCTGGAACGTGGT GAACTGGGACGAAGCAGCAGCGCGTTTCGCCGCTAAAAAATAA (SEQ ID NO: 57)
02_GTC00131_Enteritidis	ATGAGTTATACACTGCCATCCCTGCCGTACGCTTATGATGCACTGGAACCGCACTTCGATAAGCAGACGATGGAGATTCACCA CACCAAACACCATCAAACCTATGTCAACAACGCTAACGCGGCGCTGGAAAACTTGCCAGAGTTTGCTGACCTGCCGGTTGAAG AACTGATTACTAAACTGGACCAGGTGCCAGCGGACAAAAAACCGTGCTGCGTAATAACGCGGGCGGCCACGCTAACCACAGC TTCTTCTGGAAAGGCCTGAAAAAAGGCACCACCCTGCAGGGCGATCTGAAAGCGGCTATCGAGCGTGACTTCGGTTCCGTTGA CAACTTCAAAGCTGAATTCGAAAAAGCAGCAGCAACCCGTTTCGGCTCCGGCTGGGCGTGGCTGGTGCTGAAAGGCGACAAA CTGGCTGTGGTTTCTACCGCAAACCAGGATTCCCCGCTGATGGGTGAAGCCATTTCCGGCGCTTCCGGCTTCCCGATCCTGG GTCTGGACGTGTGGGAACACGCTTACTACCTGAAATTCCAGAACCGCCGCCCGGACTACATCAAAGAGTTCTGGAACGTGGTG AACTGGGACGAAGCAGCAGCGCGTTTCGCCGCTAAAAAATAA (SEQ ID NO: 69)
03_GTC09491_Enteritidis	ATGAGTTATACACTGCCATCCCTGCCGTACGCTTATGATGCACTGGAACCGCACTTCGATAAGCAGACGATGGAGATTCACCA CACCAAACACCATCAAACCTATGTCAACAACGCTAACGCGGCGCTGGAAAACTTGCCAGAGTTTGCTGACCTGCCGGTTGAAG AACTGATTACTAAACTGGACCAGGTGCCAGCGGACAAAAAACCGTGCTGCGTAATAACGCGGGCGGCCACGCTAACCACAGC TTCTTCTGGAAAGGCCTGAAAAAAGGCACCACCCTGCAGGGCGATCTGAAAGCGGCTATCGAGCGTGACTTCGGTTCCGTTGA CAACTTCAAAGCTGAATTCGAAAAAGCAGCAGCAACCCGTTTCGGCTCCGGCTGGGCGTGGCTGGTGCTGAAAGGCGACAAA CTGGCTGTGGTTTCTACCGCAAACCAGGATTCCCCGCTGATGGGTGAAGCCATTTCCGGCGCTTCCGGCTTCCCGATCCTGG GTCTGGACGTGTGGGAACACGCTTACTACCTGAAATTCCAGAACCGCCGCCCGGACTACATCAAAGAGTTCTGGAACGTGGTG AACTGGGACGAAGCAGCAGCGCGTTTCGCCGCTAAAAAATAA (SEQ ID NO: 78)
04_GTC03838_Enteritidis	ATGAGTTATACACTGCCATCCCTGCCGTACGCTTATGATGCACTGGAACCGCACTTCGATAAGCAGACGATGGAGATTCACCA CACCAAACACCATCAAACCTATGTCAACAACGCTAACGCGGCGCTGGAAAACTTGCCAGAGTTTGCTGACCTGCCGGTTGAAG AACTGATTACTAAACTGGACCAGGTGCCAGCGGACAAAAAACCGTGCTGCGTAATAACGCGGGCGGCCACGCTAACCACAGC TTCTTCTGGAAAGGCCTGAAAAAAGGCACCACCCTGCAGGGCGATCTGAAAGCGGCTATCGAGCGTGACTTCGGTTCCGTTGA CAACTTCAAAGCTGAATTCGAAAAAGCAGCAGCAACCCGTTTCGGCTCCGGCTGGGCGTGGCTGGTGCTGAAAGGCGACAAA CTGGCTGTGGTTTCTACCGCAAACCAGGATTCCCCGCTGATGGGTGAAGCCATTTCCGGCGCTTCCGGCTTCCCGATCCTGG GTCTGGACGTGTGGGAACACGCTTACTACCTGAAATTCCAGAACCGCCGCCCGGACTACATCAAAGAGTTCTGGAACGTGGTG AACTGGGACGAAGCAGCAGCGCGTTTCGCCGCTAAAAAATAA (SEQ ID NO: 85)
05_GTC08914_Enteritidis	ATGAGTTATACACTGCCATCCCTGCCGTACGCTTATGATGCACTGGAACCGCACTTCGATAAGCAGACGATGGAGATTCACCA CACCAAACACCATCAAACCTATGTCAACAACGCTAACGCGGCGCTGGAAAACTTGCCAGAGTTTGCTGACCTGCCGGTTGAAG AACTGATTACTAAACTGGACCAGGTGCCAGCGGACAAAAAACCGTGCTGCGTAATAACGCGGGCGGCCACGCTAACCACAGC TTCTTCTGGAAAGGCCTGAAAAAAGGCACCACCCTGCAGGGCGATCTGAAAGCGGCTATCGAGCGTGACTTCGGTTCCGTTGA CAACTTCAAAGCTGAATTCGAAAAAGCAGCAGCAACCCGTTTCGGCTCCGGCTGGGCGTGGCTGGTGCTGAAAGGCGACAAA CTGGCTGTGGTTTCTACCGCAAACCAGGATTCCCCGCTGATGGGTGAAGCCATTTCCGGCGCTTCCGGCTTCCCGATCCTGG GTCTGGACGTGTGGGAACACGCTTACTACCTGAAATTCCAGAACCGCCGCCCGGACTACATCAAAGAGTTCTGGAACGTGGTG AACTGGGACGAAGCAGCAGCGCGTTTCGCCGCTAAAAAATAA (SEQ ID NO: 92)
06_GTC09421_Enteritidis	ATGAGTTATACACTGCCATCCCTGCCGTACGCTTATGATGCACTGGAACCGCACTTCGATAAGCAGACGATGGAGATTCACCA CACCAAACACCATCAAACCTATGTCAACAACGCTAACGCGGCGCTGGAAAACTTGCCAGAGTTTGCTGACCTGCCGGTTGAAG AACTGATTACTAAACTGGACCAGGTGCCAGCGGACAAAAAACCGTGCTGCGTAATAACGCGGGCGGCCACGCTAACCACAGC TTCTTCTGGAAAGGCCTGAAAAAAGGCACCACCCTGCAGGGCGATCTGAAAGCGGCTATCGAGCGTGACTTCGGTTCCGTTGA CAACTTCAAAGCTGAATTCGAAAAAGCAGCAGCAACCCGTTTCGGCTCCGGCTGGGCGTGGCTGGTGCTGAAAGGCGACAAA CTGGCTGTGGTTTCTACCGCAAACCAGGATTCCCCGCTGATGGGTGAAGCCATTTCCGGCGCTTCCGGCTTCCCGATCCTGG GTCTGGACGTGTGGGAACACGCTTACTACCTGAAATTCCAGAACCGCCGCCCGGACTACATCAAAGAGTTCTGGAACGTGGTG AACTGGGACGAAGCAGCAGCGCGTTTCGCCGCTAAAAAATAA (SEQ ID NO: 102)
07_GTC09489_Enteritidis	ATGAGTTATACACTGCCATCCCTGCCGTACGCTTATGATGCACTGGAACCGCACTTCGATAAGCAGACGATGGAGATTCACCA CACCAAACACCATCAAACCTATGTCAACAACGCTAACGCGGCGCTGGAAAACTTGCCAGAGTTTGCTGACCTGCCGGTTGAAG AACTGATTACTAAACTGGACCAGGTGCCAGCGGACAAAAAACCGTGCTGCGTAATAACGCGGGCGGCCACGCTAACCACAGC TTCTTCTGGAAAGGCCTGAAAAAAGGCACCACCCTGCAGGGCGATCTGAAAGCGGCTATCGAGCGTGACTTCGGTTCCGTTGA CAACTTCAAAGCTGAATTCGAAAAAGCAGCAGCAACCCGTTTCGGCTCCGGCTGGGCGTGGCTGGTGCTGAAAGGCGACAAA CTGGCTGTGGTTTCTACCGCAAACCAGGATTCCCCGCTGATGGGTGAAGCCATTTCCGGCGCTTCCGGCTTCCCGATCCTGG GTCTGGACGTGTGGGAACACGCTTACTACCTGAAATTCCAGAACCGCCGCCCGGACTACATCAAAGAGTTCTGGAACGTGGTG AACTGGGACGAAGCAGCAGCGCGTTTCGCCGCTAAAAAATAA (SEQ ID NO: 112)

Fig. 27B

08_NBRC3163_Pullorum_Gallinarum	ATGAGTTATACACTGCCATCCCTGCCGTACGCTTATGATGCACTGGAACCGCACTTCGATAAGCAGACGATGGAGATTCACCA CACCAAACACCATCAAACCTATGTCAACAACGCTAACGCGGCGCTGGAAAACTTGCCAGAGTTTGCTGACCTGCCGGTTGAAG AACTGATTACTAAACTGGACCAGGTGCCAGCGGACAAAAAAACCGTGCTGCGTAATAACGCGGGCGGCCACGCTAACCACAGC TTCTTCTGGAAAGGCCTGAAAAAAGGCACCACCCTGCAGGGCGATCTGAAAGCGGCTATCGAGCGTGACTTCGGTTCCGTTGA CAACTTCAAAGCTGAATTCGAAAAAGCAGCAGCAACCCGTTTCGGCTCCGGCTGGGCGTGGCTGGTGCTGAAAGGCGACAAA CTGGCTGTGGTTTCTACCGCAAACCAGGATTCCCCGCTGATGGGTGAAGCCATTTCCGGCGCTTCCGGCTTCCCGATCCTGG GTCTGGACGTGTGGGAACACGCTTACTACCTGAAATTCCAGAACCGCCGCCCGGACTACATCAAAGAGTTCTGGAACGTGGTG AACTGGGACGAAGCAGCAGCGCGTTTCGCCGCTAAAAAATAA (SEQ ID NO: 119)
09_NBRC3313_Enteritidis	ATGAGTTATACACTGCCATCCCTGCCGTACGCTTATGATGCACTGGAACCGCACTTCGATAAGCAGACGATGGAGATTCACCA CACCAAACACCATCAAACCTATGTCAACAACGCTAACGCGGCGCTGGAAAACTTGCCAGAGTTTGCTGACCTGCCGGTTGAAG AACTGATTACTAAACTGGACCAGGTGCCAGCGGACAAAAAAACCGTGCTGCGTAATAACGCGGGCGGCCACGCTAACCACAGC TTCTTCTGGAAAGGCCTGAAAAAAGGCACCACCCTGCAGGGCGATCTGAAAGCGGCTATCGAGCGTGACTTCGGTTCCGTTGA CAACTTCAAAGCTGAATTCGAAAAAGCAGCAGCAACCCGTTTCGGCTCCGGCTGGGCGTGGCTGGTGCTGAAAGGCGACAAA CTGGCTGTGGTTTCTACCGCAAACCAGGATTCCCCGCTGATGGGTGAAGCCATTTCCGGCGCTTCCGGCTTCCCGATCCTGG GTCTGGACGTGTGGGAACACGCTTACTACCTGAAATTCCAGAACCGCCGCCCGGACTACATCAAAGAGTTCTGGAACGTGGTG AACTGGGACGAAGCAGCAGCGCGTTTCGCCGCTAAAAAATAA (SEQ ID NO: 129)
10_NBRC12529_Typhimurium	ATGAGTTATACACTGCCATCCCTGCCGTACGCTTATGATGCACTGGAACCGCACTTCGATAAGCAGACGATGGAGATTCACCA CACCAAACACCATCAAACCTATGTCAACAACGCTAACGCGGCGCTGGAAAACCTGCCTGAGTTTGCCAGCCTGCCGGTTGAAG AACTGATTACTAAACTGGACCAGGTGCCAGCGGACAAAAAAACTGTGCTGCGTAACAACGCGGGCGGCCATGCTAACCACAGC CTGTTCTGGAAAGGGCTGAAAAAAGGCACCACTCTGCAGGGCGATCTGAAAGCGGCTATCGAGCGTGACTTCGGTTCCGTTGA CAACTTCAAAGCTGAATTCGAAAAAGCAGCAGCAACCCGTTTCGGCTCCGGCTGGGCGTGGCTGGTGCTGAAAGGCGACAAA CTGGCTGTGGTTTCTACCGCAAACCAGGATTCCCCGCTGATGGGTGAAGCCATTTCCGGCGCTTCCGGCTTCCCGATCCTGG GCCTGGACGTGTGGGAACACGCTTACTACCTGAAATTCCAGAACCGCCGCCCGGACTACATCAAAGAGTTCTGGAACGTGGT GAACTGGGACGAAGCAGCAGCGCGTTTCGCCGCTAAAAAATAA (SEQ ID NO: 136)
11_NBRC14193_Typhimurium	ATGAGTTATACACTGCCATCCCTGCCGTACGCTTATGATGCACTGGAACCGCACTTCGATAAGCAGACGATGGAGATTCACCA CACCAAACACCATCAAACCTATGTCAACAACGCTAACGCGGCGCTGGAAAACCTGCCTGAGTTTGCCAGCCTGCCGGTTGAAG AACTGATTACTAAACTGGACCAGGTGCCAGCGGACAAAAAAACTGTGCTGCGTAACAACGCGGGCGGCCATGCTAACCACAGC CTGTTCTGGAAAGGGCTGAAAAAAGGCACCACTCTGCAGGGCGATCTGAAAGCGGCTATCGAGCGTGACTTCGGTTCCGTTGA CAACTTCAAAGCTGAATTCGAAAAAGCAGCAGCAACCCGTTTCGGCTCCGGCTGGGCGTGGCTGGTGCTGAAAGGCGACAAA CTGGCTGTGGTTTCTACCGCAAACCAGGATTCCCCGCTGATGGGTGAAGCCATTTCCGGCGCTTCCGGCTTCCCGATCCTGG GCCTGGACGTGTGGGAACACGCTTACTACCTGAAATTCCAGAACCGCCGCCCGGACTACATCAAAGAGTTCTGGAACGTGGT GAACTGGGACGAAGCAGCAGCGCGTTTCGCCGCTAAAAAATAA (SEQ ID NO: 143)
12_NBRC14194_Typhimurium	ATGAGTTATACACTGCCATCCCTGCCGTACGCTTATGATGCACTGGAACCGCACTTCGATAAGCAGACGATGGAGATTCACCA CACCAAACACCATCAAACCTATGTCAACAACGCTAACGCGGCGCTGGAAAACCTGCCTGAGTTTGCCAGCCTGCCGGTTGAAG AACTGATTACTAAACTGGACCAGGTGCCAGCGGACAAAAAAACTGTGCTGCGTAACAACGCGGGCGGCCATGCTAACCACAGC CTGTTCTGGAAAGGGCTGAAAAAAGGCACCACTCTGCAGGGCGATCTGAAAGCGGCTATCGAGCGTGACTTCGGTTCCGTTGA CAACTTCAAAGCTGAATTCGAAAAAGCAGCAGCAACCCGTTTCGGCTCCGGCTGGGCGTGGCTGGTGCTGAAAGGCGACAAA CTGGCTGTGGTTTCTACCGCAAACCAGGATTCCCCGCTGATGGGTGAAGCCATTTCCGGCGCTTCCGGCTTCCCGATCCTGG GCCTGGACGTGTGGGAACACGCTTACTACCTGAAATTCCAGAACCGCCGCCCGGACTACATCAAAGAGTTCTGGAACGTGGT GAACTGGGACGAAGCAGCAGCGCGTTTCGCCGCTAAAAAATAA (SEQ ID NO: 150)
13_NBRC14209_Typhimurium	ATGAGTTATACACTGCCATCCCTGCCGTACGCTTATGATGCACTGGAACCGCACTTCGATAAGCAGACGATGGAGATTCACCA CACCAAACACCATCAAACCTATGTCAACAACGCTAACGCGGCGCTGGAAAACCTGCCTGAGTTTGCCAGCCTGCCGGTTGAAG AACTGATTACTAAACTGGACCAGGTGCCAGCGGACAAAAAAACTGTGCTGCGTAACAACGCGGGCGGCCATGCTAACCACAGC CTGTTCTGGAAAGGGCTGAAAAAAGGCACCACTCTGCAGGGCGATCTGAAAGCGGCTATCGAGCGTGACTTCGGTTCCGTTGA CAACTTCAAAGCTGAATTCGAAAAAGCAGCAGCAACCCGTTTCGGCTCCGGCTGGGCGTGGCTGGTGCTGAAAGGCGACAAA CTGGCTGTGGTTTCTACCGCAAACCAGGATTCCCCGCTGATGGGTGAAGCCATTTCCGGCGCTTCCGGCTTCCCGATCCTGG GCCTGGACGTGTGGGAACACGCTTACTACCTGAAATTCCAGAACCGCCGCCCGGACTACATCAAAGAGTTCTGGAACGTGGT GAACTGGGACGAAGCAGCAGCGCGTTTCGCCGCTAAAAAATAA (SEQ ID NO: 157)
14_NBRC14210_Typhimurium	ATGAGTTATACACTGCCATCCCTGCCGTACGCTTATGATGCACTGGAACCGCACTTCGATAAGCAGACGATGGAGATTCACCA CACCAAACACCATCAAACCTATGTCAACAACGCTAACGCGGCGCTGGAAAACCTGCCTGAGTTTGCCAGCCTGCCGGTTGAAG AACTGATTACTAAACTGGACCAGGTGCCAGCGGACAAAAAAACTGTGCTGCGTAACAACGCGGGCGGCCATGCTAACCACAGC CTGTTCTGGAAAGGGCTGAAAAAAGGCACCACTCTGCAGGGCGATCTGAAAGCGGCTATCGAGCGTGACTTCGGTTCCGTTGA CAACTTCAAAGCTGAATTCGAAAAAGCAGCAGCAACCCGTTTCGGCTCCGGCTGGGCGTGGCTGGTGCTGAAAGGCGACAAA CTGGCTGTGGTTTCTACCGCAAACCAGGATTCCCCGCTGATGGGTGAAGCCATTTCCGGCGCTTCCGGCTTCCCGATCCTGG GCCTGGACGTGTGGGAACACGCTTACTACCTGAAATTCCAGAACCGCCGCCCGGACTACATCAAAGAGTTCTGGAACGTGGT GAACTGGGACGAAGCAGCAGCGCGTTTCGCCGCTAAAAAATAA (SEQ ID NO: 164)
15_NBRC14211_Typhimurium	ATGAGTTATACACTGCCATCCCTGCCGTACGCTTATGATGCACTGGAACCGCACTTCGATAAGCAGACGATGGAGATTCACCA CACCAAACACCATCAAACCTATGTCAACAACGCTAACGCGGCGCTGGAAAACCTGCCTGAGTTTGCCAGCCTGCCGGTTGAAG AACTGATTACTAAACTGGACCAGGTGCCAGCGGACAAAAAAACTGTGCTGCGTAACAACGCGGGCGGCCATGCTAACCACAGC CTGTTCTGGAAAGGGCTGAAAAAAGGCACCACTCTGCAGGGCGATCTGAAAGCGGCTATCGAGCGTGACTTCGGTTCCGTTGA CAACTTCAAAGCTGAATTCGAAAAAGCAGCAGCAACCCGTTTCGGCTCCGGCTGGGCGTGGCTGGTGCTGAAAGGCGACAAA CTGGCTGTGGTTTCTACCGCAAACCAGGATTCCCCGCTGATGGGTGAAGCCATTTCCGGCGCTTCCGGCTTCCCGATCCTGG GCCTGGACGTGTGGGAACACGCTTACTACCTGAAATTCCAGAACCGCCGCCCGGACTACATCAAAGAGTTCTGGAACGTGGT GAACTGGGACGAAGCAGCAGCGCGTTTCGCCGCTAAAAAATAA (SEQ ID NO: 170)
16_NBRC14212_Typhimurium	ATGAGTTATACACTGCCATCCCTGCCGTACGCTTATGATGCACTGGAACCGCACTTCGATAAGCAGACGATGGAGATTCACCA CACCAAACACCATCAAACCTATGTCAACAACGCTAACGCGGCGCTGGAAAACCTGCCTGAGTTTGCCAGCCTGCCGGTTGAAG AACTGATTACTAAACTGGACCAGGTGCCAGCGGACAAAAAAACTGTGCTGCGTAACAACGCGGGCGGCCATGCTAACCACAGC CTGTTCTGGAAAGGGCTGAAAAAAGGCACCACTCTGCAGGGCGATCTGAAAGCGGCTATCGAGCGTGACTTCGGTTCCGTTGA CAACTTCAAAGCTGAATTCGAAAAAGCAGCAGCAACCCGTTTCGGCTCCGGCTGGGCGTGGCTGGTGCTGAAAGGCGACAAA CTGGCTGTGGTTTCTACCGCAAACCAGGATTCCCCGCTGATGGGTGAAGCCATTTCCGGCGCTTCCGGCTTCCCGATCCTGG GCCTGGACGTGTGGGAACACGCTTACTACCTGAAATTCCAGAACCGCCGCCCGGACTACATCAAAGAGTTCTGGAACGTGGT GAACTGGGACGAAGCAGCAGCGCGTTTCGCCGCTAAAAAATAA (SEQ ID NO: 177)
17_NBRC15181_Typhimurium	ATGAGTTATACACTGCCATCCCTGCCGTACGCTTATGATGCACTGGAACCGCACTTCGATAAGCAGACGATGGAGATTCACCA CACCAAACACCATCAAACCTATGTCAACAACGCTAACGCGGCGCTGGAAAACCTGCCTGAGTTTGCCAGCCTGCCGGTTGAAG AACTGATTACTAAACTGGACCAGGTGCCAGCGGACAAAAAAACTGTGCTGCGTAACAACGCGGGCGGCCATGCTAACCACAGC CTGTTCTGGAAAGGGCTGAAAAAAGGCACCACTCTGCAGGGCGATCTGAAAGCGGCTATCGAGCGTGACTTCGGTTCCGTTGA CAACTTCAAAGCTGAATTCGAAAAAGCAGCAGCAACCCGTTTCGGCTCCGGCTGGGCGTGGCTGGTGCTGAAAGGCGACAAA CTGGCTGTGGTTTCTACCGCAAACCAGGATTCCCCGCTGATGGGTGAAGCCATTTCCGGCGCTTCCGGCTTCCCGATCCTGG GCCTGGACGTGTGGGAACACGCTTACTACCTGAAATTCCAGAACCGCCGCCCGGACTACATCAAAGAGTTCTGGAACGTGGT GAACTGGGACGAAGCAGCAGCGCGTTTCGCCGCTAAAAAATAA (SEQ ID NO: 184)

Fig. 27C

18_NBRC15182_Minnesota	ATGAGTTATACACTGCCATCCCTGCCGTACGCTTATGATGCACTGGAACCGCACTTCGATAAGCAGACGATGGAGATTCACCA CACCAAACACCATCAAACCTATGTCAACAACGCTAACGCGGCGCTGGAAAACCTGCCTGAGTTTGCTGACCTGCCGGTTGAAG AACTGATTACTAAACTGGACCAGGTGCCAGCGGACAAAAAAACCGTGCTGCGTAACAACGCCGGCGGCCACGCTAACCACAG CCTGTTCTGGAAAGGGCTGAAAAAAGGCACCACTCTGCAGGGCGACCTGAAAGCGGCTATCGAGCGTGACTTCGGTTCCGTT GACAACTTCAAAGCTGAATTCGAAAAAGCAGCAGCAACCCGTTTCGGCTCCGGCTGGGCGTGGCTGGTGCTGAAAGGCGACA AACTGGCTGTGGTTTCTACCGCAAACCAGGATTCCCCGCTGATGGGTGAAGCCATTTCCGGCGCTTCCGGCTTTCCGATCCT GGGTCTGGACGTGTGGGAACACGCTTACTATCTGAAATTCCAAAACCGCCGCCCGGACTACATCAAAGAGTTCTGGAACGTGG TGAACTGGGACGAAGCAGCAGCGCGTTTCGCCGCTAAAAAATAA (SEQ ID NO: 194)
19_NBRC15183_Minnesota	ATGAGTTATACACTGCCATCCCTGCCGTACGCTTATGATGCACTGGAACCGCACTTCGATAAGCAGACGATGGAGATTCACCA CACCAAACACCATCAAACCTATGTCAACAACGCTAACGCGGCGCTGGAAAACCTGCCTGAGTTTGCTGACCTGCCGGTTGAAG AACTGATTACTAAACTGGACCAGGTGCCAGCGGACAAAAAAACCGTGCTGCGTAACAACGCCGGCGGCCACGCTAACCACAG CCTGTTCTGGAAAGGGCTGAAAAAAGGCACCACTCTGCAGGGCGACCTGAAAGCGGCTATCGAGCGTGACTTCGGTTCCGTT GACAACTTCAAAGCTGAATTCGAAAAAGCAGCAGCAACCCGTTTCGGCTCCGGCTGGGCGTGGCTGGTGCTGAAAGGCGACA AACTGGCTGTGGTTTCTACCGCAAACCAGGATTCCCCGCTGATGGGTGAAGCCATTTCCGGCGCTTCCGGCTTTCCGATCCT GGGTCTGGACGTGTGGGAACACGCTTACTATCTGAAATTCCAAAACCGCCGCCCGGACTACATCAAAGAGTTCTGGAACGTGG TGAACTGGGACGAAGCAGCAGCGCGTTTCGCCGCTAAAAAATAA (SEQ ID NO: 204)
20_NBRC15184_Minnesota	ATGAGTTATACACTGCCATCCCTGCCGTACGCTTATGATGCACTGGAACCGCACTTCGATAAGCAGACGATGGAGATTCACCA CACCAAACACCATCAAACCTATGTCAACAACGCTAACGCGGCGCTGGAAAACCTGCCTGAGTTTGCTGACCTGCCGGTTGAAG AACTGATTACTAAACTGGACCAGGTGCCAGCGGACAAAAAAACCGTGCTGCGTAACAACGCCGGCGGCCACGCTAACCACAG CCTGTTCTGGAAAGGGCTGAAAAAAGGCACCACTCTGCAGGGCGACCTGAAAGCGGCTATCGAGCGTGACTTCGGTTCCGTT GACAACTTCAAAGCTGAATTCGAAAAAGCAGCAGCAACCCGTTTCGGCTCCGGCTGGGCGTGGCTGGTGCTGAAAGGCGACA AACTGGCTGTGGTTTCTACCGCAAACCAGGATTCCCCGCTGATGGGTGAAGCCATTTCCGGCGCTTCCGGCTTTCCGATCCT GGGTCTGGACGTGTGGGAACACGCTTACTATCTGAAATTCCAAAACCGCCGCCCGGACTACATCAAAGAGTTCTGGAACGTGG TGAACTGGGACGAAGCAGCAGCGCGTTTCGCCGCTAAAAAATAA (SEQ ID NO: 211)
21_NBRC15185_Minnesota	ATGAGTTATACACTGCCATCCCTGCCGTACGCTTATGATGCACTGGAACCGCACTTCGATAAGCAGACGATGGAGATTCACCA CACCAAACACCATCAAACCTATGTCAACAACGCTAACGCGGCGCTGGAAAACCTGCCTGAGTTTGCTGACCTGCCGGTTGAAG AACTGATTACTAAACTGGACCAGGTGCCAGCGGACAAAAAAACCGTGCTGCGTAACAACGCCGGCGGCCACGCTAACCACAG CCTGTTCTGGAAAGGGCTGAAAAAAGGCACCACTCTGCAGGGCGACCTGAAAGCGGCTATCGAGCGTGACTTCGGTTCCGTT GACAACTTCAAAGCTGAATTCGAAAAAGCAGCAGCAACCCGTTTCGGCTCCGGCTGGGCGTGGCTGGTGCTGAAAGGCGACA AACTGGCTGTGGTTTCTACCGCAAACCAGGATTCCCCGCTGATGGGTGAAGCCATTTCCGGCGCTTCCGGCTTTCCGATCCT GGGTCTGGACGTGTGGGAACACGCTTACTATCTGAAATTCCAAAACCGCCGCCCGGACTACATCAAAGAGTTCTGGAACGTGG TGAACTGGGACGAAGCAGCAGCGCGTTTCGCCGCTAAAAAATAA (SEQ ID NO: 218)
22_NBRC15186_Minnesota	ATGAGTTATACACTGCCATCCCTGCCGTACGCTTATGATGCACTGGAACCGCACTTCGATAAGCAGACGATGGAGATTCACCA CACCAAACACCATCAAACCTATGTCAACAACGCTAACGCGGCGCTGGAAAACCTGCCTGAGTTTGCTGACCTGCCGGTTGAAG AACTGATTACTAAACTGGACCAGGTGCCAGCGGACAAAAAAACCGTGCTGCGTAACAACGCCGGCGGCCACGCTAACCACAG CCTGTTCTGGAAAGGGCTGAAAAAAGGCACCACTCTGCAGGGCGACCTGAAAGCGGCTATCGAGCGTGACTTCGGTTCCGTT GACAACTTCAAAGCTGAATTCGAAAAAGCAGCAGCAACCCGTTTCGGCTCCGGCTGGGCGTGGCTGGTGCTGAAAGGCGACA AACTGGCTGTGGTTTCTACCGCAAACCAGGATTCCCCGCTGATGGGTGAAGCCATTTCCGGCGCTTCCGGCTTTCCGATCCT GGGTCTGGACGTGTGGGAACACGCTTACTATCTGAAATTCCAAAACCGCCGCCCGGACTACATCAAAGAGTTCTGGAACGTGG TGAACTGGGACGAAGCAGCAGCGCGTTTCGCCGCTAAAAAATAA (SEQ ID NO: 225)
23_NBRC15187_Minnesota	ATGAGTTATACACTGCCATCCCTGCCGTACGCTTATGATGCACTGGAACCGCACTTCGATAAGCAGACGATGGAGATTCACCA CACCAAACACCATCAAACCTATGTCAACAACGCTAACGCGGCGCTGGAAAACCTGCCTGAGTTTGCTGACCTGCCGGTTGAAG AACTGATTACTAAACTGGACCAGGTGCCAGCGGACAAAAAAACCGTGCTGCGTAACAACGCCGGCGGCCACGCTAACCACAG CCTGTTCTGGAAAGGGCTGAAAAAAGGCACCACTCTGCAGGGCGACCTGAAAGCGGCTATCGAGCGTGACTTCGGTTCCGTT GACAACTTCAAAGCTGAATTCGAAAAAGCAGCAGCAACCCGTTTCGGCTCCGGCTGGGCGTGGCTGGTGCTGAAAGGCGACA AACTGGCTGTGGTTTCTACCGCAAACCAGGATTCCCCGCTGATGGGTGAAGCCATTTCCGGCGCTTCCGGCTTTCCGATCCT GGGTCTGGACGTGTGGGAACACGCTTACTATCTGAAATTCCAAAACCGCCGCCCGGACTACATCAAAGAGTTCTGGAACGTGG TGAACTGGGACGAAGCAGCAGCGCGTTTCGCCGCTAAAAAATAA (SEQ ID NO: 232)
24_NBRC100797_Abony	ATGAGTTATACACTGCCATCCCTGCCGTACGCTTATGATGCACTGGAACCGCACTTCGATAAGCAGACGATGGAGATTCACCA CACCAAACACCATCAAACCTATGTCAACAACGCTAACGCGGCGCTGGAAAACCTGCCTGAGTTTGCTGACCTGCCGGTTGAAG AACTGATTACTAAACTGGATCAGGTGCCAGCGGACAAAAAAACCGTGCTGCGTAACAACGCTGGCGGCCATGCTAACCACAGC TTCTTCTGGAAGGGCCTGAAAAAAGGCACCACCCTGCAGGGTGATCTGAAAGCGGCTATCGAGCGTGACTTCGGTTCCGTTGA CAACTTCAAAGCTGAATTCGAAAAAGCAGCAGCAACCCGTTTCGGCTCCGGCTGGGCGTGGCTGGTGCTGAAAGGCGACAAA CTGGCTGTGGTTTCTACCGCAAACCAGGATTCCCCGCTGATGGGTGAAGCCATTTCCGGCGCTTCCGGCTTCCCGATCCTGG GTCTGGACGTGTGGGAACACGCTTACTATCTGAAATTCCAGAACCGCCGCCCGGACTACATCAAAGAGTTCTGGAACGTGGTG AACTGGGACGAAGCAGCAGCGCGTTTCGCCGCTAAAAAATAA (SEQ ID NO: 239)
25_NBRC105684_Choleraesuis	ATGAGTTATACACTGCCATCCCTGCCGTACGCTTATGATGCACTGGAACCGCACTTCGATAAGCAGACGATGGCGATTCACCA CACCAAACACCATCAAACCTATGTCAACAACGCTAACGCGGCGCTGGAAAACCTGCCTGAGTTTGCTGACCTGCCGGTTGAAG AACTGATTACTAAACTGGATCAGGTGCCAGCGGACAAAAAAACCGTACTGCGTAACAACGCGGGCGGCCACGCTAACCACAGC CTGTTCTGGAAAGGGCTGAAAAAAGGCACCACTCTGCAGGGCGATCTGAAAGCGGCTATCGAGCGTGACTTCGGTTCCGTTGA CAACTTCAAAGCTGAATTCGAAAAAGCAGCAGCAACCCGTTTCGGCTCCGGCTGGGCGTGGCTGGTGCTGAAAGGCGACAAA CTGGCTGTGGTTTCTACCGCAAACCAGGATTCCCCGCTGATGGGTGAAGCCATTTCCGGCGCTTCCGGCTTCCCGATCCTGG GCCTGGACGTGTGGGAACACGCTTACTACCTGAAATTCCAGAACCGCCGCCCGGACTACATCAAAGAGTTCTGGAACGTGGT GAACTGGGACGAAGCAGCAGCGCGTTTCGCCGCTAAAAAATAA (SEQ ID NO: 249)
26_NBRC105726_Typhimurium	ATGAGTTATACACTGCCATCCCTGCCGTACGCTTATGATGCACTGGAACCGCACTTCGATAAGCAGACGATGGAGATTCACCA CACCAAACACCATCAAACCTATGTCAACAACGCTAACGCGGCGCTGGAAAACCTGCCTGAGTTTGCCAGCCTGCCGGTTGAAG AACTGATTACTAAACTGGACCAGGTGCCAGCGGACAAAAAAACTGTGCTGCGTAACAACGCGGGCGGCCATGCTAACCACAGC CTGTTCTGGAAAGGGCTGAAAAAAGGCACCACTCTGCAGGGCGATCTGAAAGCGGCTATCGAGCGTGACTTCGGTTCCGTTGA CAACTTCAAAGCTGAATTCGAAAAAGCAGCAGCAACCCGTTTCGGCTCCGGCTGGGCGTGGCTGGTGCTGAAAGGCGACAAA CTGGCTGTGGTTTCTACCGCAAACCAGGATTCCCCGCTGATGGGTGAAGCCATTTCCGGCGCTTCCGGCTTCCCGATCCTGG GCCTGGACGTGTGGGAACACGCTTACTACCTGAAATTCCAGAACCGCCGCCCGGACTACATCAAAGAGTTCTGGAACGTGGT GAACTGGGACGAAGCAGCAGCGCGTTTCGCCGCTAAAAAATAA (SEQ ID NO: 259)
27_JCM3919_UN_O7_	ATGAGTTATACACTGCCATCCCTGCCGTACGCTTATGATGCACTGGAACCGCACTTCGATAAGCAGACGATGGAGATTCACCA CACCAAACACCATCAAACCTATGTCAACAACGCTAACGCGGCGCTGGAAAACCTGCCTGAGTTTGCTGACCTGCCGGTTGAAG AACTGATTACTAAACTGGATCAGGTGCCAGCGGACAAAAAAACCGTGCTGCGTAACAACGCGGGCGGCCACGCTAACCACAGC CTGTTCTGGAAAGGGCTGAAAAAAGGCACCACTCTGCAGGGCGATCTGAAAGCGGCTATCGAGCGTGACTTCGGTTCCGTTGA CAACTTCAAAGCTGAATTCGAAAAAGCAGCAGCAACCCGTTTCGGCTCCGGCTGGGCGTGGCTGGTGCTGAAAGGCGACAAA CTGGCTGTGGTTTCTACCGCAAACCAGGATTCCCCGCTGATGGGTGAAGCCATTTCCGGCGCTTCCGGCTTCCCGATCCTGG GTCTGGACGTTTGGGAACACGCTTACTACCTGAAATTCCAGAACCGCCGCCCGGACTACATCAAAGAGTTCTGGAACGTGGTG AACTGGGACGAAGCAGCAGCGCGTTTCGCCGCTAAAAAATAA (SEQ ID NO: 268)

Fig. 27D

28_NBRC15335_Minnesota	ATGAGTTATACACTGCCATCCCTGCCGTACGCTTATGATGCACTGGAACCGCACTTCGATAAGCAGACGATGGAGATTCACCA CACCAAACACCATCAAACCTATGTCAACAACGCTAACGCGGCGCTGGAAAACCTGCCTGAGTTTGCTGACCTGCCGGTTGAAG AACTGATTACTAAACTGGACCAGGTGCCAGCGGACAAAAAAACCGTGCTGCGTAACAACGCCGGCGGCCACGCTAACCACAG CCTGTTCTGGAAAGGGCTGAAAAAAGGCACCACTCTGCAGGGCGACCTGAAAGCGGCTATCGAGCGTGACTTCGGTTCCGTT GACAACTTCAAAGCTGAATTCGAAAAAGCAGCAGCAACCCGTTTCGGCTCCGGCTGGGCGTGGCTGGTGCTGAAAGGCGACA AACTGGCTGTGGTTTCTACCGCAAACCAGGATTCCCCGCTGATGGGTGAAGCCATTTCCGGCGCTTCCGGCTTTCCGATCCT GGGTCTGGACGTGTGGGAACACGCTTACTATCTGAAATTCCAAAACCGCCGCCCGGACTACATCAAAGAGTTCTGGAACGTGG TGAACTGGGACGAAGCAGCAGCGCGTTTCGCCGCTAAAAAATAA (SEQ ID NO: 275)
29_GTC09490_Enteritidis	ATGAGTTATACACTGCCATCCCTGCCGTACGCTTATGATGCACTGGAACCGCACTTCGATAAGCAGACGATGGAGATTCACCA CACCAAACACCATCAAACCTATGTCAACAACGCTAACGCGGCGCTGGAAAACTTGCCAGAGTTTGCTGACCTGCCGGTTGAAG AACTGATTACTAAACTGGACCAGGTGCCAGCGGACAAAAAAACCGTGCTGCGTAATAACGCGGGCGGCCACGCTAACCACAGC TTCTTCTGGAAAGGCCTGAAAAAAGGCACCACCCTGCAGGGCGATCTGAAAGCGGCTATCGAGCGTGACTTCGGTTCCGTTGA CAACTTCAAAGCTGAATTCGAAAAAGCAGCAGCAACCCGTTTCGGCTCCGGCTGGGCGTGGCTGGTGCTGAAAGGCGACAAA CTGGCTGTGGTTTCTACCGCAAACCAGGATTCCCCGCTGATGGGTGAAGCCATTTCCGGCGCTTCCGGCTTCCCGATCCTGG GTCTGGACGTGTGGGAACACGCTTACTACCTGAAATTCCAGAACCGCCGCCCGGACTACATCAAAGAGTTCTGGAACGTGGTG AACTGGGACGAAGCAGCAGCGCGTTTCGCCGCTAAAAAATAA (SEQ ID NO: 282)
30_GTC09492_Braenderup	ATGAGTTATACACTGCCATCCCTGCCGTACGCTTATGATGCACTGGAACCGCACTTCGATAAGCAGACGATGGAGATTCACCA CACCAAACACCATCAAACCTATGTCAACAACGCTAACGCGGCGCTGGAAAACCTGCCTGAGTTTGCTGACCTGCCGGTTGAAG AACTGATTACTAAACTGGACCAGGTGCCAGCGGACAAAAAAACCGTGCTGCGTAACAACGCCGGCGGCCACGCTAACCACAG CCTGTTCTGGAAAGGGCTGAAAAAAGGCACCACTCTGCAGGGCGATCTGAAAGCGGCTATCGAGCGTGACTTCGGTTCCGTT GACAACTTCAAAGCTGAATTCGAAAAAGCAGCAGCAACCCGTTTCGGCTCCGGCTGGGCGTGGCTGGTGCTGAAAGGCGACA AACTGGCTGTGGTTTCTACCGCAAACCAGGATTCCCCGCTGATGGGTGAAGCCGTTTCCGGCGCTTCCGGCTTCCCGATCCT GGGCCTGGACGTGTGGGAACACGCTTACTACCTGAAATTCCAGAACCGCCGCCCGGACTACATCAAAGAGTTCTGGAACGTG GTGAACTGGGACGAAGCAGCAGCGCGTTTCGCCGCTAAAAAATAA (SEQ ID NO: 289)
31_GTC09493_Pakistan	ATGAGTTATACACTGCCATCCCTGCCGTACGCTTATGATGCACTGGAACCGCACTTCGATAAGCAGACGATGGAGATTCACCA CACCAAACACCATCAAACCTATGTCAACAACGCTAACGCGGCGCTGGAAAACTTGCCAGAGTTTGCTGACCTGCCGGTTGAAG AACTGATTACTAAACTGGACCAGGTGCCAGCGGACAAAAAAACCGTGCTGCGTAACAACGCGGGCGGCCACGCTAACCACAG CTTCTTCTGGAAAGGCCTGAAAAAAGGCTCCACCCTGCAGGGCGATCTGAAAGCGGCTATCGAGCGTGACTTCGGTTCCGTT GACAACTTCAAAGCTGAATTCGAAAAAGCAGCAGCAACCCGTTTCGGCTCCGGCTGGGCGTGGCTGGTGCTGAAAGGCGACA AACTGGCTGTGGTTTCTACCGCAAACCAGGATTCCCCGCTGATGGGTGAAGCCATTTCCGGCGCTTCCGGCTTCCCGATCCT GGGTCTGGACGTGTGGGAACACGCTTACTACCTGAAATTCCAGAACCGCCGCCCGGACTACATCAAAGAGTTCTGGAACGTG GTGAACTGGGACGAAGCAGCAGCGCGTTTCGCCGCTAAAAAATAA (SEQ ID NO: 296)
32_GTC09549_Typhimurium	ATGAGTTATACACTGCCATCCCTGCCGTACGCTTATGATGCACTGGAACCGCACTTCGATAAGCAGACGATGGAGATTCACCA CACCAAACACCATCAAACCTATGTCAACAACGCTAACGCGGCGCTGGAAAACCTGCCTGAGTTTGCCAGCCTGCCGGTTGAAG AACTGATTACTAAACTGGACCAGGTGCCAGCGGACAAAAAAACTGTGCTGCGTAACAACGCGGGCGGCCATGCTAACCACAGC CTGTTCTGGAAAGGGCTGAAAAAAGGCACCACTCTGCAGGGCGATCTGAAAGCGGCTATCGAGCGTGACTTCGGTTCCGTTGA CAACTTCAAAGCTGAATTCGAAAAAGCAGCAGCAACCCGTTTCGGCTCCGGCTGGGCGTGGCTGGTGCTGAAAGGCGACAAA CTGGCTGTGGTTTCTACCGCAAACCAGGATTCCCCGCTGATGGGTGAAGCCATTTCCGGCGCTTCCGGCTTCCCGATCCTGG GCCTGGACGTGTGGGAACACGCTTACTACCTGAAATTCCAGAACCGCCGCCCGGACTACATCAAAGAGTTCTGGAACGTGGT GAACTGGGACGAAGCAGCAGCGCGTTTCGCCGCTAAAAAATAA (SEQ ID NO: 303)
33_ATCCBAA-1675_Infantis	ATGAGTTATACACTGCCATCCCTGCCGTACGCTTATGATGCACTGGAACCGCACTTCGATAAGCAGACGATGGAGATTCACCA CACCAAACACCATCAAACCTATGTCAACAACGCTAACGCGGCGCTGGAAAACCTGCCTGAGTTTGCTGACCTGCCGGTTGAAG AACTGATTACTAAACTGGACCAGGTGCCAGCGGACAAAAAAACCGTGCTGCGTAACAACGCCGGCGGCCACGCTAACCACAG CCTGTTCTGGAAAGGGCTGAAAAAAGGCACCACTCTGCAGGGCGACCTGAAAGCGGCTATCGAGCGTGACTTCGGTTCCGTT GACAACTTCAAAGCTGAATTCGAAAAAGCAGCAGCAACCCGTTTCGGCTCCGGCTGGGCGTGGCTGGTGCTGAAAGGCGACA AACTGGCTGTGGTTTCTACCGCAAACCAGGATTCCCCGCTGATGGGTGAAGCCATTTCCGGCGCTTCCGGCTTCCCGATCCT GGGTCTGGACGTGTGGGAACACGCTTACTACCTGAAATTCCAGAACCGCCGCCCGGACTACATCAAAGAGTTCTGGAACGTG GTGAACTGGGACGAAGCAGCAGCGCGTTTCGCCGCTAAAAAATAA (SEQ ID NO: 310)
34_ATCCBAA-1738_Thompson	ATGAGTTATACACTGCCATCCCTGCCGTACGCTTATGATGCACTGGAACCGCACTTCGATAAGCAGACGATGGAGATTCACCA CACCAAACACCATCAAACCTATGTCAACAACGCTAACGCGGCGCTGGAAAACCTGCCTGAGTTTGCTGACCTGCCGGTTGAAG AACTGATTACTAAACTGGATCAGGTGCCAGCGGACAAAAAAACCGTGCTGCGTAACAACGCGGGCGGCCACGCTAACCACAGC CTGTTCTGGAAAGGGCTGAAAAAAGGCACCACTCTGCAGGGCGATCTGAAAGCGGCTATCGAGCGTGACTTCGGTTCCGTTGA CAACTTCAAAGCTGAATTCGAAAAAGCAGCAGCAACCCGTTTCGGCTCCGGCTGGGCGTGGCTGGTGCTGAAAGGCGACAAA CTGGCTGTGGTTTCTACCGCAAACCAGGATTCCCCGCTGATGGGTGAAGCCATTTCCGGCGCTTCCGGCTTCCCGATCCTGG GTCTGGACGTGTGGGAACACGCTTACTATCTGAAATTCCAGAACCGCCGCCCGGACTACATCAAAGAGTTCTGGAACGTGGTG AACTGGGACGAAGCAGCAGCGCGTTTCGCCGCTAAAAAATAA (SEQ ID NO: 321)
35_ATCC9712_Saintpaul	ATGAGTTATACACTGCCATCCCTGCCGTACGCTTATGATGCACTGGAACCGCACTTCGATAAGCAGACGATGGAGATTCACCA CACCAAACACCATCAAACCTATGTCAACAACGCTAACGCGGCGCTGGAAAACCTGCCTGAGTTTGCCAGCCTGCCGGTTGAAG AACTGATTACTAAACTGGACCAGGTGCCAGCGGACAAAAAAACTGTGCTGCGTAACAACGCGGGCGGCCATGCTAACCACAGC CTGTTCTGGAAAGGGCTGAAAAAAGGCACCACTCTGCAGGGCGATCTGAAAGCGGCTATCGAGCGTGACTTCGGTTCCGTTGA CAACTTCAAAGCTGAATTCGAAAAAGCAGCAGCAACCCGTTTCGGCTCCGGCTGGGCGTGGCTGGTGCTGAAAGGCGACAAA CTGGCTGTGGTTTCTACCGCAAACCAGGATTCCCCGCTGATGGGTGAAGCCATTTCCGGCGCTTCCGGCTTCCCGATCCTGG GCCTGGACGTGTGGGAACACGCTTACTACCTGAAATTCCAGAACCGCCGCCCGGACTACATCAAAGAGTTCTGGAACGTGGT GAACTGGGACGAAGCAGCAGCGCGTTTCGCCGCTAAAAAATAA (SEQ ID NO: 331)
36_jfrlSe1402-1_Infantis	ATGAGTTATACACTGCCATCCCTGCCGTACGCTTATGATGCACTGGAACCGCACTTCGATAAGCAGACGATGGAGATTCACCA CACCAAACACCATCAAACCTATGTCAACAACGCTAACGCGGCGCTGGAAAACCTGCCTGAGTTTGCTGACCTGCCGGTTGAAG AACTGATTACTAAACTGGACCAGGTGCCAGCGGACAAAAAAACCGTGCTGCGTAACAACGCCGGCGGCCACGCTAACCACAG CCTGTTCTGGAAAGGACTGAAAAAAGGCACCACTCTGCAGGGCGATCTGAAAGCGGCTATCGAGCGTGACTTCGGTTCCGTTG ACAACTTCAAAGCTGAATTCGAAAAAGCAGCAGCAACCCGTTTCGGCTCCGGCTGGGCGTGGCTGGTGCTGAAAGGCGACAAA CTGGCTGTGGTTTCTACCGCAAACCAGGATTCCCCGCTGATGGGTGAAGCCATTTCCGGTGCTTCCGGCTTCCCGATCCTGG GTCTGGACGTGTGGGAACACGCTTACTATCTGAAATTCCAGAACCGCCGCCCGGACTACATCAAAGAGTTCTGGAACGTGGTG AACTGGGACGAAGCAGCAGCGCGTTTCGCCGCTAAAAAATAA (SEQ ID NO: 342)
38_jfrlSe1402-3_Brandenburg	ATGAGTTATACACTGCCATCCCTGCCGTACGCTTATGATGCACTGGAACCGCACTTCGATAAGCAGACGATGGAGATTCACCA CACCAAACACCATCAAACCTATGTCAACAACGCTAACGCGGCGCTGGAAAACCTGCCTGAGTTTGCTGACCTGCCGGTTGAAG AACTGATTACTAAACTGGACCAGGTGCCAGCGGACAAAAAAACCGTGCTGCGTAACAACGCCGGCGGCCACGCTAACCACAG CCTGTTCTGGAAAGGGCTGAAAAAAGGCACCACTCTGCAGGGCGACCTGAAAGCGGCTATCGAGCGTGACTTCGGTTCCGTT GACAACTTCAAAGCTGAATTCGAAAAAGCAGCAGCAACCCGTTTCGGCTCCGGCTGGGCGTGGCTGGTGCTGAAAGGCGACA AACTGGCTGTGGTTTCTACCGCAAACCAGGATTCCCCGCTGATGGGTGAAGCCATTTCCGGCGCTTCCGGCTTCCCGATCCT GGGTCTGGACGTGTGGGAACACGCTTACTACCTGAAATTCCAGAACCGCCGCCCGGACTACATCAAAGAGTTCTGGAACGTG GTGAACTGGGACGAAGCAGCAGCGCGTTTCGCCGCTAAAAAATAA (SEQ ID NO: 352)

Fig. 27E

39_jfrlSe1402-4_Infantis	ATGAGTTATACACTGCCATCCCTGCCGTACGCTTATGATGCACTGGAACCGCACTTCGATAAGCAGACGATGGAGATTCACCA CACCAAACACCATCAAACCTATGTCAACAACGCTAACGCGGCGCTGGAAAACCTGCCTGAGTTTGCTGACCTGCCGGTTGAAG AACTGATTACTAAACTGGACCAGGTGCCAGCGGACAAAAAAACCGTGCTGCGTAACAACGCCGGCGGCCACGCTAACCACAG CCTGTTCTGGAAAGGGCTGAAAAAAGGCACCACTCTGCAGGGCGATCTGAAAGCGGCTATCGAGCGTGACTTCGGTTCCGTT GACAACTTCAAAGCTGAATTCGAAAAAGCAGCAGCAACCCGTTTCGGCTCCGGCTGGGCGTGGCTGGTGCTGAAAGGCGACA AACTGGCTGTGGTTTCTACCGCAAACCAGGATTCCCCGCTGATGGGTGAAGCCATTTCCGGCGCTTCCGGCTTCCCGATCCT GGGTCTGGACGTGTGGGAACACGCTTACTATCTGAAATTCCAGAACCGCCGCCCGGACTACATCAAAGAGTTCTGGAACGTGG TGAACTGGGACGAAGCAGCAGCGCGTTTCGCCGCTAAAAAATAA (SEQ ID NO: 359)
40_jfrlSe1402-5_Brandenburg	ATGAGTTATACACTGCCATCCCTGCCGTACGCTTATGATGCACTGGAACCGCACTTCGATAAGCAGACGATGGAGATTCACCA CACCAAACACCATCAAACCTATGTCAACAACGCTAACGCGGCGCTGGAAAACCTGCCTGAGTTTGCTGACCTGCCGGTTGAAG AACTGATTACTAAACTGGACCAGGTGCCAGCGGACAAAAAAACCGTGCTGCGTAACAACGCCGGCGGCCACGCTAACCACAG CCTGTTCTGGAAAGGGCTGAAAAAAGGCACCACTCTGCAGGGCGACCTGAAAGCGGCTATCGAGCGTGACTTCGGTTCCGTT GACAACTTCAAAGCTGAATTCGAAAAAGCAGCAGCAACCCGTTTCGGCTCCGGCTGGGCGTGGCTGGTGCTGAAAGGCGACA AACTGGCTGTGGTTTCTACCGCAAACCAGGATTCCCCGCTGATGGGTGAAGCCATTTCCGGCGCTTCCGGCTTCCCGATCCT GGGTCTGGACGTGTGGGAACACGCTTACTACCTGAAATTCCAGAACCGCCGCCCGGACTACATCAAAGAGTTCTGGAACGTG GTGAACTGGGACGAAGCAGCAGCGCGTTTCGCCGCTAAAAAATAA (SEQ ID NO: 369)
41_jfrlSe1402-6_Rissen	ATGAGTTATACACTGCCATCCCTGCCGTACGCTTATGATGCACTGGAACCGCACTTCGATAAGCAGACGATGGAGATTCACCA CACCAAACACCATCAAACCTATGTCAACAACGCTAACGCGGCGCTGGAAAACCTGCCTGAGTTTGCTGACCTGCCGGTTGAAG AACTGATTACTAAACTGGACCAGGTGCCAGCGGACAAAAAAACCGTGCTGCGTAACAACGCCGGCGGCCACGCTAACCACAG CCTGTTCTGGAAAGGGCTGAAAAAAGGCACCACTCTGCAGGGCGATCTGAAAGTGGCTATCGAGCGTGACTTCGGTTCCGTTG ACAACTTCAAAGCTGAATTCGAAAAAGCAGCAGCAACCCGTTTCGGCTCCGGCTGGGCGTGGCTGGTGCTGAAAGGTGACAAA CTGGCTGTGGTTTCTACCGCAAACCAGGATTCCCCGCTGATGGGTGAAGCCATTTCCGGCGCTTCCGGCTTCCCGATCCTGG GTCTGGACGTGTGGGAACACGCTTACTACCTGAAATTCCAGAACCGCCGCCCGGACTACATCAAAGAGTTCTGGAACGTGGTG AACTGGGACGAAGCAGCAGCGCGTTTCGCCGCTAAAAAATAA (SEQ ID NO: 376)
42_jfrlSe1402-7_Orion	ATGAGTTATACACTGCCATCCCTGCCGTACGCTTATGATGCACTGGAACCGCACTTCGATAAGCAGACGATGGAGATTCACCA CACCAAACACCATCAAACCTATGTCAACAACGCTAACGCGGCGCTGGAAAACCTGCCTGAGTTTGCCAGCCTGCCGGTTGAAG AACTGATTACTAAACTGGACCAGGTGCCAGCGGACAAAAAAACTGTGCTGCGTAACAACGCGGGCGGCCATGCTAACCACAGC CTGTTCTGGAAAGGGCTGAAAAAAGGCACCACTCTGCAGGGCGATCTGAAAGCGGCTATCGAGCGTGACTTCGGTTCCGTTGA CAACTTCAAAGCTGAATTCGAAAAAGCAGCAGCAACCCGTTTCGGCTCCGGCTGGGCGTGGCTGGTGCTGAAAGGCGACAAA CTGGCTGTGGTTTCTACCGCAAACCAGGATTCCCCGCTGATGGGTGAAGCCATTTCCGGCGCTTCCGGCTTCCCGATCCTGG GCCTGGACGTGTGGGAACACGCTTACTACCTGAAATTCCAGAACCGCCGCCCGGACTACATCAAAGAGTTCTGGAACGTGGT GAACTGGGACGAAGCAGCAGCGCGTTTCGCCGCTAAAAAATAA (SEQ ID NO: 383)
43_jfrlSe1402-8_Rissen	ATGAGTTATACACTGCCATCCCTGCCGTACGCTTATGATGCACTGGAACCGCACTTCGATAAGCAGACGATGGAGATTCACCA CACCAAACACCATCAAACCTATGTCAACAACGCTAACGCGGCGCTGGAAAACCTGCCTGAGTTTGCTGACCTGCCGGTTGAAG AACTGATTACTAAACTGGACCAGGTGCCAGCGGACAAAAAAACCGTGCTGCGTAACAACGCCGGCGGCCACGCTAACCACAG CCTGTTCTGGAAAGGGCTGAAAAAAGGCACCACTCTGCAGGGCGATCTGAAAGTGGCTATCGAGCGTGACTTCGGTTCCGTTG ACAACTTCAAAGCTGAATTCGAAAAAGCAGCAGCAACCCGTTTCGGCTCCGGCTGGGCGTGGCTGGTGCTGAAAGGTGACAAA CTGGCTGTGGTTTCTACCGCAAACCAGGATTCCCCGCTGATGGGTGAAGCCATTTCCGGCGCTTCCGGCTTCCCGATCCTGG GTCTGGACGTGTGGGAACACGCTTACTACCTGAAATTCCAGAACCGCCGCCCGGACTACATCAAAGAGTTCTGGAACGTGGTG AACTGGGACGAAGCAGCAGCGCGTTTCGCCGCTAAAAAATAA (SEQ ID NO: 393)
44_jfrlSe1402-9_Rissen	ATGAGTTATACACTGCCATCCCTGCCGTACGCTTATGATGCACTGGAACCGCACTTCGATAAGCAGACGATGGAGATTCACCA CACCAAACACCATCAAACCTATGTCAACAACGCTAACGCGGCGCTGGAAAACCTGCCTGAGTTTGCTGACCTGCCGGTTGAAG AACTGATTACTAAACTGGACCAGGTGCCAGCGGACAAAAAAACCGTGCTGCGTAACAACGCCGGCGGCCACGCTAACCACAG CCTGTTCTGGAAAGGGCTGAAAAAAGGCACCACTCTGCAGGGCGATCTGAAAGTGGCTATCGAGCGTGACTTCGGTTCCGTTG ACAACTTCAAAGCTGAATTCGAAAAAGCAGCAGCAACCCGTTTCGGCTCCGGCTGGGCGTGGCTGGTGCTGAAAGGTGACAAA CTGGCTGTGGTTTCTACCGCAAACCAGGATTCCCCGCTGATGGGTGAAGCCATTTCCGGCGCTTCCGGCTTCCCGATCCTGG GTCTGGACGTGTGGGAACACGCTTACTACCTGAAATTCCAGAACCGCCGCCCGGACTACATCAAAGAGTTCTGGAACGTGGTG AACTGGGACGAAGCAGCAGCGCGTTTCGCCGCTAAAAAATAA (SEQ ID NO: 400)
45_jfrlSe1402-10_Rissen	ATGAGTTATACACTGCCATCCCTGCCGTACGCTTATGATGCACTGGAACCGCACTTCGATAAGCAGACGATGGAGATTCACCA CACCAAACACCATCAAACCTATGTCAACAACGCTAACGCGGCGCTGGAAAACCTGCCTGAGTTTGCTGACCTGCCGGTTGAAG AACTGATTACTAAACTGGACCAGGTGCCAGCGGACAAAAAAACCGTGCTGCGTAACAACGCCGGCGGCCACGCTAACCACAG CCTGTTCTGGAAAGGGCTGAAAAAAGGCACCACTCTGCAGGGCGATCTGAAAGTGGCTATCGAGCGTGACTTCGGTTCCGTTG ACAACTTCAAAGCTGAATTCGAAAAAGCAGCAGCAACCCGTTTCGGCTCCGGCTGGGCGTGGCTGGTGCTGAAAGGTGACAAA CTGGCTGTGGTTTCTACCGCAAACCAGGATTCCCCGCTGATGGGTGAAGCCATTTCCGGCGCTTCCGGCTTCCCGATCCTGG GTCTGGACGTGTGGGAACACGCTTACTACCTGAAATTCCAGAACCGCCGCCCGGACTACATCAAAGAGTTCTGGAACGTGGTG AACTGGGACGAAGCAGCAGCGCGTTTCGCCGCTAAAAAATAA (SEQ ID NO: 407)
46_jfrlSe1402-11_Rissen	ATGAGTTATACACTGCCATCCCTGCCGTACGCTTATGATGCACTGGAACCGCACTTCGATAAGCAGACGATGGAGATTCACCA CACCAAACACCATCAAACCTATGTCAACAACGCTAACGCGGCGCTGGAAAACCTGCCTGAGTTTGCTGACCTGCCGGTTGAAG AACTGATTACTAAACTGGACCAGGTGCCAGCGGACAAAAAAACCGTGCTGCGTAACAACGCCGGCGGCCACGCTAACCACAG CCTGTTCTGGAAAGGGCTGAAAAAAGGCACCACTCTGCAGGGCGATCTGAAAGTGGCTATCGAGCGTGACTTCGGTTCCGTTG ACAACTTCAAAGCTGAATTCGAAAAAGCAGCAGCAACCCGTTTCGGCTCCGGCTGGGCGTGGCTGGTGCTGAAAGGTGACAAA CTGGCTGTGGTTTCTACCGCAAACCAGGATTCCCCGCTGATGGGTGAAGCCATTTCCGGCGCTTCCGGCTTCCCGATCCTGG GTCTGGACGTGTGGGAACACGCTTACTACCTGAAATTCCAGAACCGCCGCCCGGACTACATCAAAGAGTTCTGGAACGTGGTG AACTGGGACGAAGCAGCAGCGCGTTTCGCCGCTAAAAAATAA (SEQ ID NO: 414)
47_jfrlSe1402-12_Rissen	ATGAGTTATACACTGCCATCCCTGCCGTACGCTTATGATGCACTGGAACCGCACTTCGATAAGCAGACGATGGAGATTCACCA CACCAAACACCATCAAACCTATGTCAACAACGCTAACGCGGCGCTGGAAAACCTGCCTGAGTTTGCTGACCTGCCGGTTGAAG AACTGATTACTAAACTGGACCAGGTGCCAGCGGACAAAAAAACCGTGCTGCGTAACAACGCCGGCGGCCACGCTAACCACAG CCTGTTCTGGAAAGGGCTGAAAAAAGGCACCACTCTGCAGGGCGATCTGAAAGTGGCTATCGAGCGTGACTTCGGTTCCGTTG ACAACTTCAAAGCTGAATTCGAAAAAGCAGCAGCAACCCGTTTCGGCTCCGGCTGGGCGTGGCTGGTGCTGAAAGGTGACAAA CTGGCTGTGGTTTCTACCGCAAACCAGGATTCCCCGCTGATGGGTGAAGCCATTTCCGGCGCTTCCGGCTTCCCGATCCTGG GTCTGGACGTGTGGGAACACGCTTACTACCTGAAATTCCAGAACCGCCGCCCGGACTACATCAAAGAGTTCTGGAACGTGGTG AACTGGGACGAAGCAGCAGCGCGTTTCGCCGCTAAAAAATAA (SEQ ID NO: 421)
48_jfrlSe1402-13_Mbandaka	ATGAGTTATACACTGCCATCCCTGCCGTACGCTTATGATGCACTGGAACCGCACTTCGATAAGCAGACGATGGAGATTCACCA CACCAAACACCATCAAACCTATGTCAACAACGCTAACGCGGCGCTGGAAAACCTGCCTGAGTTTGCTGACCTGCCGGTTGAAG AACTGATTACTAAACTGGATCAGGTGCCAGCGGACAAAAAAACCGTGCTGCGTAACAACGCGGGCGGCCACGCTAACCACAGC CTGTTCTGGAAAGGGCTGAAAAAAGGCACCACTCTGCAGGGCGATCTGAAAGCGGCTATCGAGCGTGACTTCGGTTCCGTTGA CAACTTCAAAGCTGAATTCGAAAAAGCAGCAGCAACCCGTTTCGGCTCCGGCTGGGCGTGGCTGGTGCTGAAAGGCGACAAA CTGGCTGTGGTTTCTACCGCAAACCAGGATTCCCCGCTGATGGGTGAAGCCATTTCCGGCGCTTCCGGCTTCCCGATCCTGG GTCTGGACGTGTGGGAACACGCTTACTACCTGAAATTCCAGAACCGCCGCCCGGACTACATCAAAGAGTTCTGGAACGTGGTG AACTGGGACGAAGCAGCAGCGCGTTTCGCCGCTAAAAAATAA (SEQ ID NO: 428)

Fig. 27F

49_jfrlSe1402-14_Mbandaka	ATGAGTTATACACTGCCATCCCTGCCGTACGCTTATGATGCACTGGAACCGCACTTCGATAAGCAGACGATGGAGATTCACCA CACCAAACACCATCAAACCTATGTCAACAACGCTAACGCGGCGCTGGAAAACCTGCCTGAGTTTGCTGACCTGCCGGTTGAAG AACTGATTACTAAACTGGATCAGGTGCCAGCGGACAAAAAACCGTGCTGCGTAACAACGCGGGCGGCCACGCTAACCACAGC CTGTTCTGGAAAGGGCTGAAAAAAGGCACCACTCTGCAGGGCGATCTGAAAGCGGCTATCGAGCGTGACTTCGGTTCCGTTGA CAACTTCAAAGCTGAATTCGAAAAAGCAGCAGCAACCCGTTTCGGCTCCGGCTGGGCGTGGCTGGTGCTGAAAGGCGACAAA CTGGCTGTGGTTTCTACCGCAAACCAGGATTCCCCGCTGATGGGTGAAGCCATTTCCGGCGCTTCCGGCTTCCCGATCCTGG GTCTGGACGTGTGGGAACACGCTTACTACCTGAAATTCCAGAACCGCCGCCCGGACTACATCAAAGAGTTCTGGAACGTGGTG AACTGGGACGAAGCAGCAGCGCGTTTCGCCGCTAAAAAATAA (SEQ ID NO: 438)
50_jfrlSe1402-15_Orion	ATGAGTTATACACTGCCATCCCTGCCGTACGCTTATGATGCACTGGAACCGCACTTCGATAAGCAGACGATGGAGATTCACCA CACCAAACACCATCAAACCTATGTCAACAACGCTAACGCGGCGCTGGAAAACCTGCCTGAGTTTGCCAGCCTGCCGGTTGAAG AACTGATTACTAAACTGGACCAGGTGCCAGCGGACAAAAAAACTGTGCTGCGTAACAACGCGGGCGGCCATGCTAACCACAGC CTGTTCTGGAAAGGGCTGAAAAAAGGCACCACTCTGCAGGGCGATCTGAAAGCGGCTATCGAGCGTGACTTCGGTTCCGTTGA CAACTTCAAAGCTGAATTCGAAAAAGCAGCAGCAACCCGTTTCGGCTCCGGCTGGGCGTGGCTGGTGCTGAAAGGCGACAAA CTGGCTGTGGTTTCTACCGCAAACCAGGATTCCCCGCTGATGGGTGAAGCCATTTCCGGCGCTTCCGGCTTCCCGATCCTGG GCCTGGACGTGTGGGAACACGCTTACTACCTGAAATTCCAGAACCGCCGCCCGGACTACATCAAAGAGTTCTGGAACGTGGT GAACTGGGACGAAGCAGCAGCGCGTTTCGCCGCTAAAAAATAA (SEQ ID NO: 448)
51_jfrlSe1409-1_Altona	ATGAGTTATACACTGCCATCCCTGCCGTACGCTTATGATGCACTGGAACCGCACTTCGATAAGCAGACGATGGAGATTCACCA CACCAAACACCATCAAACCTATGTCAACAACGCTAACGCGGCGCTGGAAAACCTGCCTGAGTTTGCTGACCTGCCGGTTGAAG AACTGATTACTAAACTGGATCAGGTGCCAGCGGACAAAAAACCGTGCTGCGTAACAACGCGGGCGGCCACGCTAACCACAGC CTGTTCTGGAAAGGGCTGAAAAAAGGCACCACTCTGCAGGGCGATCTGAAAGCGGCTATCGAGCGTGACTTCGGTTCCGTTGA CAACTTCAAAGCTGAATTCGAAAAAGCAGCAGCAACCCGTTTCGGCTCCGGCTGGGCGTGGCTGGTGCTGAAAGGCGACAAA CTGGCTGTGGTTTCTACCGCAAACCAGGATTCCCCGCTGATGGGTGAAGCCATTTCCGGCGCTTCCGGCTTCCCGATCCTGG GCCTGGACGTGTGGGAACACGCTTACTACCTGAAATTCCAGAACCGCCGCCCGGACTACATCAAAGAGTTCTGGAACGTGGT GAACTGGGACGAAGCAGCAGCGCGTTTCGCCGCTAAAAAATAA (SEQ ID NO: 457)
52_jfrlSe1409-2_Istanbul	ATGAGTTATACACTGCCATCCCTGCCGTACGCTTATGATGCACTGGAACCGCACTTCGATAAGCAGACGATGGAGATTCACCA CACCAAACACCATCAAACCTATGTCAACAACGCTAACGCGGCGCTGGAAAACCTGCCTGAGTTTGCTGACCTGCCGGTTGAAG AACTGATTACTAAACTGGACCAGGTGCCAGCGGACAAAAAAACCGTGCTGCGTAACAACGCCGGCGGCCACGCTAACCACAG CCTGTTCTGGAAAGGGCTGAAAAAAGGCACCACTCTGCAGGGCGATCTGAAAGCGGCTATCGAGCGTGACTTCGGTTCCGTT GACAACTTCAAAGCTGAATTCGAAAAAGCAGCAGCAACCCGTTTCGGCTCCGGCTGGGCGTGGCTGGTGCTGAAAGGCGACA AACTGGCTGTGGTTTCTACCGCAAACCAGGATTCCCCGCTGATGGGTGAAGCCATTTCCGGCGCTTCCGGCTTCCCGATCCT GGGTCTGGACGTATGGGAACACGCTTACTATCTGAAATTCCAGAACCGCCGCCCGGACTACATCAAAGAGTTCTGGAACGTGG TGAACTGGGACGAAGCAGCAGCGCGTTTCGCCGCTAAAAAATAA (SEQ ID NO: 465)
53_jfrlSe1409-3_Senftenberg	
54_jfrlSe1409-4_UN_O13_	ATGAGTTATACACTGCCATCCCTGCCGTACGCTTATGATGCACTGGAACCGCACTTCGATAAGCAGACGATGGAGATTCACCA CACCAAACACCATCAAACCTATGTCAACAACGCTAACGCGGCGCTGGAAAACCTGCCTGAGTTTGCTGACCTGCCGGTTGAAG AACTGATTACTAAACTGGACCAGGTGCCAGCGGACAAAAAAACCGTGCTGCGTAACAACGCCGGCGGCCACGCTAACCACAG CCTGTTCTGGAAAGGGCTGAAAAAAGGCACCACTCTGCAGGGCGATCTGAAAGCGGCTATCGAGCGTGACTTCGGTTCCGTT GACAACTTCAAAGCTGAATTCGAAAAAGCAGCAGCAACCCGTTTCGGCTCCGGCTGGGCGTGGCTGGTGCTGAAAGGCGACA AACTGGCTGTGGTTTCTACCGCAAACCAGGATTCCCCGCTGATGGGTGAAGCCATTTCCGGCGCTTCCGGCTTCCCGATCCT GGGTCTGGACGTGTGGGAACACGCTTACTACCTGAAATTCCAGAACCGCCGCCCGGACTACATCAAAGAGTTCTGGAACGTG GTGAACTGGGACGAAGCAGCAGCGCGTTTCGCCGCTAAAAAATAA (SEQ ID NO: 477)
55_jfrlSe1409-5_UN_O1,3,19_	ATGAGTTATACACTGCCATCCCTGCCGTACGCTTATGATGCACTGGAACCGCACTTCGATAAGCAGACGATGGAGATTCACCA CACCAAACACCATCAAACCTATGTCAACAACGCTAACGCGGCGCTGGAAAACCTGCCTGAGTTTGCTGACCTGCCGGTTGAAG AACTGATTACTAAACTGGACCAGGTGCCAGCGGACAAAAAAACCGTGCTGCGTAACAACGCGGGCGGCCACGCTAACCACAG CCTGTTCTGGAAAGGGCTGAAAAAAGGCACCACTCTGCAGGGCGATCTGAAAGCGGCTATCGAGCGTGACTTCGGTTCCGTT GACAACTTCAAAGCTGAATTCGAAAAAGCAGCAGCAACCCGTTTCGGCTCCGGCTGGGCGTGGCTGGTGCTGAAAGGCGACA AACTGGCTGTGGTTTCTACCGCAAACCAGGATTCCCCGCTGATGGGTGAAGCCATTTCCGGCGCTTCCGGCTTCCCGATCCT GGGTCTGGACGTGTGGGAACACGCTTACTACCTGAAATTCCAGAACCGCCGCCCGGACTACATCAAAGAGTTCTGGAACGTG GTGAACTGGGACGAAGCAGCAGCGCGTTTCGCCGCTAAAAAATAA (SEQ ID NO: 486)
56_jfrlSe1409-6_Montevideo	ATGAGTTATACACTGCCATCCCTGCCGTACGCTTATGATGCACTGGAACCGCACTTCGATAAGCAGACGATGGAGATTCACCA CACCAAACACCATCAAACCTATGTCAACAACGCTAACGCGGCGCTGGAAAACCTGCCTGAGTTTGCTGACCTGCCGGCTGAAG AACTGATTACTAAACTGGACCAGGTGCCAGCGGACAAAAAAACCGTGCTGCGTAACAACGCCGGCGGCCACGCTAACCACAG CCTGTTCTGGAAAGGGCTGAAAAAAGGCACCACTCTGCAGGGCGACCTGAAAGCGGCTATCGAGCGTGACTTCGGTTCCGTT GACAACTTCAAAGCTGAATTCGAAAAAGCAGCAGCAACCCGTTTCGGCTCCGGCTGGGCGTGGCTGGTGCTGAAAGGCGACA AACTGGCTGTGGTTTCTACCGCAAACCAGGATTCCCCGCTGATGGGTGAAGCCATTTCCGGCGCTTCCGGCTTCCCGATCCT GGGTCTGGACGTGTGGGAACACGCTTACTACCTGAAATTCCAGAACCGCCGCCCGGACTACATCAAAGAGTTCTGGAACGTG GTGAACTGGGACGAAGCAGCAGCGCGTTTCGCCGCTAAAAAATAA (SEQ ID NO: 494)
57_jfrlSe1409-7_UN_O1,3,19_	ATGAGTTATACACTGCCATCCCTGCCGTACGCTTATGATGCACTGGAACCGCACTTCGATAAGCAGACGATGGAGATTCACCA CACCAAACACCATCAAACCTATGTCAACAACGCTAACGCGGCGCTGGAAAACCTGCCTGAGTTTGCTGACCTGCCGGTTGAAG AACTGATTACTAAACTGGACCAGGTGCCAGCGGACAAAAAAACCGTGCTGCGTAACAACGCGGGCGGCCACGCTAACCACAG CCTGTTCTGGAAAGGGCTGAAAAAAGGCACCACTCTGCAGGGCGATCTGAAAGCGGCTATCGAGCGTGACTTCGGTTCCGTT GACAACTTCAAAGCTGAATTCGAAAAAGCAGCAGCAACCCGTTTCGGCTCCGGCTGGGCGTGGCTGGTGCTGAAAGGCGACA AACTGGCTGTGGTTTCTACCGCAAACCAGGATTCCCCGCTGATGGGTGAAGCCATTTCCGGCGCTTCCGGCTTCCCGATCCT GGGTCTGGACGTGTGGGAACACGCTTACTACCTGAAATTCCAGAACCGCCGCCCGGACTACATCAAAGAGTTCTGGAACGTG GTGAACTGGGACGAAGCAGCAGCGCGTTTCGCCGCTAAAAAATAA (SEQ ID NO: 502)
58_jfrlSe1409-8_UN_O18_	ATGAGTTATACACTGCCATCCCTGCCGTACGCTTATGATGCACTGGAACCGCACTTCGATAAGCAGACGATGGAGATTCACCA CACCAAACACCATCAAACCTATGTCAACAACGCTAACGCGGCGCTGGAAAACCTGCCTGAGTTTGCCAGCCTGCCGGTTGAAG AACTGATTACTAAACTGGACCAGGTGCCAGCGGACAAAAAAACCGTGCTGCGTAACAACGCGGGCGGCCACGCTAACCACAG CCTGTTCTGGAAAGGGCTGAAAAAAGGCACCACTCTGCAGGGCGACCTGAAAGCGGCTATCGAGCGTGACTTCGGTTCCGTT GACAACTTCAAAGCTGAATTCGAAAAAGCAGCAGCAACCCGTTTCGGTTCCGGCTGGGCGTGGCTGGTGCTGAAAGGTGACAA ACTGGCTGTGGTTTCTACCGCAAACCAGGATTCCCCGCTGATGGGTGAAGCCATTTCCGGCGCTTCCGGCTTCCCGATCCTG GGTCTGGACGTGTGGGAACACGCTTACTACCTGAAATTCCAGAACCGCCGCCCGGACTACATCAAAGAGTTCTGGAACGTGGT GAACTGGGACGAAGCAGCAGCGCGTTTCGCCGCTAAAAAATAA (SEQ ID NO: 511)
61_jfrlSe1409-11_UN_O1,3,19_	ATGAGTTATACACTGCCATCCCTGCCGTACGCTTATGATGCACTGGAACCGCACTTCGATAAGCAGACGATGGAGATTCACCA CACCAAACACCATCAAACCTATGTCAACAACGCTAACGCGGCGCTGGAAAACCTGCCTGAGTTTGCTGACCTGCCGGTTGAAG AACTGATTACTAAACTGGACCAGGTGCCAGCGGACAAAAAAACCGTGCTGCGTAACAACGCGGGCGGCCACGCTAACCACAG CCTGTTCTGGAAAGGGCTGAAAAAAGGCACCACTCTGCAGGGCGATCTGAAAGCGGCTATCGAGCGTGACTTCGGTTCCGTT GACAACTTCAAAGCTGAATTCGAAAAAGCAGCAGCAACCCGTTTCGGCTCCGGCTGGGCGTGGCTGGTGCTGAAAGGCGACA AACTGGCTGTGGTTTCTACCGCAAACCAGGATTCCCCGCTGATGGGTGAAGCCATTTCCGGCGCTTCCGGCTTCCCGATCCT GGGTCTGGACGTGTGGGAACACGCTTACTACCTGAAATTCCAGAACCGCCGCCCGGACTACATCAAAGAGTTCTGGAACGTG GTGAACTGGGACGAAGCAGCAGCGCGTTTCGCCGCTAAAAAATAA (SEQ ID NO: 520)

Fig. 27G

67_jfrlSe1409-17_Rissen	ATGAGTTATACACTGCCATCCCTGCCGTACGCTTATGATGCACTGGAACCGCACTTCGATAAGCAGACGATGGAGATTCACCA CACCAAACACCATCAAACCTATGTCAACAACGCTAACGCGGCGCTGGAAAACCTGCCTGAGTTTGCTGACCTGCCGGTTGAAG AACTGATTACTAAACTGGACCAGGTGCCAGCGGACAAAAAAACCGTGCTGCGTAACAACGCCGGCGGCCACGCTAACCACAG CCTGTTCTGGAAAGGGCTGAAAAAAGGCACCACTCTGCAGGGCGATCTGAAAGTGGCTATCGAGCGTGACTTCGGTTCCGTTG ACAACTTCAAAGCTGAATTCGAAAAAGCAGCAGCAACCCGTTTCGGCTCCGGCTGGGCGTGGCTGGTGCTGAAAGGTGACAAA CTGGCTGTGGTTTCTACCGCAAACCAGGATTCCCCGCTGATGGGTGAAGCCATTTCCGGCGCTTCCGGCTTCCCGATCCTGG GTCTGGACGTGTGGGAACACGCTTACTACCTGAAATTCCAGAACCGCCGCCCGGACTACATCAAAGAGTTCTGGAACGTGGTG AACTGGGACGAAGCAGCAGCGCGTTTCGCCGCTAAAAAATAA (SEQ ID NO: 529)
71_jfrlSe1409-21_Amsterdam	ATGAGTTATACACTGCCATCCCTGCCGTACGCTTATGATGCACTGGAACCGCACTTCGATAAGCAGACGATGGAGATTCACCA CACCAAACACCATCAAACCTATGTCAACAACGCTAACGCGGCGCTGGAAAACCTGCCTGAGTTTGCTGACCTGCCGGTTGAAG AACTGATTACTAAACTGGACCAGGTGCCAGCGGACAAAAAAACCGTGCTGCGTAACAACGCGGGCGGCCACGCTAACCACAG CCTGTTCTGGAAAGGGCTGAAAAAAGGCACCACTCTGCAGGGCGATCTGAAAGCGGCTATCGAGCGTGACTTCGGTTCCGTT GACAACTTCAAAGCTGAATTCGAAAAAGCAGCAGCAACCCGTTTCGGCTCCGGCTGGGCGTGGCTGGTGCTGAAAGGTGACAA ACTGGCTGTGGTTTCTACCGCAAACCAGGATTCTCCGCTGATGGGTGAAGCCATTTCCGGCGCTTCCGGCTTCCCGATCCTG GGTCTGGACGTGTGGGAACACGCTTACTACCTGAAATTCCAGAACCGCCGCCCGGACTACATCAAAGAGTTCTGGAACGTGGT GAACTGGGACGAAGCAGCAGCGCGTTTCGCCGCTAAAAAATAA (SEQ ID NO: 537)
80_HySe09_Enteritidis	ATGAGTTATACACTGCCATCCCTGCCGTACGCTTATGATGCACTGGAACCGCACTTCGATAAGCAGACGATGGAGATTCACCA CACCAAACACCATCAAACCTATGTCAACAACGCTAACGCGGCGCTGGAAAACCTGCCTGAGTTTGCTGACCTGCCGGTTGAAG AACTGATTACTAAACTGGACCAGGTGCCAGCGGACAAAAAAACCGTGCTGCGTAACAACGCCGGCGGCCACGCTAACCACAG CCTGTTCTGGAAAGGGCTGAAAAAAGGCACCACTCTGCAGGGCGATCTGAAAGCGGCTATCGAGCGTGACTTCGGTTCCGTT GACAACTTCAAAGCTGAATTCGAAAAAGCAGCAGCAACCCGTTTCGGCTCCGGCTGGGCGTGGCTGGTGCTGAAAGGCGATAA ACTGGCTGTGGTTTCTACCGCTAACCAGGATTCCCCGCTGATGGGTGAAGCCATTTCCGGCGCTTCCGGCTTCCCGATCCTG GGTCTGGACGTGTGGGAACACGCTTACTATCTGAAATTCCAGAACCGCCGCCCGGACTACATCAAAGAGTTCTGGAACGTGGT GAACTGGGACGAAGCAGCAGCGCGTTTCGCCGCTAAAAAATAA (SEQ ID NO: 547)
100_HySe29_Schwarzengrund	ATGAGTTATACACTGCCATCCCTGCCGTACGCTTATGATGCACTGGAACCGCACTTCGATAAGCAGACGATGGAGATTCACCA CACCAAACACCATCAAACCTATGTCAACAACGCTAACGCGGCGCTGGAAAACCTGCCTGAGTTTGCTGACCTGCCGGCTGAAG AACTGATTACTAAACTGGACCAGGTGCCAGCGGACAAAAAAACCGTGCTGCGTAACAACGCCGGCGGCCACGCTAACCACAG CCTGTTCTGGAAAGGGCTGAAAAAAGGCACCACTCTGCAGGGCGACCTGAAAGCGGCTATCGAGCGTGACTTCGGTTCCGTT GACAACTTCAAAGCTGAATTCGAAAAAGCAGCAGCAACCCGTTTCGGCTCCGGCTGGGCGTGGCTGGTGCTGAAAGGCGACA AACTGGCTGTGGTTTCTACCGCAAACCAGGATTCCCCGCTGATGGGTGAAGCCATTTCCGGCGCTTCCGGCTTCCCGATCCT GGGTCTGGACGTGTGGGAACACGCTTACTACCTGAAATTCCAGAACCGCCGCCCGGACTACATCAAAGAGTTCTGGAACGTG GTGAACTGGGACGAAGCAGCAGCGCGTTTCGCCGCTAAAAAATAA (SEQ ID NO: 557)
103_HySe32_Schwarzengrund	ATGAGTTATACACTGCCATCCCTGCCGTACGCTTATGATGCACTGGAACCGCACTTCGATAAGCAGACGATGGAGATTCACCA CACCAAACACCATCAAACCTATGTCAACAACGCTAACGCGGCGCTGGAAAACCTGCCTGAGTTTGCTGACCTGCCGGCTGAAG AACTGATTACTAAACTGGACCAGGTGCCAGCGGACAAAAAAACCGTGCTGCGTAACAACGCCGGCGGCCACGCTAACCACAG CCTGTTCTGGAAAGGGCTGAAAAAAGGCACCACTCTGCAGGGCGACCTGAAAGCGGCTATCGAGCGTGACTTCGGTTCCGTT GACAACTTCAAAGCTGAATTCGAAAAAGCAGCAGCAACCCGTTTCGGCTCCGGCTGGGCGTGGCTGGTGCTGAAAGGCGACA AACTGGCTGTGGTTTCTACCGCAAACCAGGATTCCCCGCTGATGGGTGAAGCCATTTCCGGCGCTTCCGGCTTCCCGATCCT GGGTCTGGACGTGTGGGAACACGCTTACTACCTGAAATTCCAGAACCGCCGCCCGGACTACATCAAAGAGTTCTGGAACGTG GTGAACTGGGACGAAGCAGCAGCGCGTTTCGCCGCTAAAAAATAA (SEQ ID NO: 565)
106_HySe35_Schwarzengrund	

Fig. 28A

01_NBRC13245T_Typhimurium	ATGTACGCGGTTTTCCAAAGTGGTGGTAAACAACACCGAGTAAGCGAAGGTCAGACCGTTCGCCTGGAAAAGCTGGACATCGCA ACTGGCGAAACTATCGAGTTCGCTGAAGTTCTGATGATCGCAAACGGTGAAGAAGTCAAAATCGGCGTTCCTTTCGTTGATGGC GGCGTAATCAAAGCTGAAGTTGTTGCCCACGGTCGTGGCGAGAAAGTTAAAATCGTTAAGTTTCGTCGCCGTAAACACTACCGT AAGCAGCAGGGCCATCGTCAGTGGTTCACTGATGTGAAAATTACCGGCATCAGCGCTTAA (SEQ ID NO: 58)
02_GTC00131_Enteritidis	ATGTACGCGGTTTTCCAAAGTGGTGGTAAACAACACCGAGTAAGCGAAGGTCAGACCGTTCGCCTGGAAAAGCTGGACATCGCA ACTGGCGAAACTATCGAGTTCGCTGAAGTTCTGATGATCGCAAACGGTGAAGAAGTCAAAATCGGCGTTCCTTTCGTTGATGGC GGCGTAATCAAAGCTGAAGTTGTTGCCCACGGTCGTGGCGAGAAAGTTAAAATCGTTAAGTTTCGTCGCCGTAAACACTACCGT AAGCAGCAGGGCCATCGTCAGTGGTTCACTGATGTGAAAATTACCGGCATCAGCGCTTAA (SEQ ID NO: 70)
03_GTC09491_Enteritidis	ATGTACGCGGTTTTCCAAAGTGGTGGTAAACAACACCGAGTAAGCGAAGGTCAGACCGTTCGCCTGGAAAAGCTGGACATCGCA ACTGGCGAAACTATCGAGTTCGCTGAAGTTCTGATGATCGCAAACGGTGAAGAAGTCAAAATCGGCGTTCCTTTCGTTGATGGC GGCGTAATCAAAGCTGAAGTTGTTGCCCACGGTCGTGGCGAGAAAGTTAAAATCGTTAAGTTTCGTCGCCGTAAACACTACCGT AAGCAGCAGGGCCATCGTCAGTGGTTCACTGATGTGAAAATTACCGGCATCAGCGCTTAA (SEQ ID NO: 79)
04_GTC03838_Enteritidis	ATGTACGCGGTTTTCCAAAGTGGTGGTAAACAACACCGAGTAAGCGAAGGTCAGACCGTTCGCCTGGAAAAGCTGGACATCGCA ACTGGCGAAACTATCGAGTTCGCTGAAGTTCTGATGATCGCAAACGGTGAAGAAGTCAAAATCGGCGTTCCTTTCGTTGATGGC GGCGTAATCAAAGCTGAAGTTGTTGCCCACGGTCGTGGCGAGAAAGTTAAAATCGTTAAGTTTCGTCGCCGTAAACACTACCGT AAGCAGCAGGGCCATCGTCAGTGGTTCACTGATGTGAAAATTACCGGCATCAGCGCTTAA (SEQ ID NO: 86)
05_GTC08914_Enteritidis	ATGTACGCGGTTTTCCAAAGTGGTGGTAAACAACACCGAGTAAGCGAAGGTCAGACCGTTCGCCTGGAAAAGCTGGACATCGCA ACTGGCGAAACTATCGAGTTCGCTGAAGTTCTGATGATCGCAAACGGTGAAGAAGTCAAAATCGGCGTTCCTTTCGTTGATGGC GGCGTAATCAAAGCTGAAGTTGTTGCCCACGGTCGTGGCGAGAAAGTTAAAATCGTTAAGTTTCGTCGCCGTAAACACTACCGT AAGCAGCAGGGCCATCGTCAGTGGTTCACTGATGTGAAAATTACCGGCATCAGCGCTTAA (SEQ ID NO: 93)
06_GTC09421_Enteritidis	ATGTACGCGGTTTTCCAAAGTGGTGGTAAACAACACCGAGTAAGCGAAGGTCAGACCGTTCGCCTGGAAAAGCTGGACATCGCA ACTGGCGAAACTATCGAGTTCGCTGAAGTTCTGATGATCGCAAACGGTGAAGAAGTCAAAATCGGCGTTCCTTTCGTTGATGGC GGCGTAATCAAAGCTGAAGTTGTTGCCCACGGTCGTGGCGAGAAAGTTAAAATCGTTAAGTTTCGTCGCCGTAAACACTACCGT AAGCAGCAGGGCCATCGTCAGTGGTTCACTGATGTGAAAATTACCGGCATCAGCGCTTAA (SEQ ID NO: 103)
07_GTC09489_Enteritidis	ATGTACGCGGTTTTCCAAAGTGGTGGTAAACAACACCGAGTAAGCGAAGGTCAGACCGTTCGCCTGGAAAAGCTGGACATCGCA ACTGGCGAAACTATCGAGTTCGCTGAAGTTCTGATGATCGCAAACGGTGAAGAAGTCAAAATCGGCGTTCCTTTCGTTGATGGC GGCGTAATCAAAGCTGAAGTTGTTGCCCACGGTCGTGGCGAGAAAGTTAAAATCGTTAAGTTTCGTCGCCGTAAACACTACCGT AAGCAGCAGGGCCATCGTCAGTGGTTCACTGATGTGAAAATTACCGGCATCAGCGCTTAA (SEQ ID NO: 113)
08_NBRC3163_Pullorum_Gallinarum	ATGTACGCGGTTTTCCAAAGTGGTGGTAAACAACACCGAGTAAGCGAAGGTCAGACCGTTCGCCTGGAAAAGCTGGACATCGCA ACTGGCGAAACTATCGAGTTCGCTGAAGTTCTGATGATCGCAAACGGTGAAGAAGTCAAAATCGGCGTTCCTTTCGTTGATGGC GGCGTAATCAAAGCTGAAGTTGTTGCCCACGGTCGTGGCGAGAAAGTTAAAATCGTTAAGTTTCGTCGCCGTAAACACTACCGT AAGCAGCAGGGCCATCGTCAGTGGTTCACTGATGTGAAAATTACCGGCATCAGCGCTTAA (SEQ ID NO: 120)
09_NBRC3313_Enteritidis	ATGTACGCGGTTTTCCAAAGTGGTGGTAAACAACACCGAGTAAGCGAAGGTCAGACCGTTCGCCTGGAAAAGCTGGACATCGCA ACTGGCGAAACTATCGAGTTCGCTGAAGTTCTGATGATCGCAAACGGTGAAGAAGTCAAAATCGGCGTTCCTTTCGTTGATGGC GGCGTAATCAAAGCTGAAGTTGTTGCCCACGGTCGTGGCGAGAAAGTTAAAATCGTTAAGTTTCGTCGCCGTAAACACTACCGT AAGCAGCAGGGCCATCGTCAGTGGTTCACTGATGTGAAAATTACCGGCATCAGCGCTTAA (SEQ ID NO: 130)
10_NBRC12529_Typhimurium	ATGTACGCGGTTTTCCAAAGTGGTGGTAAACAACACCGAGTAAGCGAAGGTCAGACCGTTCGCCTGGAAAAGCTGGACATCGCA ACTGGCGAAACTATCGAGTTCGCTGAAGTTCTGATGATCGCAAACGGTGAAGAAGTCAAAATCGGCGTTCCTTTCGTTGATGGC GGCGTAATCAAAGCTGAAGTTGTTGCCCACGGTCGTGGCGAGAAAGTTAAAATCGTTAAGTTTCGTCGCCGTAAACACTACCGT AAGCAGCAGGGCCATCGTCAGTGGTTCACTGATGTGAAAATTACCGGCATCAGCGCTTAA (SEQ ID NO: 137)

Fig. 28B

11_NBRC14193_Typhimurium	ATGTACGCGGTTTTCCAAAGTGGTGGTAAACAACACCGAGTAAGCGAAGGTCAGACCGTTCGCCTGGAAAAGCTGGACATCGCA ACTGGCGAAACTATCGAGTTCGCTGAAGTTCTGATGATCGCAAACGGTGAAGAAGTCAAAATCGGCGTTCCTTTCGTTGATGGC GGCGTAATCAAAGCTGAAGTTGTTGCCCACGGTCGTGGCGAGAAAGTTAAAATCGTTAAGTTTCGTCGCCGTAAACACTACCGT AAGCAGCAGGGCCATCGTCAGTGGTTCACTGATGTGAAAATTACCGGCATCAGCGCTTAA (SEQ ID NO: 144)
12_NBRC14194_Typhimurium	ATGTACGCGGTTTTCCAAAGTGGTGGTAAACAACACCGAGTAAGCGAAGGTCAGACCGTTCGCCTGGAAAAGCTGGACATCGCA ACTGGCGAAACTATCGAGTTCGCTGAAGTTCTGATGATCGCAAACGGTGAAGAAGTCAAAATCGGCGTTCCTTTCGTTGATGGC GGCGTAATCAAAGCTGAAGTTGTTGCCCACGGTCGTGGCGAGAAAGTTAAAATCGTTAAGTTTCGTCGCCGTAAACACTACCGT AAGCAGCAGGGCCATCGTCAGTGGTTCACTGATGTGAAAATTACCGGCATCAGCGCTTAA (SEQ ID NO: 151)
13_NBRC14209_Typhimurium	ATGTACGCGGTTTTCCAAAGTGGTGGTAAACAACACCGAGTAAGCGAAGGTCAGACCGTTCGCCTGGAAAAGCTGGACATCGCA ACTGGCGAAACTATCGAGTTCGCTGAAGTTCTGATGATCGCAAACGGTGAAGAAGTCAAAATCGGCGTTCCTTTCGTTGATGGC GGCGTAATCAAAGCTGAAGTTGTTGCCCACGGTCGTGGCGAGAAAGTTAAAATCGTTAAGTTTCGTCGCCGTAAACACTACCGT AAGCAGCAGGGCCATCGTCAGTGGTTCACTGATGTGAAAATTACCGGCATCAGCGCTTAA (SEQ ID NO: 158)
14_NBRC14210_Typhimurium	ATGTACGCGGTTTTCCAAAGTGGTGGTAAACAACACCGAGTAAGCGAAGGTCAGACCGTTCGCCTGGAAAAGCTGGACATCGCA ACTGGCGAAACTATCGAGTTCGCTGAAGTTCTGATGATCGCAAACGGTGAAGAAGTCAAAATCGGCGTTCCTTTCGTTGATGGC GGCGTAATCAAAGCTGAAGTTGTTGCCCACGGTCGTGGCGAGAAAGTTAAAATCGTTAAGTTTCGTCGCCGTAAACACTACCGT AAGCAGCAGGGCCATCGTCAGTGGTTCACTGATGTGAAAATTACCGGCATCAGCGCTTAA (SEQ ID NO: 165)
15_NBRC14211_Typhimurium	ATGTACGCGGTTTTCCAAAGTGGTGGTAAACAACACCGAGTAAGCGAAGGTCAGACCGTTCGCCTGGAAAAGCTGGACATCGCA ACTGGCGAAACTATCGAGTTCGCTGAAGTTCTGATGATCGCAAACGGTGAAGAAGTCAAAATCGGCGTTCCTTTCGTTGATGGC GGCGTAATCAAAGCTGAAGTTGTTGCCCACGGTCGTGGCGAGAAAGTTAAAATCGTTAAGTTTCGTCGCCGTAAACACTACCGT AAGCAGCAGGGCCATCGTCAGTGGTTCACTGATGTGAAAATTACCGGCATCAGCGCTTAA (SEQ ID NO: 171)
16_NBRC14212_Typhimurium	ATGTACGCGGTTTTCCAAAGTGGTGGTAAACAACACCGAGTAAGCGAAGGTCAGACCGTTCGCCTGGAAAAGCTGGACATCGCA ACTGGCGAAACTATCGAGTTCGCTGAAGTTCTGATGATCGCAAACGGTGAAGAAGTCAAAATCGGCGTTCCTTTCGTTGATGGC GGCGTAATCAAAGCTGAAGTTGTTGCCCACGGTCGTGGCGAGAAAGTTAAAATCGTTAAGTTTCGTCGCCGTAAACACTACCGT AAGCAGCAGGGCCATCGTCAGTGGTTCACTGATGTGAAAATTACCGGCATCAGCGCTTAA (SEQ ID NO: 178)
17_NBRC15181_Typhimurium	ATGTACGCGGTTTTCCAAAGTGGTGGTAAACAACACCGAGTAAGCGAAGGTCAGACCGTTCGCCTGGAAAAGCTGGACATCGCA ACTGGCGAAACTATCGAGTTCGCTGAAGTTCTGATGATCGCAAACGGTGAAGAAGTCAAAATCGGCGTTCCTTTCGTTGATGGC GGCGTAATCAAAGCTGAAGTTGTTGCCCACGGTCGTGGCGAGAAAGTTAAAATCGTTAAGTTTCGTCGCCGTAAACACTACCGT AAGCAGCAGGGCCATCGTCAGTGGTTCACTGATGTGAAAATTACCGGCATCAGCGCTTAA (SEQ ID NO: 185)
18_NBRC15182_Minnesota	ATGTACGCGGTTTTCCAAAGTGGTGGTAAACAACACCGAGTAAGCGAAGGTCAGACCGTTCGCCTGGAAAAGCTGGACATCGCA ACTGGCGAAACTATCGAGTTCGCTGAAGTTCTGATGGTCGCAAACGGTGAAGAAGTCAAAATCGGCGTTCCTTTCGTTGATGGC GGCGTAATCAAAGCTGAAGTTGTTGCCCACGGTCGTGGCGAGAAAGTTAAAATCGTTAAGTTTCGTCGCCGTAAACACTACCGT AAGCAGCAGGGCCATCGTCAGTGGTTCACTGATGTGAAAATTACCGGCATCAGCGCTTAA (SEQ ID NO: 195)
19_NBRC15183_Minnesota	ATGTACGCGGTTTTCCAAAGTGGTGGTAAACAACACCGAGTAAGCGAAGGTCAGACCGTTCGCCTGGAAAAGCTGGACATCGCA ACTGGCGAAACTATCGAGTTCGCTGAAGTTCTGATGGTCGCAAACGGTGAAGAAGTCAAAATCGGCGTTCCTTTCGTTGATGGC GGCGTAATCAAAGCTGAAGTTGTTGCCCACGGTCGTGGCGAGAAAGTTAAAATCGTTAAGTTTCGTCGCCGTAAACACTACCGT AAGCAGCAGGGCCATCGTCAGTGGTTCACTGATGTGAAAATTACCGGCATCAGCGCTTAA (SEQ ID NO: 205)
20_NBRC15184_Minnesota	ATGTACGCGGTTTTCCAAAGTGGTGGTAAACAACACCGAGTAAGCGAAGGTCAGACCGTTCGCCTGGAAAAGCTGGACATCGCA ACTGGCGAAACTATCGAGTTCGCTGAAGTTCTGATGGTCGCAAACGGTGAAGAAGTCAAAATCGGCGTTCCTTTCGTTGATGGC GGCGTAATCAAAGCTGAAGTTGTTGCCCACGGTCGTGGCGAGAAAGTTAAAATCGTTAAGTTTCGTCGCCGTAAACACTACCGT AAGCAGCAGGGCCATCGTCAGTGGTTCACTGATGTGAAAATTACCGGCATCAGCGCTTAA (SEQ ID NO: 212)
21_NBRC15185_Minnesota	ATGTACGCGGTTTTCCAAAGTGGTGGTAAACAACACCGAGTAAGCGAAGGTCAGACCGTTCGCCTGGAAAAGCTGGACATCGCA ACTGGCGAAACTATCGAGTTCGCTGAAGTTCTGATGGTCGCAAACGGTGAAGAAGTCAAAATCGGCGTTCCTTTCGTTGATGGC GGCGTAATCAAAGCTGAAGTTGTTGCCCACGGTCGTGGCGAGAAAGTTAAAATCGTTAAGTTTCGTCGCCGTAAACACTACCGT AAGCAGCAGGGCCATCGTCAGTGGTTCACTGATGTGAAAATTACCGGCATCAGCGCTTAA (SEQ ID NO: 219)
22_NBRC15186_Minnesota	ATGTACGCGGTTTTCCAAAGTGGTGGTAAACAACACCGAGTAAGCGAAGGTCAGACCGTTCGCCTGGAAAAGCTGGACATCGCA ACTGGCGAAACTATCGAGTTCGCTGAAGTTCTGATGGTCGCAAACGGTGAAGAAGTCAAAATCGGCGTTCCTTTCGTTGATGGC GGCGTAATCAAAGCTGAAGTTGTTGCCCACGGTCGTGGCGAGAAAGTTAAAATCGTTAAGTTTCGTCGCCGTAAACACTACCGT AAGCAGCAGGGCCATCGTCAGTGGTTCACTGATGTGAAAATTACCGGCATCAGCGCTTAA (SEQ ID NO: 226)
23_NBRC15187_Minnesota	ATGTACGCGGTTTTCCAAAGTGGTGGTAAACAACACCGAGTAAGCGAAGGTCAGACCGTTCGCCTGGAAAAGCTGGACATCGCA ACTGGCGAAACTATCGAGTTCGCTGAAGTTCTGATGGTCGCAAACGGTGAAGAAGTCAAAATCGGCGTTCCTTTCGTTGATGGC GGCGTAATCAAAGCTGAAGTTGTTGCCCACGGTCGTGGCGAGAAAGTTAAAATCGTTAAGTTTCGTCGCCGTAAACACTACCGT AAGCAGCAGGGCCATCGTCAGTGGTTCACTGATGTGAAAATTACCGGCATCAGCGCTTAA (SEQ ID NO: 233)
24_NBRC100797_Abony	ATGTACGCGGTTTTCCAAAGTGGTGGTAAACAACACCGAGTAAGCGAAGGTCAGACCGTTCGCCTGGAAAAGCTGGACATCGCA ACTGGCGAAACTATCGAGTTCGCTGAAGTTCTGATGATCGCAAACGGTGAAGAAGTCAAAATCGGCGTTCCTTTCGTTGATGGC GGCGTAATCAAAGCTGAAGTTGTTGCCCACGGTCGTGGCGAGAAAGTTAAAATCGTTAAGTTTCGTCGCCGTAAACACTACCGT AAGCAGCAGGGCCATCGTCAGTGGTTCACTGATGTGAAAATTACCGGCATCAGCGCTTAA (SEQ ID NO: 240)
25_NBRC105684_Choleraesuis	ATGTACGCGGTTTTCCAAAGTGGTGGTAAACAACACCGAGTAAGCGAAGGTCAGACCGTTCGCCTGGAAAAGCTGGACATCGCA ACTGGCGAAACTATCGAGTTCGCTGAAGTTCTGATGATCGCAAACGGTGAAGAAGTCAAAATCGGCGTTCCTTTCGTTGATGGC GGCGTAATCAAAGCTGAAGTTGTTGCCCACGGTCGTGGCGAGAAAGTTAAAATCGTTAAGTTTCGTCGCCGTAAACACTACCGT AAGCAGCAGGGCCATCGTCAGTGGTTCACTGATGTGAAAATTACCGGCATCAGCGCTTAA (SEQ ID NO: 250)
26_NBRC105726_Typhimurium	ATGTACGCGGTTTTCCAAAGTGGTGGTAAACAACACCGAGTAAGCGAAGGTCAGACCGTTCGCCTGGAAAAGCTGGACATCGCA ACTGGCGAAACTATCGAGTTCGCTGAAGTTCTGATGATCGCAAACGGTGAAGAAGTCAAAATCGGCGTTCCTTTCGTTGATGGC GGCGTAATCAAAGCTGAAGTTGTTGCCCACGGTCGTGGCGAGAAAGTTAAAATCGTTAAGTTTCGTCGCCGTAAACACTACCGT AAGCAGCAGGGCCATCGTCAGTGGTTCACTGATGTGAAAATTACCGGCATCAGCGCTTAA (SEQ ID NO: 260)
27_JCM3919_UN_O7_	ATGTACGCGGTTTTCCAAAGTGGTGGTAAACAACACCGAGTAAGCGAAGGTCAGACCGTTCGCCTGGAAAAGCTGGACATCGCA ACTGGCGAAACTATCGAGTTCGCTGAAGTTCTGATGATCGCAAACGGTGAAGAAGTCAAAATCGGCGTTCCTTTCGTTGATGGC GGCGTAATCAAAGCTGAAGTTGTTGCCCACGGTCGTGGCGAGAAAGTTAAAATCGTTAAGTTTCGTCGCCGTAAACACTACCGT AAGCAGCAGGGCCATCGTCAGTGGTTCACTGATGTGAAAATTACCGGCATCAGCGCTTAA (SEQ ID NO: 269)
28_NBRC15335_Minnesota	ATGTACGCGGTTTTCCAAAGTGGTGGTAAACAACACCGAGTAAGCGAAGGTCAGACCGTTCGCCTGGAAAAGCTGGACATCGCA ACTGGCGAAACTATCGAGTTCGCTGAAGTTCTGATGGTCGCAAACGGTGAAGAAGTCAAAATCGGCGTTCCTTTCGTTGATGGC GGCGTAATCAAAGCTGAAGTTGTTGCCCACGGTCGTGGCGAGAAAGTTAAAATCGTTAAGTTTCGTCGCCGTAAACACTACCGT AAGCAGCAGGGCCATCGTCAGTGGTTCACTGATGTGAAAATTACCGGCATCAGCGCTTAA (SEQ ID NO: 276)
29_GTC09490_Enteritidis	ATGTACGCGGTTTTCCAAAGTGGTGGTAAACAACACCGAGTAAGCGAAGGTCAGACCGTTCGCCTGGAAAAGCTGGACATCGCA ACTGGCGAAACTATCGAGTTCGCTGAAGTTCTGATGATCGCAAACGGTGAAGAAGTCAAAATCGGCGTTCCTTTCGTTGATGGC GGCGTAATCAAAGCTGAAGTTGTTGCCCACGGTCGTGGCGAGAAAGTTAAAATCGTTAAGTTTCGTCGCCGTAAACACTACCGT AAGCAGCAGGGCCATCGTCAGTGGTTCACTGATGTGAAAATTACCGGCATCAGCGCTTAA (SEQ ID NO: 283)
30_GTC09492_Braenderup	ATGTACGCGGTTTTCCAAAGTGGTGGTAAACAACACCGAGTAAGCGAAGGTCAGACCGTTCGCCTGGAAAAGCTGGACATCGCA ACTGGCGAAACTATCGAGTTCGCTGAAGTTCTGATGATCGCAAACGGTGAAGAAGTCAAAATCGGCGTTCCTTTCGTTGATGGC GGCGTAATCAAAGCTGAAGTTGTTGCCCACGGTCGTGGCGAGAAAGTTAAAATCGTTAAGTTTCGTCGCCGTAAACACTACCGT AAGCAGCAGGGCCATCGTCAGTGGTTCACTGATGTGAAAATTACCGGCATCAGCGCTTAA (SEQ ID NO: 290)
31_GTC09493_Pakistan	ATGTACGCGGTTTTCCAAAGTGGTGGTAAACAACACCGAGTAAGCGAAGGTCAGACCGTTCGCCTGGAAAAGCTGGACATCGCA ACTGGCGAAACTATCGAGTTCGCTGAAGTTCTGATGATCGCAAACGGTGAAGAAGTCAAAATCGGCGTTCCTTTCGTTGATGGC GGCGTAATCAAAGCTGAAGTTGTTGCCCACGGTCGTGGCGAGAAAGTTAAAATCGTTAAGTTTCGTCGCCGTAAACACTACCGT AAGCAGCAGGGCCATCGTCAGTGGTTCACTGATGTGAAAATTACCGGCATCAGCGCTTAA (SEQ ID NO: 297)

Fig. 28C

32_GTC09549_Typhimurium	ATGTACGCGGTTTTCCAAAGTGGTGGTAAACAACACCGAGTAAGCGAAGGTCAGACCGTTCGCCTGGAAAAGCTGGACATCGCA ACTGGCGAAACTATCGAGTTCGCTGAAGTTCTGATGATCGCAAACGGTGAAGAAGTCAAAATCGGCGTTCCTTTCGTTGATGGC GGCGTAATCAAAGCTGAAGTTGTTGCCCACGGTCGTGGCGAGAAAGTTAAAATCGTTAAGTTTCGTCGCCGTAAACACTACCGT AAGCAGCAGGGCCATCGTCAGTGGTTCACTGATGTGAAAATTACCGGCATCAGCGCTTAA (SEQ ID NO: 304)
33_ATCCBAA-1675_Infantis	ATGTACGCGGTTTTCCAAAGTGGTGGTAAACAACACCGAGTAAGCGAAGGTCAGACCGTTCGCCTGGAAAAGCTGGACATCGCA ACTGGCGAAACTATCGAGTTCGCTGAAGTTCTGATGGTCGCAAACGGTGAAGAAGTCAAAATCGGCGTTCCTTTCGTTGATGGC GGCGTAATCAAAGCTGAAGTTGTTGCCCACGGTCGTGGCGAGAAAGTTAAAATCGTTAAGTTTCGTCGCCGTAAACACTACCGT AAGCAGCAGGGCCATCGTCAGTGGTTCACTGATGTGAAAATTACCGGCATCAGCGCTTAA (SEQ ID NO: 311)
34_ATCCBAA-1738_Thompson	ATGTACGCGGTTTTCCAAAGTGGTGGTAAACAACACCGAGTAAGCGAAGGTCAGACCGTTCGCCTGGAAAAGCTGGACATCGCA ACTGGCGAAACTATCGAGTTCGCTGAAGTTCTGATGATCGCAAACGGTGAAGAAGTCAAAATCGGCGTTCCTTTCGTTGATGGC GGCGTAATCAAAGCTGAAGTTGTTGCCCACGGTCGTGGCGAGAAAGTTAAAATCGTTAAGTTTCGTCGCCGTAAACACTACCGT AAGCAGCAGGGCCATCGTCAGTGGTTCACTGATGTGAAAATTACCGGCATCAGCGCTTAA (SEQ ID NO: 322)
35_ATCC9712_Saintpaul	ATGTACGCGGTTTTCCAAAGTGGTGGTAAACAACACCGAGTAAGCGAAGGTCAGACCGTTCGCCTGGAAAAGCTGGACATCGCA ACTGGCGAAACTATCGAGTTCGCTGAAGTTCTGATGATCGCAAACGGTGAAGAAGTCAAAATCGGCGTTCCTTTCGTTGATGGC GGCGTAATCAAAGCTGAAGTTGTTGCCCACGGTCGTGGCGAGAAAGTTAAAATCGTTAAGTTTCGTCGCCGTAAACACTACCGT AAGCAGCAGGGCCATCGTCAGTGGTTCACTGATGTGAAAATTACCGGCATCAGCGCTTAA (SEQ ID NO: 332)
36_jfrlSe1402-1_Infantis	ATGTACGCGGTTTTCCAAAGTGGTGGTAAACAACACCGAGTAAGCGAAGGTCAGACCGTTCGCCTGGAAAAGCTGGACATCGCA ACTGGCGAAACTATCGAGTTCGCTGAAGTTCTGATGATCGCAAACGGTGAAGAAGTCAAAATCGGCGTTCCTTTCGTTGATGGC GGCGTAATCAAAGCTGAAGTTGTTGCCCACGGTCGTGGCGAGAAAGTTAAAATCGTTAAGTTTCGTCGCCGTAAACACTACCGT AAGCAGCAGGGCCATCGTCAGTGGTTCACTGATGTGAAAATTACCGGCATCAGCGCTTAA (SEQ ID NO: 343)
38_jfrlSe1402-3_Brandenburg	ATGTACGCGGTTTTCCAAAGTGGTGGTAAACAACACCGAGTAAGCGAAGGTCAGACCGTTCGCCTGGAAAAGCTGGACATCGCA ACTGGCGAAACTATCGAGTTCGCTGAAGTTCTGATGGTCGCAAACGGTGAAGAAGTCAAAATCGGCGTTCCTTTCGTTGATGGC GGCGTAATCAAAGCTGAAGTTGTTGCCCACGGTCGTGGCGAGAAAGTTAAAATCGTTAAGTTTCGTCGCCGTAAACACTACCGT AAGCAGCAGGGCCATCGTCAGTGGTTCACTGATGTGAAAATTACCGGCATCAGCGCTTAA (SEQ ID NO: 353)
39_jfrlSe1402-4_Infantis	ATGTACGCGGTTTTCCAAAGTGGTGGTAAACAACACCGAGTAAGCGAAGGTCAGACCGTTCGCCTGGAAAAGCTGGACATCGCA ACTGGCGAAACTATCGAGTTCGCTGAAGTTCTGATGATCGCAAACGGTGAAGAAGTCAAAATCGGCGTTCCTTTCGTTGATGGC GGCGTAATCAAAGCTGAAGTTGTTGCCCACGGTCGTGGCGAGAAAGTTAAAATCGTTAAGTTTCGTCGCCGTAAACACTACCGT AAGCAGCAGGGCCATCGTCAGTGGTTCACTGATGTGAAAATTACCGGCATCAGCGCTTAA (SEQ ID NO: 360)
40_jfrlSe1402-5_Brandenburg	ATGTACGCGGTTTTCCAAAGTGGTGGTAAACAACACCGAGTAAGCGAAGGTCAGACCGTTCGCCTGGAAAAGCTGGACATCGCA ACTGGCGAAACTATCGAGTTCGCTGAAGTTCTGATGGTCGCAAACGGTGAAGAAGTCAAAATCGGCGTTCCTTTCGTTGATGGC GGCGTAATCAAAGCTGAAGTTGTTGCCCACGGTCGTGGCGAGAAAGTTAAAATCGTTAAGTTTCGTCGCCGTAAACACTACCGT AAGCAGCAGGGCCATCGTCAGTGGTTCACTGATGTGAAAATTACCGGCATCAGCGCTTAA (SEQ ID NO: 370)
41_jfrlSe1402-6_Rissen	ATGTACGCGGTTTTCCAAAGTGGTGGTAAACAACACCGAGTAAGCGAAGGTCAGACCGTTCGCCTGGAAAAGCTGGACATCGCA ACTGGCGAAACTATCGAGTTCGCTGAAGTTCTGATGATCGCAAACGGTGAAGAAGTCAAAATCGGCGTTCCTTTCGTTGATGGC GGCGTAATCAAAGCTGAAGTTGTTGCCCACGGTCGTGGCGAGAAAGTTAAAATCGTTAAGTTTCGTCGCCGTAAACACTACCGT AAGCAGCAGGGCCATCGTCAGTGGTTCACTGATGTGAAAATTACCGGCATCAGCGCTTAA (SEQ ID NO: 377)
42_jfrlSe1402-7_Orion	ATGTACGCGGTTTTCCAAAGTGGTGGTAAACAACACCGAGTAAGCGAAGGTCAGACCGTTCGCCTGGAAAAGCTGGACATCGCA ACTGGCGAAACTATCGAGTTCGCTGAAGTTCTGATGATCGCAAACGGTGAAGAAGTCAAAATCGGCGTTCCTTTCGTTGATGGC GGCGTAATCAAAGCTGAAGTTGTTGCCCACGGTCGTGGCGAGAAAGTTAAAATCGTTAAGTTTCGTCGCCGTAAACACTACCGT AAGCAGCAGGGCCATCGTCAGTGGTTCACTGATGTGAAAATTACCGGCATCAGCGCTTAA (SEQ ID NO: 384)
43_jfrlSe1402-8_Rissen	ATGTACGCGGTTTTCCAAAGTGGTGGTAAACAACACCGAGTAAGCGAAGGTCAGACCGTTCGCCTGGAAAAGCTGGACATCGCA ACTGGCGAAACTATCGAGTTCGCTGAAGTTCTGATGATCGCAAACGGTGAAGAAGTCAAAATCGGCGTTCCTTTCGTTGATGGC GGCGTAATCAAAGCTGAAGTTGTTGCCCACGGTCGTGGCGAGAAAGTTAAAATCGTTAAGTTTCGTCGCCGTAAACACTACCGT AAGCAGCAGGGCCATCGTCAGTGGTTCACTGATGTGAAAATTACCGGCATCAGCGCTTAA (SEQ ID NO: 394)
44_jfrlSe1402-9_Rissen	ATGTACGCGGTTTTCCAAAGTGGTGGTAAACAACACCGAGTAAGCGAAGGTCAGACCGTTCGCCTGGAAAAGCTGGACATCGCA ACTGGCGAAACTATCGAGTTCGCTGAAGTTCTGATGATCGCAAACGGTGAAGAAGTCAAAATCGGCGTTCCTTTCGTTGATGGC GGCGTAATCAAAGCTGAAGTTGTTGCCCACGGTCGTGGCGAGAAAGTTAAAATCGTTAAGTTTCGTCGCCGTAAACACTACCGT AAGCAGCAGGGCCATCGTCAGTGGTTCACTGATGTGAAAATTACCGGCATCAGCGCTTAA (SEQ ID NO: 401)
45_jfrlSe1402-10_Rissen	ATGTACGCGGTTTTCCAAAGTGGTGGTAAACAACACCGAGTAAGCGAAGGTCAGACCGTTCGCCTGGAAAAGCTGGACATCGCA ACTGGCGAAACTATCGAGTTCGCTGAAGTTCTGATGATCGCAAACGGTGAAGAAGTCAAAATCGGCGTTCCTTTCGTTGATGGC GGCGTAATCAAAGCTGAAGTTGTTGCCCACGGTCGTGGCGAGAAAGTTAAAATCGTTAAGTTTCGTCGCCGTAAACACTACCGT AAGCAGCAGGGCCATCGTCAGTGGTTCACTGATGTGAAAATTACCGGCATCAGCGCTTAA (SEQ ID NO: 408)
46_jfrlSe1402-11_Rissen	ATGTACGCGGTTTTCCAAAGTGGTGGTAAACAACACCGAGTAAGCGAAGGTCAGACCGTTCGCCTGGAAAAGCTGGACATCGCA ACTGGCGAAACTATCGAGTTCGCTGAAGTTCTGATGATCGCAAACGGTGAAGAAGTCAAAATCGGCGTTCCTTTCGTTGATGGC GGCGTAATCAAAGCTGAAGTTGTTGCCCACGGTCGTGGCGAGAAAGTTAAAATCGTTAAGTTTCGTCGCCGTAAACACTACCGT AAGCAGCAGGGCCATCGTCAGTGGTTCACTGATGTGAAAATTACCGGCATCAGCGCTTAA (SEQ ID NO: 415)
47_jfrlSe1402-12_Rissen	ATGTACGCGGTTTTCCAAAGTGGTGGTAAACAACACCGAGTAAGCGAAGGTCAGACCGTTCGCCTGGAAAAGCTGGACATCGCA ACTGGCGAAACTATCGAGTTCGCTGAAGTTCTGATGATCGCAAACGGTGAAGAAGTCAAAATCGGCGTTCCTTTCGTTGATGGC GGCGTAATCAAAGCTGAAGTTGTTGCCCACGGTCGTGGCGAGAAAGTTAAAATCGTTAAGTTTCGTCGCCGTAAACACTACCGT AAGCAGCAGGGCCATCGTCAGTGGTTCACTGATGTGAAAATTACCGGCATCAGCGCTTAA (SEQ ID NO: 422)
48_jfrlSe1402-13_Mbandaka	ATGTACGCGGTTTTCCAAAGTGGTGGTAAACAACACCGAGTAAGCGAAGGTCAGACCGTTCGCCTGGAAAAGCTGGACATCGCA ACTGGCGAAACTATCGAGTTCGCTGAAGTTCTGATGATCGCAAACGGTGAAGAAGTCAAAATCGGCGTTCCTTTCGTTGATGGC GGCGTAATCAAAGCTGAAGTTGTTGCCCACGGTCGTGGCGAGAAAGTTAAAATCGTTAAGTTTCGTCGCCGTAAACACTACCGT AAGCAGCAGGGCCATCGTCAGTGGTTCACTGATGTGAAAATTACCGGCATCAGCGCTTAA (SEQ ID NO: 429)
49_jfrlSe1402-14_Mbandaka	ATGTACGCGGTTTTCCAAAGTGGTGGTAAACAACACCGAGTAAGCGAAGGTCAGACCGTTCGCCTGGAAAAGCTGGACATCGCA ACTGGCGAAACTATCGAGTTCGCTGAAGTTCTGATGATCGCAAACGGTGAAGAAGTCAAAATCGGCGTTCCTTTCGTTGATGGC GGCGTAATCAAAGCTGAAGTTGTTGCCCACGGTCGTGGCGAGAAAGTTAAAATCGTTAAGTTTCGTCGCCGTAAACACTACCGT AAGCAGCAGGGCCATCGTCAGTGGTTCACTGATGTGAAAATTACCGGCATCAGCGCTTAA (SEQ ID NO: 439)
50_jfrlSe1402-15_Orion	ATGTACGCGGTTTTCCAAAGTGGTGGTAAACAACACCGAGTAAGCGAAGGTCAGACCGTTCGCCTGGAAAAGCTGGACATCGCA ACTGGCGAAACTATCGAGTTCGCTGAAGTTCTGATGATCGCAAACGGTGAAGAAGTCAAAATCGGCGTTCCTTTCGTTGATGGC GGCGTAATCAAAGCTGAAGTTGTTGCCCACGGTCGTGGCGAGAAAGTTAAAATCGTTAAGTTTCGTCGCCGTAAACACTACCGT AAGCAGCAGGGCCATCGTCAGTGGTTCACTGATGTGAAAATTACCGGCATCAGCGCTTAA (SEQ ID NO: 449)
51_jfrlSe1409-1_Altona	
52_jfrlSe1409-2_Istanbul	ATGTACGCGGTTTTCCAAAGTGGTGGTAAACAACACCGAGTAAGCGAAGGTCAGACCGTTCGCCTGGAAAAGCTGGACATCGCA ACTGGCGAAACTATCGAGTTCGCTGAAGTTCTGATGATCGCAAACGGTGAAGAAGTCAAAATCGGCGTTCCTTTCGTTGATGGC GGCGTAATCAAAGCTGAAGTTGTTGCCCACGGTCGTGGCGAGAAAGTTAAAATCGTTAAGTTTCGTCGCCGTAAACACTACCGT AAGCAGCAGGGCCATCGTCAGTGGTTCACTGATGTGAAAATTACCGGCATCAGCGCTTAA (SEQ ID NO: 466)
53_jfrlSe1409-3_Senftenberg	
54_jfrlSe1409-4_UN_O13_	ATGTACGCGGTTTTCCAAAGTGGTGGTAAACAACACCGAGTAAGCGAAGGTCAGACCGTTCGCCTGGAAAAGCTGGACATCGCA ACTGGCGAAACTATCGAGTTCGCTGAAGTTCTGATGATCGCAAACGGTGAAGAAGTCAAAATCGGCGTTCCTTTCGTTGATGGC GGCGTAATCAAAGCTGAAGTTGTTGCCCACGGTCGTGGCGAGAAAGTTAAAATCGTTAAGTTTCGTCGCCGTAAACACTACCGT AAGCAGCAGGGCCATCGTCAGTGGTTCACTGATGTGAAAATTACCGGCATCAGCGCTTAA (SEQ ID NO: 478)
55_jfrlSe1409-5_UN_O1,3,19_	ATGTACGCGGTTTTCCAAAGTGGTGGTAAACAACACCGAGTAAGCGAAGGTCAGACCGTTCGCCTGGAAAAGCTGGACATCGCA ACTGGCGAAACTATCGAGTTCGCTGAAGTTCTGATGATCGCAAACGGTGAAGAAGTCAAAATCGGCGTTCCTTTCGTTGATGGC GGCGTAATCAAAGCTGAAGTTGTTGCCCACGGTCGTGGCGAGAAAGTTAAAATCGTTAAGTTTCGTCGCCGTAAACACTACCGT AAGCAGCAGGGCCATCGTCAGTGGTTCACTGATGTGAAAATTACCGGCATCAGCGCTTAA (SEQ ID NO: 487)

Fig. 28D

56_jfrISe1409-6_Montevideo	ATGTACGCGGTTTTCCAAAGTGGTGGTAAACAACACCGAGTAAGCGAAGGTCAGACCGTTCGCCTGGAAAAGCTGGACATCGCA ACTGGCGAAACTATCGAGTTCGCTGAAGTTCTGATGGTCGCAAACGGTGAAGAAGTCAAAATCGGCGTTCCTTTCGTTGATGGC GGCGTAATCAAAGCTGAAGTTGTTGCCCACGGTCGTGGCGAGAAAGTTAAAATCGTTAAGTTTCGTCGCCGTAAACACTACCGT AAGCAGCAGGGCCATCGTCAGTGGTTCACTGATGTGAAAATTACCGGCATCAGCGCTTAA (SEQ ID NO: 495)
57_jfrISe1409-7_UN_O1,3,19_	ATGTACGCGGTTTTCCAAAGTGGTGGTAAACAACACCGAGTAAGCGAAGGTCAGACCGTTCGCCTGGAAAAGCTGGACATCGCA ACTGGCGAAACTATCGAGTTCGCTGAAGTTCTGATGATCGCAAACGGTGAAGAAGTCAAAATCGGCGTTCCTTTCGTTGATGGC GGCGTAATCAAAGCTGAAGTTGTTGCCCACGGTCGTGGCGAGAAAGTTAAAATCGTTAAGTTTCGTCGCCGTAAACACTACCGT AAGCAGCAGGGCCATCGTCAGTGGTTCACTGATGTGAAAATTACCGGCATCAGCGCTTAA (SEQ ID NO: 503)
58_jfrISe1409-8_UN_O18_	ATGTACGCGGTTTTCCAAAGTGGTGGTAAACAACACCGAGTAAGCGAAGGTCAGACCGTTCGCCTGGAAAAGCTGGACATCGCA ACTGGCGAAACTATCGAGTTCGCTGAAGTTCTGATGATCGCAAACGGTGAAGAAGTCAAAATCGGCGTTCCTTTCGTTGATGGC GGCGTAATCAAAGCTGAAGTTGTTGCCCACGGTCGTGGCGAGAAAGTTAAAATCGTTAAGTTTCGTCGCCGTAAACACTACCGT AAGCAGCAGGGCCATCGTCAGTGGTTCACTGATGTGAAAATTACCGGCATCAGCGCTTAA (SEQ ID NO: 512)
61_jfrISe1409-11_UN_O1,3,19_	ATGTACGCGGTTTTCCAAAGTGGTGGTAAACAACACCGAGTAAGCGAAGGTCAGACCGTTCGCCTGGAAAAGCTGGACATCGCA ACTGGCGAAACTATCGAGTTCGCTGAAGTTCTGATGATCGCAAACGGTGAAGAAGTCAAAATCGGCGTTCCTTTCGTTGATGGC GGCGTAATCAAAGCTGAAGTTGTTGCCCACGGTCGTGGCGAGAAAGTTAAAATCGTTAAGTTTCGTCGCCGTAAACACTACCGT AAGCAGCAGGGCCATCGTCAGTGGTTCACTGATGTGAAAATTACCGGCATCAGCGCTTAA (SEQ ID NO: 521)
67_jfrISe1409-17_Rissen	
71_jfrISe1409-21_Amsterdam	ATGTACGCGGTTTTCCAAAGTGGTGGTAAACAACACCGAGTAAGCGAAGGTCAGACCGTTCGCCTGGAAAAGCTGGACATCGCA ACTGGCGAAACTATCGAGTTCGCTGAAGTTCTGATGATCGCAAACGGTGAAGAAGTCAAAATCGGCGTTCCTTTCGTTGATGGC GGCGTAATCAAAGCTGAAGTTGTTGCCCACGGTCGTGGCGAGAAAGTTAAAATCGTTAAGTTTCGTCGCCGTAAACACTACCGT AAGCAGCAGGGCCATCGTCAGTGGTTCACTGATGTGAAAATTACCGGCATCAGCGCTTAA (SEQ ID NO: 538)
80_HySe09_Enteritidis	ATGTACGCGGTTTTCCAAAGTGGTGGTAAACAACACCGAGTAAGCGAAGGTCAGACCGTTCGCCTGGAAAAGCTGGACATCGCA ACTGGCGAAACTATCGAGTTCGCTGAAGTTCTGATGATCGCAAACGGTGAAGAAGTCAAAATCGGCGTTCCTTTCGTTGATGGC GGCGTAATCAAAGCTGAAGTTGTTGCCCACGGTCGTGGCGAGAAAGTTAAAATCGTTAAGTTTCGTCGCCGTAAACACTACCGT AAGCAGCAGGGCCATCGTCAGTGGTTCACTGATGTGAAAATTACCGGCATCAGCGCTTAA (SEQ ID NO: 548)
100_HySe29_Schwarzengrund	ATGTACGCGGTTTTCCAAAGTGGTGGTAAACAACACCGAGTAAGCGAAGGTCAGACCGTTCGCCTGGAAAAGCTGGACATCGCA ACTGGCGAAACTATCGAGTTCGCTGAAGTTCTGATGGTCGCAAACGGTGAAGAAGTCAAAATCGGCGTTCCTTTCGTTGATGGC GGCGTAATCAAAGCTGAAGTTGTTGCCCACGGTCGTGGCGAGAAAGTTAAAATCGTTAAGTTTCGTCGCCGTAAACACTACCGT AAGCAGCAGGGCCATCGTCAGTGGTTCACTGATGTGAAAATTACCGGCATCAGCGCTTAA (SEQ ID NO: 558)
103_HySe32_Schwarzengrund	
106_HySe35_Schwarzengrund	ATGTACGCGGTTTTCCAAAGTGGTGGTAAACAACACCGAGTAAGCGAAGGTCAGACCGTTCGCCTGGAAAAGCTGGACATCGCA ACTGGCGAAACTATCGAGTTCGCTGAAGTTCTGATGGTCGCAAACGGTGAAGAAGTCAAAATCGGCGTTCCTTTCGTTGATGGC GGCGTAATCAAAGCTGAAGTTGTTGCCCACGGTCGTGGCGAGAAAGTTAAAATCGTTAAGTTTCGTCGCCGTAAACACTACCGT AAGCAGCAGGGCCATCGTCAGTGGTTCACTGATGTGAAAATTACCGGCATCAGCGCTTAA (SEQ ID NO: 572)

Fig. 29A

01_NBRC13245T_Typhimurium	ATGTTTACTATCAACGCAGAAGTACGTAAAGAGCAGGGTAAGGGTGCGAGCCGCCGCCTGCGCGCCGCTAACAAGTTCCCGGC AATCATCTACGGCGGTTCTGAAGCCCCGATTGCTATCGAACTGGACCACGACCAGGTGATGAACATGCAAGCTAAAGCTGAATT CTACAGCGAAGTTCTGACCCTCGTTGTTGACGGTAAAGAAGTAAAAGTTAAAGCTCAGGCTGTACAGCGTCACGCTTACAAACC GAAGCTGACTCACATCGACTTCGTTCGCGCGTAA (SEQ ID NO: 59)
02_GTC00131_Enteritidis	ATGTTTACTATCAACGCAGAAGTACGTAAAGAGCAGGGTAAGGGTGCGAGCCGCCGCCTGCGCGCCGCTAACAAGTTCCCGGC AATCATCTACGGCGGTTCTGAAGCCCCGATTGCTATCGAACTGGACCACGACCAGGTGATGAACATGCAAGCTAAAGCTGAATT CTACAGCGAAGTTCTGACCCTCGTTGTTGACGGTAAAGAAGTAAAAGTTAAAGCTCAGGCTGTACAGCGTCACGCTTACAAACC GAAGCTGACTCACATCGACTTCGTTCGCGCGTAA (SEQ ID NO: 71)
03_GTC09491_Enteritidis	ATGTTTACTATCAACGCAGAAGTACGTAAAGAGCAGGGTAAGGGTGCGAGCCGCCGCCTGCGCGCCGCTAACAAGTTCCCGGC AATCATCTACGGCGGTTCTGAAGCCCCGATTGCTATCGAACTGGACCACGACCAGGTGATGAACATGCAAGCTAAAGCTGAATT CTACAGCGAAGTTCTGACCCTCGTTGTTGACGGTAAAGAAGTAAAAGTTAAAGCTCAGGCTGTACAGCGTCACGCTTACAAACC GAAGCTGACTCACATCGACTTCGTTCGCGCGTAA (SEQ ID NO: 80)
04_GTC03838_Enteritidis	ATGTTTACTATCAACGCAGAAGTACGTAAAGAGCAGGGTAAGGGTGCGAGCCGCCGCCTGCGCGCCGCTAACAAGTTCCCGGC AATCATCTACGGCGGTTCTGAAGCCCCGATTGCTATCGAACTGGACCACGACCAGGTGATGAACATGCAAGCTAAAGCTGAATT CTACAGCGAAGTTCTGACCCTCGTTGTTGACGGTAAAGAAGTAAAAGTTAAAGCTCAGGCTGTACAGCGTCACGCTTACAAACC GAAGCTGACTCACATCGACTTCGTTCGCGCGTAA (SEQ ID NO: 87)
05_GTC08914_Enteritidis	ATGTTTACTATCAACGCAGAAGTACGTAAAGAGCAGGGTAAGGGTGCGAGCCGCCGCCTGCGCGCCGCTAACAAGTTCCCGGC AATCATCTACGGCGGTTCTGAAGCCCCGATTGCTATCGAACTGGACCACGACCAGGTGATGAACATGCAAGCTAAAGCTGAATT CTACAGCGAAGTTCTGACCCTCGTTGTTGACGGTAAAGAAGTAAAAGTTAAAGCTCAGGCTGTACAGCGTCACGCTTACAAACC GAAGCTGACTCACATCGACTTCGTTCGCGCGTAA (SEQ ID NO: 94)
06_GTC09421_Enteritidis	ATGTTTACTATCAACGCAGAAGTACGTAAAGAGCAGGGTAAGGGTGCGAGCCGCCGCCTGCGCGCCGCTAACAAGTTCCCGGC AATCATCTACGGCGGTTCTGAAGCCCCGATTGCTATCGAACTGGACCACGACCAGGTGATGAACATGCAAGCTAAAGCTGAATT CTACAGCGAAGTTCTGACCCTCGTTGTTGACGGTAAAGAAGTAAAAGTTAAAGCTCAGGCTGTACAGCGTCACGCTTACAAACC GAAGCTGACTCACATCGACTTCGTTCGCGCGTAA (SEQ ID NO: 104)
07_GTC09489_Enteritidis	ATGTTTACTATCAACGCAGAAGTACGTAAAGAGCAGGGTAAGGGTGCGAGCCGCCGCCTGCGCGCCGCTAACAAGTTCCCGGC AATCATCTACGGCGGTTCTGAAGCCCCGATTGCTATCGAACTGGACCACGACCAGGTGATGAACATGCAAGCTAAAGCTGAATT CTACAGCGAAGTTCTGACCCTCGTTGTTGACGGTAAAGAAGTAAAAGTTAAAGCTCAGGCTGTACAGCGTCACGCTTACAAACC GAAGCTGACTCACATCGACTTCGTTCGCGCGTAA (SEQ ID NO: 114)
08_NBRC3163_Pullorum_Gallinarum	ATGTTTACTATCAACGCAGAAGTACGTAAAGAGCAGGGTAAGGGTGCGAGCCGCCGCCTGCGCGCCGCTAACAAGTTCCCGGC AATCATCTACGGCGGTTCTGAAGCCCCGATTGCTATCGAACTGGACCACGACCAGGTGATGAACATGCAAGCTAAAGCTGAATT CTACAGCGAAGTTCTGACCCTCGTTGTTGACGGTAAAGAAGTAAAAGTTAAAGCTCAGGCTGTACAGCGTCACGCTTACAAACC GAAGCTGACTCACATCGACTTCGTTCGCGCGTAA (SEQ ID NO: 121)
09_NBRC3313_Enteritidis	ATGTTTACTATCAACGCAGAAGTACGTAAAGAGCAGGGTAAGGGTGCGAGCCGCCGCCTGCGCGCCGCTAACAAGTTCCCGGC AATCATCTACGGCGGTTCTGAAGCCCCGATTGCTATCGAACTGGACCACGACCAGGTGATGAACATGCAAGCTAAAGCTGAATT CTACAGCGAAGTTCTGACCCTCGTTGTTGACGGTAAAGAAGTAAAAGTTAAAGCTCAGGCTGTACAGCGTCACGCTTACAAACC GAAGCTGACTCACATCGACTTCGTTCGCGCGTAA (SEQ ID NO: 131)
10_NBRC12529_Typhimurium	ATGTTTACTATCAACGCAGAAGTACGTAAAGAGCAGGGTAAGGGTGCGAGCCGCCGCCTGCGCGCCGCTAACAAGTTCCCGGC AATCATCTACGGCGGTTCTGAAGCCCCGATTGCTATCGAACTGGACCACGACCAGGTGATGAACATGCAAGCTAAAGCTGAATT CTACAGCGAAGTTCTGACCCTCGTTGTTGACGGTAAAGAAGTAAAAGTTAAAGCTCAGGCTGTACAGCGTCACGCTTACAAACC GAAGCTGACTCACATCGACTTCGTTCGCGCGTAA (SEQ ID NO: 138)
11_NBRC14193_Typhimurium	ATGTTTACTATCAACGCAGAAGTACGTAAAGAGCAGGGTAAGGGTGCGAGCCGCCGCCTGCGCGCCGCTAACAAGTTCCCGGC AATCATCTACGGCGGTTCTGAAGCCCCGATTGCTATCGAACTGGACCACGACCAGGTGATGAACATGCAAGCTAAAGCTGAATT CTACAGCGAAGTTCTGACCCTCGTTGTTGACGGTAAAGAAGTAAAAGTTAAAGCTCAGGCTGTACAGCGTCACGCTTACAAACC GAAGCTGACTCACATCGACTTCGTTCGCGCGTAA (SEQ ID NO: 145)
12_NBRC14194_Typhimurium	ATGTTTACTATCAACGCAGAAGTACGTAAAGAGCAGGGTAAGGGTGCGAGCCGCCGCCTGCGCGCCGCTAACAAGTTCCCGGC AATCATCTACGGCGGTTCTGAAGCCCCGATTGCTATCGAACTGGACCACGACCAGGTGATGAACATGCAAGCTAAAGCTGAATT CTACAGCGAAGTTCTGACCCTCGTTGTTGACGGTAAAGAAGTAAAAGTTAAAGCTCAGGCTGTACAGCGTCACGCTTACAAACC GAAGCTGACTCACATCGACTTCGTTCGCGCGTAA (SEQ ID NO: 152)

Fig. 29B

13_NBRC14209_Typhimurium	ATGTTTACTATCAACGCAGAAGTACGTAAAGAGCAGGGTAAGGGTGCGAGCCGCCGCCTGCGCGCCGCTAACAAGTTCCCGGC AATCATCTACGGCGGTTCTGAAGCCCCGATTGCTATCGAACTGGACCACGACCAGGTGATGAACATGCAAGCTAAAGCTGAATT CTACAGCGAAGTTCTGACCCTCGTTGTTGACGGTAAAGAAGTAAAAGTTAAAGCTCAGGCTGTACAGCGTCACGCTTACAAACC GAAGCTGACTCACATCGACTTCGTTCGCGCGTAA (SEQ ID NO: 159)
14_NBRC14210_Typhimurium	ATGTTTACTATCAACGCAGAAGTACGTAAAGAGCAGGGTAAGGGTGCGAGCCGCCGCCTGCGCGCCGCTAACAAGTTCCCGGC AATCATCTACGGCGGTTCTGAAGCCCCGATTGCTATCGAACTGGACCACGACCAGGTGATGAACATGCAAGCTAAAGCTGAATT CTACAGCGAAGTTCTGACCCTCGTTGTTGACGGTAAAGAAGTAAAAGTTAAAGCTCAGGCTGTACAGCGTCACGCTTACAAACC GAAGCTGACTCACATCGACTTCGTTCGCGCGTAA (SEQ ID NO: 166)
15_NBRC14211_Typhimurium	ATGTTTACTATCAACGCAGAAGTACGTAAAGAGCAGGGTAAGGGTGCGAGCCGCCGCCTGCGCGCCGCTAACAAGTTCCCGGC AATCATCTACGGCGGTTCTGAAGCCCCGATTGCTATCGAACTGGACCACGACCAGGTGATGAACATGCAAGCTAAAGCTGAATT CTACAGCGAAGTTCTGACCCTCGTTGTTGACGGTAAAGAAGTAAAAGTTAAAGCTCAGGCTGTACAGCGTCACGCTTACAAACC GAAGCTGACTCACATCGACTTCGTTCGCGCGTAA (SEQ ID NO: 172)
16_NBRC14212_Typhimurium	ATGTTTACTATCAACGCAGAAGTACGTAAAGAGCAGGGTAAGGGTGCGAGCCGCCGCCTGCGCGCCGCTAACAAGTTCCCGGC AATCATCTACGGCGGTTCTGAAGCCCCGATTGCTATCGAACTGGACCACGACCAGGTGATGAACATGCAAGCTAAAGCTGAATT CTACAGCGAAGTTCTGACCCTCGTTGTTGACGGTAAAGAAGTAAAAGTTAAAGCTCAGGCTGTACAGCGTCACGCTTACAAACC GAAGCTGACTCACATCGACTTCGTTCGCGCGTAA (SEQ ID NO: 179)
17_NBRC15181_Typhimurium	ATGTTTACTATCAACGCAGAAGTACGTAAAGAGCAGGGTAAGGGTGCGAGCCGCCGCCTGCGCGCCGCTAACAAGTTCCCGGC AATCATCTACGGCGGTTCTGAAGCCCCGATTGCTATCGAACTGGACCACGACCAGGTGATGAACATGCAAGCTAAAGCTGAATT CTACAGCGAAGTTCTGACCCTCGTTGTTGACGGTAAAGAAGTAAAAGTTAAAGCTCAGGCTGTACAGCGTCACGCTTACAAACC GAAGCTGACTCACATCGACTTCGTTCGCGCGTAA (SEQ ID NO: 186)
18_NBRC15182_Minnesota	ATGTTTACTATCAACGCAGAAGTACGTAAAGAGCAGGGTAAGGGTGCGAGCCGCCGCCTGCGCGCCGCTAACAAGTTCCCGGC AATCATCTACGGCGGTTCTGAAGCCCCGATTGCTATCGAACTGGACCACGACCAGGTGATGAACATGCAAGCTAAAGCTGAATT CTACAGCGAAGTTCTGACCCTCGTTGTTGACGGTAAAGAAGTAAAAGTTAAAGCTCAGGCTGTACAGCGTCACGCTTACAAACC GAAGCTGACTCATATCGACTTCGTTCGCGCGTAA (SEQ ID NO: 196)
19_NBRC15183_Minnesota	ATGTTTACTATCAACGCAGAAGTACGTAAAGAGCAGGGTAAGGGTGCGAGCCGCCGCCTGCGCGCCGCTAACAAGTTCCCGGC AATCATCTACGGCGGTTCTGAAGCCCCGATTGCTATCGAACTGGACCACGACCAGGTGATGAACATGCAAGCTAAAGCTGAATT CTACAGCGAAGTTCTGACCCTCGTTGTTGACGGTAAAGAAGTAAAAGTTAAAGCTCAGGCTGTACAGCGTCACGCTTACAAACC GAAGCTGACTCATATCGACTTCGTTCGCGCGTAA (SEQ ID NO: 206)
20_NBRC15184_Minnesota	ATGTTTACTATCAACGCAGAAGTACGTAAAGAGCAGGGTAAGGGTGCGAGCCGCCGCCTGCGCGCCGCTAACAAGTTCCCGGC AATCATCTACGGCGGTTCTGAAGCCCCGATTGCTATCGAACTGGACCACGACCAGGTGATGAACATGCAAGCTAAAGCTGAATT CTACAGCGAAGTTCTGACCCTCGTTGTTGACGGTAAAGAAGTAAAAGTTAAAGCTCAGGCTGTACAGCGTCACGCTTACAAACC GAAGCTGACTCATATCGACTTCGTTCGCGCGTAA (SEQ ID NO: 213)
21_NBRC15185_Minnesota	ATGTTTACTATCAACGCAGAAGTACGTAAAGAGCAGGGTAAGGGTGCGAGCCGCCGCCTGCGCGCCGCTAACAAGTTCCCGGC AATCATCTACGGCGGTTCTGAAGCCCCGATTGCTATCGAACTGGACCACGACCAGGTGATGAACATGCAAGCTAAAGCTGAATT CTACAGCGAAGTTCTGACCCTCGTTGTTGACGGTAAAGAAGTAAAAGTTAAAGCTCAGGCTGTACAGCGTCACGCTTACAAACC GAAGCTGACTCATATCGACTTCGTTCGCGCGTAA (SEQ ID NO: 220)
22_NBRC15186_Minnesota	ATGTTTACTATCAACGCAGAAGTACGTAAAGAGCAGGGTAAGGGTGCGAGCCGCCGCCTGCGCGCCGCTAACAAGTTCCCGGC AATCATCTACGGCGGTTCTGAAGCCCCGATTGCTATCGAACTGGACCACGACCAGGTGATGAACATGCAAGCTAAAGCTGAATT CTACAGCGAAGTTCTGACCCTCGTTGTTGACGGTAAAGAAGTAAAAGTTAAAGCTCAGGCTGTACAGCGTCACGCTTACAAACC GAAGCTGACTCACATCGACTTCGTTCGCGCGTAA (SEQ ID NO: 227)
23_NBRC15187_Minnesota	ATGTTTACTATCAACGCAGAAGTACGTAAAGAGCAGGGTAAGGGTGCGAGCCGCCGCCTGCGCGCCGCTAACAAGTTCCCGGC AATCATCTACGGCGGTTCTGAAGCCCCGATTGCTATCGAACTGGACCACGACCAGGTGATGAACATGCAAGCTAAAGCTGAATT CTACAGCGAAGTTCTGACCCTCGTTGTTGACGGTAAAGAAGTAAAAGTTAAAGCTCAGGCTGTACAGCGTCACGCTTACAAACC GAAGCTGACTCATATCGACTTCGTTCGCGCGTAA (SEQ ID NO: 234)
24_NBRC100797_Abony	ATGTTTACTATCAACGCAGAAGTACGTAAAGAGCAGGGTAAGGGTGCGAGCCGCCGCCTGCGCGCCGCTAACAAGTTCCCGGC AATCATCTACGGCGGTTCTGAAGCCCCGATTGCTATCGAACTGGACCACGACCAGGTGATGAACATGCAAGCTAAAGCTGAATT CTACAGCGAAGTTCTGACCCTCGTTGTTGACGGTAAAGAAGTAAAAGTTAAAGCTCAGGCTGTACAGCGTCACGCTTACAAACC GAAGCTGACTCACATCGACTTCGTTCGCGCGTAA (SEQ ID NO: 241)
25_NBRC105684_Choleraesuis	ATGTTTACTATCAACGCAGAAGTACGTAAAGAGCAGGGTAAGGGTGCGAGCCGCCGCCTGCGCGCCGCTAACAAGTTCCCGGC AATCATCTACGGCGGTTCTGAAGCCCCGATTGCTATCGAACTGGACCACGACCAGGTGATGAACATGCAAGCTAAAGCTGAATT CTACAGCGAAGTTCTGACCCTCGTTGTTGACGGTAAAGAAGTAAAAGTTAAAGCTCAGGCTGTACAGCGTCACGCTTACAAACC GAAGCTGACTCACATCGACTTCGTTCGCGCGTAA (SEQ ID NO: 251)
26_NBRC105726_Typhimurium	ATGTTTACTATCAACGCAGAAGTACGTAAAGAGCAGGGTAAGGGTGCGAGCCGCCGCCTGCGCGCCGCTAACAAGTTCCCGGC AATCATCTACGGCGGTTCTGAAGCCCCGATTGCTATCGAACTGGACCACGACCAGGTGATGAACATGCAAGCTAAAGCTGAATT CTACAGCGAAGTTCTGACCCTCGTTGTTGACGGTAAAGAAGTAAAAGTTAAAGCTCAGGCTGTACAGCGTCACGCTTACAAACC GAAGCTGACTCACATCGACTTCGTTCGCGCGTAA (SEQ ID NO: 261)
27_JCM3919_UN_O7_	ATGTTTACTATCAACGCAGAAGTACGTAAAGAGCAGGGTAAGGGTGCGAGCCGCCGCCTGCGCGCCGCTAACAAGTTCCCGGC AATCATCTACGGCGGTTCTGAAGCCCCGATTGCTATCGAACTGGACCACGACCAGGTGATGAACATGCAAGCTAAAGCTGAATT CTACAGCGAAGTTCTGACCCTCGTTGTTGACGGTAAAGAAGTAAAAGTTAAAGCTCAGGCTGTACAGCGTCACGCTTACAAACC GAAGCTGACTCACATCGACTTCGTTCGCGCGTAA (SEQ ID NO: 270)
28_NBRC15335_Minnesota	ATGTTTACTATCAACGCAGAAGTACGTAAAGAGCAGGGTAAGGGTGCGAGCCGCCGCCTGCGCGCCGCTAACAAGTTCCCGGC AATCATCTACGGCGGTTCTGAAGCCCCGATTGCTATCGAACTGGACCACGACCAGGTGATGAACATGCAAGCTAAAGCTGAATT CTACAGCGAAGTTCTGACCCTCGTTGTTGACGGTAAAGAAGTAAAAGTTAAAGCTCAGGCTGTACAGCGTCACGCTTACAAACC GAAGCTGACTCACATCGACTTCGTTCGCGCGTAA (SEQ ID NO: 277)
29_GTC09490_Enteritidis	ATGTTTACTATCAACGCAGAAGTACGTAAAGAGCAGGGTAAGGGTGCGAGCCGCCGCCTGCGCGCCGCTAACAAGTTCCCGGC AATCATCTACGGCGGTTCTGAAGCCCCGATTGCTATCGAACTGGACCACGACCAGGTGATGAACATGCAAGCTAAAGCTGAATT CTACAGCGAAGTTCTGACCCTCGTTGTTGACGGTAAAGAAGTAAAAGTTAAAGCTCAGGCTGTACAGCGTCACGCTTACAAACC GAAGCTGACTCACATCGACTTCGTTCGCGCGTAA (SEQ ID NO: 284)
30_GTC09492_Braenderup	ATGTTTACTATCAACGCAGAAGTACGTAAAGAGCAGGGTAAGGGTGCGAGCCGCCGCCTGCGCGCCGCTAACAAGTTCCCGGC AATCATCTACGGCGGTTCTGAAGCCCCGGTTGCTATCGAACTGGACCACGACCAGGTGATGAACATGCAAGCTAAAGCTGAATT CTACAGCGAAGTTCTGACCCTCGTGGTTGACGGTAAAGAAGTAAAAGTTAAAGCTCAGGCTGTACAGCGTCACGCTTACAAACC GAAGCTGACTCACATCGACTTCGTTCGCGCGTAA (SEQ ID NO: 291)
31_GTC09493_Pakistan	ATGTTTACTATCAACGCAGAAGTACGTAAAGAGCAGGGTAAGGGTGCGAGCCGCCGCCTGCGCGCCGCTAACAAGTTCCCGGC AATCATCTACGGCGGTTCTGAAGCCCCGATTGCTATCGAACTGGACCACGACCAGGTGATGAACATGCAAGCTAAAGCTGAATT CTACAGCGAAGTTCTGACCCTCGTTGTTGACGGTAAAGAAGTAAAAGTTAAAGCTCAGGCTGTACAGCGTCACGCTTACAAACC GAAGCTGACTCACATCGACTTCGTTCGCGCGTAA (SEQ ID NO: 298)
32_GTC09549_Typhimurium	ATGTTTACTATCAACGCAGAAGTACGTAAAGAGCAGGGTAAGGGTGCGAGCCGCCGCCTGCGCGCCGCTAACAAGTTCCCGGC AATCATCTACGGCGGTTCTGAAGCCCCGATTGCTATCGAACTGGACCACGACCAGGTGATGAACATGCAAGCTAAAGCTGAATT CTACAGCGAAGTTCTGACCCTCGTTGTTGACGGTAAAGAAGTAAAAGTTAAAGCTCAGGCTGTACAGCGTCACGCTTACAAACC GAAGCTGACTCACATCGACTTCGTTCGCGCGTAA (SEQ ID NO: 305)
33_ATCCBAA-1675_Infantis	ATGTTTACTATCAACGCAGAAGTACGTAAAGAGCAGGGTAAGGGTGCGAGCCGCCGCCTGCGCGCCGCTAACAAGTTCCCGGC AATCATCTACGGCGGTTCTGAAGCCCCGATTGCTATCGAACTGGACCACGACCAGGTGATGAACATGCAAGCTAAAGCTGAATT CTACAGCGAAGTTCTGACCCTCGTTGTTGACGGTAAAGAAGTAAAAGTTAAAGCTCAGGCTGTACAGCGTCACGCTTACAAACC GAAGCTGACTCACATCGACTTCGTTCGCGCGTAA (SEQ ID NO: 312)

Fig. 29C

34_ATCCBAA-1738_Thompson	ATGTTTACTATCAACGCAGAAGTACGTAAAGAGCAGGGTAAGGGTGCGAGCCGCCGCCTGCGCGCCGCTAACAAGTTCCCGGC AATCATCTACGGCGGTTCTGAAGCCCCGATTGCTATCGAACTGGACCACGACCAGGTGATGAACATGCAAGCTAAAGCTGAATT CTACAGCGAAGTTCTGACCCTCGTTGTTGACGGTAAAGAAGTAAAAGTTAAAGCTCAGGCTGTACAGCGTCACGCTTACAAACC GAAGCTGACTCACATCGACTTCGTTCGCGCGTAA (SEQ ID NO: 323)
35_ATCC9712_Saintpaul	ATGTTTACTATCAACGCAGAAGTACGTAAAGAGCAGGGTAAGGGTGCGAGCCGCCGCCTGCGCGCCGCTAACAAGTTCCCGGC AATCATCTACGGCGGTTCTGAAGCCCCGATTGCTATCGAACTGGACCACGACCAGGTGATGAACATGCAAGCTAAAGCTGAATT CTACAGCGAAGTTCTGACCCTCGTTGTTGACGGTAAAGAAGTAAAAGTTAAAGCTCAGGCTGTACAGCGTCACGCTTACAAACC GAAGCTGACTCACATCGACTTCGTTCGCGCGTAA (SEQ ID NO: 333)
36_jfrlSe1402-1_Infantis	ATGTTTACTATCAACGCAGAAGTACGTAAAGAGCAGGGTAAGGGTGCGAGCCGCCGCCTGCGCGCCGCTAACAAGTTCCCGGC AATCATCTACGGCGGTTCTGAAGCCCCGATTGCTATCGAACTGGACCACGACCAGGTGATGAACATGCAAGCTAAAGCTGAATT CTACAGCGAAGTTCTGACCCTCGTTGTTGACGGTAAAGAAGTAAAAGTTAAAGCTCAGGCTGTACAGCGTCACGCTTACAAACC GAAGCTGACTCACATCGACTTCGTTCGCGCGTAA (SEQ ID NO: 344)
38_jfrlSe1402-3_Brandenburg	ATGTTTACTATCAACGCAGAAGTACGTAAAGAGCAGGGTAAGGGTGCGAGCCGCCGCCTGCGCGCCGCTAACAAGTTCCCGGC AATCATCTACGGCGGTTCTGAAGCCCCGATTGCTATCGAACTGGACCACGACCAGGTGATGAACATGCAAGCTAAAGCTGAATT CTACAGCGAAGTTCTGACCCTCGTTGTTGACGGTAAAGAAGTAAAAGTTAAAGCTCAGGCTGTACAGCGTCACGCTTACAAACC GAAGCTGACTCACATCGACTTCGTTCGCGCGTAA (SEQ ID NO: 354)
39_jfrlSe1402-4_Infantis	ATGTTTACTATCAACGCAGAAGTACGTAAAGAGCAGGGTAAGGGTGCGAGCCGCCGCCTGCGCGCCGCTAACAAGTTCCCGGC AATCATCTACGGCGGTTCTGAAGCCCCGATTGCTATCGAACTGGACCACGACCAGGTGATGAACATGCAAGCTAAAGCTGAATT CTACAGCGAAGTTCTGACCCTCGTTGTTGACGGTAAAGAAGTAAAAGTTAAAGCTCAGGCTGTACAGCGTCACGCTTACAAACC GAAGCTGACTCACATCGACTTCGTTCGCGCGTAA (SEQ ID NO: 361)
40_jfrlSe1402-5_Brandenburg	ATGTTTACTATCAACGCAGAAGTACGTAAAGAGCAGGGTAAGGGTGCGAGCCGCCGCCTGCGCGCCGCTAACAAGTTCCCGGC AATCATCTACGGCGGTTCTGAAGCCCCGATTGCTATCGAACTGGACCACGACCAGGTGATGAACATGCAAGCTAAAGCTGAATT CTACAGCGAAGTTCTGACCCTCGTTGTTGACGGTAAAGAAGTAAAAGTTAAAGCTCAGGCTGTACAGCGTCACGCTTACAAACC GAAGCTGACTCACATCGACTTCGTTCGCGCGTAA (SEQ ID NO: 371)
41_jfrlSe1402-6_Rissen	ATGTTTACTATCAACGCAGAAGTACGTAAAGAGCAGGGTAAGGGTGCGAGCCGCCGCCTGCGCGCCGCTAACAAGTTCCCGGC AATCATCTACGGCGGTTCTGAAGCCCCGATTGCTATCGAACTGGACCACGACCAGGTGATGAACATGCAAGCTAAAGCTGAATT CTACAGCGAAGTTCTGACCCTCGTTGTTGACGGTAAAGAAGTAAAAGTTAAAGCTCAGGCTGTACAGCGTCACGCTTACAAACC GAAGCTGACTCACATCGACTTCGTTCGCGCGTAA (SEQ ID NO: 378)
42_jfrlSe1402-7_Orion	ATGTTTACTATCAACGCAGAAGTACGTAAAGAGCAGGGTAAGGGTGCGAGCCGCCGCCTGCGCGCCGCTAACAAGTTCCCGGC AATCATCTACGGCGGTTCTGAAGCCCCGATTGCTATCGAACTGGACCACGACCAGGTGATGAACATGCAAGCTAAAGCTGAATT CTACAGCGAAGTTCTGACCCTCGTTGTTGACGGTAAAGAAGTAAAAGTTAAAGCTCAGGCTGTACAGCGTCACGCTTACAAACC GAAGCTGACTCACATCGACTTCGTTCGCGCGTAA (SEQ ID NO: 385)
43_jfrlSe1402-8_Rissen	ATGTTTACTATCAACGCAGAAGTACGTAAAGAGCAGGGTAAGGGTGCGAGCCGCCGCCTGCGCGCCGCTAACAAGTTCCCGGC AATCATCTACGGCGGTTCTGAAGCCCCGATTGCTATCGAACTGGACCACGACCAGGTGATGAACATGCAAGCTAAAGCTGAATT CTACAGCGAAGTTCTGACCCTCGTTGTTGACGGTAAAGAAGTAAAAGTTAAAGCTCAGGCTGTACAGCGTCACGCTTACAAACC GAAGCTGACTCACATCGACTTCGTTCGCGCGTAA (SEQ ID NO: 395)
44_jfrlSe1402-9_Rissen	ATGTTTACTATCAACGCAGAAGTACGTAAAGAGCAGGGTAAGGGTGCGAGCCGCCGCCTGCGCGCCGCTAACAAGTTCCCGGC AATCATCTACGGCGGTTCTGAAGCCCCGATTGCTATCGAACTGGACCACGACCAGGTGATGAACATGCAAGCTAAAGCTGAATT CTACAGCGAAGTTCTGACCCTCGTTGTTGACGGTAAAGAAGTAAAAGTTAAAGCTCAGGCTGTACAGCGTCACGCTTACAAACC GAAGCTGACTCACATCGACTTCGTTCGCGCGTAA (SEQ ID NO: 402)
45_jfrlSe1402-10_Rissen	ATGTTTACTATCAACGCAGAAGTACGTAAAGAGCAGGGTAAGGGTGCGAGCCGCCGCCTGCGCGCCGCTAACAAGTTCCCGGC AATCATCTACGGCGGTTCTGAAGCCCCGATTGCTATCGAACTGGACCACGACCAGGTGATGAACATGCAAGCTAAAGCTGAATT CTACAGCGAAGTTCTGACCCTCGTTGTTGACGGTAAAGAAGTAAAAGTTAAAGCTCAGGCTGTACAGCGTCACGCTTACAAACC GAAGCTGACTCACATCGACTTCGTTCGCGCGTAA (SEQ ID NO: 409)
46_jfrlSe1402-11_Rissen	ATGTTTACTATCAACGCAGAAGTACGTAAAGAGCAGGGTAAGGGTGCGAGCCGCCGCCTGCGCGCCGCTAACAAGTTCCCGGC AATCATCTACGGCGGTTCTGAAGCCCCGATTGCTATCGAACTGGACCACGACCAGGTGATGAACATGCAAGCTAAAGCTGAATT CTACAGCGAAGTTCTGACCCTCGTTGTTGACGGTAAAGAAGTAAAAGTTAAAGCTCAGGCTGTACAGCGTCACGCTTACAAACC GAAGCTGACTCACATCGACTTCGTTCGCGCGTAA (SEQ ID NO: 416)
47_jfrlSe1402-12_Rissen	ATGTTTACTATCAACGCAGAAGTACGTAAAGAGCAGGGTAAGGGTGCGAGCCGCCGCCTGCGCGCCGCTAACAAGTTCCCGGC AATCATCTACGGCGGTTCTGAAGCCCCGATTGCTATCGAACTGGACCACGACCAGGTGATGAACATGCAAGCTAAAGCTGAATT CTACAGCGAAGTTCTGACCCTCGTTGTTGACGGTAAAGAAGTAAAAGTTAAAGCTCAGGCTGTACAGCGTCACGCTTACAAACC GAAGCTGACTCACATCGACTTCGTTCGCGCGTAA (SEQ ID NO: 423)
48_jfrlSe1402-13_Mbandaka	ATGTTTACTATCAACGCAGAAGTACGTAAAGAGCAGGGTAAGGGTGCGAGCCGCCGCCTGCGCGCCGCTAACAAGTTCCCGGC AATCATCTACGGCGGTTCTGAAGCCCCGATTGCTATCGAACTGGACCACGACCAGGTGATGAACATGCAAGCTAAAGCTGAATT CTACAGCGAAGTTCTGACCCTCGTTGTTGACGGTAAAGAAGTAAAAGTTAAAGCTCAGGCTGTACAGCGTCACGCTTACAAACC GAAGCTGACTCACATCGACTTCGTTCGCGCGTAA (SEQ ID NO: 430)
49_jfrlSe1402-14_Mbandaka	ATGTTTACTATCAACGCAGAAGTACGTAAAGAGCAGGGTAAGGGTGCGAGCCGCCGCCTGCGCGCCGCTAACAAGTTCCCGGC AATCATCTACGGCGGTTCTGAAGCCCCGATTGCTATCGAACTGGACCACGACCAGGTGATGAACATGCAAGCTAAAGCTGAATT CTACAGCGAAGTTCTGACCCTCGTTGTTGACGGTAAAGAAGTAAAAGTTAAAGCTCAGGCTGTACAGCGTCACGCTTACAAACC GAAGCTGACTCACATCGACTTCGTTCGCGCGTAA (SEQ ID NO: 440)
50_jfrlSe1402-15_Orion	ATGTTTACTATCAACGCAGAAGTACGTAAAGAGCAGGGTAAGGGTGCGAGCCGCCGCCTGCGCGCCGCTAACAAGTTCCCGGC AATCATCTACGGCGGTTCTGAAGCCCCGATTGCTATCGAACTGGACCACGACCAGGTGATGAACATGCAAGCTAAAGCTGAATT CTACAGCGAAGTTCTGACCCTCGTTGTTGACGGTAAAGAAGTAAAAGTTAAAGCTCAGGCTGTACAGCGTCACGCTTACAAACC GAAGCTGACTCACATCGACTTCGTTCGCGCGTAA (SEQ ID NO: 450)
51_jfrlSe1409-1_Altona	
52_jfrlSe1409-2_Istanbul	
53_jfrlSe1409-3_Senftenberg	
54_jfrlSe1409-4_UN_O13_	
55_jfrlSe1409-5_UN_O1,3,19_	
56_jfrlSe1409-6_Montevideo	
57_jfrlSe1409-7_UN_O1,3,19_	
58_jfrlSe1409-8_UN_O18_	
61_jfrlSe1409-11_UN_O1,3,19_	
67_jfrlSe1409-17_Rissen	
71_jfrlSe1409-21_Amsterdam	ATGTTTACTATCAACGCAGAAGTACGTAAAGAGCAGGGTAAGGGTGCGAGCCGCCGCCTGCGCGCCGCTAACAAGTTCCCGGC AATCATCTACGGCGGTTCTGAAGCCCCGGTTGCTATCGAACTGGACCACGACCAGGTGATGAACATGCAAGCTAAAGCTGAATT CTACAGCGAAGTTCTGACCCTCGTTGTTGACGGTAAAGAAGTAAAAGTTAAAGCTCAGGCTGTACAGCGTCACGCTTACAAACC GAAGCTGACTCACATCGACTTCGTTCGCGCGTAA (SEQ ID NO: 539)
80_HySe09_Enteritidis	ATGTTTACTATCAACGCAGAAGTACGTAAAGAGCAGGGTAAGGGTGCGAGCCGCCGCCTGCGCGCCGCTAACAAGTTCCCGGC AATCATCTACGGCGGTTCTGAAGCCCCGGTTGCTATCGAACTGGACCACGACCAGGTGATGAACATGCAAGCTAAAGCTGAATT CTACAGCGAAGTTCTGACCCTCGTTGTTGACGGTAAAGAAGTAAAAGTTAAAGCTCAGGCTGTACAGCGTCACGCTTACAAACC GAAGCTGACTCACATCGACTTCGTTCGCGCGTAA (SEQ ID NO: 549)
100_HySe29_Schwarzengrund	
103_HySe32_Schwarzengrund	
106_HySe35_Schwarzengrund	

Fig. 30A

	S7
01_NBRC13245T_Typhimurium	ATGCCACGTCGTCGCGTCATTGGTCAGCGTAAAATTCTGCCGGATCCGAAGTTCGGATCAGAACTGCTGGCTAAATTTGTCAAT ATCCTGATGGTAGATGGTAAAAAATCTACTGCAGAATCAATCGTATACAGCGCGCTGGAGACCCTGGCTCAGCGTTCTGGTAAA TCTGAACTGGAAGCTTTCGAAGTCGCTCTCGAAAACGTTCGCCCGACTGTAGAAGTTAAATCCCGCCGTGTAGGTGGTTCTACT TATCAGGTACCAGTTGAAGTCCGTCCGGTCCGTCGTAATGCTCTGGCAATGCGTTGGATCGTAGAAGCTGCTCGTAAACGCGG TGATAAATCCATGGCTCTGCGCCTGGCGAACGAACTTTCTGATGCTGCAGACAACAAAGGTACTGCAGTTAAGAAACGTGAAGA CGTTCACCGTATGGCAGAAGCCAACAAGGCGTTCGCACACTACCGTTGGTAA (SEQ ID NO: 59)
02_GTC00131_Enteritidis	ATGCCACGTCGTCGCGTCATTGGTCAGCGTAAAATTCTGCCGGATCCGAAGTTCGGATCAGAACTGCTGGCTAAATTTGTCAAT ATCCTGATGGTAGATGGTAAAAAATCTACTGCAGAATCAATCGTATACAGCGCGCTGGAGACCCTGGCTCAGCGTTCTGGTAAA TCTGAACTGGAAGCTTTCGAAGTCGCTCTCGAAAACGTTCGCCCGACTGTAGAAGTTAAATCCCGCCGTGTAGGTGGTTCTACT TATCAGGTACCAGTTGAAGTCCGTCCGGTCCGTCGTAATGCTCTGGCAATGCGTTGGATCGTAGAAGCTGCTCGTAAACGCGG TGATAAATCCATGGCTCTGCGTCTGGCGAACGAACTTTCTGACGCTGCAGAAAACAAAGGTACTGCAGTTAAGAAACGTGAAGA CGTTCACCGTATGGCAGAAGCCAACAAGGCGTTCGCACACTACCGTTGGTAA (SEQ ID NO: 72)
03_GTC09491_Enteritidis	ATGCCACGTCGTCGCGTCATTGGTCAGCGTAAAATTCTGCCGGATCCGAAGTTCGGATCAGAACTGCTGGCTAAATTTGTCAAT ATCCTGATGGTAGATGGTAAAAAATCTACTGCAGAATCAATCGTATACAGCGCGCTGGAGACCCTGGCTCAGCGTTCTGGTAAA TCTGAACTGGAAGCTTTCGAAGTCGCTCTCGAAAACGTTCGCCCGACTGTAGAAGTTAAATCCCGCCGTGTAGGTGGTTCTACT TATCAGGTACCAGTTGAAGTCCGTCCGGTCCGTCGTAATGCTCTGGCAATGCGTTGGATCGTAGAAGCTGCTCGTAAACGCGG TGATAAATCCATGGCTCTGCGTCTGGCGAACGAACTTTCTGACGCTGCAGAAAACAAAGGTACTGCAGTTAAGAAACGTGAAGA CGTTCACCGTATGGCAGAAGCCAACAAGGCGTTCGCACACTACCGTTGGTAA (SEQ ID NO: 81)
04_GTC03838_Enteritidis	ATGCCACGTCGTCGCGTCATTGGTCAGCGTAAAATTCTGCCGGATCCGAAGTTCGGATCAGAACTGCTGGCTAAATTTGTCAAT ATCCTGATGGTAGATGGTAAAAAATCTACTGCAGAATCAATCGTATACAGCGCGCTGGAGACCCTGGCTCAGCGTTCTGGTAAA TCTGAACTGGAAGCTTTCGAAGTCGCTCTCGAAAACGTTCGCCCGACTGTAGAAGTTAAATCCCGCCGTGTAGGTGGTTCTACT TATCAGGTACCAGTTGAAGTCCGTCCGGTCCGTCGTAATGCTCTGGCAATGCGTTGGATCGTAGAAGCTGCTCGTAAACGCGG TGATAAATCCATGGCTCTGCGTCTGGCGAACGAACTTTCTGACGCTGCAGAAAACAAAGGTACTGCAGTTAAGAAACGTGAAGA CGTTCACCGTATGGCAGAAGCCAACAAGGCGTTCGCACACTACCGTTGGTAA (SEQ ID NO: 88)
05_GTC08914_Enteritidis	ATGCCACGTCGTCGCGTCATTGGTCAGCGTAAAATTCTGCCGGATCCGAAGTTCGGATCAGAACTGCTGGCTAAATTTGTCAAT ATCCTGATGGTAGATGGTAAAAAATCTACTGCAGAATCAATCGTATACAGCGCGCTGGAGACCCTGGCTCAGCGTTCTGGTAAA TCTGAACTGGAAGCTTTCGAAGTCGCTCTCGAAAACGTTCGCCCGACTGTAGAAGTTAAATCCCGCCGTGTAGGTGGTTCTACT TATCAGGTACCAGTTGAAGTCCGTCCGGTCCGTCGTAATGCTCTGGCAATGCGTTGGATCGTAGAAGCTGCTCGTAAACGCGG TGATAAATCCATGGCTCTGCGTCTGGCGAACGAACTTTCTGACGCTGCAGAAAACAAAGGTACTGCAGTTAAGAAACGTGAAGA CGTTCACCGTATGGCAGAAGCCAACAAGGCGTTCGCACACTACCGTTGGTAA (SEQ ID NO: 95)
06_GTC09421_Enteritidis	ATGCCACGTCGTCGCGTCATTGGTCAGCGTAAAATTCTGCCGGATCCGAAGTTCGGATCAGAACTGCTGGCTAAATTTGTCAAT ATCCTGATGGTAGATGGTAAAAAATCTACTGCAGAATCAATCGTATACAGCGCGCTGGAGACCCTGGCTCAGCGTTCTGGTAAA TCTGAACTGGAAGCTTTCGAAGTCGCTCTCGAAAACGTTCGCCCGACTGTAGAAGTTAAATCCCGCCGTGTAGGTGGTTCTACT TATCAGGTACCAGTTGAAGTCCGTCCGGTCCGTCGTAATGCTCTGGCAATGCGTTGGATCGTAGAAGCTGCTCGTAAACGCGG TGATAAATCCATGGCTCTGCGTCTGGCGAACGAACTTTCTGACGCTGCAGAAAACAAAGGTACTGCAGTTAAGAAACGTGAAGA CGTTCACCGTATGGCAGAAGCCAACAAGGCGTTCGCACACTACCGTTGGTAA (SEQ ID NO: 105)
07_GTC09489_Enteritidis	ATGCCACGTCGTCGCGTCATTGGTCAGCGTAAAATTCTGCCGGATCCGAAGTTCGGATCAGAACTGCTGGCTAAATTTGTCAAT ATCCTGATGGTAGATGGTAAAAAATCTACTGCAGAATCAATCGTATACAGCGCGCTGGAGACCCTGGCTCAGCGTTCTGGTAAA TCTGAACTGGAAGCTTTCGAAGTCGCTCTCGAAAACGTTCGCCCGACTGTAGAAGTTAAATCCCGCCGTGTAGGTGGTTCTACT TATCAGGTACCAGTTGAAGTCCGTCCGGTCCGTCGTAATGCTCTGGCAATGCGTTGGATCGTAGAAGCTGCTCGTAAACGCGG TGATAAATCCATGGCTCTGCGTCTGGCGAACGAACTTTCTGACGCTGCAGAAAACAAAGGTACTGCAGTTAAGAAACGTGAAGA CGTTCACCGTATGGCAGAAGCCAACAAGGCGTTCGCACACTACCGTTGGTAA (SEQ ID NO: 115)
08_NBRC3163_Pullorum_Gallinarum	ATGCCACGTCGTCGCGTCATTGGTCAGCGTAAAATTCTGCCGGATCCGAAGTTCGGATCAGAACTGCTGGCTAAATTTGTCAAT ATCCTGATGGTAGATGGTAAAAAATCTACTGCAGAATCAATCGTATACAGCGCGCTGGAGACCCTGGCTCAGCGTTCTGGTAAA TCTGAACTGGAAGCTTTCGAAGTCGCTCTCGAAAACGTTCGCCCGACTGTAGAAGTTAAATCCCGCCGTGTAGGTGGTTCTACT TATCAGGTACCAGTTGAAGTCCGTCCGGTCCGTCGTAATGCTCTGGCAATGCGTTGGATCGTAGAAGCTGCTCGTAAGCGCGG TGATAAATCCATGGCTCTGCGTCTGGCGAACGAACTTTCTGACGCTGCAGAAAACAAAGGTACTGCAGTTAAGAAACGTGAAGA CGTTCACCGTATGGCAGAAGCCAACAAGGCGTTCGCACACTACCGTTGGTAA (SEQ ID NO: 122)
09_NBRC3313_Enteritidis	ATGCCACGTCGTCGCGTCATTGGTCAGCGTAAAATTCTGCCGGATCCGAAGTTCGGATCAGAACTGCTGGCTAAATTTGTCAAT ATCCTGATGGTAGATGGTAAAAAATCTACTGCAGAATCAATCGTATACAGCGCGCTGGAGACCCTGGCTCAGCGTTCTGGTAAA TCTGAACTGGAAGCTTTCGAAGTCGCTCTCGAAAACGTTCGCCCGACTGTAGAAGTTAAATCCCGCCGTGTAGGTGGTTCTACT TATCAGGTACCAGTTGAAGTCCGTCCGGTCCGTCGTAATGCTCTGGCAATGCGTTGGATCGTAGAAGCTGCTCGTAAACGCGG TGATAAATCCATGGCTCTGCGTCTGGCGAACGAACTTTCTGACGCTGCAGAAAACAAAGGTACTGCAGTTAAGAAACGTGAAGA CGTTCACCGTATGGCAGAAGCCAACAAGGCGTTCGCACACTACCGTTGGTAA (SEQ ID NO: 132)
10_NBRC12529_Typhimurium	ATGCCACGTCGTCGCGTCATTGGTCAGCGTAAAATTCTGCCGGATCCGAAGTTCGGATCAGAACTGCTGGCTAAATTTGTCAAT ATCCTGATGGTAGATGGTAAAAAATCTACTGCAGAATCAATCGTATACAGCGCGCTGGAGACCCTGGCTCAGCGTTCTGGTAAA TCTGAACTGGAAGCTTTCGAAGTCGCTCTCGAAAACGTTCGCCCGACTGTAGAAGTTAAATCCCGCCGTGTAGGTGGTTCTACT TATCAGGTACCAGTTGAAGTCCGTCCGGTCCGTCGTAATGCTCTGGCAATGCGTTGGATCGTAGAAGCTGCTCGTAAACGCGG TGATAAATCCATGGCTCTGCGCCTGGCGAACGAACTTTCTGATGCTGCAGACAACAAAGGTACTGCAGTTAAGAAACGTGAAGA CGTTCACCGTATGGCAGAAGCCAACAAGGCGTTCGCACACTACCGTTGGTAA (SEQ ID NO: 139)
11_NBRC14193_Typhimurium	ATGCCACGTCGTCGCGTCATTGGTCAGCGTAAAATTCTGCCGGATCCGAAGTTCGGATCAGAACTGCTGGCTAAATTTGTCAAT ATCCTGATGGTAGATGGTAAAAAATCTACTGCAGAATCAATCGTATACAGCGCGCTGGAGACCCTGGCTCAGCGTTCTGGTAAA TCTGAACTGGAAGCTTTCGAAGTCGCTCTCGAAAACGTTCGCCCGACTGTAGAAGTTAAATCCCGCCGTGTAGGTGGTTCTACT TATCAGGTACCAGTTGAAGTCCGTCCGGTCCGTCGTAATGCTCTGGCAATGCGTTGGATCGTAGAAGCTGCTCGTAAACGCGG TGATAAATCCATGGCTCTGCGCCTGGCGAACGAACTTTCTGATGCTGCAGACAACAAAGGTACTGCAGTTAAGAAACGTGAAGA CGTTCACCGTATGGCAGAAGCCAACAAGGCGTTCGCACACTACCGTTGGTAA (SEQ ID NO: 146)
12_NBRC14194_Typhimurium	ATGCCACGTCGTCGCGTCATTGGTCAGCGTAAAATTCTGCCGGATCCGAAGTTCGGATCAGAACTGCTGGCTAAATTTGTCAAT ATCCTGATGGTAGATGGTAAAAAATCTACTGCAGAATCAATCGTATACAGCGCGCTGGAGACCCTGGCTCAGCGTTCTGGTAAA TCTGAACTGGAAGCTTTCGAAGTCGCTCTCGAAAACGTTCGCCCGACTGTAGAAGTTAAATCCCGCCGTGTAGGTGGTTCTACT TATCAGGTACCAGTTGAAGTCCGTCCGGTCCGTCGTAATGCTCTGGCAATGCGTTGGATCGTAGAAGCTGCTCGTAAACGCGG TGATAAATCCATGGCTCTGCGCCTGGCGAACGAACTTTCTGATGCTGCAGACAACAAAGGTACTGCAGTTAAGAAACGTGAAGA CGTTCACCGTATGGCAGAAGCCAACAAGGCGTTCGCACACTACCGTTGGTAA (SEQ ID NO: 153)
13_NBRC14209_Typhimurium	ATGCCACGTCGTCGCGTCATTGGTCAGCGTAAAATTCTGCCGGATCCGAAGTTCGGATCAGAACTGCTGGCTAAATTTGTCAAT ATCCTGATGGTAGATGGTAAAAAATCTACTGCAGAATCAATCGTATACAGCGCGCTGGAGACCCTGGCTCAGCGTTCTGGTAAA TCTGAACTGGAAGCTTTCGAAGTCGCTCTCGAAAACGTTCGCCCGACTGTAGAAGTTAAATCCCGCCGTGTAGGTGGTTCTACT TATCAGGTACCAGTTGAAGTCCGTCCGGTCCGTCGTAATGCTCTGGCAATGCGTTGGATCGTAGAAGCTGCTCGTAAACGCGG TGATAAATCCATGGCTCTGCGCCTGGCGAACGAACTTTCTGATGCTGCAGACAACAAAGGTACTGCAGTTAAGAAACGTGAAGA CGTTCACCGTATGGCAGAAGCCAACAAGGCGTTCGCACACTACCGTTGGTAA (SEQ ID NO: 160)
14_NBRC14210_Typhimurium	ATGCCACGTCGTCGCGTCATTGGTCAGCGTAAAATTCTGCCGGATCCGAAGTTCGGATCAGAACTGCTGGCTAAATTTGTCAAT ATCCTGATGGTAGATGGTAAAAAATCTACTGCAGAATCAATCGTATACAGCGCGCTGGAGACCCTGGCTCAGCGTTCTGGTAAA TCTGAACTGGAAGCTTTCGAAGTCGCTCTCGAAAACGTTCGCCCGACTGTAGAAGTTAAATCCCGCCGTGTAGGTGGTTCTACT TATCAGGTACCAGTTGAAGTCCGTCCGGTCCGTCGTAATGCTCTGGCAATGCGTTGGATCGTAGAAGCTGCTCGTAAACGCGG TGATAAATCCATGGCTCTGCGCCTGGCGAACGAACTTTCTGATGCTGCAGACAACAAAGGTACTGCAGTTAAGAAACGTGAAGA CGTTCACCGTATGGCAGAAGCCAACAAGGCGTTCGCACACTACCGTTGGTAA (SEQ ID NO: 167)

Fig. 30B

15_NBRC14211_Typhimurium	ATGCCACGTCGTCGCGTCATTGGTCAGCGTAAAATTCTGCCGGATCCGAAGTTCGGATCAGAACTGCTGGCTAAATTTGTCAAT ATCCTGATGGTAGATGGTAAAAAATCTACTGCAGAATCAATCGTATACAGCGCGCTGGAGACCCTGGCTCAGCGTTCTGGTAAA TCTGAACTGGAAGCTTTCGAAGTCGCTCTCGAAAACGTTCGCCCGACTGTAGAAGTTAAATCCCGCCGTGTAGGTGGTTCTACT TATCAGGTACCAGTTGAAGTCCGTCCGGTCCGTCGTAATGCTCTGGCAATGCGTTGGATCGTAGAAGCTGCTCGTAAACGCGG TGATAAATCCATGGCTCTGCGCCTGGCGAACGAACTTTCTGATGCTGCAGACAACAAAGGTACTGCAGTTAAGAAACGTGAAGA CGTTCACCGTATGGCAGAAGCCAACAAGGCGTTCGCACACTACCGTTGGTAA (SEQ ID NO: 173)
16_NBRC14212_Typhimurium	ATGCCACGTCGTCGCGTCATTGGTCAGCGTAAAATTCTGCCGGATCCGAAGTTCGGATCAGAACTGCTGGCTAAATTTGTCAAT ATCCTGATGGTAGATGGTAAAAAATCTACTGCAGAATCAATCGTATACAGCGCGCTGGAGACCCTGGCTCAGCGTTCTGGTAAA TCTGAACTGGAAGCTTTCGAAGTCGCTCTCGAAAACGTTCGCCCGACTGTAGAAGTTAAATCCCGCCGTGTAGGTGGTTCTACT TATCAGGTACCAGTTGAAGTCCGTCCGGTCCGTCGTAATGCTCTGGCAATGCGTTGGATCGTAGAAGCTGCTCGTAAACGCGG TGATAAATCCATGGCTCTGCGCCTGGCGAACGAACTTTCTGATGCTGCAGACAACAAAGGTACTGCAGTTAAGAAACGTGAAGA CGTTCACCGTATGGCAGAAGCCAACAAGGCGTTCGCACACTACCGTTGGTAA (SEQ ID NO: 180)
17_NBRC15181_Typhimurium	ATGCCACGTCGTCGCGTCATTGGTCAGCGTAAAATTCTGCCGGATCCGAAGTTCGGATCAGAACTGCTGGCTAAATTTGTCAAT ATCCTGATGGTAGATGGTAAAAAATCTACTGCAGAATCAATCGTATACAGCGCGCTGGAGACCCTGGCTCAGCGTTCTGGTAAA TCTGAACTGGAAGCTTTCGAAGTCGCTCTCGAAAACGTTCGCCCGACTGTAGAAGTTAAATCCCGCCGTGTAGGTGGTTCTACT TATCAGGTACCAGTTGAAGTCCGTCCGGTCCGTCGTAATGCTCTGGCAATGCGTTGGATCGTAGAAGCTGCTCGTAAACGCGG TGATAAATCCATGGCTCTGCGCCTGGCGAACGAACTTTCTGATGCTGCAGACAACAAAGGTACTGCAGTTAAGAAACGTGAAGA CGTTCACCGTATGGCAGAAGCCAACAAGGCGTTCGCACACTACCGTTGGTAA (SEQ ID NO: 187)
18_NBRC15182_Minnesota	ATGCCACGTCGTCGCGTCATTGGTCAGCGTAAAATTCTGCCGGATCCGAAGTTCGGATCAGAACTGCTGGCTAAATTTGTCAAT ATCCTGATGGTAGATGGTAAAAAATCTACTGCAGAATCAATCGTATACAGCGCGCTGGAGACCCTGGCTCAGCGTTCTGGTAAA TCTGAACTGGAAGCTTTCGAAGTCGCTCTCGAAAACGTTCGCCCGACTGTAGAAGTTAAATCCCGCCGTGTAGGTGGTTCTACT TATCAGGTACCAGTTGAAGTCCGTCCGGTCCGTCGTAATGCTCTGGCAATGCGTTGGATCGTAGAAGCTGCTCGTAAACGCGG TGATAAATCCATGGCTCTGCGCCTGGCGAACGAACTTTCTGATGCTGCAGACAACAAAGGTACTGCAGTTAAGAAACGTGAAGA CGTTCACCGTATGGCAGAAGCCAACAAGGCGTTCGCACACTACCGTTGGTAA (SEQ ID NO: 197)
19_NBRC15183_Minnesota	ATGCCACGTCGTCGCGTCATTGGTCAGCGTAAAATTCTGCCGGATCCGAAGTTCGGATCAGAACTGCTGGCTAAATTTGTCAAT ATCCTGATGGTAGATGGTAAAAAATCTACTGCAGAATCAATCGTATACAGCGCGCTGGAGACCCTGGCTCAGCGTTCTGGTAAA TCTGAACTGGAAGCTTTCGAAGTCGCTCTCGAAAACGTTCGCCCGACTGTAGAAGTTAAATCCCGCCGTGTAGGTGGTTCTACT TATCAGGTACCAGTTGAAGTCCGTCCGGTCCGTCGTAATGCTCTGGCAATGCGTTGGATCGTAGAAGCTGCTCGTAAACGCGG TGATAAATCCATGGCTCTGCGCCTGGCGAACGAACTTTCTGATGCTGCAGACAACAAAGGTACTGCAGTTAAGAAACGTGAAGA CGTTCACCGTATGGCAGAAGCCAACAAGGCGTTCGCACACTACCGTTGGTAA (SEQ ID NO: 207)
20_NBRC15184_Minnesota	ATGCCACGTCGTCGCGTCATTGGTCAGCGTAAAATTCTGCCGGATCCGAAGTTCGGATCAGAACTGCTGGCTAAATTTGTCAAT ATCCTGATGGTAGATGGTAAAAAATCTACTGCAGAATCAATCGTATACAGCGCGCTGGAGACCCTGGCTCAGCGTTCTGGTAAA TCTGAACTGGAAGCTTTCGAAGTCGCTCTCGAAAACGTTCGCCCGACTGTAGAAGTTAAATCCCGCCGTGTAGGTGGTTCTACT TATCAGGTACCAGTTGAAGTCCGTCCGGTCCGTCGTAATGCTCTGGCAATGCGTTGGATCGTAGAAGCTGCTCGTAAACGCGG TGATAAATCCATGGCTCTGCGCCTGGCGAACGAACTTTCTGATGCTGCAGACAACAAAGGTACTGCAGTTAAGAAACGTGAAGA CGTTCACCGTATGGCAGAAGCCAACAAGGCGTTCGCACACTACCGTTGGTAA (SEQ ID NO: 214)
21_NBRC15185_Minnesota	ATGCCACGTCGTCGCGTCATTGGTCAGCGTAAAATTCTGCCGGATCCGAAGTTCGGATCAGAACTGCTGGCTAAATTTGTCAAT ATCCTGATGGTAGATGGTAAAAAATCTACTGCAGAATCAATCGTATACAGCGCGCTGGAGACCCTGGCTCAGCGTTCTGGTAAA TCTGAACTGGAAGCTTTCGAAGTCGCTCTCGAAAACGTTCGCCCGACTGTAGAAGTTAAATCCCGCCGTGTAGGTGGTTCTACT TATCAGGTACCAGTTGAAGTCCGTCCGGTCCGTCGTAATGCTCTGGCAATGCGTTGGATCGTAGAAGCTGCTCGTAAACGCGG TGATAAATCCATGGCTCTGCGCCTGGCGAACGAACTTTCTGATGCTGCAGACAACAAAGGTACTGCAGTTAAGAAACGTGAAGA CGTTCACCGTATGGCAGAAGCCAACAAGGCGTTCGCACACTACCGTTGGTAA (SEQ ID NO: 221)
22_NBRC15186_Minnesota	ATGCCACGTCGTCGCGTCATTGGTCAGCGTAAAATTCTGCCGGATCCGAAGTTCGGATCAGAACTGCTGGCTAAATTTGTCAAT ATCCTGATGGTAGATGGTAAAAAATCTACTGCAGAATCAATCGTATACAGCGCGCTGGAGACCCTGGCTCAGCGTTCTGGTAAA TCTGAACTGGAAGCTTTCGAAGTCGCTCTCGAAAACGTTCGCCCGACTGTAGAAGTTAAATCCCGCCGTGTAGGTGGTTCTACT TATCAGGTACCAGTTGAAGTCCGTCCGGTCCGTCGTAATGCTCTGGCAATGCGTTGGATCGTAGAAGCTGCTCGTAAACGCGG TGATAAATCCATGGCTCTGCGCCTGGCGAACGAACTTTCTGATGCTGCAGACAACAAAGGTACTGCAGTTAAGAAACGTGAAGA CGTTCACCGTATGGCAGAAGCCAACAAGGCGTTCGCACACTACCGTTGGTAA (SEQ ID NO: 228)
23_NBRC15187_Minnesota	ATGCCACGTCGTCGCGTCATTGGTCAGCGTAAAATTCTGCCGGATCCGAAGTTCGGATCAGAACTGCTGGCTAAATTTGTCAAT ATCCTGATGGTAGATGGTAAAAAATCTACTGCAGAATCAATCGTATACAGCGCGCTGGAGACCCTGGCTCAGCGTTCTGGTAAA TCTGAACTGGAAGCTTTCGAAGTCGCTCTCGAAAACGTTCGCCCGACTGTAGAAGTTAAATCCCGCCGTGTAGGTGGTTCTACT TATCAGGTACCAGTTGAAGTCCGTCCGGTCCGTCGTAATGCTCTGGCAATGCGTTGGATCGTAGAAGCTGCTCGTAAACGCGG TGATAAATCCATGGCTCTGCGCCTGGCGAACGAACTTTCTGATGCTGCAGACAACAAAGGTACTGCAGTTAAGAAACGTGAAGA CGTTCACCGTATGGCAGAAGCCAACAAGGCGTTCGCACACTACCGTTGGTAA (SEQ ID NO: 235)
24_NBRC100797_Abony	ATGCCACGTCGTCGCGTCATTGGTCAGCGTAAAATTCTGCCGGATCCGAAGTTCGGATCAGAACTGCTGGCTAAATTTGTCAAT ATCCTGATGGTAGATGGTAAAAAATCTACTGCAGAATCAATCGTATACAGCGCGCTGGAGACCCTGGCTCAGCGTTCTGGTAAA TCTGAACTGGAAGCTTTCGAAGTCGCTCTCGAAAACGTTCGCCCGACTGTAGAAGTTAAATCCCGCCGTGTAGGTGGTTCTACT TATCAGGTACCAGTTGAAGTCCGTCCGGTCCGTCGTAATGCTCTGGCAATGCGTTGGATCGTAGAAGCTGCTCGTAAACGCGG TGATAAATCCATGGCTCTGCGCCTGGCGAACGAACTTTCTGACGCTGCAGACAACAAAGGTACTGCAGTTAAGAAACGTGAAGA CGTTCACCGTATGGCAGAAGCCAACAAGGCGTTCGCACACTACCGTTGGTAA (SEQ ID NO: 242)
25_NBRC105684_Choleraesuis	ATGCCACGTCGTCGCGTCATTGGTCAGCGTAAAATTCTGCCGGATCCGAAGTTCGGATCAGAACTGCTGGCTAAATTTGTCAAT ATCCTGATGGTAGATGGTAAAAAATCTACTGCAGAATCAATCGCATACAGCGCGCTGGAGACCCTGGCTCAGCGTTCTGGTAAA TCTGAACTGGAAGCTTTCGAAGTCGCTCTCGAAAACGTTCGCCCGACTGTAGAAGTTAAATCCCGCCGTGTAGGTGGTTCTACT TATCAGGTACCAGTTGAAGTCCGTCCGGTCCGTCGTAATGCTCTGGCAATGCGTTGGATCGTAGAAGCTGCTCGTAAACGCGG TGATAAATCCATGGCTCTGCGCCTGGCGAACGAACTTTCTGACGCTGCAGACAACAAAGGTACTGCAGTTAAGAAACGTGAAGA CGTTCACCGTATGGCAGAAGCCAACAAGGCGTTCGCACACTACCGTTGGTAA (SEQ ID NO: 252)
26_NBRC105726_Typhimurium	
27_JCM3919_UN_O7_	ATGCCACGTCGTCGCGTCATTGGTCAGCGTAAAATTCTGCCGGATCCGAAGTTCGGATCAGAACTGCTGGCTAAATTTGTCAAT ATCCTGATGGTAGATGGTAAAAAATCTACTGCAGAATCAATCGTATACAGCGCGCTGGAGACCCTGGCTCAGCGTTCTGGTAAA TCTGAACTGGAAGCTTTCGAAGTCGCTCTCGAAAACGTTCGCCCGACTGTAGAAGTTAAATCCCGCCGTGTAGGTGGTTCTACT TATCAGGTACCAGTTGAAGTCCGTCCGGTCCGTCGTAATGCTCTGGCAATGCGTTGGATCGTAGAAGCTGCTCGTAAACGCGG TGATAAATCCATGGCTCTGCGCCTGGCGAACGAACTTTCTGACGCTGCAGACAACAAAGGTACTGCAGTTAAGAAACGTGAAGA CGTTCACCGTATGGCAGAAGCCAACAAGGCGTTCGCACACTACCGTTGGTAA (SEQ ID NO: 271)
28_NBRC15335_Minnesota	ATGCCACGTCGTCGCGTCATTGGTCAGCGTAAAATTCTGCCGGATCCGAAGTTCGGATCAGAACTGCTGGCTAAATTTGTCAAT ATCCTGATGGTAGATGGTAAAAAATCTACTGCAGAATCAATCGTATACAGCGCGCTGGAGACCCTGGCTCAGCGTTCTGGTAAA TCTGAACTGGAAGCTTTCGAAGTCGCTCTCGAAAACGTTCGCCCGACTGTAGAAGTTAAATCCCGCCGTGTAGGTGGTTCTACT TATCAGGTACCAGTTGAAGTCCGTCCGGTCCGTCGTAATGCTCTGGCAATGCGTTGGATCGTAGAAGCTGCTCGTAAACGCGG TGATAAATCCATGGCTCTGCGCCTGGCGAACGAACTTTCTGATGCTGCAGACAACAAAGGTACTGCAGTTAAGAAACGTGAAGA CGTTCACCGTATGGCAGAAGCCAACAAGGCGTTCGCACACTACCGTTGGTAA (SEQ ID NO: 278)
29_GTC09490_Enteritidis	ATGCCACGTCGTCGCGTCATTGGTCAGCGTAAAATTCTGCCGGATCCGAAGTTCGGATCAGAACTGCTGGCTAAATTTGTCAAT ATCCTGATGGTAGATGGTAAAAAATCTACTGCAGAATCAATCGTATACAGCGCGCTGGAGACCCTGGCTCAGCGTTCTGGTAAA TCTGAACTGGAAGCTTTCGAAGTCGCTCTCGAAAACGTTCGCCCGACTGTAGAAGTTAAATCCCGCCGTGTAGGTGGTTCTACT TATCAGGTACCAGTTGAAGTCCGTCCGGTCCGTCGTAATGCTCTGGCAATGCGTTGGATCGTAGAAGCTGCTCGTAAACGCGG TGATAAATCCATGGCTCTGCGTCTGGCGAACGAACTTTCTGACGCTGCAGAAAACAAAGGTACTGCAGTTAAGAAACGTGAAGA CGTTCACCGTATGGCAGAAGCCAACAAGGCGTTCGCACACTACCGTTGGTAA (SEQ ID NO: 285)

Fig. 30C

30_GTC09492_Braenderup	ATGCCACGTCGTCGCGTCATTGGTCAGCGTAAAATTCTGCCGGATCCGAAGTTCGGATCAGAACTGCTGGCTAAATTTGTCAAT ATCCTGATGGTAGATGGTAAAAAATCTACTGCAGAATCAATCGTATACAGCGCGCTGGAGACCCTGGCTCAGCGTTCTGGTAAA TCTGAACTGGAAGCTTTCGAAGTCGCTCTCGAAAACGTTCGCCCGACTGTAGAAGTTAAATCCCGCCGTGTAGGTGGTTCTACT TATCAGGTACCAGTTGAAGTCCGTCCGGTCCGTCGTAATGCTCTGGCAATGCGTTGGATCGTAGAAGCTGCTCGTAAACGCGG TGATAAATCCATGGCTCTGCGCCTGGCGAACGAACTTTCTGACGCTGCAGAAAACAAAGGTACTGCAGTTAAGAAACGTGAAGA CGTTCACCGTATGGCAGAAGCCAACAAGGCGTTCGCACACTACCGTTGGTAA (SEQ ID NO: 292)
31_GTC09493_Pakistan	ATGCCACGTCGTCGCGTCATTGGTCAGCGTAAAATTCTGCCGGATCCGAAGTTCGGATCAGAACTGCTGGCTAAATTTGTCAAT ATCCTGATGGTAGATGGTAAAAAATCTACTGCAGAATCAATCGTATACAGCGCGCTGGAGACCCTGGCTCAGCGTTCTGGTAAA TCTGAACTGGAAGCTTTCGAAGTCGCTCTCGAAAACGTTCGCCCGACTGTAGAAGTTAAATCCCGCCGTGTAGGTGGTTCTACT TATCAGGTACCAGTTGAAGTCCGTCCGGTCCGTCGTAATGCTCTGGCAATGCGTTGGATCGTAGAAGCTGCTCGTAAACGCGG TGATAAATCCATGGCTCTGCGCCTGGCGAACGAACTTTCTGACGCTGCAGACAACAAAGGTACTGCAGTTAAGAAACGTGAAGA CGTTCACCGTATGGCAGAAGCCAACAAGGCGTTCGCACACTACCGTTGGTAA (SEQ ID NO: 299)
32_GTC09549_Typhimurium	ATGCCACGTCGTCGCGTCATTGGTCAGCGTAAAATTCTGCCGGATCCGAAGTTCGGATCAGAACTGCTGGCTAAATTTGTCAAT ATCCTGATGGTAGATGGTAAAAAATCTACTGCAGAATCAATCGTATACAGCGCGCTGGAGACCCTGGCTCAGCGTTCTGGTAAA TCTGAACTGGAAGCTTTCGAAGTCGCTCTCGAAAACGTTCGCCCGACTGTAGAAGTTAAATCCCGCCGTGTAGGTGGTTCTACT TATCAGGTACCAGTTGAAGTCCGTCCGGTCCGTCGTAATGCTCTGGCAATGCGTTGGATCGTAGAAGCTGCTCGTAAACGCGG TGATAAATCCATGGCTCTGCGCCTGGCGAACGAACTTTCTGATGCTGCAGACAACAAAGGTACTGCAGTTAAGAAACGTGAAGA CGTTCACCGTATGGCAGAAGCCAACAAGGCGTTCGCACACTACCGTTGGTAA (SEQ ID NO: 306)
33_ATCCBAA-1675_Infantis	ATGCCACGTCGTCGCGTCATTGGTCAGCGTAAAATTCTGCCGGATCCGAAGTTCGGATCAGAACTGCTGGCTAAATTTGTCAAT ATCCTGATGGTAGATGGTAAAAAATCTACTGCAGAATCAATCGTATACAGCGCGCTGGAGACCCTGGCTCAGCGTTCTGGTAAA TCTGAACTGGAAGCTTTCGAAGTCGCTCTCGAAAACGTTCGCCCGACTGTAGAAGTTAAATCCCGCCGTGTAGGTGGTTCTACT TATCAGGTACCAGTTGAAGTCCGTCCGGTCCGTCGTAATGCTCTGGCAATGCGTTGGATCGTAGAAGCTGCTCGTAAACGCGG TGATAAATCCATGGCTCTGCGCCTGGCGAACGAACTTTCTGACGCTGCAGACAACAAAGGTACTGCAGTTAAGAAACGTGAAGA CGTTCACCGTATGGCAGAAGCCAACAAGGCGTTCGCACACTACCGTTGGTAA (SEQ ID NO: 313)
34_ATCCBAA-1738_Thompson	ATGCCACGTCGTCGCGTCATTGGTCAGCGTAAAATTCTGCCGGATCCGAAGTTCGGATCAGAACTGCTGGCTAAATTTGTCAAT ATCCTGATGGTAGATGGTAAAAAATCTACTGCAGAATCAATCGTATACAGCGCGCTGGAGACCCTGGCTCAGCGTTCTGGTAAA TCTGAACTGGAAGCTTTCGAAGTCGCTCTCGAAAACGTTCGCCCGACTGTAGAAGTTAAATCCCGCCGTGTAGGTGGTTCTACT TATCAGGTACCAGTTGAAGTCCGTCCGGTCCGTCGTAATGCTCTGGCAATGCGTTGGATCGTAGAAGCTGCTCGTAAACGCGG TGATAAATCCATGGCTCTGCGCCTGGCGAACGAACTTTCTGACGCTGCAGAAAACAAAGGTACTGCAGTTAAGAAACGTGAAGA CGTTCACCGTATGGCAGAAGCCAACAAGGCGTTCGCACACTACCGTTGGTAA (SEQ ID NO: 324)
35_ATCC9712_Saintpaul	ATGCCACGTCGTCGCGTCATTGGTCAGCGTAAAATTCTGCCGGATCCGAAGTTCGGATCAGAACTGCTGGCTAAATTTGTCAAT ATCCTGATGGTAGATGGTAAAAAATCTACTGCAGAATCAATCGTATACAGCGCGCTGGAGACCCTGGCTCAGCGTTCTGGTAAA TCTGAACTGGAAGCTTTCGAAGTCGCTCTCGAAAACGTTCGCCCGACTGTAGAAGTTAAATCCCGCCGTGTAGGTGGTTCTACT TATCAGGTACCAGTTGAAGTCCGTCCGGTCCGTCGTAATGCTCTGGCAATGCGTTGGATCGTAGAAGCTGCTCGTAAACGCGG TGATAAATCCATGGCTCTGCGCCTGGCGAACGAACTTTCTGATGCTGCAGACAACAAAGGTACTGCAGTTAAGAAACGTGAAGA CGTTCACCGTATGGCAGAAGCCAACAAGGCGTTCGCACACTACCGTTGGTAA (SEQ ID NO: 334)
36_jfrlSe1402-1_Infantis	ATGCCACGTCGTCGCGTCATTGGTCAGCGTAAAATTCTGCCGGATCCGAAGTTCGGATCAGAACTGCTGGCTAAATTTGTCAAT ATCCTGATGGTAGATGGTAAAAAATCTACTGCAGAATCAATCGTATACAGCGCGCTGGAGACCCTGGCTCAGCGTTCTGGTAAA TCTGAACTGGAAGCTTTCGAAGTCGCTCTCGAAAACGTTCGCCCGACTGTAGAAGTTAAATCCCGCCGTGTAGGTGGTTCTACT TATCAGGTACCAGTTGAAGTCCGTCCGGTCCGTCGTAATGCTCTGGCAATGCGTTGGATCGTAGAAGCTGCTCGTAAACGCGG TGATAAATCCATGGCTCTGCGCCTGGCGAACGAACTTTCTGACGCTGCAGACAACAAAGGTACTGCAGTTAAGAAACGTGAAGA CGTTCACCGTATGGCAGAAGCCAACAAGGCGTTCGCACACTACCGTTGGTAA (SEQ ID NO: 345)
38_jfrlSe1402-3_Brandenburg	ATGCCACGTCGTCGCGTCATTGGTCAGCGTAAAATTCTGCCGGATCCGAAGTTCGGATCAGAACTGCTGGCTAAATTTGTCAAT ATCCTGATGGTAGATGGTAAAAAATCTACTGCAGAATCAATCGTATACAGCGCGCTGGAGACCCTGGCTCAGCGTTCTGGTAAA TCTGAACTGGAAGCTTTCGAAGTCGCTCTCGAAAACGTTCGCCCGACTGTAGAAGTTAAATCCCGCCGTGTAGGTGGTTCTACT TATCAGGTACCAGTTGAAGTCCGTCCGGTCCGTCGTAATGCTCTGGCAATGCGTTGGATCGTAGAAGCTGCTCGTAAACGCGG TGATAAATCCATGGCTCTGCGCCTGGCGAACGAACTTTCTGACGCTGCAGACAACAAAGGTACTGCAGTTAAGAAACGTGAAGA CGTTCACCGTATGGCAGAAGCCAACAAGGCGTTCGCACACTACCGTTGGTAA (SEQ ID NO: 355)
39_jfrlSe1402-4_Infantis	ATGCCACGTCGTCGCGTCATTGGTCAGCGTAAAATTCTGCCGGATCCGAAGTTCGGATCAGAACTGCTGGCTAAATTTGTCAAT ATCCTGATGGTAGATGGTAAAAAATCTACTGCAGAATCAATCGTATACAGCGCGCTGGAGACCCTGGCTCAGCGTTCTGGTAAA TCTGAACTGGAAGCTTTCGAAGTCGCTCTCGAAAACGTTCGCCCGACTGTAGAAGTTAAATCCCGCCGTGTAGGTGGTTCTACT TATCAGGTACCAGTTGAAGTCCGTCCGGTCCGTCGTAATGCTCTGGCAATGCGTTGGATCGTAGAAGCTGCTCGTAAACGCGG TGATAAATCCATGGCTCTGCGCCTGGCGAACGAACTTTCTGACGCTGCAGACAACAAAGGTACTGCAGTTAAGAAACGTGAAGA CGTTCACCGTATGGCAGAAGCCAACAAGGCGTTCGCACACTACCGTTGGTAA (SEQ ID NO: 362)
40_jfrlSe1402-5_Brandenburg	ATGCCACGTCGTCGCGTCATTGGTCAGCGTAAAATTCTGCCGGATCCGAAGTTCGGATCAGAACTGCTGGCTAAATTTGTCAAT ATCCTGATGGTAGATGGTAAAAAATCTACTGCAGAATCAATCGTATACAGCGCGCTGGAGACCCTGGCTCAGCGTTCTGGTAAA TCTGAACTGGAAGCTTTCGAAGTCGCTCTCGAAAACGTTCGCCCGACTGTAGAAGTTAAATCCCGCCGTGTAGGTGGTTCTACT TATCAGGTACCAGTTGAAGTCCGTCCGGTCCGTCGTAATGCTCTGGCAATGCGTTGGATCGTAGAAGCTGCTCGTAAACGCGG TGATAAATCCATGGCTCTGCGCCTGGCGAACGAACTTTCTGACGCTGCAGACAACAAAGGTACTGCAGTTAAGAAACGTGAAGA CGTTCACCGTATGGCAGAAGCCAACAAGGCGTTCGCACACTACCGTTGGTAA (SEQ ID NO: 372)
41_jfrlSe1402-6_Rissen	ATGCCACGTCGTCGCGTCATTGGTCAGCGTAAAATTCTGCCGGATCCGAAGTTCGGATCAGAACTGCTGGCTAAATTTGTCAAT ATCCTGATGGTAGATGGTAAAAAATCTACTGCAGAATCAATCGTATACAGCGCGCTGGAGACCCTGGCTCAGCGTTCTGGTAAA TCTGAACTGGAAGCTTTCGAAGTCGCTCTCGAAAACGTTCGCCCGACTGTAGAAGTTAAATCCCGCCGTGTAGGTGGTTCTACT TATCAGGTACCAGTTGAAGTCCGTCCGGTCCGTCGTAATGCTCTGGCAATGCGTTGGATCGTAGAAGCTGCTCGTAAACGCGG TGATAAATCCATGGCTCTGCGCCTGGCGAACGAACTTTCTGACGCTGCAGAAAACAAAGGTACTGCAGTTAAGAAACGTGAAGA CGTTCACCGTATGGCAGAAGCCAACAAGGCGTTCGCACACTACCGTTGGTAA (SEQ ID NO: 379)
42_jfrlSe1402-7_Orion	ATGCCACGTCGTCGCGTCATTGGTCAGCGTAAAATTCTGCCGGATCCGAAGTTCGGATCAGAACTGCTGGCTAAATTTGTCAAT ATCCTGATGGTAGATGGTAAAAAATCTACTGCAGAATCAATCGTATACAGCGCGCTGGAGACCCTGGCTCAGCGTTCTGGTAAA TCTGAACTGGAAGCTTTCGAAGTCGCTCTCGAAAACGTTCGCCCGACTGTAGAAGTTAAATCCCGCCGTGTAGGTGGTTCTACT TATCAGGTACCAGTTGAAGTCCGTCCGGTCCGTCGTAATGCTCTGGCAATGCGTTGGATCGTAGAAGCTGCTCGTAAACGCGG TGATAAATCCATGGCTCTGCGCCTGGCGAACGAACTTTCTGATGCTGCAGACAACAAAGGTACTGCAGTTAAGAAACGTGAAGA CGTTCACCGTATGGCAGAAGCCAACAAGGCGTTCGCACACTACCGTTGGTAA (SEQ ID NO: 386)
43_jfrlSe1402-8_Rissen	ATGCCACGTCGTCGCGTCATTGGTCAGCGTAAAATTCTGCCGGATCCGAAGTTCGGATCAGAACTGCTGGCTAAATTTGTCAAT ATCCTGATGGTAGATGGTAAAAAATCTACTGCAGAATCAATCGTATACAGCGCGCTGGAGACCCTGGCTCAGCGTTCTGGTAAA TCTGAACTGGAAGCTTTCGAAGTCGCTCTCGAAAACGTTCGCCCGACTGTAGAAGTTAAATCCCGCCGTGTAGGTGGTTCTACT TATCAGGTACCAGTTGAAGTCCGTCCGGTCCGTCGTAATGCTCTGGCAATGCGTTGGATCGTAGAAGCTGCTCGTAAACGCGG TGATAAATCCATGGCTCTGCGCCTGGCGAACGAACTTTCTGACGCTGCAGAAAACAAAGGTACTGCAGTTAAGAAACGTGAAGA CGTTCACCGTATGGCAGAAGCCAACAAGGCGTTCGCACACTACCGTTGGTAA (SEQ ID NO: 396)
44_jfrlSe1402-9_Rissen	ATGCCACGTCGTCGCGTCATTGGTCAGCGTAAAATTCTGCCGGATCCGAAGTTCGGATCAGAACTGCTGGCTAAATTTGTCAAT ATCCTGATGGTAGATGGTAAAAAATCTACTGCAGAATCAATCGTATACAGCGCGCTGGAGACCCTGGCTCAGCGTTCTGGTAAA TCTGAACTGGAAGCTTTCGAAGTCGCTCTCGAAAACGTTCGCCCGACTGTAGAAGTTAAATCCCGCCGTGTAGGTGGTTCTACT TATCAGGTACCAGTTGAAGTCCGTCCGGTCCGTCGTAATGCTCTGGCAATGCGTTGGATCGTAGAAGCTGCTCGTAAACGCGG TGATAAATCCATGGCTCTGCGCCTGGCGAACGAACTTTCTGACGCTGCAGAAAACAAAGGTACTGCAGTTAAGAAACGTGAAGA CGTTCACCGTATGGCAGAAGCCAACAAGGCGTTCGCACACTACCGTTGGTAA (SEQ ID NO: 403)

Fig. 30D

45_jfrlSe1402-10_Rissen	ATGCCACGTCGTCGCGTCATTGGTCAGCGTAAAATTCTGCCGGATCCGAAGTTCGGATCAGAACTGCTGGCTAAATTTGTCAAT ATCCTGATGGTAGATGGTAAAAAATCTACTGCAGAATCAATCGTATACAGCGCGCTGGAGACCCTGGCTCAGCGTTCTGGTAAA TCTGAACTGGAAGCTTTCGAAGTCGCTCTCGAAAACGTTCGCCCGACTGTAGAAGTTAAATCCCGCCGTGTAGGTGGTTCTACT TATCAGGTACCAGTTGAAGTCCGTCCGGTCCGTCGTAATGCTCTGGCAATGCGTTGGATCGTAGAAGCTGCTCGTAAACGCGG TGATAAATCCATGGCTCTGCGCCTGGCGAACGAACTTTCTGACGCTGCAGAAAACAAAGGTACTGCAGTTAAGAAACGTGAAGA CGTTCACCGTATGGCAGAAGCCAACAAGGCGTTCGCACACTACCGTTGGTAA (SEQ ID NO: 410)
46_jfrlSe1402-11_Rissen	ATGCCACGTCGTCGCGTCATTGGTCAGCGTAAAATTCTGCCGGATCCGAAGTTCGGATCAGAACTGCTGGCTAAATTTGTCAAT ATCCTGATGGTAGATGGTAAAAAATCTACTGCAGAATCAATCGTATACAGCGCGCTGGAGACCCTGGCTCAGCGTTCTGGTAAA TCTGAACTGGAAGCTTTCGAAGTCGCTCTCGAAAACGTTCGCCCGACTGTAGAAGTTAAATCCCGCCGTGTAGGTGGTTCTACT TATCAGGTACCAGTTGAAGTCCGTCCGGTCCGTCGTAATGCTCTGGCAATGCGTTGGATCGTAGAAGCTGCTCGTAAACGCGG TGATAAATCCATGGCTCTGCGCCTGGCGAACGAACTTTCTGACGCTGCAGAAAACAAAGGTACTGCAGTTAAGAAACGTGAAGA CGTTCACCGTATGGCAGAAGCCAACAAGGCGTTCGCACACTACCGTTGGTAA (SEQ ID NO: 417)
47_jfrlSe1402-12_Rissen	ATGCCACGTCGTCGCGTCATTGGTCAGCGTAAAATTCTGCCGGATCCGAAGTTCGGATCAGAACTGCTGGCTAAATTTGTCAAT ATCCTGATGGTAGATGGTAAAAAATCTACTGCAGAATCAATCGTATACAGCGCGCTGGAGACCCTGGCTCAGCGTTCTGGTAAA TCTGAACTGGAAGCTTTCGAAGTCGCTCTCGAAAACGTTCGCCCGACTGTAGAAGTTAAATCCCGCCGTGTAGGTGGTTCTACT TATCAGGTACCAGTTGAAGTCCGTCCGGTCCGTCGTAATGCTCTGGCAATGCGTTGGATCGTAGAAGCTGCTCGTAAACGCGG TGATAAATCCATGGCTCTGCGCCTGGCGAACGAACTTTCTGACGCTGCAGAAAACAAAGGTACTGCAGTTAAGAAACGTGAAGA CGTTCACCGTATGGCAGAAGCCAACAAGGCGTTCGCACACTACCGTTGGTAA (SEQ ID NO: 424)
48_jfrlSe1402-13_Mbandaka	ATGCCACGTCGTCGCGTCATTGGTCAGCGTAAAATTCTGCCGGATCCGAAGTTCGGATCAGAACTGCTGGCTAAATTTGTCAAT ATCCTGATGGTAGATGGTAAAAAATCTACTGCAGAATCAATCGTATACAGCGCGCTGGAGACCCTGGCTCAGCGTTCTGGTAAA TCTGAACTGGAAGCTTTCGAAGTCGCTCTCGAAAACGTTCGCCCGACTGTAGAAGTTAAATCCCGCCGTGTAGGTGGTTCTACT TATCAGGTACCAGTTGAAGTCCGTCCGGTCCGTCGTAATGCTCTGGCAATGCGTTGGATCGTAGAAGCTGCTCGTAAACGCGG TGATAAATCCATGGCTCTGCGCCTGGCGAACGAACTTTCTGACGCTGCAGAAAACAAAGGTACTGCAGTTAAGAAACGTGAAGA CGTTCACCGTATGGCAGAAGCCAACAAGGCGTTCGCACACTACCGTTGGTAA (SEQ ID NO: 431)
49_jfrlSe1402-14_Mbandaka	ATGCCACGTCGTCGCGTCATTGGTCAGCGTAAAATTCTGCCGGATCCGAAGTTCGGATCAGAACTGCTGGCTAAATTTGTCAAT ATCCTGATGGTAGATGGTAAAAAATCTACTGCAGAATCAATCGTATACAGCGCGCTGGAGACCCTGGCTCAGCGTTCTGGTAAA TCTGAACTGGAAGCTTTCGAAGTCGCTCTCGAAAACGTTCGCCCGACTGTAGAAGTTAAATCCCGCCGTGTAGGTGGTTCTACT TATCAGGTACCAGTTGAAGTCCGTCCGGTCCGTCGTAATGCTCTGGCAATGCGTTGGATCGTAGAAGCTGCTCGTAAACGCGG TGATAAATCCATGGCTCTGCGCCTGGCGAACGAACTTTCTGACGCTGCAGAAAACAAAGGTACTGCAGTTAAGAAACGTGAAGA CGTTCACCGTATGGCAGAAGCCAACAAGGCGTTCGCACACTACCGTTGGTAA (SEQ ID NO: 441)
50_jfrlSe1402-15_Orion	ATGCCACGTCGTCGCGTCATTGGTCAGCGTAAAATTCTGCCGGATCCGAAGTTCGGATCAGAACTGCTGGCTAAATTTGTCAAT ATCCTGATGGTAGATGGTAAAAAATCTACTGCAGAATCAATCGTATACAGCGCGCTGGAGACCCTGGCTCAGCGTTCTGGTAAA TCTGAACTGGAAGCTTTCGAAGTCGCTCTCGAAAACGTTCGCCCGACTGTAGAAGTTAAATCCCGCCGTGTAGGTGGTTCTACT TATCAGGTACCAGTTGAAGTCCGTCCGGTCCGTCGTAATGCTCTGGCAATGCGTTGGATCGTAGAAGCTGCTCGTAAACGCGG TGATAAATCCATGGCTCTGCGCCTGGCGAACGAACTTTCTGATGCTGCAGACAACAAAGGTACTGCAGTTAAGAAACGTGAAGA CGTTCACCGTATGGCAGAAGCCAACAAGGCGTTCGCACACTACCGTTGGTAA (SEQ ID NO: 451)
51_jfrlSe1409-1_Altona	ATGCCACGTCGTCGCGTCATTGGTCAGCGTAAAATTCTGCCGGATCCGAAGTTCGGATCAGAACTGCTGGCTAAATTTGTCAAT ATCCTGATGGTAGATGGTAAAAAATCTACTGCAGAATCAATCGTATACAGCGCGCTGGAGACCCTGGCTCAGCGTTCTGGTAAA TCTGAACTGGAAGCTTTCGAAGTCGCTCTCGAAAACGTTCGCCCGACTGTAGAAGTTAAATCCCGCCGTGTAGGTGGTTCTACT TATCAGGTACCAGTTGAAGTCCGTCCGGTCCGTCGTAATGCTCTGGCAATGCGTTGGATCGTAGAAGCTGCTCGTAAACGCGG TGATAAATCCATGGCTCTGCGCCTGGCGAACGAACTTTCTGACGCTGCAGAAAACAAAGGTACTGCAGTTAAGAAACGTGAAGA CGTTCACCGTATGGCAGAAGCCAACAAGGCGTTCGCACACTACCGTTGGTAA (SEQ ID NO: 458)
52_jfrlSe1409-2_Istanbul	ATGCCACGTCGTCGCGTCATTGGTCAGCGTAAAATTCTGCCGGATCCGAAGTTCGGATCAGAACTGCTGGCTAAATTTGTCAAT ATCCTGATGGTAGATGGTAAAAAATCTACTGCAGAATCAATCGTATACAGCGCGCTGGAGACCCTGGCTCAGCGTTCTGGTAAA TCTGAACTGGAAGCTTTCGAAGTCGCTCTCGAAAACGTTCGCCCGACTGTAGAAGTTAAATCCCGCCGTGTAGGTGGTTCTACT TATCAGGTACCAGTTGAAGTCCGTCCGGTCCGTCGTAATGCTCTGGCAATGCGTTGGATCGTAGAAGCTGCTCGTAAACGCGG TGATAAATCCATGGCTCTGCGCCTGGCGAACGAACTTTCTGACGCTGCAGAAAACAAAGGTACTGCAGTTAAGAAACGTGAAGA CGTTCACCGTATGGCAGAAGCCAACAAGGCGTTCGCACACTACCGTTGGTAA (SEQ ID NO: 467)
53_jfrlSe1409-3_Senftenberg	
54_jfrlSe1409-4_UN_O13_	ATGCCACGTCGTCGCGTCATTGGTCAGCGTAAAATTCTGCCGGATCCGAAGTTCGGATCAGAACTGCTGGCTAAATTTGTCAAT ATCCTGATGGTAGATGGTAAAAAATCTACTGCAGAATCAATCGTATACAGCGCGCTGGAGACCCTGGCTCAGCGTTCTGGTAAA TCTGAACTGGAAGCTTTCGAAGTCGCTCTCGAAAACGTTCGCCCGACTGTAGAAGTTAAATCCCGCCGTGTAGGTGGTTCTACT TATCAGGTACCAGTTGAAGTCCGTCCGGTCCGTCGTAATGCTCTGGCAATGCGTTGGATCGTAGAAGCTGCTCGTAAACGCGG TGATAAATCCATGGCTCTGCGCCTGGCGAACGAACTTTCTGACGCTGCAGAAAACAAAGGTACTGCAGTTAAGAAACGTGAAGA CGTTCACCGTATGGCAGAAGCCAACAAGGCGTTCGCACACTACCGTTGGTAA (SEQ ID NO: 479)
55_jfrlSe1409-5_UN_O1,3,19_	ATGCCACGTCGTCGCGTCATTGGTCAGCGTAAAATTCTGCCGGATCCGAAGTTCGGATCAGAACTGCTGGCTAAATTTGTCAAT ATCCTGATGGTAGATGGTAAAAAATCTACTGCAGAATCAATCGTATACAGCGCGCTGGAGACCCTGGCTCAGCGTTCTGGTAAA TCTGAACTGGAAGCTTTCGAAGTCGCTCTCGAAAACGTTCGCCCGACTGTAGAAGTTAAATCCCGCCGTGTAGGTGGTTCTACT TATCAGGTACCAGTTGAAGTCCGTCCGGTCCGTCGTAATGCTCTGGCAATGCGTTGGATCGTAGAAGCTGCTCGTAAACGCGG TGATAAATCCATGGCTCTGCGCCTGGCGAACGAACTTTCTGACGCTGCAGAAAACAAAGGTACTGCAGTTAAGAAACGTGAAGA CGTTCACCGTATGGCAGAAGCCAACAAGGCGTTCGCACACTACCGTTGGTAA (SEQ ID NO: 488)
56_jfrlSe1409-6_Montevideo	
57_jfrlSe1409-7_UN_O1,3,19_	ATGCCACGTCGTCGCGTCATTGGTCAGCGTAAAATTCTGCCGGATCCGAAGTTCGGATCAGAACTGCTGGCTAAATTTGTCAAT ATCCTGATGGTAGATGGTAAAAAATCTACTGCAGAATCAATCGTATACAGCGCGCTGGAGACCCTGGCTCAGCGTTCTGGTAAA TCTGAACTGGAAGCTTTCGAAGTCGCTCTCGAAAACGTTCGCCCGACTGTAGAAGTTAAATCCCGCCGTGTAGGTGGTTCTACT TATCAGGTACCAGTTGAAGTCCGTCCGGTCCGTCGTAATGCTCTGGCAATGCGTTGGATCGTAGAAGCTGCTCGTAAACGCGG TGATAAATCCATGGCTCTGCGCCTGGCGAACGAACTTTCTGACGCTGCAGAAAACAAAGGTACTGCAGTTAAGAAACGTGAAGA CGTTCACCGTATGGCAGAAGCCAACAAGGCGTTCGCACACTACCGTTGGTAA (SEQ ID NO: 504)
58_jfrlSe1409-8_UN_O18_	ATGCCACGTCGTCGCGTCATTGGTCAGCGTAAAATTCTGCCGGATCCGAAGTTCGGATCAGAACTGCTGGCTAAATTTGTCAAT ATCCTGATGGTAGATGGTAAAAAATCTACTGCAGAATCAATCGTATACAGCGCGCTGGAGACCCTGGCTCAGCGTTCTGGTAAA TCTGAACTGGAAGCTTTCGAAGTCGCTCTCGAAAACGTTCGCCCGACTGTAGAAGTTAAATCCCGCCGTGTAGGTGGTTCTACT TATCAGGTACCAGTTGAAGTCCGTCCGGTCCGTCGTAATGCTCTGGCAATGCGTTGGATCGTAGAAGCTGCTCGTAAACGCGG TGATAAATCCATGGCTCTGCGCCTGGCGAACGAACTTTCTGACGCTGCAGAAAACAAAGGTACTGCAGTTAAGAAACGTGAAGA CGTTCACCGTATGGCAGAAGCCAACAAGGCGTTCGCACACTACCGTTGGTAA (SEQ ID NO: 513)
61_jfrlSe1409-11_UN_O1,3,19_	ATGCCACGTCGTCGCGTCATTGGTCAGCGTAAAATTCTGCCGGATCCGAAGTTCGGATCAGAACTGCTGGCTAAATTTGTCAAT ATCCTGATGGTAGATGGTAAAAAATCTACTGCAGAATCAATCGTATACAGCGCGCTGGAGACCCTGGCTCAGCGTTCTGGTAAA TCTGAACTGGAAGCTTTCGAAGTCGCTCTCGAAAACGTTCGCCCGACTGTAGAAGTTAAATCCCGCCGTGTAGGTGGTTCTACT TATCAGGTACCAGTTGAAGTCCGTCCGGTCCGTCGTAATGCTCTGGCAATGCGTTGGATCGTAGAAGCTGCTCGTAAACGCGG TGATAAATCCATGGCTCTGCGCCTGGCGAACGAACTTTCTGACGCTGCAGAAAACAAAGGTACTGCAGTTAAGAAACGTGAAGA CGTTCACCGTATGGCAGAAGCCAACAAGGCGTTCGCACACTACCGTTGGTAA (SEQ ID NO: 522)
67_jfrlSe1409-17_Rissen	ATGCCACGTCGTCGCGTCATTGGTCAGCGTAAAATTCTGCCGGATCCGAAGTTCGGATCAGAACTGCTGGCTAAATTTGTCAAT ATCCTGATGGTAGATGGTAAAAAATCTACTGCAGAATCAATCGTATACAGCGCGCTGGAGACCCTGGCTCAGCGTTCTGGTAAA TCTGAACTGGAAGCTTTCGAAGTCGCTCTCGAAAACGTTCGCCCGACTGTAGAAGTTAAATCCCGCCGTGTAGGTGGTTCTACT TATCAGGTACCAGTTGAAGTCCGTCCGGTCCGTCGTAATGCTCTGGCAATGCGTTGGATCGTAGAAGCTGCTCGTAAACGCGG TGATAAATCCATGGCTCTGCGCCTGGCGAACGAACTTTCTGACGCTGCAGAAAACAAAGGTACTGCAGTTAAGAAACGTGAAGA CGTTCACCGTATGGCAGAAGCCAACAAGGCGTTCGCACACTACCGTTGGTAA (SEQ ID NO: 530)

Fig. 30E

71_jfrlSe1409-21_Amsterdam	ATGGCAACAGTTAACCAGCTGGTACGCAAACCACGTGCTCGCAAAGTTGCGAAAAGCAACGTGCCTGCGCTGGAAGCATGCCC GCAAAAACGTGGCGTATGTACTCGTGTATATACTACCACTCCTAAAAAACCGAACTCCGCACTGCGTAAAGTTTGCCGTGTTCG TCTGACTAACGGTTTTGAAGTGACTTCCTACATCGGTGGTGAAGGTCACAACCTGCAGGAGCACTCCGTGATCCTGATCCGTG GCGGTCGTGTTAAAGACCTCCCGGGTGTTCGTTACCACACCGTTCGTGGCGCGCTTGACTGCTCCGGCGTTAAAGACCGTAAG CAAGCTCGTTCTAAGTACGGCGTGAAGCGTCCTAAGGCTTAA (SEQ ID NO: 540)
80_HySe09_Enteritidis	ATGCCACGTCGTCGCGTCATTGGTCAGCGTAAAATTCTGCCGGATCCGAAGTTCGGATCAGAACTGCTGGCTAAATTTGTCAAT ATCCTGATGGTAGATGGTAAAAAATCTACTGCAGAATCAATCGTATACAGCGCGCTGGAGACCCTGGCTCAGCGTTCTGGTAAA TCTGAACTGGAAGCTTTCGAAGTCGCTCTCGAAAACGTTCGCCCGACTGTAGAAGTTAAATCCCGCCGTGTAGGTGGTTCTACT TATCAGGTACCAGTTGAAGTCCGTCCGGTCCGTCGTAATGCTCTGGCAATGCGTTGGATCGTAGAAGCTGCTCGTAAACGCGG TGATAAATCCATGGCTCTGCGCCTGGCGAACGAACTTTCTGACGCTGCAGAAAACAAAGGTACTGCAGTTAAGAAACGTGAAGA CGTTCACCGTATGGCAGAAGCCAACAAGGCGTTCGCACACTACCGTTGGTAA (SEQ ID NO: 550)
100_HySe29_Schwarzengrund	ATGCCACGTCGTCGCGTCATTGGTCAGCGTAAAATTCTGCCGGATCCGAAGTTCGGATCAGAACTGCTGGCTAAATTTGTCAAT ATCCTGATGGTAGATGGTAAAAAATCTACTGCAGAATCAATCGTATACAGCGCGCTGGAGACCCTGGCTCAGCGTTCTGGTAAA TCTGAACTGGAAGCTTTCGAAGTCGCTCTCGAAAACGTTCGCCCGACTGTAGAAGTTAAATCCCGCCGTGTAGGTGGTTCTACT TATCAGGTACCAGTTGAAGTCCGTCCGGTCCGTCGTAATGCTCTGGCAATGCGTTGGATCGTAGAAGCTGCTCGTAAACGCGG TGATAAATCCATGGCTCTGCGCCTGGCGAACGAACTTTCTGACGCTGCAGAAAACAAAGGTACTGCAGTTAAGAAACGTGAAGA CGTTCACCGTATGGCAGAAGCCAACAAGGCGTTCGCACACTACCGTTGGTAA (SEQ ID NO: 559)
103_HySe32_Schwarzengrund	ATGCCACGTCGTCGCGTCATTGGTCAGCGTAAAATTCTGCCGGATCCGAAGTTCGGATCAGAACTGCTGGCTAAATTTGTCAAT ATCCTGATGGTAGATGGTAAAAAATCTACTGCAGAATCAATCGTATACAGCGCGCTGGAGACCCTGGCTCAGCGTTCTGGTAAA TCTGAACTGGAAGCTTTCGAAGTCGCTCTCGAAAACGTTCGCCCGACTGTAGAAGTTAAATCCCGCCGTGTAGGTGGTTCTACT TATCAGGTACCAGTTGAAGTCCGTCCGGTCCGTCGTAATGCTCTGGCAATGCGTTGGATCGTAGAAGCTGCTCGTAAACGCGG TGATAAATCCATGGCTCTGCGCCTGGCGAACGAACTTTCTGACGCTGCAGAAAACAAAGGTACTGCAGTTAAGAAACGTGAAGA CGTTCACCGTATGGCAGAAGCCAACAAGGCGTTCGCACACTACCGTTGGTAA (SEQ ID NO: 566)
106_HySe35_Schwarzengrund	ATGCCACGTCGTCGCGTCATTGGTCAGCGTAAAATTCTGCCGGATCCGAAGTTCGGATCAGAACTGCTGGCTAAATTTGTCAAT ATCCTGATGGTAGATGGTAAAAAATCTACTGCAGAATCAATCGTATACAGCGCGCTGGAGACCCTGGCTCAGCGTTCTGGTAAA TCTGAACTGGAAGCTTTCGAAGTCGCTCTCGAAAACGTTCGCCCGACTGTAGAAGTTAAATCCCGCCGTGTAGGTGGTTCTACT TATCAGGTACCAGTTGAAGTCCGTCCGGTCCGTCGTAATGCTCTGGCAATGCGTTGGATCGTAGAAGCTGCTCGTAAACGCGG TGATAAATCCATGGCTCTGCGCCTGGCGAACGAACTTTCTGACGCTGCAGAAAACAAAGGTACTGCAGTTAAGAAACGTGAAGA CGTTCACCGTATGGCAGAAGCCAACAAGGCGTTCGCACACTACCGTTGGTAA (SEQ ID NO: 573)

Fig. 31A

	gns
01_NBRC13245T_Typhimurium	ATGAACAGCGAAGAGTTGACACATAAAGCAGAAGAGGAAATCGCGGCACTCATTAGCAAAAAGGTTGCCGAACTACGGAAGAAA ACCGGGCAAGAGGTTTCCGAAATAGAGTTCGCGCCGCGAGAAACGATGAAAGGGCTTGAGGGATACCACGTTAAAATTAAACTA CTCTAG (SEQ ID NO: 59)
02_GTC00131_Enteritidis	
03_GTC09491_Enteritidis	
04_GTC03838_Enteritidis	
05_GTC08914_Enteritidis	ATGAACAGCGAAGAGTTGACACATAAAGCAGAAGAGGAAATCGCGGCACTCATTAGCAAAAAGGTCGCCGAACTACGGAAGAAA ACCGGGCAAGAGGTTTCCGAAATAGAGTTCGCGCCGCGAGAAACGATGAAAGGGCTTGAGGGATACCACGTTAAAATTAAACTA CTCTAG (SEQ ID NO: 96)
06_GTC09421_Enteritidis	ATGAACAGCGAAGAGTTGACACATAAAGCAGAAGAGGAAATCGCGGCACTCATTAGCAAAAAGGTCGCCGAACTACGGAAGAAA ACCGGGCAAGAGGTTTCCGAAATAGAGTTCGCGCCGCGAGAAACGATGAAAGGGCTTGAGGGATACCACGTTAAAATTAAACTA CTCTAG (SEQ ID NO: 106)
07_GTC09489_Enteritidis	
08_NBRC3163_Pullorum_Gallinarum	ATGAACAGCGAAGAGTTGACACATAAAGCAGAAGAGGAAATCGCGGCACTCATTAGCAAAAAGGTCGCAGAACTACGGAAGAAA ACCGGGCAAGAGGTTTCCGAAATAGAGTTCGCGCCGCGAGAAACGATGAAAGGGCTTGAGGGATACCACGTTAAAATTAAACTA CTCTAG (SEQ ID NO: 123)
09_NBRC3313_Enteritidis	
10_NBRC12529_Typhimurium	
11_NBRC14193_Typhimurium	
12_NBRC14194_Typhimurium	
13_NBRC14209_Typhimurium	
14_NBRC14210_Typhimurium	
15_NBRC14211_Typhimurium	
16_NBRC14212_Typhimurium	
17_NBRC15181_Typhimurium	ATGAACAGCGAAGAGTTGACACATAAAGCAGAAGAGGAAATCGCGGCACTCATTAGCAAAAAGGTCGCCGAACTACGGAAGAAA ACCGGGCAAGAGGTTTCCGAAATAGAGTTCGTGCCGCGAGAAACGATGAAAGGGCTTGAGGGATACCACGTTAAAATTAAACTA CTCTAG (SEQ ID NO: 188)
18_NBRC15182_Minnesota	ATGAACAGCGAAGAGTTGACACATAAAGCAGAAGAGGAAATCGCGGCACTCATTAGCAAAAAGGTTGCCGAACTACGGAAGAAA ACCGGGCAAGAGGTTTCCGAAATAGAGTTCGCGCCGCGAGAAACGATGAAAGGGCTTGAGGGATACCACGTTAAAATTAAACTA CTCTAG (SEQ ID NO: 198)
19_NBRC15183_Minnesota	
20_NBRC15184_Minnesota	
21_NBRC15185_Minnesota	
22_NBRC15186_Minnesota	
23_NBRC15187_Minnesota	
24_NBRC100797_Abony	ATGAACAGCGAAGAGTTGACACATAAAGCAGAAGAGGAAATCGCGGCACTCATTAGCAAAAAGGTTGCCGAACTACGGAAGAAA ACCGGGCAAGAGGTTTCCGAAATAGAGTTCGCGCCGCGAGAAACGATGAAAGGGCTTGAGGGATACCACGTTAAAATTAAACTA CTCTAG (SEQ ID NO: 243)
25_NBRC105684_Choleraesuis	ATGAACAGCGAAGAGTTGACACATAAAGCAGAAGAGGAAATCGCGGCACTCATTAGCAAAAAGGTTGCCGAACTACGGAAGAAA ACCGGGCAAGAGGTTTCCGAAATAGAGTTCGCGCCGCGAGAAACGATGAAAGGGCTTGAGGGATACCACGTTAAAATTAAACTA CTCTAG (SEQ ID NO: 253)
26_NBRC105726_Typhimurium	ATGAACAGCGAAGAGTTGACACATAAAGCAGAAGAGGAAATCGCGGCACTCATTAGCAAAAAGGTTGCCGAACTACGGAAGAAA ACCGGGCAAGAGGTTTCCGAAATAGAGTTCGCGCCGCGAGAAACGATGAAAGGGCTTGAGGGATACCACGTTAAAATTAAACTA CTCTAG (SEQ ID NO: 262)
27_JCM3919_UN_O7_	
28_NBRC15335_Minnesota	
29_GTC09490_Enteritidis	
30_GTC09492_Braenderup	
31_GTC09493_Pakistan	
32_GTC09549_Typhimurium	
33_ATCCBAA-1675_Infantis	ATGAACAGCGAAGAGTTGACACATAAAGCAGAAGAGGAAATCGCGGCACTCATTAGCAAAAAGGTTGCCGAACTACGGAAGAAA ACCGGGCAAGAGGTTTCCGAAATAGAGTTCGCGCCGCGAGAAACGATGAAAGGGCTTGAGGGATACCACGTTAAAATTAAACTA CTCTAG (SEQ ID NO: 314)

Fig. 31B

34_ATCCBAA-1738_Thompson	ATGAACAGCGAAGAGTTGACACATAAAGCAGAAGAGGAAATCGCGGCACTCATTAGCAAAAAGGTTGCCGAACTACGGAAGAAA ACCGGGCAAGAGGTTTCCGAAATAGAGTTCGCGCCGCGAGAAACGATGAAAGGGCTTGAGGGATACCACGTTAAAATTAAACTA CTCTAG (SEQ ID NO: 325)
35_ATCC9712_Saintpaul	ATGAACAGCGAAGAGTTGACACATAAAGCAGAAGAGGAAATCGCGGCACTCATTAGCAAAAAGGTTGCCGAACTACGGAAGAAA ACCGGGCAAGAGGTTTCCGAAATAGAGTTCGCGCCGCGAGAAACGATGAAAGGGCTTGAGGGATACCACGTTAAAATTAAACTA CTCTAG (SEQ ID NO: 335)
36_jfrlSe1402-1_Infantis	ATGAACAGCGAAGAGTTGACACATAAAGCAGAAGAGGAAATCGCGGCACTCATTAGCAAAAAGGTTGCCGAACTACGGAAGAAA ACCGGGCAAGAGGTTTCCGAAATAGAGTTCGTGCCGCGAGAAACGATGAAAGGGCTTGAGGGATACCACGTTAAAATTAAACTA CTCTAG (SEQ ID NO: 346)
38_jfrlSe1402-3_Brandenburg	
39_jfrlSe1402-4_Infantis	ATGAACAGCGAAGAGTTGACACATAAAGCAGAAGAGGAAATCGCGGCACTCATTAGCAAAAAGGTTGCCGAACTACGGAAGAAA ACCGGGCAAGAGGTTTCCGAAATAGAGTTCGTGCCGCGAGAAACGATGAAAGGGCTTGAGGGATACCACGTTAAAATTAAACTA CTCTAG (SEQ ID NO: 363)
40_jfrlSe1402-5_Brandenburg	
41_jfrlSe1402-6_Rissen	
42_jfrlSe1402-7_Orion	ATGAACAGCGAAGAGTTGACACATAAAGCAGAAGAGGAAATCGCGGCACTCATTAGCAAAAAGGTTGCCGAACTACGGAAGAAA ACCGGGCAAGAGGTTTCCGAAATAGAGTTCGCGCCGCGAGAAACGATGAAAGGGCTTGAGGGATACCACGTTAAAATTAAACTA CTCTAG (SEQ ID NO: 387)
43_jfrlSe1402-8_Rissen	
44_jfrlSe1402-9_Rissen	
45_jfrlSe1402-10_Rissen	
46_jfrlSe1402-11_Rissen	
47_jfrlSe1402-12_Rissen	
48_jfrlSe1402-13_Mbandaka	ATGAACAGCGAAGAGTTGACACATAAAGCAGAAGAGGAAATCGCGGCACTCATTAGCAAAAAGGTCGCCGAACTACGGAAGAAA ACCGGGCAAGAGGTTTCCGAAATAGAGTTCGTGCCGCGAGAAACGATGAAAGGGCTTGAGGGATACCACGTTAAAATTAAACTA CTCTAG (SEQ ID NO: 432)
49_jfrlSe1402-14_Mbandaka	ATGAACAGCGAAGAGTTGACACATAAAGCAGAAGAGGAAATCGCGGCACTCATTAGCAAAAAGGTCGCCGAACTACGGAAGAAA ACCGGGCAAGAGGTTTCCGAAATAGAGTTCGTGCCGCGAGAAACGATGAAAGGGCTTGAGGGATACCACGTTAAAATTAAACTA CTCTAG (SEQ ID NO: 442)
50_jfrlSe1402-15_Orion	ATGAACAGCGAAGAGTTGACACATAAAGCAGAAGAGGAAATCGCGGCACTCATTAGCAAAAAGGTTGCCGAACTACGGAAGAAA ACCGGGCAAGAGGTTTCCGAAATAGAGTTCGCGCCGCGAGAAACGATGAAAGGGCTTGAGGGATACCACGTTAAAATTAAACTA CTCTAG (SEQ ID NO: 452)
51_jfrlSe1409-1_Altona	ATGAACAGCGAAGAGTTGACACATAAAGCAGAAGAGGAAATCGCGGCACTCATTAGCAAAAAGGTCGCCGAACTACGGAAGAAA ACCGGGCAAGAGGTTTCCGAAATAGAGTTCGCGCCGCGAGAAACGATGAAAGGGCTTGAGGGATACCACGTTAAAATTAAACTA CTCTAG (SEQ ID NO: 459)
52_jfrlSe1409-2_Istanbul	ATGAACAGCGAAGAGTTGACACATAAAGCAGAAGAGGAAATCGCGGCACTCATTAGCAAAAAGGTTGCCGAACTACGGAAGAAA ACCGGGCAAGAGGTTTCCGAAATAGAGTTCGCGCCGCGAGAAACGATGAAAGGGCTTGAGGGATACCACGTTAAAATTAAACTA CTCTAG (SEQ ID NO: 468)
53_jfrlSe1409-3_Senftenberg	ATGAACAGCGAAGAGTTGACACATAAAGCAGAAGAGGAAATCGCGGCACTCATTAGCAAAAAGGTTGCCGAACTACGGAAGAAA ACCGGGCAAGAGGTTTCCGAAATAGAGTTCGTGCCGCGAGAAACGATGAAAGGGCTTGAGGGATACCACGTTAAAATTAAACTA CTCTAG (SEQ ID NO: 471)
54_jfrlSe1409-4_UN_O13_	ATGAACAGCGAAGAGTTGACACATAAAGCAGAAGAGGAAATCGCGGCACTCATTAGCAAAAAGGTCGCCGAACTACGGAAGAAA ACCGGGCAAGAGGTTTCCGAAATAGAGTTCGCGCCGCGAGAAACGATGAAAGGGCTTGAGGGATACCACGTTAAAATTAAACTA CTCTAG (SEQ ID NO: 480)
55_jfrlSe1409-5_UN_O1,3,19_	ATGAACAGCGAAGAGTTGACACATAAAGCAGAAGAGGAAATCGCGGCACTCATTAGCAAAAAGGTTGCCGAACTACGGAAGAAA ACCGGGCAAGAGGTTTCCGAAATAGAGTTCGTGCCGCGAGAAACGATGAAAGGGCTTGAGGGATACCACGTTAAAATTAAACTA CTCTAG (SEQ ID NO: 489)
56_jfrlSe1409-6_Montevideo	ATGAACAGCGAAGAGTTGACACATAAAGCAGAAGAGGAAATCGCGGCACTCATTAGCAAAAAGGTTGCCGAACTACGGAAGAAA ACCGGGCAAGAGGTTTCCGAAATAGAGTTCGCGCCGCGAGAAACGATGAAAGGGCTTGAGGGATACCACGTTAAAATTAAACTA CTCTAG (SEQ ID NO: 496)
57_jfrlSe1409-7_UN_O1,3,19_	ATGAACAGCGAAGAGTTGACACATAAAGCAGAAGAGGAAATCGCGGCACTCATTAGCAAAAAGGTTGCCGAACTACGGAAGAAA ACCGGGCAAGAGGTTTCCGAAATAGAGTTCGTGCCGCGAGAAACGATGAAAGGGCTTGAGGGATACCACGTTAAAATTAAACTA CTCTAG (SEQ ID NO: 505)
58_jfrlSe1409-8_UN_O18_	ATGAACAGCGAAGAGTTGACACATAAAGCAGAAGAGGAAATCGCGGCACTCATAAGCAAAAAGGTCGCCGAACTACGGAAGAAA ACCGGGCAAGAGGTTTCCGAAATAGAGTTCGCGCCGCGAGAAACGATGAAAGGGCTTGAGGGATACCACGTTAAAATTAAACTA CTCTAG (SEQ ID NO: 514)
61_jfrlSe1409-11_UN_O1,3,19_	ATGAACAGCGAAGAGTTGACACATAAAGCAGAAGAGGAAATCGCGGCACTCATTAGCAAAAAGGTTGCCGAACTACGGAAGAAA ACCGGGCAAGAGGTTTCCGAAATAGAGTTCGTGCCGCGAGAAACGATGAAAGGGCTTGAGGGATACCACGTTAAAATTAAACTA CTCTAG (SEQ ID NO: 523)
67_jfrlSe1409-17_Rissen	ATGAACAGCGAAGAGTTGACACATAAAGCAGAAGAGGAAATCGCGGCACTCATTAGCAAAAAGGTTGCCGAACTACGGAAGAAA ACTGGGCAAGAGGTTTCCGAAATAGAGTTCGCGCCGCGAGAAACGATGAAAGGGCTTGAGGGATACCACGTTAAAATTAAACTA CTCTAG (SEQ ID NO: 531)
71_jfrlSe1409-21_Amsterdam	ATGAACAGCGAAGAGTTGACACATAAAGCAGAAGAGGAAATCGCGGCACTCATTAGCAAAAAAGTCGCCGAACTACGGAAGAAA ACCGGGCAAGAGGTTTCCGAAATAGAGTTCGTGCCGCGAGAAACGATGAAAGGGCTTGAGGGCTACCACGTTAAAATTAAACTA CTCTAG (SEQ ID NO: 541)
80_HySe09_Enteritidis	ATGAACAGCGAAGAGTTGACACATAAAGCAGAAGAGGAAATCGCGGCACTCATTAGCAAAAAGGTCGCCGAACTACGGAAGAAA ACCGGGCAAGAGGTTTCCGAAATAGAGTTCGTGCCGCGAGAAACGATGAAAGGGCTTGAGGGCTACCACGTTAAAATTAAACTA CTCTAG (SEQ ID NO: 551)
100_HySe29_Schwarzengrund	ATGAACAGCGAAGAGTTGACACATAAAGCAGAAGAGGAAATCGCGGCACTCATTAGCAAAAAGGTTGCCGAACTACGGAAGAAA ACCGGGCAAGAGGTTTCCGAAATAGAGTTCGCGCCGCGAGAAACGATGAAAGGGCTTGAGGGATACCACGTTAAAATTAAACTA CTCTAG (SEQ ID NO: 560)
103_HySe32_Schwarzengrund	ATGAACAGCGAAGAGTTGACACATAAAGCAGAAGAGGAAATCGCGGCACTCATTAGCAAAAAGGTTGCCGAACTACGGAAGAAA ACCGGGCAAGAGGTTTCCGAAATAGAGTTCGCGCCGCGAGAAACGATGAAAGGGCTTGAGGGATACCACGTTAAAATTAAACTA CTCTAG (SEQ ID NO: 567)
106_HySe35_Schwarzengrund	ATGAACAGCGAAGAGTTGACACATAAAGCAGAAGAGGAAATCGCGGCACTCATTAGCAAAAAGGTTGCCGAACTACGGAAGAAA ACCGGGCAAGAGGTTTCCGAAATAGAGTTCGCGCCGCGAGAAACGATGAAAGGGCTTGAGGGATACCACGTTAAAATTAAACTA CTCTAG (SEQ ID NO: 574)

Fig. 32A

	yibT
01_NBRC13245T_Typhimurium	ATGGCAAAAATAGGTGAGAACGTACCACTTCTTATTGATAAAGCCGTCGATTTTATGGCGTCCAGCCAGGCGTTCCGTGAATAC CTGAATAAAACGCCGCCGCGTGATTATGTTCCTTCCGAGGTGCCGTCGGAGAGCGCGCCCATTTATTTGCAGCGTCTGGAATA TTACCGGCGGCTGTACCGACCAAAAGAAGAAGAAAGAGGTTAA (SEQ ID NO: 62)
02_GTC00131_Enteritidis	
03_GTC09491_Enteritidis	
04_GTC03838_Enteritidis	
05_GTC08914_Enteritidis	ATGGCAAAAATAGGTGAGAACGTACCACTTCTTATTGATAAAGCCGTCGATTTTATGGCGTCCAGCCAGGCGTTCCGTGAATAC CTGAATAAAACGCCGCCGCGTGATTATGTTCCTTCCGAGGTGCCGTCGGAGAGCGCGCCCATTTATTTGCAGCGTCTGGAATA TTACCGGCGGCTGTACCGACCAAAAGAAGAAGAAAGAGGTTAA (SEQ ID NO: 97)
06_GTC09421_Enteritidis	ATGGCAAAAATAGGTGAGAACGTACCACTTCTTATTGATAAAGCCGTCGATTTTATGGCGTCCAGCCAGGCGTTCCGTGAATAC CTGAATAAAACGCCGCCGCGTGATTATGTTCCTTCCGAGGTGCCGTCGGAGAGCGCGCCCATTTATTTGCAGCGTCTGGAATA TTACCGGCGGCTGTACCGACCAAAAGAAGAAGAAAGAGGTTAA (SEQ ID NO: 107)
07_GTC09489_Enteritidis	
08_NBRC3163_Pullorum_Gallinarum	ATGGCAAAAATAGGTGAGAACGTACCACTTCTTATTGATAAAGCCGTCGATTTTAAGGCGTCCAGCCAGGCGTTCCGTGAATAC CTGAATAAAACGCCGCCGCGTGATTATGTTCCTTCCGAGGTGCCGTCGGAGAGCGCGCCCATTTATTTGCAGCGTCTGGAATA TTACCGGCGGCTGTACCGACCAAAAGAAGAAGAAAGAGGTTAA (SEQ ID NO: 124)
09_NBRC3313_Enteritidis	
10_NBRC12529_Typhimurium	
11_NBRC14193_Typhimurium	
12_NBRC14194_Typhimurium	
13_NBRC14209_Typhimurium	
14_NBRC14210_Typhimurium	
15_NBRC14211_Typhimurium	
16_NBRC14212_Typhimurium	
17_NBRC15181_Typhimurium	ATGGCAAAAATAGGTGAGAACGTACCACTTCTTATTGATAAAGCCGTCGATTTTATGGCGTCCAGCCAGGCGTTCCGTGAATAC CTGAATAAAACGCCGCCGCGTGATTATGTTCCTTCCGAGGTGCCGTCGGAGAGCGCGCCCATTTATTTGCAGCGTCTGGAATA TTACCGGCGGCTGTACCGACCAAAAGAAGAAGAAAGAGGTTAA (SEQ ID NO: 189)
18_NBRC15182_Minnesota	ATGGCAAAAATAGGTGAGAACGTACCACTTCTTATTGATAAAGCCGTCGATTTTATGGCGTCCAGCCAGGCGTTCCGTGAATAC CTGAATAAAACGCCGCCGCGTGATTATGTTCCTTCCGAGGTGCCGTCGGAGAGCGCGCCCATTTATTTGCAGCGTCTGGAATA TTACCGGCGGCTGTACCGACCAAAAGAAGAAGAAAGAGGTTAA (SEQ ID NO: 199)
19_NBRC15183_Minnesota	
20_NBRC15184_Minnesota	
21_NBRC15185_Minnesota	
22_NBRC15186_Minnesota	
23_NBRC15187_Minnesota	
24_NBRC100797_Abony	ATGGCAAAAATAGGTGAGAACGTACCACTTCTTATTGATAAAGCCGTCGATTTTATGGCGTCCAGCCAGGCGTTCCGTGAGTAC TTGAATAAAACGCCGCCGCGTGATTATGTTCCTTCCGAAGTGCCGTCGGAGAGCGCGCCCATTTATTTGCAGCGTCTGGAATAT TACCGGCGGCTGTACCGACCAAAAGAAGAAGAAAGAGGTTAA (SEQ ID NO: 244)
25_NBRC105684_Choleraesuis	ATGGCAAAAATAGGTGAGAACGTACCACTTCTTATTGATAAAGCCGTCGATTTTATGGCGTCCAGCCAGGCGTTCCGTGAATAC CTGAATAAAACGCCGCCGCGTGATTATGTTCCTTCCGAGGTGCCGTCGGAGAGCGCGCCCATTTATTTGCAGCGTCTGGAATA TTACCGGCGGCTGTACCGACCAAAAGAAGAAGAAAGAGGTTAA (SEQ ID NO: 254)
26_NBRC105726_Typhimurium	ATGGCAAAAATAGGTGAGAACGTACCACTTCTTATTGATAAAGCCGTCGATTTTATGGCGTCCAGCCAGGCGTTCCGTGAATAC CTGAATAAAACGCCGCCGCGTGATTATGTTCCTTCCGAGGTGCCGTCGGAGAGCGCGCCCATTTATTTGCAGCGTCTGGAATA TTACCGGCGGCTGTACCGACCAAAAGAAGAAGAAAGAGGTTAA (SEQ ID NO: 263)
27_JCM3919_UN_O7_	
28_NBRC15335_Minnesota	
29_GTC09490_Enteritidis	
30_GTC09492_Braenderup	
31_GTC09493_Pakistan	
32_GTC09549_Typhimurium	
33_ATCCBAA-1675_Infantis	ATGGCAAAAATAGGTGAGAACGTACCACTTCTTATTGATAAAGCCGTCGATTTTATGGCGTCCAGCCAGGCGTTCCGTGAATAC CTGAATAAAACGCCGCCGCGTGATTATGTTCCTTCCGAGGTGCCGTCGGAGAGCGCGCCCATTTATTTGCAGCGTCTGGAATA TTACCGGCGGCTGTACCGACCAAAAGAAGAAGAAAGAGGTTAA (SEQ ID NO: 315)
34_ATCCBAA-1738_Thompson	ATGGCAAAAATAGGTGAGAACGTACCACTTCTTATTGATAAAGCCGTCGATTTTATGGCGTCCAGCCAGGCGTTCCGTGAATAC CTGAATAAAACGCCGCCGCGTGATTATGTTCCTTCCGAGGTGCCGTCGGAGAGCGCGCCCATTTATTTGCAGCGTCTGGAATA TTACCGGCGGCTGTACCGACCAAAAGAAGAAGAAAGGAGTTAA (SEQ ID NO: 326)
35_ATCC9712_Saintpaul	ATGGCAAAAATAGGTGAGAACGTACCACTTCTTATTGATAAAGCCGTCGATTTTATGGCGTCCAGCCAGGCGTTCCGTGAATAC CTGAATAAAACGCCGCCGCGTGATTATGTTCCTTCCGAAGTGCCGTCGGAGAGCGCGCCCATTTATTTGCAGCGTCTGGCATA TTACCGACGGTTGTACCGACCAAAAGAAGAAGAAAGAGGTTAA (SEQ ID NO: 336)
36_jfrlSe1402-1_Infantis	ATGGCAAAAATAGGTGAGAACGTACCACTTCTTATTGATAAAGCCGTCGATTTTATGGCGTCCAGCCAGGCGTTCCGTGAATAC CTGAATAAAACGCCGCCGCGTGATTATGTTCCTTCCGAGGTGCCGTCGGAGAGCGCGCCCATTTATTTGCAGCGTCTGGAATA TTACCGGCGGCTGTACCGACCAAAAGAAGAAGAAAGAGGTTAA (SEQ ID NO: 347)
38_jfrlSe1402-3_Brandenburg	
39_jfrlSe1402-4_Infantis	ATGGCAAAAATAGGTGAGAACGTACCACTTCTTATTGATAAAGCCGTCGATTTTATGGCGTCCAGCCAGGCGTTCCGTGAATAC CTGAATAAAACGCCGCCGCGTGATTATGTTCCTTCCGAGGTGCCGTCGGAGAGCGCGCCCATTTATTTGCAGCGTCTGGAATA TTACCGGCGGCTGTACCGACCAAAAGAAGAAGAAAGAGGTTAA (SEQ ID NO: 364)
40_jfrlSe1402-5_Brandenburg	
41_jfrlSe1402-6_Rissen	
42_jfrlSe1402-7_Orion	ATGGCAAAAATAGGTGAGAACGTACCACTTCTTATTGATAAAGCCGTCGATTTTATGGCGTCCAGCCAGGCGTTCCGTGAATAC TTGAATAAAACGCCGCCGCGTGATTATGTTCCTTCCGAAGTGCCGTCGGAGAGCGCGCCCATTTATTTGCAGCGTCTGGAATAT TACCGGCGGCTGTACCGACCAAAAGAAGAAGAAAGAGGTTAA (SEQ ID NO: 388)
43_jfrlSe1402-8_Rissen	
44_jfrlSe1402-9_Rissen	
45_jfrlSe1402-10_Rissen	
46_jfrlSe1402-11_Rissen	
47_jfrlSe1402-12_Rissen	
48_jfrlSe1402-13_Mbandaka	
49_jfrlSe1402-14_Mbandaka	ATGGCAAAAATAGGTGAGAACGTACCACTTCTTATTGATAAAGCCGTCGATTTTATGGCGTCCAGCCAGGCGTTCCGTGAATAC CTGAATAAAACGCCGCCGCGTGATTATGTTCCTTCCGAAGTGCCGTCGGAGAGCGCGCCCATTTATTTGCAGCGTCTGGAATAT TACCGACGGCTGTACCGACCAAAAGAAGAAGAAAGAGGTTAA (SEQ ID NO: 443)
50_jfrlSe1402-15_Orion	ATGGCAAAAATAGGTGAGAACGTACCACTTCTTATTGATAAAGCCGTCGATTTTATGGCGTCCAGCCAGGCGTTCCGTGAATAC TTGAATAAAACGCCGCCGCGTGATTATGTTCCTTCCGAAGTGCCGTCGGAGAGCGCGCCCATTTATTTGCAGCGTCTGGAATAT TACCGGCGGCTGTACCGACCAAAAGAAGAAGAAAGAGGTTAA (SEQ ID NO: 453)
51_jfrlSe1409-1_Altona	ATGGCAAAAATAGGTGAGAACGTACCACTTCTTATTGATAAAGCCGTCGATTTTATGGCGTCCAGCCAGGCGTTCCGTGAATAC CTGAATAAAACGCCGCCGCGTGATTATGTTCCTTCCGAAGTGCCGTCGGAGAGCGCGCCCATTTATTTGCAGCGTCTGGAATAT TACCGACGGCTGTACCGACCAAAAGAAGAAGAAAGAGGTTAACGGCGGGATTT (SEQ ID NO: 460)

Fig. 32B

52_jfrlSe1409-2_Istanbul	ATGGCAAAAATAGGTGAGAACGTACCACTTCTTATTGATAAAGCCGTCGATTTTATGGCGTCCAGCCAGGCGTTCCGTGAGTAC TTGAATAAAACGCCGCCGCGTGACTATGTTCCTTCCGAAGTGCCGTCGGAGAGCGCGCCCATTTATTTGCAGCGTCTGGAATAT TACCGGCGGCTGTACCGACCAAAAGAAGAAAGAGGTTAACGGCGG (SEQ ID NO: 469)
53_jfrlSe1409-3_Senftenberg	ATGGCAAAAATAGGTGAGAACGTACCACTTCTTATTGATAAAGCCGTCGATTTTATGGCGTCCAGCCAGGCGTTCCGTGAATAC CTGAATAAAACGCCGCCGCGTGATTATGTTCCTTCCGAAGTGCCGTCGGAGAGCGCGCCCATTTATTTGCAGCGTCTGGCATA TTACCGACGGTTGTACCGACCAAAAGAAGAAGAAAGAGGTTAA (SEQ ID NO: 472)
54_jfrlSe1409-4_UN_O13_	ATGGCAAAAATAGGTGAGAACGTACCACTTCTTATTGATAAAGCCGTCGATTTTATGGCGTCCAGCCAGGCGTTCCGTGAATAC CTGAATAAAACGCCGCCGCGTGATTATGTTCCTTCCGAAGTGCCGTCGGAGAGCGCGCCCATTTATTTGCAGCGTCTGGAATAT TACCGACGGCTGTACCGACCAAAAGAAGAAGAAAGAGGTTAA (SEQ ID NO: 481)
55_jfrlSe1409-5_UN_O1,3,19_	ATGGCAAAAATAGGTGAGAACGTACCACTTCTTATTGATAAAGCCGTCGATTTTATGGCGTCCAGCCAGGCGTTCCGTGAATTC AAGTATAAAACGCCGCCGCGTGATTATGTTCCTTCCGAAGTGCCGTCGGAGAGCGCGCCCATTTATTTGCAGCGTCTGGCATAT TACCGACGGTTGTACCGACCAAAAGAAGAAGAAAGAGGTTAA (SEQ ID NO: 490)
56_jfrlSe1409-6_Montevideo	ATGGCAAAAATAGGTGAGAACGTACCACTTCTTATTGATAAAGCCGTCGATTTTATGGCGTCCAGCCAGGCGTTCCGTGAATAC CTGAATAAAACGCCGCCGCGTGATTATGTTCCTTCCGAGGTGCCGTCGGAGAGCGCGCCCATTTATTTGCAGCGTCTGGAATA TTACCGGCGGCTGTACCGACCAAAAGAAGAAGAAAGAGGTTAA (SEQ ID NO: 497)
57_jfrlSe1409-7_UN_O1,3,19_	ATGGCAAAAATAGGTGAGAACGTACCACTTCTTATTGATAAAGCCGTCGATTTTATGGCGTCCAGCCAGGCGTTCCGTGAATAC CTGAATAAAACGCCGCCGCGTGATTATGTTCCTTCCGAAGTGCCGTCGGAGAGCGCGCCCATTTATTTGCAGCGTCTGGCATA TTACCGACGGTTGTACCGACCAAAAGAAGAAGAAAGAGGTTAA (SEQ ID NO: 506)
58_jfrlSe1409-8_UN_O18_	ATGGCAAAAATAGGTGAGAACGTACCACTTCTTATTGATAAAGCCGTCGATTTTATGGCGTCCAGCCAGGCGTTCCGTGAGTAC TTGAATAAAACGCCGCCGCGTGATTATGTTCCTTCCGAAGTGCCGTCGGAGAGCGCGCCCATTTATTTGCAGCGTCTGGAATAT TACCGGCGGCTGTACCGACCAAAAGAAGAAGAAAGAGGTTAA (SEQ ID NO: 515)
61_jfrlSe1409-11_UN_O1,3,19_	ATGGCAAAAATAGGTGAGAACGTACCACTTCTTATTGATAAAGCCGTCGATTTTATGGCGTCCAGCCAGGCGTTCCGTGAATAC CTGAATAAAACGCCGCCGCGTGATTATGTTCCTTCCGAAGTGCCGTCGGAGAGCGCGCCCATTTATTTGCAGCGTCTGGCATA TTACCGACGGTTGTACCGACCAAAAGAAGAAGAAAGAGGTTAA (SEQ ID NO: 524)
67_jfrlSe1409-17_Rissen	ATGGCAAAAATAGGTGAGAACGTACCACTTCTTATTGATAAAGCCGTCGATTTTATGGCGTCCAGCCAGGCGTTCCGTGAATAC CTGAATAAAACGCCGCCGCGTGATTATGTTCCTTCCGAAGTGCCGTCGGAGAGCGCGCCCATTTATTTGCAGCGTCTGGAATAT TACCGACGGCTGTACCGACCAAAAGAAGAAGAAAGAGGTTAA (SEQ ID NO: 532)
71_jfrlSe1409-21_Amsterdam	ATGGCAAAAATAGGTGAGAACGTACCACTTCTTATTGATAAAGCCGTCGATTTTATGGCGTCCAGCCAGGCGTTCCGTGAATAC TTGAATAAAACGCCGCCGCGTGATTATGTTCCTTCCGAAGTGCCGTCGGAGAGCGCGCCCATTTATTTGCAGCGTCTGGAATAT TACCGGCGGCTGTACCGACCAAAAGAAGAAGAAAGAGGTTAA (SEQ ID NO: 542)
80_HySe09_Enteritidis	ATGGCAAAAATAGGTGAGAACGTACCACTTCTTATTGATAAAGCCGTCGATTTTATGGCGTCCAGCCAGGCGTTCCGTGAATAC CTGAATAAAACGCCGCCGCGTGATTATGTTCCTTCCGAGGTGCCGTCGGAGAGCGCGCCCATTTATTTGCAGCGTCTGGAATA TTACCGGCGGCTGTACCGACCAAAAGAAGAAGAAAGAGGTTAA (SEQ ID NO: 552)
100_HySe29_Schwarzengrund	
103_HySe32_Schwarzengrund	
106_HySe35_Schwarzengrund	

Fig. 33A

	ppiC
01_NBRC13245T_Typhimurium	ATGGCAAAAATGGCAGCAGCACTGCATATTCTTGTAAAAGAAGAGAAACTGGCTTTAGATCTTCTGGAGCAAATTAAAAACGGC GGCGATTTTGAGAAGCTGGCGAAGAAGCATTCTATCTGCCCATCCGGTAAAAAAGGCGGTCATTTAGGCGAATTTCGTCAGGG CCAGATGGTTCCGGCATTCGATAAAGTAGTCTTTTCCTGCCCGGTACTGGAGCCAACCGGCCCGCTGCATACCCAGTTCGGT TACCACATCATTAAGGTATTGTATCGTAAATAA (SEQ ID NO: 63)
02_GTC00131_Enteritidis	
03_GTC09491_Enteritidis	
04_GTC03838_Enteritidis	
05_GTC08914_Enteritidis	ATGGCAAAAATGGCAGCAGCACTGCATATCCTTGTAAAAGAAGAGAAACTGGCTTTAGATCTTCTGGAGCAAATTAAAAACGGC GGCGATTTTGAGAAGCTGGCGAAGAAGCATTCTATCTGCCCATCCGGTAAAAAAGGCGGTCATTTAGGCGAATTTCGTCAGGG CCAGATGGTTCCGGCATTCGATAAAGTAGTCTTTTCCTGCCCGGTACTGGAGCCAACCGGCCCGCTGCATACCCAGTTCGGT TACCACATCATTAAGGTATTGTATCGCAAATAA (SEQ ID NO: 98)
06_GTC09421_Enteritidis	ATGGCAAAAATGGCAGCAGCACTGCATATCCTTGTAAAAGAAGAGAAACTGGCTTTAGATCTTCTGGAGCAAATTAAAAACGGC GGCGATTTTGAGAAGCTGGCGAAGAAGCATTCTATCTGCCCATCCGGTAAAAAAGGCGGTCATTTAGGCGAATTTCGTCAGGG CCAGATGGTTCCGGCATTCGATAAAGTAGTCTTTTCCTGCCCGGTACTGGAGCCAACCGGCCCGCTGCATACCCAGTTCGGT TACCACATCATTAAAGTATTGTATCGCAAATAA (SEQ ID NO: 108)
07_GTC09489_Enteritidis	
08_NBRC3163_Pullorum_Gallinarum	ATGGCAAAAATGGCAGCAGCACTGCATATCCTTGTAAAAGAAGAGAAACTGGCTTTAGATCTTCTGGAGCAAATTAAAAACGGC GGCGATTTTGAGAAGCTGGCGAAGAAGCATTCTATCTGCCCATCCGGTAAAAAAGGCGGTCATTTAGGCGAATTTCGTCAGGG CCAGATGGTTCCGGCATTCGATAAAGTAGTCTTTTCCTGCCCGGTACTGGAGCCAACCGGCCCGCTGCATACCCAGTTCGGT TACCACATCATTAAAGTATTGTATCGCAAATAA (SEQ ID NO: 125)
09_NBRC3313_Enteritidis	
10_NBRC12529_Typhimurium	
11_NBRC14193_Typhimurium	
12_NBRC14194_Typhimurium	
13_NBRC14209_Typhimurium	
14_NBRC14210_Typhimurium	
15_NBRC14211_Typhimurium	
16_NBRC14212_Typhimurium	
17_NBRC15181_Typhimurium	ATGGCAAAAATGGCAGCAGCACTGCATATTCTTGTAAAAGAAGAGAAACTGGCTTTAGATCTTCTGGAGCAAATTAAAAACGGC GGCGATTTTGAGAAGCTGGCGAAGAAGCATTCTATCTGCCCATCCGGTAAAAAAGGCGGTCATTTAGGCGAATTTCGTCAGGG CCAGATGGTTCCGGCATTCGATAAAGTAGTCTTTTCCTGCCCGGTACTGGAGCCAACCGGCCCGCTGCATACCCAGTTCGGT TACCACATCATTAAGGTATTGTATCGTAAATAA (SEQ ID NO: 190)
18_NBRC15182_Minnesota	ATGGCAAAAATGGCAGCAGCACTGCATATTCTTGTAAAAGAAGAGAAACTGGCTTTAGATCTTCTGGAGCAAATTAAAAACGGC GGCGATTTTGAGAAGCTGGCGAAGAAGCATTCTATCTGCCCATCCGGTAAAAAAGGCGGTCATTTAGGCGAATTTCGTCAGGG CCAGATGGTTCCGGCATTCGATAAAGTAGTCTTTTCCTGCCCGGTACTGGAGCCAACCGGCCCGCTGCATACCCAGTTCGGT TACCACATCATTAAAGTATTGTATCGCAAATAA (SEQ ID NO: 200)
19_NBRC15183_Minnesota	
20_NBRC15184_Minnesota	
21_NBRC15185_Minnesota	
22_NBRC15186_Minnesota	
23_NBRC15187_Minnesota	
24_NBRC100797_Abony	ATGGCAAAAATGGCAGCAGCACTGCATATCCTTGTAAAAGAAGAGAAACTGGCATTGGATCTTCTGGAGCAAATTAAAAACGGC GGCGATTTTGAGAAGCTGGCGAAGAAGCATTCTATCTGCCCATCCGGTAAAAAAGGCGGTCATTTAGGCGAATTTCGTCAGGG CCAGATGGTTCCGGCATTCGATAAAGTAGTCTTTTCCTGCCCGGTACTGGAGCCAACCGGCCCATTGCACACCCAGTTCGGT TATCACATCATTAAGGTATTGTATCGCAAATAA (SEQ ID NO: 245)
25_NBRC105684_Choleraesuis	ATGGCAAAAATGGCAGCAGCACTGCATATTCTTGTAAAAGAAGAGAAACTGGCTTTAGATCTTCTGGAGCAAATTAAAAACGGC GGCGATTTTGAGAAGCTGGCGAAGAAGCATTCTATCTGCCCATCCGGTAAAAAAGGCGGTCATTTAGGCGAATTTCGTCAGGG CCAGATGGTTCCGGCATTCGATAAAGTAGTCTTTTCCTGCCCGGTACTGGAGCCAACCGGCCCATTGCACACCCAGTTCGGT TATCACATCATTAAGGTATTGTATCGCAAATAA (SEQ ID NO: 255)
26_NBRC105726_Typhimurium	ATGGCAAAAATGGCAGCAGCACTGCATATTCTTGTAAAAGAAGAGAAACTGGCTTTAGATCTTCTGGAGCAAATTAAAAACGGC GGCGATTTTGAGAAGCTGGCGAAGAAGCATTCTATCTGCCCATCCGGTAAAAAAGGCGGTCATTTAGGCGAATTTCGTCAGGG CCAGATGGTTCCGGCATTCGATAAAGTAGTCTTTTCCTGCCCGGTACTGGAGCCAACCGGCCCGCTGCATACCCAGTTCGGT TACCACATCATTAAGGTATTGTATCGTAAATAA (SEQ ID NO: 264)
27_JCM3919_UN_O7_	
28_NBRC15335_Minnesota	
29_GTC09490_Enteritidis	
30_GTC09492_Braenderup	
31_GTC09493_Pakistan	
32_GTC09549_Typhimurium	
33_ATCCBAA-1675_Infantis	ATGGCAAAAATGGCAGCAGCACTGCATATTCTTGTAAAAGAAGAGAAACTGGCTTTAGATCTTCTGGAGCAAATTAAAAACGGC GGCGATTTTGAGAAGCTGGCGAAGAAGCATTCTATCTGCCCATCCGGTAAAAAAGGCGGTCATTTAGGCGAATTTCGTCAGGG CCAGATGGTTCCGGCATTCGATAAAGTAGTCTTTTCCTGCCCGGTACTGGAGCCAACCGGCCCGCTGCATACCCAGTTCGGT TACCACATCATTAAAGTATTGTATCGCAAATAA (SEQ ID NO: 316)
34_ATCCBAA-1738_Thompson	ATGGCAAAAATGGCAGCAGCACTGCATATCCTTGTAAAAGAAGAGAAACTGGCTTTAGATCTTCTGGAGCAAATTAAAAACGGC GGCGATTTTGAGAAGCTGGCGAAGAAGCATTCTATCTGCCCATCCGGTAAAAAAGGCGGTCATTTAGGCGAATTTCGTCAGGG CCAGATGGTTCCGGCATTCGATAAAGTAGTCTTTTCCTGCCCGGTACTGGAGCCAACCGGCCCGCTGCATACCCAGTTCGGT TACCACATCATTAAGGTATTGTATCGTAAATAA (SEQ ID NO: 327)
35_ATCC9712_Saintpaul	ATGGCAAAAATGGCAGCAGCACTGCATATCCTTGTAAAAGAAGAGAAACTGGCTTTAGATCTTCTGGAGCAAATTAAAAACGGC GGCGATTTTGAGAAGCTGGCGAAGAAGCATTCTATCTGCCCATCCGGTAAAAAAGGCGGTCATTTAGGCGAATTTCGTCAGGG CCAGATGGTTCCGGCATTCGATAAAGTAGTCTTTTCCTGCCCGGTACTGGAGCCAACCGGCCCGCTGCATACCCAGTTCGGT TACCACATCATTAAGGTATTGTATCGTAAATAA (SEQ ID NO: 337)
36_jfrlSe1402-1_Infantis	ATGGCAAAAATGGCAGCAGCACTGCATATTCTTGTAAAAGAAGAGAAACTGGCTTTAGATCTTCTGGAGCAAATTAAAAACGGC GGCGATTTTGAGAAGCTGGCGAAGAAGCATTCTATCTGCCCATCCGGTAAAAAAGGCGGTCATTTAGGCGAATTTCGTCAGGG CCAGATGGTTCCGGCATTCGATAAAGTAGTCTTTTCCTGCCCGGTACTGGAGCCAACCGGCCCGCTGCATACTCAGTTCGGT TACCACATCATTAAAGTATTGTATCGCAAATAA (SEQ ID NO: 348)
38_jfrlSe1402-3_Brandenburg	
39_jfrlSe1402-4_Infantis	ATGGCAAAAATGGCAGCAGCACTGCATATTCTTGTAAAAGAAGAGAAACTGGCTTTAGATCTTCTGGAGCAAATTAAAAACGGC GGCGATTTTGAGAAGCTGGCGAAGAAGCATTCTATCTGCCCATCCGGTAAAAAAGGCGGTCATTTAGGCGAATTTCGTCAGGG CCAGATGGTTCCGGCATTCGATAAAGTAGTCTTTTCCTGCCCGGTACTGGAGCCAACCGGCCCGCTGCATACTCAGTTCGGT TACCACATCATTAAAGTATTGTATCGCAAATAA (SEQ ID NO: 365)
40_jfrlSe1402-5_Brandenburg	
41_jfrlSe1402-6_Rissen	
42_jfrlSe1402-7_Orion	ATGGCAAAAATGGCAGCAGCACTGCATATCCTTGTAAAAGAAGAGAAAATGGCATTGGATCTTCTGGAGCAAATTAAAAACGGC GGCGATTTTGAGAAGCTGGCGAAGAAGCATTCTATCTGCCCATCCGGTAAAAAAGGCGGTCATTTAGGCGAATTTCGTCAGGG CCAGATGGTTCCGGCATTCGATAAAGTAGTCTTTTCCTGCCCGGTACTGGAGCCAACCGGCCCATTGCACACCCAGTTCGGT TATCACATCATTAAGGTATTGTATCGCAAATAA (SEQ ID NO: 389)

Fig. 33B

43_jfrlSe1402-8_Rissen	
44_jfrlSe1402-9_Rissen	
45_jfrlSe1402-10_Rissen	
46_jfrlSe1402-11_Rissen	
47_jfrlSe1402-12_Rissen	
48_jfrlSe1402-13_Mbandaka	ATGGCAAAAATGGCAGCAGCACTGCATATTCTTGTAAAAGAAGAGAAACTGGCTTTAGATCTTCTGGAGCAAATTAAAAACGGC GGCGATTTTGAGAAGCTGGCGAAGAAGCATTCTATCTGCCCATCCGGTAAAAAAGGCGGTCATTTAGGCGAATTTCGTCAGGG CCAGATGGTTCCGGCATTCGATAAAGTAGTCTTTTCCTGCCCGGTACTGGAGCCAACCGGCCCGCTGCATACTCAGTTCGGT TACCACATCATTAAAGTATTGTATCGCAAATAA (SEQ ID NO: 433)
49_jfrlSe1402-14_Mbandaka	ATGGCAAAAATGGCAGCAGCACTGCATATTCTTGTAAAAGAAGAGAAACTGGCTTTAGATCTTCTGGAGCAAATTAAAAACGGC GGCGATTTTGAGAAGCTGGCGAAGAAGCATTCTATCTGCCCATCCGGTAAAAAAGGCGGTCATTTAGGCGAATTTCGTCAGGG CCAGATGGTTCCGGCATTCGATAAAGTAGTCTTTTCCTGCCCGGTACTGGAGCCAACCGGCCCGCTGCATACTCAGTTCGGT TACCACATCATTAAAGTATTGTATCGCAAATAA (SEQ ID NO: 444)
50_jfrlSe1402-15_Orion	ATGGCAAAAATGGCAGCAGCACTGCATATCCTTGTAAAAGAAGAGAAAATGGCATTGGATCTTCTGGAGCAAATTAAAAACGGC GGCGATTTTGAGAAGCTGGCGAAGAAGCATTCTATCTGCCCATCCGGTAAAAAAGGCGGTCATTTAGGCGAATTTCGTCAGGG CCAGATGGTTCCGGCATTCGATAAAGTAGTCTTTTCCTGCCCGGTACTGGAGCCAACCGGCCCATTGCACACCCAGTTCGGT TATCACATCATTAAGGTATTGTATCGCAAATAA (SEQ ID NO: 454)
51_jfrlSe1409-1_Altona	ATGGCAAAAATGGCAGCAGCACTGCATATCCTTGTAAAAGAAGAGAAACTGGCATTGGATCTTCTGGAGCAAATTAAAAACGGC GGCGATTTTGAGAAGCTGGCGAAGAAACATTCTATCTGCCCATCCGGTAAAAAAGGCGGTCATTTAGGCGAATTTCGTCAGGG CCAGATGGTTCCGGCATTCGATAAAGTAGTCTTTTCCTGCCCGGTACTGGAGCCAACCGGCCCATTGCACACCCAGTTCGGT TATCACATCATTAAGGTATTGTATCGCAAATAA (SEQ ID NO: 461)
52_jfrlSe1409-2_Istanbul	ATGGCAAAAATGGCAGCAGCACTGCATATTCTTGTAAAAGAAGAGAAACTGGCTTTAGATCTTCTGGAGCAAATTAAAAACGGC GGCGATTTTGAGAAGCTGGCGAAGAAGCATTCTATCTGCCCATCCGGTAAAAAAGGCGGTCATTTAGGCGAATTTCGTCAGGG CCAGATGGTTCCGGCATTCGATAAAGTAGTCTTTTCCTGCCCGGTACTGGAGCCAACCGGCCCGCTGCATACTCAGTTCGGT TACCACATCATTAAAGTATTGTATCGTAAATAA (SEQ ID NO: 470)
53_jfrlSe1409-3_Senftenberg	ATGGCAAAAATGGCAGCAGCACTGCATATCCTTGTAAAAGAAGAGAAACTGGCTTTAGATCTTCTGGAGCAAATTAAAAACGGC GGCGATTTTGAGAAGCTGGCGAAGAAGCATTCTATCTGCCCATCCGGTAAAAAAGGCGGTCATTTAGGCGAATTTCGTCAGGG CCAGATGGTTCCGGCATTCGATAAAGTAGTCTTTTCCTGCCCGGTACTGGAGCCAACCGGCCCGCTGCATACCCAGTTCGGT TACCACATCATTAAAGTATTGTATCGCAAATAA (SEQ ID NO: 473)
54_jfrlSe1409-4_UN_O13_	ATGGCAAAAATGGCAGCAGCACTGCATATTCTTGTAAAAGAAGAGAAACTGGCTTTAGATCTTCTGGAGCAAATTAAAAACGGC GGCGATTTTGAGAAGCTGGCGAAGAAGCATTCTATCTGCCCATCCGGTAAAAAAGGCGGTCATTTAGGCGAATTTCGTCAGGG CCAGATGGTTCCGGCATTCGATAAAGTAGTCTTTTCCTGCCCGGTACTGGAGCCAACCGGCCCGCTGCATACTCAGTTCGGT TACCACATCATTAAAGTATTGTATCGCAAATAA (SEQ ID NO: 482)
55_jfrlSe1409-5_UN_O1,3,19_	ATGGCAAAAATGGCAGCAGCACTGCATATCCTTGTAAAAGAAGAGAAACTGGCTTTAGATCTTCTGGAGCAAATTAAAAACGGC GGCGATTTTGAGAAGCTGGCGAAGAAGCATTCTATCTGCCCATCCGGTAAAAAAGGCGGTCATTTAGGCGAATTTCGTCAGGG CCAGATGGTTCCGGCATTCGATAAAGTAGTCTTTTCCTGCCCGGTACTGGAGCCAACCGGCCCGCTGCATACCCAGTTCGGT TACCACATCATTAAAGTATTGTATCGCAAATAA (SEQ ID NO: 491)
56_jfrlSe1409-6_Montevideo	ATGGCAAAAATGGCAGCAGCACTGCATATTCTTGTAAAAGAAGAGAAACTGGCTTTAGATCTTCTGGAGCAAATTAAAAACGGC GGCGATTTTGAGAAGCTGGCGAAGAAGCATTCTATCTGCCCATCCGGTAAAAAAGGCGGTCATTTAGGCGAATTTCGTCAGGG CCAGATGGTTCCGGCATTCGATAAAGTAGTCTTTTCCTGCCCGGTACTGGAGCCAACCGGCCCGCTGCATACCCAGTTCGGT TACCACATCATTAAAGTATTGTATCGCAAATAA (SEQ ID NO: 498)
57_jfrlSe1409-7_UN_O1,3,19_	ATGGCAAAAATGGCAGCAGCACTGCATATCCTTGTAAAAGAAGAGAAACTGGCTTTAGATCTTCTGGAGCAAATTAAAAACGGC GGCGATTTTGAGAAGCTGGCGAAGAAGCATTCTATCTGCCCATCCGGTAAAAAAGGCGGTCATTTAGGCGAATTTCGTCAGGG CCAGATGGTTCCGGCATTCGATAAAGTAGTCTTTTCCTGCCCGGTACTGGAGCCAACCGGCCCGCTGCATACCCAGTTCGGT TACCACATCATTAAAGTATTGTATCGCAAATAA (SEQ ID NO: 507)
58_jfrlSe1409-8_UN_O18_	ATGGCAAAAATGGCAGCAGCACTGCATATCCTTGTAAAAGAAGAGAAACTGGCTTTAGATCTTCTGGAGCAAATTAAAAACGGC GGCGATTTTGAGAAGCTGGCGAAGAAGCATTCTATCTGCCCATCCGGTAAAAAAGGCGGTCATTTAGGCGAATTTCGTCAGGG CCAGATGGTTCCGGCATTCGATAAAGTAGTCTTTTCCTGCCCGGTACTGGAGCCAACCGGCCCGCTGCATACTCAGTTCGGT TACCACATCATTAAAGTATTGTATCGCAAATAA (SEQ ID NO: 516)
61_jfrlSe1409-11_UN_O1,3,19_	ATGGCAAAAATGGCAGCAGCACTGCATATCCTTGTAAAAGAAGAGAAACTGGCTTTAGATCTTCTGGAGCAAATTAAAAACGGC GGCGATTTTGAGAAGCTGGCGAAGAAGCATTCTATCTGCCCATCCGGTAAAAAAGGCGGTCATTTAGGCGAATTTCGTCAGGG CCAGATGGTTCCGGCATTCGATAAAGTAGTCTTTTCCTGCCCGGTACTGGAGCCAACCGGCCCGCTGCATACCCAGTTCGGT TACCACATCATTAAAGTATTGTATCGCAAATAA (SEQ ID NO: 525)
67_jfrlSe1409-17_Rissen	ATGGCAAAAATGGCAGCAGCACTGCATATTCTTGTAAAAGAAGAGAAACTGGCTTTAGATCTTCTGGAGCAAATTAAAAACGGC GGCGATTTTGAGAAGCTGGCGAAGAAGCATTCTATCTGCCCATCCGGTAAAAAAGGCGGTCATTTAGGCGAATTTCGTCAGGG CCAGATGGTTCCGGCATTCGATAAAGTAGTCTTTTCCTGCCCGGTACTGGAGCCAACCGGCCCGCTGCATACTCAGTTCGGT TACCACATCATTAAAGTATTGTATCGCAAATAA (SEQ ID NO: 533)
71_jfrlSe1409-21_Amsterdam	ATGGCAAAAATGGCAGCAGCACTGCATATCCTTGTAAAAGAAGAGAAACTGGCTTTAGATCTTCTGGAGCAAATTAAAAACGGC GGCGATTTTGAGAAGCTGGCGAAGAAGCATTCTATCTGCCCATCCGGTAAAAAAGGCGGTCATTTAGGCGAATTTCGTCAGGG CCAGATGGTTCCGGCATTCGATAAAGTAGTCTTTTCCTGCCCGGTACTGGAGCCAACCGGCCCGCTGCATACCCAGTTCGGT TACCACATCATTAAGGTATTGTATCGTAAATAA (SEQ ID NO: 543)
80_HySe09_Enteritidis	ATGGCAAAAATGGCAGCAGCACTGCATATTCTTGTAAAAGAAGAGAAACTAGCTTTAGATCTTCTGGAGCAAATTAAAAACGGC GGCGATTTTGAGAAGCTGGCGAAGAAGCATTCTATCTGCCCATCCGGTAAAAAAGGCGGTCATTTAGGCGAATTTCGTCAGGG CCAGATGGTTCCGGCATTCGATAAAGTAGTCTTTTCCTGCCCGGTACTGGAGCCAACCGGCCCGCTGCATACCCAGTTCGGT TACCACATCATTAAAGTATTGTATCGCAAATAA (SEQ ID NO: 553)
100_HySe29_Schwarzengrund	ATGGCAAAAATGGCAGCAGCACTGCATATTCTTGTAAAAGAAGAGAAACTGGCTTTAGATCTTCTGGAGCAAATTAAAAACGGC GGCGATTTTGAGAAGCTGGCGAAGAAGCATTCTATCTGCCCATCCGGTAAAAAAGGCGGTCATTTAGGCGAATTTCGTCAGGG CCAGATGGTTCCGGCATTCGATAAAGTAGTATTTTCCTGCCCGGTACTGGAGCCAACCGGCCCGCTGCATACCCAGTTCGGT TACCACATCATTAAAGTATTGTATCGCAAATAA (SEQ ID NO: 561)
103_HySe32_Schwarzengrund	ATGGCAAAAATGGCAGCAGCACTGCATATTCTTGTAAAAGAAGAGAAACTGGCTTTAGATCTTCTGGAGCAAATTAAAAACGGC GGCGATTTTGAGAAGCTGGCGAAGAAGCATTCTATCTGCCCATCCGGTAAAAAAGGCGGTCATTTAGGCGAATTTCGTCAGGG CCAGATGGTTCCGGCATTCGATAAAGTAGTATTTTCCTGCCCGGTACTGGAGCCAACCGGCCCGCTGCATACCCAGTTCGGT TACCACATCATTAAAGTATTGTATCGCAAATAA (SEQ ID NO: 568)
106_HySe35_Schwarzengrund	ATGGCAAAAATGGCAGCAGCACTGCATATTCTTGTAAAAGAAGAGAAACTGGCTTTAGATCTTCTGGAGCAAATTAAAAACGGC GGCGATTTTGAGAAGCTGGCGAAGAAGCATTCTATCTGCCCATCCGGTAAAAAAGGCGGTCATTTAGGCGAATTTCGTCAGGG CCAGATGGTTCCGGCATTCGATAAAGTAGTATTTTCCTGCCCGGTACTGGAGCCAACCGGCCCGCTGCATACCCAGTTCGGT TACCACATCATTAAAGTATTGTATCGCAAATAA (SEQ ID NO: 575)

Fig. 34

	yaiA
01_NBRC13245T_Typhimurium	ATGCCAACCAGACCACCTTATCCGCGGGAAGCTTATATCGTCACCATTGAAAAAGGCACGCCGGGCCAGACGGTGACGTGGTATCAGCTACGGGCTGACCATCCGAAACCTGATTCGCTCATCAGCGAGCATCCGACCGCAGAAGAAGCGATGGATGCGAAAAATCGTTACGAAGATCCGGATAAATCATAG (SEQ ID NO: 64)
02_GTC00131_Enteritidis	ATGCCAACCAGACCACCTTATCCGCGGGAAGCTTATATCGTCACCATTGAAAAAGGCACGCCGGGCCAGACGGTGACGTGGTATCAGCTACGGGCTGACCATCCGAAACCTGATTCGCTCATCAGCGAGCATCCGACCGCAGAAGAAGCGATGGATGCGAAAAAACGTTACGAAGATCCGGATAAATCATAG (SEQ ID NO: 73)
03_GTC09491_Enteritidis	
04_GTC03838_Enteritidis	
05_GTC08914_Enteritidis	
06_GTC09421_Enteritidis	
07_GTC09489_Enteritidis	
08_NBRC3163_Pullorum_Gallinarum	
09_NBRC3313_Enteritidis	
10_NBRC12529_Typhimurium	
11_NBRC14193_Typhimurium	
12_NBRC14194_Typhimurium	
13_NBRC14209_Typhimurium	
14_NBRC14210_Typhimurium	
15_NBRC14211_Typhimurium	
16_NBRC14212_Typhimurium	
17_NBRC15181_Typhimurium	
18_NBRC15182_Minnesota	
19_NBRC15183_Minnesota	
20_NBRC15184_Minnesota	
21_NBRC15185_Minnesota	
22_NBRC15186_Minnesota	
23_NBRC15187_Minnesota	
24_NBRC100797_Abony	
25_NBRC105684_Choleraesuis	
26_NBRC105726_Typhimurium	
27_JCM3919_UN_O7_	
28_NBRC15335_Minnesota	
29_GTC09490_Enteritidis	
30_GTC09492_Braenderup	
31_GTC09493_Pakistan	
32_GTC09549_Typhimurium	
33_ATCCBAA-1675_Infantis	
34_ATCCBAA-1738_Thompson	
35_ATCC9712_Saintpaul	ATGCCAACCAGACCACCTTATCCGCGGGAAGCTTATATCGTCACCATTGAAAAAGGCACGCCGGGCCAGACGGTGACGTGGTATCAGCTACGGGCTGACCATCCGAAACCTGATTCGCTCATCAGCGAGCATCCGACCGCAGAAGAAGCGATGGATGCGAAAAAACGTTACGAAGATCCGGATAAATCATAG (SEQ ID NO: 338)
36_jfrlSe1402-1_Infantis	
38_jfrlSe1402-3_Brandenburg	
39_jfrlSe1402-4_Infantis	
40_jfrlSe1402-5_Brandenburg	
41_jfrlSe1402-6_Rissen	
42_jfrlSe1402-7_Orion	
43_jfrlSe1402-8_Rissen	
44_jfrlSe1402-9_Rissen	
45_jfrlSe1402-10_Rissen	
46_jfrlSe1402-11_Rissen	
47_jfrlSe1402-12_Rissen	
48_jfrlSe1402-13_Mbandaka	
49_jfrlSe1402-14_Mbandaka	
50_jfrlSe1402-15_Orion	
51_jfrlSe1409-1_Altona	
52_jfrlSe1409-2_Istanbul	
53_jfrlSe1409-3_Senftenberg	
54_jfrlSe1409-4_UN_O13_	
55_jfrlSe1409-5_UN_O1,3,19_	
56_jfrlSe1409-6_Montevideo	
57_jfrlSe1409-7_UN_O1,3,19_	
58_jfrlSe1409-8_UN_O18_	
61_jfrlSe1409-11_UN_O1,3,19_	
67_jfrlSe1409-17_Rissen	
71_jfrlSe1409-21_Amsterdam	
80_HySe09_Enteritidis	
100_HySe29_Schwarzengrund	
103_HySe32_Schwarzengrund	
106_HySe35_Schwarzengrund	

Fig. 35A

	yciF
01_NBRC13245T_Typhimurium	ATGAATATCAAAACCGTTGAAGACCTTTTTATCCATCTACTTTCAGATACCTATAGTGCAGAAAAACAATTAACCAAGGCTCTTCCT AAACTTGCCAGAGCCACGTCCAATGAAAAATTAAGCCAGGCCTTTCAATCTCATCTTGAAGAAACCCAGGGTCAGATTGAACGTAT TGATCAGATCGTCGAATCTGAATCTGGCATTAAACTGAAAAGAATGAAATGCGTCGCTATGGAAGGGCTGATTGAAGAAGCCAATG AAGTCATCGAAAGTACGGAGAAAAACGAAGTACGCGATGCAGCGCTTATCGCCGCGGCGCAAAAAGTCGAGCATTACGAAATCGCC AGCTACGGCACGCTAGCCACCCTGGCCGAGCAGCTCGGCTATAGCAAAGCATTAAAACTGCTCAAAGAAACCCTCGACGAGGAAA AACAAACTGATTTAAAACTTACCGATTTAGCAGTCAGCAATGTTAATAAAAGTGCTGAACGCAAATCGAAATAA (SEQ ID NO: 65)
02_GTC00131_Enteritidis	ATGAATATCAAAACCGTTGAAGACCTTTTTATCCATCTACTTTCAGATACCTATAGTGCAGAAAAACAATTAACCAAGGCTCTTTCT AAACTAGCCAGAGCCACGTCCAATGAAAAATTAAGCCAGGCCTTTCAATCTCATCTTGAAGAAACCCAGGGTCAGATTGAACGTAT TGATCAGATCGTCGAATCTGAATCTGGCATTAAACTGAAAAGAATGAAATGCGTCGCTATGGAAGGGCTGATCGAAGAAGCCAATG AAGTCATCGAAAGTACGGAGAAAAACGAAGTACGCGATGCAGCGCTTATCGCCGCGGCGCAAAAAGTCGAGCATTACGAAATCGCC AGCTACGGCACGCTAGCCACCCTGGCCGAGCAGCTCGGCTATAGCAAGGCATTAAAACTGCTCAAAGAAACCCTCGACGAGGAAA AACAAACTGATTTAAAACTTACCGATTTAGCAGTCAGCAATGTTAATAAAAGTGCTGAACGCAAATCGAAATAA (SEQ ID NO: 74)
03_GTC09491_Enteritidis	
04_GTC03838_Enteritidis	
05_GTC08914_Enteritidis	
06_GTC09421_Enteritidis	
07_GTC09489_Enteritidis	
08_NBRC3163_Pullorum_Gallinarum	
09_NBRC3313_Enteritidis	
10_NBRC12529_Typhimurium	
11_NBRC14193_Typhimurium	
12_NBRC14194_Typhimurium	
13_NBRC14209_Typhimurium	
14_NBRC14210_Typhimurium	
15_NBRC14211_Typhimurium	
16_NBRC14212_Typhimurium	
17_NBRC15181_Typhimurium	
18_NBRC15182_Minnesota	
19_NBRC15183_Minnesota	
20_NBRC15184_Minnesota	
21_NBRC15185_Minnesota	
22_NBRC15186_Minnesota	
23_NBRC15187_Minnesota	
24_NBRC100797_Abony	
25_NBRC105684_Choleraesuis	
26_NBRC105726_Typhimurium	
27_JCM3919_UN_O7_	
28_NBRC15335_Minnesota	
29_GTC09490_Enteritidis	
30_GTC09492_Braenderup	
31_GTC09493_Pakistan	
32_GTC09549_Typhimurium	
33_ATCCBAA-1675_Infantis	ATGAATATTAAAACCGTTGAAGACCTTTTTATCCATCTACTTTCAGATACCTATAGTGCAGAAAAACAATTAACCAAGGCTCTTTCT AAACTAGCCAGAGCCACGTCCAATGAAAAATTAAGCCAGGCCTTTCAATCTCATCTTGAAGAAACCCAGGGTCAGATTGAACGTAT TGATCAGATCGTCGAATCTGAATCTGGCATTAAACTGAAAAGAATGAAATGCGTCGCTATGGAAGGGCTGATTGAAGAAGCCAATG AAGTCATCGAAAGTACGGAGAAAAACGAAGTACGCGATGCAGCGCTTATCGCCGCGGCGCAAAAAGTCGAGCATTACGAAATCGCC AGCTACGGCACGCTTGCCACCCTGGCCGAGCAGCTCGGCTATAGCAAAGCATTAAAACTGCTCAAAGAAACCCTCGACGAGGAAA AACAAACTGATTTAAAACTTACCGATTTAGCAGTCAGCAATGTTAATAAAAGTGCTGAACGCAAATCGAAATAA (SEQ ID NO: 317)
34_ATCCBAA-1738_Thompson	
35_ATCC9712_Saintpaul	
36_jfrlSe1402-1_Infantis	
38_jfrlSe1402-3_Brandenburg	
39_jfrlSe1402-4_Infantis	
40_jfrlSe1402-5_Brandenburg	
41_jfrlSe1402-6_Rissen	
42_jfrlSe1402-7_Orion	
43_jfrlSe1402-8_Rissen	
44_jfrlSe1402-9_Rissen	
45_jfrlSe1402-10_Rissen	
46_jfrlSe1402-11_Rissen	
47_jfrlSe1402-12_Rissen	
48_jfrlSe1402-13_Mbandaka	ATGAATATCAAAACCGTTGAAGACCTTTTTATCCATCTACTTTCAGATACCTATAGTGCAGAAAAACAATTAACCAAGGCTCTTCCT AAACTTGCCAGAGCCACGTCCAATGAAAAATTAAGCCAGGCCTTTCAATCTCATCTTGAAGAAACCCAGGGTCAGATTGAACGTAT TGATCAGATCGTCGAATCTGAATCTGGCATTAAACTGAAAAGAATGAAATGCGTCGCTATGGAAGGGCTGATTGAAGAAGCCAATG AAGTCATCGAAAGTACGGAGAAAAACGAAGTACGCGATGCAGCGCTTATCGCCGCGGCGCAAAAAGTCGAGCATTACGAAATCGCC AGCTACGGCACGCTAGCCACCCTGGCCGAGCAGCTCGGCTATAGCAAAGCATTAAAACTGCTCAAAGAAACCCTCGACGAGGAAA AACAAACTGATTTAAAACTTACCGATTTAGCAGTCAGCAATGTTAATAAAAGTGCTGAACGCAAATCGAAATAA (SEQ ID NO: 434)
49_jfrlSe1402-14_Mbandaka	
50_jfrlSe1402-15_Orion	
51_jfrlSe1409-1_Altona	
52_jfrlSe1409-2_Istanbul	
53_jfrlSe1409-3_Senftenberg	ATGAATATCAAAACCGTTGAAGACCTTTTTATCCATCTACTTTCAGATACCTATAGTGCAGAAAAACAATTAACCAAGGCTCTTCCT AAACTTGCCAGAGCCACGTCCAATGAAAAATTAAGCCAGGCCTTTCAATCTCATCTTGAAGAAACCCAGGGTCAGATTGAACGTAT TGATCAGATCGTCGAATCTGAATCTGGCATTAAACTGAAAAGAATAAAATGCGTCGCTATGGAAGGGCTGATTGAAGAAGCCAATGA AGTCATCGAAAGTACGGAGAAAAACGAAGTACGCGATGCAGCGCTTATCGCCGCGGCGCAAAAAGTCGAGCATTACGAAATCGCCA GCTATGGCACGCTAGCCACCCTGGCCGAGCAGCTCGGCTATGGCAAGGCATTAAAACTGCTTAAAGAAACCCTCGACGAGGAAAA ACAAACTGATTTAAAACTTACCGATTTAGCAGTCAGCAATGTTAATAAAAGTGCTGAACGCAAATCGAAATAA (SEQ ID NO: 474)
54_jfrlSe1409-4_UN_O13_	
55_jfrlSe1409-5_UN_O1,3,19_	
56_jfrlSe1409-6_Montevideo	
57_jfrlSe1409-7_UN_O1,3,19_	
58_jfrlSe1409-8_UN_O18_	
61_jfrlSe1409-11_UN_O1,3,19_	
67_jfrlSe1409-17_Rissen	
71_jfrlSe1409-21_Amsterdam	

Fig. 35B

80_HySe09_Enteritidis	
100_HySe29_Schwarzengrund	
103_HySe32_Schwarzengrund	
106_HySe35_Schwarzengrund	

Fig. 36A

	SOD
01_NBRC13245T_Typhimurium	MSYTLPSLPYAYDALEPHFDKQTMEIHHTKHHQTYVNNANAALENLPEFASLPVEELITKLDQVPADKKTVLRNNAGGHANHSLFWKGLKKG TTLQGDLKAAIERDFGSVDNFKAEFEKAAATRFGSGWAWLVLKGDKLAVVSTANQDSPLMGEAISGASGFPILGLDVWEHAYYLKFQNRRPD YIKEFWNVVNWDEAAARFAAKK (SEQ ID NO:576)
02_GTC00131_Enteritidis	MSYTLPSLPYAYDALEPHFDKQTMEIHHTKHHQTYVNNANAALENLPEFADLPVEELITKLDQVPADKKTVLRNNAGGHANHSFFWKGLKKG TTLQGDLKAAIERDFGSVDNFKAEFEKAAATRFGSGWAWLVLKGDKLAVVSTANQDSPLMGEAISGASGFPILGLDVWEHAYYLKFQNRRPD YIKEFWNVVNWDEAAARFAAKK (SEQ ID NO:588)
03_GTC09491_Enteritidis	MSYTLPSLPYAYDALEPHFDKQTMEIHHTKHHQTYVNNANAALENLPEFADLPVEELITKLDQVPADKKTVLRNNAGGHANHSFFWKGLKKG TTLQGDLKAAIERDFGSVDNFKAEFEKAAATRFGSGWAWLVLKGDKLAVVSTANQDSPLMGEAISGASGFPILGLDVWEHAYYLKFQNRRPD YIKEFWNVVNWDEAAARFAAKK (SEQ ID NO:597)
04_GTC03838_Enteritidis	MSYTLPSLPYAYDALEPHFDKQTMEIHHTKHHQTYVNNANAALENLPEFADLPVEELITKLDQVPADKKTVLRNNAGGHANHSFFWKGLKKG TTLQGDLKAAIERDFGSVDNFKAEFEKAAATRFGSGWAWLVLKGDKLAVVSTANQDSPLMGEAISGASGFPILGLDVWEHAYYLKFQNRRPD YIKEFWNVVNWDEAAARFAAKK (SEQ ID NO:604)
05_GTC08914_Enteritidis	MSYTLPSLPYAYDALEPHFDKQTMEIHHTKHHQTYVNNANAALENLPEFADLPVEELITKLDQVPADKKTVLRNNAGGHANHSFFWKGLKKG TTLQGDLKAAIERDFGSVDNFKAEFEKAAATRFGSGWAWLVLKGDKLAVVSTANQDSPLMGEAISGASGFPILGLDVWEHAYYLKFQNRRPD YIKEFWNVVNWDEAAARFAAKK (SEQ ID NO:611)
06_GTC09421_Enteritidis	MSYTLPSLPYAYDALEPHFDKQTMEIHHTKHHQTYVNNANAALENLPEFADLPVEELITKLDQVPADKKTVLRNNAGGHANHSFFWKGLKKG TTLQGDLKAAIERDFGSVDNFKAEFEKAAATRFGSGWAWLVLKGDKLAVVSTANQDSPLMGEAISGASGFPILGLDVWEHAYYLKFQNRRPD YIKEFWNVVNWDEAAARFAAKK (SEQ ID NO:621)
07_GTC09489_Enteritidis	MSYTLPSLPYAYDALEPHFDKQTMEIHHTKHHQTYVNNANAALENLPEFADLPVEELITKLDQVPADKKTVLRNNAGGHANHSFFWKGLKKG TTLQGDLKAAIERDFGSVDNFKAEFEKAAATRFGSGWAWLVLKGDKLAVVSTANQDSPLMGEAISGASGFPILGLDVWEHAYYLKFQNRRPD YIKEFWNVVNWDEAAARFAAKK (SEQ ID NO:630)
08_NBRC3163_Pullorum_Gallinarum	MSYTLPSLPYAYDALEPHFDKQTMEIHHTKHHQTYVNNANAALENLPEFADLPVEELITKLDQVPADKKTVLRNNAGGHANHSFFWKGLKKG TTLQGDLKAAIERDFGSVDNFKAEFEKAAATRFGSGWAWLVLKGDKLAVVSTANQDSPLMGEAISGASGFPILGLDVWEHAYYLKFQNRRPD YIKEFWNVVNWDEAAARFAAKK (SEQ ID NO:637)
09_NBRC3313_Enteritidis	MSYTLPSLPYAYDALEPHFDKQTMEIHHTKHHQTYVNNANAALENLPEFADLPVEELITKLDQVPADKKTVLRNNAGGHANHSFFWKGLKKG TTLQGDLKAAIERDFGSVDNFKAEFEKAAATRFGSGWAWLVLKGDKLAVVSTANQDSPLMGEAISGASGFPILGLDVWEHAYYLKFQNRRPD YIKEFWNVVNWDEAAARFAAKK (SEQ ID NO:647)
10_NBRC12529_Typhimurium	MSYTLPSLPYAYDALEPHFDKQTMEIHHTKHHQTYVNNANAALENLPEFASLPVEELITKLDQVPADKKTVLRNNAGGHANHSLFWKGLKKG TTLQGDLKAAIERDFGSVDNFKAEFEKAAATRFGSGWAWLVLKGDKLAVVSTANQDSPLMGEAISGASGFPILGLDVWEHAYYLKFQNRRPD YIKEFWNVVNWDEAAARFAAKK (SEQ ID NO:654)
11_NBRC14193_Typhimurium	MSYTLPSLPYAYDALEPHFDKQTMEIHHTKHHQTYVNNANAALENLPEFASLPVEELITKLDQVPADKKTVLRNNAGGHANHSLFWKGLKKG TTLQGDLKAAIERDFGSVDNFKAEFEKAAATRFGSGWAWLVLKGDKLAVVSTANQDSPLMGEAISGASGFPILGLDVWEHAYYLKFQNRRPD YIKEFWNVVNWDEAAARFAAKK (SEQ ID NO:661)
12_NBRC14194_Typhimurium	MSYTLPSLPYAYDALEPHFDKQTMEIHHTKHHQTYVNNANAALENLPEFASLPVEELITKLDQVPADKKTVLRNNAGGHANHSLFWKGLKKG TTLQGDLKAAIERDFGSVDNFKAEFEKAAATRFGSGWAWLVLKGDKLAVVSTANQDSPLMGEAISGASGFPILGLDVWEHAYYLKFQNRRPD YIKEFWNVVNWDEAAARFAAKK (SEQ ID NO:668)
13_NBRC14209_Typhimurium	MSYTLPSLPYAYDALEPHFDKQTMEIHHTKHHQTYVNNANAALENLPEFASLPVEELITKLDQVPADKKTVLRNNAGGHANHSLFWKGLKKG TTLQGDLKAAIERDFGSVDNFKAEFEKAAATRFGSGWAWLVLKGDKLAVVSTANQDSPLMGEAISGASGFPILGLDVWEHAYYLKFQNRRPD YIKEFWNVVNWDEAAARFAAKK (SEQ ID NO:675)
14_NBRC14210_Typhimurium	MSYTLPSLPYAYDALEPHFDKQTMEIHHTKHHQTYVNNANAALENLPEFASLPVEELITKLDQVPADKKTVLRNNAGGHANHSLFWKGLKKG TTLQGDLKAAIERDFGSVDNFKAEFEKAAATRFGSGWAWLVLKGDKLAVVSTANQDSPLMGEAISGASGFPILGLDVWEHAYYLKFQNRRPD YIKEFWNVVNWDEAAARFAAKK (SEQ ID NO:682)
15_NBRC14211_Typhimurium	MSYTLPSLPYAYDALEPHFDKQTMEIHHTKHHQTYVNNANAALENLPEFASLPVEELITKLDQVPADKKTVLRNNAGGHANHSLFWKGLKKG TTLQGDLKAAIERDFGSVDNFKAEFEKAAATRFGSGWAWLVLKGDKLAVVSTANQDSPLMGEAISGASGFPILGLDVWEHAYYLKFQNRRPD YIKEFWNVVNWDEAAARFAAKK (SEQ ID NO:689)
16_NBRC14212_Typhimurium	MSYTLPSLPYAYDALEPHFDKQTMEIHHTKHHQTYVNNANAALENLPEFASLPVEELITKLDQVPADKKTVLRNNAGGHANHSLFWKGLKKG TTLQGDLKAAIERDFGSVDNFKAEFEKAAATRFGSGWAWLVLKGDKLAVVSTANQDSPLMGEAISGASGFPILGLDVWEHAYYLKFQNRRPD YIKEFWNVVNWDEAAARFAAKK (SEQ ID NO:695)
17_NBRC15181_Typhimurium	MSYTLPSLPYAYDALEPHFDKQTMEIHHTKHHQTYVNNANAALENLPEFASLPVEELITKLDQVPADKKTVLRNNAGGHANHSLFWKGLKKG TTLQGDLKAAIERDFGSVDNFKAEFEKAAATRFGSGWAWLVLKGDKLAVVSTANQDSPLMGEAISGASGFPILGLDVWEHAYYLKFQNRRPD YIKEFWNVVNWDEAAARFAAKK (SEQ ID NO:702)
18_NBRC15182_Minnesota	MSYTLPSLPYAYDALEPHFDKQTMEIHHTKHHQTYVNNANAALENLPEFADLPVEELITKLDQVPADKKTVLRNNAGGHANHSLFWKGLKKG TTLQGDLKAAIERDFGSVDNFKAEFEKAAATRFGSGWAWLVLKGDKLAVVSTANQDSPLMGEAISGASGFPILGLDVWEHAYYLKFQNRRPD YIKEFWNVVNWDEAAARFAAKK (SEQ ID NO:712)
19_NBRC15183_Minnesota	MSYTLPSLPYAYDALEPHFDKQTMEIHHTKHHQTYVNNANAALENLPEFADLPVEELITKLDQVPADKKTVLRNNAGGHANHSLFWKGLKKG TTLQGDLKAAIERDFGSVDNFKAEFEKAAATRFGSGWAWLVLKGDKLAVVSTANQDSPLMGEAISGASGFPILGLDVWEHAYYLKFQNRRPD YIKEFWNVVNWDEAAARFAAKK (SEQ ID NO:722)
20_NBRC15184_Minnesota	MSYTLPSLPYAYDALEPHFDKQTMEIHHTKHHQTYVNNANAALENLPEFADLPVEELITKLDQVPADKKTVLRNNAGGHANHSLFWKGLKKG TTLQGDLKAAIERDFGSVDNFKAEFEKAAATRFGSGWAWLVLKGDKLAVVSTANQDSPLMGEAISGASGFPILGLDVWEHAYYLKFQNRRPD YIKEFWNVVNWDEAAARFAAKK (SEQ ID NO:729)
21_NBRC15185_Minnesota	MSYTLPSLPYAYDALEPHFDKQTMEIHHTKHHQTYVNNANAALENLPEFADLPVEELITKLDQVPADKKTVLRNNAGGHANHSLFWKGLKKG TTLQGDLKAAIERDFGSVDNFKAEFEKAAATRFGSGWAWLVLKGDKLAVVSTANQDSPLMGEAISGASGFPILGLDVWEHAYYLKFQNRRPD YIKEFWNVVNWDEAAARFAAKK (SEQ ID NO:736)
22_NBRC15186_Minnesota	MSYTLPSLPYAYDALEPHFDKQTMEIHHTKHHQTYVNNANAALENLPEFADLPVEELITKLDQVPADKKTVLRNNAGGHANHSLFWKGLKKG TTLQGDLKAAIERDFGSVDNFKAEFEKAAATRFGSGWAWLVLKGDKLAVVSTANQDSPLMGEAISGASGFPILGLDVWEHAYYLKFQNRRPD YIKEFWNVVNWDEAAARFAAKK (SEQ ID NO:743)
23_NBRC15187_Minnesota	MSYTLPSLPYAYDALEPHFDKQTMEIHHTKHHQTYVNNANAALENLPEFADLPVEELITKLDQVPADKKTVLRNNAGGHANHSLFWKGLKKG TTLQGDLKAAIERDFGSVDNFKAEFEKAAATRFGSGWAWLVLKGDKLAVVSTANQDSPLMGEAISGASGFPILGLDVWEHAYYLKFQNRRPD YIKEFWNVVNWDEAAARFAAKK (SEQ ID NO:750
24_NBRC100797_Abony	MSYTLPSLPYAYDALEPHFDKQTMEIHHTKHHQTYVNNANAALENLPEFADLPVEELITKLDQVPADKKTVLRNNAGGHANHSFFWKGLKKG TTLQGDLKAAIERDFGSVDNFKAEFEKAAATRFGSGWAWLVLKGDKLAVVSTANQDSPLMGEAISGASGFPILGLDVWEHAYYLKFQNRRPD YIKEFWNVVNWDEAAARFAAKK (SEQ ID NO:757)
25_NBRC105684_Choleraesuis	MSYTLPSLPYAYDALEPHFDKQTMAIHHTKHHQTYVNNANAALENLPEFADLPVEELITKLDQVPADKKTVLRNNAGGHANHSLFWKGLKKG TTLQGDLKAAIERDFGSVDNFKAEFEKAAATRFGSGWAWLVLKGDKLAVVSTANQDSPLMGEAISGASGFPILGLDVWEHAYYLKFQNRRPD YIKEFWNVVNWDEAAARFAAKK (SEQ ID NO:767)
26_NBRC105726_Typhimurium	MSYTLPSLPYAYDALEPHFDKQTMEIHHTKHHQTYVNNANAALENLPEFASLPVEELITKLDQVPADKKTVLRNNAGGHANHSLFWKGLKKG TTLQGDLKAAIERDFGSVDNFKAEFEKAAATRFGSGWAWLVLKGDKLAVVSTANQDSPLMGEAISGASGFPILGLDVWEHAYYLKFQNRRPD YIKEFWNVVNWDEAAARFAAKK (SEQ ID NO:777)
27_JCM3919_UN_O7_	MSYTLPSLPYAYDALEPHFDKQTMEIHHTKHHQTYVNNANAALENLPEFADLPVEELITKLDQVPADKKTVLRNNAGGHANHSLFWKGLKKG TTLQGDLKAAIERDFGSVDNFKAEFEKAAATRFGSGWAWLVLKGDKLAVVSTANQDSPLMGEAISGASGFPILGLDVWEHAYYLKFQNRRPD YIKEFWNVVNWDEAAARFAAKK (SEQ ID NO:786)
28_NBRC15335_Minnesota	MSYTLPSLPYAYDALEPHFDKQTMEIHHTKHHQTYVNNANAALENLPEFADLPVEELITKLDQVPADKKTVLRNNAGGHANHSLFWKGLKKG TTLQGDLKAAIERDFGSVDNFKAEFEKAAATRFGSGWAWLVLKGDKLAVVSTANQDSPLMGEAISGASGFPILGLDVWEHAYYLKFQNRRPD YIKEFWNVVNWDEAAARFAAKK (SEQ ID NO:793)
29_GTC09490_Enteritidis	MSYTLPSLPYAYDALEPHFDKQTMEIHHTKHHQTYVNNANAALENLPEFADLPVEELITKLDQVPADKKTVLRNNAGGHANHSFFWKGLKKG TTLQGDLKAAIERDFGSVDNFKAEFEKAAATRFGSGWAWLVLKGDKLAVVSTANQDSPLMGEAISGASGFPILGLDVWEHAYYLKFQNRRPD YIKEFWNVVNWDEAAARFAAKK (SEQ ID NO:800)

Fig. 36B

30_GTC09492_Braenderup	MSYTLPSLPYAYDALEPHFDKQTMEIHHTKHHQTYVNNANAALENLPEFADLPVEELITKLDQVPADKKTVLRNNAGGHANHSLFWKGLKKG TTLQGDLKAAIERDFGSVDNFKAEFEKAAATRFGSGWAWLVLKGDKLAVVSTANQDSPLMGEAVSGASGFPILGLDVWEHAYYLKFQNRRPD YIKEFWNVVNWDEAAARFAAKK (SEQ ID NO:807)
31_GTC09493_Pakistan	MSYTLPSLPYAYDALEPHFDKQTMEIHHTKHHQTYVNNANAALENLPEFADLPVEELITKLDQVPADKKTVLRNNAGGHANHSFFWKGLKKG STLQGDLKAAIERDFGSVDNFKAEFEKAAATRFGSGWAWLVLKGDKLAVVSTANQDSPLMGEAISGASGFPILGLDVWEHAYYLKFQNRRPD YIKEFWNVVNWDEAAARFAAKK (SEQ ID NO:814)
32_GTC09549_Typhimurium	MSYTLPSLPYAYDALEPHFDKQTMEIHHTKHHQTYVNNANAALENLPEFASLPVEELITKLDQVPADKKTVLRNNAGGHANHSLFWKGLKKG TTLQGDLKAAIERDFGSVDNFKAEFEKAAATRFGSGWAWLVLKGDKLAVVSTANQDSPLMGEAISGASGFPILGLDVWEHAYYLKFQNRRPD YIKEFWNVVNWDEAAARFAAKK (SEQ ID NO:821)
33_ATCCBAA-1675_Infantis	MSYTLPSLPYAYDALEPHFDKQTMEIHHTKHHQTYVNNANAALENLPEFADLPVEELITKLDQVPADKKTVLRNNAGGHANHSLFWKGLKKG TTLQGDLKAAIERDFGSVDNFKAEFEKAAATRFGSGWAWLVLKGDKLAVVSTANQDSPLMGEAISGASGFPILGLDVWEHAYYLKFQNRRPD YIKEFWNVVNWDEAAARFAAKK (SEQ ID NO:828)
34_ATCCBAA-1738_Thompson	MSYTLPSLPYAYDALEPHFDKQTMEIHHTKHHQTYVNNANAALENLPEFADLPVEELITKLDQVPADKKTVLRNNAGGHANHSLFWKGLKKG TTLQGDLKAAIERDFGSVDNFKAEFEKAAATRFGSGWAWLVLKGDKLAVVSTANQDSPLMGEAISGASGFPILGLDVWEHAYYLKFQNRRPD YIKEFWNVVNWDEAAARFAAKK (SEQ ID NO:839)
35_ATCC9712_Saintpaul	MSYTLPSLPYAYDALEPHFDKQTMEIHHTKHHQTYVNNANAALENLPEFASLPVEELITKLDQVPADKKTVLRNNAGGHANHSLFWKGLKKG TTLQGDLKAAIERDFGSVDNFKAEFEKAAATRFGSGWAWLVLKGDKLAVVSTANQDSPLMGEAISGASGFPILGLDVWEHAYYLKFQNRRPD YIKEFWNVVNWDEAAARFAAKK (SEQ ID NO:849)
36_jfrlSe1402-1_Infantis	MSYTLPSLPYAYDALEPHFDKQTMEIHHTKHHQTYVNNANAALENLPEFADLPVEELITKLDQVPADKKTVLRNNAGGHANHSLFWKGLKKG TTLQGDLKAAIERDFGSVDNFKAEFEKAAATRFGSGWAWLVLKGDKLAVVSTANQDSPLMGEAISGASGFPILGLDVWEHAYYLKFQNRRPD YIKEFWNVVNWDEAAARFAAKK (SEQ ID NO:860)
38_jfrlSe1402-3_Brandenburg	MSYTLPSLPYAYDALEPHFDKQTMEIHHTKHHQTYVNNANAALENLPEFADLPVEELITKLDQVPADKKTVLRNNAGGHANHSLFWKGLKKG TTLQGDLKAAIERDFGSVDNFKAEFEKAAATRFGSGWAWLVLKGDKLAVVSTANQDSPLMGEAISGASGFPILGLDVWEHAYYLKFQNRRPD YIKEFWNVVNWDEAAARFAAKK (SEQ ID NO:870)
39_jfrlSe1402-4_Infantis	MSYTLPSLPYAYDALEPHFDKQTMEIHHTKHHQTYVNNANAALENLPEFADLPVEELITKLDQVPADKKTVLRNNAGGHANHSLFWKGLKKG TTLQGDLKAAIERDFGSVDNFKAEFEKAAATRFGSGWAWLVLKGDKLAVVSTANQDSPLMGEAISGASGFPILGLDVWEHAYYLKFQNRRPD YIKEFWNVVNWDEAAARFAAKK (SEQ ID NO:877)
40_jfrlSe1402-5_Brandenburg	MSYTLPSLPYAYDALEPHFDKQTMEIHHTKHHQTYVNNANAALENLPEFADLPVEELITKLDQVPADKKTVLRNNAGGHANHSLFWKGLKKG TTLQGDLKAAIERDFGSVDNFKAEFEKAAATRFGSGWAWLVLKGDKLAVVSTANQDSPLMGEAISGASGFPILGLDVWEHAYYLKFQNRRPD YIKEFWNVVNWDEAAARFAAKK (SEQ ID NO:887)
41_jfrlSe1402-6_Rissen	MSYTLPSLPYAYDALEPHFDKQTMEIHHTKHHQTYVNNANAALENLPEFADLPVEELITKLDQVPADKKTVLRNNAGGHANHSLFWKGLKKG TTLQGDLKVAIERDFGSVDNFKAEFEKAAATRFGSGWAWLVLKGDKLAVVSTANQDSPLMGEAISGASGFPILGLDVWEHAYYLKFQNRRPD YIKEFWNVVNWDEAAARFAAKK (SEQ ID NO:894)
42_jfrlSe1402-7_Orion	MSYTLPSLPYAYDALEPHFDKQTMEIHHTKHHQTYVNNANAALENLPEFASLPVEELITKLDQVPADKKTVLRNNAGGHANHSLFWKGLKKG TTLQGDLKAAIERDFGSVDNFKAEFEKAAATRFGSGWAWLVLKGDKLAVVSTANQDSPLMGEAISGASGFPILGLDVWEHAYYLKFQNRRPD YIKEFWNVVNWDEAAARFAAKK (SEQ ID NO:901)
43_jfrlSe1402-8_Rissen	MSYTLPSLPYAYDALEPHFDKQTMEIHHTKHHQTYVNNANAALENLPEFADLPVEELITKLDQVPADKKTVLRNNAGGHANHSLFWKGLKKG TTLQGDLKVAIERDFGSVDNFKAEFEKAAATRFGSGWAWLVLKGDKLAVVSTANQDSPLMGEAISGASGFPILGLDVWEHAYYLKFQNRRPD YIKEFWNVVNWDEAAARFAAKK (SEQ ID NO:911)
44_jfrlSe1402-9_Rissen	MSYTLPSLPYAYDALEPHFDKQTMEIHHTKHHQTYVNNANAALENLPEFADLPVEELITKLDQVPADKKTVLRNNAGGHANHSLFWKGLKKG TTLQGDLKVAIERDFGSVDNFKAEFEKAAATRFGSGWAWLVLKGDKLAVVSTANQDSPLMGEAISGASGFPILGLDVWEHAYYLKFQNRRPD YIKEFWNVVNWDEAAARFAAKK (SEQ ID NO:918)
45_jfrlSe1402-10_Rissen	MSYTLPSLPYAYDALEPHFDKQTMEIHHTKHHQTYVNNANAALENLPEFADLPVEELITKLDQVPADKKTVLRNNAGGHANHSLFWKGLKKG TTLQGDLKVAIERDFGSVDNFKAEFEKAAATRFGSGWAWLVLKGDKLAVVSTANQDSPLMGEAISGASGFPILGLDVWEHAYYLKFQNRRPD YIKEFWNVVNWDEAAARFAAKK (SEQ ID NO:925)
46_jfrlSe1402-11_Rissen	MSYTLPSLPYAYDALEPHFDKQTMEIHHTKHHQTYVNNANAALENLPEFADLPVEELITKLDQVPADKKTVLRNNAGGHANHSLFWKGLKKG TTLQGDLKVAIERDFGSVDNFKAEFEKAAATRFGSGWAWLVLKGDKLAVVSTANQDSPLMGEAISGASGFPILGLDVWEHAYYLKFQNRRPD YIKEFWNVVNWDEAAARFAAKK (SEQ ID NO:932)
47_jfrlSe1402-12_Rissen	MSYTLPSLPYAYDALEPHFDKQTMEIHHTKHHQTYVNNANAALENLPEFADLPVEELITKLDQVPADKKTVLRNNAGGHANHSLFWKGLKKG TTLQGDLKVAIERDFGSVDNFKAEFEKAAATRFGSGWAWLVLKGDKLAVVSTANQDSPLMGEAISGASGFPILGLDVWEHAYYLKFQNRRPD YIKEFWNVVNWDEAAARFAAKK (SEQ ID NO:939)
48_jfrlSe1402-13_Mbandaka	MSYTLPSLPYAYDALEPHFDKQTMEIHHTKHHQTYVNNANAALENLPEFADLPVEELITKLDQVPADKKTVLRNNAGGHANHSLFWKGLKKG TTLQGDLKAAIERDFGSVDNFKAEFEKAAATRFGSGWAWLVLKGDKLAVVSTANQDSPLMGEAISGASGFPILGLDVWEHAYYLKFQNRRPD YIKEFWNVVNWDEAAARFAAKK (SEQ ID NO:946)
49_jfrlSe1402-14_Mbandaka	MSYTLPSLPYAYDALEPHFDKQTMEIHHTKHHQTYVNNANAALENLPEFADLPVEELITKLDQVPADKKTVLRNNAGGHANHSLFWKGLKKG TTLQGDLKAAIERDFGSVDNFKAEFEKAAATRFGSGWAWLVLKGDKLAVVSTANQDSPLMGEAISGASGFPILGLDVWEHAYYLKFQNRRPD YIKEFWNVVNWDEAAARFAAKK (SEQ ID NO:956)
50_jfrlSe1402-15_Orion	MSYTLPSLPYAYDALEPHFDKQTMEIHHTKHHQTYVNNANAALENLPEFASLPVEELITKLDQVPADKKTVLRNNAGGHANHSLFWKGLKKG TTLQGDLKAAIERDFGSVDNFKAEFEKAAATRFGSGWAWLVLKGDKLAVVSTANQDSPLMGEAISGASGFPILGLDVWEHAYYLKFQNRRPD YIKEFWNVVNWDEAAARFAAKK (SEQ ID NO:966)
51_jfrlSe1409-1_Altona	MSYTLPSLPYAYDALEPHFDKQTMEIHHTKHHQTYVNNANAALENLPEFADLPVEELITKLDQVPADKKTVLRNNAGGHANHSLFWKGLKKG TTLQGDLKAAIERDFGSVDNFKAEFEKAAATRFGSGWAWLVLKGDKLAVVSTANQDSPLMGEAISGASGFPILGLDVWEHAYYLKFQNRRPD YIKEFWNVVNWDEAAARFAAKK (SEQ ID NO:976)
52_jfrlSe1409-2_Istanbul	MSYTLPSLPYAYDALEPHFDKQTMEIHHTKHHQTYVNNANAALENLPEFADLPVEELITKLDQVPADKKTVLRNNAGGHANHSLFWKGLKKG TTLQGDLKAAIERDFGSVDNFKAEFEKAAATRFGSGWAWLVLKGDKLAVVSTANQDSPLMGEAISGASGFPILGLDVWEHAYYLKFQNRRPD YIKEFWNVVNWDEAAARFAAKK (SEQ ID NO:983)
53_jfrlSe1409-3_Senftenberg	
54_jfrlSe1409-4_UN_O13_	MSYTLPSLPYAYDALEPHFDKQTMEIHHTKHHQTYVNNANAALENLPEFADLPVEELITKLDQVPADKKTVLRNNAGGHANHSLFWKGLKKG TTLQGDLKAAIERDFGSVDNFKAEFEKAAATRFGSGWAWLVLKGDKLAVVSTANQDSPLMGEAISGASGFPILGLDVWEHAYYLKFQNRRPD YIKEFWNVVNWDEAAARFAAKK (SEQ ID NO:996)
55_jfrlSe1409-5_UN_O1,3,19_	MSYTLPSLPYAYDALEPHFDKQTMEIHHTKHHQTYVNNANAALENLPEFADLPVEELITKLDQVPADKKTVLRNNAGGHANHSLFWKGLKKG TTLQGDLKAAIERDFGSVDNFKAEFEKAAATRFGSGWAWLVLKGDKLAVVSTANQDSPLMGEAISGASGFPILGLDVWEHAYYLKFQNRRPD YIKEFWNVVNWDEAAARFAAKK (SEQ ID NO:1004)
56_jfrlSe1409-6_Montevideo	MSYTLPSLPYAYDALEPHFDKQTMEIHHTKHHQTYVNNANAALENLPEFADLPAEELITKLDQVPADKKTVLRNNAGGHANHSLFWKGLKKG TTLQGDLKAAIERDFGSVDNFKAEFEKAAATRFGSGWAWLVLKGDKLAVVSTANQDSPLMGEAISGASGFPILGLDVWEHAYYLKFQNRRPD YIKEFWNVVNWDEAAARFAAKK (SEQ ID NO:1013)
57_jfrlSe1409-7_UN_O1,3,19_	MSYTLPSLPYAYDALEPHFDKQTMEIHHTKHHQTYVNNANAALENLPEFADLPVEELITKLDQVPADKKTVLRNNAGGHANHSLFWKGLKKG TTLQGDLKAAIERDFGSVDNFKAEFEKAAATRFGSGWAWLVLKGDKLAVVSTANQDSPLMGEAISGASGFPILGLDVWEHAYYLKFQNRRPD YIKEFWNVVNWDEAAARFAAKK (SEQ ID NO:1020)
58_jfrlSe1409-8_UN_O18_	MSYTLPSLPYAYDALEPHFDKQTMEIHHTKHHQTYVNNANAALENLPEFASLPVEELITKLDQVPADKKTVLRNNAGGHANHSLFWKGLKKG TTLQGDLKAAIERDFGSVDNFKAEFEKAAATRFGSGWAWLVLKGDKLAVVSTANQDSPLMGEAISGASGFPILGLDVWEHAYYLKFQNRRPD YIKEFWNVVNWDEAAARFAAKK (SEQ ID NO:1029)
61_jfrlSe1409-11_UN_O1,3,19_	MSYTLPSLPYAYDALEPHFDKQTMEIHHTKHHQTYVNNANAALENLPEFADLPVEELITKLDQVPADKKTVLRNNAGGHANHSLFWKGLKKG TTLQGDLKAAIERDFGSVDNFKAEFEKAAATRFGSGWAWLVLKGDKLAVVSTANQDSPLMGEAISGASGFPILGLDVWEHAYYLKFQNRRPD YIKEFWNVVNWDEAAARFAAKK (SEQ ID NO:1038)
67_jfrlSe1409-17_Rissen	MSYTLPSLPYAYDALEPHFDKQTMEIHHTKHHQTYVNNANAALENLPEFADLPVEELITKLDQVPADKKTVLRNNAGGHANHSLFWKGLKKG TTLQGDLKVAIERDFGSVDNFKAEFEKAAATRFGSGWAWLVLKGDKLAVVSTANQDSPLMGEAISGASGFPILGLDVWEHAYYLKFQNRRPD YIKEFWNVVNWDEAAARFAAKK (SEQ ID NO:1047)
71_jfrlSe1409-21_Amsterdam	MSYTLPSLPYAYDALEPHFDKQTMEIHHTKHHQTYVNNANAALENLPEFADLPVEELITKLDQVPADKKTVLRNNAGGHANHSLFWKGLKKG TTLQGDLKAAIERDFGSVDNFKAEFEKAAATRFGSGWAWLVLKGDKLAVVSTANQDSPLMGEAISGASGFPILGLDVWEHAYYLKFQNRRPD YIKEFWNVVNWDEAAARFAAKK (SEQ ID NO:1055)
80_HySe09_Enteritidis	MSYTLPSLPYAYDALEPHFDKQTMEIHHTKHHQTYVNNANAALENLPEFADLPVEELITKLDQVPADKKTVLRNNAGGHANHSLFWKGLKKG TTLQGDLKAAIERDFGSVDNFKAEFEKAAATRFGSGWAWLVLKGDKLAVVSTANQDSPLMGEAISGASGFPILGLDVWEHAYYLKFQNRRPD YIKEFWNVVNWDEAAARFAAKK (SEQ ID NO:1065)

Fig. 36C

100_HySe29_Schwarzengrund	MSYTLPSLPYAYDALEPHFDKQTMEIHHTKHHQTYVNNANAALENLPEFADLPAEELITKLDQVPADKKTVLRNNAGGHANHSLFWKGLKKG TTLQGDLKAAIERDFGSVDNFKAEFEKAAATRFGSGWAWLVLKGDKLAVVSTANQDSPLMGEAISGASGFPILGLDVWEHAYYLKFQNRRPD YIKEFWNVVNWDEAAARFAAKK (SEQ ID NO:1075)
103_HySe32_Schwarzengrund	MSYTLPSLPYAYDALEPHFDKQTMEIHHTKHHQTYVNNANAALENLPEFADLPAEELITKLDQVPADKKTVLRNNAGGHANHSLFWKGLKKG TTLQGDLKAAIERDFGSVDNFKAEFEKAAATRFGSGWAWLVLKGDKLAVVSTANQDSPLMGEAISGASGFPILGLDVWEHAYYLKFQNRRPD YIKEFWNVVNWDEAAARFAAKK (SEQ ID NO:1083)
106_HySe35_Schwarzengrund	

Fig. 37A

	L17
01_NBRC13245T_Typhimurium	MRHRKSGRQLNRNSSHRQAMFRNMAGSLVRHEIIKTTLPKAKELRRVVEPLITLAKTDSVANRRLAFARTRDNEIVAKLFN ELGPRFASRAGGYTRILKCGFRAGDNAPMAYIELVDRSEKTEAAAE (SEQ ID NO:577)
02_GTC00131_Enteritidis	MRHRKSGRQLNRNSSHRQAMFRNMAGSLVRHEIIKTTLPKAKELRRVVEPLITLAKTDSVANRRLAFARTRDNEIVAKLFN ELGPRFASRAGGYTRILKCGFRAGDNAPMAYIELVDRSEKTEAAAE (SEQ ID NO:589)
03_GTC09491_Enteritidis	MRHRKSGRQLNRNSSHRQAMFRNMAGSLVRHEIIKTTLPKAKELRRVVEPLITLAKTDSVANRRLAFARTRDNEIVAKLFN ELGPRFASRAGGYTRILKCGFRAGDNAPMAYIELVDRSEKTEAAAE (SEQ ID NO:598)
04_GTC03838_Enteritidis	MRHRKSGRQLNRNSSHRQAMFRNMAGSLVRHEIIKTTLPKAKELRRVVEPLITLAKTDSVANRRLAFARTRDNEIVAKLFN ELGPRFASRAGGYTRILKCGFRAGDNAPMAYIELVDRSEKTEAAAE (SEQ ID NO:605)
05_GTC08914_Enteritidis	MRHRKSGRQLNRNSSHRQAMFRNMAGSLVRHEIIKTTLPKAKELRRVVEPLITLAKTDSVANRRLAFARTRDNEIVAKLFN ELGPRFASRAGGYTRILKCGFRAGDNAPMAYIELVDRSEKTEAAAE (SEQ ID NO:612)
06_GTC09421_Enteritidis	MRHRKSGRQLNRNSSHRQAMFRNMAGSLVRHEIIKTTLPKAKELRRVVEPLITLAKTDSVANRRLAFARTRDNEIVAKLFN ELGPRFASRAGGYTRILKCGFRAGDNAPMAYIELVDRSEKTEAAAE (SEQ ID NO:622)
07_GTC09489_Enteritidis	MRHRKSGRQLNRNSSHRQAMFRNMAGSLVRHEIIKTTLPKAKELRRVVEPLITLAKTDSVANRRLAFARTRDNEIVAKLFN ELGPRFASRAGGYTRILKCGFRAGDNAPMAYIELVDRSEKTEAAAE (SEQ ID NO:631)
08_NBRC3163_Pullorum_Gallinarum	MRHRKSGRQLNRNSSHRQAMFRNMAGSLVRHEIIKTTLPKAKELRRVVEPLITLAKTDSVANRRLAFARTRDNEIVAKLFN DLGPRFASRAGGYTRILKCGFRAGDNAPMAYIELVDRSEKTEAAAE (SEQ ID NO:638)
09_NBRC3313_Enteritidis	MRHRKSGRQLNRNSSHRQAMFRNMAGSLVRHEIIKTTLPKAKELRRVVEPLITLAKTDSVANRRLAFARTRDNEIVAKLFN ELGPRFASRAGGYTRILKCGFRAGDNAPMAYIELVDRSEKTEAAAE (SEQ ID NO:648)
10_NBRC12529_Typhimurium	MRHRKSGRQLNRNSSHRQAMFRNMAGSLVRHEIIKTTLPKAKELRRVVEPLITLAKTDSVANRRLAFARTRDNEIVAKLFN ELGPRFASRAGGYTRILKCGFRAGDNAPMAYIELVDRSEKTEAAAE (SEQ ID NO:655)
11_NBRC14193_Typhimurium	MRHRKSGRQLNRNSSHRQAMFRNMAGSLVRHEIIKTTLPKAKELRRVVEPLITLAKTDSVANRRLAFARTRDNEIVAKLFN ELGPRFASRAGGYTRILKCGFRAGDNAPMAYIELVDRSEKTEAAAE (SEQ ID NO:662)
12_NBRC14194_Typhimurium	MRHRKSGRQLNRNSSHRQAMFRNMAGSLVRHEIIKTTLPKAKELRRVVEPLITLAKTDSVANRRLAFARTRDNEIVAKLFN ELGPRFASRAGGYTRILKCGFRAGDNAPMAYIELVDRSEKTEAAAE (SEQ ID NO:669)
13_NBRC14209_Typhimurium	MRHRKSGRQLNRNSSHRQAMFRNMAGSLVRHEIIKTTLPKAKELRRVVEPLITLAKTDSVANRRLAFARTRDNEIVAKLFN ELGPRFASRAGGYTRILKCGFRAGDNAPMAYIELVDRSEKTEAAAE (SEQ ID NO:676)
14_NBRC14210_Typhimurium	MRHRKSGRQLNRNSSHRQAMFRNMAGSLVRHEIIKTTLPKAKELRRVVEPLITLAKTDSVANRRLAFARTRDNEIVAKLFN ELGPRFASRAGGYTRILKCGFRAGDNAPMAYIELVDRSEKTEAAAE (SEQ ID NO:683)
15_NBRC14211_Typhimurium	MRHRKSGRQLNRNSSHRQAMFRNMAGSLVRHEIIKTTLPKAKELRRVVEPLITLAKTDSVANRRLAFARTRDNEIVAKLFN ELGPRFASRAGGYTRILKCGFRAGDNAPMAYIELVDRSEKTEAAAE (SEQ ID NO:690)
16_NBRC14212_Typhimurium	MRHRKSGRQLNRNSSHRQAMFRNMAGSLVRHEIIKTTLPKAKELRRVVEPLITLAKTDSVANRRLAFARTRDNEIVAKLFN ELGPRFASRAGGYTRILKCGFRAGDNAPMAYIELVDRSEKTEAAAE (SEQ ID NO:696)
17_NBRC15181_Typhimurium	MRHRKSGRQLNRNSSHRQAMFRNMAGSLVRHEIIKTTLPKAKELRRVVEPLITLAKTDSVANRRLAFARTRDNEIVAKLFN ELGPRFASRAGGYTRILKCGFRAGDNAPMAYIELVDRSEKTEAAAE (SEQ ID NO:703)
18_NBRC15182_Minnesota	MRHRKSGRQLNRNSSHRQAMFRNMAGSLVRHEIIKTTLPKAKELRRVVEPLITLAKTDSVANRRLAFARTRDNEIVAKLFN ELGPRFASRAGGYTRILKCGFRAGDNAPMAYIELVDRSEKTEAAAE (SEQ ID NO:713)
19_NBRC15183_Minnesota	MRHRKSGRQLNRNSSHRQAMFRNMAGSLVRHEIIKTTLPKAKELRRVVEPLITLAKTDSVANRRLAFARTRDNEIVAKLFN ELGPRFASRAGGYTRILKCGFRAGDNAPMAYIELVDRSEKTEAAAE (SEQ ID NO:723)
20_NBRC15184_Minnesota	MRHRKSGRQLNRNSSHRQAMFRNMAGSLVRHEIIKTTLPKAKELRRVVEPLITLAKTDSVANRRLAFARTRDNEIVAKLFN ELGPRFASRAGGYTRILKCGFRAGDNAPMAYIELVDRSEKTEAAAE (SEQ ID NO:730)
21_NBRC15185_Minnesota	MRHRKSGRQLNRNSSHRQAMFRNMAGSLVRHEIIKTTLPKAKELRRVVEPLITLAKTDSVANRRLAFARTRDNEIVAKLFN ELGPRFASRAGGYTRILKCGFRAGDNAPMAYIELVDRSEKTEAAAE (SEQ ID NO:737)
22_NBRC15186_Minnesota	MRHRKSGRQLNRNSSHRQAMFRNMAGSLVRHEIIKTTLPKAKELRRVVEPLITLAKTDSVANRRLAFARTRDNEIVAKLFN ELGPRFASRAGGYTRILKCGFRAGDNAPMAYIELVDRSEKTEAAAE (SEQ ID NO:744)
23_NBRC15187_Minnesota	MRHRKSGRQLNRNSSHRQAMFRNMAGSLVRHEIIKTTLPKAKELRRVVEPLITLAKTDSVANRRLAFARTRDNEIVAKLFN ELGPRFASRAGGYTRILKCGFRAGDNAPMAYIELVDRSEKTEAAAE (SEQ ID NO:751)
24_NBRC100797_Abony	MRHRKSGRQLNRNSSHRQAMFRNMAGSLVRHEIIKTTLPKAKELRRVVEPLITLAKTDSVANRRLAFARTRDNEIVAKLFN ELGPRFASRAGGYTRILKCGFRAGDNAPMAYIELVDRSEKTEAAAE (SEQ ID NO:758)
25_NBRC105684_Choleraesuis	MRHRKSGRQLNRNSSHRQAMFRNMAGSLVRHEIIKTTLPKAKELRRVVEPLITLAKTDSVANRRLAFARTRDNEIVAKLFN ELGPRFASRAGGYTRILKCGFRAGDNAPMAYIELVDRSEKTEAAAE (SEQ ID NO:768)
26_NBRC105726_Typhimurium	MRHRKSGRQLNRNSSHRQAMFRNMAGSLVRHEIIKTTLPKAKELRRVVEPLITLAKTDSVANRRLAFARTRDNEIVAKLFN ELGPRFASRAGGYTRILKCGFRAGDNAPMAYIELVDRSEKTEAAAE (SEQ ID NO:778)
27_JCM3919_UN_O7_	MRHRKSGRQLNRNSSHRQAMFRNMAGSLVRHEIIKTTLPKAKELRRVVEPLITLAKTDSVANRRLAFARTRDNEIVAKLFN ELGPRFASRAGGYTRILKCGFRAGDNAPMAYIELVDRSEKTEAAAE (SEQ ID NO:787)
28_NBRC15335_Minnesota	MRHRKSGRQLNRNSSHRQAMFRNMAGSLVRHEIIKTTLPKAKELRRVVEPLITLAKTDSVANRRLAFARTRDNEIVAKLFN ELGPRFASRAGGYTRILKCGFRAGDNAPMAYIELVDRSEKTEAAAE (SEQ ID NO:794)
29_GTC09490_Enteritidis	MRHRKSGRQLNRNSSHRQAMFRNMAGSLVRHEIIKTTLPKAKELRRVVEPLITLAKTDSVANRRLAFARTRDNEIVAKLFN ELGPRFASRAGGYTRILKCGFRAGDNAPMAYIELVDRSEKTEAAAE (SEQ ID NO:801)
30_GTC09492_Braenderup	MRHRKSGRQLNRNSSHRQAMFRNMAGSLVRHEIIKTTLPKAKELRRVVEPLITLAKTDSVANRRLAFARTRDNEIVAKLFN ELGPRFASRAGGYTRILKCGFRAGDNAPMAYIELVDRSEKTEAAAE (SEQ ID NO:808)
31_GTC09493_Pakistan	MRHRKSGRQLNRNSSHRQAMFRNMAGSLVRHEIIKTTLPKAKELRRVVEPLITLAKTDSVANRRLAFARTRDNEIVAKLFN ELGPRFASRAGGYTRILKCGFRAGDNAPMAYIELVDRSEKTEAAAE (SEQ ID NO:815)
32_GTC09549_Typhimurium	MRHRKSGRQLNRNSSHRQAMFRNMAGSLVRHEIIKTTLPKAKELRRVVEPLITLAKTDSVANRRLAFARTRDNEIVAKLFN ELGPRFASRAGGYTRILKCGFRAGDNAPMAYIELVDRSEKTEAAAE (SEQ ID NO:822)
33_ATCCBAA-1675_Infantis	MRHRKSGRQLNRNSSHRQAMFRNMAGSLVRHEIIKTTLPKAKELRRVVEPLITLAKTDSVANRRLAFARTRDNEIVAKLFN ELGPRFASRAGGYTRILKCGFRAGDNAPMAYIELVDRSEKTEAAAE (SEQ ID NO:829)
34_ATCCBAA-1738_Thompson	MRHRKSGRQLNRNSSHRQAMFRNMAGSLVRHEIIKTTLPKAKELRRVVEPLITLAKTDSVANRRLAFARTRDNEIVAKLFN ELGPRFASRAGGYTRILKCGFRAGDNAPMAYIELVDRSEKTEAAAE (SEQ ID NO:840)
35_ATCC9712_Saintpaul	MRHRKSGRQLNRNSSHRQAMFRNMAGSLVRHEIIKTTLPKAKELRRVVEPLITLAKTDSVANRRLAFARTRDNEIVAKLFN ELGPRFASRAGGYTRILKCGFRAGDNAPMAYIELVDRSEKTEAAAE (SEQ ID NO:850)
36_jfrlSe1402-1_Infantis	MRHRKSGRQLNRNSSHRQAMFRNMAGSLVRHEIIKTTLPKAKELRRVVEPLITLAKTDSVANRRLAFARTRDNEIVAKLFN ELGPRFASRAGGYTRILKCGFRAGDNAPMAYIELVDRSEKTEAAAE (SEQ ID NO:861)
38_jfrlSe1402-3_Brandenburg	MRHRKSGRQLNRNSSHRQAMFRNMAGSLVRHEIIKTTLPKAKELRRVVEPLITLAKTDSVANRRLAFARTRDNEIVAKLFN ELGPRFASRAGGYTRILKCGFRAGDNAPMAYIELVDRSEKTEAAAE (SEQ ID NO:871)
39_jfrlSe1402-4_Infantis	MRHRKSGRQLNRNSSHRQAMFRNMAGSLVRHEIIKTTLPKAKELRRVVEPLITLAKTDSVANRRLAFARTRDNEIVAKLFN ELGPRFASRAGGYTRILKCGFRAGDNAPMAYIELVDRSEKTEAAAE (SEQ ID NO:878)
40_jfrlSe1402-5_Brandenburg	MRHRKSGRQLNRNSSHRQAMFRNMAGSLVRHEIIKTTLPKAKELRRVVEPLITLAKTDSVANRRLAFARTRDNEIVAKLFN ELGPRFASRAGGYTRILKCGFRAGDNAPMAYIELVDRSEKTEAAAE (SEQ ID NO:888)
41_jfrlSe1402-6_Rissen	MRHRKSGRQLNRNSSHRQAMFRNMAGSLVRHEIIKTTLPKAKELRRVVEPLITLAKTDSVANRRLAFARTRDNEIVAKLFN ELGPRFASRAGGYTRILKCGFRAGDNAPMAYIELVDRSEKTEAAAE (SEQ ID NO:895)
42_jfrlSe1402-7_Orion	MRHRKSGRQLNRNSSHRQAMFRNMAGSLVRHEIIKTTLPKAKELRRVVEPLITLAKTDSVANRRLAFARTRDNEIVAKLFN ELGPRFASRAGGYTRILKCGFRAGDNAPMAYIELVDRSEKTEAAAE (SEQ ID NO:902)

Fig. 37B

43_jfrlSe1402-8_Rissen	MRHRKSGRQLNRNSSHRQAMFRNMAGSLVRHEIIKTTLPKAKELRRVVEPLITLAKTDSVANRRLAFARTRDNEIVAKLFN ELGPRFASRAGGYTRILKCGFRAGDNAPMAYIELVDRSEKTEAAAE (SEQ ID NO:912)
44_jfrlSe1402-9_Rissen	MRHRKSGRQLNRNSSHRQAMFRNMAGSLVRHEIIKTTLPKAKELRRVVEPLITLAKTDSVANRRLAFARTRDNEIVAKLFN ELGPRFASRAGGYTRILKCGFRAGDNAPMAYIELVDRSEKTEAAAE (SEQ ID NO:919)
45_jfrlSe1402-10_Rissen	MRHRKSGRQLNRNSSHRQAMFRNMAGSLVRHEIIKTTLPKAKELRRVVEPLITLAKTDSVANRRLAFARTRDNEIVAKLFN ELGPRFASRAGGYTRILKCGFRAGDNAPMAYIELVDRSEKTEAAAE (SEQ ID NO:926)
46_jfrlSe1402-11_Rissen	MRHRKSGRQLNRNSSHRQAMFRNMAGSLVRHEIIKTTLPKAKELRRVVEPLITLAKTDSVANRRLAFARTRDNEIVAKLFN ELGPRFASRAGGYTRILKCGFRAGDNAPMAYIELVDRSEKTEAAAE (SEQ ID NO:933)
47_jfrlSe1402-12_Rissen	MRHRKSGRQLNRNSSHRQAMFRNMAGSLVRHEIIKTTLPKAKELRRVVEPLITLAKTDSVANRRLAFARTRDNEIVAKLFN ELGPRFASRAGGYTRILKCGFRAGDNAPMAYIELVDRSEKTEAAAE (SEQ ID NO:940)
48_jfrlSe1402-13_Mbandaka	MRHRKSGRQLNRNSSHRQAMFRNMAGSLVRHEIIKTTLPKAKELRRVVEPLITLAKTDSVANRRLAFARTRDNEIVAKLFN ELGPRFASRAGGYTRILKCGFRAGDNAPMAYIELVDRSEKTEAAAE (SEQ ID NO:947)
49_jfrlSe1402-14_Mbandaka	MRHRKSGRQLNRNSSHRQAMFRNMAGSLVRHEIIKTTLPKAKELRRVVEPLITLAKTDSVANRRLAFARTRDNEIVAKLFN ELGPRFASRAGGYTRILKCGFRAGDNAPMAYIELVDRSEKTEAAAE (SEQ ID NO:957)
50_jfrlSe1402-15_Orion	MRHRKSGRQLNRNSSHRQAMFRNMAGSLVRHEIIKTTLPKAKELRRVVEPLITLAKTDSVANRRLAFARTRDNEIVAKLFN ELGPRFASRAGGYTRILKCGFRAGDNAPMAYIELVDRSEKTEAAAE (SEQ ID NO:967)
51_jfrlSe1409-1_Altona	
52_jfrlSe1409-2_Istanbul	MRHRKSGRQLNRNSSHRQAMFRNMAGSLVRHEIIKTTLPKAKELRRVVEPLITLAKTDSVANRRLAFARTRDNEIVAKLFN ELGPRFASRAGGYTRILKCGFRAGDNAPMAYIELVDRSEKTEAAAE (SEQ ID NO:984)
53_jfrlSe1409-3_Senftenberg	
54_jfrlSe1409-4_UN_O13_	
55_jfrlSe1409-5_UN_O1,3,19_	MRHRKSGRQLNRNSSHRQAMFRNMAGSLVRHEIIKTTLPKAKELRRVVEPLITLAKTDSVANRRLAFARTRDNEIVAKLFN ELGPRFASRAGGYTRILKCGFRAGDNAPMAYIELVDRSEKTEAAAE (SEQ ID NO:1005)
56_jfrlSe1409-6_Montevideo	
57_jfrlSe1409-7_UN_O1,3,19_	MRHRKSGRQLNRNSSHRQAMFRNMAGSLVRHEIIKTTLPKAKELRRVVEPLITLAKTDSVANRRLAFARTRDNEIVAKLFN ELGPRFASRAGGYTRILKCGFRAGDNAPMAYIELVDRSEKTEAAAE (SEQ ID NO:1021)
58_jfrlSe1409-8_UN_O18_	MRHRKSGRQLNRNSSHRQAMFRNMAGSLVRHEIIKTTLPKAKELRRVVEPLITLAKTDSVANRRLAFARTRDNEIVAKLFN ELGPRFASRAGGYTRILKCGFRAGDNAPMAYIELVDRSEKTEAAAE (SEQ ID NO:1030)
61_jfrlSe1409-11_UN_O1,3,19_	MRHRKSGRQLNRNSSHRQAMFRNMAGSLVRHEIIKTTLPKAKELRRVVEPLITLAKTDSVANRRLAFARTRDNEIVAKLFN ELGPRFASRAGGYTRILKCGFRAGDNAPMAYIELVDRSEKTEAAAE (SEQ ID NO:1039)
67_jfrlSe1409-17_Rissen	MRHRKSGRQLNRNSSHRQAMFRNMAGSLVRHEIIKTTLPKAKELRRVVEPLITLAKTDSVANRRLAFARTRDNEIVAKLFN ELGPRFASRAGGYTRILKCGFRAGDNAPMAYIELVDRSEKTEAAAE (SEQ ID NO:1048)
71_jfrlSe1409-21_Amsterdam	MRHRKSGRQLNRNSSHRQAMFRNMAGSLVRHEIIKTTLPKAKELRRVVEPLITLAKTDSVANRRLAFARTRDNEIVAKLFN ELGPRFASRAGGYTRILKCGFRAGDNAPMAYIELVDRSEKTEAAAE (SEQ ID NO:1056)
80_HySe09_Enteritidis	MRHRKSGRQLNRNSSHRQAMFRNMAGSLVRHEIIKTTLPKAKELRRVVEPLITLAKTDSVANRRLAFARTRDNEIVAKLFN ELGPRFASRAGGYTRILKCGFRAGDNAPMAYIELVDRSEKTEAAAE (SEQ ID NO:1066)
100_HySe29_Schwarzengrund	MKEEKPEFDPICCACRSGIYWSALLTASRQKLSTISVIWYSVPRLSFLRRLTWVKNLLPRLKTCWLPVDCLWVCAWKTGHR QASLTSNRITGGFTEKDKVMRHRKSGRQLNRNSSHRQAMFRNMAGSLVRHEIIKTTLPKAKELRRVVEPLITLAKTDSVAN RRLAFARTRDNEIVAKLFNELGPRFASRAGGYTRILKCGFRAGDNAPMAYIELVDRSEKTEAAAESVVTKNPLRRVFLYPPE PHVSTIIVFFSX (SEQ ID NO:1076)
103_HySe32_Schwarzengrund	MRHRKSGRQLNRNSSHRQAMFRNMAGSLVRHEIIKTTLPKAKELRRVVEPLITLAKTDSVANRRLAFARTRDNEIVAKLFN ELGPRFASRAGGYTRILKCGFRAGDNAPMAYIELVDRSEKTEAAAE (SEQ ID NO:1084)
106_HySe35_Schwarzengrund	MRHRKSGRQLNRNSSHRQAMFRNMAGSLVRHEIIKTTLPKAKELRRVVEPLITLAKTDSVANRRLAFARTRDNEIVAKLFN ELGPRFASRAGGYTRILKCGFRAGDNAPMAYIELVDRSEKTEAAAE (SEQ ID NO:1090)

Fig. 38A

	L21
01_NBRC13245T_Typhimurium	MYAVFQSGGKQHRVSEGQTVRLEKLDIATGETIEFAEVLMIANGEEVKIGVPFVDGGVIKAEVVAHGRGEKVKIVKFR RRKHYRKQQGHRQWFTDVKITGISA (SEQ ID NO:578)
02_GTC00131_Enteritidis	MYAVFQSGGKQHRVSEGQTVRLEKLDIATGETIEFAEVLMIANGEEVKIGVPFVDGGVIKAEVVAHGRGEKVKIVKFR RRKHYRKQQGHRQWFTDVKITGISA (SEQ ID NO:590)
03_GTC09491_Enteritidis	MYAVFQSGGKQHRVSEGQTVRLEKLDIATGETIEFAEVLMIANGEEVKIGVPFVDGGVIKAEVVAHGRGEKVKIVKFR RRKHYRKQQGHRQWFTDVKITGISA (SEQ ID NO:599)
04_GTC03838_Enteritidis	MYAVFQSGGKQHRVSEGQTVRLEKLDIATGETIEFAEVLMIANGEEVKIGVPFVDGGVIKAEVVAHGRGEKVKIVKFR RRKHYRKQQGHRQWFTDVKITGISA (SEQ ID NO:606)
05_GTC08914_Enteritidis	MYAVFQSGGKQHRVSEGQTVRLEKLDIATGETIEFAEVLMIANGEEVKIGVPFVDGGVIKAEVVAHGRGEKVKIVKFR RRKHYRKQQGHRQWFTDVKITGISA (SEQ ID NO:613)
06_GTC09421_Enteritidis	MYAVFQSGGKQHRVSEGQTVRLEKLDIATGETIEFAEVLMIANGEEVKIGVPFVDGGVIKAEVVAHGRGEKVKIVKFR RRKHYRKQQGHRQWFTDVKITGISA (SEQ ID NO:623)
07_GTC09489_Enteritidis	MYAVFQSGGKQHRVSEGQTVRLEKLDIATGETIEFAEVLMIANGEEVKIGVPFVDGGVIKAEVVAHGRGEKVKIVKFR RRKHYRKQQGHRQWFTDVKITGISA (SEQ ID NO:632)
08_NBRC3163_Pullorum_Gallinarum	MYAVFQSGGKQHRVSEGQTVRLEKLDIATGETIEFAEVLMIANGEEVKIGVPFVDGGVIKAEVVAHGRGEKVKIVKFR RRKHYRKQQGHRQWFTDVKITGISA (SEQ ID NO:639)
09_NBRC3313_Enteritidis	MYAVFQSGGKQHRVSEGQTVRLEKLDIATGETIEFAEVLMIANGEEVKIGVPFVDGGVIKAEVVAHGRGEKVKIVKFR RRKHYRKQQGHRQWFTDVKITGISA (SEQ ID NO:649)
10_NBRC12529_Typhimurium	MYAVFQSGGKQHRVSEGQTVRLEKLDIATGETIEFAEVLMIANGEEVKIGVPFVDGGVIKAEVVAHGRGEKVKIVKFR RRKHYRKQQGHRQWFTDVKITGISA (SEQ ID NO:656)
11_NBRC14193_Typhimurium	MYAVFQSGGKQHRVSEGQTVRLEKLDIATGETIEFAEVLMIANGEEVKIGVPFVDGGVIKAEVVAHGRGEKVKIVKFR RRKHYRKQQGHRQWFTDVKITGISA (SEQ ID NO:663)
12_NBRC14194_Typhimurium	MYAVFQSGGKQHRVSEGQTVRLEKLDIATGETIEFAEVLMIANGEEVKIGVPFVDGGVIKAEVVAHGRGEKVKIVKFR RRKHYRKQQGHRQWFTDVKITGISA (SEQ ID NO:670)
13_NBRC14209_Typhimurium	MYAVFQSGGKQHRVSEGQTVRLEKLDIATGETIEFAEVLMIANGEEVKIGVPFVDGGVIKAEVVAHGRGEKVKIVKFR RRKHYRKQQGHRQWFTDVKITGISA (SEQ ID NO:677)
14_NBRC14210_Typhimurium	MYAVFQSGGKQHRVSEGQTVRLEKLDIATGETIEFAEVLMIANGEEVKIGVPFVDGGVIKAEVVAHGRGEKVKIVKFR RRKHYRKQQGHRQWFTDVKITGISA (SEQ ID NO:684)
15_NBRC14211_Typhimurium	MYAVFQSGGKQHRVSEGQTVRLEKLDIATGETIEFAEVLMIANGEEVKIGVPFVDGGVIKAEVVAHGRGEKVKIVKFR RRKHYRKQQGHRQWFTDVKITGISA (SEQ ID NO:691)
16_NBRC14212_Typhimurium	MYAVFQSGGKQHRVSEGQTVRLEKLDIATGETIEFAEVLMIANGEEVKIGVPFVDGGVIKAEVVAHGRGEKVKIVKFR RRKHYRKQQGHRQWFTDVKITGISA (SEQ ID NO:697)
17_NBRC15181_Typhimurium	MYAVFQSGGKQHRVSEGQTVRLEKLDIATGETIEFAEVLMIANGEEVKIGVPFVDGGVIKAEVVAHGRGEKVKIVKFR RRKHYRKQQGHRQWFTDVKITGISA (SEQ ID NO:704)
18_NBRC15182_Minnesota	MYAVFQSGGKQHRVSEGQTVRLEKLDIATGETIEFAEVLMVANGEEVKIGVPFVDGGVIKAEVVAHGRGEKVKIVKFR RRKHYRKQQGHRQWFTDVKITGISA (SEQ ID NO:714)
19_NBRC15183_Minnesota	MYAVFQSGGKQHRVSEGQTVRLEKLDIATGETIEFAEVLMVANGEEVKIGVPFVDGGVIKAEVVAHGRGEKVKIVKFR RRKHYRKQQGHRQWFTDVKITGISA (SEQ ID NO:724)
20_NBRC15184_Minnesota	MYAVFQSGGKQHRVSEGQTVRLEKLDIATGETIEFAEVLMVANGEEVKIGVPFVDGGVIKAEVVAHGRGEKVKIVKFR RRKHYRKQQGHRQWFTDVKITGISA (SEQ ID NO:731)
21_NBRC15185_Minnesota	MYAVFQSGGKQHRVSEGQTVRLEKLDIATGETIEFAEVLMVANGEEVKIGVPFVDGGVIKAEVVAHGRGEKVKIVKFR RRKHYRKQQGHRQWFTDVKITGISA (SEQ ID NO:738)
22_NBRC15186_Minnesota	MYAVFQSGGKQHRVSEGQTVRLEKLDIATGETIEFAEVLMVANGEEVKIGVPFVDGGVIKAEVVAHGRGEKVKIVKFR RRKHYRKQQGHRQWFTDVKITGISA (SEQ ID NO:745)
23_NBRC15187_Minnesota	MYAVFQSGGKQHRVSEGQTVRLEKLDIATGETIEFAEVLMVANGEEVKIGVPFVDGGVIKAEVVAHGRGEKVKIVKFR RRKHYRKQQGHRQWFTDVKITGISA (SEQ ID NO:752)

Fig. 38B

24_NBRC100797_Abony	MYAVFQSGGKQHRVSEGQTVRLEKLDIATGETIEFAEVLMIANGEEVKIGVPFVDGGVIKAEVVAHGRGEKVKIVKFR RRKHYRKQQGHRQWFTDVKITGISA (SEQ ID NO:759)
25_NBRC105684_Choleraesuis	MYAVFQSGGKQHRVSEGQTVRLEKLDIATGETIEFAEVLMIANGEEVKIGVPFVDGGVIKAEVVAHGRGEKVKIVKFR RRKHYRKQQGHRQWFTDVKITGISA (SEQ ID NO:769)
26_NBRC105726_Typhimurium	MYAVFQSGGKQHRVSEGQTVRLEKLDIATGETIEFAEVLMIANGEEVKIGVPFVDGGVIKAEVVAHGRGEKVKIVKFR RRKHYRKQQGHRQWFTDVKITGISA (SEQ ID NO:779)
27_JCM3919_UN_O7_	MYAVFQSGGKQHRVSEGQTVRLEKLDIATGETIEFAEVLMIANGEEVKIGVPFVDGGVIKAEVVAHGRGEKVKIVKFR RRKHYRKQQGHRQWFTDVKITGISA (SEQ ID NO:788)
28_NBRC15335_Minnesota	MYAVFQSGGKQHRVSEGQTVRLEKLDIATGETIEFAEVLMVANGEEVKIGVPFVDGGVIKAEVVAHGRGEKVKIVKFR RRKHYRKQQGHRQWFTDVKITGISA (SEQ ID NO:795)
29_GTC09490_Enteritidis	MYAVFQSGGKQHRVSEGQTVRLEKLDIATGETIEFAEVLMIANGEEVKIGVPFVDGGVIKAEVVAHGRGEKVKIVKFR RRKHYRKQQGHRQWFTDVKITGISA (SEQ ID NO:802)
30_GTC09492_Braenderup	MYAVFQSGGKQHRVSEGQTVRLEKLDIATGETIEFAEVLMIANGEEVKIGVPFVDGGVIKAEVVAHGRGEKVKIVKFR RRKHYRKQQGHRQWFTDVKITGISA (SEQ ID NO:809)
31_GTC09493_Pakistan	MYAVFQSGGKQHRVSEGQTVRLEKLDIATGETIEFAEVLMIANGEEVKIGVPFVDGGVIKAEVVAHGRGEKVKIVKFR RRKHYRKQQGHRQWFTDVKITGISA (SEQ ID NO:816)
32_GTC09549_Typhimurium	MYAVFQSGGKQHRVSEGQTVRLEKLDIATGETIEFAEVLMIANGEEVKIGVPFVDGGVIKAEVVAHGRGEKVKIVKFR RRKHYRKQQGHRQWFTDVKITGISA (SEQ ID NO:823)
33_ATCCBAA-1675_Infantis	MYAVFQSGGKQHRVSEGQTVRLEKLDIATGETIEFAEVLMVANGEEVKIGVPFVDGGVIKAEVVAHGRGEKVKIVKFR RRKHYRKQQGHRQWFTDVKITGISA (SEQ ID NO:830)
34_ATCCBAA-1738_Thompson	MYAVFQSGGKQHRVSEGQTVRLEKLDIATGETIEFAEVLMIANGEEVKIGVPFVDGGVIKAEVVAHGRGEKVKIVKFR RRKHYRKQQGHRQWFTDVKITGISA (SEQ ID NO:841)
35_ATCC9712_Saintpaul	MYAVFQSGGKQHRVSEGQTVRLEKLDIATGETIEFAEVLMIANGEEVKIGVPFVDGGVIKAEVVAHGRGEKVKIVKFR RRKHYRKQQGHRQWFTDVKITGISA (SEQ ID NO:851)
36_jfrlSe1402-1_Infantis	MYAVFQSGGKQHRVSEGQTVRLEKLDIATGETIEFAEVLMIANGEEVKIGVPFVDGGVIKAEVVAHGRGEKVKIVKFR RRKHYRKQQGHRQWFTDVKITGISA (SEQ ID NO:862)
38_jfrlSe1402-3_Brandenburg	MYAVFQSGGKQHRVSEGQTVRLEKLDIATGETIEFAEVLMVANGEEVKIGVPFVDGGVIKAEVVAHGRGEKVKIVKFR RRKHYRKQQGHRQWFTDVKITGISA (SEQ ID NO:872)
39_jfrlSe1402-4_Infantis	MYAVFQSGGKQHRVSEGQTVRLEKLDIATGETIEFAEVLMIANGEEVKIGVPFVDGGVIKAEVVAHGRGEKVKIVKFR RRKHYRKQQGHRQWFTDVKITGISA (SEQ ID NO:879)
40_jfrlSe1402-5_Brandenburg	MYAVFQSGGKQHRVSEGQTVRLEKLDIATGETIEFAEVLMVANGEEVKIGVPFVDGGVIKAEVVAHGRGEKVKIVKFR RRKHYRKQQGHRQWFTDVKITGISA (SEQ ID NO:889)
41_jfrlSe1402-6_Rissen	MYAVFQSGGKQHRVSEGQTVRLEKLDIATGETIEFAEVLMIANGEEVKIGVPFVDGGVIKAEVVAHGRGEKVKIVKFR RRKHYRKQQGHRQWFTDVKITGISA (SEQ ID NO:896)
42_jfrlSe1402-7_Orion	MYAVFQSGGKQHRVSEGQTVRLEKLDIATGETIEFAEVLMIANGEEVKIGVPFVDGGVIKAEVVAHGRGEKVKIVKFR RRKHYRKQQGHRQWFTDVKITGISA (SEQ ID NO:903)
43_jfrlSe1402-8_Rissen	MYAVFQSGGKQHRVSEGQTVRLEKLDIATGETIEFAEVLMIANGEEVKIGVPFVDGGVIKAEVVAHGRGEKVKIVKFR RRKHYRKQQGHRQWFTDVKITGISA (SEQ ID NO:913)
44_jfrlSe1402-9_Rissen	MYAVFQSGGKQHRVSEGQTVRLEKLDIATGETIEFAEVLMIANGEEVKIGVPFVDGGVIKAEVVAHGRGEKVKIVKFR RRKHYRKQQGHRQWFTDVKITGISA (SEQ ID NO:920)
45_jfrlSe1402-10_Rissen	MYAVFQSGGKQHRVSEGQTVRLEKLDIATGETIEFAEVLMIANGEEVKIGVPFVDGGVIKAEVVAHGRGEKVKIVKFR RRKHYRKQQGHRQWFTDVKITGISA (SEQ ID NO:927)
46_jfrlSe1402-11_Rissen	MYAVFQSGGKQHRVSEGQTVRLEKLDIATGETIEFAEVLMIANGEEVKIGVPFVDGGVIKAEVVAHGRGEKVKIVKFR RRKHYRKQQGHRQWFTDVKITGISA (SEQ ID NO:934)
47_jfrlSe1402-12_Rissen	MYAVFQSGGKQHRVSEGQTVRLEKLDIATGETIEFAEVLMIANGEEVKIGVPFVDGGVIKAEVVAHGRGEKVKIVKFR RRKHYRKQQGHRQWFTDVKITGISA (SEQ ID NO:941)
48_jfrlSe1402-13_Mbandaka	MYAVFQSGGKQHRVSEGQTVRLEKLDIATGETIEFAEVLMIANGEEVKIGVPFVDGGVIKAEVVAHGRGEKVKIVKFR RRKHYRKQQGHRQWFTDVKITGISA (SEQ ID NO:948)
49_jfrlSe1402-14_Mbandaka	MYAVFQSGGKQHRVSEGQTVRLEKLDIATGETIEFAEVLMIANGEEVKIGVPFVDGGVIKAEVVAHGRGEKVKIVKFR RRKHYRKQQGHRQWFTDVKITGISA (SEQ ID NO:958)
50_jfrlSe1402-15_Orion	MYAVFQSGGKQHRVSEGQTVRLEKLDIATGETIEFAEVLMIANGEEVKIGVPFVDGGVIKAEVVAHGRGEKVKIVKFR RRKHYRKQQGHRQWFTDVKITGISA (SEQ ID NO:968)
51_jfrlSe1409-1_Altona	
52_jfrlSe1409-2_Istanbul	MYAVFQSGGKQHRVSEGQTVRLEKLDIATGETIEFAEVLMIANGEEVKIGVPFVDGGVIKAEVVAHGRGEKVKIVKFR RRKHYRKQQGHRQWFTDVKITGISA (SEQ ID NO:985)
53_jfrlSe1409-3_Senftenberg	
54_jfrlSe1409-4_UN_O13_	MYAVFQSGGKQHRVSEGQTVRLEKLDIATGETIEFAEVLMIANGEEVKIGVPFVDGGVIKAEVVAHGRGEKVKIVKFR RRKHYRKQQGHRQWFTDVKITGISA (SEQ ID NO:997)
55_jfrlSe1409-5_UN_O1,3,19_	MYAVFQSGGKQHRVSEGQTVRLEKLDIATGETIEFAEVLMIANGEEVKIGVPFVDGGVIKAEVVAHGRGEKVKIVKFR RRKHYRKQQGHRQWFTDVKITGISA (SEQ ID NO:1006)
56_jfrlSe1409-6_Montevideo	MYAVFQSGGKQHRVSEGQTVRLEKLDIATGETIEFAEVLMVANGEEVKIGVPFVDGGVIKAEVVAHGRGEKVKIVKFR RRKHYRKQQGHRQWFTDVKITGISA (SEQ ID NO:1014)
57_jfrlSe1409-7_UN_O1,3,19_	MYAVFQSGGKQHRVSEGQTVRLEKLDIATGETIEFAEVLMIANGEEVKIGVPFVDGGVIKAEVVAHGRGEKVKIVKFR RRKHYRKQQGHRQWFTDVKITGISA (SEQ ID NO:1022)
58_jfrlSe1409-8_UN_O18_	MYAVFQSGGKQHRVSEGQTVRLEKLDIATGETIEFAEVLMIANGEEVKIGVPFVDGGVIKAEVVAHGRGEKVKIVKFR RRKHYRKQQGHRQWFTDVKITGISA (SEQ ID NO:1031)
61_jfrlSe1409-11_UN_O1,3,19_	MYAVFQSGGKQHRVSEGQTVRLEKLDIATGETIEFAEVLMIANGEEVKIGVPFVDGGVIKAEVVAHGRGEKVKIVKFR RRKHYRKQQGHRQWFTDVKITGISA (SEQ ID NO:1040)
67_jfrlSe1409-17_Rissen	
71_jfrlSe1409-21_Amsterdam	MYAVFQSGGKQHRVSEGQTVRLEKLDIATGETIEFAEVLMIANGEEVKIGVPFVDGGVIKAEVVAHGRGEKVKIVKFR RRKHYRKQQGHRQWFTDVKITGISA (SEQ ID NO:1057)
80_HySe09_Enteritidis	MYAVFQSGGKQHRVSEGQTVRLEKLDIATGETIEFAEVLMIANGEEVKIGVPFVDGGVIKAEVVAHGRGEKVKIVKFR RRKHYRKQQGHRQWFTDVKITGISA (SEQ ID NO:1067)
100_HySe29_Schwarzengrund	MYAVFQSGGKQHRVSEGQTVRLEKLDIATGETIEFAEVLMVANGEEVKIGVPFVDGGVIKAEVVAHGRGEKVKIVKFR RRKHYRKQQGHRQWFTDVKITGISA (SEQ ID NO:1077)
103_HySe32_Schwarzengrund	
106_HySe35_Schwarzengrund	

Fig. 39A

	S8
01_NBRC13245T_Typhimurium	MSMQDPIADMLTRIRNGQAANKAAVTMPSSKLKVAIANVLKEEGFIEDFKVEGDTKPELELTLKYFQGKAVVESI QRVSRPGLRIYKRKDELPKVMAGLGIAVVSTSKGVMTDRAARQAGLGGEIICYVA (SEQ ID NO:579)
02_GTC00131_Enteritidis	MSMQDPIADMLTRIRNGQAANKAAVTMPSSKLKVAIANVLKEEGFIEDFKVEGDTKPELELTLKYFQGKAVVESI QRVSRPGLRIYKRKDELPKVMAGLGIAVVSTSKGVMTDRAARQAGLGGEIICYVA (SEQ ID NO:591)
03_GTC09491_Enteritidis	MSMQDPIADMLTRIRNGQAANKAAVTMPSSKLKVAIANVLKEEGFIEDFKVEGDTKPELELTLKYFQGKAVVESI QRVSRPGLRIYKRKDELPKVMAGLGIAVVSTSKGVMTDRAARQAGLGGEIICYVA (SEQ ID NO:600)
04_GTC03838_Enteritidis	MSMQDPIADMLTRIRNGQAANKAAVTMPSSKLKVAIANVLKEEGFIEDFKVEGDTKPELELTLKYFQGKAVVESI QRVSRPGLRIYKRKDELPKVMAGLGIAVVSTSKGVMTDRAARQAGLGGEIICYVA (SEQ ID NO:607)
05_GTC08914_Enteritidis	MSMQDPIADMLTRIRNGQAANKAAVTMPSSKLKVAIANVLKEEGFIEDFKVEGDTKPELELTLKYFQGKAVVESI QRVSRPGLRIYKRKDELPKVMAGLGIAVVSTSKGVMTDRAARQAGLGGEIICYVA (SEQ ID NO:614)
06_GTC09421_Enteritidis	MSMQDPIADMLTRIRNGQAANKAAVTMPSSKLKVAIANVLKEEGFIEDFKVEGDTKPELELTLKYFQGKAVVESI QRVSRPGLRIYKRKDELPKVMAGLGIAVVSTSKGVMTDRAARQAGLGGEIICYVA (SEQ ID NO:624)
07_GTC09489_Enteritidis	MSMQDPIADMLTRIRNGQAANKAAVTMPSSKLKVAIANVLKEEGFIEDFKVEGDTKPELELTLKYFQGKAVVESI QRVSRPGLRIYKRKDELPKVMAGLGIAVVSTSKGVMTDRAARQAGLGGEIICYVA (SEQ ID NO:633)
08_NBRC3163_Pullorum_Gallinarum	MSMQDPIADMLTRIRNGQAANKAAVTMPSSKLKVAIANVLKEEGFIEDFKVEGDTKPELELTLKYFQGKAVVESI QRVSRPGLRIYKRKDELPKVMAGLGIAVVSTSKGVMTDRAARQAGLGGEIICYVA (SEQ ID NO:640)
09_NBRC3313_Enteritidis	MSMQDPIADMLTRIRNGQAANKAAVTMPSSKLKVAIANVLKEEGFIEDFKVEGDTKPELELTLKYFQGKAVVESI QRVSRPGLRIYKRKDELPKVMAGLGIAVVSTSKGVMTDRAARQAGLGGEIICYVA (SEQ ID NO:650)
10_NBRC12529_Typhimurium	MSMQDPIADMLTRIRNGQAANKAAVTMPSSKLKVAIANVLKEEGFIEDFKVEGDTKPELELTLKYFQGKAVVESI QRVSRPGLRIYKRKDELPKVMAGLGIAVVSTSKGVMTDRAARQAGLGGEIICYVA (SEQ ID NO:657)
11_NBRC14193_Typhimurium	MSMQDPIADMLTRIRNGQAANKAAVTMPSSKLKVAIANVLKEEGFIEDFKVEGDTKPELELTLKYFQGKAVVESI QRVSRPGLRIYKRKDELPKVMAGLGIAVVSTSKGVMTDRAARQAGLGGEIICYVA (SEQ ID NO:664)
12_NBRC14194_Typhimurium	MSMQDPIADMLTRIRNGQAANKAAVTMPSSKLKVAIANVLKEEGFIEDFKVEGDTKPELELTLKYFQGKAVVESI QRVSRPGLRIYKRKDELPKVMAGLGIAVVSTSKGVMTDRAARQAGLGGEIICYVA (SEQ ID NO:671)
13_NBRC14209_Typhimurium	MSMQDPIADMLTRIRNGQAANKAAVTMPSSKLKVAIANVLKEEGFIEDFKVEGDTKPELELTLKYFQGKAVVESI QRVSRPGLRIYKRKDELPKVMAGLGIAVVSTSKGVMTDRAARQAGLGGEIICYVA (SEQ ID NO:678)
14_NBRC14210_Typhimurium	MSMQDPIADMLTRIRNGQAANKAAVTMPSSKLKVAIANVLKEEGFIEDFKVEGDTKPELELTLKYFQGKAVVESI QRVSRPGLRIYKRKDELPKVMAGLGIAVVSTSKGVMTDRAARQAGLGGEIICYVA (SEQ ID NO:685)
15_NBRC14211_Typhimurium	MSMQDPIADMLTRIRNGQAANKAAVTMPSSKLKVAIANVLKEEGFIEDFKVEGDTKPELELTLKYFQGKAVVESI QRVSRPGLRIYKRKDELPKVMAGLGIAVVSTSKGVMTDRAARQAGLGGEIICYVA (SEQ ID NO:692)
16_NBRC14212_Typhimurium	MSMQDPIADMLTRIRNGQAANKAAVTMPSSKLKVAIANVLKEEGFIEDFKVEGDTKPELELTLKYFQGKAVVESI QRVSRPGLRIYKRKDELPKVMAGLGIAVVSTSKGVMTDRAARQAGLGGEIICYVA (SEQ ID NO:698)
17_NBRC15181_Typhimurium	MSMQDPIADMLTRIRNGQAANKAAVTMPSSKLKVAIANVLKEEGFIEDFKVEGDTKPELELTLKYFQGKAVVESI QRVSRPGLRIYKRKDELPKVMAGLGIAVVSTSKGVMTDRAARQAGLGGEIICYVA (SEQ ID NO:705)
18_NBRC15182_Minnesota	MSMQDPIADMLTRIRNGQAANKAAVTMPSSKLKVAIANVLKEEGFIEDFKVEGDTKPELELTLKYFQGKAVVESI QRVSRPGLRIYKRKDELPKVMAGLGIAVVSTSKGVMTDRAARQAGLGGEIICYVA (SEQ ID NO:715)
19_NBRC15183_Minnesota	MSMQDPIADMLTRIRNGQAANKAAVTMPSSKLKVAIANVLKEEGFIEDFKVEGDTKPELELTLKYFQGKAVVESI QRVSRPGLRIYKRKDELPKVMAGLGIAVVSTSKGVMTDRAARQAGLGGEIICYVA (SEQ ID NO:725)
20_NBRC15184_Minnesota	MSMQDPIADMLTRIRNGQAANKAAVTMPSSKLKVAIANVLKEEGFIEDFKVEGDTKPELELTLKYFQGKAVVESI QRVSRPGLRIYKRKDELPKVMAGLGIAVVSTSKGVMTDRAARQAGLGGEIICYVA (SEQ ID NO:732)
21_NBRC15185_Minnesota	MSMQDPIADMLTRIRNGQAANKAAVTMPSSKLKVAIANVLKEEGFIEDFKVEGDTKPELELTLKYFQGKAVVESI QRVSRPGLRIYKRKDELPKVMAGLGIAVVSTSKGVMTDRAARQAGLGGEIICYVA (SEQ ID NO:739)
22_NBRC15186_Minnesota	MSMQDPIADMLTRIRNGQAANKAAVTMPSSKLKVAIANVLKEEGFIEDFKVEGDTKPELELTLKYFQGKAVVESI QRVSRPGLRIYKRKDELPKVMAGLGIAVVSTSKGVMTDRAARQAGLGGEIICYVA (SEQ ID NO:746)
23_NBRC15187_Minnesota	MSMQDPIADMLTRIRNGQAANKAAVTMPSSKLKVAIANVLKEEGFIEDFKVEGDTKPELELTLKYFQGKAVVESI QRVSRPGLRIYKRKDELPKVMAGLGIAVVSTSKGVMTDRAARQAGLGGEIICYVA (SEQ ID NO:753)
24_NBRC100797_Abony	MSMQDPIADMLTRIRNGQAANKAAVTMPSSKLKVAIANVLKEEGFIEDFKVEGDTKPELELTLKYFQGKAVVESI QRVSRPGLRIYKRKDELPKVMAGLGIAVVSTSKGVMTDRAARQAGLGGEIICYVA (SEQ ID NO:760)
25_NBRC105684_Choleraesuis	MSMQDPIADMLTRIRNGQAANKAAVTMPSSKLKVAIANVLKEEGFIEDFKVEGDTKPELELTLKYFQGKAVVESI QRVSRPGLRIYKRKDELPKVMAGLGIAVVSTSKGVMTDRAARQAGLGGEIICYVA (SEQ ID NO:770)
26_NBRC105726_Typhimurium	MSMQDPIADMLTRIRNGQAANKAAVTMPSSKLKVAIANVLKEEGFIEDFKVEGDTKPELELTLKYFQGKAVVESI QRVSRPGLRIYKRKDELPKVMAGLGIAVVSTSKGVMTDRAARQAGLGGEIICYVA (SEQ ID NO:780)
27_JCM3919_UN_O7_	MSMQDPIADMLTRIRNGQAANKAAVTMPSSKLKVAIANVLKEEGFIEDFKVEGDTKPELELTLKYFQGKAVVESI QRVSRPGLRIYKRKDELPKVMAGLGIAVVSTSKGVMTDRAARQAGLGGEIICYVA (SEQ ID NO:789)
28_NBRC15335_Minnesota	MSMQDPIADMLTRIRNGQAANKAAVTMPSSKLKVAIANVLKEEGFIEDFKVEGDTKPELELTLKYFQGKAVVESI QRVSRPGLRIYKRKDELPKVMAGLGIAVVSTSKGVMTDRAARQAGLGGEIICYVA (SEQ ID NO:796)
29_GTC09490_Enteritidis	MSMQDPIADMLTRIRNGQAANKAAVTMPSSKLKVAIANVLKEEGFIEDFKVEGDTKPELELTLKYFQGKAVVESI QRVSRPGLRIYKRKDELPKVMAGLGIAVVSTSKGVMTDRAARQAGLGGEIICYVA (SEQ ID NO:803)
30_GTC09492_Braenderup	MSMQDPIADMLTRIRNGQAANKAAVTMPSSKLKVAIANVLKEEGFIEDFKVEGDTKPELELTLKYFQGKAVVESI QRVSRPGLRIYKRKDELPKVMAGLGIAVVSTSKGVMTDRAARQAGLGGEIICYVA (SEQ ID NO:810)
31_GTC09493_Pakistan	MSMQDPIADMLTRIRNGQAANKAAVTMPSSKLKVAIANVLKEEGFIEDFKVEGDTKPELELTLKYFQGKAVVESI QRVSRPGLRIYKRKDELPKVMAGLGIAVVSTSKGVMTDRAARQAGLGGEIICYVA (SEQ ID NO:818)
32_GTC09549_Typhimurium	MSMQDPIADMLTRIRNGQAANKAAVTMPSSKLKVAIANVLKEEGFIEDFKVEGDTKPELELTLKYFQGKAVVESI QRVSRPGLRIYKRKDELPKVMAGLGIAVVSTSKGVMTDRAARQAGLGGEIICYVA (SEQ ID NO:824)
33_ATCCBAA-1675_Infantis	MSMQDPIADMLTRIRNGQAANKAAVTMPSSKLKVAIANVLKEEGFIEDFKVEGDTKPELELTLKYFQGKAVVESI QRVSRPGLRIYKRKDELPKVMAGLGIAVVSTSKGVMTDRAARQAGLGGEIICYVA (SEQ ID NO:831)
34_ATCCBAA-1738_Thompson	MSMQDPIADMLTRIRNGQAANKAAVTMPSSKLKVAIANVLKEEGFIEDFKVEGDTKPELELTLKYFQGKAVVESI QRVSRPGLRIYKRKDELPKVMAGLGIAVVSTSKGVMTDRAARQAGLGGEIICYVA (SEQ ID NO:842)
35_ATCC9712_Saintpaul	MSMQDPIADMLTRIRNGQAANKAAVTMPSSKLKVAIANVLKEEGFIEDFKVEGDTKPELELTLKYFQGKAVVESI QRVSRPGLRIYKRKDELPKVMAGLGIAVVSTSKGVMTDRAARQAGLGGEIICYVA (SEQ ID NO:852)
36_jfrlSe1402-1_Infantis	MSMQDPIADMLTRIRNGQAANKAAVTMPSSKLKVAIANVLKEEGFIEDFKVEGDTKPELELTLKYFQGKAVVESI QRVSRPGLRIYKRKDELPKVMAGLGIAVVSTSKGVMTDRAARQAGLGGEIICYVA (SEQ ID NO:863)
38_jfrlSe1402-3_Brandenburg	MSMQDPIADMLTRIRNGQAANKAAVTMPSSKLKVAIANVLKEEGFIEDFKVEGDTKPELELTLKYFQGKAVVESI QRVSRPGLRIYKRKDELPKVMAGLGIAVVSTSKGVMTDRAARQAGLGGEIICYVA (SEQ ID NO:873)
39_jfrlSe1402-4_Infantis	MSMQDPIADMLTRIRNGQAANKAAVTMPSSKLKVAIANVLKEEGFIEDFKVEGDTKPELELTLKYFQGKAVVESI QRVSRPGLRIYKRKDELPKVMAGLGIAVVSTSKGVMTDRAARQAGLGGEIICYVA (SEQ ID NO:880)
40_jfrlSe1402-5_Brandenburg	MSMQDPIADMLTRIRNGQAANKAAVTMPSSKLKVAIANVLKEEGFIEDFKVEGDTKPELELTLKYFQGKAVVESI QRVSRPGLRIYKRKDELPKVMAGLGIAVVSTSKGVMTDRAARQAGLGGEIICYVA (SEQ ID NO:890)
41_jfrlSe1402-6_Rissen	MSMQDPIADMLTRIRNGQAANKAAVTMPSSKLKVAIANVLKEEGFIEDFKVEGDTKPELELILKYFQGKAVVESI QRVSRPGLRIYKRKDELPKVMAGLGIAVVSTSKGVMTDRAARQAGLGGEIICYVA (SEQ ID NO:897)
42_jfrlSe1402-7_Orion	MSMQDPIADMLTRIRNGQAANKAAVTMPSSKLKVAIANVLKEEGFIEDFKVEGDTKPELELTLKYFQGKAVVESI QRVSRPGLRIYKRKDELPKVMAGLGIAVVSTSKGVMTDRAARQAGLGGEIICYVA (SEQ ID NO:904)
43_jfrlSe1402-8_Rissen	MSMQDPIADMLTRIRNGQAANKAAVTMPSSKLKVAIANVLKEEGFIEDFKVEGDTKPELELILKYFQGKAVVESI QRVSRPGLRIYKRKDELPKVMAGLGIAVVSTSKGVMTDRAARQAGLGGEIICYVA (SEQ ID NO:914)
44_jfrlSe1402-9_Rissen	MSMQDPIADMLTRIRNGQAANKAAVTMPSSKLKVAIANVLKEEGFIEDFKVEGDTKPELELILKYFQGKAVVESI QRVSRPGLRIYKRKDELPKVMAGLGIAVVSTSKGVMTDRAARQAGLGGEIICYVA (SEQ ID NO:921)
45_jfrlSe1402-10_Rissen	MSMQDPIADMLTRIRNGQAANKAAVTMPSSKLKVAIANVLKEEGFIEDFKVEGDTKPELELILKYFQGKAVVESI QRVSRPGLRIYKRKDELPKVMAGLGIAVVSTSKGVMTDRAARQAGLGGEIICYVA (SEQ ID NO:928)

Fig. 39B

46_jfrlSe1402-11_Rissen	MSMQDPIADMLTRIRNGQAANKAAVTMPSSKLKVAIANVLKEEGFIEDFKVEGDTKPELELILKYFQGKAVVESI QRVSRPGLRIYKRKDELPKVMAGLGIAVVSTSKGVMTDRAARQAGLGGEIICYVA (SEQ ID NO:935)
47_jfrlSe1402-12_Rissen	MSMQDPIADMLTRIRNGQAANKAAVTMPSSKLKVAIANVLKEEGFIEDFKVEGDTKPELELILKYFQGKAVVESI QRVSRPGLRIYKRKDELPKVMAGLGIAVVSTSKGVMTDRAARQAGLGGEIICYVA (SEQ ID NO:942)
48_jfrlSe1402-13_Mbandaka	MSMQDPIADMLTRIRNGQAANKAAVTMPSSKLKVAIANVLKEEGFIEDFKVEGDTKPELELTLKYFQGKAVVESI QRVSRPGLRIYKRKDELPKVMAGLGIAVVSTSKGVMTDRAARQAGLGGEIICYVA (SEQ ID NO:949)
49_jfrlSe1402-14_Mbandaka	MSMQDPIADMLTRIRNGQAANKAAVTMPSSKLKVAIANVLKEEGFIEDFKVEGDTKPELELTLKYFQGKAVVESI QRVSRPGLRIYKRKDELPKVMAGLGIAVVSTSKGVMTDRAARQAGLGGEIICYVA (SEQ ID NO:959)
50_jfrlSe1402-15_Orion	MSMQDPIADMLTRIRNGQAANKAAVTMPSSKLKVAIANVLKEEGFIEDFKVEGDTKPELELTLKYFQGKAVVESI QRVSRPGLRIYKRKDELPKVMAGLGIAVVSTSKGVMTDRAARQAGLGGEIICYVA (SEQ ID NO:969)
51_jfrlSe1409-1_Altona	MSMQDPIADMLTRIRNGQAANKAAVTMPSSKLKVAIANVLKEEGFIEDFKVEGDTKPELELTLKYFQGKAVVESI QRVSRPGLRIYKRKDELPKVMAGLGIAVVSTSKGVMTDRAARQAGLGGEIICYVA (SEQ ID NO:977)
52_jfrlSe1409-2_Istanbul	MSMQDPIADMLTRIRNGQAANKAAVTMPSSKLKVAIANVLKEEGFIEDFKVEGDTKPELELTLKYFQGKAVVESI QRVSRPGLRIYKRKDELPKVMAGLGIAVVSTSKGVMTDRAARQAGLGGEIICYVA (SEQ ID NO:986)
53_jfrlSe1409-3_Senftenberg	
54_jfrlSe1409-4_UN_O13_	MSMQDPIADMLTRIRNGQAANKAAVTMPSSKLKVAIANVLKEEGFIEDFKVEGDTKPELELTLKYFQGKAVVESI QRVSRPGLRIYKRKDELPKVMAGLGIAVVSTSKGVMTDRAARQAGLGGEIICYVA (SEQ ID NO:998)
55_jfrlSe1409-5_UN_O1,3,19_	MSMQDPIADMLTRIRNGQAANKAAVTMPSSKLKVAIANVLKEEGFIEDFKVEGDTKPELELTLKYFQGKAVVESI QRVSRPGLRIYKRKDELPKVMAGLGIAVVSTSKGVMTDRAARQAGLGGEIICYVA (SEQ ID NO:1007)
56_jfrlSe1409-6_Montevideo	MSMQDPIADMLTRIRNGQAANKAAVTMPSSKLKVAIANVLKEEGFIEDFKVEGDTKPELELTLKYFQGKAVVESI QRVSRPGLRIYKRKDELPKVMAGLGIAVVSTSKGVMTDRAARQAGLGGEIICYVA (SEQ ID NO:1015)
57_jfrlSe1409-7_UN_O1,3,19_	MSMQDPIADMLTRIRNGQAANKAAVTMPSSKLKVAIANVLKEEGFIEDFKVEGDTKPELELTLKYFQGKAVVESI QRVSRPGLRIYKRKDELPKVMAGLGIAVVSTSKGVMTDRAARQAGLGGEIICYVA (SEQ ID NO:1023)
58_jfrlSe1409-8_UN_O18_	MSMQDPIADMLTRIRNGQAANKAAVTMPSSKLKVAIANVLKEEGFIEDFKVEGDTKPELELTLKYFQGKAVVESI QRVSRPGLRIYKRKDELPKVMAGLGIAVVSTSKGVMTDRAARQAGLGGEIICYVA (SEQ ID NO:1032)
61_jfrlSe1409-11_UN_O1,3,19_	MSMQDPIADMLTRIRNGQAANKAAVTMPSSKLKVAIANVLKEEGFIEDFKVEGDTKPELELTLKYFQGKAVVESI QRVSRPGLRIYKRKDELPKVMAGLGIAVVSTSKGVMTDRAARQAGLGGEIICYVA (SEQ ID NO:1041)
67_jfrlSe1409-17_Rissen	MSMQDPIADMLTRIRNGQAANKAAVTMPSSKLKVAIANVLKEEGFIEDFKVEGDTKPELELILKYFQGKAVVESI QRVSRPGLRIYKRKDELPKVMAGLGIAVVSTSKGVMTDRAARQAGLGGEIICYVA (SEQ ID NO:1049)
71_jfrlSe1409-21_Amsterdam	MSMQDPIADMLTRIRNGQAANKAAVTMPSSKLKVAIANVLKEEGFIEDFKVEGDTKPELELTLKYFQGKAVVESI QRVSRPGLRIYKRKDELPKVMAGLGIAVVSTSKGVMTDRAARQAGLGGEIICYVA (SEQ ID NO:1058)
80_HySe09_Enteritidis	MSMQDPIADMLTRIRNGQAANKAAVTMPSSKLKVAIANVLKEEGFIEDFKVEGDTKPELELTLKYFQGKAVVESI QRVSRPGLRIYKRKDELPKVMAGLGIAVVSTSKGVMTDRAARQAGLGGEIICYVA (SEQ ID NO:1068)
100_HySe29_Schwarzengrund	MSMQDPIADMLTRIRNGQAANKAAVTMPSSKLKVAIANVLKEEGFIEDFKVEGDTKPELELTLKYFQGKAVVESI QRVSRPGLRIYKRKDELPKVMAGLGIAVVSTSKGVMTDRAARQAGLGGEIICYVA (SEQ ID NO:1078)
103_HySe32_Schwarzengrund	MSMQDPIADMLTRIRNGQAANKAAVTMPSSKLKVAIANVLKEEGFIEDFKVEGDTKPELELTLKYFQGKAVVESI QRVSRPGLRIYKRKDELPKVMAGLGIAVVSTSKGVMTDRAARQAGLGGEIICYVA (SEQ ID NO:1085)
106_HySe35_Schwarzengrund	MSMQDPIADMLTRIRNGQAANKAAVTMPSSKLKVAIANVLKEEGFIEDFKVEGDTKPELELTLKYFQGKAVVESI QRVSRPGLRIYKRKDELPKVMAGLGIAVVSTSKGVMTDRAARQAGLGGEIICYVA (SEQ ID NO:1091)

Fig. 40A

	L15
01_NBRC13245T_Typhimurium	MRLNTLSPAEGSKKAGKRLGRGIGSGLGKTGGRGHKGQKSRSGGGVRRGFEGGQMPLYRRLPKFGFTSRKAAITAEVR LSDLAKVEGGVVDLNTLKAANIIGIQIEFAKVILAGEVTTPVTVRGLRVTKGARAAIEAAGGKIEE (SEQ ID NO:580)
02_GTC00131_Enteritidis	MRLNTLSPAEGSKKAGKRLGRGIGSGLGKTGGRGHKGQKSRSGGGVRRGFEGGQMPLYRRLPKFGFTSRKAAITAEVR LSDLAKVEGGVVDLNTLKAANIIGIQIEFAKVILAGEVTTPVTVRGLRVTKGARAAIEAAGGKIEE (SEQ ID NO:592)
03_GTC09491_Enteritidis	MRLNTLSPAEGSKKAGKRLGRGIGSGLGKTGGRGHKGQKSRSGGGVRRGFEGGQMPLYRRLPKFGFTSRKAAITAEVR LSDLAKVEGGVVDLNTLKAANIIGIQIEFAKVILAGEVTTPVTVRGLRVTKGARAAIEAAGGKIEE (SEQ ID NO:601)
04_GTC03838_Enteritidis	MRLNTLSPAEGSKKAGKRLGRGIGSGLGKTGGRGHKGQKSRSGGGVRRGFEGGQMPLYRRLPKFGFTSRKAAITAEVR LSDLAKVEGGVVDLNTLKAANIIGIQIEFAKVILAGEVTTPVTVRGLRVTKGARAAIEAAGGKIEE (SEQ ID NO:608)
05_GTC08914_Enteritidis	MRLNTLSPAEGSKKAGKRLGRGIGSGLGKTGGRGHKGQKSRSGGGVRRGFEGGQMPLYRRLPKFGFTSRKAAITAEVR LSDLAKVEGGVVDLNTLKAANIIGIQIEFAKVILAGEVTTPVTVRGLRVTKGARAAIEAAGGKIEE (SEQ ID NO:615)
06_GTC09421_Enteritidis	MRLNTLSPAEGSKKAGKRLGRGIGSGLGKTGGRGHKGQKSRSGGGVRRGFEGGQMPLYRRLPKFGFTSRKAAITAEVR LSDLAKVEGGVVDLNTLKAANIIGIQIEFAKVILAGEVTTPVTVRGLRVTKGARAAIEAAGGKIEE (SEQ ID NO:625)
07_GTC09489_Enteritidis	MRLNTLSPAEGSKKAGKRLGRGIGSGLGKTGGRGHKGQKSRSGGGVRRGFEGGQMPLYRRLPKFGFTSRKAAITAEVR LSDLAKVEGGVVDLNTLKAANIIGIQIEFAKVILAGEVTTPVTVRGLRVTKGARAAIEAAGGKIEE (SEQ ID NO:634)
08_NBRC3163_Pullorum_Gallinarum	MRLNTLSPAEGSKKAGKRLGRGIGSGLGKTGGRGHKGQKSRSGGGVRRGFEGGQMPLYRRLPKFGFTSRKAAITAEVR LSDLAKVEGGVVDLNTLKAANIIGIQIEFAKVILAGEVTTPVTVRGLRVTKGARAAIEAAGGKIEE (SEQ ID NO:641)
09_NBRC3313_Enteritidis	MRLNTLSPAEGSKKAGKRLGRGIGSGLGKTGGRGHKGQKSRSGGGVRRGFEGGQMPLYRRLPKFGFTSRKAAITAEVR LSDLAKVEGGVVDLNTLKAANIIGIQIEFAKVILAGEVTTPVTVRGLRVTKGARAAIEAAGGKIEE (SEQ ID NO:651)
10_NBRC12529_Typhimurium	MRLNTLSPAEGSKKAGKRLGRGIGSGLGKTGGRGHKGQKSRSGGGVRRGFEGGQMPLYRRLPKFGFTSRKAAITAEVR LSDLAKVEGGVVDLNTLKAANIIGIQIEFAKVILAGEVTTPVTVRGLRVTKGARAAIEAAGGKIEE (SEQ ID NO:658)
11_NBRC14193_Typhimurium	MRLNTLSPAEGSKKAGKRLGRGIGSGLGKTGGRGHKGQKSRSGGGVRRGFEGGQMPLYRRLPKFGFTSRKAAITAEVR LSDLAKVEGGVVDLNTLKAANIIGIQIEFAKVILAGEVTTPVTVRGLRVTKGARAAIEAAGGKIEE (SEQ ID NO:665)
12_NBRC14194_Typhimurium	MRLNTLSPAEGSKKAGKRLGRGIGSGLGKTGGRGHKGQKSRSGGGVRRGFEGGQMPLYRRLPKFGFTSRKAAITAEVR LSDLAKVEGGVVDLNTLKAANIIGIQIEFAKVILAGEVTTPVTVRGLRVTKGARAAIEAAGGKIEE (SEQ ID NO:672)
13_NBRC14209_Typhimurium	MRLNTLSPAEGSKKAGKRLGRGIGSGLGKTGGRGHKGQKSRSGGGVRRGFEGGQMPLYRRLPKFGFTSRKAAITAEVR LSDLAKVEGGVVDLNTLKAANIIGIQIEFAKVILAGEVTTPVTVRGLRVTKGARAAIEAAGGKIEE (SEQ ID NO:679)
14_NBRC14210_Typhimurium	MRLNTLSPAEGSKKAGKRLGRGIGSGLGKTGGRGHKGQKSRSGGGVRRGFEGGQMPLYRRLPKFGFTSRKAAITAEVR LSDLAKVEGGVVDLNTLKAANIIGIQIEFAKVILAGEVTTPVTVRGLRVTKGARAAIEAAGGKIEE (SEQ ID NO:686)
15_NBRC14211_Typhimurium	
16_NBRC14212_Typhimurium	MRLNTLSPAEGSKKAGKRLGRGIGSGLGKTGGRGHKGQKSRSGGGVRRGFEGGQMPLYRRLPKFGFTSRKAAITAEVR LSDLAKVEGGVVDLNTLKAANIIGIQIEFAKVILAGEVTTPVTVRGLRVTKGARAAIEAAGGKIEE (SEQ ID NO:699)
17_NBRC15181_Typhimurium	MRLNTLSPAEGSKKAGKRLGRGIGSGLGKTGGRGHKGQKSRSGGGVRRGFEGGQMPLYRRLPKFGFTSRKAAITAEVR LSDLAKVEGGVVDLNTLKAANIIGIQIEFAKVILAGEVTTPVTVRGLRVTKGARAAIEAAGGKIEE (SEQ ID NO:706)
18_NBRC15182_Minnesota	MRLNTLSPAEGSKKAGKRLGRGIGSGLGKTGGRGHKGQKSRSGGGVRRGFEGGQMPLYRRLPKFGFTSRKAAITAEVR LSDLAKVEGGVVDLNTLKAANIIGIQIEFAKVILAGEVTTPVTVRGLRVTKGARAAIEAAGGKIEE (SEQ ID NO:716)
19_NBRC15183_Minnesota	MRLNTLSPAEGSKKAGKRLGRGIGSGLGKTGGRGHKGQKSRSGGGVRRGFEGGQMPLYRRLPKFGFTSRKAAITAEVR LSDLAKVEGGVVDLNTLKAANIIGIQIEFAKVILAGEVTTPVTVRGLRVTKGARAAIEAAGGKIEE (SEQ ID NO:726)
20_NBRC15184_Minnesota	MRLNTLSPAEGSKKAGKRLGRGIGSGLGKTGGRGHKGQKSRSGGGVRRGFEGGQMPLYRRLPKFGFTSRKAAITAEVR LSDLAKVEGGVVDLNTLKAANIIGIQIEFAKVILAGEVTTPVTVRGLRVTKGARAAIEAAGGKIEE (SEQ ID NO:733)
21_NBRC15185_Minnesota	MRLNTLSPAEGSKKAGKRLGRGIGSGLGKTGGRGHKGQKSRSGGGVRRGFEGGQMPLYRRLPKFGFTSRKAAITAEVR LSDLAKVEGGVVDLNTLKAANIIGIQIEFAKVILAGEVTTPVTVRGLRVTKGARAAIEAAGGKIEE (SEQ ID NO:740)
22_NBRC15186_Minnesota	MRLNTLSPAEGSKKAGKRLGRGIGSGLGKTGGRGHKGQKSRSGGGVRRGFEGGQMPLYRRLPKFGFTSRKAAITAEVR LSDLAKVEGGVVDLNTLKAANIIGIQIEFAKVILAGEVTTPVTVRGLRVTKGARAAIEAAGGKIEE (SEQ ID NO:747)
23_NBRC15187_Minnesota	MRLNTLSPAEGSKKAGKRLGRGIGSGLGKTGGRGHKGQKSRSGGGVRRGFEGGQMPLYRRLPKFGFTSRKAAITAEVR LSDLAKVEGGVVDLNTLKAANIIGIQIEFAKVILAGEVTTPVTVRGLRVTKGARAAIEAAGGKIEE (SEQ ID NO:754)
24_NBRC100797_Abony	MRLNTLSPAEGSKKAGKRLGRGIGSGLGKTGGRGHKGQKSRSGGGVRRGFEGGQMPLYRRLPKFGFTSRKAAITAEVR LSDLAKVEGGVVDLNTLKAANIIGIQIEFAKVILAGEVTTPVTVRGLRVTKGARAAIEAAGGKIEE (SEQ ID NO:761)
25_NBRC105684_Choleraesuis	MRLNTLSPAEGSKKAGKRLGRGIGSGLGKTGGRGHKGQKSRSGGGVRRGFEGGQMPLYRRLPKFGFTSRKAAITAEVR LSDLAKVEGGVVDLNTLKAANIIGIQIEFAKVILAGEVTTPVTVRGLRVTKGARAAIEAAGGKIEE (SEQ ID NO:771)

Fig. 40B

26_NBRC105726_Typhimurium	MRLNTLSPAEGSKKAGKRLGRGIGSGLGKTGGRGHKGQKSRSGGGVRRGFEGGQMPLYRRLPKFGFTSRKAAITAEVR LSDLAKVEGGVVDLNTLKAANIIGIQIEFAKVILAGEVTTPVTIRGLRVTKGARAAIEAAGGKIEE (SEQ ID NO:781)
27_JCM3919_UN_O7_	MRLNTLSPAEGSKKAGKRLGRGIGSGLGKTGGRGHKGQKSRSGGGVRRGFEGGQMPLYRRLPKFGFTSRKAAITAEVR LSDLAKVEGGVVDLNTLKAANIIGIQIEFAKVILAGEVTTPVTVRGLRVTKGARAAIEAAGGKIEE (SEQ ID NO:790)
28_NBRC15335_Minnesota	MRLNTLSPAEGSKKAGKRLGRGIGSGLGKTGGRGHKGQKSRSGGGVRRGFEGGQMPLYRRLPKFGFTSRKAAITAEVR LSDLAKVEGGVVDLNTLKAANIIGIQIEFAKVILAGEVTTPVTVRGLRVTKGARAAIEAAGGKIEE (SEQ ID NO:797)
29_GTC09490_Enteritidis	MRLNTLSPAEGSKKAGKRLGRGIGSGLGKTGGRGHKGQKSRSGGGVRRGFEGGQMPLYRRLPKFGFTSRKAAITAEVR LSDLAKVEGGVVDLNTLKAANIIGIQIEFAKVILAGEVTTPVTVRGLRVTKGARAAIEAAGGKIEE (SEQ ID NO:804)
30_GTC09492_Braenderup	MRLNTLSPAEGSKKAGKRLGRGIGSGLGKTGGRGHKGQKSRSGGGVRRGFEGGQMPLYRRLPKFGFTSRKAAITAEVR LSDLAKVEGGVVDLNTLKAANIIGIQIEFAKVILAGEVTTPVTVRGLRVTKGARAAIEAAGGKIEE (SEQ ID NO:811)
31_GTC09493_Pakistan	MRLNTLSPAEGSKKAGKRLGRGIGSGLGKTGGRGHKGQKSRSGGGVRRGFEGGQMPLYRRLPKFGFTSRKAAITAEVR LSDLAKVEGGVVDLNTLKAANIIGIQIEFAKVILAGEVTTPVTVRGLRVTKGARAAIEAAGGKIEE (SEQ ID NO:819)
32_GTC09549_Typhimurium	MRLNTLSPAEGSKKAGKRLGRGIGSGLGKTGGRGHKGQKSRSGGGVRRGFEGGQMPLYRRLPKFGFTSRKAAITAEVR LSDLAKVEGGVVDLNTLKAANIIGIQIEFAKVILAGEVTTPVTVRGLRVTKGARAAIEAAGGKIEE (SEQ ID NO:825)
33_ATCCBAA-1675_Infantis	MRLNTLSPAEGSKKAGKRLGRGIGSGLGKTGGRGHKGQKSRSGGGVRRGFEGGQMPLYRRLPKFGFTSRKAAITAEVR LSDLAKVEGGVVDLNTLKAANIIGIQIEFAKVILAGEVTTPVTVRGLRVTKGARAAIEAAGGKIEE (SEQ ID NO:832)
34_ATCCBAA-1738_Thompson	MRLNTLSPAEGSKKAGKRLGRGIGSGLGKTGGRGHKGQKSRSGGGVRRGFEGGQMPLYRRLPKFGFTSRKAAITAEVR LSDLAKVEGGVVDLNTLKAANIIGIQIEFAKVILAGEVTTPVTVRGLRVTKGARAAIEAAGGKIEE (SEQ ID NO:843)
35_ATCC9712_Saintpaul	MRLNTLSPAEGSKKAGKRLGRGIGSGLGKTGGRGHKGQKSRSGGGVRRGFEGGQMPLYRRLPKFGFTSRKAAITAEVR LSDLAKVEGGVVDLNTLKAANIIGIQIEFAKVILAGEVTTPVTVRGLRVTKGARAAIEAAGGKIEE (SEQ ID NO:853)
36_jfrlSe1402-1_Infantis	MRLNTLSPAEGSKKAGKRLGRGIGSGLGKTGGRGHKGQKSRSGGGVRRGFEGGQMPLYRRLPKFGFTSRKAAITAEVR LSDLAKVEGGVVDLNTLKAANIIGIQIEFAKVILAGEVTTPVTVRGLRVTKGARAAIEAAGGKIEE (SEQ ID NO:864)
38_jfrlSe1402-3_Brandenburg	MRLNTLSPAEGSKKAGKRLGRGIGSGLGKTGGRGHKGQKSRSGGGVRRGFEGGQMPLYRRLPKFGFTSRKAAITAEVR LSDLAKVEGGVVDLNTLKAANIIGIQIEFAKVILAGEVTTPVTVRGLRVTKGARAAIEAAGGKIEE (SEQ ID NO:874)
39_jfrlSe1402-4_Infantis	MRLNTLSPAEGSKKAGKRLGRGIGSGLGKTGGRGHKGQKSRSGGGVRRGFEGGQMPLYRRLPKFGFTSRKAAITAEVR LSDLAKVEGGVVDLNTLKAANIIGIQIEFAKVILAGEVTTPVTVRGLRVTKGARAAIEAAGGKIEE (SEQ ID NO:881)
40_jfrlSe1402-5_Brandenburg	MRLNTLSPAEGSKKAGKRLGRGIGSGLGKTGGRGHKGQKSRSGGGVRRGFEGGQMPLYRRLPKFGFTSRKAAITAEVR LSDLAKVEGGVVDLNTLKAANIIGIQIEFAKVILAGEVTTPVTVRGLRVTKGARAAIEAAGGKIEE (SEQ ID NO:891)
41_jfrlSe1402-6_Rissen	MRLNTLSPAEGSKKAGKRLGRGIGSGLGKTGGRGHKGQKSRSGGGVRRGFEGGQMPLYRRLPKFGFTSRKAAITAEVR LSDLAKVEGGVVDLNTLKAANIIGIQIEFAKVILAGEVTTPVTVRGLRVTKGARAAIEAAGGKIEE (SEQ ID NO:898)
42_jfrlSe1402-7_Orion	MRLNTLSPAEGSKKAGKRLGRGIGSGLGKTGGRGHKGQKSRSGGGVRRGFEGGQMPLYRRLPKFGFTSRKAAITAEVR LSDLAKVEGGVVDLNTLKAANIIGIQIEFAKVILAGEVTTPVTVRGLRVTKGARAAIEAAGGKIEE (SEQ ID NO:905)
43_jfrlSe1402-8_Rissen	MRLNTLSPAEGSKKAGKRLGRGIGSGLGKTGGRGHKGQKSRSGGGVRRGFEGGQMPLYRRLPKFGFTSRKAAITAEVR LSDLAKVEGGVVDLNTLKAANIIGIQIEFAKVILAGEVTTPVTVRGLRVTKGARAAIEAAGGKIEE (SEQ ID NO:915)
44_jfrlSe1402-9_Rissen	MRLNTLSPAEGSKKAGKRLGRGIGSGLGKTGGRGHKGQKSRSGGGVRRGFEGGQMPLYRRLPKFGFTSRKAAITAEVR LSDLAKVEGGVVDLNTLKAANIIGIQIEFAKVILAGEVTTPVTVRGLRVTKGARAAIEAAGGKIEE (SEQ ID NO:922)
45_jfrlSe1402-10_Rissen	MRLNTLSPAEGSKKAGKRLGRGIGSGLGKTGGRGHKGQKSRSGGGVRRGFEGGQMPLYRRLPKFGFTSRKAAITAEVR LSDLAKVEGGVVDLNTLKAANIIGIQIEFAKVILAGEVTTPVTVRGLRVTKGARAAIEAAGGKIEE (SEQ ID NO:929)
46_jfrlSe1402-11_Rissen	MRLNTLSPAEGSKKAGKRLGRGIGSGLGKTGGRGHKGQKSRSGGGVRRGFEGGQMPLYRRLPKFGFTSRKAAITAEVR LSDLAKVEGGVVDLNTLKAANIIGIQIEFAKVILAGEVTTPVTVRGLRVTKGARAAIEAAGGKIEE (SEQ ID NO:936)
47_jfrlSe1402-12_Rissen	MRLNTLSPAEGSKKAGKRLGRGIGSGLGKTGGRGHKGQKSRSGGGVRRGFEGGQMPLYRRLPKFGFTSRKAAITAEVR LSDLAKVEGGVVDLNTLKAANIIGIQIEFAKVILAGEVTTPVTVRGLRVTKGARAAIEAAGGKIEE (SEQ ID NO:943)
48_jfrlSe1402-13_Mbandaka	MRLNTLSPAEGSKKAGKRLGRGIGSGLGKTGGRGHKGQKSRSGGGVRRGFEGGQMPLYRRLPKFGFTSRKAAITAEVR LSDLAKVEGGVVDLNTLKAANIIGIQIEFAKVILAGEVTTPVTVRGLRVTKGARAAIEAAGGKIEE (SEQ ID NO:950)
49_jfrlSe1402-14_Mbandaka	MRLNTLSPAEGSKKAGKRLGRGIGSGLGKTGGRGHKGQKSRSGGGVRRGFEGGQMPLYRRLPKFGFTSRKAAITAEVR LSDLAKVEGGVVDLNTLKAANIIGIQIEFAKVILAGEVTTPVTVRGLRVTKGARAAIEAAGGKIEE (SEQ ID NO:960)
50_jfrlSe1402-15_Orion	MRLNTLSPAEGSKKAGKRLGRGIGSGLGKTGGRGHKGQKSRSGGGVRRGFEGGQMPLYRRLPKFGFTSRKAAITAEVR LSDLAKVEGGVVDLNTLKAANIIGIQIEFAKVILAGEVTTPVTVRGLRVTKGARAAIEAAGGKIEE (SEQ ID NO:970)
51_jfrlSe1409-1_Altona	MHLNTLSPAEGSKKAGKRLGRGIGSGLGKTGGRGHKGQKSRSGGGVRRGFEGGQMPLYRRLPKFGFTSRKAAITAEVR LSDLAKVEGGVVDLNTLKAANIIGIQIEFAKVILAGEVTTPVTVRGLRVTKGARAAIEAAGGKIEE (SEQ ID NO:978)
52_jfrlSe1409-2_Istanbul	MRLNTLSPAEGSKKAGKRLGRGIGSGLGKTGGRGHKGQKSRSGGGVRRGFEGGQMPLYRRLPKFGFTSRKAAITAEVR LSDLAKVEGGVVDLNTLKAANIIGIQIEFAKVILAGEVTTPVTVRGLRVTKGARAAIEAAGGKIEE (SEQ ID NO:987)
53_jfrlSe1409-3_Senftenberg	
54_jfrlSe1409-4_UN_O13_	MRLNTLSPAEGSKKAGKRLGRGIGSGLGKTGGRGHKGQKSRSGGGVRRGFEGGQMPLYRRLPKFGFTSRKAAITAEVR LSDLAKVEGGVVDLNTLKAANIIGIQIEFAKVILAGEVTTPVTVRGLRVTKGARAAIEAAGGKIEE (SEQ ID NO:999)
55_jfrlSe1409-5_UN_O1,3,19_	MRLNTLSPAEGSKKAGKRLGRGIGSGLGKTGGRGHKGQKSRSGGGVRRGFEGGQMPLYRRLPKFGFTSRKAAITAEVR LSDLAKVEGGVVDLNTLKAANIIGIQIEFAKVILAGEVTTPVTVRGLRVTKGARAAIEAAGGKIEE (SEQ ID NO:1008)
56_jfrlSe1409-6_Montevideo	MRLNTLSPAEGSKKAGKRLGRGIGSGLGKTGGRGHKGQKSRSGGGVRRGFEGGQMPLYRRLPKFGFTSRKAAITAEVR LSDLAKVEGGVVDLNTLKAANIIGIQIEFAKVILAGEVTTPVTVRGLRVTKGARAAIEAAGGKIEE (SEQ ID NO:1016)
57_jfrlSe1409-7_UN_O1,3,19_	MRLNTLSPAEGSKKAGKRLGRGIGSGLGKTGGRGHKGQKSRSGGGVRRGFEGGQMPLYRRLPKFGFTSRKAAITAEVR LSDLAKVEGGVVDLNTLKAANIIGIQIEFAKVILAGEVTTPVTVRGLRVTKGARAAIEAAGGKIEE (SEQ ID NO:1024)
58_jfrlSe1409-8_UN_O18_	MRLNTLSPAEGSKKAGKRLGRGIGSGLGKTGGRGHKGQKSRSGGGVRRGFEGGQMPLYRRLPKFGFTSRKAAITAEVR LSDLAKVEGGVVDLNTLKAANIIGIQIEFAKVILAGEVTTPVTVRGLRVTKGARAAIEAAGGKIEE (SEQ ID NO:1033)
61_jfrlSe1409-11_UN_O1,3,19_	MRLNTLSPAEGSKKAGKRLGRGIGSGLGKTGGRGHKGQKSRSGGGVRRGFEGGQMPLYRRLPKFGFTSRKAAITAEVR LSDLAKVEGGVVDLNTLKAANIIGIQIEFAKVILAGEVTTPVTVRGLRVTKGARAAIEAAGGKIEE (SEQ ID NO:1042)
67_jfrlSe1409-17_Rissen	MRLNTLSPAEGSKKAGKRLGRGIGSGLGKTGGRGHKGQKSRSGGGVRRGFEGGQMPLYRRLPKFGFTSRKAAITAEVR LSDLAKVEGGVVDLNTLKAANIIGIQIEFAKVILAGEVTTPVTVRGLRVTKGARAAIEAAGGKIEE (SEQ ID NO:1050)
71_jfrlSe1409-21_Amsterdam	MRLNTLSPAEGSKKAGKRLGRGIGSGLGKTGGRGHKGQKSRSGGGVRRGFEGGQMPLYRRLPKFGFTSRKAAITAEVR LSDLAKVEGGVVDLNTLKAANIIGIQIEFAKVILAGEVTTPVTVRGLRVTKGARAAIEAAGGKIEE (SEQ ID NO:1059)
80_HySe09_Enteritidis	MRLNTLSPAEGSKKAGKRLGRGIGSGLGKTGGRGHKGQKSRSGGGVRRGFEGGQMPLYRRLPKFGFTSRKAAITAEVR LSDLAKVEGGVVDLNTLKAANIIGIQIEFAKVILAGEVTTPVTVRGLRVTKGARAAIEAAGGKIEE (SEQ ID NO:1069)
100_HySe29_Schwarzengrund	MRLNTLSPAEGSKKAGKRLGRGIGSGLGKTGGRGHKGQKSRSGGGVRRGFEGGQMPLYRRLPKFGFTSRKAAITAEVR LSDLAKVEGGVVDLNTLKAANIIGIQIEFAKVILAGEVTTPVTVRGLRVTKGARAAIEAAGGKIEE (SEQ ID NO:1079)
103_HySe32_Schwarzengrund	MRLNTLSPAEGSKKAGKRLGRGIGSGLGKTGGRGHKGQKSRSGGGVRRGFEGGQMPLYRRLPKFGFTSRKAAITAEVR LSDLAKVEGGVVDLNTLKAANIIGIQIEFAKVILAGEVTTPVTVRGLRVTKGARAAIEAAGGKIEE (SEQ ID NO:1086)
106_HySe35_Schwarzengrund	MRLNTLSPAEGSKKAGKRLGRGIGSGLGKTGGRGHKGQKSRSGGGVRRGFEGGQMPLYRRLPKFGFTSRKAAITAEVR LSDLAKVEGGVVDLNTLKAANIIGIQIEFAKVILAGEVTTPVTVRGLRVTKGARAAIEAAGGKIEE (SEQ ID NO:1092)

Fig. 41A

	S7
01_NBRC13245T_Typhimurium	MPRRRVIGQ RKILPDPKFGSELLAKFVNILMVDGKKSTAESIVYSALETLAQRSGKSELEAFEVALENVRPTVEVKSRRVGGSTYQVP VEVRPVRRNALAMRWIVEAARKRGDKSMALRLANELSDAADNKGTAVKKREDVHRMAEANKAFAHYRW (SEQ ID NO:581)
02_GTC00131_Enteritidis	MPRRRVIGQ RKILPDPKFGSELLAKFVNILMVDGKKSTAESIVYSALETLAQRSGKSELEAFEVALENVRPTVEVKSRRVGGSTYQVP VEVRPVRRNALAMRWIVEAARKRGDKSMALRLANELSDAAENKGTAVKKREDVHRMAEANKAFAHYRW (SEQ ID NO:593)
03_GTC09491_Enteritidis	MPRRRVIGQ RKILPDPKFGSELLAKFVNILMVDGKKSTAESIVYSALETLAQRSGKSELEAFEVALENVRPTVEVKSRRVGGSTYQVP VEVRPVRRNALAMRWIVEAARKRGDKSMALRLANELSDAAENKGTAVKKREDVHRMAEANKAFAHYRW (SEQ ID NO:602)
04_GTC03838_Enteritidis	MPRRRVIGQ RKILPDPKFGSELLAKFVNILMVDGKKSTAESIVYSALETLAQRSGKSELEAFEVALENVRPTVEVKSRRVGGSTYQVP VEVRPVRRNALAMRWIVEAARKRGDKSMALRLANELSDAAENKGTAVKKREDVHRMAEANKAFAHYRW (SEQ ID NO:609)
05_GTC08914_Enteritidis	MPRRRVIGQ RKILPDPKFGSELLAKFVNILMVDGKKSTAESIVYSALETLAQRSGKSELEAFEVALENVRPTVEVKSRRVGGSTYQVP VEVRPVRRNALAMRWIVEAARKRGDKSMALRLANELSDAAENKGTAVKKREDVHRMAEANKAFAHYRW (SEQ ID NO:616)
06_GTC09421_Enteritidis	MPRRRVIGQ RKILPDPKFGSELLAKFVNILMVDGKKSTAESIVYSALETLAQRSGKSELEAFEVALENVRPTVEVKSRRVGGSTYQVP VEVRPVRRNALAMRWIVEAARKRGDKSMALRLANELSDAAENKGTAVKKREDVHRMAEANKAFAHYRW (SEQ ID NO:626)
07_GTC09489_Enteritidis	MPRRRVIGQ RKILPDPKFGSELLAKFVNILMVDGKKSTAESIVYSALETLAQRSGKSELEAFEVALENVRPTVEVKSRRVGGSTYQVP VEVRPVRRNALAMRWIVEAARKRGDKSMALRLANELSDAAENKGTAVKKREDVHRMAEANKAFAHYRW (SEQ ID NO:635)
08_NBRC3163_Pullorum_Gallinarum	MPRRRVIGQ RKILPDPKFGSELLAKFVNILMVDGKKSTAESIVYSALETLAQRSGKSELEAFEVALENVRPTVEVKSRRVGGSTYQVP VEVRPVRRNALAMRWIVEAARKRGDKSMALRLANELSDAAENKGTAVKKREDVHRMAEANKAFAHYRW (SEQ ID NO:642)
09_NBRC3313_Enteritidis	MPRRRVIGQ RKILPDPKFGSELLAKFVNILMVDGKKSTAESIVYSALETLAQRSGKSELEAFEVALENVRPTVEVKSRRVGGSTYQVP VEVRPVRRNALAMRWIVEAARKRGDKSMALRLANELSDAAENKGTAVKKREDVHRMAEANKAFAHYRW (SEQ ID NO:652)
10_NBRC12529_Typhimurium	MPRRRVIGQ RKILPDPKFGSELLAKFVNILMVDGKKSTAESIVYSALETLAQRSGKSELEAFEVALENVRPTVEVKSRRVGGSTYQVP VEVRPVRRNALAMRWIVEAARKRGDKSMALRLANELSDAADNKGTAVKKREDVHRMAEANKAFAHYRW (SEQ ID NO:659)
11_NBRC14193_Typhimurium	MPRRRVIGQ RKILPDPKFGSELLAKFVNILMVDGKKSTAESIVYSALETLAQRSGKSELEAFEVALENVRPTVEVKSRRVGGSTYQVP VEVRPVRRNALAMRWIVEAARKRGDKSMALRLANELSDAADNKGTAVKKREDVHRMAEANKAFAHYRW (SEQ ID NO:666)
12_NBRC14194_Typhimurium	MPRRRVIGQ RKILPDPKFGSELLAKFVNILMVDGKKSTAESIVYSALETLAQRSGKSELEAFEVALENVRPTVEVKSRRVGGSTYQVP VEVRPVRRNALAMRWIVEAARKRGDKSMALRLANELSDAADNKGTAVKKREDVHRMAEANKAFAHYRW (SEQ ID NO:673)
13_NBRC14209_Typhimurium	MPRRRVIGQ RKILPDPKFGSELLAKFVNILMVDGKKSTAESIVYSALETLAQRSGKSELEAFEVALENVRPTVEVKSRRVGGSTYQVP VEVRPVRRNALAMRWIVEAARKRGDKSMALRLANELSDAADNKGTAVKKREDVHRMAEANKAFAHYRW (SEQ ID NO:680)
14_NBRC14210_Typhimurium	MPRRRVIGQ RKILPDPKFGSELLAKFVNILMVDGKKSTAESIVYSALETLAQRSGKSELEAFEVALENVRPTVEVKSRRVGGSTYQVP VEVRPVRRNALAMRWIVEAARKRGDKSMALRLANELSDAADNKGTAVKKREDVHRMAEANKAFAHYRW (SEQ ID NO:687)
15_NBRC14211_Typhimurium	MPRRRVIGQ RKILPDPKFGSELLAKFVNILMVDGKKSTAESIVYSALETLAQRSGKSELEAFEVALENVRPTVEVKSRRVGGSTYQVP VEVRPVRRNALAMRWIVEAARKRGDKSMALRLANELSDAADNKGTAVKKREDVHRMAEANKAFAHYRW (SEQ ID NO:693)
16_NBRC14212_Typhimurium	MPRRRVIGQ RKILPDPKFGSELLAKFVNILMVDGKKSTAESIVYSALETLAQRSGKSELEAFEVALENVRPTVEVKSRRVGGSTYQVP VEVRPVRRNALAMRWIVEAARKRGDKSMALRLANELSDAADNKGTAVKKREDVHRMAEANKAFAHYRW (SEQ ID NO:700)
17_NBRC15181_Typhimurium	MPRRRVIGQ RKILPDPKFGSELLAKFVNILMVDGKKSTAESIVYSALETLAQRSGKSELEAFEVALENVRPTVEVKSRRVGGSTYQVP VEVRPVRRNALAMRWIVEAARKRGDKSMALRLANELSDAADNKGTAVKKREDVHRMAEANKAFAHYRW (SEQ ID NO:707)
18_NBRC15182_Minnesota	MPRRRVIGQ RKILPDPKFGSELLAKFVNILMVDGKKSTAESIVYSALETLAQRSGKSELEAFEVALENVRPTVEVKSRRVGGSTYQVP VEVRPVRRNALAMRWIVEAARKRGDKSMALRLANELSDAADNKGTAVKKREDVHRMAEANKAFAHYRW (SEQ ID NO:717)
19_NBRC15183_Minnesota	MPRRRVIGQ RKILPDPKFGSELLAKFVNILMVDGKKSTAESIVYSALETLAQRSGKSELEAFEVALENVRPTVEVKSRRVGGSTYQVP VEVRPVRRNALAMRWIVEAARKRGDKSMALRLANELSDAADNKGTAVKKREDVHRMAEANKAFAHYRW (SEQ ID NO:727)
20_NBRC15184_Minnesota	MPRRRVIGQ RKILPDPKFGSELLAKFVNILMVDGKKSTAESIVYSALETLAQRSGKSELEAFEVALENVRPTVEVKSRRVGGSTYQVP VEVRPVRRNALAMRWIVEAARKRGDKSMALRLANELSDAADNKGTAVKKREDVHRMAEANKAFAHYRW (SEQ ID NO:734)
21_NBRC15185_Minnesota	MPRRRVIGQ RKILPDPKFGSELLAKFVNILMVDGKKSTAESIVYSALETLAQRSGKSELEAFEVALENVRPTVEVKSRRVGGSTYQVP VEVRPVRRNALAMRWIVEAARKRGDKSMALRLANELSDAADNKGTAVKKREDVHRMAEANKAFAHYRW (SEQ ID NO:741)
22_NBRC15186_Minnesota	MPRRRVIGQ RKILPDPKFGSELLAKFVNILMVDGKKSTAESIVYSALETLAQRSGKSELEAFEVALENVRPTVEVKSRRVGGSTYQVP VEVRPVRRNALAMRWIVEAARKRGDKSMALRLANELSDAADNKGTAVKKREDVHRMAEANKAFAHYRW (SEQ ID NO:748)
23_NBRC15187_Minnesota	MPRRRVIGQ RKILPDPKFGSELLAKFVNILMVDGKKSTAESIVYSALETLAQRSGKSELEAFEVALENVRPTVEVKSRRVGGSTYQVP VEVRPVRRNALAMRWIVEAARKRGDKSMALRLANELSDAADNKGTAVKKREDVHRMAEANKAFAHYRW (SEQ ID NO:755)
24_NBRC100797_Abony	MPRRRVIGQ RKILPDPKFGSELLAKFVNILMVDGKKSTAESIVYSALETLAQRSGKSELEAFEVALENVRPTVEVKSRRVGGSTYQVP VEVRPVRRNALAMRWIVEAARKRGDKSMALRLANELSDAADNKGTAVKKREDVHRMAEANKAFAHYRW (SEQ ID NO:762)
25_NBRC105684_Choleraesuis	MPRRRVIGQ RKILPDPKFGSELLAKFVNILMVDGKKSTAESIAYSALETLAQRSGKSELEAFEVALENVRPTVEVKSRRVGGSTYQVP VEVRPVRRNALAMRWIVEAARKRGDKSMALRLANELSDAADNKGTAVKKREDVHRMAEANKAFAHYRW (SEQ ID NO:772)
26_NBRC105726_Typhimurium	
27_JCM3919_UN_O7_	MPRRRVIGQ RKILPDPKFGSELLAKFVNILMVDGKKSTAESIVYSALETLAQRSGKSELEAFEVALENVRPTVEVKSRRVGGSTYQVP VEVRPVRRNALAMRWIVEAARKRGDKSMALRLANELSDAADNKGTAVKKREDVHRMAEANKAFAHYRW (SEQ ID NO:791)
28_NBRC15335_Minnesota	MPRRRVIGQ RKILPDPKFGSELLAKFVNILMVDGKKSTAESIVYSALETLAQRSGKSELEAFEVALENVRPTVEVKSRRVGGSTYQVP VEVRPVRRNALAMRWIVEAARKRGDKSMALRLANELSDAADNKGTAVKKREDVHRMAEANKAFAHYRW (SEQ ID NO:798)
29_GTC09490_Enteritidis	MPRRRVIGQ RKILPDPKFGSELLAKFVNILMVDGKKSTAESIVYSALETLAQRSGKSELEAFEVALENVRPTVEVKSRRVGGSTYQVP VEVRPVRRNALAMRWIVEAARKRGDKSMALRLANELSDAAENKGTAVKKREDVHRMAEANKAFAHYRW (SEQ ID NO:805)
30_GTC09492_Braenderup	MPRRRVIGQ RKILPDPKFGSELLAKFVNILMVDGKKSTAESIVYSALETLAQRSGKSELEAFEVALENVRPTVEVKSRRVGGSTYQVP VEVRPVRRNALAMRWIVEAARKRGDKSMALRLANELSDAAENKGTAVKKREDVHRMAEANKAFAHYRW (SEQ ID NO:812)
31_GTC09493_Pakistan	MPRRRVIGQ RKILPDPKFGSELLAKFVNILMVDGKKSTAESIVYSALETLAQRSGKSELEAFEVALENVRPTVEVKSRRVGGSTYQVP VEVRPVRRNALAMRWIVEAARKRGDKSMALRLANELSDAADNKGTAVKKREDVHRMAEANKAFAHYRW (SEQ ID NO:820)
32_GTC09549_Typhimurium	MPRRRVIGQ RKILPDPKFGSELLAKFVNILMVDGKKSTAESIVYSALETLAQRSGKSELEAFEVALENVRPTVEVKSRRVGGSTYQVP VEVRPVRRNALAMRWIVEAARKRGDKSMALRLANELSDAADNKGTAVKKREDVHRMAEANKAFAHYRW (SEQ ID NO:826)
33_ATCCBAA-1675_Infantis	MPRRRVIGQ RKILPDPKFGSELLAKFVNILMVDGKKSTAESIVYSALETLAQRSGKSELEAFEVALENVRPTVEVKSRRVGGSTYQVP VEVRPVRRNALAMRWIVEAARKRGDKSMALRLANELSDAADNKGTAVKKREDVHRMAEANKAFAHYRW (SEQ ID NO:833)
34_ATCCBAA-1738_Thompson	MPRRRVIGQ RKILPDPKFGSELLAKFVNILMVDGKKSTAESIVYSALETLAQRSGKSELEAFEVALENVRPTVEVKSRRVGGSTYQVP VEVRPVRRNALAMRWIVEAARKRGDKSMALRLANELSDAAENKGTAVKKREDVHRMAEANKAFAHYRW (SEQ ID NO:844)
35_ATCC9712_Saintpaul	MPRRRVIGQ RKILPDPKFGSELLAKFVNILMVDGKKSTAESIVYSALETLAQRSGKSELEAFEVALENVRPTVEVKSRRVGGSTYQVP VEVRPVRRNALAMRWIVEAARKRGDKSMALRLANELSDAADNKGTAVKKREDVHRMAEANKAFAHYRW (SEQ ID NO:854)
36_jfrlSe1402-1_Infantis	MPRRRVIGQ RKILPDPKFGSELLAKFVNILMVDGKKSTAESIVYSALETLAQRSGKSELEAFEVALENVRPTVEVKSRRVGGSTYQVP VEVRPVRRNALAMRWIVEAARKRGDKSMALRLANELSDAADNKGTAVKKREDVHRMAEANKAFAHYRW (SEQ ID NO:865)
38_jfrlSe1402-3_Brandenburg	MPRRRVIGQ RKILPDPKFGSELLAKFVNILMVDGKKSTAESIVYSALETLAQRSGKSELEAFEVALENVRPTVEVKSRRVGGSTYQVP VEVRPVRRNALAMRWIVEAARKRGDKSMALRLANELSDAADNKGTAVKKREDVHRMAEANKAFAHYRW (SEQ ID NO:875)
39_jfrlSe1402-4_Infantis	MPRRRVIGQ RKILPDPKFGSELLAKFVNILMVDGKKSTAESIVYSALETLAQRSGKSELEAFEVALENVRPTVEVKSRRVGGSTYQVP VEVRPVRRNALAMRWIVEAARKRGDKSMALRLANELSDAADNKGTAVKKREDVHRMAEANKAFAHYRW (SEQ ID NO:882)
40_jfrlSe1402-5_Brandenburg	MPRRRVIGQ RKILPDPKFGSELLAKFVNILMVDGKKSTAESIVYSALETLAQRSGKSELEAFEVALENVRPTVEVKSRRVGGSTYQVP VEVRPVRRNALAMRWIVEAARKRGDKSMALRLANELSDAADNKGTAVKKREDVHRMAEANKAFAHYRW (SEQ ID NO:892)
41_jfrlSe1402-6_Rissen	MPRRRVIGQ RKILPDPKFGSELLAKFVNILMVDGKKSTAESIVYSALETLAQRSGKSELEAFEVALENVRPTVEVKSRRVGGSTYQVP VEVRPVRRNALAMRWIVEAARKRGDKSMALRLANELSDAAENKGTAVKKREDVHRMAEANKAFAHYRW (SEQ ID NO:899)
42_jfrlSe1402-7_Orion	MPRRRVIGQ RKILPDPKFGSELLAKFVNILMVDGKKSTAESIVYSALETLAQRSGKSELEAFEVALENVRPTVEVKSRRVGGSTYQVP VEVRPVRRNALAMRWIVEAARKRGDKSMALRLANELSDAADNKGTAVKKREDVHRMAEANKAFAHYRW (SEQ ID NO:906)
43_jfrlSe1402-8_Rissen	MPRRRVIGQ RKILPDPKFGSELLAKFVNILMVDGKKSTAESIVYSALETLAQRSGKSELEAFEVALENVRPTVEVKSRRVGGSTYQVP VEVRPVRRNALAMRWIVEAARKRGDKSMALRLANELSDAAENKGTAVKKREDVHRMAEANKAFAHYRW (SEQ ID NO:916)
44_jfrlSe1402-9_Rissen	MPRRRVIGQ RKILPDPKFGSELLAKFVNILMVDGKKSTAESIVYSALETLAQRSGKSELEAFEVALENVRPTVEVKSRRVGGSTYQVP VEVRPVRRNALAMRWIVEAARKRGDKSMALRLANELSDAAENKGTAVKKREDVHRMAEANKAFAHYRW (SEQ ID NO:923)
45_jfrlSe1402-10_Rissen	MPRRRVIGQ RKILPDPKFGSELLAKFVNILMVDGKKSTAESIVYSALETLAQRSGKSELEAFEVALENVRPTVEVKSRRVGGSTYQVP VEVRPVRRNALAMRWIVEAARKRGDKSMALRLANELSDAAENKGTAVKKREDVHRMAEANKAFAHYRW (SEQ ID NO:930)
46_jfrlSe1402-11_Rissen	MPRRRVIGQ RKILPDPKFGSELLAKFVNILMVDGKKSTAESIVYSALETLAQRSGKSELEAFEVALENVRPTVEVKSRRVGGSTYQVP VEVRPVRRNALAMRWIVEAARKRGDKSMALRLANELSDAAENKGTAVKKREDVHRMAEANKAFAHYRW (SEQ ID NO:937)
47_jfrlSe1402-12_Rissen	MPRRRVIGQ RKILPDPKFGSELLAKFVNILMVDGKKSTAESIVYSALETLAQRSGKSELEAFEVALENVRPTVEVKSRRVGGSTYQVP VEVRPVRRNALAMRWIVEAARKRGDKSMALRLANELSDAAENKGTAVKKREDVHRMAEANKAFAHYRW (SEQ ID NO:944)
48_jfrlSe1402-13_Mbandaka	MPRRRVIGQ RKILPDPKFGSELLAKFVNILMVDGKKSTAESIVYSALETLAQRSGKSELEAFEVALENVRPTVEVKSRRVGGSTYQVP VEVRPVRRNALAMRWIVEAARKRGDKSMALRLANELSDAAENKGTAVKKREDVHRMAEANKAFAHYRW (SEQ ID NO:951)

Fig. 41B

49_jfrlSe1402-14_Mbandaka	MPRRRVIGQRKILPDPKFGSELLAKFVNILMVDGKKSTAESIVYSALETLAQRSGKSELEAFEVALENVRPTVEVKSRRVGGSTYQVP VEVRPVRRNALAMRWIVEAARKRGDKSMALRLANELSDAAENKGTAVKKREDVHRMAEANKAFAHYRW (SEQ ID NO:961)
50_jfrlSe1402-15_Orion	MPRRRVIGQRKILPDPKFGSELLAKFVNILMVDGKKSTAESIVYSALETLAQRSGKSELEAFEVALENVRPTVEVKSRRVGGSTYQVP VEVRPVRRNALAMRWIVEAARKRGDKSMALRLANELSDAADNKGTAVKKREDVHRMAEANKAFAHYRW (SEQ ID NO:971)
51_jfrlSe1409-1_Altona	MPRRRVIGQRKILPDPKFGSELLAKFVNILMVDGKKSTAESIVYSALETLAQRSGKSELEAFEVALENVRPTVEVKSRRVGGSTYQVP VEVRPVRRNALAMRWIVEAARKRGDKSMALRLANELSDAAENKGTAVKKREDVHRMAEANKAFAHYRW (SEQ ID NO:979)
52_jfrlSe1409-2_Istanbul	MPRRRVIGQRKILPDPKFGSELLAKFVNILMVDGKKSTAESIVYSALETLAQRSGKSELEAFEVALENVRPTVEVKSRRVGGSTYQVP VEVRPVRRNALAMRWIVEAARKRGDKSMALRLANELSDAAENKGTAVKKREDVHRMAEANKAFAHYRW (SEQ ID NO:988)
53_jfrlSe1409-3_Senftenberg	
54_jfrlSe1409-4_UN_O13_	MPRRRVIGQRKILPDPKFGSELLAKFVNILMVDGKKSTAESIVYSALETLAQRSGKSELEAFEVALENVRPTVEVKSRRVGGSTYQVP VEVRPVRRNALAMRWIVEAARKRGDKSMALRLANELSDAAENKGTAVKKREDVHRMAEANKAFAHYRW (SEQ ID NO:1000)
55_jfrlSe1409-5_UN_O1,3,19_	MPRRRVIGQRKILPDPKFGSELLAKFVNILMVDGKKSTAESIVYSALETLAQRSGKSELEAFEVALENVRPTVEVKSRRVGGSTYQVP VEVRPVRRNALAMRWIVEAARKRGDKSMALRLANELSDAAENKGTAVKKREDVHRMAEANKAFAHYRW (SEQ ID NO:1009)
56_jfrlSe1409-6_Montevideo	
57_jfrlSe1409-7_UN_O1,3,19_	MPRRRVIGQRKILPDPKFGSELLAKFVNILMVDGKKSTAESIVYSALETLAQRSGKSELEAFEVALENVRPTVEVKSRRVGGSTYQVP VEVRPVRRNALAMRWIVEAARKRGDKSMALRLANELSDAAENKGTAVKKREDVHRMAEANKAFAHYRW (SEQ ID NO:1025)
58_jfrlSe1409-8_UN_O18_	MPRRRVIGQRKILPDPKFGSELLAKFVNILMVDGKKSTAESIVYSALETLAQRSGKSELEAFEVALENVRPTVEVKSRRVGGSTYQVP VEVRPVRRNALAMRWIVEAARKRGDKSMALRLANELSDAAENKGTAVKKREDVHRMAEANKAFAHYRW (SEQ ID NO:1034)
61_jfrlSe1409-11_UN_O1,3,19_	MPRRRVIGQRKILPDPKFGSELLAKFVNILMVDGKKSTAESIVYSALETLAQRSGKSELEAFEVALENVRPTVEVKSRRVGGSTYQVP VEVRPVRRNALAMRWIVEAARKRGDKSMALRLANELSDAAENKGTAVKKREDVHRMAEANKAFAHYRW (SEQ ID NO:1043)
67_jfrlSe1409-17_Rissen	MPRRRVIGQRKILPDPKFGSELLAKFVNILMVDGKKSTAESIVYSALETLAQRSGKSELEAFEVALENVRPTVEVKSRRVGGSTYQVP VEVRPVRRNALAMRWIVEAARKRGDKSMALRLANELSDAAENKGTAVKKREDVHRMAEANKAFAHYRW (SEQ ID NO:1051)
71_jfrlSe1409-21_Amsterdam	MATVNQLVRKPRARKVAKSNVPALEACPQKRGVCTRVYTTTPKKPNSALRKVCRVRLTNGFEVTSYIGGEGHNLQEHSVILIRGGR VKDLPGVRYHTVRGALDCSGVKDRKQARSKYGVKRPKA (SEQ ID NO:1060)
80_HySe09_Enteritidis	MPRRRVIGQRKILPDPKFGSELLAKFVNILMVDGKKSTAESIVYSALETLAQRSGKSELEAFEVALENVRPTVEVKSRRVGGSTYQVP VEVRPVRRNALAMRWIVEAARKRGDKSMALRLANELSDAAENKGTAVKKREDVHRMAEANKAFAHYRW (SEQ ID NO:1070)
100_HySe29_Schwarzengrund	MPRRRVIGQRKILPDPKFGSELLAKFVNILMVDGKKSTAESIVYSALETLAQRSGKSELEAFEVALENVRPTVEVKSRRVGGSTYQVP VEVRPVRRNALAMRWIVEAARKRGDKSMALRLANELSDAAENKGTAVKKREDVHRMAEANKAFAHYRW (SEQ ID NO:1080)
103_HySe32_Schwarzengrund	MPRRRVIGQRKILPDPKFGSELLAKFVNILMVDGKKSTAESIVYSALETLAQRSGKSELEAFEVALENVRPTVEVKSRRVGGSTYQVP VEVRPVRRNALAMRWIVEAARKRGDKSMALRLANELSDAAENKGTAVKKREDVHRMAEANKAFAHYRW (SEQ ID NO:1087)
106_HySe35_Schwarzengrund	MPRRRVIGQRKILPDPKFGSELLAKFVNILMVDGKKSTAESIVYSALETLAQRSGKSELEAFEVALENVRPTVEVKSRRVGGSTYQVP VEVRPVRRNALAMRWIVEAARKRGDKSMALRLANELSDAAENKGTAVKKREDVHRMAEANKAFAHYRW (SEQ ID NO:1093)

Fig.42

	gns
01_NBRC13245T_Typhimurium	MNSEELTHKAEEEIAALISKKVAELRKKTGQEVSEIEFAPRETMKGLEGYHVKIKLL (SEQ ID NO:582)
02_GTC00131_Enteritidis	
03_GTC09491_Enteritidis	
04_GTC03838_Enteritidis	
05_GTC08914_Enteritidis	MNSEELTHKAEEEIAALISKKVAELRKKTGQEVSEIEFAPRETMKGLEGYHVKIKLL (SEQ ID NO:617)
06_GTC09421_Enteritidis	MNSEELTHKAEEEIAALISKKVAELRKKTGQEVSEIEFAPRETMKGLEGYHVKIKLL (SEQ ID NO:627)
07_GTC09489_Enteritidis	
08_NBRC3163_Pullorum_Gallinarum	MNSEELTHKAEEEIAALISKKVAELRKKTGQEVSEIEFAPRETMKGLEGYHVKIKLL (SEQ ID NO:643)
09_NBRC3313_Enteritidis	
10_NBRC12529_Typhimurium	
11_NBRC14193_Typhimurium	
12_NBRC14194_Typhimurium	
13_NBRC14209_Typhimurium	
14_NBRC14210_Typhimurium	
15_NBRC14211_Typhimurium	
16_NBRC14212_Typhimurium	
17_NBRC15181_Typhimurium	MNSEELTHKAEEEIAALISKKVAELRKKTGQEVSEIEFVPRETMKGLEGYHVKIKLL (SEQ ID NO:708)
18_NBRC15182_Minnesota	MNSEELTHKAEEEIAALISKKVAELRKKTGQEVSEIEFAPRETMKGLEGYHVKIKLL (SEQ ID NO:718)
19_NBRC15183_Minnesota	
20_NBRC15184_Minnesota	
21_NBRC15185_Minnesota	
22_NBRC15186_Minnesota	
23_NBRC15187_Minnesota	
24_NBRC100797_Abony	MNSEELTHKAEEEIAALISKKVAELRKKTGQEVSEIEFAPRETMKGLEGYHVKIKLL (SEQ ID NO:763)
25_NBRC105684_Choleraesuis	MNSEELTHKAEEEIAALISKKVAELRKKTGQEVSEIEFAPRETMKGLEGYHVKIKLL (SEQ ID NO:773)
26_NBRC105726_Typhimurium	MNSEELTHKAEEEIAALISKKVAELRKKTGQEVSEIEFAPRETMKGLEGYHVKIKLL (SEQ ID NO:782)
27_JCM3919_UN_O7_	
28_NBRC15335_Minnesota	
29_GTC09490_Enteritidis	
30_GTC09492_Braenderup	
31_GTC09493_Pakistan	
32_GTC09549_Typhimurium	
33_ATCCBAA-1675_Infantis	MNSEELTHKAEEEIAALISKKVAELRKKTGQEVSEIEFAPRETMKGLEGYHVKIKLL (SEQ ID NO:834)
34_ATCCBAA-1738_Thompson	MNSEELTHKAEEEIAALISKKVAELRKKTGQEVSEIEFAPRETMKGLEGYHVKIKLL (SEQ ID NO:845)
35_ATCC9712_Saintpaul	MNSEELTHKAEEEIAALISKKVAELRKKTGQEVSEIEFAPRETMKGLEGYHVKIKLL (SEQ ID NO:855)
36_jfrlSe1402-1_Infantis	MNSEELTHKAEEEIAALISKKVAELRKKTGQEVSEIEFVPRETMKGLEGYHVKIKLL (SEQ ID NO:866)
38_jfrlSe1402-3_Brandenburg	
39_jfrlSe1402-4_Infantis	MNSEELTHKAEEEIAALISKKVAELRKKTGQEVSEIEFVPRETMKGLEGYHVKIKLL (SEQ ID NO:883)
40_jfrlSe1402-5_Brandenburg	
41_jfrlSe1402-6_Rissen	
42_jfrlSe1402-7_Orion	MNSEELTHKAEEEIAALISKKVAELRKKTGQEVSEIEFAPRETMKGLEGYHVKIKLL (SEQ ID NO:907)
43_jfrlSe1402-8_Rissen	
44_jfrlSe1402-9_Rissen	
45_jfrlSe1402-10_Rissen	
46_jfrlSe1402-11_Rissen	
47_jfrlSe1402-12_Rissen	
48_jfrlSe1402-13_Mbandaka	MNSEELTHKAEEEIAALISKKVAELRKKTGQEVSEIEFVPRETMKGLEGYHVKIKLL (SEQ ID NO:952)
49_jfrlSe1402-14_Mbandaka	MNSEELTHKAEEEIAALISKKVAELRKKTGQEVSEIEFVPRETMKGLEGYHVKIKLL (SEQ ID NO:962)
50_jfrlSe1402-15_Orion	MNSEELTHKAEEEIAALISKKVAELRKKTGQEVSEIEFAPRETMKGLEGYHVKIKLL (SEQ ID NO:972)
51_jfrlSe1409-1_Altona	MNSEELTHKAEEEIAALISKKVAELRKKTGQEVSEIEFAPRETMKGLEGYHVKIKLL (SEQ ID NO:980)
52_jfrlSe1409-2_Istanbul	MNSEELTHKAEEEIAALISKKVAELRKKTGQEVSEIEFAPRETMKGLEGYHVKIKLL (SEQ ID NO:989)
53_jfrlSe1409-3_Senftenberg	MNSEELTHKAEEEIAALISKKVAELRKKTGQEVSEIEFVPRETMKGLEGYHVKIKLL (SEQ ID NO:992)
54_jfrlSe1409-4_UN_O13_	MNSEELTHKAEEEIAALISKKVAELRKKTGQEVSEIEFAPRETMKGLEGYHVKIKLL (SEQ ID NO:1001)
55_jfrlSe1409-5_UN_O1,3,19_	MNSEELTHKAEEEIAALISKKVAELRKKTGQEVSEIEFVPRETMKGLEGYHVKIKLL (SEQ ID NO:1010)
56_jfrlSe1409-6_Montevideo	MNSEELTHKAEEEIAALISKKVAELRKKTGQEVSEIEFAPRETMKGLEGYHVKIKLL (SEQ ID NO:1017)
57_jfrlSe1409-7_UN_O1,3,19_	MNSEELTHKAEEEIAALISKKVAELRKKTGQEVSEIEFVPRETMKGLEGYHVKIKLL (SEQ ID NO:1026)
58_jfrlSe1409-8_UN_O18_	MNSEELTHKAEEEIAALISKKVAELRKKTGQEVSEIEFAPRETMKGLEGYHVKIKLL (SEQ ID NO:1035)
61_jfrlSe1409-11_UN_O1,3,19_	MNSEELTHKAEEEIAALISKKVAELRKKTGQEVSEIEFVPRETMKGLEGYHVKIKLL (SEQ ID NO:1044)
67_jfrlSe1409-17_Rissen	MNSEELTHKAEEEIAALISKKVAELRKKTGQEVSEIEFAPRETMKGLEGYHVKIKLL (SEQ ID NO:1052)
71_jfrlSe1409-21_Amsterdam	MNSEELTHKAEEEIAALISKKVAELRKKTGQEVSEIEFVPRETMKGLEGYHVKIKLL (SEQ ID NO:1061)
80_HySe09_Enteritidis	MNSEELTHKAEEEIAALISKKVAELRKKTGQEVSEIEFVPRETMKGLEGYHVKIKLL (SEQ ID NO:1071)
100_HySe29_Schwarzengrund	MNSEELTHKAEEEIAALISKKVAELRKKTGQEVSEIEFAPRETMKGLEGYHVKIKLL (SEQ ID NO:1081)
103_HySe32_Schwarzengrund	MNSEELTHKAEEEIAALISKKVAELRKKTGQEVSEIEFAPRETMKGLEGYHVKIKLL (SEQ ID NO:1088)
106_HySe35_Schwarzengrund	MNSEELTHKAEEEIAALISKKVAELRKKTGQEVSEIEFAPRETMKGLEGYHVKIKLL (SEQ ID NO:1094)

Fig.43

	YibT
01_NBRC13245T_Typhimurium	MAKIGENVPLLIDKAVDFMASSQAFREYLNKTPPRDYVPSEVPSESAPIYLQRLEYYRRLYRPKEEERG (SEQ ID NO:583)
02_GTC00131_Enteritidis	
03_GTC09491_Enteritidis	
04_GTC03838_Enteritidis	
05_GTC08914_Enteritidis	MAKIGENVPLLIDKAVDFMASSQAFREYLNKTPPRDYVPSEVPSESAPIYLQRLEYYRRLYRPKEEERG (SEQ ID NO:618)
06_GTC09421_Enteritidis	MAKIGENVPLLIDKAVDFMASSQAFREYLNKTPPRDYVPSEVPSESAPIYLQRLEYYRRLYRPKEEERG (SEQ ID NO:628)
07_GTC09489_Enteritidis	
08_NBRC3163_Pullorum_Gallinarum	MAKIGENVPLLIDKAVDFKASSQAFREYLNKTPPRDYVPSEVPSESAPIYLQRLEYYRRLYRPKEEERG (SEQ ID NO:644)
09_NBRC3313_Enteritidis	
10_NBRC12529_Typhimurium	
11_NBRC14193_Typhimurium	
12_NBRC14194_Typhimurium	
13_NBRC14209_Typhimurium	
14_NBRC14210_Typhimurium	
15_NBRC14211_Typhimurium	
16_NBRC14212_Typhimurium	
17_NBRC15181_Typhimurium	MAKIGENVPLLIDKAVDFMASSQAFREYLNKTPPRDYVPSEVPSESAPIYLQRLEYYRRLYRPKEEERG (SEQ ID NO:709)
18_NBRC15182_Minnesota	MAKIGENVPLLIDKAVDFMASSQAFREYLNKTPPRDYVPSEVPSESAPIYLQRLEYYRRLYRPKEEERG (SEQ ID NO:719)
19_NBRC15183_Minnesota	
20_NBRC15184_Minnesota	
21_NBRC15185_Minnesota	
22_NBRC15186_Minnesota	
23_NBRC15187_Minnesota	
24_NBRC100797_Abony	MAKIGENVPLLIDKAVDFMASSQAFREYLNKTPPRDYVPSEVPSESAPIYLQRLEYYRRLYRPKEEERG (SEQ ID NO:764)
25_NBRC105684_Choleraesuis	MAKIGENVPLLIDKAVDFMASSQAFREYLNKTPPRDYVPSEVPSESAPIYLQRLEYYRRLYRPKEEERG (SEQ ID NO:774)
26_NBRC105726_Typhimurium	MAKIGENVPLLIDKAVDFMASSQAFREYLNKTPPRDYVPSEVPSESAPIYLQRLEYYRRLYRPKEEERG (SEQ ID NO:783)
27_JCM3919_UN_O7_	
28_NBRC15335_Minnesota	
29_GTC09490_Enteritidis	
30_GTC09492_Braenderup	
31_GTC09493_Pakistan	
32_GTC09549_Typhimurium	
33_ATCCBAA-1675_Infantis	MAKIGENVPLLIDKAVDFMASSQAFREYLNKTPPRDYVPSEVPSESAPIYLQRLEYYRRLYRPKEEERG (SEQ ID NO:835)
34_ATCCBAA-1738_Thompson	MAKIGENVPLLIDKAVDFMASSQAFREYLNKTPPRDYVPSEVPSESAPIYLQRLEYYRRLYRPKEEERS (SEQ ID NO:846)
35_ATCC9712_Saintpaul	MAKIGENVPLLIDKAVDFMASSQAFREYLNKTPPRDYVPSEVPSESAPIYLQRLAYYRRLYRPKEEERG (SEQ ID NO:856)
36_jfrlSe1402-1_Infantis	MAKIGENVPLLIDKAVDFMASSQAFREYLNKTPPRDYVPSEVPSESAPIYLQRLEYYRRLYRPKEEERG (SEQ ID NO:867)
38_jfrlSe1402-3_Brandenburg	
39_jfrlSe1402-4_Infantis	MAKIGENVPLLIDKAVDFMASSQAFREYLNKTPPRDYVPSEVPSESAPIYLQRLEYYRRLYRPKEEERG (SEQ ID NO:884)
40_jfrlSe1402-5_Brandenburg	
41_jfrlSe1402-6_Rissen	
42_jfrlSe1402-7_Orion	MAKIGENVPLLIDKAVDFMASSQAFREYLNKTPPRDYVPSEVPSESAPIYLQRLEYYRRLYRPKEEERG (SEQ ID NO:908)
43_jfrlSe1402-8_Rissen	
44_jfrlSe1402-9_Rissen	
45_jfrlSe1402-10_Rissen	
46_jfrlSe1402-11_Rissen	
47_jfrlSe1402-12_Rissen	
48_jfrlSe1402-13_Mbandaka	
49_jfrlSe1402-14_Mbandaka	MAKIGENVPLLIDKAVDFMASSQAFREYLNKTPPRDYVPSEVPSESAPIYLQRLEYYRRLYRPKEEERG (SEQ ID NO:963)
50_jfrlSe1402-15_Orion	MAKIGENVPLLIDKAVDFMASSQAFREYLNKTPPRDYVPSEVPSESAPIYLQRLEYYRRLYRPKEEERG (SEQ ID NO:973)
51_jfrlSe1409-1_Altona	MAKIGENVPLLIDKAVDFMASSQAFREYLNKTPPRDYVPSEVPSESAPIYLQRLEYYRRLYRPKEEERGRRDX (SEQ ID NO:981)
52_jfrlSe1409-2_Istanbul	MAKIGENVPLLIDKAVDFMASSQAFREYLNKTPPRDYVPSEVPSESAPIYLQRLEYYRRLYRPKEERGRR (SEQ ID NO:990)
53_jfrlSe1409-3_Senftenberg	MAKIGENVPLLIDKAVDFMASSQAFREYLNKTPPRDYVPSEVPSESAPIYLQRLAYYRRLYRPKEEERG (SEQ ID NO:993)
54_jfrlSe1409-4_UN_O13_	MAKIGENVPLLIDKAVDFMASSQAFREYLNKTPPRDYVPSEVPSESAPIYLQRLEYYRRLYRPKEEERG (SEQ ID NO:1002)
55_jfrlSe1409-5_UN_O1,3,19_	MAKIGENVPLLIDKAVDFMASSQAFREFKYKTPPRDYVPSEVPSESAPIYLQRLAYYRRLYRPKEEERG (SEQ ID NO:1011)
56_jfrlSe1409-6_Montevideo	MAKIGENVPLLIDKAVDFMASSQAFREYLNKTPPRDYVPSEVPSESAPIYLQRLEYYRRLYRPKEEERG (SEQ ID NO:1018)
57_jfrlSe1409-7_UN_O1,3,19_	MAKIGENVPLLIDKAVDFMASSQAFREYLNKTPPRDYVPSEVPSESAPIYLQRLAYYRRLYRPKEEERG (SEQ ID NO:1027)
58_jfrlSe1409-8_UN_O18_	MAKIGENVPLLIDKAVDFMASSQAFREYLNKTPPRDYVPSEVPSESAPIYLQRLEYYRRLYRPKEEERG (SEQ ID NO:1036)
61_jfrlSe1409-11_UN_O1,3,19_	MAKIGENVPLLIDKAVDFMASSQAFREYLNKTPPRDYVPSEVPSESAPIYLQRLAYYRRLYRPKEEERG (SEQ ID NO:1045)
67_jfrlSe1409-17_Rissen	MAKIGENVPLLIDKAVDFMASSQAFREYLNKTPPRDYVPSEVPSESAPIYLQRLEYYRRLYRPKEEERG (SEQ ID NO:1053)
71_jfrlSe1409-21_Amsterdam	MAKIGENVPLLIDKAVDFMASSQAFREYLNKTPPRDYVPSEVPSESAPIYLQRLEYYRRLYRPKEEERG (SEQ ID NO:1062)
80_HySe09_Enteritidis	MAKIGENVPLLIDKAVDFMASSQAFREYLNKTPPRDYVPSEVPSESAPIYLQRLEYYRRLYRPKEEERG (SEQ ID NO:1072)
100_HySe29_Schwarzengrund	
103_HySe32_Schwarzengrund	
106_HySe35_Schwarzengrund	

Fig. 44

	ppic
01_NBRC13245T_Typhimurium	MAKMAAALHILVKEEKLALDLLEQIKNGGDFEKLAKKHSICPSGKKGGHLGEFRQGQMVPAFDK VVFSCPVLEPTGPLHTQFGYHIIKVLYRK (SEQ ID NO:584)
02_GTC00131_Enteritidis	
03_GTC09491_Enteritidis	
04_GTC03838_Enteritidis	
05_GTC08914_Enteritidis	MAKMAAALHILVKEEKLALDLLEQIKNGGDFEKLAKKHSICPSGKKGGHLGEFRQGQMVPAFDK VVFSCPVLEPTGPLHTQFGYHIIKVLYRK (SEQ ID NO:619)
06_GTC09421_Enteritidis	MAKMAAALHILVKEEKLALDLLEQIKNGGDFEKLAKKHSICPSGKKGGHLGEFRQGQMVPAFDK VVFSCPVLEPTGPLHTQFGYHIIKVLYRK (SEQ ID NO:629)
07_GTC09489_Enteritidis	
08_NBRC3163_Pullorum_Gallinarum	MAKMAAALHILVKEEKLALDLLEQIKNGGDFEKLAKKHSICPSGKKGGHLGEFRQGQMVPAFDK VVFSCPVLEPTGPLHTQFGYHIIKVLYRK (SEQ ID NO:645)
09_NBRC3313_Enteritidis	
10_NBRC12529_Typhimurium	
11_NBRC14193_Typhimurium	
12_NBRC14194_Typhimurium	
13_NBRC14209_Typhimurium	
14_NBRC14210_Typhimurium	
15_NBRC14211_Typhimurium	
16_NBRC14212_Typhimurium	
17_NBRC15181_Typhimurium	MAKMAAALHILVKEEKLALDLLEQIKNGGDFEKLAKKHSICPSGKKGGHLGEFRQGQMVPAFDK VVFSCPVLEPTGPLHTQFGYHIIKVLYRK (SEQ ID NO:710)
18_NBRC15182_Minnesota	MAKMAAALHILVKEEKLALDLLEQIKNGGDFEKLAKKHSICPSGKKGGHLGEFRQGQMVPAFDK VVFSCPVLEPTGPLHTQFGYHIIKVLYRK (SEQ ID NO:720)
19_NBRC15183_Minnesota	
20_NBRC15184_Minnesota	
21_NBRC15185_Minnesota	
22_NBRC15186_Minnesota	
23_NBRC15187_Minnesota	
24_NBRC100797_Abony	MAKMAAALHILVKEEKLALDLLEQIKNGGDFEKLAKKHSICPSGKKGGHLGEFRQGQMVPAFDK VVFSCPVLEPTGPLHTQFGYHIIKVLYRK (SEQ ID NO:765)
25_NBRC105684_Choleraesuis	MAKMAAALHILVKEEKLALDLLEQIKNGGDFEKLAKKHSICPSGKKGGHLGEFRQGQMVPAFDK VVFSCPVLEPTGPLHTQFGYHIIKVLYRK (SEQ ID NO:775)
26_NBRC105726_Typhimurium	MAKMAAALHILVKEEKLALDLLEQIKNGGDFEKLAKKHSICPSGKKGGHLGEFRQGQMVPAFDK VVFSCPVLEPTGPLHTQFGYHIIKVLYRK (SEQ ID NO:784)
27_JCM3919_UN_O7	
28_NBRC15335_Minnesota	
29_GTC09490_Enteritidis	
30_GTC09492_Braenderup	
31_GTC09493_Pakistan	
32_GTC09549_Typhimurium	
33_ATCCBAA-1675_Infantis	MAKMAAALHILVKEEKLALDLLEQIKNGGDFEKLAKKHSICPSGKKGGHLGEFRQGQMVPAFDK VVFSCPVLEPTGPLHTQFGYHIIKVLYRK (SEQ ID NO:836)
34_ATCCBAA-1738_Thompson	MAKMAAALHILVKEEKLALDLLEQIKNGGDFEKLAKKHSICPSGKKGGHLGEFRQGQMVPAFDK VVFSCPVLEPTGPLHTQFGYHIIKVLYRK (SEQ ID NO:847)
35_ATCC9712_Saintpaul	MAKMAAALHILVKEEKLALDLLEQIKNGGDFEKLAKKHSICPSGKKGGHLGEFRQGQMVPAFDK VVFSCPVLEPTGPLHTQFGYHIIKVLYRK (SEQ ID NO:857)
36_jfrlSe1402-1_Infantis	MAKMAAALHILVKEEKLALDLLEQIKNGGDFEKLAKKHSICPSGKKGGHLGEFRQGQMVPAFDK VVFSCPVLEPTGPLHTQFGYHIIKVLYRK (SEQ ID NO:868)
38_jfrlSe1402-3_Brandenburg	
39_jfrlSe1402-4_Infantis	MAKMAAALHILVKEEKLALDLLEQIKNGGDFEKLAKKHSICPSGKKGGHLGEFRQGQMVPAFDK VVFSCPVLEPTGPLHTQFGYHIIKVLYRK (SEQ ID NO:885)
40_jfrlSe1402-5_Brandenburg	
41_jfrlSe1402-6_Rissen	
42_jfrlSe1402-7_Orion	MAKMAAALHILVKEEKMALDLLEQIKNGGDFEKLAKKHSICPSGKKGGHLGEFRQGQMVPAFD KVVFSCPVLEPTGPLHTQFGYHIIKVLYRK (SEQ ID NO:909)
43_jfrlSe1402-8_Rissen	
44_jfrlSe1402-9_Rissen	
45_jfrlSe1402-10_Rissen	
46_jfrlSe1402-11_Rissen	
47_jfrlSe1402-12_Rissen	
48_jfrlSe1402-13_Mbandaka	MAKMAAALHILVKEEKLALDLLEQIKNGGDFEKLAKKHSICPSGKKGGHLGEFRQGQMVPAFDK VVFSCPVLEPTGPLHTQFGYHIIKVLYRK (SEQ ID NO:953)
49_jfrlSe1402-14_Mbandaka	MAKMAAALHILVKEEKLALDLLEQIKNGGDFEKLAKKHSICPSGKKGGHLGEFRQGQMVPAFDK VVFSCPVLEPTGPLHTQFGYHIIKVLYRK (SEQ ID NO:964)
50_jfrlSe1402-15_Orion	MAKMAAALHILVKEEKMALDLLEQIKNGGDFEKLAKKHSICPSGKKGGHLGEFRQGQMVPAFD KVVFSCPVLEPTGPLHTQFGYHIIKVLYRK (SEQ ID NO:974)
51_jfrlSe1409-1_Altona	MAKMAAALHILVKEEKLALDLLEQIKNGGDFEKLAKKHSICPSGKKGGHLGEFRQGQMVPAFDK VVFSCPVLEPTGPLHTQFGYHIIKVLYRK (SEQ ID NO:982)

Fig. 45

	L25
01_NBRC13245T_Typhimurium	MFTINAEVRKEQGKGASRRLRAANKFPAIIYGGSEAPIAIELDHDQVMNMQAKAEFYSEVLTLVVDGKEVKVKAQAVQRHAYKPKLTHIDFVRA (SEQ ID NO:585)
02_GTC00131_Enteritidis	MFTINAEVRKEQGKGASRRLRAANKFPAIIYGGSEAPIAIELDHDQVMNMQAKAEFYSEVLTLVVDGKEVKVKAQAVQRHAYKPKLTHIDFVRA (SEQ ID NO:594)
03_GTC09491_Enteritidis	MFTINAEVRKEQGKGASRRLRAANKFPAIIYGGSEAPIAIELDHDQVMNMQAKAEFYSEVLTLVVDGKEVKVKAQAVQRHAYKPKLTHIDFVRA (SEQ ID NO:603)
04_GTC03838_Enteritidis	MFTINAEVRKEQGKGASRRLRAANKFPAIIYGGSEAPIAIELDHDQVMNMQAKAEFYSEVLTLVVDGKEVKVKAQAVQRHAYKPKLTHIDFVRA (SEQ ID NO:610)
05_GTC08914_Enteritidis	MFTINAEVRKEQGKGASRRLRAANKFPAIIYGGSEAPIAIELDHDQVMNMQAKAEFYSEVLTLVVDGKEVKVKAQAVQRHAYKPKLTHIDFVRA (SEQ ID NO:620)
06_GTC09421_Enteritidis	MFTINAEVRKEQGKGASRRLRAANKFPAIIYGGSEAPIAIELDHDQVMNMQAKAEFYSEVLTLVVDGKEVKVKAQAVQRHAYKPKLTHIDFVRA
07_GTC09489_Enteritidis	MFTINAEVRKEQGKGASRRLRAANKFPAIIYGGSEAPIAIELDHDQVMNMQAKAEFYSEVLTLVVDGKEVKVKAQAVQRHAYKPKLTHIDFVRA (SEQ ID NO:636)
08_NBRC3163_Pullorum_Gallinarum	MFTINAEVRKEQGKGASRRLRAANKFPAIIYGGSEAPIAIELDHDQVMNMQAKAEFYSEVLTLVVDGKEVKVKAQAVQRHAYKPKLTHIDFVRA (SEQ ID NO:646)
09_NBRC3313_Enteritidis	MFTINAEVRKEQGKGASRRLRAANKFPAIIYGGSEAPIAIELDHDQVMNMQAKAEFYSEVLTLVVDGKEVKVKAQAVQRHAYKPKLTHIDFVRA (SEQ ID NO:653)
10_NBRC12529_Typhimurium	MFTINAEVRKEQGKGASRRLRAANKFPAIIYGGSEAPIAIELDHDQVMNMQAKAEFYSEVLTLVVDGKEVKVKAQAVQRHAYKPKLTHIDFVRA (SEQ ID NO:660)
11_NBRC14193_Typhimurium	MFTINAEVRKEQGKGASRRLRAANKFPAIIYGGSEAPIAIELDHDQVMNMQAKAEFYSEVLTLVVDGKEVKVKAQAVQRHAYKPKLTHIDFVRA (SEQ ID NO:667)
12_NBRC14194_Typhimurium	MFTINAEVRKEQGKGASRRLRAANKFPAIIYGGSEAPIAIELDHDQVMNMQAKAEFYSEVLTLVVDGKEVKVKAQAVQRHAYKPKLTHIDFVRA (SEQ ID NO:674)
13_NBRC14209_Typhimurium	MFTINAEVRKEQGKGASRRLRAANKFPAIIYGGSEAPIAIELDHDQVMNMQAKAEFYSEVLTLVVDGKEVKVKAQAVQRHAYKPKLTHIDFVRA (SEQ ID NO:681)
14_NBRC14210_Typhimurium	MFTINAEVRKEQGKGASRRLRAANKFPAIIYGGSEAPIAIELDHDQVMNMQAKAEFYSEVLTLVVDGKEVKVKAQAVQRHAYKPKLTHIDFVRA (SEQ ID NO:688)
15_NBRC14211_Typhimurium	MFTINAEVRKEQGKGASRRLRAANKFPAIIYGGSEAPIAIELDHDQVMNMQAKAEFYSEVLTLVVDGKEVKVKAQAVQRHAYKPKLTHIDFVRA (SEQ ID NO:694)
16_NBRC14212_Typhimurium	MFTINAEVRKEQGKGASRRLRAANKFPAIIYGGSEAPIAIELDHDQVMNMQAKAEFYSEVLTLVVDGKEVKVKAQAVQRHAYKPKLTHIDFVRA (SEQ ID NO:701)
17_NBRC15181_Typhimurium	MFTINAEVRKEQGKGASRRLRAANKFPAIIYGGSEAPIAIELDHDQVMNMQAKAEFYSEVLTLVVDGKEVKVKAQAVQRHAYKPKLTHIDFVRA (SEQ ID NO:711)
18_NBRC15182_Minnesota	MFTINAEVRKEQGKGASRRLRAANKFPAIIYGGSEAPIAIELDHDQVMNMQAKAEFYSEVLTLVVDGKEVKVKAQAVQRHAYKPKLTHIDFVRA (SEQ ID NO:721)
19_NBRC15183_Minnesota	MFTINAEVRKEQGKGASRRLRAANKFPAIIYGGSEAPIAIELDHDQVMNMQAKAEFYSEVLTLVVDGKEVKVKAQAVQRHAYKPKLTHIDFVRA (SEQ ID NO:728)
20_NBRC15184_Minnesota	MFTINAEVRKEQGKGASRRLRAANKFPAIIYGGSEAPIAIELDHDQVMNMQAKAEFYSEVLTLVVDGKEVKVKAQAVQRHAYKPKLTHIDFVRA (SEQ ID NO:735)
21_NBRC15185_Minnesota	MFTINAEVRKEQGKGASRRLRAANKFPAIIYGGSEAPIAIELDHDQVMNMQAKAEFYSEVLTLVVDGKEVKVKAQAVQRHAYKPKLTHIDFVRA (SEQ ID NO:742)
22_NBRC15186_Minnesota	MFTINAEVRKEQGKGASRRLRAANKFPAIIYGGSEAPIAIELDHDQVMNMQAKAEFYSEVLTLVVDGKEVKVKAQAVQRHAYKPKLTHIDFVRA (SEQ ID NO:749)
23_NBRC15187_Minnesota	MFTINAEVRKEQGKGASRRLRAANKFPAIIYGGSEAPIAIELDHDQVMNMQAKAEFYSEVLTLVVDGKEVKVKAQAVQRHAYKPKLTHIDFVRA (SEQ ID NO:756)
24_NBRC100797_Abony	MFTINAEVRKEQGKGASRRLRAANKFPAIIYGGSEAPIAIELDHDQVMNMQAKAEFYSEVLTLVVDGKEVKVKAQAVQRHAYKPKLTHIDFVRA (SEQ ID NO:766)
25_NBRC105684_Choleraesuis	MFTINAEVRKEQGKGASRRLRAANKFPAIIYGGSEAPIAIELDHDQVMNMQAKAEFYSEVLTLVVDGKEVKVKAQAVQRHAYKPKLTHIDFVRA (SEQ ID NO:776)
26_NBRC105726_Typhimurium	MFTINAEVRKEQGKGASRRLRAANKFPAIIYGGSEAPIAIELDHDQVMNMQAKAEFYSEVLTLVVDGKEVKVKAQAVQRHAYKPKLTHIDFVRA (SEQ ID NO:785)
27_JCM3919_UN_O7_	MFTINAEVRKEQGKGASRRLRAANKFPAIIYGGSEAPIAIELDHDQVMNMQAKAEFYSEVLTLVVDGKEVKVKAQAVQRHAYKPKLTHIDFVRA (SEQ ID NO:792)
28_NBRC15335_Minnesota	MFTINAEVRKEQGKGASRRLRAANKFPAIIYGGSEAPIAIELDHDQVMNMQAKAEFYSEVLTLVVDGKEVKVKAQAVQRHAYKPKLTHIDFVRA (SEQ ID NO:799)
29_GTC09490_Enteritidis	MFTINAEVRKEQGKGASRRLRAANKFPAIIYGGSEAPIAIELDHDQVMNMQAKAEFYSEVLTLVVDGKEVKVKAQAVQRHAYKPKLTHIDFVRA (SEQ ID NO:806)
30_GTC09492_Braenderup	MFTINAEVRKEQGKGASRRLRAANKFPAIIYGGSEAPVAIELDHDQVMNMQAKAEFYSEVLTLVVDGKEVKVKAQAVQRHAYKPKLTHIDFVRA (SEQ ID NO:813)
31_GTC09493_Pakistan	MFTINAEVRKEQGKGASRRLRAANKFPAIIYGGSEAPIAIELDHDQVMNMQAKAEFYSEVLTLVVDGKEVKVKAQAVQRHAYKPKLTHIDFVRA (SEQ ID NO:817)
32_GTC09549_Typhimurium	MFTINAEVRKEQGKGASRRLRAANKFPAIIYGGSEAPIAIELDHDQVMNMQAKAEFYSEVLTLVVDGKEVKVKAQAVQRHAYKPKLTHIDFVRA (SEQ ID NO:827)
33_ATCCBAA-1675_Infantis	MFTINAEVRKEQGKGASRRLRAANKFPAIIYGGSEAPIAIELDHDQVMNMQAKAEFYSEVLTLVVDGKEVKVKAQAVQRHAYKPKLTHIDFVRA (SEQ ID NO:837)
34_ATCCBAA-1738_Thompson	MFTINAEVRKEQGKGASRRLRAANKFPAIIYGGSEAPIAIELDHDQVMNMQAKAEFYSEVLTLVVDGKEVKVKAQAVQRHAYKPKLTHIDFVRA (SEQ ID NO:848)
35_ATCC9712_Saintpaul	MFTINAEVRKEQGKGASRRLRAANKFPAIIYGGSEAPIAIELDHDQVMNMQAKAEFYSEVLTLVVDGKEVKVKAQAVQRHAYKPKLTHIDFVRA (SEQ ID NO:858)
36_frISe1402-1_Infantis	MFTINAEVRKEQGKGASRRLRAANKFPAIIYGGSEAPIAIELDHDQVMNMQAKAEFYSEVLTLVVDGKEVKVKAQAVQRHAYKPKLTHIDFVRA (SEQ ID NO:869)
38_frISe1402-3_Brandenburg	MFTINAEVRKEQGKGASRRLRAANKFPAIIYGGSEAPIAIELDHDQVMNMQAKAEFYSEVLTLVVDGKEVKVKAQAVQRHAYKPKLTHIDFVRA (SEQ ID NO:876)
39_frISe1402-4_Infantis	MFTINAEVRKEQGKGASRRLRAANKFPAIIYGGSEAPIAIELDHDQVMNMQAKAEFYSEVLTLVVDGKEVKVKAQAVQRHAYKPKLTHIDFVRA (SEQ ID NO:886)
40_frISe1402-5_Brandenburg	MFTINAEVRKEQGKGASRRLRAANKFPAIIYGGSEAPIAIELDHDQVMNMQAKAEFYSEVLTLVVDGKEVKVKAQAVQRHAYKPKLTHIDFVRA (SEQ ID NO:893)
41_frISe1402-6_Rissen	MFTINAEVRKEQGKGASRRLRAANKFPAIIYGGSEAPIAIELDHDQVMNMQAKAEFYSEVLTLVVDGKEVKVKAQAVQRHAYKPKLTHIDFVRA (SEQ ID NO:900)
42_frISe1402-7_Orion	MFTINAEVRKEQGKGASRRLRAANKFPAIIYGGSEAPIAIELDHDQVMNMQAKAEFYSEVLTLVVDGKEVKVKAQAVQRHAYKPKLTHIDFVRA (SEQ ID NO:910)
43_frISe1402-8_Rissen	MFTINAEVRKEQGKGASRRLRAANKFPAIIYGGSEAPIAIELDHDQVMNMQAKAEFYSEVLTLVVDGKEVKVKAQAVQRHAYKPKLTHIDFVRA (SEQ ID NO:917)
44_frISe1402-9_Rissen	MFTINAEVRKEQGKGASRRLRAANKFPAIIYGGSEAPIAIELDHDQVMNMQAKAEFYSEVLTLVVDGKEVKVKAQAVQRHAYKPKLTHIDFVRA (SEQ ID NO:924)
45_frISe1402-10_Rissen	MFTINAEVRKEQGKGASRRLRAANKFPAIIYGGSEAPIAIELDHDQVMNMQAKAEFYSEVLTLVVDGKEVKVKAQAVQRHAYKPKLTHIDFVRA (SEQ ID NO:931)
46_frISe1402-11_Rissen	MFTINAEVRKEQGKGASRRLRAANKFPAIIYGGSEAPIAIELDHDQVMNMQAKAEFYSEVLTLVVDGKEVKVKAQAVQRHAYKPKLTHIDFVRA (SEQ ID NO:938)
47_frISe1402-12_Rissen	MFTINAEVRKEQGKGASRRLRAANKFPAIIYGGSEAPIAIELDHDQVMNMQAKAEFYSEVLTLVVDGKEVKVKAQAVQRHAYKPKLTHIDFVRA (SEQ ID NO:945)
48_frISe1402-13_Mbandaka	MFTINAEVRKEQGKGASRRLRAANKFPAIIYGGSEAPIAIELDHDQVMNMQAKAEFYSEVLTLVVDGKEVKVKAQAVQRHAYKPKLTHIDFVRA (SEQ ID NO:954)
49_frISe1402-14_Mbandaka	MFTINAEVRKEQGKGASRRLRAANKFPAIIYGGSEAPIAIELDHDQVMNMQAKAEFYSEVLTLVVDGKEVKVKAQAVQRHAYKPKLTHIDFVRA (SEQ ID NO:965)
50_frISe1402-15_Orion	MFTINAEVRKEQGKGASRRLRAANKFPAIIYGGSEAPIAIELDHDQVMNMQAKAEFYSEVLTLVVDGKEVKVKAQAVQRHAYKPKLTHIDFVRA (SEQ ID NO:975)
51_frISe1409-1_Altona	
52_frISe1409-2_Istanbul	
53_frISe1409-3_Senftenberg	
54_frISe1409-4_UN_O13_	
55_frISe1409-5_UN_O1,3,19_	
56_frISe1409-6_Montevideo	
57_frISe1409-7_UN_O1,3,19_	
58_frISe1409-8_UN_O18_	
61_frISe1409-11_UN_O1,3,19_	
67_frISe1409-17_Rissen	
71_frISe1409-21_Amsterdam	MFTINAEVRKEQGKGASRRLRAANKFPAIIYGGSEAPVAIELDHDQVMNMQAKAEFYSEVLTLVVDGKEVKVKAQAVQRHAYKPKLTHIDFVRA (SEQ ID NO:1064)
80_HySe09_Enteritidis	MFTINAEVRKEQGKGASRRLRAANKFPAIIYGGSEAPVAIELDHDQVMNMQAKAEFYSEVLTLVVDGKEVKVKAQAVQRHAYKPKLTHIDFVRA (SEQ ID NO:1074)
100_HySe29_Schwarzengrund	
103_HySe32_Schwarzengrund	
106_HySe35_Schwarzengrund	

Fig. 46

	YaiA
01_NBRC13245T_Typhimurium	MPTRPPYPREAYIVTIEKGTPGQTVTWYQLRADHPKPDSLISEHPTAEEAMDAKNRYEDPDKS (SEQ ID NO:586)
02_GTC00131_Enteritidis	MPTRPPYPREAYIVTIEKGTPGQTVTWYQLRADHPKPDSLISEHPTAEEAMDAKKRYEDPDKS (SEQ ID NO:595)
03_GTC09491_Enteritidis	
04_GTC03838_Enteritidis	
05_GTC08914_Enteritidis	
06_GTC09421_Enteritidis	
07_GTC09489_Enteritidis	
08_NBRC3163_Pullorum_Gallinarum	
09_NBRC3313_Enteritidis	
10_NBRC12529_Typhimurium	
11_NBRC14193_Typhimurium	
12_NBRC14194_Typhimurium	
13_NBRC14209_Typhimurium	
14_NBRC14210_Typhimurium	
15_NBRC14211_Typhimurium	
16_NBRC14212_Typhimurium	
17_NBRC15181_Typhimurium	
18_NBRC15182_Minnesota	
19_NBRC15183_Minnesota	
20_NBRC15184_Minnesota	
21_NBRC15185_Minnesota	
22_NBRC15186_Minnesota	
23_NBRC15187_Minnesota	
24_NBRC100797_Abony	
25_NBRC105684_Choleraesuis	
26_NBRC105726_Typhimurium	
27_JCM3919_UN_O7_	
28_NBRC15335_Minnesota	
29_GTC09490_Enteritidis	
30_GTC09492_Braenderup	
31_GTC09493_Pakistan	
32_GTC09549_Typhimurium	
33_ATCCBAA-1675_Infantis	
34_ATCCBAA-1738_Thompson	
35_ATCC9712_Saintpaul	MPTRPPYPREAYIVTIEKGTPGQTVTWYQLRADHPKPDSLISEHPTAEEAMDAKKRYEDPDKS (SEQ ID NO:859)
36_jfrlSe1402-1_Infantis	
38_jfrlSe1402-3_Brandenburg	
39_jfrlSe1402-4_Infantis	
40_jfrlSe1402-5_Brandenburg	
41_jfrlSe1402-6_Rissen	
42_jfrlSe1402-7_Orion	
43_jfrlSe1402-8_Rissen	
44_jfrlSe1402-9_Rissen	
45_jfrlSe1402-10_Rissen	
46_jfrlSe1402-11_Rissen	
47_jfrlSe1402-12_Rissen	
48_jfrlSe1402-13_Mbandaka	
49_jfrlSe1402-14_Mbandaka	
50_jfrlSe1402-15_Orion	
51_jfrlSe1409-1_Altona	
52_jfrlSe1409-2_Istanbul	
53_jfrlSe1409-3_Senftenberg	
54_jfrlSe1409-4_UN_O13_	
55_jfrlSe1409-5_UN_O1,3,19_	
56_jfrlSe1409-6_Montevideo	
57_jfrlSe1409-7_UN_O1,3,19_	
58_jfrlSe1409-8_UN_O18_	
61_jfrlSe1409-11_UN_O1,3,19_	
67_jfrlSe1409-17_Rissen	
71_jfrlSe1409-21_Amsterdam	
80_HySe09_Enteritidis	
100_HySe29_Schwarzengrund	
103_HySe32_Schwarzengrund	
106_HySe35_Schwarzengrund	

Fig. 47

	YciF
01_NBRC13245T_Typhimurium	MNIKTVEDLFIHLLSDTYSAEKQLTKALPKLARATSNEKLSQAFQSHLEETQGQIERIDQIVESESGIKLKRMKCVAMEGLIEEANEVIESTE KNEVRDAALIAAAQKVEHYEIASYGTLATLAEQLGYSKALKLLKETLDEEKQTDLKLTDLAVSNVNKSAERKSK (SEQ ID NO:587)
02_GTC00131_Enteritidis	MNIKTVEDLFIHLLSDTYSAEKQLTKALSKLARATSNEKLSQAFQSHLEETQGQIERIDQIVESESGIKLKRMKCVAMEGLIEEANEVIESTE KNEVRDAALIAAAQKVEHYEIASYGTLATLAEQLGYSKALKLLKETLDEEKQTDLKLTDLAVSNVNKSAERKSK (SEQ ID NO:596)
03_GTC09491_Enteritidis	
04_GTC03838_Enteritidis	
05_GTC08914_Enteritidis	
06_GTC09421_Enteritidis	
07_GTC09489_Enteritidis	
08_NBRC3163_Pullorum_Gallinarum	
09_NBRC3313_Enteritidis	
10_NBRC12529_Typhimurium	
11_NBRC14193_Typhimurium	
12_NBRC14194_Typhimurium	
13_NBRC14209_Typhimurium	
14_NBRC14210_Typhimurium	
15_NBRC14211_Typhimurium	
16_NBRC14212_Typhimurium	
17_NBRC15181_Typhimurium	
18_NBRC15182_Minnesota	
19_NBRC15183_Minnesota	
20_NBRC15184_Minnesota	
21_NBRC15185_Minnesota	
22_NBRC15186_Minnesota	
23_NBRC15187_Minnesota	
24_NBRC100797_Abony	
25_NBRC105684_Choleraesuis	
26_NBRC105726_Typhimurium	
27_JCM3919_UN_O7_	
28_NBRC15335_Minnesota	
29_GTC09490_Enteritidis	
30_GTC09492_Braenderup	
31_GTC09493_Pakistan	
32_GTC09549_Typhimurium	
33_ATCCBAA-1675_Infantis	MNIKTVEDLFIHLLSDTYSAEKQLTKALSKLARATSNEKLSQAFQSHLEETQGQIERIDQIVESESGIKLKRMKCVAMEGLIEEANEVIESTE KNEVRDAALIAAAQKVEHYEIASYGTLATLAEQLGYSKALKLLKETLDEEKQTDLKLTDLAVSNVNKSAERKSK (SEQ ID NO:838)
34_ATCCBAA-1738_Thompson	
35_ATCC9712_Saintpaul	
36_jfrlSe1402-1_Infantis	
38_jfrlSe1402-3_Brandenburg	
39_jfrlSe1402-4_Infantis	
40_jfrlSe1402-5_Brandenburg	
41_jfrlSe1402-6_Rissen	
42_jfrlSe1402-7_Orion	
43_jfrlSe1402-8_Rissen	
44_jfrlSe1402-9_Rissen	
45_jfrlSe1402-10_Rissen	
46_jfrlSe1402-11_Rissen	
47_jfrlSe1402-12_Rissen	
48_jfrlSe1402-13_Mbandaka	MNIKTVEDLFIHLLSDTYSAEKQLTKALPKLARATSNEKLSQAFQSHLEETQGQIERIDQIVESESGIKLKRMKCVAMEGLIEEANEVIESTE KNEVRDAALIAAAQKVEHYEIASYGTLATLAEQLGYSKALKLLKETLDEEKQTDLKLTDLAVSNVNKSAERKSK (SEQ ID NO:955)
49_jfrlSe1402-14_Mbandaka	
50_jfrlSe1402-15_Orion	
51_jfrlSe1409-1_Altona	
52_jfrlSe1409-2_Istanbul	
53_jfrlSe1409-3_Senftenberg	MNIKTVEDLFIHLLSDTYSAEKQLTKALPKLARATSNEKLSQAFQSHLEETQGQIERIDQIVESESGIKLKRIKCVAMEGLIEEANEVIESTEK NEVRDAALIAAAQKVEHYEIASYGTLATLAEQLGYGKALKLLKETLDEEKQTDLKLTDLAVSNVNKSAERKSK (SEQ ID NO:995)
54_jfrlSe1409-4_UN_O13_	
55_jfrlSe1409-5_UN_O1,3,19_	
56_jfrlSe1409-6_Montevideo	
57_jfrlSe1409-7_UN_O1,3,19_	
58_jfrlSe1409-8_UN_O18_	
61_jfrlSe1409-11_UN_O1,3,19_	
67_jfrlSe1409-17_Rissen	
71_jfrlSe1409-21_Amsterdam	
80_HySe09_Enteritidis	
100_HySe29_Schwarzengrund	
103_HySe32_Schwarzengrund	
106_HySe35_Schwarzengrund	

MICROORGANISM IDENTIFICATION METHOD

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a Rule 53(b) Continuation application of U.S. application Ser. No. 16/089,836 filed Sep. 28, 2018, which is a National Stage of International Application No. PCT/JP2016/060865 filed Mar. 31, 2016, the respective disclosures of all of the above of which are incorporated herein by reference in their entirety.

STATEMENT OF JOINT RESEARCH AGREEMENT

Pursuant to 35 U.S.C. § 102 (c), the undersigned states on behalf of SHIMADZU CORPORATION AND MEIJO UNIVERSITY that the subject matter of U.S. Pat. No. 11,774,448 and the claimed invention were made by or on the behalf of parties to a joint research agreement within the meaning of 35 U.S.C. §§ 100(h) and 102(c) and 37 C.F.R. § 1.9(e), that the joint research agreement was in effect on or before the effective filing date of the claimed invention, and that the claimed invention was made as a result of activities undertaken within the scope of the joint research agreement.

INCORPORATION

The content of the electronically submitted sequence listing, file name: Q289660_sequence listing as filed; size 1,232,443 bytes; and date of creation: Nov. 1, 2023, filed herewith, is incorporated herein by reference in its entirety.

TECHNICAL FIELD

The present invention relates to a microorganism identification method using mass spectrometry.

BACKGROUND ART

Salmonella belongs to the family of enterobacteriaceae of gram-negative facultative anaerobic bacilli, and three species of *Salmonella enterica, Salmonella bongori* and *Salmonella subterranea* belong to the genus *Salmonella*. Further, *Salmonella enterica* is classified into six subspecies (*Salmonella* (sometimes abbreviated as "S.") *enterica* subsp. *enterica, S. enterica* subsp. *salamae, S. enterica* subsp. *arizonae, S. enterica* subsp. *diarizonae, S. enterica* subsp. *houtenae, S. enterica* subsp. *indica*).

There are about 2,500 serovars in the genus *Salmonella*, which are decided by the Kauffmann-White classification based on the difference in combination of a cell wall lipopolysaccharide O antigen, and a flagellar protein H antigen. Pathogenic *Salmonella* such as *Salmonella* causing food poisoning belongs mostly to *S. enterica* subsp. *enterica*. This subspecies is also classified into about 1,500 types of serovars (Non Patent Literature 1). Currently, in order to decide the serovar, an agglutination test with antisera is used. It is an O type test by slide agglutination and an H type test by test tube agglutination, and the H type test increases mobility and performs phase induction for first phase and second phase decision, thus requires time and proficient skills for serovar decision.

Some serovars have determined pathogenic hosts. For example, *typhi, choleraesuis*, Dublin and *gallinarum* cause systemic infection specifically in humans, pigs, cattle, and chickens. However, many other serovars infect multiple hosts like humans, domestic animals, pets and wild animals and become pathogens of nontyphoidal acute gastroenteritis (food poisoning). Infection routes of nontyphoidal *Salmonella* range widely such as environments such as rivers, wild animals, pets, and foods (including secondary pollution as well as primary pollution such as through rodents and insects). Serovar decision is important for infection prevention and epidemiological analysis and has been used for more than 80 years (Non Patent Literature 2).

Highly detected serovars of nontyphoidal *Salmonella* infections in recent years are Enteritidis, Thompson, Infantis, Typhimurium, Saintpaul, Braenderup, Schwarzengrund, Litchfield, and Montevideo (IASR HP (Reference Document 1)). In the Act on Domestic Animal Infectious Diseases Control in Japan, when livestock is infected with Dublin, Enteritidis, Typhimurium or *choleraesuis*, notification to the Ministry of Agriculture, Forestry and Fisheries is mandatory.

As methods for detecting *Salmonella* and deciding serovars, multiplex PCR (Non Patent Literatures 3 and 4), pulsed field gel electrophoresis (Non Patent Literature 5), multilocus sequence typing method (Non Patent Literature 6) and the like have been reported so far. However, with multiplex PCR, there are problems that only a few serovars are decided, or only a part of the O antigen and H antigen is decided, and the other methods require a complicated operation and take time.

On the other hand, in recent years, the microorganism identification technique by matrix-assisted laser desorption/ionization time-of-flight mass-spectrometry (MALDI-TOF MS) has spread rapidly in clinical and food fields. This method is a method of identifying microorganisms based on a mass spectral pattern obtained using a very small amount of microorganism sample, which can obtain an analysis result in a short time and also easily perform continuous analysis of multiple specimens. Therefore, easy and rapid microorganism identification is possible. So far, attempts have been made to identify *Salmonella* using MALDI-TOF MS by multiple research groups (Non Patent Literatures 7, 8, 9, 10)

Non Patent Literature 10 distinguishes subspecies of *Salmonella enterica* subsp. *enterica* and five major serovars by selecting a biomarker and preparing a decision tree. While the research by Dieckmann et al. scrutinizes protein peaks very minutely, there are strains in which biomarker peak is present or absent, and it takes time to confirm the peak.

CITATION LIST

Patent Literature

Patent Literature 1: JP 2006-191922 A
Patent Literature 2: JP 2013-085517 A

Non Patent Literature

Non Patent Literature 1: ANTIGENIC FORMULAE OF THE SALMONELLA SEROVARS 2007 9th edition WHO Collaborating Center for Reference and Research on Salmonella Patrick A. D. Grimont, François-Xavier Weill Institut Pasteur, 28 rue du Dr. Roux, 75724 Paris Cedex 15, France

Non Patent Literature 2: Winfield&Groisman, 2003, Fukuoka Institute of Health and Environmental Sciences

Non Patent Literature 3: M Akiba Et. al., Microbiological Methods, 2011, 85, 9-15

Non Patent Literature 4: Y Hong et al., BMC microbiology 2008, 8: 178

Non Patent Literature 5: F Tenover, et al. Journal of clinical microbiology 33.9 (1995): 2233.

Non Patent Literature 6: M Achtman, et al. PLoS Pathog 8.6 (2012): e1002776.

Non Patent Literature 7: Seng, Piseth, et al. Future microbiology 5.11 (2010): 1733-1754.

Non Patent Literature 8: M Kuhns et al. PLoS One 7.6 (2012): e40004.

Non Patent Literature 9: R Dieckmann et al. AEM, 74.24 (2008): 7767-7778.

Non Patent Literature 10: R Dieckmann, et al. (2011): AEM-02418.

Non Patent Literature 11: T, Ojima-Kato, et al. PLOS one 2014: e113458.

SUMMARY OF INVENTION

Technical Problem

On the other hand, Patent Literature 1 shows that a method (S10-GERMS method) of attributing the type of protein to be the origin of the peak by associating the mass-to-charge ratio of the peak obtained by mass spectrometry with a calculated mass estimated from the amino acid sequence obtained by translating the base sequence information of the ribosomal protein gene, utilizing the fact that about half of the peaks obtained by subjecting microbial cells to mass spectrometry is derived from ribosomal proteins, is useful (Patent Literature 1). According to this method, it is possible to perform highly reliable microorganism identification based on a theoretical basis using mass spectrometry and software attached thereto (Patent Literature 2).

An object to be solved by the present invention is to provide a highly reliable biomarker based on genetic information that can rapidly and easily identify the serovar of *Salmonella enterica* subsp. *enterica*.

Solution to Problem

As a result of extensive studies, the present inventors have found that two types of ribosomal proteins S8 and Peptidylpropyl isomerase are useful as marker proteins used for identifying which species of serovar of *Salmonella* genus bacteria is contained in a sample by mass spectrometry, and it is possible to identify the serovar of *Salmonella* genus bacteria reproducibly and quickly by using at least one of these ribosomal proteins, and have reached the present invention.

More specifically, a microorganism identification method according to the present invention, which has been made to solve the above problems, includes

a) a step of subjecting a sample containing microorganisms to mass spectrometry to obtain a mass spectrum,

b) a step of reading a mass-to-charge ratio m/z of a peak derived from a marker protein from the mass spectrum, and

c) an identification step of identifying which bacteria of serovar of *Salmonella* genus bacteria the microorganisms contained in the sample contain, based on the mass-to-charge ratio m/z, in which at least one of two types of ribosomal proteins S8 and Peptidylpropyl isomerase is used as the marker protein.

In the above microorganism identification method, it is preferable that the serovars of *Salmonella* genus bacteria are classified using cluster analysis using as an index the mass-to-charge ratio m/z derived from at least 12 types of ribosomal proteins S8, L15, L17, L21, L25, S7, SODa, Peptidylpropyl isomerase, gns, YibT, YaiA and YciF as the marker protein.

In this case, it is preferable to further include a step of generating a dendrogram representing an identification result by the cluster analysis.

In addition, in the above microorganism identification method, when the serovar of *Salmonella* genus bacteria is Orion, at least Peptidylpropyl isomerase is preferably contained as the marker protein.

Moreover, when the serovar of *Salmonella* genus bacteria is Rissen, at least S8 is preferably contained as the marker protein.

Also, when the serovar of *Salmonella* genus bacteria is Saintpaul, at least L21, S7, YaiA and YciF are preferably contained as the marker protein.

Further, when the serovar of *Salmonella* genus bacteria is Braenderup, at least the group consisting of SOD, or gns and L25 is preferably contained as the marker protein.

Furthermore, when the serovar of *Salmonella* genus bacteria is Montevideo or Schwarzengrund, at least one of SOD and L21, and S7 are preferably contained as the marker protein.

Also, when the serovar of *Salmonella* genus bacteria is Enteritidis, at least SOD, L17 and S7 are preferably contained as the marker protein.

Further, when the serovar of *Salmonella* genus bacteria is *infantis*, at least SOD, L21, S7, YibT and YciF are preferably contained as the marker protein.

Advantageous Effects of Invention

According to the present invention, since a ribosomal protein showing a mutation peculiar to the serovar of *Salmonella* genus bacteria is used as the marker protein, the serovar of *Salmonella* genus bacteria can be reproducibly and quickly identified.

Also, by using a ribosomal protein showing a mutation peculiar to the serovar of *Salmonella* genus bacteria as the marker protein and performing a cluster analysis using the mass-to-charge ratio m/z of the peak derived from the marker protein on the mass spectrum as an index, the serovars of *Salmonella* genus bacteria contained in a plurality of samples can be collectively identified.

BRIEF DESCRIPTION OF DRAWINGS

FIG. **1** is a configuration diagram showing a main part of a microorganism identification system used for a microorganism identification method according to the present invention.

FIG. **2** is a flowchart showing an example of a procedure of a microorganism identification method according to the present invention.

FIG. **3** shows a list of species name, subspecies name and serovar of *Salmonella* genus bacteria used in examples.

FIG. **4** shows relationships between a combination of an agglutinated immune serum and a serovar.

FIG. **5** shows a list of primers used in examples.

FIG. **6** shows a mass of each amino acid.

FIG. **7**A shows a list of theoretical mass values of each ribosomal protein of *Salmonella* genus bacteria used in examples and measured values by MALDI-TOF MS (part 1).

FIG. 7B shows a list of theoretical mass values of each ribosomal protein of *Salmonella* genus bacteria used in examples and measured values by MALDI-TOF MS (part 2).

FIG. 7C shows a list of theoretical mass values of each ribosomal protein of *Salmonella* genus bacteria used in examples and measured values by MALDI-TOF MS (part 3).

FIG. 7D shows a list of theoretical mass values of each ribosomal protein of *Salmonella* genus bacteria used in examples and measured values by MALDI-TOF MS (part 4).

FIG. 7E shows a list of theoretical mass values of each ribosomal protein of *Salmonella* genus bacteria used in examples and measured values by MALDI-TOF MS (part 5).

FIG. 7F shows a list of theoretical mass values of each ribosomal protein of *Salmonella* genus bacteria used in examples and measured values by MALDI-TOF MS (part 6).

FIG. 7G shows a list of theoretical mass values of each ribosomal protein of *Salmonella* genus bacteria used in examples and measured values by MALDI-TOF MS (part 7).

FIG. **8**A is attribution results based on measured values of 12 types of ribosomal proteins (part 1).

FIG. **8**B is attribution results based on measured values of 12 types of ribosomal proteins (part 2).

FIG. **8**C is attribution results based on measured values of 12 types of ribosomal proteins (part 3).

FIG. **8**D is attribution results based on measured values of 12 types of ribosomal proteins (part 4).

FIG. **9** is a chart obtained by MALDI-TOF MS measurement.

FIG. **10**A is identification results by SARAMIS (part 1).

FIG. **10**B is identification results by SARAMIS (part 2).

FIG. **11** is a peak chart of ribosomal protein SOD.

FIG. **12** is a peak chart of ribosomal protein L17.

FIG. **13** is a peak chart of ribosomal protein L21.

FIG. **14** is a peak chart of ribosomal protein S8.

FIG. **15** is a peak chart of ribosomal protein L15.

FIG. **16** is a peak chart of ribosomal protein S7.

FIG. **17** is a peak chart of ribosomal protein gns.

FIG. **18** is a peak chart of ribosomal protein YibT.

FIG. **19** is a peak chart of ribosomal protein ppic.

FIG. **20** is a peak chart of ribosomal protein L25.

FIG. **21** is a peak chart of ribosomal protein YaiA.

FIG. **22** is a peak chart of ribosomal protein YciF.

FIG. **23** is a dendrogram generated using 12 types of ribosomal proteins.

FIG. **24**A is DNA sequences of ribosomal protein S8 (part 1).

FIG. **24**B is DNA sequences of ribosomal protein S8 (part 2).

FIG. **24**C is DNA sequences of ribosomal protein S8 (part 3).

FIG. **24**D is DNA sequences of ribosomal protein S8 (part 4).

FIG. **25**A is DNA sequences of ribosomal protein L15 (part 1).

FIG. **25**B is DNA sequences of ribosomal protein L15 (part 2).

FIG. **25**C is DNA sequences of ribosomal protein L15 (part 3).

FIG. **25**D is DNA sequences of ribosomal protein L15 (part 4).

FIG. **25**E is DNA sequences of ribosomal protein L15 (part 5).

FIG. **26**A is DNA sequences of ribosomal protein L17 (part 1).

FIG. **26**B is DNA sequences of ribosomal protein L17 (part 2).

FIG. **26**C is DNA sequences of ribosomal protein L17 (part 3).

FIG. **26**D is DNA sequences of ribosomal protein L17 (part 4).

FIG. **26**E is DNA sequences of ribosomal protein L17 (part 5).

FIG. **27**A is DNA sequences of ribosomal protein sodA (part 1).

FIG. **27**B is DNA sequences of ribosomal protein sodA (part 2).

FIG. **27**C is DNA sequences of ribosomal protein sodA (part 3).

FIG. **27**D is DNA sequences of ribosomal protein sodA (part 4).

FIG. **27**E is DNA sequences of ribosomal protein sodA (part 5).

FIG. **27**F is DNA sequences of ribosomal protein sodA (part 6).

FIG. **27**G is DNA sequences of ribosomal protein sodA (part 7).

FIG. **28**A is DNA sequences of ribosomal protein L21 (part 1).

FIG. **28**B is DNA sequences of ribosomal protein L21 (part 2).

FIG. **28**C is DNA sequences of ribosomal protein L21 (part 3).

FIG. **28**D is DNA sequences of ribosomal protein L21 (part 4).

FIG. **29**A is DNA sequences of ribosomal protein L25 (part 1).

FIG. **29**B is DNA sequences of ribosomal protein L25 (part 2).

FIG. **29**C is DNA sequences of ribosomal protein L25 (part 3).

FIG. **30**A is DNA sequences of ribosomal protein S7 (part 1).

FIG. **30**B is DNA sequences of ribosomal protein S7 (part 2).

FIG. **30**C is DNA sequences of ribosomal protein S7 (part 3).

FIG. **30**D is DNA sequences of ribosomal protein S7 (part 4).

FIG. **30**E is DNA sequences of ribosomal protein S7 (part 5).

FIG. **31**A is DNA sequences of ribosomal protein gns (part 1).

FIG. **31**B is DNA sequences of ribosomal protein gns (part 2).

FIG. **32**A is DNA sequences of ribosomal protein yibT (part 1).

FIG. **32**B is DNA sequences of ribosomal protein yibT (part 2).

FIG. **33**A is DNA sequences of ribosomal protein ppiC (part 1).

FIG. **33**B is DNA sequences of ribosomal protein ppiC (part 2).

FIG. **34** is DNA sequences of ribosomal protein yaiA.

FIG. **35**A is DNA sequences of ribosomal protein yciF (part 1).

FIG. **35**B is DNA sequences of ribosomal protein yciF (part 2).

FIG. **36**A is amino acid sequences of ribosomal protein SOD (part 1).

FIG. **36**B is amino acid sequences of ribosomal protein SOD (part 2).

FIG. **36**C is amino acid sequences of ribosomal protein SOD (part 3).

FIG. **37**A is amino acid sequences of ribosomal protein L17 (part 1).

FIG. **37**B is amino acid sequences of ribosomal protein L17 (part 2).

FIG. **38**A is amino acid sequences of ribosomal protein L21 (part 1).

FIG. **38**B is amino acid sequences of ribosomal protein L21 (part 2).

FIG. **39**A is amino acid sequences of ribosomal protein S8 (part 1).

FIG. **39**B is amino acid sequences of ribosomal protein S8 (part 2).

FIG. **40**A is amino acid sequences of ribosomal protein L15 (part 1).

FIG. **40**B is amino acid sequences of ribosomal protein L15 (part 2).

FIG. **41**A is amino acid sequences of ribosomal protein S7 (part 1).

FIG. **41**B is amino acid sequences of ribosomal protein S7 (part 2).

FIG. **42** is amino acid sequences of ribosomal protein gns.

FIG. **43** is amino acid sequences of the ribosomal protein YibT.

FIG. **44** is amino acid sequences of the ribosomal protein ppic.

FIG. **45** is amino acid sequences of ribosomal protein L25.

FIG. **46** is amino acid sequences of ribosomal protein YaiA.

FIG. **47** is amino acid sequences of ribosomal protein YciF.

DESCRIPTION OF EMBODIMENTS

Hereinafter, a specific embodiment of the microorganism identification method according to the present invention will be described.

FIG. **1** is an overview of a microorganism identification system used for a microorganism identification method according to the present invention. This microorganism identification system is roughly composed of a mass spectrometry unit **10** and a microorganism discrimination unit **20**. The mass spectrometry unit **10** includes an ionization section **11** for ionizing molecules and atoms in a sample by a matrix-assisted laser desorption ionization (MALDI) method, a time-of-flight mass separator (TOF) **12** for separating various kinds of ions emitted from the ionization section **11** according to the mass-to-charge ratio.

The TOF **12** includes an extraction electrode **13** for extracting ions from the ionization section **11** and leading the ions to an ion flight space in the TOF **12**, and a detector **14** for detecting ions mass-separated in the ion flight space.

The substance of the microorganism discrimination unit **20** is a computer such as a workstation or a personal computer, in which a Central Processing Unit (CPU) **21** that is a central processing unit, a memory **22**, a display section **23** consisting of a Liquid Crystal Display (LCD) and the like, an input section **24** consisting of a keyboard, a mouse and the like, and a storage section **30** consisting of a mass storage device such as a hard disk and a SSD (Solid State Drive) are connected to each other. In the storage section **30**, an Operating System (OS) **31**, a spectrum generation program **32**, a genus/species decision program **33**, and a subclass decision program **35** (program according to the present invention) are stored, and also a first database **34** and a second database **36** are housed. The microorganism discrimination unit **20** further includes an interface (I/F) **25** for direct connection with an external device and for controlling connection with an external device or the like via a network such as a LAN (Local Area Network), and is connected to the mass spectrometry unit **10** via a network cable NW (or wireless LAN) from the interface **25**.

In FIG. **1**, the spectrum acquisition part **37**, the m/z reading part **38**, the subclass determination part **39**, the cluster analysis part **40**, and the dendrogram (system diagram) generation part **41** are shown as related with the subclass decision program **35**. Basically, these are all functional means realized by software by the CPU **21** executing the subclass decision program **35**. The subclass decision program **35** is not necessarily a single program but may be a function incorporated in a part of a program for controlling the genus/species decision program **33** or the mass spectrometry unit **10**, for example, and its form is not particularly limited. As the genus/species decision program **33**, for example, a program for performing microorganism identification by a conventional fingerprint method or the like can be used.

Also, in FIG. **1**, a configuration in which the spectrum generation program **32**, the genus/species decision program **33**, and the subclass decision program **35**, the first database **34**, and the second database **36** are mounted on the terminal operated by the user is shown. However, a configuration in which at least part or all of them is provided in another device connected to the terminal via the computer network, and processing according to a program provided in the another device and/or access to the database is executed according to an instruction from the terminal may be used.

A large number of mass lists related to known microorganisms are registered in the first database **34** of the storage section **30**. This mass list lists the mass-to-charge ratios of ions detected upon mass spectrometry of certain microbial cells. In addition to the information of the mass-to-charge ratio, at least, information (classification information) of the classification group (family, genus, species, etc.) to which the microbial cells belong is contained. Such mass list is desirably created on the basis of data (measured data) obtained by actually subjecting various microbial cells to mass spectrometry in advance by the same ionization method and mass separation method as those by the mass spectrometry unit **10**.

When creating a mass list from the measured data, first, a peak appearing in a predetermined mass-to-charge ratio range is extracted from the mass spectrum acquired as the measured data. At this time, by setting the mass-to-charge ratio range to about 2,000 to 35,000, it is possible to mainly extract a protein-derived peak. Also, by extracting only peaks whose height (relative intensity) is equal to or greater than a predetermined threshold, undesirable peaks (noise) can be excluded. Since the ribosomal protein group is expressed in a large amount in the cell, most of the mass-to-charge ratio described in the mass list can be derived from the ribosomal protein by appropriately setting the threshold. Then, the mass-to-charge ratios (m/z) of the peaks extracted as above are listed for each cell and registered in the first database **34** after adding the classification information and the like. In order to suppress variations in gene expression due to culture conditions, it is desirable to standardize culture conditions in advance for each microbial cell used for collecting the measured data.

In the second database **36** of the storage section **30**, information on marker proteins for identifying known microorganisms by a classification (subspecies, pathotype, serovar, strain, etc.) lower than the species is registered. Information on the marker protein includes at least information on the mass-to-charge ratio (m/z) of the marker protein in the known microorganisms. In the second database **36** in the present embodiment, the values of mass-to-charge ratio m/z derived from at least 12 types of ribosomal proteins S8, L15, L17, L21, L25, S7, SODa, Peptidylpropyl isomerase, gns, YibT, YaiA and YciF are stored, as information on a marker protein for determining which serovar of *Salmonella* genus bacteria a test microorganism is. The values of mass-to-charge ratio of these ribosomal proteins will be described later.

It is desirable that the values of mass-to-charge ratio of the marker protein stored in the second database **36** are selected by comparing the calculated mass obtained by translating the base sequence of each marker protein into an amino acid sequence with the mass-to-charge ratio detected by actual measurement. The base sequence of the marker protein can be decided by sequence, or also can use a public database, for example, one acquired from a database of NCBI (National Center for Biotechnology Information) or the like. When obtaining the calculated mass from the above amino acid sequence, it is desirable to consider cleavage of the N-terminal methionine residue as a post-translational modification. Specifically, when the penultimate amino acid residue is Gly, Ala, Ser, Pro, Val, Thr or Cys, the theoretical value is calculated assuming that the N-terminal methionine is cleaved. In addition, since molecules added with protons are actually observed by MALDI-TOF MS, it is desirable to obtain the calculated mass also considering the protons (that is, the theoretical value of mass-to-charge ratio of ions obtained when each protein is analyzed by MALDI-TOF MS).

The procedure for identifying the serovar of *Salmonella* genus bacteria using the microorganism identification system according to this embodiment will be described with reference to a flowchart.

First, the user prepares a sample containing constituents of test microorganism, sets the sample in the mass spectrometry unit **10**, and performs mass spectrometry. At this time, as the sample, in addition to a cell extract, or a cellular constituent such as a ribosomal protein purified from a cell extract, a bacterial cell or a cell suspension can be also used as it is.

The spectrum generation program **32** acquires a detection signal acquired from the detector **14** of the mass spectrometry unit **10** via the interface **25**, and generates a mass spectrum of the test microorganism based on the detection signal (Step S**101**).

Next, the species decision program **33** collates the mass spectrum of the test microorganism with the mass lists of the known microorganisms recorded in the first database **34**, and extracts a mass list of the test microorganism having a mass-to-charge ratio pattern similar to the mass spectrum of the test microorganism, for example, a mass list containing many peaks that coincide with each peak in the mass spectrum of the test microorganism in a predetermined error range (Step S**102**). The species decision program **33** subsequently refers to the classification information stored in the first database **34** in association with the mass list extracted in Step S**102** to specify a species to which the known microorganism corresponding to the mass list belongs (Step S**103**). Then, when this species is not *Salmonella* genus bacteria (No in Step S**104**), the species is outputted to the display section **23** as a species of the test microorganism (Step S**116**), and the identification processing is terminated. On the other hand, when the species is *Salmonella* genus bacteria (Yes in Step S**104**), then the process proceeds to the identification processing by the subclass decision program **35**. When it is determined in advance that the sample contains *Salmonella* genus bacteria by other methods, the process may proceeds to the subclass decision program **35** without utilizing the species decision program using the mass spectrum.

In the subclass decision program **35**, first, the subclass determination part **39** reads out each of the values of mass-to-charge ratio of 12 types of ribosomal proteins S8, L15, L17, L21, L25, S7, SODa, Peptidylpropyl isomerase, gns, YibT, YaiA and YciF from the second database **36** (Step S**105**). Subsequently, the spectrum acquisition part **37** acquires the mass spectrum of the test microorganism generated in Step S**101**. Then, the m/z reading part **38** selects peaks appearing in the mass-to-charge ratio range stored in the second database **36** in association with each marker protein on the mass spectrum as peaks corresponding to each marker protein, and reads the mass-to-charge ratio (Step S**106**). And, cluster analysis using the read mass-to-charge ratio as an index is performed. Specifically, the subclass determination part **39** compares the mass-to-charge ratio with the values of mass-to-charge ratio of each marker protein read out from the second database **36** and decides attribution of the protein with respect to the read mass-to-charge ratio (Step S**107**). Then, cluster analysis is performed based on the decided attribution to determine the serovar of the test microorganism (Step S**108**), and the result is output to the display section **23** as the identification result of the test microorganism (Step S**109**).

Although the embodiments for carrying out the present invention have been described above with reference to the drawings, the present invention is not limited to the above-described embodiments, and appropriate modifications are permitted within the scope of the gist of the present invention.

EXAMPLES

(1) Strains Used

As described in FIG. **3**, a total of 64 strains of *Salmonella* available from the National Institute of Technology and Evaluation Nite Biological Resource Center (NBRC), Microbe Division/Japan Collection of Microorganisms (JCM) RIKEN BioResource Research Center (Tsukuba), National Bioresource Project GTC Collection (Gifu) and the American Type Culture Collection (Manassas, VA, USA) that are strain culture collection, isolates from Japan Food Research Laboratories and isolates from Hyogo Prefectural Institute of Public Health science were used for analysis. The serovar of *Salmonella enterica* subsp. *enterica* was decided by multiplex PCR method reported by *Salmonella* immune serum "Seiken" (DENKA SEIKEN Co., Ltd.) and Non Patent Literatures 3 and 4. The strains were classified into 22 serovars by this method. FIG. **4** shows relationships between O-antigen immune serum and a serovar.

(2) Analysis of DNA

Among the primers used in *Escherichia coli* database creation (Non Patent Literature 11), those which cannot be shared with *Salmonella* genus bacteria were designed based on consensus sequences. The designed primers are shown in FIG. **5**. Using these primers, DNA sequences of S10-spc-alpha operon and protein genes that could be biomarkers were analyzed. Specifically, genomic extraction was performed from each strain by a conventional method, and PCR was carried out using KOD plus as a template to amplify a target gene region. The obtained PCR product was purified and used as a template for sequence analysis. Sequence analysis was performed using Big Dye ver. 3.1 Cycle Sequencing Kit (Applied Biosystems, Foster City, California, USA). The DNA sequence of the gene was converted to the amino acid sequence of each gene, and the mass-to-charge ratio was calculated based on the amino acid mass in FIG. **6** to obtain a theoretical mass value.

(3) Analysis by MALDI-TOF MS

Bacterial cells grown in Luria Agar medium (Sigma-Aldrich Japan, Tokyo, Japan) were recovered and approximately 2 colonies of bacterial cells were added in 10 μL of a sinapinic matrix agent (25 mg/mL sinapinic acid (Wako Pure Chemical Industries, Ltd., Osaka, Japan) in 50 v/v % acetonitrile and 0.6 v/v % trifluoroacetic acid solution) and stirred well, and 1.2 μL out of the solution was loaded on a sample plate and air-dried. For MALDI-TOF MS measurement, the sample was measured in positive linear mode, at spectral range of 2000 m/z to 35000 m/z using AXIMA microorganism identification system (Shimadzu Corporation, Kyoto City, Japan). The above-described calculated mass was matched with the measured mass-to-charge ratio with a tolerance of 500 ppm, and proper modification was made. The calibration of the mass spectrometer was performed according to the instruction manual, using *Escherichia coli* DH5a strain.

(4) Construction of *Salmonella enterica* Subsp. *enterica* Database

By comparing the theoretical mass values of the ribosomal proteins obtained in the above (2) with the peak chart by MALDI-TOF MS obtained in (3), it was confirmed that there was no difference between the theoretical values obtained from gene sequences and the measured values, regarding the protein which could be detected by actual measurement. The theoretical and measured values of the ribosomal proteins in the S10-spc-α operon and proteins that can be other biomarkers showing different masses depending on the strain are summarized as a database as shown in FIGS. **7**A to **7**G.

The numbers shown in FIGS. **7**A to **7**G are the theoretical mass of the mass-to-charge ratio (m/z) obtained from genes. In addition, symbols "○", "Δ", and "x" represent mass peak detection results in actual measurement. Specifically, the symbol "○" indicates that it was detected as a peak within the 500 ppm range of the theoretical value at the default peak processing setting (threshold offset; 0.015 mV, threshold response; 1.200) of AXIMA microorganism identification system, and the symbol "x" indicates that there was a case where a peak could not be detected. In addition, the symbol "Δ" means that the theoretical mass difference in each strain or the difference from other protein peaks was 500 ppm or less, respectively, and the mass difference could not be identified even when a peak was detected.

As can be seen from FIGS. **7**A to **7**G, it was showed that the theoretical mass values of the ribosomal proteins L23, L16, L24, S8, L6, S5, L15 and L17 encoded in the s10-spc-alpha operon and L21, L25, S7, SODa, gns, YibT, Peptidylpropyl isomerase, YaiA and YciF outside the operon (total 17 types) differ depending on the strain of *Salmonella enterica* subsp. *enterica*, thus are possibly useful protein markers that can be used for serovar identification of *Salmonella enterica* subsp. *enterica*.

However, while it can be seen that L23, L16, L24, L6 and S5 have strains whose theoretical mass differences are separated by 500 ppm or more and can be a powerful biomarker for identification of these strains, there was a strain that could not be detected in actual measurement.

On the other hand, a total of seven types of proteins, S8, L15, L17, L21, L25, S7 and Peptidylpropyl isomerase, were stably detected irrespective of the strains, and the mass difference by the strains was also 500 ppm or more. Therefore, these proteins were found useful as biomarkers for serovar identification of *Salmonella enterica* subsp. *enterica* in MALDI-TOF MS.

SODa is an important biomarker for serovar identification of *Salmonella enterica* subsp. *enterica*, but the genotypes were varied and seven different mass-to-charge ratios were confirmed. All of these mass-to-charge ratios are as large as m/z around 23000, and in this region, the analysis accuracy of currently provided MALDI-TOF MS is low unless the difference between the other mass-to-charge ratios is 800 ppm or more, thus SODa cannot identify the serovars. Therefore, four types that can identify the serovar at this time were used as biomarkers. Regarding gns, YibT, YaiA and YciF, contamination peaks exist in one of the theoretical mass values, but since serovars *infantis*, Thompson and Typhimurium are proteins that are mutated specifically, only the theoretical mass value without contamination peak was used as a biomarker. Therefore, 12 types of proteins were used as biomarkers for *Salmonella enterica* subsp. *enterica* serovar identification.

(5) Attribution of Measured Values of MALDI-TOFMS by Software

Based on the above, using a total of 12 types of proteins, 8 types of proteins S8, L15, L17, L21, L25, S7, SODa and Peptidylpropyl isomerase that are stably detected regardless of the strain and 4 types of proteins gns, YibT, YaiA and YciF, as biomarkers, their theoretical mass values were registered in the software as shown in Patent Literature 2.

5: 22962.8 that was within the mass difference of 800 ppm of SODa was registered as the closest 1: 22948.82, and 6: 22996.82 and 7: 23004.88 as 2: 23010.84. In addition, gns, YibT, YaiA and YciF in which contamination peaks exist are registered as 6483.51, 8023.08, 7110.89 and 18643.13/18653.16, respectively.

Next, measured data in MALDI-TOF MS was analyzed with this software, and whether each biomarker was correctly attributed as a registered mass peak was examined. As a result, as shown in FIGS. **8**A to **8**G, all biomarker mass peaks of all the strains were attributed as registered mass numbers. Each attribution mass pattern was classified into groups 1 to 31, and compared with the serovar of each strain. Then, it was found that Typhimurium belongs to 1, 2 and 3, O4 group with unknown serovar to 4 and 5, Saintpaul to 6, O18 group with unknown serovar to 7, Orion to 8, Braenderup to 9, Montevideo and Schwarzengrund to 10, Schwarzengrund to 11, Abony and Pakistan to 12, Enteritidis to 13 and 14, Rissen to 15, *Gallinarum pullorum* to 16, Altona to 17, Amsterdam to 18, *infantis* to 19 and 20, Istanbul to 21, O4 group with unknown serovar to 22, Manhattan to 23, Mbandaka to 24, Senftenberg and O1, 3 and 19 groups with unknown serovar to 25, Thompson to 26, O4 group with unknown serovar to 27, O7 group with unknown serovar to 28, Brandenburg, Minnesota and Saintpaul to 29, Brandenburg and Saintpaul to 30, and *choleraesuis* strain to 31.

Based on the above, it was found that use of the mass of S8 (m/z 13996.36 or 14008.41), L15 (m/z 14967.38, 14981.41 or 14948.33), L17 (m/z 14395.61 or 14381.59), L21 (m/z 11579.36 or 11565.33), L25 (m/z 10542.19 or 10528.17), S7 (m/z 17460.15, 17474.18 or 17432.1), SODa (m/z 22948.82, 23010.84, 22976.83 or 22918.79), Peptidylpropyl isomerase (m/z 10198.07 or 10216.11), gns (m/z6483.51), YibT (m/z 8023.08), YaiA (m/z 7110.89) and YciF (m/z 18643.13) as biomarkers for MALDI-TOF MS analysis is useful for serovar identification of *Salmonella enterica* subsp. *enterica*.

Among the biomarkers found out this time, 10 types except S8 and Peptidylpropyl isomerase have been reported in Non Patent Literature 10. However, Non Patent Literature 10 requires confirmation of each peak one by one, thus takes time for spectral analysis of MALDI-TOF MS for identifying serovar. Also, as to the mass-to-charge ratio m/z 6036 reported to be an important peak for identification of Enteriridis in Non Patent Literature 10, a peak was not confirmed in 5 strains out of 32 strains in Non Patent Literature 10, and in this example, a peak could not be confirmed in 8 strains out of 35 strains. Therefore, it was not used as a biomarker for serovar identification of *Salmonella enterica* subsp. *enterica*.

By adding S8 and Peptidylpropyl isomerase to the biomarkers and using 12 types of carefully selected proteins as biomarkers, it became possible to provide a database that automatically identifies *Salmonella enterica* subsp. *enterica* to 31 groups for the first time.

(6) Comparison with Fingerprint Method (SARAMIS)

In fact, the identification result by the existing fingerprint method (SARAMIS) was compared with the identification result using the biomarker theoretical mass value shown in Table 6 as indices. First, in actual measurement in MALDI-TOF MS, a chart as shown in FIG. **9** was obtained. This result was analyzed by SARAMIS according to the instruction manual of AXIMA microorganism identification system. Results thus obtained are shown in FIG. **10**A and FIG. **10**B. As can be seen from these figures, all *Salmonella* genus bacteria used in the sample were identified as *Salmonella enterica* subsp. *enterica* in 91% to 99.9%, and species identification and serovar identification were not performed.

Therefore, whether measurement results of strains of different subspecies can be identified based on the theoretical mass database shown in FIG. **8**A was attempted. FIGS. **11** to **22** are enlarged views of 12 types of biomarker peak portions of the charts of FIG. **8**. As can be seen from FIGS. **11** to **22**, peaks can be distinguished since each biomarker mass is shifted. When compared with the measured values of 12 types of biomarkers and attributed, they agreed with the results shown in FIGS. **8**A to **8**D.

Next, cluster analysis was performed using the attribution results of 12 types of ribosomal proteins, and dendrogram was generated. The results are shown in FIG. **23**. In this method, although serovars of *infantis*, Brandenburg, Minnesota and Saintpaul could not be identified, other serovars could be almost identified.

Based on the above, the following can be seen.

SODa, S7 and gns are involved in the identification of multiple serovars and are particularly important as biomarkers for serovar identification of *Salmonella enterica* subsp. *enterica*.

Moreover, Enteritidis, Mbandaka and *choleraesuis* can be identified from other serovars by combination of SODa and S7 mutation.

Furthermore, *infantis* is identified, and Enteritidis and Mbandaka are identified by gns.

Typhimurium, which is the top of serovar responsible for nontyphoidal *Salmonella* infections, is separated by YaiA, and Thompson by YibT. Also, Pullorm (*gallinarum*) is identified by L17, Rissen by S8, Orion by Peptidylpropyl isomerase, and Altona by L15. L25 separates *infantis* and Amsterdam, and L21 is important to identify Montevideo and Shwarzengrund, Minnesota. YciF is important for identification of *infantis*.

(7) Gene Sequence and Amino Acid Sequence of Biomarkers

DNA sequences and amino acid sequences in each strain of a total of 12 types of ribosomal proteins, S8, L15 and L17 encoded in the S10-spc-alpha operon and SODa, L21, L25, S7, gns, YibT, Peptidylpropyl isomerase and YciF outside the operon, which exhibit theoretical mass values different depending on the strain of *Salmonella enterica* subsp. *enterica*, are summarized in FIGS. **24** to **47**.

REFERENCE SIGNS LIST

10 . . . Mass Spectrometry Unit
11 . . . Ionization Section
12 . . . TOF
13 . . . Extraction Electrode
14 . . . Detector
20 . . . Microorganism Discrimination Unit
21 . . . CPU
22 . . . Memory
23 . . . Display Section
24 . . . Input Section
25 . . . I/F
30 . . . Storage Section
31 . . . OS
32 . . . Spectrum Generation Program
33 . . . Genus/Species Decision Program
34 . . . First Database
35 . . . Subclass Decision Program
36 . . . Second Database
37 . . . Spectrum Acquisition Part
38 . . . m/z Reading Part
39 . . . Subclass Determination Part
40 . . . Cluster Analysis Part
41 . . . Dendrogram Generation Part

SEQUENCE LISTING

The patent contains a lengthy sequence listing. A copy of the sequence listing is available in electronic form from the USPTO web site (https://seqdata.uspto.gov/?pageRequest=docDetail&DocID=US12416641B2). An electronic copy of the sequence listing will also be available from the USPTO upon request and payment of the fee set forth in 37 CFR 1.19(b)(3).

The invention claimed is:

1. A method of identifying a serovar of *Salmonella* bacteria comprising

a) a step of subjecting a sample containing microorganisms to mass spectrometry to obtain a mass spectrum,

b) a step of reading a mass-to-charge ratio m/z of a peak derived from a marker protein from the mass spectrum, and

c) an identification step of identifying a serovar of *Salmonella* bacteria in the sample, based on the mass-to-charge ratio m/z,

wherein the serovars of *Salmonella* bacteria are classified using cluster analysis using as an index the mass-to-charge ratio m/z derived from at least 12 types of ribosomal proteins S8, L15, L17, L21, L25, S7, SODa, peptidylprolyl isomerase, gns, YibT, YaiA and YciF as the marker proteins.

2. The method according to claim **1**, further comprising a step of generating a dendrogram representing an identification result by the cluster analysis.

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